MAgent Theory 2.0

Gauge-Covariant Agents
and Inherited Organization

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October 2026

Abstract

An agent holds beliefs and generative models, receives records, makes predictions, and changes its laws. We formulate these objects on associated statistical bundles, with independent directed interactions that may depend on the current belief and model laws. On regular statistical families, the joint-law Fisher metric and a connection induce a covariant pullback tensor on the contextual base. A collective least-squares construction selects a connection class from declared comparison data. Exact identities distinguish information lost under a channel from changes caused by incompatible connections, and characterize when a parent retention preserves the collective connection.

Interacting agents admit three objectives, which we keep separate. In a global free energy all agents are parts of one normalized joint, and local updates are coordinate descent of one functional. In per-agent free energies each agent's generative model treats its peers' communicated laws as competing sources of prior information. A reciprocal law potential generates coupled Fisher dynamics in which sources also move toward their receivers; we prove that it is not the free energy of any fixed normalized model whose recognition laws are the agent laws. Conditional-information identities describe communication, common recovery, and the cost of discarding record history. Parent dynamics and boundary responses must descend through the retained configuration.

A formation standard requires selection, inheritance, compression, interaction necessity checked against a null flow, and a second stage that depends on the first. Under the reciprocal potential, a profile construction preserves coupled identity-channel dynamics and a declared response interface through arbitrary finite binary depth on a supplied family; it is an equivariant symmetry reduction. A four-agent Wishart construction selects two parents and then their union with exact original-time inheritance, but its membership schedule is the centroid dendrogram of the prepared initial shapes and is the same at zero coupling. A six-agent construction admits a full open set of positive bit-circle laws, with preferential pairs, a later parent join, inherited moving-representative dynamics, adaptive response histories, and nonflat rank-two contextual geometry on a declared finite window. Its orientation loss is relative to the supplied interface, and in the proof regime the null flow reproduces its membership history. A nonsmooth extension gives finite-time fusion whose grouping requires the interaction, but fused parents compress nothing.

For the normalized per-agent free energies, a separate construction proves persistent interaction-dependent grouping against conflicting records. An angular extension admits a physically noninjective parent preserving the original-time port dynamics, every admitted finite adaptive response-record history, and the actual collective connection and contextual tensor on a nonflat finite-window family. Its phase-insensitive interface is available before grouping; no new second compressive handoff is established. These results are single-stage or two-stage selections with exact inheritance at their stated scopes. A construction meeting every clause of the formation standard remains open.

A collective phase quotient preserves fixed relative probes and admits exact local two-jet evolution with finite reference memory. A finite preparation generator gives another conditional realization. A causal local phase accumulator obeys the original-time native dynamics and preserves every

admitted history; actual geometric inheritance additionally requires initial collective-phase coherence across context. Scoped obstructions exclude rotation-invariant local value-path memory and smooth bounded-jet causal memories on the rich open preparation class. The unrestricted causal-local-parent question remains inconclusive; neither interaction-created invisibility nor a second compressive handoff is proved.

A separate contextual extension couples neighboring laws through an auxiliary alignment energy. On finite graphs, a regular minimizing branch gives an explicit Fisher force and local uniqueness. On a supplied flat torus, a free proper isometric action reduces the contextual cost to a quotient Dirichlet energy. The resulting shape PDE and frame and source-row ODEs have a local Sobolev solution with exact total-energy dissipation. These results retain fixed-reference forces that need not descend to the quotient. They establish no global regularity, singular-orbit passage, independent row or link diffusion, or graph-refinement limit.

A separate geometric construction derives base transport from regular contextual law, interface, and directed-link data. A spanning message matrix gives displacement identity, affine connection covariance, and an explicit curvature formula; local alignment also gives a curved example built only from reciprocal rotation links. A single marked square frame instead gives flat local transport. Comparison metrics remain constitutive, continuous invisible gauge sectors obstruct a link-only selector, and no new evolution or inheritance result is asserted.

Selective communication introduces a further distinction between the transmitted code and the invertible dictionary used to interpret it. Conditional-information identities quantify reconstruction loss for general laws. Grounded messages identify dictionaries modulo code stabilizers, while an anchored loop prior couples their unread directions. A Gaussian tangent model gives exact recognition formulas and separates global, receiver-owned, and reciprocal link dynamics. Integrating out Gaussian contents produces an affine cost on coded reinterpretations. Its relation to directed holonomy requires explicit cycle hypotheses. Supplied rigid clusters admit a restricted parent description under normalization and connectivity conditions; this reduction supplies no new formation theorem.

Reading the construction

The mathematical objects are introduced in the order in which they are needed: agent laws, generative models and records, interactions, statistical and contextual geometry, free energies and dynamics, predictive compression, inherited organization, a contextual-coupling extension, and base transport derived from contextual agent data. The final three technical chapters develop selective communication, latent dictionary learning, and curvature costs in supplied hierarchies. The main results include their decisive proofs. The appendices give the internal history construction, conditional prediction and generative memory, directional links and base connections, an optional determinant-metric and circle example, and notation. Secondary calculations are collected in a separate supplement.

Definitions specify the model's objects; constitutive choices specify how those objects interact. A theorem states a consequence of explicit hypotheses. An open problem identifies a further construction that is not supplied by those hypotheses. General probability measures are used whenever measurability suffices. Finite alphabets, smooth statistical families, and vector fibers

enter only where an example or a differential calculation requires them. The general definitions assume no Gaussian closure; the fusion example states its lognormal restriction explicitly.

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Chapter 1

Agents, Possibilities, and Records

The program behind this theory asks whether interacting agents can produce further agents. This text supplies the objects needed to pose that question exactly, the conditions under which a parent description inherits the dynamics of its children, and constructions measured against a formation standard (definition 6.31); a construction meeting every clause of that standard remains open (open problem 6.32). The first distinction is between an agent's probability law and the possibilities represented by that law. The second is between comparing two descriptions and specifying an interaction that changes them. A parent construction must preserve enough of both to continue the dynamics.

1.1 Statistical sections

Let $\mathcal{C}$ be a second-countable smooth base of contexts and let $P \to \mathcal{C}$ be a principal G-bundle, with G a second-countable Lie group. A context labels where a description applies; it is not identified with physical position or time. For each agent i let K_i and M_i be standard Borel spaces with jointly Borel actions $G \times \mathsf{K}_i \to \mathsf{K}_i$ and $G \times \mathsf{M}_i \to \mathsf{M}_i$. Their elements k and m represent state contents and model presentations. The actions may be different representations of the same group. They act on probability measures by pushforward. When no agent index is needed we write K, M.

Throughout, one structure group acts on both sectors. A gauge transformation of an agent is a single element of G, acting on state contents and on model presentations through the two actions; there is no separate belief group and model group. The represented contents form two associated bundles over the same principal bundle,

$$\mathcal{E}_q = P \times_G \mathsf{K}, \qquad \mathcal{E}_s = P \times_G \mathsf{M}, \tag{1.1}$$

which in a linear realization are vector bundles $P \times_{\varrho_q} V_q$ and $P \times_{\varrho_s} V_s$ for representations $\varrho_q : G \to \mathrm{GL}(V_q)$ and $\varrho_s : G \to \mathrm{GL}(V_s)$, of ranks $K = \dim V_q$ and $\tilde{K} = \dim V_s$. The statistical bundle defined next carries laws over the fiber product of $\mathcal{E}_q$ and $\mathcal{E}_s$, on which G acts diagonally.

Definition 1.1 (Statistical agent section). Set $\mathsf{Y}_i = \mathsf{K}_i \times \mathsf{M}_i$, with the product action, and give $\mathcal{P}(\mathsf{Y}_i)$ the sigma-algebra generated by evaluation on measurable sets. The associated statistical bundle of agent i is

$$\mathcal{B}_i = P \times_G \mathcal{P}(\mathsf{Y}_i), \qquad (ug, \zeta) \sim (u, g_{\#}\zeta). \tag{1.2}$$

The statistical component of agent i is a Borel section ζ_i of $\mathcal{B}_i$ over an open domain $\mathcal{U}_i \subseteq \mathcal{C}$. Smooth sections are defined only into admitted smooth subbundles. The full state of the agent also includes the internal variables and access mechanisms read by its update rule.

The associated-bundle construction is that of Kobayashi and Nomizu [87, Section I.5]. Its measurable form needs the joint measurability of the action. If each g acts bimeasurably but $(g, y) \mapsto gy$ is not jointly Borel, for example a translation of $\mathbb{R}$ by a discontinuous additive function of $g \in \mathbb{R}$, the Borel class of a section can depend on the trivialization in which it is written.

Lemma 1.2 (Measurable structure of the statistical bundle). Under the stated hypotheses, $(g, \zeta) \mapsto g_{\#}\zeta$ is Borel on $G \times \mathcal{P}(\mathsf{Y}_i)$, and $\mathcal{B}_i$ carries a standard Borel structure, independent of the atlas, for which every local trivialization is a Borel isomorphism onto its image. Transition maps preserve fiberwise mixtures, null sets, mutual absolute continuity, and relative entropy.

Proof. For a Borel set A, $(g_{\#}\zeta)(A) = \int \mathbf{1}_A(gy)\zeta(dy)$. The integrand is Borel in (g, y) by joint measurability, and the integral of a bounded jointly Borel function against ζ is Borel in (g, ζ): this holds for products $f(g)h(y)$ and extends by the monotone class theorem. Since $\mathcal{C}$ is second countable, countably many trivializing patches cover it, and their transition functions are Borel. The gluing maps $(c, \zeta) \mapsto (c, g_{\alpha\beta}(c)_{\#}\zeta)$ are therefore Borel bijections with Borel inverses. The space $\mathcal{P}(\mathsf{Y}_i)$ is standard Borel [84], so $\mathcal{B}_i$ is a countable Borel union of standard Borel pieces glued by Borel isomorphisms; a second atlas yields the same structure because its transitions to the first are Borel by the same argument. Pushforward is affine and preserves null sets and absolute continuity, and proposition 1.5 gives invariance of relative entropy. □

At a fixed context, disintegration gives

$$\zeta_i(dk, dm) = q_i(dk \mid m)s_i(dm), \qquad q_i(dk) = \int q_i(dk \mid m)s_i(dm). \tag{1.3}$$

Here q_i is the state marginal and s_i is the law over model presentations. Regular conditional laws exist on the stated spaces, with versions defined up to the appropriate marginal null sets [82]. Disintegration commutes with the product action: the conditional law of $g_{\#}\zeta_i$ given m' is $g_{\#}q_i(\cdot \mid g^{-1}m')$ for $g_{\#}s_i$-almost every m', so eq. (1.3) descends to bundle points. A model presentation acquires predictive content through a normalized kernel; its name or coordinate value alone supplies no prediction (section 1.4).

The product restriction $\zeta_i = q_i \otimes s_i$ removes dependence between the two sectors within an agent. A product over agents makes a further restriction on represented contents. Neither restriction prevents the update of one law from depending on other laws. Coupling is a property of the dynamics; factorization is a property of a chosen probability description. The coupled population flows developed below impose the product restriction. The general joint law enters through its Fisher metric in chapter 3 and through the local conditional-prediction construction of section 5.2 and appendix B. The restriction is not neutral, as chapter 4 shows: it changes the local optimum of a belief, and the joint-law flow does not preserve it.

Smoothness is an additional hypothesis. When derivatives of laws are required, use a declared regular statistical family stable under the relevant coordinate changes. The whole space $\mathcal{P}(\mathsf{Y}_i)$ is not assigned a smooth manifold structure by notation. Chapter 3 equips an admitted family with the Fisher metric of the full joint law and pulls that metric back to contextual directions through a covariant derivative. The product restriction has a simpler metric, but marginal metrics alone do not measure changes of dependence in a general joint family. A collection of pointwise marginals also

does not determine a probability law on fields over $\mathcal{C}$; section 1.5 supplies such a law as a separate object.

The population at a context consists of the agents whose domains contain that context. Most of the dynamics below is formulated at a common c_*. This retains the ambient bundle and local sections without requiring parallel transport between distinct base points. Independent interactions between agents at c_* are defined in chapter 2. Contextual statistical geometry is developed in chapter 3; appendix C records the transformation law of connection data in a moving presentation. Chapter 17 constructs such data from regular contextual laws, interfaces, and links with declared comparison rules; selection of those rules and continuation through singular regions remain open. The potentials of chapter 4 contain no coupling between contexts. The formation constructions retain these context-local flows and compute connections as diagnostics. The separate extension in chapter 16 adds an aligned contextual energy and derives its neighboring-context Fisher force on declared regular families.

Parents in later chapters are maps between statistical descriptions of different agents. The following definition supplies the corresponding morphism of bundles.

Definition 1.3 (Morphism of statistical bundles). Let $\tilde{f} : P \to \bar{P}$ be a principal bundle map covering $f : \mathcal{C} \to \bar{\mathcal{C}}$ with homomorphism $\phi : G \to \bar{G}$, so that $\tilde{f}(ug) = \tilde{f}(u)\phi(g)$. Let $\kappa : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be a Markov kernel with $\kappa(gy, \cdot) = \phi(g)_{\#}\kappa(y, \cdot)$. The induced morphism is

$$\Psi_{\tilde{f},\kappa}[u, \zeta] = [\tilde{f}(u), \zeta\kappa]. \tag{1.4}$$

Proposition 1.4 (Statistical bundle morphisms). The map eq. (1.4) is well defined, fiberwise affine, and contracts relative entropy within each fiber. In a linear realization, let $A : V_q \to \bar{V}_q$ and $\tilde{A} : V_s \to \bar{V}_s$ be linear maps with $A\varrho_q(g) = \bar{\varrho}_q(\phi(g))A$ and $\tilde{A}\varrho_s(g) = \bar{\varrho}_s(\phi(g))\tilde{A}$ for all $g \in G$. Then the same principal map induces well-defined morphisms of the content bundles,

$$\Phi[u, k] = [\tilde{f}(u), Ak], \qquad \tilde{\Phi}[u, m] = [\tilde{f}(u), \tilde{A}m], \tag{1.5}$$

and the deterministic kernel $\kappa((k, m), \cdot) = \delta_{(Ak,\tilde{A}m)}$ induces the corresponding statistical morphism.

Proof. Equivariance of κ gives $(g_{\#}\zeta)\kappa = \phi(g)_{\#}(\zeta\kappa)$. Hence $[u, g_{\#}\zeta] \mapsto [\tilde{f}(u), \phi(g)_{\#}(\zeta\kappa)] = [\tilde{f}(u)\phi(g), \zeta\kappa]$, which is the image of the equivalent representative $[ug, \zeta]$. The map $\zeta \mapsto \zeta\kappa$ is affine, and $D_{\mathrm{KL}}(\zeta\kappa\|\zeta'\kappa) \le D_{\mathrm{KL}}(\zeta\|\zeta')$ is the data-processing inequality [35]. For the content morphisms, the representative $[ug, \varrho_q(g)^{-1}k]$ is sent to $[\tilde{f}(u)\phi(g), A\varrho_q(g)^{-1}k] = [\tilde{f}(u)\phi(g), \bar{\varrho}_q(\phi(g))^{-1}Ak] = [\tilde{f}(u), Ak]$, and likewise for $\tilde{\Phi}$. The deterministic kernel is equivariant by the two intertwining relations. □

The belief and model morphisms share the principal map $\tilde{f}$. A pair of sector morphisms covering different principal maps would act with independent groups in the two sectors, and is not admitted.

A parent is an agent of a declared type when its retained state is again a section of such a bundle, carries an admitted family, a register with a covariant readout, and an evaluator, and interacts with exterior agents through channels of definition 2.1. Proving that a retention is closed in this sense is part of the formation problem of chapter 6, not a consequence of the definition.

1.2 Presentation and probability

Fix c_* and a reference point $\sigma \in P_{c_*}$. A frame of agent i is a point $u_i \in P_{c_*}$, and $u_i = \sigma U_i^{-1}$ defines $U_i \in G$. The point u_i identifies the fiber of $\mathcal{B}_i$ at c_* with $\mathcal{P}(\mathsf{Y}_i)$, and the law displayed by agent i is the coordinate expression of its bundle point in that frame. In a linear realization $\mathsf{K}_i = V_q$ and $\mathsf{M}_i = V_s$ carry linear representations ϱ_q, ϱ_s of G, the sector frames are $U_i^q = \varrho_q(U_i)$ and $U_i^s = \varrho_s(U_i)$, and E_i^q, E_i^s denote agent i's coordinate copies of V_q, V_s. Laws written on E_i^a, as in chapter 2, are the displayed laws. The reference laws are

$$\bar{q}_i = ((U_i^q)^{-1})_{\#} q_i, \qquad \bar{s}_i = ((U_i^s)^{-1})_{\#} s_i. \tag{1.6}$$

They are the coordinate expressions in the reference frame σ, since $[\sigma U_i^{-1}, \zeta] = [\sigma, (U_i^{-1})_{\#}\zeta]$. A passive change sends u_i to $u_i g_i^{-1}$, hence q_i to $(g_i^q)_{\#} q_i$ and U_i^q to $g_i^q U_i^q$, with the corresponding model-sector rule. The reference laws are unchanged. This is a change of description, not an inference step. One frame induces both sector frames, and a passive change is one element g_i with $g_i^q = \varrho_q(g_i)$ and $g_i^s = \varrho_s(g_i)$. Independent sector changes would require the product group $G_q \times G_s$; the theory does not use them.

Because every frame-dependent object can be written through its reference law, covariance under independent passive changes at each agent holds for any rule expressed in reference laws and constrains nothing by itself. The frames act here as a dressing field in the sense of Fournel et al. [58]. A substantive symmetry requirement is a statement about the reference rules themselves, such as equivariance of an interaction under a common change of reference at one context. Chapter 2 states it as eq. (2.13), and a result uses it only where it says so.

An active transformation changes the reference laws at fixed frames, for example $\bar{q}_i \mapsto a_{\#}\bar{q}_i$ for a bimeasurable bijection a of K_i. Whether two configurations related in this way are distinguishable depends on a declared observable algebra: the rows, channels, and record kernels of the agents, together with any declared external response protocol. A distinction is called gauge relative to such an algebra when no element of the algebra separates the configurations. When only an external protocol separates them, the text names that protocol rather than calling the distinction physical without qualification.

For probability measures μ, ν on one measurable space, relative entropy is

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log \frac{d\mu}{d\nu} d\mu, \tag{1.7}$$

when $\mu \ll \nu$, and is $+\infty$ otherwise. In this formula and below, a displayed integral is understood with its declared measure; no common Lebesgue density is required. Relative entropy compares distinguishability of laws [89].

Proposition 1.5 (Coordinate invariance). For an injective Borel map f between standard Borel spaces, in particular a bimeasurable bijection, including divergences with value $+\infty$,

$$D_{\mathrm{KL}}(f_{\#}\mu\|f_{\#}\nu) = D_{\mathrm{KL}}(\mu\|\nu). \tag{1.8}$$

Proof. By the Lusin–Souslin theorem [84, Theorem 15.1] the image of f is Borel and f is a Borel isomorphism onto it, so it suffices to treat a bimeasurable bijection onto a Borel set carrying both pushforwards. Absolute continuity is preserved in both directions. If $h = d\mu/d\nu$, the transformed density is $h \circ f^{-1}$. Integrating its logarithm against $f_{\#}\mu$ gives the original integral. If absolute continuity fails, a set of zero ν-measure and positive μ-measure is carried to a set with the same property. □

Definition 1.6 (Normalized channel). A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ to each x, measurably in x. Its action on a law is

$$(\mu K)(A) = \int K(x, A)\mu(dx). \tag{1.9}$$

A deterministic measurable map $f : X \to Y$ is represented by $K_f(x, \cdot) = \delta_{f(x)}$. Then $\mu K_f = f_{\#}\mu$.

An invertible frame change preserves all distinctions. A channel that discards information or adds noise need not. Both are well-defined operations on general probability laws, but their roles in an agent interaction differ.

1.3 Internal records

At a fixed context, let S_i be the admitted internal-state space of agent i, and let S be the complete configuration space. It contains the statistical laws, the source rows defined in chapter 4, interfaces, and memory variables that the dynamics reads. Write $\mathrm{pr}_i : \mathsf{S} \to \mathsf{S}_i$ for the local state projection. For the measurable constructions, take these spaces to be standard Borel. A finite collection of probability-law spaces and finite or Euclidean memory spaces supplies such a realization. A flow, retention, or response used in a measurable construction must be Borel for these structures. Smoothness in a Banach chart of log densities does not by itself ensure this, because the norm-Borel sigma-algebra of $L^\infty(\lambda)$ for nonatomic λ is strictly finer than the trace of the evaluation sigma-algebra; where such flows are used, their measurability is a stated hypothesis.

Definition 1.7 (Record and readout). A record readout is a measurable map

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i, \tag{1.10}$$

where O_i is a standard Borel alphabet of internally distinguishable outcomes. If an interaction with source j produces configuration Z_{t+1}, its record variable is

$$O_{ij,t} = \rho_i(\mathrm{pr}_i\, Z_{t+1}). \tag{1.11}$$

For stochastic dynamics this is a measurable function from the probability space of realizations to O_i. A realized value is denoted $o_{ij,t}$. The source label is part of the record only when the internal state retains it.

A record can be transient. Persistence means that the distinction remains accessible through a specified later readout. A list kept outside the agent does not supply this accessibility. A memory

coordinate is therefore an explicit part of the model unless the required distinction is already represented in its state.

Proposition 1.8 (Induced record law). Let $\Gamma_{ij} : \mathsf{S} \rightsquigarrow \mathsf{S}$ be a normalized interaction transition. Then

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\left(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\right), \tag{1.12}$$

is a normalized record kernel. If $\Gamma_{ij}(x, \cdot) = \delta_{F_{ij}(x)}$, the record is the deterministic function $\rho_i \circ \mathrm{pr}_i \circ F_{ij}$.

Proof. The right side is the pushforward of a probability measure through the measurable map $\rho_i \circ \mathrm{pr}_i$. Its total mass is one, and measurability in x follows from the defining property of a kernel. The deterministic case follows by evaluating a Dirac measure. □

A readout that reads frame-dependent coordinates can change under a mere change of description. For example, a readout of the first coordinate of the displayed mean of q_i returns 1 for the law $\mathcal{N}(e_1, I)$ in one frame and 0 after a passive rotation by $\pi/2$, although the configuration is the same. Records therefore carry a covariance requirement. A passive change γ acts on S by a Borel automorphism, on laws by pushforward, on frames by left multiplication, and on interface data by their declared rule, with $\mathrm{pr}_i \circ \gamma = \gamma_i \circ \mathrm{pr}_i$. A readout is invariant if $\rho_i \circ \gamma_i = \rho_i$, and equivariant if $\rho_i \circ \gamma_i = \tau_i(\gamma) \circ \rho_i$ for a declared measurable action τ_i on O_i. A transition is covariant if $\Gamma_{ij}(\gamma x, \gamma B) = \Gamma_{ij}(x, B)$ for all measurable B. Every readout, record kernel, and response kernel of this text is required to be invariant or declared equivariant, and every interaction transition covariant.

Proposition 1.9 (Covariant record laws). If Γ_{ij} is covariant and ρ_i is equivariant with action τ_i, then $\mathcal{R}_{ij}(\gamma x, A) = \mathcal{R}_{ij}(x, \tau_i(\gamma)^{-1} A)$. For an invariant readout, $\mathcal{R}_{ij}(\gamma x, A) = \mathcal{R}_{ij}(x, A)$.

Proof. Write $B_A = \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}$. Covariance gives $\mathcal{R}_{ij}(\gamma x, A) = \Gamma_{ij}(x, \gamma^{-1} B_A)$, and $\gamma^{-1} B_A = \{x' : \tau_i(\gamma)\rho_i(\mathrm{pr}_i\, x') \in A\} = B_{\tau_i(\gamma)^{-1} A}$. □

This definition introduces no observing agent outside the population. It specifies where an outcome is available and how it can enter later dynamics. It does not derive the transition Γ_{ij} or decide whether its stochasticity represents objective chance. Those are further modeling choices. A response protocol queried from outside the population, as in chapter 13, is a separate declared device and is named wherever it is the only element of the observable algebra that resolves a distinction.

For example, a vector comparison can be represented by

$$r_{ij} : E_i^q \times E_j^q \times \mathbb{R}^r \to \mathbb{R}^r, \qquad r_{ij}(k_i, k_j, \epsilon) = A_i k_i - B_{ij} k_j + \epsilon. \tag{1.13}$$

Under passive changes $k_i \mapsto g_i k_i$ and $k_j \mapsto g_j k_j$ the comparison matrices transform as $A_i \mapsto A_i g_i^{-1}$ and $B_{ij} \mapsto B_{ij} g_j^{-1}$, so the comparison value is invariant. A noise law turns this map into a record kernel on represented contents. A joint content law determines its predicted outcomes. A transition on agent states must still specify how an outcome is registered and how the agent changes. A predictive likelihood and an internal update thus have different domains.

1.4 Beliefs and generative models

An agent holds beliefs and generative models. The joint law ζ_i describes the first; the second is the mechanism that gives a model presentation predictive content.

Definition 1.10 (Generative model of an agent). At a fixed context, a generative model of agent i consists of a proper reference law r_i on M_i, a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$, and a normalized record kernel $\ell_i : \mathsf{K}_i \times \mathsf{M}_i \rightsquigarrow \mathsf{O}_i$. They define the normalized joint

$$P_i(dm, dk, do) = r_i(dm)G_i(m, dk)\ell_i(k, m; do). \tag{1.14}$$

Each presentation m indexes the generative model $G_i(m, dk)\ell_i(k, m; do)$: a prior over represented contents together with a likelihood for records. The reference r_i is the prior over presentations. When ℓ_i does not depend on m, it is the record kernel H_i of chapter 5.

Given a record value o at which a declared density of ℓ_i yields $0 < P_i(o) < \infty$, the recognition law ζ_i is a candidate for the posterior $P_i(dm, dk \mid o)$, and

$$\mathcal{F}_i(\zeta_i; o) = D_{\mathrm{KL}}\left(\zeta_i \| P_i(\cdot \mid o)\right) - \log P_i(o) \tag{1.15}$$

is its variational free energy [18, 59]. In this reading q_i is the agent's belief about represented contents, its perception, and s_i is its belief over generative models. The model law alone predicts nothing: by eq. (1.14), prediction requires the evaluator and the record kernel. An evaluator that does not depend on m makes the model sector predictively inert. Without records, the unique minimizer of $D_{\mathrm{KL}}(\zeta_i \| r_i G_i)$ is the reference joint itself, whose belief conditional $q_i(dk \mid m)$ equals the evaluator $G_i(m, dk)$; the conditional in eq. (1.3) is the agent's current departure from that prior.

Definition 1.11 (Shared generative model). Agents i and j share a generative model up to frame if $\mathsf{M}_i = \mathsf{M}_j$, $\mathsf{K}_i = \mathsf{K}_j$, $\mathsf{O}_i = \mathsf{O}_j$, and there are bimeasurable bijections ϕ of M_i and ψ of K_i with

$$r_j = \phi_{\#} r_i, \qquad G_j(\phi(m), \cdot) = \psi_{\#} G_i(m, \cdot), \qquad \ell_j(\psi(k), \phi(m); \cdot) = \ell_i(k, m; \cdot). \tag{1.16}$$

A population whose members pairwise share a generative model is called a species.

Sharing a generative model is a statement about mechanisms. Agreement of beliefs over models, $D_{\mathrm{KL}}(s_j \| \phi_{\#} s_i) = 0$, is a statement about states. The two are independent. Agents with a shared model can hold different beliefs, and equal beliefs over presentations predict equal records only when the evaluators and record kernels agree as in eq. (1.16) and the agents condition on the same information. A record registered by one agent changes its posterior, and therefore its predictions, without changing those of an agent that did not register it.

The free energy eq. (1.15) is a single-agent object. Interactions add a further choice. In one choice all agents are parts of one normalized joint over their contents and records. In another each agent's generative model treats the communicated laws of its peers as data. Chapter 4 develops both choices and a third, a direct interaction potential between laws, and proves that the third is not the free energy of any fixed normalized model in which the agent laws are recognition laws.

1.5 Content laws on fields

The global choice needs a joint law over the contents of all agents and, where contexts are compared, over contents at different contexts. Write $E_i = P \times_G \mathsf{Y}_i$ for the associated sample bundle. A point $[u, \zeta]$ of $\mathcal{B}_i$ at c is identified with the law $(y \mapsto [u, y])_{\#}\zeta$ on the fiber of E_i at c; this identification does not depend on the representative.

Definition 1.12 (Content law on fields). A content model consists of a standard Borel coefficient space Z, a proper prior P_0 on Z, and jointly measurable sample-section maps $f_i : \mathsf{Z} \times \mathcal{U}_i \to E_i$ with $f_i(z, c)$ in the fiber over c. For a recognition law Q on Z, the agent sections and finite-design laws are

$$\zeta_i^Q(c) = (f_{i,c})_{\#}Q, \qquad Q_D = (F_D)_{\#}Q, \qquad F_D(z) = \big(f_i(z, c)\big)_{(i,c)\in D}, \tag{1.17}$$

where $f_{i,c} = f_i(\cdot, c)$ and D is a finite set of pairs with $c \in \mathcal{U}_i$. A record mechanism is a normalized kernel $K_a(do_a \mid z)$ that may read field evaluations, model presentations, communication maps, and parallel transports along declared paths. A record kernel must be specified through bundle points, equivalently it must transform with every fiber datum it reads. With conditionally independent records, $\mathbb{P}(dz, do) = P_0(dz) \prod_a K_a(do_a \mid z)$.

Proposition 1.13 (Consistency and projected free energy). For $D \subseteq D'$ let $r_{D'D}$ delete the added coordinates. Then $(r_{D'D})_{\#}Q_{D'} = Q_D$, and the marginal of Q_D at (i, c) is $\zeta_i^Q(c)$. For a record value with likelihood ℓ_o and posterior Π_o as in eq. (4.18), and whenever the free energy is finite,

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q_D \| (F_D)_{\#}\Pi_o) - \log Z_o + \int D_{\mathrm{KL}}\big(Q(dz \mid F_D = w) \| \Pi_o(dz \mid F_D = w)\big) Q_D(dw). \tag{1.18}$$

Proof. Since $F_D = r_{D'D} \circ F_{D'}$, pushforwards compose, which gives consistency, and the marginal statement is the case of one pair. The second identity is the chain rule for relative entropy applied to Q and Π_o disintegrated along the measurable map F_D, which is the deterministic-kernel case of theorem 5.1, combined with $\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o$. □

The last term of eq. (1.18) is the exact cost of the information that a finite description omits. It need not vanish even when D contains every evaluation the records read, because recognition can have the wrong conditional law in unobserved directions. The law-valued configuration of the agents at the contexts in D is the list of marginals of Q_D; it does not determine Q_D. The reciprocal potential of chapter 4 is a separate constitutive object defined on such lists, and proposition 4.21 shows that it is not $\mathcal{F}(Q; o)$ for any fixed normalized model with these marginals as recognition laws.

1.6 What correlations describe

Suppose two represented bits have the independent uniform law. Register the event that they agree. Their conditional joint law assigns probability $1/2$ to each of $(0, 0)$ and $(1, 1)$. Each marginal remains uniform, while the conditional expectation of their product changes from $1/4$ to $1/2$. An exact joint posterior has acquired dependence that its two marginals cannot report.

This example identifies a concrete need for a joint law: predicting an observable that reads both bits after the shared record. It does not imply that every admissible agent dynamics must propagate that posterior. A coupled product-family dynamics is another precise model, provided it is declared, but it does not preserve the marginal symmetry. Against the hard agreement record, a product recognition law has finite free energy only when it is a point mass on $(0,0)$ or on $(1,1)$, at relative entropy $\log 2$ from the posterior. For a soft record $\Pi \propto e^{J\sigma_1\sigma_2}$ in spin variables, the symmetric product optimum has magnetization solving $m = \tanh(Jm)$, which is zero for $J \leq 1$ and nonzero for $J > 1$. Similarly, two routes that inspect the same source must share its identity; replacing it by independent redraws specifies a different experiment.

The same distinction governs what a comparison of laws can detect along a loop of routes.

Proposition 1.14 (Law comparisons and shared samples). Let H, H' be bimeasurable bijections of a standard Borel space E, for example two loop transports, and let $\nu \in \mathcal{P}(E)$.

(i) If $H_\#\nu = H'_\#\nu$, every functional of the pair of laws $(\nu, H_\#\nu)$ takes the same value for H and H'. In particular $D_{\mathrm{KL}}(\nu \| H_\#\nu) = 0$ exactly when H preserves ν.

(ii) The law $(\mathrm{id}, H)_\#\nu$ of the shared-sample pair (Y, HY), $Y \sim \nu$, determines H up to ν-null sets.

Proof. Part (i) is immediate because the pair of laws is the same. For (ii), suppose $(\mathrm{id}, H)_\#\nu = (\mathrm{id}, H')_\#\nu$. The complement of the graph of H' is a Borel set B. Its measure under the first law is $\nu\{H \neq H'\}$ and under the second law is zero. □

Every interaction cost of chapters 2 and 4 compares laws, so the dynamics defined there cannot respond to a loop transport that lies in the stabilizer of the transported law. A rotation by a fixed angle, for instance, leaves an isotropic Gaussian law unchanged. A record that is sensitive to such a transport must read a coupling of the same sample along the routes, as the route mark of the supplement does; under a frame change g at the base point the pair transforms diagonally, $(Y, HY) \mapsto (gY, gHg^{-1}gY)$, so invariant records are diagonal-invariant functions of the pair. Such records require a content law in the sense of definition 1.12.

The configuration x of law-valued agents, a sample tuple $(k_i, m_i)_i$ used in a probabilistic model, and an ensemble of possible configurations are three different objects. A deterministic dynamics on x needs no probability law over configurations. A preparation ensemble can be introduced to assess predictions or compression, but it is then stated explicitly. These distinctions will matter when a parent retains a law over contents while discarding information needed to evolve that law.

Chapter 2

Law-Dependent Interactions

An interaction compares or transmits information between agents. Its law may depend on the agents' current statistical states. The general object is a normalized Markov channel between represented spaces; exponential links of matrix generators are one constitutive example of such a channel, not the definition of an interaction. Gauge covariance specifies how this dependence is represented when coordinates change; it does not require the interaction to be a frame identification or select a unique mechanism. We first construct these interactions at one context. Transport between different contexts is a separate datum, whose transformation law is recorded in appendix C.

2.1 Two sectors and their comparison spaces

Fix a context c and a finite set of agents defined there. Write

$$\nu_i^q = q_i, \qquad \nu_i^s = s_i, \tag{2.1}$$

for the belief and model marginals of agent i. The sector index a takes values in $\{q, s\}$. These laws live on standard Borel spaces E_i^a. No product assumption between the two sectors, or between different agents, is required to define an interaction between their marginals. Such a product is a separate choice when a joint recognition law is needed.

In a vector realization choose reference spaces V_q, V_s and invertible presentation maps

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{q}_i = (U_i^q)_\#^{-1} q_i, \qquad \bar{s}_i = (U_i^s)_\#^{-1} s_i. \tag{2.2}$$

We also write $\bar{\nu}_i^a = (U_i^a)_\#^{-1} \nu_i^a$ and $\bar{\boldsymbol{\nu}}^a = (\bar{\nu}_i^a)_i$ for the tuple of reference laws in sector a. Here E_i^a is agent i's displayed coordinate copy of V_a. At a common context all agents' statistical sections take values in one fiber of the bundle of eq. (1.2); the per-agent presentations are points of the frame torsor at that context, used to display each agent's law, and are not separate fibers. One structure group G acts through the representations ϱ_q, ϱ_s of eq. (1.1), so a passive change at agent i is a single element g_i, with $g_i^a = \varrho_a(g_i)$. Independent sector changes would require the product group $G_q \times G_s$, which the theory does not use; the collective connection of proposition 3.9 depends on this choice, because the group enters its normal equation. The two presentations do not identify the belief and model interaction maps. Their domains, parameters, and dependence on the laws remain separately specified.

A passive re-expression $y_i' = g_i^a y_i$ sends ν_i^a to $(g_i^a)_\# \nu_i^a$ and U_i^a to $g_i^a U_i^a$. Therefore the reference laws in eq. (2.2) are unchanged. A common reference change $U_i^a \mapsto U_i^a h_a$ instead sends every reference law to $(h_a^{-1})_\# \bar{\nu}_i^a$. The first operation changes an agent's displayed coordinates; the second changes the common coordinates in which interaction parameters are written.

Definition 2.1 (Normalized interaction channel). Let S be the standard Borel configuration space of chapter 1. A sector channel from source j to receiver i is a Markov kernel

$$\mathcal{K}^a_{ij} : \mathsf{S} \times E^a_j \rightsquigarrow E^a_i, \tag{2.3}$$

that is, $(x, y) \mapsto \mathcal{K}^a_{ij}[x](y, A)$ is jointly measurable for every measurable A and is a probability measure in A. At configuration x it sends a law to $(\nu^a_j \mathcal{K}^a_{ij}[x])(A) = \int \mathcal{K}^a_{ij}[x](y, A)\nu^a_j(dy)$. When the channel does not depend on the configuration we omit x. Covariance requires

$$\mathcal{K}^{a\prime}_{ij}(g^a_j y, A) = \mathcal{K}^a_{ij}(y, (g^a_i)^{-1}A). \tag{2.4}$$

Composition of sending, transmission, and receiving kernels produces another normalized channel by iterated integration. Invertibility is not required. Alternatively, two channels can send the receiver and source laws into a common message space, where their communicated predictions are compared. The primitive self channel $\mathcal{K}^a_{ii}$ is constitutive data. The identity channel is one choice; later constructions may declare a nonidentity self channel, and they state which.

Joint measurability makes $x \mapsto \nu^a_j \mathcal{K}^a_{ij}[x]$ a measurable map into $\mathcal{P}(E^a_i)$ by a monotone-class argument, so the divergences, rows, and potentials built from these channels in chapter 4 are measurable functions of the configuration.

Proposition 2.2 (Covariant law comparison). Under eq. (2.4), the divergence

$$D^a_{ij} = D_{\mathrm{KL}}(\nu^a_i \| \nu^a_j \mathcal{K}^a_{ij}), \tag{2.5}$$

is unchanged by passive bimeasurable coordinate changes, including when its value is infinite.

Proof. Changing the integration variable in the kernel gives $\nu^{a\prime}_j \mathcal{K}^{a\prime}_{ij} = (g^a_i)_{\#}(\nu^a_j \mathcal{K}^a_{ij})$. For any bimeasurable bijection g, absolute continuity is preserved and the transformed Radon–Nikodym derivative is $(d\mu/d\eta) \circ g^{-1}$. Its logarithmic integral is therefore unchanged. Failure of absolute continuity is also preserved, giving infinity on both sides. □

This is the coordinate invariance of relative entropy [89]. It makes the comparison intrinsic without asserting that either law is a posterior under a fixed population model.

2.2 Reference consistency

Per-agent covariance is automatic for any rule written in reference coordinates. If a channel is specified by a reference kernel $\bar{\mathcal{K}}^a_{ij} : V_a \rightsquigarrow V_a$ and displayed as $\mathcal{K}^a_{ij}(y, A) = \bar{\mathcal{K}}^a_{ij}((U^a_j)^{-1}y, (U^a_i)^{-1}A)$, then eq. (2.4) holds identically, and proposition 1.5 with $f = (U^a_i)^{-1}$ gives

$$D_{\mathrm{KL}}(\nu^a_i \| \nu^a_j \mathcal{K}^a_{ij}) = D_{\mathrm{KL}}(\bar{\nu}^a_i \| \bar{\nu}^a_j \bar{\mathcal{K}}^a_{ij}). \tag{2.6}$$

The frames therefore act as a dressing field, as stated after eq. (1.6): per-agent covariance constrains no reference rule. The substantive symmetry requirement concerns the common reference change.

Hypothesis 2.3 (Reference consistency). A reference rule is reference consistent if, for every common reference change h_a in the declared structure group, re-expressing all reference laws and supplied interface data leaves the displayed channel unchanged. For a reference kernel this means

$$\bar{\mathcal{K}}^{a\prime}_{ij}\big[(h_a^{-1})_{\#}\bar{\boldsymbol{\nu}}^a\big](h_a^{-1}v, h_a^{-1}B) = \bar{\mathcal{K}}^a_{ij}[\bar{\boldsymbol{\nu}}^a](v, B), \tag{2.7}$$

and for the exponential generators below it is eq. (2.13). Without it, a displayed interaction depends on the choice of reference point in the frame torsor and is not a function of the bundle data; on a nontrivial principal bundle no global choice of that point exists. Every result below whose conclusion concerns bundle data, rather than one fixed reference presentation, assumes reference consistency and says so.

2.3 Exponential links: one constitutive example

Choose a message space H_a and reference interfaces

$$T^a_j : V_a \longrightarrow H_a, \qquad O^a_i : H_a \longrightarrow V_a. \tag{2.8}$$

Their direct content map is $M^a_{ij} = U^a_i O^a_i T^a_j (U^a_j)^{-1}$. A deterministic channel can push the entire law through this map even when the image is lower dimensional. A density formula for a nonsingular law cannot then be used on the original ambient space without accounting for that change of support.

Definition 2.4 (Law-dependent reference generator). Let $\mathcal{L}^a_{ij}$ be a measurable function of the tuple of reference laws $\bar{\boldsymbol{\nu}}^a$, and of any explicitly declared interface data, with values in $\mathrm{End}(V_a)$. Define

$$L^a_{ij} = \mathcal{L}^a_{ij}(\bar{\boldsymbol{\nu}}^a), \qquad \Theta^a_{ij} = U^a_i \exp(L^a_{ij})(U^a_j)^{-1}. \tag{2.9}$$

Dependence on the two endpoint laws alone is the special case $\mathcal{L}^a_{ij}(\bar{\nu}^a_i, \bar{\nu}^a_j)$. For a receiving–sending factorization, $L^a_{ij} = O^a_i(\bar{\boldsymbol{\nu}}^a)T^a_j(\bar{\boldsymbol{\nu}}^a)$, provided both maps refer to the same H_a and V_a. The associated deterministic channel is $\mathcal{K}^a_{ij}(y,\cdot) = \delta_{\Theta^a_{ij} y}$, which is an instance of definition 2.1. For exponential links the primitive self link is the identity, $L^a_{ii} = 0$.

The exponential is taken on the reference endomorphism, not on a map with different domain and codomain. Its values lie in $\exp(\mathfrak{gl}(V_a))$, which is contained in the identity component $\mathrm{GL}^+(V_a)$ and, for $\dim V_a \geq 2$, is a proper subset of it: the matrix $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ has determinant one and no real logarithm, because its single Jordan block for a negative eigenvalue occurs an odd number of times [39]. Orientation-reversing and discrete interactions are therefore not exponential links, although they are channels in the sense of definition 2.1. A smaller group acting on the frames can still re-express exponential links covariantly. Interpreting them as parallel transports of a connection for that smaller group would additionally require generators in its represented Lie algebra.

Definition 2.5 (Group-valued links). Let $\mathfrak{g}$ be the Lie algebra of G, and identify frames with elements $U_i \in G$ as in eq. (1.6). Group-valued links assign to each directed edge $j \to i$ a belief generator $\omega_{ij} \in \mathfrak{g}$ and a model generator $\tilde{\omega}_{ij} \in \mathfrak{g}$, either of which may depend on the laws as in

definition 2.4, and set

$$\theta_{ij} = U_i \exp(\omega_{ij}) U_j^{-1}, \qquad \tilde{\theta}_{ij} = U_i \exp(\tilde{\omega}_{ij}) U_j^{-1}, \qquad \Theta^q_{ij} = \varrho_q(\theta_{ij}), \qquad \Theta^s_{ij} = \varrho_s(\tilde{\theta}_{ij}). \tag{2.10}$$

The reference generators of eq. (2.9) are then $L^q_{ij} = d\varrho_q(\omega_{ij})$ and $L^s_{ij} = d\varrho_s(\tilde{\omega}_{ij})$. The links are tied when $\theta_{ij} = \tilde{\theta}_{ij}$, that is, when one G-valued link acts on both sectors through the two representations. Opposite edges remain independent: θ_{ji} is not required to equal θ_{ij}^{-1}, in either sector.

Proposition 2.6 (The sector mismatch is covariant). Under a passive change $U_i \mapsto g_i U_i$ at every agent, $\theta_{ij} \mapsto g_i \theta_{ij} g_j^{-1}$ and $\tilde{\theta}_{ij} \mapsto g_i \tilde{\theta}_{ij} g_j^{-1}$. The mismatch

$$\Xi_{ij} = \tilde{\theta}_{ij} \theta_{ij}^{-1} \in G \tag{2.11}$$

transforms by conjugation at the receiver, $\Xi_{ij} \mapsto g_i \Xi_{ij} g_i^{-1}$. Its conjugacy class is therefore independent of the frames, and the tied case $\Xi_{ij} = e$ is a frame-independent condition.

Proof. The reference generators are unchanged by a passive change, so $\theta'_{ij} = g_i U_i \exp(\omega_{ij}) U_j^{-1} g_j^{-1} = g_i \theta_{ij} g_j^{-1}$, and likewise for $\tilde{\theta}_{ij}$. Then $\Xi'_{ij} = g_i \tilde{\theta}_{ij} g_j^{-1} g_j \theta_{ij}^{-1} g_i^{-1} = g_i \Xi_{ij} g_i^{-1}$. □

The general theory keeps both link fields. Results that use one G-valued link for both sectors state the tied hypothesis $\Xi_{ij} = e$. Because Ξ_{ij} is covariant, a nonzero mismatch is a frame-independent property of the interaction: the belief and model transports along the edge disagree.

Proposition 2.7 (Endpoint covariance and reference consistency). The links in eq. (2.9) obey

$$\Theta^{a\prime}_{ij} = g_i^a \Theta^a_{ij} (g_j^a)^{-1}, \tag{2.12}$$

and their associated channels satisfy eq. (2.4). The links are also independent of a common reference rechoice if

$$\mathcal{L}^{a\prime}_{ij}\big((h_a^{-1})_{\#}\bar{\boldsymbol{\nu}}^a\big) = h_a^{-1} L^a_{ij} h_a, \tag{2.13}$$

where the prime includes the re-expression of any supplied interface data.

Proof. Passive site changes leave the reference laws and generator unchanged, so multiplication by the transformed endpoint frames proves eq. (2.12). For the channel, $\delta_{g_i \Theta g_j^{-1} g_j y}(A) = \delta_{\Theta y}(g_i^{-1} A)$. Under a reference rechoice, the exponential series gives $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$. Substitution cancels both occurrences of h_a in eq. (2.9). □

For factored generators, $O_i^a \mapsto h_a^{-1} O_i^a$ and $T_j^a \mapsto T_j^a h_a$ suffice. Message-coordinate changes $T_j^a \mapsto r_a T_j^a$, $O_i^a \mapsto O_i^a r_a^{-1}$ cancel within the product. By eq. (2.6), eq. (2.12) holds for every generator rule and constrains nothing; eq. (2.13) is the constraint, and it is the form in which hypothesis 2.3 is imposed on exponential links.

Proposition 2.8 (Low-rank generator and invertible transport). For finite matrices $O : H \to V$ and $T : V \to H$, the map OT has rank at most $\dim H$, whereas e^{OT} is invertible. Moreover,

$$e^{OT} v = v \quad (v \in \ker T), \qquad \operatorname{im}(e^{OT} - I) \subseteq \operatorname{im} O. \tag{2.14}$$

Proof. The rank statement follows from factorization. The exponential series and $e^{OT}e^{-OT} = I$ prove invertibility. Every positive power of OT vanishes on $\ker T$, and every such power has image in $\operatorname{im} O$. Summing the convergent series proves the remaining assertions. □

Thus the exponential link retains an identity action in unread directions. It is a different operation from sending only the selective content OTv. Because the link is invertible, proposition 1.5 gives $D_{\mathrm{KL}}(\nu_i\|(\Theta_{ij})_{\#}\nu_j) = D_{\mathrm{KL}}((\Theta_{ij}^{-1})_{\#}\nu_i\|\nu_j)$: an exponential link of a low-rank generator compares complete laws and discards nothing. Information-discarding, or selective, channels are noninvertible kernels of definition 2.1. The identity self-link also differs from a coarse internal loop: a parent may retain nonidentity loop information inherited from several child interactions.

2.4 An explicit constitutive family for general laws

The preceding definition does not select the functional $\mathcal{L}$. The following choice shows that nonconstant, reference-consistent generators can depend directly on laws without restricting them to a Gaussian family.

Definition 2.9 (Moment-dependent receiving and sending). For each sector let $V_a = \mathbb{R}^{d_a}$ and admit laws with finite second moments and positive-definite covariance. In reference coordinates write their means and covariances as μ_i^a and Σ_i^a. For a supplied scalar λ_{ij}^a on each nonself directed edge, put

$$L_{ij}^a = \lambda_{ij}^a O_i^a T_j^a, \qquad O_i^a(t) = \mu_i^a t, \qquad T_j^a(v) = (\mu_j^a)^T(\Sigma_j^a)^{-1}v. \tag{2.15}$$

Here $H_a = \mathbb{R}$, and $L_{ij}^a = \lambda_{ij}^a \mu_i^a (\mu_j^a)^T(\Sigma_j^a)^{-1}$ has rank at most one. Set $L_{ii}^a = 0$ separately. The edge coefficient is constitutive data, not a consequence of covariance.

Proposition 2.10 (Equivariance of the moment family). The family eq. (2.15) satisfies eq. (2.13) under every invertible linear reference change.

Proof. Under $U_i^a \mapsto U_i^a h_a$, the moments become $\mu_i^{a\prime} = h_a^{-1}\mu_i^a$ and $\Sigma_j^{a\prime} = h_a^{-1}\Sigma_j^a h_a^{-T}$. Consequently

$$(\mu_j^{a\prime})^T(\Sigma_j^{a\prime})^{-1} = (\mu_j^a)^T(\Sigma_j^a)^{-1}h_a. \tag{2.16}$$

Multiplication with the transformed receiving column gives $L_{ij}^{a\prime} = h_a^{-1}L_{ij}^a h_a$. □

No fixed Euclidean metric was inserted into this comparison: the source covariance supplies the inverse bilinear form and transforms with the law. Finite moments and invertible covariance are assumptions of this example alone. Centered laws give zero generators in this family, so it does not encode all distinctions between laws.

For a non-Gaussian example in two dimensions, take independent centered Laplace coordinates of variance one and translate their joint law to means e_1 and e_2 for two agents. Both covariances equal I. With identity frames and unit edge coefficients,

$$L_{12} = E_{12}, \qquad L_{21} = E_{21}, \qquad \Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.17}$$

The loop is nontrivial even though both interfaces have rank one. This is a communication loop at one context, not the curvature of a base connection. Chapter 17 constructs base transport from smooth contextual fields of law and link data with a declared comparison rule.

2.5 Loops, reciprocity, and what a law-level cost can see

Opposite attending edges are independent constitutive choices. In particular, Θ_{ji} need not equal Θ_{ij}^{-1}, as eq. (2.17) demonstrates. Geometric reversal of the *same* transport is instead its inverse. Endpoint covariance and conjugation of loop products are the familiar transformation laws of lattice link variables [143], here holding in display coordinates only; neither imposes an inverse relation between two independently modeled interactions. Directed non-inverse links at one context form a representation of the interaction quiver. Under independent changes at each vertex its polynomial invariants are generated by traces of products along oriented cycles [94]. Under the common reference change alone, the group is one copy of $\mathrm{GL}(V_a)$ acting by simultaneous conjugation on all reference links, and the invariants include traces of arbitrary words in those links [122], for example $\mathrm{tr}(e^{L_{12}}e^{L_{34}})$, which is not a path product.

Proposition 2.11 (Reciprocal and backtrack parts of a directed pair). Let Θ_{ij} and Θ_{ji} be invertible links of one sector, and suppose the backtrack loop $N_j = \Theta_{ji}\Theta_{ij}$ at j has no eigenvalue in $(-\infty, 0]$. With the principal square root, put

$$\Omega_{ij} = \Theta_{ij}N_j^{-1/2}, \qquad \Theta_{ij} = \Omega_{ij}N_j^{1/2}. \tag{2.18}$$

Then $\Omega_{ji} = \Omega_{ij}^{-1}$, and under passive endpoint changes $\Omega'_{ij} = g_i\Omega_{ij}g_j^{-1}$ and $N'_j = g_jN_jg_j^{-1}$.

Proof. The loop $N_i = \Theta_{ij}\Theta_{ji} = \Theta_{ij}N_j\Theta_{ij}^{-1}$ is similar to N_j, so it also has no eigenvalue in $(-\infty, 0]$. For a primary matrix function f defined on the common spectrum of AB and BA, $Bf(AB) = f(BA)B$ [76, Chapter 1]. With $A = \Theta_{ij}$, $B = \Theta_{ji}$ and $f(z) = z^{-1/2}$,

$$\Omega_{ji}\Omega_{ij} = \Theta_{ji}N_i^{-1/2}\Theta_{ij}N_j^{-1/2} = N_j^{-1/2}\Theta_{ji}\Theta_{ij}N_j^{-1/2} = I. \tag{2.19}$$

For the transformation law, $N'_j = g_jN_jg_j^{-1}$ by eq. (2.12), and the principal square root commutes with similarity, so $(N'_j)^{-1/2} = g_jN_j^{-1/2}g_j^{-1}$. □

The reciprocal parts Ω form a representation in which reversal is inversion, as for the transport of appendix C, while N_j records nonreciprocity. When $\Theta_{ji} = \Theta_{ij}^T$, eq. (2.18) is the polar decomposition. In eq. (2.17), $N_2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ is positive definite and $\Omega_{12} = 5^{-1/2}\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$ is a rotation. The nontrivial two-cycle of that example is therefore pure nonreciprocity, $\Omega_{12}\Omega_{21} = I$; cycle products of the reciprocal parts on cycles of length at least three can carry reciprocal holonomy.

Proposition 2.12 (Law-level comparisons and the stabilizer). Let H, H' be bimeasurable bijections of a standard Borel space E and let $\nu \in \mathcal{P}(E)$. Every functional of the pair $(\nu, H_\#\nu)$, in particular $D_{\mathrm{KL}}(\nu\|H_\#\nu)$, takes the same value for H and H' whenever $H_\#\nu = H'_\#\nu$, and $D_{\mathrm{KL}}(\nu\|H_\#\nu) = 0$ exactly when $H_\#\nu = \nu$. In contrast, the shared-sample law $(\mathrm{id}, H)_\#\nu$ of (Y, HY) with $Y \sim \nu$ determines H up to a ν-null set.

Proof. The first two statements follow from the definition of a functional of two laws and from $D_{\mathrm{KL}}(\mu\|\eta) = 0$ if and only if $\mu = \eta$. For the last, the graph of the Borel map H' is a Borel subset of $E \times E$, so its complement $C = \{(y, z) : z \neq H'y\}$ is Borel. If $(\mathrm{id}, H)_{\#}\nu = (\mathrm{id}, H')_{\#}\nu$, evaluating both on C gives $\nu\{y : Hy \neq H'y\} = \nu\{y : H'y \neq H'y\} = 0$. □

The peer-law comparison terms introduced above and in chapter 4 have the form $D_{\mathrm{KL}}(\nu_i\|\nu_j\mathcal{K}_{ij})$. A loop map H entering a comparison only through $H_{\#}\nu$ cannot be distinguished from another map with the same transported law; a rotation loop, for example, is invisible against a centered isotropic Gaussian and visible against an anisotropic one. Under a frame change g at the base vertex the shared-sample pair transforms diagonally, $(Y, HY) \mapsto (gY, (gHg^{-1})gY)$. A direct shared-sample loop experiment therefore requires a joint law coupling the routes. This is one way to identify the loop map. Grounded edge messages can also identify dictionaries, and hence their products, to the extent allowed by their codes and referents; chapter 18 states that experiment and its identifiability conditions. These observations use more information than the two marginal laws in the original comparison.

Proposition 2.13 (Law-dressed loop defect). Let $W = \Theta_{ij}\Theta_{ji}$ be a two-cycle based at i in a linear sector, and let Σ_i be the covariance of ν_i in the same displayed coordinates, assumed positive definite. Then

$$Q_{ij} = \tfrac{1}{2}\operatorname{tr}\big(\Sigma_i(I - W)^T\Sigma_i^{-1}(I - W)\big) = \tfrac{1}{2}\big\|\Sigma_i^{-1/2}(I - W)\Sigma_i^{1/2}\big\|_F^2, \tag{2.20}$$

is unchanged by passive changes and by every common invertible reference change, is nonnegative, and vanishes exactly when $W = I$.

Proof. Under $g \in \mathrm{GL}(E_i)$, $W \mapsto gWg^{-1}$ and $\Sigma_i \mapsto g\Sigma_i g^T$. Substitution gives $\frac{1}{2}\operatorname{tr}(g\Sigma_i(I - W)^T\Sigma_i^{-1}(I - W)g^{-1})$, which equals the original trace by cyclicity. A common reference change acts on the reference loop and reference covariance in the same way. The second expression is a squared Frobenius norm of a similarity transform of $I - W$, which proves the last two assertions. □

Continuous conjugation-invariant functions of W alone cannot detect a unipotent loop such as $I + E_{12}$, whose conjugacy class contains I in its closure, whereas $Q = \frac{1}{2}$ for that loop with $\Sigma_i = I$. For eq. (2.17), $Q_{12} = \frac{3}{2}$. This link defect remains positive for a nonidentity map that fixes the law, so it differs from a transported-law divergence. It is not a term of the original potential of chapter 4. Chapters 18 and 19 separately introduce latent generators with an anchored loop prior in a declared tangent model. Their recognition dynamics does not alter the fixed-link formation constructions.

2.6 Dependence, direction, and inherited responses

Covariance constrains which law-only generators are possible. If a reference transformation fixes all input laws and all supplied data, eq. (2.13) requires its matrix to commute with the resulting generator. Identical centered laws invariant under every orthogonal transformation therefore cannot select a distinguished off-diagonal direction through a fully orthogonally equivariant law-only rule. Directional interfaces can carry additional information, but that information must be declared.

When links depend on the current laws, differentiation must include that dependence. At fixed frames a variation of an objective has the form

$$d\mathcal{E} = d_\nu \mathcal{E} + \sum_{a,i,j} \langle d_{L^a_{ij}} \mathcal{E}, D\mathcal{L}^a_{ij}[d\bar{\boldsymbol{\nu}}^a] \rangle. \tag{2.21}$$

This statement uses a differentiable statistical family and differentiable constitutive maps. It includes both the sender and receiver occurrences, and every other law on which the generator depends. Freezing the link during an update defines another declared dynamics; it is not generally the derivative of the substituted objective. Nor does substituting live recognition laws into a likelihood make that likelihood part of a fixed generative model during recognition variation.

An inherited parent's state must determine its future boundary response, including the law dependence of its generators. If two child configurations with identical retained data give different future responses, those data do not support autonomous closure.

Chapter 3

Geometry of Statistical Sections

A section's statistical change induces contextual geometry relative to a connection. We construct that tensor, a collective rule for choosing the connection, and conditions for their inheritance.

The role of these objects in the rest of the manuscript is limited and should be stated at the outset. No vector field, source row, channel, or record kernel in chapter 4 through chapter 13 depends on a connection, on the collective connection selected below, or on their curvature; each context evolves by a context-local flow. The formation chapters compute connections, pullback tensors, and curvature as diagnostics of those evolved sections. The separate construction in chapter 16 adds the minimized contextual energy to the reciprocal potential and proves local well-posedness for finite graphs and a regular periodic continuum chart. The remaining extensions are recorded in open problem 3.11.

3.1 Regular law families

Definition 3.1 (Fisher geometry on an admitted family). Let $\mathcal{N}$ be a finite-dimensional smooth manifold of probability laws on a standard Borel space Y. In each parameter chart $\theta \mapsto P_\theta$, assume differentiability in quadratic mean with centered square-integrable score $\ell_\theta : \mathsf{Y} \to \mathbb{R}^d$. At a fixed θ, this means that the Lebesgue decomposition $P_{\theta+\delta} = p_\delta P_\theta + P_\delta^\perp$ satisfies

$$P_\delta^\perp(\mathsf{Y}) = o(|\delta|^2), \qquad \left\| \sqrt{p_\delta} - 1 - \tfrac{1}{2}\delta^\top \ell_\theta \right\|_{L^2(P_\theta)} = o(|\delta|). \tag{3.1}$$

For a tangent vector v, its scalar score ℓ_v is the corresponding directional derivative in this expansion. Require the Fisher tensor

$$g_P^F(v, w) = \mathbb{E}_P[\ell_v \ell_w] \tag{3.2}$$

to be smooth and positive definite on the admitted tangent space. Redundant parameter directions must be removed or treated as null directions before imposing this last requirement.

Condition eq. (3.1) is Le Cam's differentiability in quadratic mean [95]. These family-specific hypotheses allow parameter-dependent supports: the laws of the family need not share null sets, and no common positive density is required [6]. A finite-dimensional family satisfying eq. (3.1) is nevertheless dominated. Differentiability in quadratic mean makes $\theta \mapsto P_\theta$ continuous in total variation, and for a countable dense set $\{\theta_n\}$ the mixture $\sum_n 2^{-n} P_{\theta_n}$ dominates every P_θ. The choice of the Fisher tensor is not arbitrary: up to a constant factor it is the only Riemannian metric on finite statistical simplices invariant under Markov congruent embeddings [28], with extensions to general sample spaces under the hypotheses of Ay et al. [5]. That characterization concerns the metric on one law space; it does not fix the relative weights of separate factors, which remain constitutive below.

Suppose the finite-dimensional Lie group G acts smoothly on $\mathcal{N}$ by the sample-space pushforwards in definition 1.1. A fixed bimeasurable bijection carries a score ℓ_v to $\ell_v \circ g^{-1}$. Change of variables in eq. (3.2) therefore makes the action a Fisher isometry. Its infinitesimal action is

$$J_P\xi = \left.\frac{d}{dt}\right|_{t=0} (\exp(t\xi))_{\#}P, \qquad \xi \in \mathfrak{g}. \tag{3.3}$$

The regular family must be stable under every group element used in these comparisons.

Proposition 3.2 (Fisher metric of the joint agent law). Let $\zeta_\theta(dk, dm) = q_\theta(dk \mid m)s_\theta(dm)$ be a regular family. Assume the marginal and conditional derivatives obey the product rule, that the joint score is $\ell^\zeta_v(k, m) = \ell^s_v(m) + \ell^{q|m}_v(k)$, and that these scores are square-integrable. Assume differentiation of conditional normalization is valid, so $\int \ell^{q|m}_v(k)q_\theta(dk \mid m) = 0$ for s_θ-almost every m. Then

$$g^F_\zeta(v, w) = g^F_s(v, w) + \int g^F_{q(\cdot|m)}(v, w)s(dm). \tag{3.4}$$

Proof. Expand $\mathbb{E}_\zeta[(\ell^s_v + \ell^{q|m}_v)(\ell^s_w + \ell^{q|m}_w)]$. Each mixed term vanishes after conditioning on m. The two remaining terms are exactly those displayed. □

This is the chain rule for Fisher information [145], stated for a joint law with differentiable conditional normalization.

For the actual product family $q_\theta \otimes s_\theta$, eq. (3.4) reduces to $g^F_q + g^F_s$. The two marginal metrics do not in general recover the joint metric. For example, the law

$$\zeta_\rho(k, m) = \frac{1 + \rho km}{4}, \qquad k, m \in \{-1, 1\}, \qquad |\rho| < 1, \tag{3.5}$$

has two fixed uniform marginals, whereas its joint score is $km/(1 + \rho km)$. Summing its squared score over the four outcomes gives

$$g^F_{\zeta_\rho}(\partial_\rho, \partial_\rho) = \frac{1}{1 - \rho^2}. \tag{3.6}$$

A declared marginal metric omits this dependence direction. We use the joint law unless a restriction is stated.

3.2 Covariant statistical change

Restrict the statistical bundle of eq. (1.2) to the admitted smooth fiber $\mathcal{N}$, writing $E = P \times_G \mathcal{N}$ and $\pi : E \to \mathcal{C}$. A principal connection ω induces a horizontal lift $H^\omega_e : T_c\mathcal{C} \to T_eE$ and vertical projection $\mathrm{ver}^\omega : T_eE \to V_eE$, where $V_eE = \ker T_e\pi$ and $c = \pi(e)$. The vertical Fisher metric is well defined because the transition action is isometric.

Definition 3.3 (Covariant pullback). For a smooth statistical section $\zeta : \mathcal{U} \to E$, define

$$D^\omega\zeta = \mathrm{ver}^\omega \circ T\zeta, \qquad h^\omega_\zeta(X, Y) = g^F_\zeta(D^\omega\zeta X, D^\omega\zeta Y). \tag{3.7}$$

The vertical projection removes the base component of $T\zeta$ forced by $\pi \circ \zeta = \mathrm{id}$. The resulting base tensor is positive semidefinite.

In a local frame write A for the connection form and $p(c)$ for the law coordinates. With the convention that the horizontal lift is $(X, -J_p A(X))$,

$$D^A p(X) = dp(X) + J_p A(X). \tag{3.8}$$

Here $J_p A(X)$ is a statistical tangent vector.

Proposition 3.4 (Gauge covariance and connection dependence). Under a passive change $p' = g_{\#}p$ and $A' = \mathrm{Ad}_g A - (dg)g^{-1}$,

$$D^{A'} p' = T(g_{\#}) D^A p, \qquad h^{A'}_{p'} = h^A_p. \tag{3.9}$$

For two connections on the same bundle, let $\alpha = \omega' - \omega$ denote their adjoint-bundle-valued difference. Along one fixed section put $b_X = J_\zeta \alpha(X)$ and $u_X = D^\omega \zeta X$. Then

$$D^{\omega'} \zeta X = u_X + b_X, \tag{3.10}$$

$$h^{\omega'}_\zeta(X,Y) = h^\omega_\zeta(X,Y) + g^F(u_X, b_Y) + g^F(b_X, u_Y) + g^F(b_X, b_Y). \tag{3.11}$$

Proof. Differentiation of $p' = g_{\#}p$ gives $dp' = T(g_{\#})dp + J_{p'}((dg)g^{-1})$. Equivariance of the fundamental action gives $J_{p'} \mathrm{Ad}_g = T(g_{\#}) J_p$. Substitution in eq. (3.8) cancels the inhomogeneous terms. Fisher invariance proves eq. (3.9). Replacing A by $A + \alpha$ in that same derivative adds b; expanding the bilinear pairing gives eq. (3.11). □

Changing the connection changes the comparison. Appendix C records its transformation law in a moving presentation. Chapter 17 derives connections from regular contextual law, interface, and link data with declared comparison rules. Those selectors need not equal the Fisher selector of proposition 3.9, and passive covariance does not remove their constitutive dependence.

Passive invariance does not make the tensor a property of the section alone. Suppose G acts transitively on $\mathcal{N}$. The orbit map $g \mapsto g_{\#}p_0$ is then a submersion and has local sections, so a local gauge makes the section constant, $p(c) = p_0$. In that gauge eq. (3.8) gives $D^A p = J_{p_0} A$, and

$$h^A_p(X,Y) = \mathfrak{k}_{p_0}(A(X), A(Y)), \qquad \mathfrak{k}_{p_0}(\xi,\eta) = g^F(J_{p_0}\xi, J_{p_0}\eta). \tag{3.12}$$

As A ranges over local connection forms, this ranges over every positive semidefinite form of rank at most $\dim \mathcal{N}$. For a homogeneous fiber the per-agent contextual tensor therefore records the supplied connection and no gauge-invariant property of the section; this is the situation of a single Gaussian family under the affine group, which acts transitively.

Proposition 3.5 (Transported relative entropy). Fix a smooth curve γ through c with $\dot\gamma(0) = X$, and transport its laws back to the fiber at c:

$$\mu_t = (T^\omega_{\gamma(t)\to c})_{\#} \zeta(\gamma(t)). \tag{3.13}$$

Assume, in addition to definition 3.1, that μ_t has positive densities f_t with respect to one measure near $t = 0$, that $\log f_t$ is twice differentiable, and that both derivatives can be interchanged with the integrals of f_t and $f_0 \log f_t$. Assume their second derivatives are continuous at zero. Then

$$D_{\mathrm{KL}}(\zeta(c)\|\mu_t) = \frac{t^2}{2} h^{\omega}_{\zeta}(X, X) + o(t^2). \tag{3.14}$$

Proof. The derivative of the transported law is $D^{\omega}\zeta X$ by the horizontal-lift convention. Write $\ell = \partial_t \log f_t|_0$ for its score. Differentiating normalization once gives $\mathbb{E}_{\mu_0}\ell = 0$. Differentiating twice gives $\mathbb{E}_{\mu_0}[\partial_t^2 \log f_t|_0 + \ell^2] = 0$. The first derivative of the displayed relative entropy is therefore zero and its second derivative is $\mathbb{E}_{\mu_0}\ell^2 = h^{\omega}_{\zeta}(X, X)$. Taylor's theorem proves the claim. □

This is the classical second-order relation between relative entropy and Fisher information [88], applied to the transported law. Quadratic-mean differentiability alone does not justify this second-order KL expansion.

3.3 Null directions and quotients

Proposition 3.6 (Rank and quotient geometry). For the positive fiber Fisher metric in definition 3.1,

$$\operatorname{rad} h^{\omega}_{\zeta} = \ker D^{\omega}\zeta, \qquad \operatorname{rank} h^{\omega}_{\zeta} = \operatorname{rank} D^{\omega}\zeta. \tag{3.15}$$

If the latter rank is constant, $K = \ker D^{\omega}\zeta$ is a smooth subbundle and $T\mathcal{U}/K$ carries the positive-definite metric $\bar{h}([X], [Y]) = h^{\omega}_{\zeta}(X, Y)$.

Proof. The diagonal value of h^{ω}_{ζ} is the squared Fisher norm of $D^{\omega}\zeta X$, so it vanishes exactly on the derivative's kernel. For a positive semidefinite form, its diagonal null set is its radical. Rank-nullity gives the second identity, and the constant-rank theorem for bundle maps makes K a smooth subbundle. The pairing is independent of representatives because it annihilates K, and it is positive on every nonzero quotient class. □

A quotient vector bundle is not automatically the tangent bundle of a quotient manifold. Suppose a smooth Hausdorff manifold $\bar{\mathcal{U}}$ and a surjective submersion $q : \mathcal{U} \to \bar{\mathcal{U}}$ with connected fibers satisfy $\ker Tq = K$. The tensor descends to $\bar{\mathcal{U}}$ exactly when it is basic: besides annihilating K, it obeys $\mathcal{L}_Z h^{\omega}_{\zeta} = 0$ for every vertical field $Z \in \Gamma(K)$. Necessity follows by differentiating a pulled-back tensor along a fiber. For sufficiency, use submersion coordinates (y, z): the annihilation condition leaves only $dy^a dy^b$ components, and the Lie derivatives in the z directions make their coefficients independent of z. Connected fibers make these local definitions agree across each fiber. The resulting tensor is positive definite by eq. (3.15). A basic semidefinite tensor of this kind is a bundle-like transverse metric for the foliation by the fibers [108, 123].

Integrability of the radical and basicness are both measured by the curvature. In a local frame write $V = D^A p$, $j(p, \xi) = J_p \xi$, $d_p j$ for its derivative in p in a chart of $\mathcal{N}$, and $F = dA + [A \wedge A]$ with $[A \wedge A](X, Y) = [A(X), A(Y)]$.

Proposition 3.7 (Curvature criteria for the radical). For base vector fields X, Y,

$$V([X, Y]) = \big(X(VY) + d_p j(VY, A(X))\big) - \big(Y(VX) + d_p j(VX, A(Y))\big) - J_p F(X, Y). \tag{3.16}$$

Consequently, where the rank of V is constant, the radical $K = \ker V$ is involutive exactly when $J_pF(X,Y) = 0$ for all $X, Y \in K$. For $Z \in \Gamma(K)$ and arbitrary X, Y,

$$(\mathcal{L}_Z h^A_p)(X,Y) = g^F\big(J_pF(Z,X), VY\big) + g^F\big(VX, J_pF(Z,Y)\big), \tag{3.17}$$

so an involutive radical carries a basic tensor exactly when the right side vanishes. Both conditions hold for every flat connection.

Proof. In chart coordinates, $dp([X,Y]) = X(dp(Y)) - Y(dp(X))$ and $A([X,Y]) = X(A(Y)) - Y(A(X)) - dA(X,Y)$. Differentiating $VY = dp(Y) + J_pA(Y)$ along X gives $X(VY) = X(dp(Y)) + d_pj(dp(X), A(Y)) + J_pX(A(Y))$. Combining these identities and substituting $dp = V - J_pA$ leaves the terms of eq. (3.16) together with $d_pj(J_pA(X), A(Y)) - d_pj(J_pA(Y), A(X))$. That expression is the bracket of the fundamental fields of $A(X)$ and $A(Y)$, which for a left action equals $-J_p[A(X), A(Y)]$; with the $-J_pdA(X,Y)$ term it gives $-J_pF(X,Y)$. If $X, Y \in \Gamma(K)$, then $VX = VY = 0$ identically, and eq. (3.16) reduces to $V([X,Y]) = -J_pF(X,Y)$, which proves the involutivity criterion.

For eq. (3.17), write $(\mathcal{L}_Z h)(X,Y) = Z(g_p(VX,VY)) - h([Z,X],Y) - h(X,[Z,Y])$, expand the first term with $dp(Z) = -J_pA(Z)$, and insert eq. (3.16) with $VZ = 0$. The terms containing $\xi = A(Z)$ combine into $-(\mathcal{L}_{\xi^*}g^F)(VX,VY)$, the Lie derivative of the Fisher metric along the fundamental field of ξ, which vanishes because the action is isometric. The curvature terms remain. If $F = 0$, both right sides vanish. □

The criteria are sharp. On the trivial translation bundle over $\mathbb{R}^3$, use the unit-variance normal location family, the constant section of mean zero, and $A = dz - xdy$. Its location Fisher metric is one, so

$$D^Ap = dz - xdy, \qquad h^A_p = (dz - xdy)^2. \tag{3.18}$$

This has constant rank one, but its null fields ∂_x and $\partial_y + x\partial_z$ have bracket ∂_z, which is not null. Here $F = -dx \wedge dy$ and $V([\partial_x, \partial_y + x\partial_z]) = 1 = -F(\partial_x, \partial_y + x\partial_z)$, as eq. (3.16) requires. The connection in this example is supplied, not selected: for a single location family under translations the minimizing connection of proposition 3.9 makes the tensor vanish, and proposition 3.10 shows that the minimal tensor always has an involutive radical. Gaussian laws are not required for the preceding results.

A contextual tangent quotient differs from a group-orbit quotient or measurable parent retention, which may discard information having no Fisher tangent.

3.4 A collective choice of connection

Several law sections can determine a common comparison rule by minimizing their shared frame motion. Choose smooth comparisons S_i placing the participating laws in compatible associated fibers, and write $p_i = (\mathsf{S}_i)_\#\zeta_i$. For bimeasurable comparisons, independent source changes g_i and a common target change g act by

$$\zeta_i' = (g_i)_\#\zeta_i, \qquad \mathsf{S}_i' = g\mathsf{S}_ig_i^{-1}, \qquad p_i' = g_\#p_i. \tag{3.19}$$

The compositions in this formula mean composition of represented sample maps. More general comparisons may be declared, but their target laws must still form an admitted smooth equivariant family.

Definition 3.8 (Collective Fisher data). Let $\boldsymbol{p} = (p_i)_i$ lie in a finite product of regular statistical manifolds with diagonal G-action. Choose positive constants w_i and define

$$\mathsf{G}_{\boldsymbol{p}}(v, w) = \sum_i w_i g^F_{p_i}(v_i, w_i), \tag{3.20}$$

$$J_{\boldsymbol{p}}\xi = (J_{p_i}\xi)_i, \qquad \mathfrak{h}_{\boldsymbol{p}} = \ker J_{\boldsymbol{p}}, \qquad H_{\boldsymbol{p}} = (\operatorname{im} J_{\boldsymbol{p}})^{\perp_{\mathsf{G}}}, \tag{3.21}$$

$$\mathfrak{k}_{\boldsymbol{p}}(\xi, \eta) = \mathsf{G}_{\boldsymbol{p}}(J_{\boldsymbol{p}}\xi, J_{\boldsymbol{p}}\eta). \tag{3.22}$$

The weights and comparisons are constitutive data. A different declared joint statistical family may replace the product if its metric and group action satisfy the same hypotheses.

Proposition 3.9 (Collective minimizing connection class). Suppose $J_{\boldsymbol{p}(c)}$ has constant rank on $\mathcal{U}$. For each contextual direction X, the minimizers of

$$a \longmapsto \|d\boldsymbol{p}(X) + J_{\boldsymbol{p}}a\|^2_{\mathsf{G}} \tag{3.23}$$

form an affine class $A^F(X) + \mathfrak{h}_{\boldsymbol{p}}$, characterized by

$$\mathfrak{k}(A^F(X), \xi) = -\mathsf{G}(d\boldsymbol{p}(X), J\xi), \tag{3.24}$$

$$r_X := d\boldsymbol{p}(X) + JA^F(X) = P_H d\boldsymbol{p}(X). \tag{3.25}$$

Smooth local representatives can be chosen, linear in X. With $h^{\min}(X, Y) = \mathsf{G}(r_X, r_Y)$, any other local connection form A satisfies

$$h^A(X, Y) = h^{\min}(X, Y) + \mathfrak{k}((A - A^F)(X), (A - A^F)(Y)). \tag{3.26}$$

The minimizing class transforms by the affine connection law, and r transforms by the tangent action.

Proof. Project $-d\boldsymbol{p}(X)$ orthogonally onto $\operatorname{im} J$. Its preimages under J give all minimizing a, and two preimages differ exactly by $\ker J$. Orthogonality to every $J\xi$ gives the normal equation. At constant rank, choose a smooth local complement to $\ker J$; the positive form $\mathfrak{k}$ restricted to it has a smooth inverse, providing the stated representative. Writing $d\boldsymbol{p} + JA = r + J(A - A^F)$ and using orthogonality gives eq. (3.26). Under a passive target change, the derivative and connection terms cancel exactly as in eq. (3.9). Isometry of the action carries minimizers to minimizers and residuals to residuals. □

If infinitesimal isotropy is nonzero, the construction leaves stabilizer-valued components undetermined. An arbitrarily chosen pseudoinverse representative need not obey the affine law. Global minimizing representatives nevertheless exist on any paracompact domain of constant rank. Being minimizing at c is a frame-independent property of a connection, and at each c the minimizing values form an affine subspace. Given local minimizing connections ω_α and a subordinate partition of unity ϕ_α, the connection $\sum_\alpha \phi_\alpha \omega_\alpha$ is an affine combination of minimizers at every point and is

therefore minimizing. What requires $\ker J = 0$ is uniqueness: then the local forms are unique, obey the affine law, and agree on overlaps; discrete stabilizers do not obstruct this. Curvature statements about a class should be made for $J_{\boldsymbol{p}}F$ on the radical, which by eq. (3.16) is the same for every representative, or after a representative has been fixed, since the full curvature depends on that choice.

Proposition 3.10 (Shape-space form of the minimal tensor). Assume the hypotheses of proposition 3.9 with $\ker J_{\boldsymbol{p}(c)} = 0$ on $\mathcal{U}$. Let $\theta(v) = \mathfrak{k}^{-1}\mathsf{G}(v, J\cdot)$ denote the mechanical connection of the diagonal action, defined where the action is locally free. Then $A^F = -\boldsymbol{p}^*\theta$ and $F^{A^F} = -\boldsymbol{p}^*\Omega$, where $\Omega = d\theta - [\theta \wedge \theta]$ is horizontal. Hence $J_{\boldsymbol{p}}F^{A^F}(Z, \cdot) = 0$ for Z in the radical of $h^{\min}$; where $h^{\min}$ has constant rank, its radical is involutive and $h^{\min}$ is basic. If the action is free and proper, so that $\pi : \mathcal{N}^n \to \mathcal{N}^n/G$ is a Riemannian submersion for the quotient metric g_{sh}, then

$$h^{\min} = (\pi \circ \boldsymbol{p})^* g_{\mathrm{sh}}, \qquad \operatorname{rad} h^{\min} = \ker d(\pi \circ \boldsymbol{p}). \tag{3.27}$$

Proof. The normal equation eq. (3.25) reads $A^F(X) = -\theta(d\boldsymbol{p}(X))$. Pullback commutes with d and with the bracket of Lie-algebra-valued forms, so $F^{A^F} = -\boldsymbol{p}^*(d\theta - [\theta \wedge \theta])$. For a left action $[\xi^*, \eta^*] = -[\xi, \eta]^*$, and $\theta(\xi^*) = \xi$ is constant, so $\Omega(\xi^*, \eta^*) = -\theta([\xi^*, \eta^*]) - [\xi, \eta] = 0$. For a horizontal field v, $\Omega(\xi^*, v) = -\theta([\xi^*, v]) = (\mathcal{L}_{\xi^*}\theta)(v)$, and equivariance of θ makes this Lie derivative a commutator of ξ with $\theta(v) = 0$, so it vanishes. Thus Ω is horizontal. If Z lies in the radical, then $r_Z = 0$ and $d\boldsymbol{p}(Z) = -JA^F(Z)$ is an orbit direction, so $F^{A^F}(Z, \cdot) = 0$; proposition 3.7 gives involutivity and basicness. Under a free proper action, $d\pi$ restricted to H is an isometry onto the tangent space of the orbit space, and $d\pi$ annihilates $\operatorname{im} J$. Since $r_X = P_H d\boldsymbol{p}(X)$, $h^{\min}(X, Y) = g_{\mathrm{sh}}(d\pi(d\boldsymbol{p}(X)), d\pi(d\boldsymbol{p}(Y)))$. □

The collective construction is therefore classical. The minimizing connection is the mechanical connection of Smale [133], the orthogonal separation of rotations from shape change studied for molecules by Guichardet [70] and as a gauge field of the n-body problem by Littlejohn and Reinsch [101]; eq. (3.26) is the split of kinetic energy into shape and locked-rotation parts, and $h^{\min}$ is the pullback of the quotient metric of the Riemannian submersion onto shape space [113]. What is new here is the use of the Fisher metric of the declared collective family as the kinetic metric. Rank is bounded by the shape dimension, $\operatorname{rank} h^{\min} \le \dim H = \dim \mathcal{N}^n - \dim(\text{orbit})$. A single transitive family has $H = 0$ and zero minimizing residual; a collective family need not.

The constant-rank hypothesis fails where infinitesimal isotropy increases. Agreement need not cause such an increase: any number of unit-covariance planar normal laws with common nonzero mean e_1 has a free diagonal $SO(2)$ action and positive locked inertia. If the infinitesimal stabilizer does increase, the rank of J drops; a smooth extension of the minimizing Lie-algebra-valued representative is then not guaranteed [99]. For example, let $SO(2)$ act on the unit-covariance normal family on $\mathbb{R}^2$ and take the section $\mu(c) = (\epsilon, c)$ for one agent. Then $J\xi = \xi(-c, \epsilon)$, $\mathfrak{k} = c^2 + \epsilon^2$, and

$$A^F(\partial_c) = -\frac{\epsilon}{c^2 + \epsilon^2}, \qquad \int_{\mathbb{R}} A^F(\partial_c) dc = -\pi, \tag{3.28}$$

so as $\epsilon \to 0$ the connection concentrates to $-\pi\delta(c)$ at the isotropic point. The minimal tensor stays bounded, $h^{\min}(X, X) \le \|d\boldsymbol{p}(X)\|^2_{\mathsf{G}}$ by the choice $a = 0$. This one-dimensional example exhibits connection concentration, not curvature blowup. A smooth collective connection and its curvature

require the relevant regularity hypotheses; the displayed bound for $h^{\min}$ alone does not control derivatives of a connection near a change of orbit type.

The constant weights w_i in definition 3.8 have a probabilistic reading. With $\sum_i w_i = 1$, the weighted metric is the Fisher metric, in the $\boldsymbol{p}$ directions, of the labeled family $\sum_i w_i \delta_i \otimes p_i$. If the weights depend on the context, its Fisher metric acquires the term $\sum_i dw_i \otimes dw_i / w_i$. The group acts on samples and not on labels, so the weight scores have zero Fisher pairing with every orbit direction; the normal equation is unchanged and $h^{\min}$ gains that connection-independent term. A correlated population law, whose Fisher metric sees dependence directions invisible to every marginal (proposition 3.2), is not of this weighted product form and is not treated here.

The collective connection is also the auxiliary field of a gauge-invariant contextual energy. For a Riemannian metric γ on a domain of $\mathcal{C}$ with volume form dvol_γ, put

$$\mathcal{E}_{\mathrm{ctx}}[\boldsymbol{p}, A] = \tfrac{1}{2} \int \gamma^{\mu\nu} \mathsf{G}\big(D^A_\mu \boldsymbol{p}, D^A_\nu \boldsymbol{p}\big) \mathrm{dvol}_\gamma. \tag{3.29}$$

By eq. (3.9) it is gauge invariant. Its integrand is a sum over directions of the costs minimized in proposition 3.9, so at fixed $\boldsymbol{p}$ its minimizers over A are the minimizing connections, and the minimum is $\frac{1}{2} \int \mathrm{tr}_\gamma\, h^{\min} \mathrm{dvol}_\gamma$. This is the structure of the $\mathbb{CP}^{N-1}$ sigma model, whose gauge field has no kinetic term and is eliminated algebraically [40]; the analogy is structural and does not import that model's quantum dynamics.

Open problem 3.11 (Extensions of the contextual Fisher flow). Chapter 16 supplies a finite-graph construction and a continuum local theorem on a flat torus. Auxiliary elimination diffuses the law shape and leaves frame and source-row equations without independent diffusion. Global regularity, changes of orbit type, physical boundaries, independently evolving rows and communication links, and a graph-refinement theorem remain open. The quotient target can be open and positively curved, so the local theorem alone gives no global continuation. Deriving a kinetic term for the connection from a declared ensemble of contextual sections is a separate problem; that ensemble is not supplied.

Proposition 3.12 (Aggregate covariant derivative). Let $\mathcal{M}$ be a smooth equivariant map from the collective statistical family to a finite-dimensional representation W of G. For $M = \mathcal{M}(\boldsymbol{p})$ define

$$e(X) = d\mathcal{M}_{\boldsymbol{p}} r_X = dM(X) + \rho_*(A^F(X))M. \tag{3.30}$$

This W-valued one-form is independent of the representative of the minimizing class. If $\dim \mathcal{C} = \dim W$, it is an isomorphism exactly when the actual map $d\mathcal{M}_{\boldsymbol{p}} \circ r$ has full rank.

Proof. Differentiating equivariance gives $d\mathcal{M}_{\boldsymbol{p}} J\xi = \rho_*(\xi)M$. Applying this identity to eq. (3.25) proves eq. (3.30). A change in A^F belonging to $\ker J$ has zero image under $d\mathcal{M}$, proving representative independence. The rank statement is the criterion for a linear map between equal-dimensional spaces to be invertible. □

Surjectivity of $d\mathcal{M}|_H$ establishes that a full-rank first jet can be chosen at the law tuple: choose horizontal preimages of a basis of W and realize those as derivatives in a local statistical chart. It does not establish full rank for the supplied section or its preservation under inference. The

determinant specialization in appendix D uses an additional form on W; the Fisher metric above remains positive. Surjectivity itself is a restriction on the number and configuration of the agents; proposition D.2 computes it for proper complex Gaussian targets.

3.5 Information retained by a channel

Theorem 3.13 (Score projection and exact Fisher loss). Let $\{P_\theta\}$ satisfy eq. (3.1) at θ_0, and let $K : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be one normalized Markov kernel independent of θ. Under the joint law $P_{\theta_0}(dy)K(y, d\bar{y})$, the output family is differentiable in quadratic mean with score and Fisher matrix

$$\bar{\ell}(\bar{Y}) = \mathbb{E}[\ell(Y) \mid \bar{Y}], \tag{3.31}$$

$$I(P_{\theta_0}) - I(P_{\theta_0}K) = \mathbb{E}\,\mathrm{Cov}(\ell(Y) \mid \bar{Y}) \succeq 0. \tag{3.32}$$

Equality in direction v holds exactly when $v^\top \ell(Y)$ is measurable from $\bar{Y}$ up to a joint null set.

Proof. Lift the decomposition in eq. (3.1) to the joint space. Its absolutely continuous square root is $V_\delta = \sqrt{p_\delta(Y)}$ with the same L^2 remainder, and its singular mass is still $o(|\delta|^2)$. For its output, put $m_\delta = \mathbb{E}[V_\delta \mid \bar{Y}]$ and $v_\delta = \mathbb{E}[(V_\delta - m_\delta)^2 \mid \bar{Y}]$. The projected density of this absolutely continuous part is $q_\delta = m_\delta^2 + v_\delta$. Conditional expectation is an L^2 contraction, so $m_\delta = 1 + \frac{1}{2}\delta^\top \mathbb{E}[\ell \mid \bar{Y}] + o_{L^2}(|\delta|)$.

To replace m_δ by $\sqrt{q_\delta}$, observe that $m_\delta \geq 0$ and

$$(\sqrt{q_\delta} - m_\delta)^2 = v_\delta \frac{v_\delta}{(\sqrt{q_\delta} + m_\delta)^2} \leq v_\delta. \tag{3.33}$$

Set the fraction to zero where its numerator and denominator both vanish. It tends to zero in probability. Moreover $v_\delta/|\delta|^2$ is uniformly integrable as $\delta \to 0$: the square-root expansion expresses $(V_\delta - 1)/|\delta|$ as a bounded finite-dimensional combination of the fixed L^2 scores plus an L^2 remainder tending to zero, and conditional variance preserves the resulting uniform integrability. Thus the last display has expectation $o(|\delta|^2)$. The projected singular part may acquire an absolutely continuous density a_δ, but its total mass is $o(|\delta|^2)$. The inequality $(\sqrt{q_\delta + a_\delta} - \sqrt{q_\delta})^2 \leq a_\delta$ shows that this change does not affect the score; any remaining singular output mass has the same small bound. This proves output differentiability and its score formula. Total covariance gives the matrix identity. Its quadratic form in v is the expected conditional variance of $v^\top \ell$, which vanishes exactly under the stated measurability condition. □

This score-projection mechanism is the one in Pollard [120, Theorem 7]; the Markov formulation is also treated in Ay et al. [6]. It identifies local score sufficiency. It does not identify a kernel recovering every law in the experiment; that an insufficient statistic can preserve Fisher information is the phenomenon of Kagan and Shepp [81]. For example, take independent Bernoulli variables with probabilities $1/2 + \theta/4$ and $1/2 + \theta^2/4$, and discard the second. At $\theta = 0$ its score vanishes, so no Fisher information is lost. A fixed recovery kernel applied to the first Bernoulli law makes every reconstructed event probability affine in θ, whereas the required second-coordinate probability is quadratic. Such a kernel cannot recover the family on a neighborhood.

Proposition 3.14 (Moving-channel correction). Let the channel instead have density $k_\theta(\bar{y} \mid y)$ with respect to a fixed output measure. Assume the joint family is differentiable in quadratic mean

with square-integrable score $\ell_\theta(Y) + b_\theta(Y, \bar{Y})$, where $b_\theta = \partial_\theta \log k_\theta$, and assume differentiation of channel normalization gives $\mathbb{E}[b_\theta \mid Y] = 0$. Then

$$\bar{\ell}_\theta(\bar{Y}) = \mathbb{E}_\theta[\ell_\theta(Y) + b_\theta(Y, \bar{Y}) \mid \bar{Y}], \tag{3.34}$$

$$I(\bar{Y}) = I(Y) + \mathbb{E}_\theta I(K_\theta(Y, \cdot)) - \mathbb{E}_\theta \operatorname{Cov}(\ell_\theta + b_\theta \mid \bar{Y}). \tag{3.35}$$

Proof. Apply theorem 3.13 to the fixed output projection of the joint experiment. Its score is the stated sum. Conditional centering of b_θ eliminates the mixed terms in its Fisher matrix, leaving the input matrix plus the average conditional channel matrix. Subtracting the conditional covariance proves the formula. □

Consequently input-to-output Fisher change has no fixed sign for a moving channel. A constant input followed by a Bernoulli output with success probability $(1 + e^{-\theta})^{-1}$ has zero input Fisher information and output information $e^{-\theta}/(1 + e^{-\theta})^2 > 0$. A singular moving deterministic statistic need not have a channel log density at all. Its actual weak derivative, schematically $d(PK) = (dP)K + P(dK)$ when these derivatives exist, must replace that expression. These distinctions apply to the law-dependent channels used in the interaction potential.

3.6 Geometry across a retention

Let $\Psi : E \to \bar{E}$ be a smooth bundle map covering $f : \mathcal{C} \to \bar{\mathcal{C}}$, where both statistical bundles carry positive vertical Fisher metrics and connections $\omega, \bar{\omega}$. Assume the sections are related,

$$\Psi \circ \zeta = \bar{\zeta} \circ f. \tag{3.36}$$

This is a hypothesis: a map over f does not automatically make every fine section descend through f.

Such maps arise from equivariant kernels. Let $\tilde{f} : P \to \bar{P}$ be a principal bundle map covering f with homomorphism $\phi : G \to \bar{G}$, and let $\kappa : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be a Markov kernel with $\kappa(gy, \cdot) = \phi(g)_\# \kappa(y, \cdot)$. Then $\Psi[u, \zeta] = [\tilde{f}(u), \zeta\kappa]$ is well defined, since $[ug, \zeta]$ and $[u, g_\#\zeta]$ are sent to $[\tilde{f}(u)\phi(g), \zeta\kappa]$ and $[\tilde{f}(u), \phi(g)_\#(\zeta\kappa)]$, which coincide. It is fiberwise affine, contracts relative entropy by data processing, and for a fixed kernel gives $\Delta_F \succeq 0$ below by theorem 3.13.

Definition 3.15 (Horizontal transport defect). For $X \in T_c\mathcal{C}$, define

$$a_\Psi(X) = T\Psi(H^\omega_{\zeta(c)} X) - H^{\bar{\omega}}_{\Psi(\zeta(c))}(Tf X). \tag{3.37}$$

Both terms project to $Tf X$, so their difference lies in the same coarse vertical tangent space. Set $u_X = D^\omega \zeta X$, $\Psi_* = T^V \Psi$, and $\Delta_F = g^F - \Psi_*^* \bar{g}^F$ along the section.

Proposition 3.16 (Signed comparison of base tensors). Under eq. (3.36),

$$D^{\bar{\omega}} \bar{\zeta}(Tf X) = \Psi_* u_X + a_\Psi(X), \tag{3.38}$$

$$h^\omega_\zeta - f^* h^{\bar{\omega}}_{\bar{\zeta}} = u^* \Delta_F - \mathcal{X}_\Psi - \mathcal{Q}_\Psi, \tag{3.39}$$

where

$$\mathcal{X}_\Psi(X,Y) = \bar{g}^F(\Psi_* u_X, a_\Psi(Y)) + \bar{g}^F(a_\Psi(X), \Psi_* u_Y), \tag{3.40}$$

$$\mathcal{Q}_\Psi(X,Y) = \bar{g}^F(a_\Psi(X), a_\Psi(Y)). \tag{3.41}$$

If each fiber map is realized by a fixed parameter-independent Markov kernel, then $\Delta_F \succeq 0$. In that case base contraction holds exactly when, for every X,

$$2\bar{g}^F(\Psi_* u_X, a_\Psi(X)) + \|a_\Psi(X)\|^2_{\bar{g}^F} \le \Delta_F(u_X, u_X). \tag{3.42}$$

Proof. Differentiate eq. (3.36), decompose $T\zeta X = H^\omega X + u_X$, and take coarse vertical parts. This proves the first identity. Expand the coarse Fisher pairing of $\Psi_* u + a_\Psi$ and subtract it from the fine pairing to obtain the second. For a fixed Markov fiber map, theorem 3.13 gives $\Delta_F \succeq 0$. Evaluating the signed identity on the diagonal proves the exact contraction criterion. □

Zero horizontal defect is sufficient, but is not necessary, for eq. (3.42). For a witness, take the unit-variance normal location family under translations on both levels, a fiber map adding independent unit-variance normal noise, so that the coarse location Fisher metric is $\frac{1}{2}$ and $\Delta_F = \frac{1}{2}$, and connections differing by a constant c, so that $a_\Psi = c$ in the location coordinate. For $u_X = 1$, the base tensors are 1 and $\frac{1}{2}(1+c)^2$. The difference $1 - \frac{1}{2}(1+c)^2$ equals $\frac{79}{200}$ at $c = \frac{1}{10}$, which is contraction with a nonzero defect, and $-\frac{1}{8}$ at $c = \frac{1}{2}$, where contraction fails. Information discarded in the fiber and mismatch of horizontal transports enter different terms. For two successive smooth statistical maps R_{01}, R_{12}, the vertical defects also compose exactly:

$$\Delta_{02} = \Delta_{01} + R^*_{01}\Delta_{12}, \qquad \Delta_{ab} = g_a - R^*_{ab} g_b. \tag{3.43}$$

This follows by inserting and subtracting $R^*_{01} g_1$ and using contravariance of pullback. Both terms are nonnegative when the two maps are fixed Markov morphisms.

Proposition 3.17 (Retention of the collective connection). Let $R : \mathcal{N} \to \bar{\mathcal{N}}$ be a smooth G-equivariant map between positive-metric collective statistical families, independent of the contextual coordinate. Denote their metrics by $g, \bar{g}$, their infinitesimal orbit maps by $J, \bar{J}$, and put $\Delta = g - R^* \bar{g}$. Let $\boldsymbol{p}(c)$ be a smooth fine section, and let A^F and $r_X = d\boldsymbol{p}(X) + JA^F(X)$ be a fine minimizing representative and residual. The same representative minimizes the cost at $\bar{\boldsymbol{p}} = R(\boldsymbol{p})$ exactly when

$$\Delta(r_X, J\xi) = 0 \quad \text{for all } X, \xi. \tag{3.44}$$

Under this condition, equality of the entire fine and coarse affine minimizing classes holds exactly when $\ker J = \ker \bar{J}$. For every possible fine first jet, the condition is $\Delta(H, \operatorname{im} J) = 0$. If $\Delta \succeq 0$, vanishing loss on every orbit direction is sufficient. This includes a fixed Markov morphism equipped with the Fisher metrics of its input and output experiments in the same normalization; arbitrary independently reweighted metrics need not satisfy that premise.

Proof. Equivariance gives $dRJ = \bar{J}$. Thus the coarse residual using A^F is dRr_X. Its normal equation is $\bar{g}(dRr_X, \bar{J}\xi) = 0$. Fine orthogonality gives $g(r_X, J\xi) = 0$, so the coarse equation is equivalent to eq. (3.44). Once the representative minimizes both costs, their classes are $A^F + \ker J$

and $A^F + \ker \bar{J}$, proving the equality criterion. Every horizontal vector is realizable as a residual by choosing the first jet equal to that vector, which gives the all-jets statement. When $\Delta \succeq 0$, Cauchy–Schwarz for this semidefinite form makes $\Delta(v, J\xi) = 0$ whenever $\Delta(J\xi, J\xi) = 0$. □

Contraction alone is insufficient. Two independent unit-variance normal location laws with $J\xi = (\xi, \xi)$ and means $(x, -x)$ have Fisher matrix I and minimizing coefficient zero. Adding independent unit normal noise only to the first sample is a fixed equivariant channel. Its coarse Fisher matrix is $\mathrm{diag}(1/2, 1)$, whose cost $\frac{1}{2}(1+a)^2 + (-1+a)^2$ is minimized at $a = 1/3$. The minimizing connection has changed despite Fisher contraction. Its preservation is an additional requirement beyond the dynamical and response conditions in chapter 6. The minimal tensor itself nevertheless contracts, and the connection shift is measured exactly.

Corollary 3.18 (Exact contraction of the minimal tensor). In the setting of proposition 3.17, assume that the coarse orbit map $\bar{J}$ also has constant rank along $\bar{\boldsymbol{p}} = R(\boldsymbol{p})$, and let $\bar{A}^F$ be a coarse minimizing representative with minimal tensor $\bar{h}^{\min}$ and locked inertia $\bar{\mathfrak{k}}$. Then

$$h^{\min}(X, X) - \bar{h}^{\min}(X, X) = \Delta(r_X, r_X) + \bar{\mathfrak{k}}\big((A^F - \bar{A}^F)(X), (A^F - \bar{A}^F)(X)\big). \tag{3.45}$$

The right side does not depend on the coarse representative. If $\Delta \succeq 0$, in particular for a fixed Markov morphism with Fisher metrics in the same normalization, both terms are nonnegative, so $\bar{h}^{\min} \leq h^{\min}$; the second term vanishes exactly when eq. (3.44) holds.

Proof. By equivariance, the coarse residual of the fine representative is $dR(r_X)$. Applying eq. (3.26) at the coarse level to the connection A^F gives $\bar{g}(dR(r_X), dR(r_X)) = \bar{h}^{\min}(X, X) + \bar{\mathfrak{k}}((A^F - \bar{A}^F)(X), (A^F - \bar{A}^F)(X))$. The left side equals $g(r_X, r_X) - \Delta(r_X, r_X) = h^{\min}(X, X) - \Delta(r_X, r_X)$. The form $\bar{\mathfrak{k}}$ annihilates $\ker \bar{J}$, which gives representative independence. The last statement is the criterion of proposition 3.17. □

In the two-location example above, $h^{\min} = 2$, $\bar{h}^{\min} = \frac{4}{3}$, $\Delta(r, r) = \frac{1}{2}$, and the inertia term is $\frac{3}{2}(\frac{1}{3})^2 = \frac{1}{6}$. For successive equivariant retentions the identity telescopes, each step contributing its own pair of nonnegative terms. Combined with proposition 3.10, when $\Delta \succeq 0$ the map induced by R between shape spaces does not increase the quotient metric along the section.

Chapter 4

Free Energies and Coupled Dynamics of Agent Laws

Interacting agents can be given three different objectives, and this chapter keeps them apart. In the global picture all agents are parts of one normalized joint over their contents and records, and each agent's local free energy is that global free energy viewed as a function of its own recognition factor. In the per-agent picture each agent has its own normalized generative model in the sense of definition 1.10, in which its own evaluator and the communicated laws of its peers are competing sources of prior information. The third object is a direct interaction potential between laws, specified without an observation likelihood. Its gradient flow moves each source toward the receivers that attend to it. We first develop this reciprocal potential, its source rows, and its Fisher dynamics, because the constructions of chapter 7 to chapter 13 use it. We then give the two free energies and prove that the reciprocal potential is not the free energy of any fixed normalized model in which the agent laws are recognition laws.

4.1 A potential for the two interacting sectors

Fix a context and a finite set of agents. Agent i has state and model spaces $\mathsf{K}_i, \mathsf{M}_i$, both standard Borel. In this chapter its joint law is restricted explicitly to $\zeta_i = q_i \otimes s_i$. Choose a proper model reference r_i and a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$. The reference joint is

$$R_i(dk, dm) = r_i(dm)G_i(m, dk), \qquad \mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i). \tag{4.1}$$

The evaluator couples the sectors even though their recognition law is a product. Where the terms are finite, disintegration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot))s_i(dm). \tag{4.2}$$

This is a conditional relative-entropy identity: the model marginal contributes the first term and the state conditional contributes the second. In particular, the expectation of a log evaluator in this objective is different from the log of the predictive mixture $s_i G_i$.

Remark 4.1 (What the product restriction removes). Over unrestricted joint laws the unique minimizer of $D_{\mathrm{KL}}(\zeta_i \| R_i)$ is $\zeta_i = R_i$, whose belief conditional is the evaluator. Under the product restriction and an evaluator density g_i, the belief minimizer at fixed s_i is instead $q_i \propto \exp \int \log g_i(m, \cdot)s_i(dm)$, which differs from the model prediction $s_i G_i$. The restriction therefore carries part of the distinction between belief and model. It is also not invariant under the

joint-law dynamics. For binary $k, m \in \{-1, 1\}$, let the joint Fisher flow be $\partial_t \log \zeta_i = -(h - \mathbb{E}_{\zeta_i} h)$ with $h(k, m) = \log(\zeta_i / R_i)(k, m) + h^q(k) + h^s(m)$, where the peer terms h^q, h^s depend only on the marginals. The additive terms cancel in the log odds ratio $\mathrm{LOR}(\zeta) = \sum_{k,m} km \log \zeta(k, m)$, so $\partial_t \mathrm{LOR}(\zeta_i) = -(\mathrm{LOR}(\zeta_i) - \mathrm{LOR}(R_i))$. Dependence between the sectors is generated at unit rate whenever R_i is not a product. The product flow used below is the restriction of the dynamics to the product family, not a consequence of it.

Let $\mathbf{q} = (q_i)_i$ and $\mathbf{s} = (s_i)_i$. For each allowed directed edge $j \to i$, choose normalized channels $\mathcal{K}^q_{ij}[\mathbf{q}] : \mathsf{K}_j \rightsquigarrow \mathsf{K}_i$ and $\mathcal{K}^s_{ij}[\mathbf{s}] : \mathsf{M}_j \rightsquigarrow \mathsf{M}_i$. Their dependence on the laws is part of the specified interaction. A law-dependent channel is required to be a kernel $\mathcal{K}^q_{ij} : \mathsf{S} \times \mathsf{K}_j \rightsquigarrow \mathsf{K}_i$, jointly in the configuration and the source value, and likewise in the model sector. Then $x \mapsto \nu^q_{ij}(x)$ is Borel by a monotone-class argument, relative entropy is Borel on pairs of laws through its supremum over finite partitions from a countable generating algebra [68], and the costs, Gibbs rows, and reduced potential below are Borel functions of the configuration, as proposition 1.8 and the record appendix require. Define

$$\begin{aligned} \nu^q_{ij} &= q_j \mathcal{K}^q_{ij}[\mathbf{q}], \quad D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij}), \\ \nu^s_{ij} &= s_j \mathcal{K}^s_{ij}[\mathbf{s}], \quad D^s_{ij} = D_{\mathrm{KL}}(s_i \| \nu^s_{ij}). \end{aligned} \tag{4.3}$$

The source sets $\mathcal{N}^q_i, \mathcal{N}^s_i$ are finite and nonempty, with positive normalized reference rows π^q_i, π^s_i. Belief routing uses β_i and model routing uses γ_i on their respective sets.

Definition 4.2 (Reciprocal law potential). For fixed $T_q, T_s > 0$, set

$$\begin{aligned} \mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_{j \in \mathcal{N}^q_i} \beta_{ij} D^q_{ij} + T_q D_{\mathrm{KL}}(\beta_i \| \pi^q_i) \right] \\ &+ \sum_i \left[\sum_{j \in \mathcal{N}^s_i} \gamma_{ij} D^s_{ij} + T_s D_{\mathrm{KL}}(\gamma_i \| \pi^s_i) \right]. \end{aligned} \tag{4.4}$$

The reference laws, evaluators, source priors, and temperatures are held fixed in the variations below. The admissible domain consists of laws for which all displayed divergences are finite.

This potential is a constitutive choice at the level of agent laws. No distribution over these law configurations is required to define it. It contains no record; records enter its dynamics only through the assimilation events of definition 4.24. Its gradient is reciprocal: a law that serves as a source appears in the cost of every receiver that attends to it, so the flow moves sources as well as receivers (section 4.3). Section 4.6 gives the per-agent alternative in which peers' laws are data, and proposition 4.21 shows that this potential is not a global variational free energy. General probability laws enter eq. (4.4); differentiability will require stronger hypotheses. An identity self-channel has $D^q_{ii} = 0$ throughout its domain, with the same convention in the model sector. Independent opposite attending edges remain independent data.

The receiver side of eq. (4.4) is a pooling rule. With the other agents, rows, and channels frozen, and with the appearances of q_i as a source ignored, stationarity of q_i gives $\log q_i = (1 + \sum_j \beta_{ij})^{-1} \big(\int \log g_i(m, \cdot) s_i(dm) + \sum_j \beta_{ij} \log \nu^q_{ij} \big) + \mathrm{const}$, a logarithmic opinion pool of the evaluator and the compared laws [1, 63]. Iterated relative-entropy averaging over a network is the consensus

scheme of Battistelli and Chisci [12], with DeGroot [44] as its linear antecedent. The source side, which proposition 4.5 identifies, is not part of these pooling rules.

The correlated local inference of section 5.2 and proposition B.2 makes a different use of the same generative-model definition: it conditions a declared joint exactly before any output projection. Its component evidence weights and finite conditioning step do not identify it with the product-family population flow considered here.

4.2 Source selection and reduction

Proposition 4.3 (Gibbs rows and the exact reduced potential). Hold the agent laws fixed and suppose every source cost is finite. Define

$$\begin{aligned} Z_i^q &= \sum_j \pi_{ij}^q e^{-D_{ij}^q/T_q}, \quad \beta_{ij}^* = \frac{\pi_{ij}^q e^{-D_{ij}^q/T_q}}{Z_i^q}, \\ Z_i^s &= \sum_j \pi_{ij}^s e^{-D_{ij}^s/T_s}, \quad \gamma_{ij}^* = \frac{\pi_{ij}^s e^{-D_{ij}^s/T_s}}{Z_i^s}. \end{aligned} \tag{4.5}$$

These are the unique minimizing rows. With

$$\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z_i^q - T_s \sum_i \log Z_i^s, \tag{4.6}$$

the exact identity is

$$\mathcal{E} = \mathcal{E}_{\mathrm{red}} + T_q \sum_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*) + T_s \sum_i D_{\mathrm{KL}}(\gamma_i \| \gamma_i^*). \tag{4.7}$$

Proof. For one row, $\log \beta_j^* = \log \pi_j - D_j/T - \log Z$. Substitute this expression into $T D_{\mathrm{KL}}(\beta \| \beta^*)$ to obtain the row objective plus $T \log Z$. Summing proves eq. (4.7). Relative entropy is nonnegative and vanishes exactly when its two probability laws agree, which proves uniqueness, including competitors on the simplex boundary. □

Proposition 4.3 is the finite case of the Gibbs variational principle, the convex duality between relative entropy and the log moment generating function [49, 50]. What is specific here is that the costs D_{ij} are themselves divergences between agent laws, so the rows move with the laws they select.

On a differentiable family, direct differentiation of the finite sums in eq. (4.5) gives

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta_{ij}^* dD_{ij}^q + \sum_{ij} \gamma_{ij}^* dD_{ij}^s. \tag{4.8}$$

Indeed, $-T_q d \log Z_i^q = \sum_j \beta_{ij}^* dD_{ij}^q$, and the other sector is identical. There is no derivative of an optimized row to add. Every dependence of its cost on the laws still belongs in dD. If a source prior were varied, this formula would also contain $-T_q \sum_{ij} \beta_{ij}^* d \log \pi_{ij}^q$ and its model counterpart.

Remark 4.4 (Identity self sources and learned source priors). Suppose the source set of i contains i with an identity self-channel, so $D^q_{ii} = 0$. Since every cost is nonnegative, $\pi^q_{ii} \le Z^q_i \le 1$, hence $\beta^*_{ii} \ge \pi^q_{ii}$ and $\beta^*_{ij} \le (\pi^q_{ij}/\pi^q_{ii})e^{-D^q_{ij}/T_q}$ for $j \ne i$. A peer whose divergence exceeds $T_q \log(\pi^q_{ij}/(\pi^q_{ii}\varepsilon))$ receives row weight below ε, so at fixed priors the reduced interaction is a smooth form of bounded-confidence opinion dynamics [74]. If instead the prior row π^q_i is made a dynamical table of eq. (4.4) with unit Fisher mobility, then $\partial\mathcal{E}/\partial\pi^q_{ij} = -T_q\beta_{ij}/\pi^q_{ij}$ and its Fisher flow is exactly

$$\dot\pi^q_i = T_q(\beta_i - \pi^q_i). \tag{4.9}$$

With Gibbs rows this is $\dot\pi^q_{ij} = (T_q/Z^q_i)\pi^q_{ij}(e^{-D^q_{ij}/T_q} - Z^q_i)$, a replicator equation with fitness e^{-D/T_q} [78], slow at small T_q without an inserted multiplier. An identity self source then has $\dot\pi^q_{ii} = (T_q/Z^q_i)\pi^q_{ii}(1 - Z^q_i) \ge 0$, and the infimum of the row terms over priors and rows is approached in the limit of isolation, where each agent attends only to itself. A learned-prior mechanism therefore needs self excluded from the learnable rows or its self prior held fixed. Whether learned rows without self concentrate on a pairing is open.

4.3 The derivative includes every interaction role

To display the variation, choose densities against fixed references on a regular dominated family. Write a for the density of a receiver law and b for that of its compared law. For mass-preserving variations and justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b)da - \int (a/b)db. \tag{4.10}$$

Here the notation $\int f da$ denotes integration against the signed variation of the law, rather than differentiation of the sample coordinate. The formula follows by differentiating $a\log(a/b)$; the additional integral of da is zero.

For a belief interaction the compared-law variation is

$$d\nu^q_{ij} = (dq_j)\mathcal{K}^q_{ij}[\mathbf{q}] + q_j D\mathcal{K}^q_{ij}[\mathbf{q}](d\mathbf{q}). \tag{4.11}$$

The second term differentiates the prescribed channel itself. Thus an agent contributes when it is a receiver, when it is a source, and whenever its law changes a channel elsewhere. For fixed discrete kernels, the incoming contribution from $j \to i$ to a variation at source value v is

$$-\beta_{ij}\sum_u \frac{q_i(u)}{\nu^q_{ij}(u)}\mathcal{K}^q_{ij}(v,u). \tag{4.12}$$

The model sector has the same expression with q, β replaced by s, γ. If the channels are constructed from law-dependent moments or generators, eq. (4.11) also differentiates those moments and generators. Keeping their numerical values fixed computes a different vector field.

An identity self-channel illustrates why the variable identification matters. Here $a = b$ and $da = db$, so eq. (4.10) is zero throughout the state space. Freezing the source as an old snapshot while moving the receiver replaces this zero functional by a divergence from that snapshot. Its first derivative

still vanishes at equality, but its curvature and its force after displacement differ. More generally, directed source selection does not make the gradient of eq. (4.4) one-way: a source appears in the energy of every row that attends to it. Each directed edge $j \to i$ therefore carries influence in both directions. The force on the source is the receiver likelihood ratio q_i/ν_{ij}^q pulled back through the adjoint of the channel, so it is supported on what the channel reads; for a deterministic readout T it is a function of $T(v)$, and only the message marginal of the source moves. The communication model of chapter 5 sends messages from source to receiver only. A population of communicating agents realizes the reciprocal flow only if this reverse message is also modeled; otherwise each source must be able to read the laws of every receiver that attends to it. Every agent descends the same total $\mathcal{E}$, so the population plays an identical-interest game in the sense of Monderer and Shapley [109]. Section 4.6 shows what is lost when each agent descends only its own objective.

Proposition 4.5 (Receiver and source pools). Let the receivers $i \in I_j$ of a source j use a common fixed channel K_j, with weights $\beta_{ij} > 0$, $W_j = \sum_{i\in I_j} \beta_{ij}$, and pooled receiver law $\bar{q}_{\to j} = W_j^{-1} \sum_{i\in I_j} \beta_{ij} q_i$. If all displayed divergences are finite,

$$\sum_{i\in I_j} \beta_{ij} D_{\mathrm{KL}}(q_i \| q_j K_j) = \sum_{i\in I_j} \beta_{ij} D_{\mathrm{KL}}(q_i \| \bar{q}_{\to j}) + W_j D_{\mathrm{KL}}(\bar{q}_{\to j} \| q_j K_j). \tag{4.13}$$

Proof. Expand both sides with $\int \log$ of densities against a common reference. The receiver entropies agree. On the right, $-\sum_i \beta_{ij} \int \log \bar{q}_{\to j} dq_i + W_j \int \log \bar{q}_{\to j} d\bar{q}_{\to j} = 0$ because $\sum_i \beta_{ij} q_i = W_j \bar{q}_{\to j}$, and the remaining cross term equals the left cross term for the same reason. □

The first term on the right does not contain q_j. At fixed rows the complete source-role force on q_j is therefore the force of $W_j D_{\mathrm{KL}}(\bar{q}_{\to j} \| q_j K_j)$, which fits one source law to the arithmetic pool of its receivers in the mass-covering direction, while the receiver-role force is the logarithmic pool above. In a Gaussian covariance sector with identity channel, the source force on the variance v_j from one receiver $\mathcal{N}(m_i, v_i)$ is $(v_j - v_i - (m_i - m_j)^2)/(2v_j^2)$, which vanishes at $v_j = v_i + (m_i - m_j)^2$: a source more confident than its receivers' spread is widened.

4.4 Fisher dynamics with equal mobilities

The kinetic metric in this chapter uses its declared product recognition family. It is the product specialization of proposition 3.2, with separate Fisher blocks for the source rows. The contextual tensor in definition 3.3 instead evaluates changes along the base through a connection; it does not replace the vertical metric used for inference at a fixed context.

Theorem 4.6 (Local flow and exact dissipation on positive tables). Suppose the state and model alphabets are finite, all r_i and evaluator entries $G_i(m, k)$ are strictly positive, and the channel entries are twice continuously differentiable functions of the interior probability tables. Assume every compared-law entry is positive there. Let u^a range over all tables $q_i, s_i, \beta_i, \gamma_i$. Equip their product with the Fisher metric

$$g_u(v, w) = \sum_a \sum_z \frac{v_z^a w_z^a}{u_z^a}, \qquad \sum_z v_z^a = \sum_z w_z^a = 0. \tag{4.14}$$

For every interior initial condition there is a unique maximal interior solution of

$$\dot{u}_z^a = -u_z^a \left(h_z^a - \sum_w u_w^a h_w^a \right), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}. \tag{4.15}$$

Along this solution,

$$\frac{d\mathcal{E}}{dt} = -\sum_a \mathrm{Var}_{u^a}(h^a) \le 0. \tag{4.16}$$

Proof. The stated positivity and regularity make $\mathcal{E}$ twice continuously differentiable in interior simplex coordinates. The right side of eq. (4.15) is therefore locally Lipschitz, giving a unique local solution and its maximal continuation in the interior. Its entries sum to zero within each table. Centering h^a removes any ambiguity from extending $\mathcal{E}$ off the normalized tables. For every tangent vector v, the metric pairing of $u^a(h^a - \mathbb{E}_{u^a} h^a)$ with v^a equals $\sum_z h_z^a v_z^a$. This identifies the Fisher gradient. Finally, substitute eq. (4.15) into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ to obtain the sum of negative variances. Interior continuation does not assert that the trajectory stays away from the boundary for all time. □

On a single simplex eq. (4.15) is the replicator equation with fitness $-h$, the metric eq. (4.14) is the Shahshahani metric, and its identification as a gradient flow is Shahshahani's theorem [78, 128]; eq. (4.16) is the corresponding form of Fisher's fundamental theorem. Theorem 4.6 is that statement on a product of simplices with law-dependent channels. For fixed channels the local statement extends to all time.

Proposition 4.7 (Global flow for fixed channels). In addition to the hypotheses of theorem 4.6, suppose every channel is independent of the laws and has no null column: $c_{\min} = \min_{i,j,z} \sum_v \mathcal{K}_{ij}(v,z) > 0$ in each sector. Then every interior solution of eq. (4.15) exists for all $t \ge 0$, its entries are bounded below by a positive constant uniformly in time, $\int_0^\infty \sum_a \mathrm{Var}_{u^a}(h^a) dt \le \mathcal{E}(u(0))$, and it converges to a single critical point of $\mathcal{E}$.

Proof. Consider the belief tables. Constants in h cancel on centering, so replace the belief force of agent i by $\tilde{h}_z = \log q_i(z) - \int \log G_i(m,z) s_i(dm) + \sum_j \beta_{ij} \log(q_i(z)/\nu_{ij}(z)) + \sigma_z$, where $\sigma_z \le 0$ collects the incoming terms eq. (4.12). Let $m(t)$ be the minimum of $\log q_i(z)$ over agents and states, attained at (i,z). Every entry of every belief table is at least e^m, so $\nu_{ij}(z) \ge c_{\min} e^m$ and $\tilde{h}_z \le m + C_G + \log^+(1/c_{\min})$, with $C_G = \max(-\log G)$. On the other hand $\mathbb{E}_{q_i} \tilde{h} \ge -\log |\mathsf{K}_i| - N$: the entropy term is at least $-\log |\mathsf{K}_i|$, the evaluator and receiver terms are nonnegative, and each incoming term averages to $-\beta_{ki}$ under q_i. Hence the lower Dini derivative of m is at least $-m - C$ with $C = C_G + \log^+(1/c_{\min}) + \log \max_i |\mathsf{K}_i| + N$, and $m(t) \ge \min(m(0), -C)$. The model tables are treated in the same way, with $-\log r_i$ added to the bound. With these floors every D_{ij} is bounded by a constant $\bar{D}$, and the same argument applied to $h_j = D_{ij} + T(\log(\beta_{ij}/\pi_{ij}) + 1)$ bounds the rows below by $\min(\log \beta_{ij}(0), -\bar{D}/T - \log(1/\min \pi))$ in logarithm. The trajectory therefore stays in a compact subset of the interior, so it is global. Since $\mathcal{E} \ge 0$, integrating eq. (4.16) bounds the dissipated variance. With fixed channels $\mathcal{E}$ is real analytic on the interior of the product of simplices, and on the compact set the Fisher metric is uniformly equivalent to the Euclidean one, so the Łojasiewicz gradient inequality gives convergence of the bounded trajectory to one critical point [2, 102]. □

For law-dependent channels the channel-derivative term of eq. (4.11) can be unbounded near the boundary, and no statement of this kind follows without further hypotheses.

Proposition 4.8 (Row relaxation). Hold all costs of one row fixed and write $l_j = \log(\beta_j/\beta_j^*)$ with β^* from eq. (4.5). Under eq. (4.15), $\frac{d}{dt}(l_j - l_k) = -T(l_j - l_k)$ for every pair of sources.

Proof. The row force is $h_j = D_j + T(\log(\beta_j/\pi_j) + 1) = Tl_j + T(1 - \log Z)$, and eq. (4.15) gives $\frac{d}{dt}\log\beta_j = -(h_j - \mathbb{E}_\beta h)$. □

All blocks have unit mobility, and proposition 4.8 shows its price: under unit mobility, log-odds contrasts within a row relax at rate exactly T, so rows that select sharply are slow. Exact Gibbs rows are the singular limit of infinite row mobility, not a property of the unit-mobility flow. One may optimize the source rows at each state and apply the same construction to $\mathcal{E}_{\mathrm{red}}$ and the matter tables alone; eq. (4.8) gives its complete derivative. The two flows are different vector fields with different transients, but they share their critical points and their stability (proposition 4.9). Equal mobility has a justification beyond convention. Unit mobilities are the Fisher metric of the single product law of all tables, regarded as one law of all represented variables and source labels, because the Fisher metric of a product law restricted to the product family is the sum of the factor metrics. If the kinetic metric is required to be the restriction of a metric on the law space of all these variables jointly that is invariant under Markov congruent embeddings, Čencov's theorem [28] forces it to be a constant multiple of the Fisher metric, so equal mobility is fixed up to one global time scale. Imposing invariance on each table separately allows an independent constant for each table, equivalent to replicating that table. A persistent slow model sector would have to follow from the interactions and their state, rather than from an inserted small multiplier in its equation.

Proposition 4.9 (Full and reduced flows have the same critical structure). Under the hypotheses of theorem 4.6, $(u, \beta^*(u), \gamma^*(u))$ is a critical point of $\mathcal{E}$ exactly when u is a critical point of $\mathcal{E}_{\mathrm{red}}$, and at such a point the Hessians of $\mathcal{E}$ and $\mathcal{E}_{\mathrm{red}}$ on the tangent spaces have the same number of negative and zero eigenvalues.

Proof. At fixed laws the row terms are $\sum_j \beta_j D_j + TD_{\mathrm{KL}}(\beta\|\pi)$, whose Hessian on the tangent space of the simplex is T times the Fisher metric, hence positive definite; the row optimum is β^*, and eq. (4.8) identifies the critical points. The Hessian of $\mathcal{E}_{\mathrm{red}}$ is the Schur complement of the positive-definite row block in the Hessian of $\mathcal{E}$, and Haynsworth's inertia additivity [73] gives the claim. □

For a regular finite-dimensional family with positive-definite Fisher tensor $g_{ab}(\theta)$, the corresponding equation is $\dot\theta^a = -(g^{-1})^{ab}\partial_b\mathcal{E}$ [3]. Whenever the derivatives exist, contraction gives $\dot{\mathcal{E}} = -(\partial\mathcal{E})^T g^{-1}(\partial\mathcal{E})$. Bimeasurable re-expression of samples, carrying all kernels and reference laws with it, preserves relative entropy and Fisher score inner products. On a family preserved by that re-expression, the gradient therefore transforms as a tangent vector; this is passive covariance. Active equivariance would further need equivariant constitutive maps for the law-dependent channels, as in eq. (2.13), and transported references and evaluators. These statements concern fixed coordinate changes; coordinates that themselves evolve require their coordinate-velocity term.

Definition 4.10 (Restricted-family flow). Let $\theta \mapsto p_\theta$ be an admitted regular finite-dimensional family for some of the laws, with Fisher tensor $g(\theta)$. The restricted-family flow is $\dot\theta = -g(\theta)^{-1}\partial_\theta\mathcal{E}(p_\theta)$.

For a Gaussian law $\mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^d$, with $\nabla_\mu \mathcal{E}$ and the symmetric gradient $\nabla_\Sigma \mathcal{E}$, it reads

$$\dot{\mu} = -\Sigma \nabla_\mu \mathcal{E}, \qquad \dot{\Sigma} = -2\Sigma(\nabla_\Sigma \mathcal{E})\Sigma. \tag{4.17}$$

The restricted flow is the Fisher-orthogonal projection of the full velocity onto the tangent space of the family. It is the full flow only on families that the full flow leaves invariant, and the source role prevents this for Gaussian agents.

Proposition 4.11 (Gaussian agents are not an invariant family). Let all belief laws be Gaussian on $\mathbb{R}^d$ with identity channels and fixed rows, let the evaluators be Gaussian with means affine in m and fixed covariance, and let the model laws have finite second moments. Suppose exactly one receiver $k \neq i$ attends to i, with $\beta_{ki} > 0$ and $q_k \neq q_i$. Then the Fisher velocity $-(\varphi - \mathbb{E}_{q_i}\varphi)$ of q_i, with φ the first variation of $\mathcal{E}$ in q_i, is not the logarithmic derivative of any differentiable curve of Gaussian laws.

Proof. For a curve of Gaussian laws, $\partial_t \log q$ is a polynomial of degree at most two. The local term contributes $\log q_i - \int \log g_i(m, \cdot) s_i(dm)$, and each receiver term $\beta_{ij} \log(q_i/q_j)$; both are polynomials of degree at most two. By eq. (4.12) with an identity channel, the source term is $-\beta_{ki} q_k/q_i = -\beta_{ki} e^Q$ with Q a nonconstant polynomial of degree at most two. Along a line on which Q is nonconstant, e^Q is not a polynomial, since a polynomial $p = e^Q$ would satisfy $p' = Q'p$ with $\deg(Q'p) \geq \deg p > \deg p'$. Hence φ is not a polynomial of degree at most two. □

The inheritance and formation results of chapter 8 to chapter 13 are proved for full density charts. They do not transfer to the restricted Gaussian flow without a separate proof. Channels that are constant in their input, as in chapter 12, remove the source term and with it this obstruction.

The density-level flow used in those chapters needs its own hypotheses. For a law q on a standard Borel space with probability reference λ, write $q_l = e^l \lambda / \mathbb{E}_\lambda e^l$ for l in the chart $L^\infty_0(\lambda) = \{l \in L^\infty(\lambda) : \mathbb{E}_\lambda l = 0\}$, and use log-odds coordinates for rows. Suppose $\mathcal{E}$ has a first variation $\varphi_q \in L^\infty(\lambda)$ in each law, in the sense that $\frac{d}{dt}\mathcal{E}(q_{l(t)}) = \sum \int \varphi_q d\dot{q}$ along every continuously differentiable chart curve.

Definition 4.12 (Fisher flow in bounded log-density charts). The chart Fisher flow is $\dot{l} = -(\varphi_{q_l} - \mathbb{E}_\lambda \varphi_{q_l})$ for every law, together with eq. (4.15) for the rows.

Proposition 4.13 (Local well-posedness in charts). Suppose the map from chart coordinates to the first variations $\varphi_q \in L^\infty(\lambda)$ is Lipschitz on bounded sets. Then for every initial point there is a unique maximal continuously differentiable solution on $[0, t^*)$, and if $t^* < \infty$ the chart coordinates are unbounded as $t \to t^*$. Along it, $\partial_t \log q = -(\varphi_q - \mathbb{E}_q \varphi_q)$ and $d\mathcal{E}/dt = -\sum \mathrm{Var}_q(\varphi_q) \leq 0$, the sum running over all laws and rows. The hypothesis holds for relative-entropy terms when the reference laws and evaluators have bounded log densities against λ and every channel is either an identity between agents sharing λ or a fixed kernel with densities bounded above and below by positive constants; for law-dependent channels it is an additional hypothesis on the first variation of the channel.

Proof. The maps $l \mapsto \mathbb{E}_{q_l} f = \mathbb{E}_\lambda[f e^l]/\mathbb{E}_\lambda e^l$ are locally Lipschitz in $(l, f) \in L^\infty \times L^\infty$, so the vector field is Lipschitz on bounded sets, and the Picard–Lindelöf theorem in a Banach space gives

local existence, uniqueness, and the continuation alternative [47]. Since $\log q_l = l - \log \mathbb{E}_\lambda e^l$ and $\frac{d}{dt} \log \mathbb{E}_\lambda e^l = \mathbb{E}_q \dot{l}$, the logarithmic velocity is $-(\varphi_q - \mathbb{E}_q \varphi_q)$; integrating it against $\varphi_q dq$ gives the variance. Under the stated bounds, the receiver terms $\log(q_i/\nu_{ij})$ and evaluator terms are bounded, the source terms $-\int (q_i/\nu_{ij})(w) \mathcal{K}_{ij}(v, dw)$ are bounded, and each is a composition of exponentials, logarithms on bounded ranges, and integrals against fixed bounded kernels, hence Lipschitz on bounded sets. □

Global statements on general spaces need maximum-principle estimates for particular channel classes and are stated where they are used. The Fisher–Rao metric used here is, up to scale, the metric singled out by invariance under sufficient statistics [5, 28].

4.5 A global free energy and its local free energies

Choose a proper prior $P_0(dy, dJ)$ on represented contents and source labels, and a normalized record kernel $\mathscr{R}(do \mid y, J)$. Then $\mathbb{P} = P_0 \mathscr{R}$ is normalized by integrating the record first. At a declared record density $\ell_o(y, J)$ with $0 < Z_o = \int \ell_o dP_0 < \infty$, its posterior is $\Pi_o = \ell_o P_0 / Z_o$. Substitution proves

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o, \tag{4.18}$$

when the split terms are defined. The right side supplies the extended definition otherwise. This is the variational free energy of a fixed model [59, 80, 112]. A product family for Q is permitted and its block updates remain coupled through the likelihood factors. Each block derivative contains every factor incident on it.

In the global picture the represented contents of all agents are variables of one such model. Write $y = (y_1, \dots, y_N)$ with $y_i \in \mathsf{Y}_i$, where Y_i collects the state and model coordinates of agent i, and let λ_i be a sigma-finite reference on Y_i. Peer interactions are likelihood factors: for example, a comparison record eq. (1.13) between agents i and j contributes a factor that depends on (y_i, y_j).

Proposition 4.14 (Local free energies of a global model). Let Π_o have a density p with respect to $\bigotimes_j \lambda_j$, and restrict recognition to products $Q = \bigotimes_j Q_j$ with $Q_j = a_j \lambda_j$. Assume $\int |a_j \log a_j| d\lambda_j < \infty$ for every j and $\int |\log p| dQ < \infty$. Fix i and the factors $Q_{-i} = \bigotimes_{j \neq i} Q_j$, and put

$$\bar{e}_i(y_i) = \int \log p(y_i, y_{-i}) Q_{-i}(dy_{-i}), \qquad Z_i = \int e^{\bar{e}_i} d\lambda_i. \tag{4.19}$$

If $0 < Z_i < \infty$, then with $Q_i^* = e^{\bar{e}_i} \lambda_i / Z_i$,

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q_i \| Q_i^*) - \log Z_i + \sum_{j \neq i} \int a_j \log a_j d\lambda_j - \log Z_o. \tag{4.20}$$

Consequently the local free energy of agent i, defined as $\mathcal{F}$ viewed as a function of Q_i alone, has the unique minimizer Q_i^*, every update of one factor to its minimizer lowers $\mathcal{F}$ or leaves it unchanged, and $\mathcal{F}$ is a Lyapunov function for every schedule of such block updates. The peers enter $\bar{e}_i$ only through expectations, so $\bar{e}_i$ is affine in each Q_j with $j \neq i$.

Proof. By the assumed integrability, $\mathcal{F}(Q;o) = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o = \sum_j \int a_j \log a_j d\lambda_j - \int \log p dQ - \log Z_o$. Fubini's theorem gives $\int \log p dQ = \int \bar{e}_i dQ_i$. Since $\log(dQ_i/dQ_i^*) = \log a_i - \bar{e}_i + \log Z_i$, the terms involving Q_i equal $D_{\mathrm{KL}}(Q_i\|Q_i^*) - \log Z_i$, which proves eq. (4.20). The remaining statements follow because relative entropy is nonnegative and vanishes only at equality, and because $\bar{e}_i$ is an integral of a fixed function against the product Q_{-i}. □

This is the coordinate-ascent identity of mean-field variational inference [15, 18], here read as a statement about agents: each agent minimizing its own local free energy, with its peers' current factors given, performs coordinate descent of one collective functional. The identification requires that every agent's model be a conditional of the same joint and that peers act through expectations of their contents. The latter property is what the two other objectives of this chapter lack.

With independent source priors and one likelihood factor reading each label, a factorized source row has cost $C_{ij} = \mathbb{E}_{Q_Y}[-\log \ell_i(o_i \mid Y, j)]$ and optimizer $\beta_{ij} \propto \pi_{ij} e^{-C_{ij}}$. This expected log likelihood differs from both the peer divergence and the likelihood integrated over a prior. For an isolated source record, exact source inference gives $\mathbb{P}(J = j \mid o) \propto \pi_j \mathbb{E}_{P_Y} \ell_j(o \mid Y)$. Equality of these different rows needs additional hypotheses; proposition 4.16 and theorem 4.22 give two exact settings.

Proposition 4.15 (Entropy and the likelihood Jensen gap). On standard Borel spaces, let $k(w \mid v)$ be a density of a normalized kernel into a receiver space with sigma-finite reference λ. Let the receiver law be $q = a\lambda$ and the source law be p. Put $b(w) = \int k(w \mid v) p(dv)$. Assume $0 < b(w) < \infty$ for q-almost every w, $k > 0$ almost everywhere under $q \otimes p$, and finiteness of $\mathbb{E}_q|\log a|$, $\mathbb{E}_q|\log b|$, and $\mathbb{E}_{q\otimes p}|\log k|$. Define $p^w(dv) = k(w \mid v)p(dv)/b(w)$ on this full-measure set. Then

$$\mathbb{E}_{q\otimes p}[-\log k(W \mid V)] = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK) + \int D_{\mathrm{KL}}(p\|p^w) q(dw). \tag{4.21}$$

Proof. The density ratio gives $D_{\mathrm{KL}}(p\|p^w) = \log b(w) - \int \log k(w \mid v) p(dv)$. Integrating against q and using $-\int \log b dq = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK)$ proves the identity. The last term vanishes exactly when $p^w = p$ for q-almost every w. □

The entropy here is relative to the declared reference. A record likelihood normalized in o need not be normalized in the receiver coordinate w, so proposition 4.15 requires that additional kernel interpretation. Even when it applies, the entropy and conditional divergence can affect the dynamics.

The identity gives the exact relation between the Gibbs rows and a normalized source-label model in one sector.

Proposition 4.16 (Gibbs rows and a normalized label model). Let all alphabets be finite with counting reference, let the represented contents $Y_i \sim r_i$ be independent, let labels $J_i \sim \pi_i$ range over nonself sources, and let each receiver register one binary record whose success probability given $J_i = j$ is $k_{ij}(Y_i \mid Y_j)^{1/T}$, where $k_{ij}(w \mid v) = \mathcal{K}_{ij}(v, \{w\}) > 0$. With all records registered and product recognition $\bigotimes_i q_i \otimes \bigotimes_i \beta_i$, the free energy $\mathcal{F}$ of this model satisfies

$$T\mathcal{F} = \mathcal{E} + \sum_i \big[H(q_i) + (T-1) D_{\mathrm{KL}}(q_i\|r_i)\big] + \sum_{ij} \beta_{ij} G_{ij}, \qquad G_{ij} = \int D_{\mathrm{KL}}(q_j\|q_j^w) q_i(dw), \tag{4.22}$$

where $\mathcal{E}$ is eq. (4.4) in this sector with local term $\sum_i D_{\mathrm{KL}}(q_i\|r_i)$, and q_j^w is the source conditional of proposition 4.15. The recognition-optimal rows are $\beta_{ij} \propto \pi_{ij} e^{-(D_{ij}+G_{ij})/T}$, and they equal the Gibbs rows eq. (4.5) exactly when G_{ij} does not depend on j.

Proof. Since $k_{ij} \le 1$ on a finite alphabet, the record probabilities are normalized. Substitution gives $T\mathcal{F} = T\sum_i D_{\mathrm{KL}}(q_i\|r_i) + T\sum_i D_{\mathrm{KL}}(\beta_i\|\pi_i) + \sum_{ij}\beta_{ij}\mathbb{E}_{q_i\otimes q_j}[-\log k_{ij}]$. Apply eq. (4.21) to each expectation and use $\sum_j \beta_{ij} = 1$. The row statement is proposition 4.3 with costs $D_{ij} + G_{ij}$. □

This generalizes the Gaussian calculation of the supplement. A channel whose kernel vanishes on a set charged by $q_i \otimes q_j$, such as a deterministic bit-transmitting channel, makes the expected log likelihood infinite and has no reading as a label likelihood of this kind.

There is also an exact obstruction to universal equality. Freeze a channel K and receiver q. Suppose p_1K, p_2K have positive densities that differ on a set of positive q-measure, with both divergences finite. Strict convexity of $-\log$ gives

$$D_{\mathrm{KL}}(q\|(tp_1 + (1-t)p_2)K) < tD_{\mathrm{KL}}(q\|p_1K) + (1-t)D_{\mathrm{KL}}(q\|p_2K), \qquad 0 < t < 1. \tag{4.23}$$

This follows pointwise on the differing-density set and then by integration. By contrast, the expectation $\mathbb{E}_{q\otimes p}e(W,V)$ of any fixed integrable sample energy is affine in p. It cannot equal the divergence at both endpoints and their strict mixture. The obstruction concerns a fixed sample energy; it does not exclude a direct law potential, a frozen-template episode, or an explicitly different probability model whose variables are law points.

4.6 Per-agent free energies

In the per-agent picture each agent has its own normalized generative model, and the communicated laws of its peers enter that model as data. The agent's own evaluator and reference become one source among several, and a latent label records which source supplies the prior.

Definition 4.17 (Per-agent mixture-source model). Fix agent i and the current laws of its peers. Let the finite source sets $\mathcal{N}_i^q, \mathcal{N}_i^s$ contain the self label i, with positive normalized priors π_i^q, π_i^s. Define content sources and presentation sources by

$$\kappa_{ii}(m,\cdot) = G_i(m,\cdot), \quad \kappa_{ij}(m,\cdot) = \nu_{ij}^q \ (j \ne i), \qquad \rho_{ii} = r_i, \quad \rho_{il} = \nu_{il}^s \ (l \ne i), \tag{4.24}$$

with ν_{ij}^q, ν_{il}^s as in eq. (4.3). The per-agent model on (m,k,J,L,o) is

$$P_i(dm,dk,J=j,L=l,do) = \pi_{il}^s \rho_{il}(dm)\pi_{ij}^q \kappa_{ij}(m,dk)\ell_i(k,m;do). \tag{4.25}$$

It is normalized for every value of the peer laws. Those laws are parameters of agent i's model, not variables over which it infers.

Proposition 4.18 (Per-agent free energy). In eq. (4.25), fix a record value o with a declared density of ℓ_i and $0 < P_i(o) < \infty$. Take recognition $s_i \otimes q_i \otimes \beta_i \otimes \gamma_i$ on (m,k,J,L), and assume that

every term below is finite. Then

$$\begin{aligned}\mathcal{F}_i = &\sum_l \gamma_{il} D_{\mathrm{KL}}(s_i \| \rho_{il}) + D_{\mathrm{KL}}(\gamma_i \| \pi_i^s) + \sum_j \beta_{ij} \int D_{\mathrm{KL}}(q_i \| \kappa_{ij}(m,\cdot)) s_i(dm) \\ &+ D_{\mathrm{KL}}(\beta_i \| \pi_i^q) + \iint -\log \ell_i(o \mid k, m) q_i(dk) s_i(dm).\end{aligned} \tag{4.26}$$

For fixed laws the minimizing rows are the Gibbs rows of proposition 4.3 at unit temperature, with self costs $D_{\mathrm{KL}}(s_i \| r_i)$ and $\int D_{\mathrm{KL}}(q_i \| G_i(m,\cdot)) s_i(dm)$ and peer costs D^s_{il}, D^q_{ij}.

Proof. Write densities of ρ_{il} and $\kappa_{ij}(m,\cdot)$ against the references of s_i and q_i. Then $-\log$ of eq. (4.25) is the sum of $-\log \pi^s_{il}$, $-\log \rho_{il}(m)$, $-\log \pi^q_{ij}$, $-\log \kappa_{ij}(m,k)$, and $-\log \ell_i(o \mid k, m)$. Since $\mathcal{F}_i = \mathbb{E}_Q[\log Q - \log P_i(\cdot, o)]$, group each term with the entropy of the recognition factor it shares. The label terms give $D_{\mathrm{KL}}(\gamma_i \| \pi^s_i) + D_{\mathrm{KL}}(\beta_i \| \pi^q_i)$. Conditioning on $L = l$ and integrating $\log s_i - \log \rho_{il}$ against s_i gives $\sum_l \gamma_{il} D_{\mathrm{KL}}(s_i \| \rho_{il})$. Conditioning on $J = j$ and m and integrating $\log q_i - \log \kappa_{ij}(m,\cdot)$ against q_i gives the third term. The record term remains. The row statement is proposition 4.3 with $T_q = T_s = 1$. □

At unit temperatures the peer terms of eq. (4.26) are exactly the peer terms of the reciprocal potential, and the self terms reproduce the split eq. (4.2) of $\mathcal{E}_0$, now weighted by the self probabilities β_{ii}, γ_{ii}. The agent's own model competes with its peers for the role of prior, and the rows infer which source to trust. Records add a likelihood term that persists in the objective. A temperature $T \neq 1$ multiplies the label divergences; dividing by T shows that it corresponds to tempered peer and self factors, a generalized-Bayes objective in the sense of Bissiri et al. [16] rather than a normalized model.

Definition 4.19 (Per-agent dynamics). On finite positive tables satisfying the hypotheses of theorem 4.6, let each table u^a belong to one agent $i(a)$. The per-agent dynamics is eq. (4.15) with $h^a_z = \partial \mathcal{F}_{i(a)} / \partial u^a_z$, where the peer laws inside $\mathcal{F}_{i(a)}$ take their current values.

The proof of theorem 4.6 applies without change and gives a unique maximal interior solution. The dissipation identity does not: each agent lowers its own free energy at the rate of its own variances, while the free energies of the receivers that attend to it change through terms it does not control. Let $\mathcal{F}_{\mathrm{tot}} = \sum_i \mathcal{F}_i$, regarded as a function of all tables. The per-agent force on u^a omits from $\partial \mathcal{F}_{\mathrm{tot}} / \partial u^a$ exactly the derivatives of the other agents' free energies, namely the source-role terms eq. (4.12) and the channel derivatives eq. (4.11) of edges that $i(a)$ does not receive. The Fisher gradient flow of $\mathcal{F}_{\mathrm{tot}}$ keeps those terms and is the reciprocal flow of a potential of the form eq. (4.4), with the agent's own model entering as the self source instead of as the additive term $\mathcal{E}_0$. Results proved for the reciprocal flow therefore do not transfer to the per-agent dynamics without a separate argument.

Proposition 4.20 (Per-agent dynamics need not be a gradient flow). Take three agents with Bernoulli belief laws of log odds l_1, l_2, l_3, a trivial model sector, evaluators equal to Bernoulli references of log odds ϱ_i, identity channels, and rows held fixed at $\beta_{ii} = 1/(1+w)$ and $\beta_{i,i+1} = w/(1+w)$ with $w > 0$ and indices modulo three. The per-agent dynamics is

$$(1+w)\dot{l}_i = -(l_i - \varrho_i) - w(l_i - l_{i+1}), \tag{4.27}$$

whose linear part has eigenvalues $-1/(1+w)$ and $(-1-3w/2 \pm i\sqrt{3}w/2)/(1+w)$. No Riemannian metric makes eq. (4.27) the gradient flow of a function. By contrast, the linearization of the Fisher gradient flow of eq. (4.4) at any interior critical point is self-adjoint for the Fisher metric and has real spectrum.

Proof. For a Bernoulli law the Fisher flow in log odds is $\dot{l} = -\partial F/\partial p$, and $\partial_p D_{\mathrm{KL}}(\mathrm{Ber}(p)\|\,\mathrm{Ber}(p')) = l - l'$, which gives eq. (4.27). Its linear part is circulant, with eigenvalues $(-1-w+w\omega)/(1+w)$ for the cube roots of unity ω. At an equilibrium of a gradient flow $\dot{x} = -g^{-1}\nabla F$ the linearization is $-g^{-1}\operatorname{Hess} F$, which is self-adjoint for g and hence has real eigenvalues; the same argument applies to eq. (4.4). □

The cross-derivative criterion of Monderer and Shapley [109] says the same in game-theoretic terms: agent-wise objectives $D_{\mathrm{KL}}(q_1\|q_2)$ and $D_{\mathrm{KL}}(q_2\|q_1)$ have unequal mixed partial derivatives unless $q_1 = q_2$, so they admit no exact potential. Damped oscillation on directed cycles is the characteristic behavior of non-reciprocal interactions [62]. The reciprocal potential excludes it by construction, and an exact inheritance of a Fisher gradient flow cannot create it, since differentiating an exact intertwining by a submersion at an equilibrium shows that the parent spectrum is contained in the real spectrum of the fine linearization.

4.7 The reciprocal potential is not a global free energy

Proposition 4.21 (No global free-energy identification). Hold the rows of eq. (4.4) fixed and positive, and suppose agent i attends to a source $j \neq i$ through an identity channel.

(a) Let $\mathsf{K}_i = \mathsf{K}_j$ be finite with at least two points. There is no normalized joint $\mathbb{P}$ over finitely many finite-valued variables, record value o, rule $x \mapsto Q(x)$ assigning recognition laws, and constants $a > 0$, b with $\mathcal{E}(x) = a\mathcal{F}(Q(x); o) + b$ on the interior of the configuration space.

(b) Suppose in addition that the channels are fixed. There is no normalized joint $\mathbb{P}$ and record value o such that, with the product recognition $Q = \bigotimes_k \zeta_k$ whose factors are the agent laws themselves, $\mathcal{E} = a\mathcal{F}(Q; o) + b$ with $a > 0$ on an open set of interior configurations.

Proof. For (a), every finite value of $\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o$ satisfies $D_{\mathrm{KL}}(Q\|\Pi_o) = \sum Q \log Q - \sum Q \log \Pi_o \leq \max_{\operatorname{supp}\Pi_o}(-\log \Pi_o)$, a finite bound independent of Q. Every term of eq. (4.4) is nonnegative, so $\mathcal{E} \geq \beta_{ij} D_{\mathrm{KL}}(q_i\|q_j)$. Let q_i tend to the point mass at u and q_j to the point mass at $v \neq u$ through interior laws. Then $\mathcal{E}$ is finite along the path and tends to infinity, which no bounded function can match.

For (b), $\mathcal{F}(Q; o)$ is a sum of single-factor terms $\int \zeta_k \log \zeta_k$ and the integral of a fixed function against $\bigotimes_k \zeta_k$, which is affine in each factor. Hence the third derivative in a mass-preserving direction w of q_i and twice in a mass-preserving direction v of q_j vanishes identically for $i \neq j$. For $\mathcal{E}$ the same derivative is that of $\beta_{ij} D_{\mathrm{KL}}(q_i\|q_j)$, namely $\beta_{ij} \int w(v/q_j)^2$ in densities against a common reference. For two-point alphabets with $w = v = (v_1, -v_1)$, this equals $\beta_{ij} v_1^3 [q_j(1)^{-2} - q_j(2)^{-2}]$, which is nonzero whenever $v_1 \neq 0$ and q_j is not uniform. An affine identity on an open set would equate the two derivatives there. □

The obstruction concerns models in which the agent laws are recognition laws. The per-agent picture evades it because each $\mathcal{F}_i$ is a fixed model only with its peers' laws held as data. Two further identifications place the agent laws elsewhere. In the first, agent i receives n independent messages from a source $J \sim \pi_i^q$ with law ν_{iJ}^q and registers whether their empirical type equals q_i, a type with denominator n. The method of types gives the registration probability $|T_{q_i}| \exp\{-n[H(q_i) + D_{\mathrm{KL}}(q_i \| \nu_{ij}^q)]\}$ given $J = j$ [35, Theorem 11.1.2], so the posterior of J given registration is the Gibbs row eq. (4.5) with $T_q = 1/n$. There the laws are parameters of message laws. In the second, a prior proportional to $e^{-\mathcal{E}/\tau}$ against the Fisher volume on finite positive tables is normalizable because $\mathcal{E} \geq 0$ and that volume is finite; records then add a likelihood, and the mode follows the Fisher flow of $\mathcal{E} - \tau \log \ell$. There the laws are latent variables of a hierarchical model. Neither construction makes q_i or s_i a recognition law.

The first identification is exact on a type lattice. Fix $n \geq 1$, set $T_q = T_s = T = 1/n$, and let $\mathcal{X}_n$ be a finite set of configurations $x = (q_i, s_i)_i$ on finite alphabets in which every q_i, s_i, and $q_i \otimes s_i$ is a type with denominator n and every compared law $\nu_{ij}^q(x), \nu_{ij}^s(x)$ and reference joint R_i is positive; law-dependent channels are allowed. In the registered-agreement model, X is uniform on $\mathcal{X}_n$, the labels $J_i^q \sim \pi_i^q$ and $J_i^s \sim \pi_i^s$ are independent, agent i receives conditionally independent batches $Y_i^0 \sim R_i^{\otimes n}$, $Y_i^q \sim \nu_{iJ_i^q}^q(X)^{\otimes n}$, and $Y_i^s \sim \nu_{iJ_i^s}^s(X)^{\otimes n}$, and it registers the binary records that the empirical types of these batches equal $q_i \otimes s_i$, q_i, and s_i. Write o for the event that every record is registered, $|T_p|$ for the size of the type class of p, and $c_n(p) = H(p) - n^{-1} \log |T_p|$.

Theorem 4.22 (Registered agreement). In the registered-agreement model the following hold.

(i) $\mathbb{P}(o_i^q \mid x, J_i^q = j) = |T_{q_i}| \exp\{-n[H(q_i) + D_{ij}^q(x)]\}$, and likewise in the model sector and for the reference batch with $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i)$.

(ii) The posterior of J_i^q given x and o is the Gibbs row $\beta_i^*(x)$ of eq. (4.5), and likewise for J_i^s.

(iii) For recognition $\delta_x \otimes \bigotimes_i \beta_i \otimes \bigotimes_i \gamma_i$ of (X, J), with the batches integrated out, the free energy satisfies $T\mathcal{F} = \mathcal{E}(x, \beta, \gamma) + C_n(x) + T \log |\mathcal{X}_n|$, where $C_n(x)$ is the sum of c_n over the registered targets and $0 \leq c_n(p) \leq |\operatorname{supp} p| \log(n+1)/n$.

(iv) $\mathcal{E}_{\mathrm{red}}(x) = -T \log \mathbb{P}(o \mid x) - C_n(x)$, and $\mathcal{E}_{\mathrm{red}}(x) + C_n(x) + T \log |\mathcal{X}_n| \geq -T \log \mathbb{P}(o)$.

Proof. A sequence of type p has probability $\exp\{-n[H(p) + D_{\mathrm{KL}}(p \| \nu)]\}$ under $\nu^{\otimes n}$ [35, Theorem 11.1.2]; summing over the type class gives (i). In the posterior of a label the factor $|T_{q_i}| e^{-nH(q_i)}$ is common to all sources and cancels, which gives (ii). For (iii), $D_{\mathrm{KL}}(\delta_x \| \mathrm{Unif}) = \log |\mathcal{X}_n|$, the label terms are $D_{\mathrm{KL}}(\beta_i \| \pi_i^q)$ and $D_{\mathrm{KL}}(\gamma_i \| \pi_i^s)$, and by (i) each expected negative log likelihood is n times a divergence of eq. (4.4) plus nc_n of its target; multiplying by $T = 1/n$ gives the identity. The bounds on c_n are those on the size of a type class [35, Theorem 11.1.3]. For (iv), $-T \log \mathbb{P}(o \mid x)$ equals $\mathcal{E}_0(x) + C_n(x) - T \sum_i \log Z_i^q - T \sum_i \log Z_i^s$ by (i), and the free energy at the optimal rows bounds the negative log evidence. □

In this model the row temperature is the reciprocal number of exchanged messages, the Gibbs rows are exact posteriors of which source was heard, and the reduced potential is, up to the receiver-only term C_n, a negative log evidence at a point recognition of the configuration. The identification needs equal temperatures and one batch size; unequal temperatures give a weighted composite likelihood in the sense of Bissiri et al. [16]. A nonuniform configuration prior ρ adds $-T \log \rho(x)$.

On general standard Borel spaces, registration of an empirical law in a neighborhood of the target gives the divergence as a large-deviation rate through Sanov's theorem [38]; the double limit that would make (iii) exact there is not established here. Resampling n messages from a table p has covariance $(\operatorname{diag} p - pp^T)/n$, which is T times the inverse of the Fisher metric eq. (4.14), so the same message count would set the temperature of a stochastic version of the flow; that version and its metastability are open.

At frozen sources the same device needs no type lattice.

Corollary 4.23 (Frozen-source episodes). Hold the compared laws ν_{ij} of agent i fixed, let $T = 1/N$ with N a positive integer, and consider the normalized model $J \sim \pi_i$, $X^{1:N} \mid J \sim \nu_{iJ}^{\otimes N}$. For the tied-copy recognition $\beta_i \otimes q_i^{\otimes N}$,

$$T D_{\mathrm{KL}}\big(\beta_i \otimes q_i^{\otimes N} \| \pi_i(dJ)\nu_{iJ}^{\otimes N}\big) = \sum_j \beta_{ij} D_{ij} + T D_{\mathrm{KL}}(\beta_i \| \pi_i), \tag{4.28}$$

and for a sequence of type q_i the posterior of J is the Gibbs row at temperature T. Replicating the reference joint in the same way represents the local term, so at $T_q = T_s = 1/N$ the terms of one agent at frozen sources are T times the record-free free energy of a fixed normalized model, and records attached to the copies add a likelihood with an evidence for each frozen episode.

Proof. The chain rule gives $D_{\mathrm{KL}}(\beta_i \| \pi_i) + \sum_j \beta_{ij} D_{\mathrm{KL}}(q_i^{\otimes N} \| \nu_{ij}^{\otimes N})$, and relative entropy is additive over independent copies. The posterior statement is (ii) of theorem 4.22 for one row. □

Freezing the sources removes their source-role forces, so a sequence of such episodes is a per-agent dynamics in the sense of section 4.6, not the reciprocal flow.

Definition 4.24 (Record-interrupted reciprocal dynamics). Let $0 < t_1 < t_2 < \cdots$ be declared record times and let $A_o : \mathsf{S} \to \mathsf{S}$ be declared measurable assimilation maps, for example the conditioning updates of chapter 5. The record-interrupted dynamics follows the reciprocal Fisher flow on each interval $[t_k, t_{k+1})$ and applies A_{o_k} at t_k, with right-continuous paths.

This is a piecewise-deterministic process in the sense of Davis [41]. The reciprocal potential decreases along each flow interval but need not decrease at an assimilation, since conditioning on a record can move an agent away from the minimizer of eq. (4.4). In the free-energy pictures the same record is a likelihood term, and the objective it enters is fixed between records.

Relative to a fixed normalized model, the reciprocal potential therefore gives up four things: an evidence for model comparison, a posterior that the dynamics targets (its stationary points are compromises among references and peers), one objective that spans both interaction and data and is monotone across records, and a learning criterion for references, evaluators, priors, and temperatures. Its stationary set does not depend on records, so after an assimilation event the flow relaxes back toward record-independent states. A persistent record dependence must be declared.

Definition 4.25 (Evidence-augmented reciprocal potential). For registered records $o = (o_i)_i$ with declared densities of the record kernels ℓ_i of definition 1.10, set

$$\mathcal{E}_o = \mathcal{E} + \sum_i \iint -\log \ell_i(o_i \mid k, m) q_i(dk) s_i(dm). \tag{4.29}$$

The added term is the record term of eq. (4.26), so $\mathcal{E}_o$ is the reciprocal potential with the per-agent record likelihoods kept in the objective. Its stationary points depend on the records. The constructions of chapter 7 to chapter 13 are proved for $\mathcal{E}$ with records entering only as assimilation events, and they are not claimed for $\mathcal{E}_o$.

Open problem 4.26 (Relating the three objectives). Characterize the configurations and channels for which the per-agent models eq. (4.25) of a population are conditionals of one normalized joint, so that the per-agent dynamics is coordinate descent of a global free energy as in proposition 4.14. Determine the families, such as linear Gaussian comparisons with fixed covariances, on which the reciprocal, per-agent and global objectives have the same stationary points. Decide whether a species in the sense of definition 1.11, with matched evaluators and information, admits such a common joint. For each construction of chapter 7 to chapter 13, determine whether its conclusions hold under the per-agent dynamics or under a global free energy.

Chapter 5

Prediction and Selective Communication

A model law is useful through the records it predicts. Communication can preserve those predictions while discarding other distinctions, and a receiver can combine a message with information already present in its model. This chapter gives exact identities for both operations. The retained prediction is always named; agreement about one output is not automatically agreement about every future interaction.

5.1 Model presentations and their evaluators

Let $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$ be a normalized local model and let $H_i : \mathsf{K}_i \rightsquigarrow \mathsf{O}_i$ be a normalized record kernel. Their composition

$$K_i^{\mathrm{rec}}(m, A) = \int H_i(k, A) G_i(m, dk), \qquad p_i^{\mathrm{rec}} = s_i K_i^{\mathrm{rec}}, \tag{5.1}$$

gives the model's predictive record law. An evaluator may also take a preparation, accessible history, or chosen interaction as input; these are held fixed when two predictions are compared. More general normalized evaluators need not factor through G_i and H_i.

Suppose a model link $\Theta_{ij}^s : \mathsf{M}_j \to \mathsf{M}_i$ is a bimeasurable bijection. Exact compatibility of the two evaluators, after identifying their record alphabets, means

$$K_i^{\mathrm{rec}}(\Theta_{ij}^s m, A) = K_j^{\mathrm{rec}}(m, A). \tag{5.2}$$

Consequently, $s_i = (\Theta_{ij}^s)_{\#} s_j$ implies equal predictive record laws. The compatibility is additional information: two equal laws on a one-point model space can predict opposite deterministic bits through different evaluators.

Theorem 5.1 (Exact predictive information identity). Let s, t be probability laws on a standard Borel model space and let K be a common normalized record kernel. Assume $D_{\mathrm{KL}}(s\|t) < \infty$. Set $\mathbb{P}_s(dm, do) = s(dm)K(m, do)$ and $\mathbb{P}_t(dm, do) = t(dm)K(m, do)$, and denote their conditional model laws given o by s^o, t^o. Then

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do). \tag{5.3}$$

Proof. The common kernel cancels in the joint Radon–Nikodym derivative, giving $D_{\mathrm{KL}}(\mathbb{P}_s\|\mathbb{P}_t) = D_{\mathrm{KL}}(s\|t)$. Disintegrating the same joint laws by their record marginal factors their density ratio into

the record ratio and the conditional model ratio. The logarithm is their sum. Integrating the first term gives $D_{\mathrm{KL}}(sK\|tK)$, and integrating the second first over m gives the displayed conditional term. Finiteness makes the decomposition unambiguous. □

Theorem 5.1 is the chain rule for relative entropy [35, 89], written for a joint law built from a common kernel. Without the finiteness assumption the identity still holds in $[0,\infty]$, both sides being infinite together [68]. With a deterministic kernel $K(m,\cdot)=\delta_{T(m)}$ it decomposes a divergence into the divergence of the pushforwards and the expected divergence of the conditional laws given T, the form used in corollary 5.3.

The second term is the model distinction that the record leaves unresolved. Its vanishing is an exact sufficiency condition: $D_{\mathrm{KL}}(sK\|tK)=D_{\mathrm{KL}}(s\|t)$ exactly when the Bayesian inverse of K computed at t returns s from sK, the classical case of Petz's recovery theorem [119]. A constant evaluator can make all predictions agree while retaining no model distinction at all. Useful agreement therefore requires a specified predictive task as well as a comparison of laws. For a common history kernel, the same identity holds with the full finite record history in place of o.

If the model link depends on the current laws, theorem 5.1 applies to the resulting fixed comparison at that configuration, provided eq. (5.2) holds there. Preservation of compatibility during an update is a separate condition on the dynamics and the evaluator.

5.2 Conditional prediction within an agent

An evaluator can relate a cue to a different result, so that a record of one changes predictions for the other. On standard Borel presentation, cue, result, and record spaces, choose normalized kernels $\mu_m(du)$, $K_m(u,dv)$, and $\mathcal{L}(u,do)$, with a proper presentation reference $r(dm)$. The joint

$$P(dm,du,dv,do)=r(dm)\mu_m(du)K_m(u,dv)\mathcal{L}(u,do) \tag{5.4}$$

is an instance of definition 1.10, with represented contents (u,v) and evaluator $G(m,d(u,v))=\mu_m(du)K_m(u,dv)$. For a record likelihood $l(o\mid u)$ with finite positive evidence, the posterior has the correlated form $w^o(dm)\Pi^o_m(du,dv)$ of eq. (B.3). The result prediction averages the component-conditional results using the posterior model belief. The presentation selects a mechanism; its posterior probability is distinct from the parameters defining that mechanism.

The local free energy retains this dependence. Under the finiteness conditions of proposition B.2, a candidate $s(dm)q_m(dk)$ satisfies

$$\mathcal{F}=D_{\mathrm{KL}}(s\|w^o)+\int D_{\mathrm{KL}}(q_m\|\Pi^o_m)s(dm)-\log Z(o). \tag{5.5}$$

Exact conditioning minimizes this functional. Requiring one content law independent of the presentation changes the optimization, as in remark 4.1. Thus the general joint agent law supports a concrete local inference operation beyond the product-family flows.

For an additive Gaussian cue record $O=X+\epsilon$, $\epsilon\sim\mathcal{N}(0,C)$, the fixed-model response is

$$D_\xi\mathbb{E}[Y\mid O=\xi]=\mathrm{Cov}(Y,X\mid O=\xi)C^{-1}. \tag{5.6}$$

Proposition B.4 states the moment and differentiation hypotheses and proves the identity. The cross-covariance permits a negative or rotating response. This is conditional prediction through a statistical relation, while the Fisher tensor measures distinguishability of the posterior law. Gaussian mixture regression gives an explicit realization [24].

Returning only a Gaussian with the result moments is a further operation: it minimizes $D_{\mathrm{KL}}(p^o_Y \| q_G)$, not in general the reverse divergence of variational free energy. The exact loss of a product carrier consisting of model weights and that Gaussian is

$$D_{\mathrm{KL}}(\Pi^o_{M,Y} \| w^o \otimes q_G) = I_{\Pi^o}(M;Y) + D_{\mathrm{KL}}(p^o_Y \| q_G), \tag{5.7}$$

whenever these terms are finite. The chain rule proves the split; proposition B.5 supplies the Gaussian projection proof. This local moment-matching operation is familiar from assumed-density filtering [107]. A later Gaussian observation can distinguish the exact mixture from its carrier (proposition B.6), so the retained interface must name which predictions it preserves. Appendix B gives the self-contained construction, its frame laws, and an exact sequential filtering case. Repeated use with communication-generated records additionally requires a specified history-conditioned likelihood and retained state, as in appendix A.

5.3 What the receiver reconstructs

A sending map $T_j : E_j \to \mathsf{H}$ induces the message law $\mu_j = (T_j)_{\#} q_j$. Different laws can have the same message law. Messages travel from source to receiver. The reciprocal flow of chapter 4 also moves each source by the likelihood ratio of its receivers, so a communication model that realizes that flow must include this reverse message; the per-agent dynamics of definition 4.19 needs only the forward one. For linear maps $T_j : E_j \to \mathsf{H}$ and $O_i : \mathsf{H} \to E_i$,

$$\mathrm{rank}(O_i T_j) \le \mathrm{rank}(T_j). \tag{5.8}$$

Writing a message into a larger space cannot by itself recover distinctions discarded by T_j. A richer reconstruction uses the receiver's conditional model.

For a received value h and a declared likelihood $\ell_i(h \mid k)$, Bayesian assimilation is

$$q_i^+(dk) = \frac{\ell_i(h \mid k) q_i^-(dk)}{\int \ell_i(h \mid k') q_i^-(dk')}, \tag{5.9}$$

when the denominator is finite and positive. An unmeasured coordinate can change because it is conditionally related to the measured one in q_i^-. The observation selects among possibilities already related by the receiver's model. Under the reciprocal flow such an update is an assimilation event in the sense of definition 4.24; its effect persists in the objective only in the free-energy pictures or in the evidence-augmented potential eq. (4.29).

Proposition 5.2 (Exact conditional reconstruction). Let $T : E \to \mathsf{H}$ be measurable on standard Borel spaces. Let q, p be laws with $q \ll p$, and set $\mu = T_{\#} q$. Choose a regular conditional law

$p(dk \mid h)$ for p given $T = h$. Define

$$\widehat{q}(dk) = \int p(dk \mid h)\mu(dh). \tag{5.10}$$

Then $T_{\#}\widehat{q} = \mu$ and, in $[0, \infty]$,

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\mu(dh). \tag{5.11}$$

In particular, $\widehat{q} = q$ exactly when the two conditional laws agree for μ-almost every h.

Proof. Absolute continuity gives $\mu \ll T_{\#}p$, so the conditional version of p is defined at every h needed up to a μ-null set. That conditional is supported on $T^{-1}(h)$ almost surely. Hence its mixture has pushforward μ. The laws q and $\widehat{q}$ have the same h-marginal and respective conditional laws $q(\cdot \mid h)$ and $p(\cdot \mid h)$. Factoring their density ratio by h proves eq. (5.11). Nonnegativity and the equality condition of relative entropy give the last claim. □

The reconstruction eq. (5.10) is Jeffrey's rule of conditioning the prior p on the marginal μ [46]. It preserves the communicated marginal and supplies the discarded conditional from p. It makes precise the sense in which a receiver contributes structure that the message alone does not specify. The proposition assumes a common content presentation; a link or a separately declared correspondence is needed before applying it across different fibers.

Corollary 5.3 (Pythagorean identity and information projection). Under the hypotheses of proposition 5.2, in $[0, \infty]$,

$$D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\|\widehat{q}) + D_{\mathrm{KL}}(\widehat{q}\|p), \qquad D_{\mathrm{KL}}(\widehat{q}\|p) = D_{\mathrm{KL}}(\mu\|T_{\#}p). \tag{5.12}$$

Among all laws r with $T_{\#}r = \mu$ and $D_{\mathrm{KL}}(\mu\|T_{\#}p) < \infty$, the reconstruction $\widehat{q}$ is the unique minimizer of $D_{\mathrm{KL}}(r\|p)$.

Proof. Apply the deterministic-kernel case of the chain rule to (q, p), to $(\widehat{q}, p)$, and to (r, p). The pushforwards are μ and $T_{\#}p$ in each case. The conditional term vanishes for $\widehat{q}$, equals the right side of eq. (5.11) for q, and is nonnegative for r, with equality only when r has the conditional laws of p, that is, when $r = \widehat{q}$. □

This is the Pythagorean identity for information projections [37], and the minimization property is the characterization of Jeffrey's rule by minimum relative entropy [46]. What proposition 5.2 adds to these sources is its use as a receiver mechanism: the loss term measures exactly what a receiver's prior conditional fails to restore.

Proposition 5.4 (Common recovery by likelihood descent). Let $C : \mathsf{X} \to \mathsf{Z}$ be a measurable map between standard Borel spaces. Let P_0 be a probability law and let $P_\alpha = L_\alpha P_0$ be a family of probability laws dominated by it. Choose the reference conditional $R_0(z, dx) = P_0(dx \mid C = z)$, and write $\bar{L}_\alpha(z) = \mathbb{E}_{P_0}[L_\alpha \mid C = z]$. Then

$$(C_{\#}P_\alpha)R_0(dx) = \bar{L}_\alpha(C(x))P_0(dx). \tag{5.13}$$

Consequently this one kernel recovers every P_α exactly when each L_α has a $\sigma(C)$-measurable version, that is, exactly when C is sufficient for the family $\{P_0\} \cup \{P_\alpha\}$. No finite-divergence assumption is needed.

Proof. For each measurable $A \subseteq \mathsf{Z}$, $P_\alpha(C^{-1}A) = \int_A \bar{L}_\alpha(z)(C_\# P_0)(dz)$, by the defining property of conditional expectation. For a measurable $B \subseteq \mathsf{X}$, disintegration therefore gives

$$\begin{aligned}((C_\# P_\alpha)R_0)(B) &= \int \bar{L}_\alpha(z) R_0(z, B)(C_\# P_0)(dz) \\ &= \int_B \bar{L}_\alpha(C(x)) P_0(dx).\end{aligned} \tag{5.14}$$

Equality with P_α is equivalent to equality of their Radon–Nikodym densities, $L_\alpha = \bar{L}_\alpha \circ C$ almost surely. Conditional expectation fixes exactly the integrable $\sigma(C)$-measurable functions, proving both directions. □

A posterior-specific conditional always reconstructs that posterior from its own marginal. It need not reconstruct the reference or other posteriors. The common kernel in proposition 5.4 makes the stronger requirement explicit. Regular conditional laws and their almost-sure uniqueness are understood in the standard Borel sense [82]. The equivalence in proposition 5.4 is the Halmos–Savage factorization criterion for a family dominated by one of its members [71]; the common reference conditional is the kernel that defines sufficiency, compare the adequate subfields of Skibinsky [132], and the statement is an instance of abstract sufficiency in the Markov category of standard Borel spaces [61]. What is used below is the explicit density eq. (5.13) of the reconstruction when descent fails.

Corollary 5.5 (Recovery without reading the exterior). Let P_0 now be a joint reference law of (X, E) on standard Borel spaces, and let $Z = C(X)$. Retain (Z, E) but reconstruct X using only $R_0(z, dx) = P_0(dx \mid Z = z)$, leaving E unchanged. This reconstructs P_0 exactly if and only if X and E are conditionally independent given Z. Under that condition it also reconstructs a dominated family $P_\alpha = L_\alpha P_0$ exactly if and only if each likelihood ratio descends to (Z, E).

Proof. The lifted reference has conditional law $R_0(z, dx)$ given (z, e). Because R_0 is supported on $C^{-1}(z)$ almost surely, its joint law with (Z, E) agrees with the true law exactly when $P_0(dx \mid Z, E) = P_0(dx \mid Z)$ almost surely. This is the stated conditional independence. In that case the exterior-preserving lift is the reference conditional for the map $(x, e) \mapsto (C(x), e)$. Apply proposition 5.4 to that map. □

Descent of the likelihood ratios alone does not suffice. Let P_0 be uniform on $\{(0, 0), (1, 1)\}$ for (X, E) and let C be constant. The family $\{P_0\}$ has a trivially descending likelihood ratio, but the lift draws X uniformly and independently of E, so it does not reconstruct P_0; here X and E are dependent given Z.

5.4 A predictive coding criterion

Lossy source coding makes fidelity part of the problem definition [130]. Decoder side information is a realized variable correlated with the source, as in Wyner–Ziv coding [144]; knowing a prior without such a realization does not supply that variable. A few noiseless real numbers also do not specify a finite-bit channel. A communication rate requires a declared coding, precision, or noise model.

For a predictive criterion, let $P(X, Y, Z)$ be a declared preparation law. The source X may be represented content or a complete fine configuration; the choice is fixed in each application. Let Z be accessible receiver information and let Y be a response to be preserved. An encoder $\mathcal{T}(du \mid X)$ gives the conditional independence

$$U - X - (Y, Z). \tag{5.15}$$

The decoder predicts Y from (U, Z). This is the setting of an information bottleneck, with the relevant variable explicitly named [135].

Theorem 5.6 (Predictive distortion and the optimal decoder). On standard Borel spaces, assume the displayed information quantities are finite and eq. (5.15) holds. For any predictive decoder $\Delta(dy \mid u, z)$ with finite expected relative entropy,

$$\begin{aligned} &\mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid X, Z) \| \Delta(Y \mid U, Z)\big) \\ &\quad = I(Y; X \mid U, Z) + \mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid U, Z) \| \Delta(Y \mid U, Z)\big). \end{aligned} \tag{5.16}$$

The optimal decoder is $P(dy \mid u, z)$. Its distortion is

$$\mathcal{D}(\mathcal{T}) = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{5.17}$$

Proof. Conditional independence gives $P(Y \mid X, U, Z) = P(Y \mid X, Z)$. In the logarithmic ratio defining the left side of eq. (5.16), insert the intermediate conditional law $P(Y \mid U, Z)$. The first resulting expectation is the definition of $I(Y; X \mid U, Z)$. In the second, averaging over X gives the conditional law $P(Y \mid U, Z)$, yielding the remaining divergence. The second term vanishes at the stated decoder and is nonnegative for every decoder. Finally, expand $I(Y; X, U \mid Z)$ in both orders and use $I(Y; U \mid X, Z) = 0$. □

The decomposition eq. (5.16) is the logarithmic-loss identity of source coding with soft reconstructions, in which the optimal reconstruction is a conditional law and the distortion is a conditional mutual information [34].

The functional $I(X; U \mid Z) + \lambda \mathcal{D}(\mathcal{T})$, for $\lambda \geq 0$, is one possible criterion for selecting a representation. Its first term is an information cost, not automatically the length of an individual message. An operational coding theorem requires its own source and block-coding assumptions. Transmitting a sample from $(T_j)_{\#} q_j$ and transmitting that whole probability law are also different protocols.

For parent formation, choose X to include the fine configuration and choose Y to be a future boundary response over a specified horizon and interaction. Zero distortion means that U and Z are sufficient for this target under this preparation. A subsequent interaction may read information that this target ignored. Exact inheritance therefore requires the dynamical condition developed in chapter 6, not only predictive sufficiency for one chosen response.

Chapter 6

Inherited Parents and Recursive Formation

A collection becomes a parent when its retained state supports inherited interaction and evolution. A probability law obtained by compression is a first part of this construction. It does not determine how that law will change, what the parent can communicate, or which information its children must retain. We give exact conditions for these further steps, classify the mechanisms that make them hold, and give a finite drive–response example in which exact lumping works through two levels. The example supplies its reductions; formation of the reductions remains a separate question.

Most exactness criteria below are instances of classical mathematics, and we attribute them where they are stated: factor maps of dynamical systems, exact lumping of differential equations, strong lumpability and Markov functions of Markov chains, Riemannian submersions, equivariant reduction, and conditional independence. What this chapter adds is their assembly into one notion of an inherited parent with declared protocol and preparation classes, the counterexamples showing which natural retentions fail, and the restrictions that these criteria place on the formation constructions of later chapters.

6.1 Two objects that a parent must retain

Definition 6.1 (Sample compression and configuration retention). Let $x \in \mathsf{S}$ be a complete agent configuration and let Q_x be its declared law on a standard Borel space Y of represented contents. A normalized sample channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ gives the retained law

$$Q_x^c(A) = \int C(y, A) Q_x(dy). \tag{6.1}$$

A configuration retention is a measurable map $R : \mathsf{S} \to \mathsf{S}^c$. Its value includes the parent laws and the additional data needed for its declared interactions and update. Proposition 6.12 identifies the smallest such data for a declared interface.

The distinction applies to both sectors. A parent carrying q_I and s_I also needs the evaluator that makes s_I predictive. Boundary channels, source rows, source marks, and memory can be further components of $R(x)$. Equal marginal laws do not imply equality of these components. When a sample channel reads random links or source variables, those variables belong to the joint input law of the channel. Substitution of a mean link or a selected source is a different retention.

A law of represented contents and a probability law over complete configurations are different objects. Coupled deterministic evolution of law-valued agents requires no probability distribution over x. If such a distribution is used below, its preparation or transition mechanism is stated. All

constructions admit general probability laws. A Gaussian parent is one possible family, subject to its own closure conditions.

Throughout the chapter, $\circ$ composes maps and juxtaposition composes kernels in the order in which they act: for kernels $K : \mathsf{X} \rightsquigarrow \mathsf{O}$ and $D : \mathsf{O} \rightsquigarrow \mathsf{O}'$, $(KD)(x, A) = \int D(o, A)K(x, do)$. A measurable map used in a kernel product stands for its deterministic kernel. With this convention every square below is a commutative diagram of Markov kernels between standard Borel spaces, and the composition results are the pasting of such diagrams in the Markov category of these spaces [61].

6.2 Exact deterministic inheritance

Proposition 6.2 (An update descends precisely when it respects retained states). Let S, S^c be standard Borel spaces, let $R : \mathsf{S} \to \mathsf{S}^c$ be a Borel surjection, and let $F : \mathsf{S} \to \mathsf{S}$ be Borel. There is a Borel parent update F^c satisfying

$$R \circ F = F^c \circ R, \tag{6.2}$$

if and only if

$$R(x) = R(x') \quad \Longrightarrow \quad R(F(x)) = R(F(x')). \tag{6.3}$$

The parent update is unique. The same conclusion holds on arbitrary measurable spaces when R has a measurable right inverse.

Proof. Equation (6.2) implies the implication in (6.3). Conversely, set $F^c(z) = R(F(x))$ for any x with $R(x) = z$. The implication makes this independent of the choice of x, and surjectivity defines it on all of S^c. Its graph is the image of S under the Borel map $x \mapsto (R(x), R(F(x)))$ into the standard Borel space $\mathsf{S}^c \times \mathsf{S}^c$, hence an analytic set. A function between standard Borel spaces with analytic graph is Borel [84, Thm. 14.12]. With a measurable right inverse ι, the formula $F^c = R \circ F \circ \iota$ gives measurability directly. Every parent state has a preimage, so (6.2) fixes F^c uniquely. □

This is the factor-map (semiconjugacy) criterion of dynamical systems. The analytic-graph argument matters because a Borel surjection need not have a Borel right inverse; the uniformization theorem supplies only universally measurable ones [84, Sec. 18.A]. Repeated substitution gives $R \circ F^n = (F^c)^n \circ R$ for every nonnegative integer n. This is an exact statement about the chosen update, including its order and retained parameters. Replacing that update by descent of a newly defined parent objective requires another argument.

For a differentiable version, suppose R is continuously differentiable and the fine and parent vector fields B, B^c have unique local solutions. Their flows satisfy $R \circ \Phi_t = \Phi^c_t \circ R$ on their common existence interval if

$$DR_x B(x) = B^c(R(x)). \tag{6.4}$$

Indeed, the chain rule makes $R(\Phi_t(x))$ a parent solution with initial value $R(x)$, and uniqueness identifies it. Conversely, differentiation of the flow identity at zero gives (6.4). For a moving map R_t, the left side becomes $\partial_t R_t(x) + DR_t(x)B_t(x)$. Membership changes therefore need an explicit handoff map; a differential identity on one fixed partition does not specify them. Condition (6.4) is the exact-lumping condition for differential equations, introduced for linear kinetics by Wei and Kuo [142], for nonlinear kinetics by Li and Rabitz [100], and for flows on manifolds by Horstmeyer

and Atay [79]; flow intertwining has also been proposed as a criterion of macroscopic emergence [9]. The differentiable statement needs R to be continuously differentiable between manifolds on which both flows are defined. A retention into spaces of laws that is differentiable only weakly, with a parent law of motion given by a weak continuity equation, is not covered, and its intertwining must be proved directly.

Proposition 6.3 (Projectable vector fields). Let $R : \mathsf{S} \to \mathsf{S}^c$ be a C^{k+1} surjective submersion between finite-dimensional manifolds, $k \geq 1$, whose fibers are connected, and let B be a C^k vector field on S. A C^k parent field B^c satisfying (6.4) exists if and only if, in every submersion chart $R(a, b) = a$ with $B = (B_A, B_V)$,

$$\partial_b B_A(a, b) = 0. \tag{6.5}$$

For a linear surjection $R = M$ on $\mathbb{R}^n$ this is the condition $M(DB(x)v) = 0$ for all x and all $v \in \ker M$.

Proof. In a submersion chart, $DR_x B(x) = B_A(a, b)$. Condition (6.5) says that this map is locally constant along each fiber; a locally constant function on a connected fiber is constant, so $x \mapsto DR_x B(x)$ is constant on fibers, and conversely constancy on fibers gives (6.5). A submersion has smooth local sections σ through every point, and $B^c = (DR \cdot B) \circ \sigma$ is then well defined and C^k. Since $k \geq 1$, both fields have unique local solutions. In the linear case the fibers are affine, hence connected, and the derivative of MB along $v \in \ker M$ is $M(DB(x)v)$. □

Remark 6.4 (Why inheritance is tested on a set of configurations). Dynamical inheritance checked only along one realized trajectory carries no information. If $x(t)$ is a smooth solution on $[0, T]$ and R is smooth with $z = R \circ x$ an embedded arc, the tangent field of the arc extends to a smooth field B^c on a neighborhood of the arc, and $R(\Phi_t x_0) = \Phi^c_t(Rx_0)$ holds with unique parent solutions. The same holds on a prepared family on which R is injective. The fiber test (6.3) constrains a retention only over a set containing distinct configurations with the same retained value. For this reason the inheritance identities are imposed below on a declared invariant set Ω_I of post-handoff states (definition 6.16), and the formation standard requires an admitted class that is open or a declared invariant family.

Proposition 6.5 (Gradient descent through a horizontally conformal submersion). Let $R : (\mathsf{S}, g) \to (\mathsf{S}^c, g_c)$ be a smooth surjective submersion between finite-dimensional Riemannian manifolds, with horizontal spaces $H_x = (\ker DR_x)^\perp$. Assume there is a smooth positive function d such that

$$g_c(DR_x u, DR_x v) = d(x)^2 g(u, v), \qquad u, v \in H_x. \tag{6.6}$$

Let E, E_c be smooth potentials and a a positive function. Then

$$DR_x[-\operatorname{grad}_g E(x)] = a(x)[-\operatorname{grad}_{g_c} E_c](R(x)) \tag{6.7}$$

holds at x if and only if

$$d(x)^2 dE_x(v) = a(x)(dE_c)_{R(x)}(DR_x v) \qquad \text{for every } v \in H_x. \tag{6.8}$$

In particular, if $E = \chi \circ E_c \circ R$ with $\chi' > 0$, then (6.7) holds with $a(x) = \chi'(E_c(R(x)))d(x)^2$, and a Riemannian submersion with $E = E_c \circ R$ gives same-time inheritance.

Proof. Let $w = \operatorname{grad}_g E(x)$ and let w^H be its horizontal part. Since $DR_x w = DR_x w^H$, for any coarse tangent z with horizontal lift v,

$$g_c(DR_x w, z) = d(x)^2 g(w^H, v) = d(x)^2 g(w, v) = d(x)^2 dE_x(v). \tag{6.9}$$

Nondegeneracy of g_c shows that $DR_x w = a(x) \operatorname{grad}_{g_c} E_c(R(x))$ exactly when $d(x)^2 dE_x(v) = a(x)(dE_c)_{R(x)}(z)$ for every z, which is (6.8) because horizontal lifting is a bijection $T_{R(x)}\mathsf{S}^c \to H_x$. If $E = \chi \circ E_c \circ R$, then $dE_x(v) = \chi'(E_c(R(x)))(dE_c)_{R(x)}(DR_x v)$ for every v. When $a = 1$, eq. (6.4) applies. □

This is the gradient pushforward for Riemannian submersions [113] and horizontally conformal submersions [7]. The potential hypothesis $E = \chi \circ E_c \circ R$ is much stronger than the criterion. It forces dE to vanish on $\ker DR$, so the fine gradient is horizontal and every fine trajectory is the horizontal lift of its projection: for a product metric and $R(a, b) = a$, the discarded coordinate does not move. Criterion (6.8) constrains only horizontal derivatives and leaves the discarded dynamics free. On Euclidean $\mathbb{R}^2$ with $R(x_1, x_2) = (x_1 + x_2)/\sqrt{2}$, the potential

$$E = \tfrac{1}{2}x_1^2 + \tfrac{1}{2}x_2^2 + W(x_1 - x_2), \qquad E_c(z) = \tfrac{1}{2}z^2, \tag{6.10}$$

with any smooth W, satisfies (6.8) with $a = 1$, so the center of mass descends at the same time, although $E \neq \chi \circ E_c \circ R$ whenever $W' \neq 0$ and the separation $x_1 - x_2$ evolves under W. The metric hypothesis is necessary in a uniform sense: if (6.7) holds with $a \equiv 1$ for every potential of the form $E_c \circ R$, then $DR_x \circ g^{-1} \circ DR_x^* = g_c^{-1}$ at $R(x)$, because $DR(\operatorname{grad}_g(E_c \circ R)) = (DR \circ g^{-1} \circ DR^*)(dE_c)$ must equal $g_c^{-1} dE_c$ for every covector, and this identity says that R is a Riemannian submersion. For a single potential the metric hypothesis is not necessary.

For $a > 0$, a fine trajectory projects onto the same oriented coarse gradient orbit after the time change $d\tau/dt = a(x(t))$. An autonomous field in the original time additionally requires the product $a(x) \operatorname{grad}_{g_c} E_c(R(x))$ to be constant on retention fibers. Factorization $a = \bar{a} \circ R$ suffices; dependence on discarded state need not satisfy this requirement. The proposition is one sufficient route to inheritance. Fisher contraction and preservation of the collective connection in chapter 3 are different conditions.

Fiber minimization alone does not supply the criterion. For $R(u, v) = v$ on Euclidean $\mathbb{R}^2$, let

$$E(u, v) = \tfrac{1}{2}(u - v)^2 + \tfrac{1}{2}v^2, \qquad E_c(v) = \inf_u E(u, v) = \tfrac{1}{2}v^2. \tag{6.11}$$

Equality $E = E_c \circ R$ holds only on $u = v$. The horizontal direction is ∂_v, and $dE(\partial_v) - dE_c(DR(\partial_v)) = v - u$, so the same-time criterion fails off the diagonal; the pushed field $u - 2v$ is not a function of v. Since $dE(\partial_u) = u - v$ does not vanish on any open set, there is also no identity $E = \chi(E_c \circ R)$ on an open set.

Proposition 6.6 (Equivariant descent). Let a Lie group G act smoothly, freely, and properly on a finite-dimensional manifold S, and let $\pi : \mathsf{S} \to \mathsf{S}/G$ be the quotient map.

(i) If a smooth vector field B is G-equivariant, there is a unique smooth field B^c on S/G with $D\pi \cdot B = B^c \circ \pi$, and the flows intertwine. Every G-invariant measurable response kernel or channel on S factors through π.

(ii) If $B = -\operatorname{grad}_g E$ with g and E both G-invariant, then S/G carries a unique metric g_c making π a Riemannian submersion, $E = E_c \circ \pi$ for a smooth E_c, and $B^c = -\operatorname{grad}_{g_c} E_c$.

Proof. By the quotient manifold theorem [96, Thm. 21.10], S/G is a smooth manifold and π a surjective submersion whose fibers are the orbits. Since $\pi \circ \phi_h = \pi$ for the action ϕ_h, $D\pi_{hx} D\phi_h = D\pi_x$, and equivariance gives $D\pi_{hx} B(hx) = D\pi_{hx} D\phi_h B(x) = D\pi_x B(x)$. Thus $D\pi \cdot B$ is constant on fibers, and $B^c = (D\pi \cdot B) \circ \sigma$ for local smooth sections σ is well defined and smooth. Invariant kernels are constant on orbits and descend by proposition 6.2 applied to the kernel-valued map. For (ii), invariance of E makes dE vanish on orbit tangents, which for a free action span $\ker D\pi$. Isometries carry horizontal spaces to horizontal spaces, so $g_c(D\pi u, D\pi v) = g(u, v)$ for horizontal u, v is well defined [113]. Proposition 6.5 with $d = 1$ and $\chi = \mathrm{id}$ gives the gradient identity. □

Equivariant descent holds on the whole configuration space, so the fiber test needs no restriction to an admitted class. It forgets only orbit data. Without freeness the quotient is stratified by orbit type and the statement holds on each stratum. On infinite-dimensional law charts, (ii) requires in addition a closed complement to the orbit tangent in the Fisher metric and a quotient Banach manifold, which must be verified for the chart in use.

Proposition 6.7 (Coordinate projections are exact only for isolated blocks). Let $\mathsf{S} = \mathsf{S}_A \times \mathsf{S}_B$ be a product of connected manifolds with metric $g = g_A(a) \oplus g_B(b)$, and let $E \in C^2$. The projection $R(a, b) = a$ intertwines $-\operatorname{grad}_g E$ with an autonomous field if and only if $\partial_a \partial_b E \equiv 0$, that is, $E(a, b) = E_A(a) + E_B(b)$. The discarded block is then autonomous as well.

Proof. The retained velocity is $\dot{a} = -g_A(a)^{-1} \partial_a E(a, b)$. Since the fibers $\{a\} \times \mathsf{S}_B$ are connected and $g_A(a)$ is invertible, it is independent of b exactly when $\partial_b \partial_a E = 0$, that is, when $\partial_a E(a, b) = \phi(a)$. Integrating along paths in the connected factors gives $E = E_A(a) + E_B(b)$, and then $\dot{b} = -g_B(b)^{-1} \partial_b E_B(b)$. □

The Fisher flow of chapter 4 with product recognition has agent-local metric blocks and one potential, and a source appears in the cost of every receiver that attends to it. Discarding whole agents or whole sectors is therefore exact in that dynamics only for energetically isolated blocks, with no peer cost in either direction between the retained and the discarded agents. Exact compression of reciprocally coupled agents must come from symmetry (proposition 6.6) or from horizontal energy structure (proposition 6.5), as in the center-of-mass example.

Proposition 6.8 (Exact inheritance of a gradient flow creates no oscillatory mode). Let $B = -\operatorname{grad}_g E$ with $E \in C^2$, and let x^* be a critical point of E. Then $L = DB(x^*) = -g(x^*)^{-1} \operatorname{Hess} E(x^*)$ is self-adjoint for $g(x^*)$ and has real spectrum. If a C^2 submersion R intertwines B with a C^1 field B^c near x^* in the sense of (6.4), then $R(x^*)$ is an equilibrium of B^c and $\operatorname{spec} DB^c(R(x^*)) \subseteq \operatorname{spec} L \subset \mathbb{R}$.

Proof. At a critical point the Hessian is a symmetric bilinear form, so L is $g(x^*)$-self-adjoint. Evaluating (6.4) at x^* gives $B^c(R(x^*)) = 0$. Differentiating it at x^*, the term containing the second derivative of R is multiplied by $B(x^*) = 0$, leaving $DR \circ L = L^c \circ DR$ with $L^c = DB^c(R(x^*))$. Hence $\ker DR$ is L-invariant, and since DR is surjective, L^c is conjugate through DR to the map that L induces on $T_{x^*}\mathsf{S}/\ker DR$. The characteristic polynomial of L is the product of those of its restriction and of the induced quotient map, so every eigenvalue of L^c is an eigenvalue of L. □

An exactly inherited parent of a Fisher gradient flow therefore has no underdamped linear mode at an inherited equilibrium. Oscillatory collective modes require nonreciprocal fields, explicit driving, or declared second-order kinetics.

Proposition 6.9 (Loss of dynamical roles under product recognition). Take $Q_x = \mathrm{Ber}(p_1) \otimes \mathrm{Ber}(p_2)$, with $x = (p_1, p_2) \in [0,1]^2$, and the coupled deterministic law

$$\dot{p}_1 = -p_1 + \lambda(p_2 - p_1), \qquad \dot{p}_2 = -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \tag{6.12}$$

Retain only the law of $K_1 + K_2$. No autonomous update of that retained law reproduces every trajectory of (6.12).

Proof. The linear system has a unique global solution. Its vector field does not point outward on any face of the square: at $p_1 = 0$, $\dot{p}_1 = \lambda p_2 \geq 0$, at $p_1 = 1$, $\dot{p}_1 \leq -1$, and similarly for p_2. The square is therefore forward invariant; the field is tangent to a face when $\lambda = 0$ and vanishes at the corner $(0,0)$. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for the values $0, 1, 2$. However,

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2), \tag{6.13}$$

equals -1.2 and -1.8 respectively. Their retained laws therefore have different derivatives, and no autonomous update of the retained law can reproduce both trajectories. □

This obstruction occurs while recognition remains a product. The missing datum is which probability belongs to which dynamical role. Correlated recognition is neither a prerequisite for this counterexample nor its automatic remedy. Coupled mean-field dynamics are admissible; the proposed retention must still pass its own inheritance test.

6.3 Exact stochastic inheritance

Proposition 6.10 (Finite strong lumpability). Let T be a transition matrix on a finite configuration space S and let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective. A single parent transition matrix reproduces the retained process for every initial law if and only if, for every $z' \in \mathsf{S}^c$,

$$\sum_{y : R(y) = z'} T(x, y) = \sum_{y : R(y) = z'} T(x', y) \quad \text{whenever } R(x) = R(x'). \tag{6.14}$$

In that case the common sum defines $T^c(R(x), z')$ uniquely.

Proof. The common sums are nonnegative and sum to one, so they define a transition matrix. Conditional on any retained history ending at z, the hidden configuration lies in $R^{-1}(z)$. Every state in that fiber gives the same next retained distribution $T^c(z, \cdot)$; averaging over its conditional law leaves this distribution unchanged. Induction gives the parent path law from the pushed initial law. Conversely, start the fine process at each point mass δ_x. A common parent kernel must give the same one-step distribution whenever these initial states have the same image. This is (6.14). □

This is the finite strong-lumpability criterion [85]; its general state-space form is Dynkin's criterion for a function of a Markov process [51]. On standard Borel spaces the same argument applies whenever $x \mapsto T(x, R^{-1}(\cdot))$ is constant on fibers: the analytic-graph argument of proposition 6.2,

applied to this map into the standard Borel space of probability laws on S^c, makes the parent kernel measurable. Closure for every initial law is stronger than closure for one chosen preparation. The latter is weak lumpability [85], and the question whether a function of a Markov chain is itself Markov goes back to Burke and Rosenblatt [23]; Buchholz [21] compares these notions with exact lumpability.

An equilibrium conditional average of fine transitions defines a normalized candidate parent kernel. It does not prove (6.14): the hidden conditional law during a transient can differ from the equilibrium conditional. Likewise, agreement of invariant distributions does not establish agreement of histories.

Proposition 6.11 (Intertwined handoffs). Let T be a transition matrix on a finite space S, let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective, and let Φ_R be its deterministic kernel. Let Λ be a stochastic matrix from S^c to S with $\Lambda(z, R^{-1}\{z\}) = 1$ for every z, and suppose

$$\Lambda T = T^c \Lambda \tag{6.15}$$

for some matrix T^c. Then $T^c = \Lambda T \Phi_R$ is a transition matrix. For every law ν on S^c, under the initial law $\nu\Lambda$ the retained process $Z_n = R(X_n)$ is Markov with initial law ν and kernel T^c, and on every retained history of positive probability the conditional law of X_n is $\Lambda(Z_n, \cdot)$. For a response kernel K_a on S, the predictive response given the retained history is $(\Lambda K_a)(Z_n, \cdot)$.

Proof. The support condition gives $\Lambda\Phi_R = I$, so multiplying (6.15) on the right by Φ_R yields $T^c = \Lambda T \Phi_R$, a product of stochastic matrices. Let P_z be the diagonal selector of $R^{-1}\{z\}$. The support condition also gives $\Lambda(w, \cdot)P_z = \delta_{wz}\Lambda(z, \cdot)$. Let f_n be the row vector whose x entry is the probability of the history $(z_0, \dots, z_n)$ jointly with $X_n = x$. Then $f_0 = \nu\Lambda P_{z_0} = \nu(z_0)\Lambda(z_0, \cdot)$. If $f_n = c_n\Lambda(z_n, \cdot)$, then

$$f_{n+1} = f_n T P_{z_{n+1}} = c_n (T^c\Lambda)(z_n, \cdot)P_{z_{n+1}} = c_n T^c(z_n, z_{n+1})\Lambda(z_{n+1}, \cdot). \tag{6.16}$$

By induction $c_n = \nu(z_0)\prod_{k<n} T^c(z_k, z_{k+1})$ is the probability of the history and $\Lambda(z_n, \cdot)$ is the normalized conditional law. The predictive response follows by integrating K_a against it. □

This is the Markov-function theorem of Rogers and Pitman [124], who also give the continuous-time and general state-space versions. It describes exactly what a handoff "from a declared conditional law" must supply for the parent to be memoryless: the kernel Λ, and an initial law of the form $\nu\Lambda$. It does not require strong lumpability. On $\mathsf{S} = \{1, 2, 3\}$ with $R(1) = R(2) = a$, $R(3) = b$, rows $(0.3, 0.5, 0.2)$, $(0.4, 0.2, 0.4)$, $(0.25, 0.25, 0.5)$, and $\Lambda(a, \cdot) = (\frac{1}{2}, \frac{1}{2}, 0)$, $\Lambda(b, \cdot) = \delta_3$, condition (6.14) fails ($0.2 \neq 0.4$) while (6.15) holds with $T^c = \left(\begin{smallmatrix} 7/10 & 3/10 \\ 1/2 & 1/2 \end{smallmatrix}\right)$. From the initial law δ_1, which is not of the form $\nu\Lambda$, the retained path laws of length five differ from those of T^c by $1/10$ in total variation.

Proposition 6.12 (The coarsest exact parent containing an interface). Let $F : \mathsf{S} \to \mathsf{S}$ and an interface $Q : \mathsf{S} \to \mathsf{Y}$ be measurable. The interface collects what the parent must reproduce: its laws, the responses of a declared protocol class, and the channels and evaluators it reads. Define

$$\pi(x) = (Q(F^n x))_{n \geq 0} \in \mathsf{Y}^{\mathbb{N}}. \tag{6.17}$$

Then π satisfies (6.3), with parent update the left shift on $\pi(\mathsf{S})$, and Q is its zeroth coordinate. If a retention R satisfies (6.3) and $Q = Q^c \circ R$ for some Q^c, then $R(x) = R(x')$ implies $\pi(x) = \pi(x')$.

Thus π is, up to relabeling of its values, the unique coarsest exact retention through which the interface factors. For a forward-complete flow, replace F^n by Φ_t, $t \geq 0$. If B and Q are real analytic, $\pi(x) = \pi(x')$ exactly when $L_B^k Q(x) = L_B^k Q(x')$ for every $k \geq 0$.

Proof. The identity $\pi(F(x)) = (Q(F^{n+1}x))_{n\geq 0}$ is the shift of $\pi(x)$, so equal values have equal successors. If R satisfies (6.3), induction gives $R(F^n x) = R(F^n x')$ whenever $R(x) = R(x')$, and applying Q^c gives $\pi(x) = \pi(x')$. In the analytic case $t \mapsto Q(\Phi_t x)$ is real analytic on its interval of existence and its derivatives at zero are the Lie derivatives $L_B^k Q(x)$, so the identity theorem gives the stated equivalence. □

The analytic case is the observability construction of Hermann and Krener [75]. For finite Markov chains the corresponding object is computed by partition refinement.

Proposition 6.13 (The coarsest strongly lumpable refinement). Let T be a transition matrix on a finite space S and let $\mathcal{B}_0$ be the partition of S into level sets of an interface Q. Define $\mathcal{B}_{k+1}$ by splitting each block of $\mathcal{B}_k$ according to the vector $(T(x, B))_{B\in\mathcal{B}_k}$. The sequence stabilizes at a partition $\mathcal{B}_*$, which is the unique coarsest partition refining $\mathcal{B}_0$ that satisfies (6.14).

Proof. Each step refines the previous partition, so the sequence stabilizes after finitely many steps. At the fixed point, states in one block have equal masses into every block, which is (6.14). Let $\mathcal{B}$ be any strongly lumpable partition refining $\mathcal{B}_0$, and suppose it refines $\mathcal{B}_k$. Each block of $\mathcal{B}_k$ is a union of blocks of $\mathcal{B}$, so two states in one block of $\mathcal{B}$ have equal masses into every block of $\mathcal{B}_k$ and lie in one block of $\mathcal{B}_{k+1}$. By induction $\mathcal{B}$ refines $\mathcal{B}_*$. □

This is probabilistic bisimulation [91]; an efficient algorithm is given by Derisavi et al. [45]. The two propositions make the phrase “the additional data needed” in definition 6.1 a theorem: given the update and the interface, the canonical parent is determined. A larger interface gives a finer canonical parent, so the protocol class decides how much a parent can forget. Unrestricted identity-channel probes make it injective on exposed child laws (section 10.3, proposition 11.13). The compression clause of the formation standard asks whether this canonical parent, for the interface fixed before the retention, is noninjective on the post-handoff invariant set.

6.4 Boundary responses and composition

A parent must inherit what it does to the rest of the population. Fix a declared boundary protocol a, including the preparation and receiving conditions. Let $K_a : \mathsf{S} \rightsquigarrow \mathsf{O}_a$ be its normalized response kernel and let $D_a : \mathsf{O}_a \rightsquigarrow \mathsf{O}_a^c$ retain the response of interest. An inherited kernel K_a^c satisfies

$$(K_a D_a)(x, A) = K_a^c(R(x), A). \tag{6.18}$$

On standard Borel spaces, such a kernel exists exactly when $x \mapsto (K_a D_a)(x, \cdot)$ is constant on each fiber of R; the analytic-graph argument of proposition 6.2 gives measurability, and substitution proves necessity. When several outputs are used jointly, this condition applies to their joint kernel. Preserving each output marginal alone need not preserve their dependence.

The same applies to a response and the next retained state. For a stochastic update, response inheritance concerns the joint kernel $x \mapsto \mathrm{law}(D_a(O_a), R(X'))$ on $\mathsf{O}_a^c \times \mathsf{S}^c$, which must factor through R. Separate descent of the response kernel and of the retained transition does not imply it. Let

$x = (u, v) \in \{0, 1\}^2$, $R(x) = u$, draw a fresh fair bit ξ, set $u' = u \oplus \xi$, draw v' freshly, and respond with $O = \xi \oplus v$. Both the response and the next retained state are uniform from every configuration, but at $(0, 0)$ the joint law of (O, u') is uniform on $\{(0, 0), (1, 1)\}$ and at $(0, 1)$ it is uniform on $\{(1, 0), (0, 1)\}$. If the randomness of the response is conditionally independent of the transition randomness given x, separate descent suffices. For adaptive protocols, in which a receiver chooses its next action from past outputs, the joint condition is required for each admitted action.

Proposition 6.14 (Exact elimination through a conditional separator). Let Z, U, V have a declared joint reference law $\omega(dz, du, dv) = \omega_Z(dz)\omega_{UV}(du, dv \mid z)$ on standard Borel spaces. Denote its conditional marginals by ω_U, ω_V. For bounded nonnegative likelihood factors $L_U(u, z), L_V(v, z)$, jointly measurable, define

$$F_U(z) = \int L_U(u, z)\omega_U(du \mid z), \qquad F_V(z) = \int L_V(v, z)\omega_V(dv \mid z). \tag{6.19}$$

The identity

$$\int L_U(u, z)L_V(v, z)\omega_{UV}(du, dv \mid z) = F_U(z)F_V(z) \tag{6.20}$$

holds ω_Z-almost everywhere for every such pair precisely when U and V are conditionally independent given Z. Under that condition, a nonnegative factor $L_Z(z)$ with finite positive total evidence gives the unnormalized retained measure

$$m_Z(dz) = L_Z(z)F_U(z)F_V(z)\omega_Z(dz). \tag{6.21}$$

Proof. Conditional independence factors ω_{UV} as $\omega_U \otimes \omega_V$ almost surely. Tonelli's theorem then gives eqs. (6.20) and (6.21). Conversely, take indicator factors from countable generating families closed under finite intersections for the two standard Borel spaces. Outside one ω_Z-null set, the conditional joint and product laws agree on every generating rectangle. Uniqueness of probability measures extends this equality to the product sigma algebra, proving conditional independence. □

The shared factor L_Z appears once. If the hidden blocks are not conditionally independent, their joint conditional factor must be retained. This elimination statement concerns the declared reference and likelihood; it does not replace the law-dependent interaction potential by a fixed joint model. It is the separator factorization of Dawid [42] that underlies junction-tree propagation [93].

The proposition presupposes the joint law. A gluing statement holds for two blocks: laws of (Z, U) and (Z, V) with a common Z marginal have exactly one joint law with U and V conditionally independent given Z, namely the conditional product. Elimination of $k \geq 3$ blocks needs mutual conditional independence, not pairwise independence: with U, V independent fair bits, $W = U \oplus V$, and Z constant, every pair is independent, but $\mathbb{E}[1_{U=1}1_{V=1}1_{W=1}] = 0 \neq 1/8 = F_U F_V F_W$. Gluing over a family of overlapping separators requires a junction tree. Edge laws on a four-cycle forcing $A = B$, $B = C$, $C = D$, and $D \neq A$ with uniform single-site marginals are pairwise consistent but have no joint law; existence of a joint law for every consistent family characterizes the decomposable (running-intersection) hypergraphs [140].

For a child channel $\mathcal{K}_{ij} : \mathsf{Y}_j \rightsquigarrow \mathsf{Y}_i$ and retentions $C_j : \mathsf{Y}_j \rightsquigarrow \mathsf{Z}_J$ and $C_i : \mathsf{Y}_i \rightsquigarrow \mathsf{Z}_I$, the corresponding square is

$$\mathcal{K}_{ij} C_i = C_j \mathcal{K}^c_{IJ}, \qquad \mathcal{K}^c_{IJ} : \mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I. \tag{6.22}$$

The right side means first retain the sender and then apply the parent channel. A parent channel exists exactly when $\mathcal{K}_{ij} C_i$ is a garbling of C_j, that is, when it lies in the set $C_j \mathcal{M}$ with $\mathcal{M}$ ranging over Markov kernels $\mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I$; for finite alphabets this is a linear feasibility problem. For a deterministic retention C_j it reduces to constancy of $\mathcal{K}_{ij} C_i$ on the fibers of C_j: the retained sender must determine every distinction visible to the retained receiver. For a stochastic C_j the condition is stronger than any linear solvability. If C_j is the binary symmetric channel with flip probability $1/4$ and $\mathcal{K}_{ij} = C_i = \mathrm{id}$, the only linear solution is $C_j^{-1} = \left(\begin{smallmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{smallmatrix}\right)$, which is not a Markov kernel, so no parent channel exists. For a law-dependent channel $\mathcal{K}_{ij}[x]$, the square must hold at each configuration and the map $x \mapsto \mathcal{K}_{ij}[x] C_i$ must also factor through the configuration retention. In a deterministic linear realization the square reads $C_i \circ M_{ij} = M^c_{IJ} \circ C_j$; the kernel-inclusion criterion and its proof are given in the supplement.

Proposition 6.15 (Inheritance composes). Suppose $R_{10} : \mathsf{S}_0 \to \mathsf{S}_1$ and $R_{21} : \mathsf{S}_1 \to \mathsf{S}_2$ intertwine deterministic updates at successive levels. Then $R_{20} = R_{21} \circ R_{10}$ intertwines the first and third updates. The analogous assertion holds for finite lumped Markov transitions. Let $K^\ell_a : \mathsf{S}_\ell \rightsquigarrow \mathsf{O}_\ell$ be boundary kernels, with response retentions $D_{10} : \mathsf{O}_0 \rightsquigarrow \mathsf{O}_1$ and $D_{21} : \mathsf{O}_1 \rightsquigarrow \mathsf{O}_2$. If

$$K^0_a D_{10} = R_{10} K^1_a, \qquad K^1_a D_{21} = R_{21} K^2_a, \tag{6.23}$$

then $K^0_a D_{10} D_{21} = R_{10} R_{21} K^2_a$.

Proof. For deterministic updates, $R_{21} \circ R_{10} \circ F^0 = R_{21} \circ F^1 \circ R_{10} = F^2 \circ R_{21} \circ R_{10}$. For Markov transitions and their deterministic retention kernels, in kernel order, $T^0 R_{10} R_{21} = R_{10} T^1 R_{21} = R_{10} R_{21} T^2$, and the kernel $R_{10} R_{21}$ is the deterministic kernel of $R_{21} \circ R_{10}$. For boundary kernels, associate the successive integrals and substitute (6.23). All integrands for events are nonnegative, so interchanging the integration order is valid. □

When changes of gauge presentation act on the retained spaces, the retentions and boundary kernels must transform compatibly with those actions. Re-expressing a commuting square then gives the same square in the new coordinates. Neither a parent frame chosen for this purpose nor the square itself determines a base connection. If the parent is also required to retain the collective Fisher connection, the additional condition in proposition 3.17 must be checked.

Definition 6.16 (Inherited parent). An inherited parent of a block I consists of the following data, declared in this order: a protocol class A_I of boundary protocols, a preparation class $\mathcal{P}_I$ of initial laws (for stochastic dynamics), an invariant set Ω_I containing every post-handoff configuration that arises from the admitted class with the block I, and a measurable retention $R_I : \Omega_I \to \mathsf{S}^c_I$. It must satisfy four conditions.

(D) Dynamics. In deterministic dynamics, (6.3) holds for all $x, x' \in \Omega_I$; in stochastic dynamics, a single parent kernel reproduces the retained process for every initial law in $\mathcal{P}_I$ (strong lumpability when $\mathcal{P}_I$ contains all laws, the intertwining of proposition 6.11 when $\mathcal{P}_I = \{\nu\Lambda\}$).

(B) Responses. For every $a \in A_I$, the joint kernel of the retained response and the next retained state factors through R_I on Ω_I, for each admitted receiver action of an adaptive protocol.

(C) Channels. Every channel and evaluator read by a protocol in A_I or by the parent update factors through R_I, including its dependence on the configuration.

(G) Geometry, when declared. The condition of proposition 3.17 holds.

This is the sense of inheritance in clause (F3) of definition 6.31. Fixing A_I before R_I is essential. An empty protocol class makes (B) vacuous, and an unrestricted one forces the retention to be injective on exposed child laws. Dynamical, response, and connection inheritance are separate conditions.

This notion differs from a Markov-blanket hierarchy [60, 86, 115]. A Markov blanket is a conditional-independence property of a declared joint law over the states of a system. A parent here is a retention that intertwines the given dynamics and the declared responses; no joint law over configurations is assumed or derived, and conditional independence enters only where a separator is eliminated as in proposition 6.14.

Definition 6.17 (Type closure). Fix a constitutive class $\mathcal{C}$ of agent dynamics, for example Fisher gradient flows of reciprocal law potentials with normalized, possibly law-dependent channels and evaluators. An inherited parent is type-closed for $\mathcal{C}$ if its descended update, channels, and responses again belong to $\mathcal{C}$, with the parent laws as agent laws and their Fisher metric as the metric.

Type closure is what allows a result proved for primitive agents to be applied to parents without change, and hence by induction at every level. For the gradient class, proposition 6.5 gives the test under horizontal conformality: the retention must be horizontally conformal from the fine Fisher metric to the Fisher metric of the parent laws, and the fine potential must satisfy (6.8) with a reciprocal potential of the parent laws.

Open problem 6.18 (Type-closed parents). Determine which of the exact parents of the later chapters are type-closed for the class of Fisher gradient flows of reciprocal law potentials. A first case is the additive common-precision Wishart parent of chapter 12, where the first step is to test horizontal conformality of matrix addition with respect to the two Fisher metrics.

6.5 A finite two-level construction

Let a complete configuration be $x = (a, b, c, r) \in \{0, 1\}^4$. The coordinates are labels of internal configurations, with r an internal binary register and $\rho(x) = r$ its readout. To attach actual agent laws, choose

$$q_a = \mathrm{Ber}\left(\frac{1+2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2+a}{4}\right), \qquad \zeta_a = q_a \otimes s_a. \tag{6.24}$$

Thus both sectors vary, with tied parameters in this supplied family; b, c are additional internal marks. A model presentation $m \in \{0, 1\}$ has evaluator

$$K(m, \{1\}) = \frac{1}{4} + \frac{m}{2}. \tag{6.25}$$

The corresponding predictive law is $\nu_a = s_a K = \mathrm{Ber}(1/2 + a/8)$, which distinguishes the two values of a.

At each step draw independent bits $\xi \sim \mathrm{Ber}(1/4)$, $\eta \sim \mathrm{Ber}(1/3)$, and $\upsilon \sim \mathrm{Ber}(1/5)$, independently of preceding draws, and set

$$a' = a \oplus \xi, \qquad b' = b \oplus a \oplus \eta, \qquad c' = c \oplus b \oplus \upsilon. \tag{6.26}$$

Independently of these bit draws, conditional on the current configuration, give r' law ν_a. This supplies a transition of an internal register; the evaluator alone did not force this choice. The symbol $\oplus$ denotes addition modulo two. The dynamics are coupled: a affects the next b, and b affects the next c.

Retain $R_{10}(a, b, c, r) = (a, b, r)$ and then $R_{21}(a, b, r) = (a, r)$. For $f_p(0) = 1 - p$ and $f_p(1) = p$, the first parent kernel is

$$\begin{aligned} T^1((a, b, r), (a', b', r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\}) \\ \cdot f_{1/3}(b' \oplus b \oplus a). \end{aligned} \tag{6.27}$$

Summing the fine transition over c' gives this expression, independently of c. Summing it over b' gives $T^2((a, r), (a', r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\})$, independently of b. Both reductions satisfy proposition 6.10. Their configuration spaces have sizes sixteen, eight, and four, and both parents retain the laws in (6.24), their evaluator, and the register. Its readout descends by the identity on record values. The first parent additionally retains b.

Both parents reproduce the entire finite record history as well as the retained configuration history. Indeed, write the fine path probability as its initial mass times its transition factors. Summing over hidden coordinates gives the parent path law by proposition 6.10; the record at each step is already a coordinate of that path. The next record has probability $1/2 + a/8$ of being one, so the inherited response is nonconstant. No independence of the initial coordinates is needed.

This construction has nontrivial reductions and coupled dynamics, but its retentions were specified in advance. It proves exact two-level inheritance, without proving that the dynamics choose either reduction. It is exact lumping of a drive–response chain: the fine configuration carries one law pair and internal marks, so neither level groups two or more agents, and each retained block is autonomous because the update of a reads only a and the update of (a, b) reads only (a, b). For any coordinate projection, (6.14) is precisely such autonomy of the retained block in law. In the reciprocal gradient dynamics of chapter 4, proposition 6.7 shows that this one-way mechanism is unavailable between energetically coupled agents.

6.6 Averaged laws under smooth peer dynamics

The arithmetic mixture of child laws is a normalized candidate parent law, the linear opinion pool [63], which minimizes the average divergence $\sum_j w_j D_{\mathrm{KL}}(p_j \| q)$ [1]. It need not determine either its own derivative or its boundary responses. The following example establishes both failures within the smooth peer-KL family of chapter 4, with identical agents, a coupled evaluator, and equal Fisher mobilities. It concerns this specified retention; it is not an obstruction to every possible parent state.

Definition 6.19 (A coupled binary model with law-dependent channels). Use counting measure on $k, m \in \{-1, 1\}$ and the strictly positive laws

$$q_x(k) = \frac{1 + kx}{2}, \qquad s_y(m) = \frac{1 + my}{2}, \qquad x, y \in (-1, 1). \tag{6.28}$$

Choose $J = 1/2$, reference $r(m) = 1/2$, and normalized evaluator $G_J(k \mid m) = e^{Jkm}/(2\cosh J)$. For the declared product recognition family, the local variational term is exactly

$$\begin{aligned} E_0(x, y) &= \sum_{k,m} q_x(k) s_y(m) \log \frac{q_x(k) s_y(m)}{r(m) G_J(k \mid m)} \\ &= \psi(x) + \psi(y) - Jxy + \log\cosh J, \\ \psi(x) &= \frac{1 + x}{2} \log(1 + x) + \frac{1 - x}{2} \log(1 - x). \end{aligned} \tag{6.29}$$

All expectations are finite. The mixed derivative is $-J$, so changing either sector changes the force on the other.

In either sector, let a sender with parameter x use the normalized kernel

$$K_x(\ell \mid k) = \frac{1 + \vartheta(x) k\ell}{2}, \qquad \vartheta(x) = \frac{1 + x^2}{4}, \qquad q_x K_x = q_{f(x)}, \quad f(x) = \frac{x + x^3}{4}. \tag{6.30}$$

Every entry exceeds $1/4$. The channel transmits the sampled sender bit, and its transmission coefficient ϑ depends nontrivially on the sender law. For passive frame covariance, present each alphabet as the atoms $\{\pm u_i\}$ of a one-dimensional fiber, with $u_i \neq 0$, and define $\mathcal{K}_{ij}(u_j k, \{u_i \ell\}) = K_{x_j}(\ell \mid k)$. Under $g_i \in \mathrm{GL}(1, \mathbb{R})$, push the laws forward and replace u_i by $g_i u_i$. The reference parameters remain fixed, and

$$\mathcal{K}'_{ij}(g_j v, g_i A) = \mathcal{K}_{ij}(v, A). \tag{6.31}$$

The model sector, with its own representation of the same group, gives the same construction for s; pushing G_J through its model and belief frames preserves its normalization. These frames are per-agent coordinates on each agent's fiber; the identity is a passive re-expression and constrains nothing in the dynamics. For two identical agents, set

$$\begin{aligned} d(x, z) &= D_{\mathrm{KL}}(q_x \| q_z), \\ P(x_1, x_2) &= d(x_1, f(x_2)) + d(x_2, f(x_1)), \\ E &= \sum_{i=1}^{2} E_0(x_i, y_i) + P(x_1, x_2) + P(y_1, y_2). \end{aligned} \tag{6.32}$$

Each source row contains only the other agent, so its normalized weight is one and its row-relative-entropy term vanishes. This is a specialization of the declared peer potential. The local term is the displayed KL expectation; no further identification of the entire law-dependent potential with a fixed-joint-law ELBO is assumed. The product Fisher metric and its negative gradient flow are

$$g = \sum_i \left(\frac{dx_i^2}{1 - x_i^2} + \frac{dy_i^2}{1 - y_i^2} \right), \qquad \dot{x}_i = -(1 - x_i^2) \partial_{x_i} E, \quad \dot{y}_i = -(1 - y_i^2) \partial_{y_i} E. \tag{6.33}$$

Both mobilities are one. The vector field is smooth on $(-1,1)^4$.

Proposition 6.20 (The mixture retention fails the dynamical inheritance test). For the flow in (6.33), the retention

$$R(x_1,y_1,x_2,y_2)=\left(\frac{q_{x_1}+q_{x_2}}{2},\frac{s_{y_1}+s_{y_2}}{2}\right)=(q_{\bar{x}},s_{\bar{y}}) \tag{6.34}$$

does not support an autonomous parent vector field on the whole interior. The same failure holds if the averaged joint law $\frac{1}{2}\sum_i q_{x_i}\otimes s_{y_i}$ is also retained.

Proof. For $0<|a|<1$ and $0<|c|<1$, take $A=(0,c,0,c)$ and $B=(a,c,-a,c)$ in the coordinate order of (6.34). Both retain (q_0,s_c) and the same averaged joint law $q_0\otimes s_c$. These are equal full binary probability laws, not merely equal estimates of their means. The identities $f(-x)=-f(x)$ and $d(-x,-z)=d(x,z)$ imply

$$P(x_1,x_2)=P(x_2,x_1)=P(-x_1,-x_2),\qquad \partial_1P(a,-a)=-\partial_2P(a,-a),\quad \nabla P(0,0)=0. \tag{6.35}$$

At B the two Fisher prefactors are equal. Consequently the peer forces and the two atanh terms cancel in $\dot{\bar{x}}$, giving

$$\dot{\bar{x}}(A)=Jc,\qquad \dot{\bar{x}}(B)=Jc(1-a^2). \tag{6.36}$$

Both model-sector derivatives can also be written exactly. Define

$$h(c)=2\operatorname{atanh}c-\operatorname{atanh}f(c)+f'(c)\frac{f(c)-c}{1-f(c)^2}. \tag{6.37}$$

Indeed, direct differentiation gives $\partial_x d(x,z)=\operatorname{atanh}x-\operatorname{atanh}z$ and $\partial_z d(x,z)=(z-x)/(1-z^2)$. Thus

$$\begin{aligned}(\dot{y}_1,\dot{y}_2)(A)&=-(1-c^2)(h(c),h(c)),\\ (\dot{y}_1,\dot{y}_2)(B)&=-(1-c^2)(h(c)-Ja,h(c)+Ja),\\ \dot{\bar{y}}(A)=\dot{\bar{y}}(B)&=-(1-c^2)h(c).\end{aligned} \tag{6.38}$$

The retained belief derivatives nevertheless differ by $-Jca^2\neq 0$. For $a=1/2,c=1/4$, they are $1/8$ and $3/32$. Hence $DR_xB(x)$ is not constant on a retention fiber, contradicting the necessary condition (6.4). The identical retained joint laws establish the additional assertion. □

The parameter a may be arbitrarily small. A neighborhood of the synchronized configurations therefore does not make this retention exact. Unlike proposition 6.9, the example uses identical dynamical roles.

Proposition 6.21 (The same retention loses a boundary response). Choose a child uniformly, draw its belief bit, and transmit that bit through (6.30) to an external receiver. This declared protocol has a normalized binary response law with mean $b(x_1,x_2)=(f(x_1)+f(x_2))/2$. For any m,a with $|m|+|a|<1$, compare belief parameters (m,m) and $(m+a,m-a)$, keeping both model parameters equal to c. The retained marginal and averaged joint laws agree, but

$$b(m+a,m-a)-b(m,m)=\frac{3}{4}ma^2. \tag{6.39}$$

For $ma \neq 0$, their full binary response laws differ.

Proof. Expand $f(x) = (x + x^3)/4$ and use $((m + a)^3 + (m - a)^3)/2 = m^3 + 3ma^2$. At $m = a = 1/4$, the two response means are $20/256$ and $17/256$. Thus the response kernel is not constant on a retention fiber, violating (6.18). □

Proposition 6.22 (Interior synchronization cannot first occur at finite time). For (6.33), the diagonal $\Delta = \{x_1 = x_2,\ y_1 = y_2\}$ is invariant. A solution starting outside Δ cannot enter it at a finite time while remaining in the interior.

Proof. Agent exchange preserves the potential and metric, hence the vector field. On its fixed set Δ, the corresponding component derivatives agree, which proves invariance. The smooth field has unique local solutions in both time directions. If an interior solution meets Δ, uniqueness from that meeting point identifies it locally, backward as well as forward, with the solution in Δ. Along the original existence interval, the set of times at which it lies in Δ is therefore open; it is also closed by continuity. Connectedness of that interval makes it the entire interval, contradicting the initial condition. □

The argument is the invariance of fixed-point subspaces of equivariant vector fields [67] combined with backward uniqueness.

This result permits asymptotic synchronization, state-dependent threshold selection, and exact quotients that do not require synchronization. It does not prove convergence. An alternative retained state or a restricted invariant family must be tested against both inheritance conditions; the calculations above do not exclude such constructions.

6.7 Memory and formation

When a proposed retention fails, one exact option is to retain more of the configuration. For a declared stochastic process, another is to retain the conditional law of the hidden configuration given the accessible retained history. That conditional law determines future predictions. In a finite model it is a probability vector with an exact recursive update, derived in the supplement. Its attainable values need not form a finite set. Introducing such a filter presupposes a process and a preparation law; it is not compulsory for deterministic coupled agents.

The conditional law is not a configuration retention in the sense of definition 6.1. It is a function of the retained history, and it evolves as a Markov process on the simplex of laws on S, an information state, moving from $\widehat{p}$ to $\widehat{p}TP_z/(\widehat{p}TP_z\mathbf{1})$ with probability $\widehat{p}TP_z\mathbf{1}$ in the notation of the supplement. It compresses only when noise or nonobservability prevents the history from determining the fine state. For deterministic dynamics observed through R, the filter is supported on the image under F^n of the initial configurations consistent with the history, a set that decreases to the observability class of proposition 6.12 with interface R; when that class is a point, the filter becomes the fine state. The intertwined handoffs of proposition 6.11 are the case in which the filter is a function of the current retained value.

Theorem 6.23 (Information lost by a retained history). Let $\mathbb{P}$ be a probability law on $\mathsf{Z}_0 \times \cdots \times \mathsf{Z}_{n-1}$, where each record space is standard Borel and the horizon n is finite. Set $H_t = (Z_0, \ldots, Z_{t-1})$, with H_0 a point. For measurable maps into standard Borel statistic spaces, let $M_t = f_t(H_t)$. Write

$P_t^M(\cdot \mid m)$ for a regular conditional law of Z_t given $M_t = m$. Let $\widetilde{T}_t$ be any normalized kernel from the statistic space to Z_t, and let $\widetilde{\mathbb{P}}$ be the causal path law whose next-record kernel is $\widetilde{T}_t(f_t(h), \cdot)$ at history h. Then, in $[0, \infty]$,

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P} \| \widetilde{\mathbb{P}}) = &\sum_{t=0}^{n-1} I_{\mathbb{P}}(Z_t; H_t \mid M_t) \\ &+ \sum_{t=0}^{n-1} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(P_t^M(\cdot \mid M_t) \| \widetilde{T}_t(M_t, \cdot)\big). \end{aligned} \tag{6.40}$$

The first sum is the minimum path divergence among these predictors, attained by choosing $\widetilde{T}_t = P_t^M$. Its t-th term vanishes exactly when the true next-record law has a version depending on H_t only through M_t.

Proof. Let $P_t(\cdot \mid h)$ be the true conditional law given $H_t = h$. The relative-entropy chain rule expresses the path divergence as

$$\sum_{t=0}^{n-1} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(P_t(\cdot \mid H_t) \| \widetilde{T}_t(M_t, \cdot)\big). \tag{6.41}$$

To decompose its t-th summand, compare the true joint law of (H_t, Z_t) with the law obtained by keeping its H_t marginal and drawing Z_t from $\widetilde{T}_t(M_t, \cdot)$. Disintegrate these two laws by (M_t, Z_t). The divergence of their marginals is $\mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(P_t^M(\cdot \mid M_t) \| \widetilde{T}_t(M_t, \cdot))$. In the second joint law the conditional history given (M_t, Z_t) is the true conditional history given M_t alone. The conditional divergence term is consequently $I_{\mathbb{P}}(Z_t; H_t \mid M_t)$.

The chain rule on standard Borel spaces is an equality of nonnegative extended quantities. A singular marginal or conditional contributes infinite divergence; no subtraction of infinite terms is used. Applying it to each summand of eq. (6.41) gives eq. (6.40). Choosing the true statistic-conditional rows makes every term in the second sum zero. Conditional mutual information vanishes exactly when the corresponding conditional joint law is the product of its conditional marginals. Since M_t is a function of H_t, this is equivalent to $P_t(\cdot \mid H_t) = P_t^M(\cdot \mid M_t)$ almost surely. □

The finite-window case is the Markov approximation of a record process [68, Thm. 7.3.1]; the theorem permits any measurable history statistic. Its two terms separate forgetting information from predicting incorrectly with the information kept. Vanishing of the first sum is predictive sufficiency of the statistic [129], and for a statistic of a coarse process it is the stochastic autonomy criterion of Barnett and Seth [9, Sec. 2.3]. A statistic alone need not have a compact recursive realization. To update from the retained state and new record only, one additionally needs $M_{t+1} = U_t(M_t, Z_t)$, or a sufficient enlargement of M_t; this is the determinism of causal states in Shalizi and Crutchfield [129]. The theorem concerns one preparation and protocol. Sufficiency for every admitted intervention requires the corresponding conditional-law identity for each such protocol.

A handoff initializes the parent from the actual fine state or its declared conditional law, including the source marks and memory read by future interactions. Proposition 6.11 gives the class of declared conditional laws for which the parent is memoryless. When evidence is shared across levels, its likelihood belongs to the joint model once. Reusing a child's record through a complete same-

episode parent posterior requires the appropriate conditional or cavity accounting. A subsequent predictive prior is a different operation.

No parent in the later constructions acts as a source for its children. A parent law can enter a child's cost as a law-dependent source, but the exact reverse channel exerts no force: if ζ_I is the pushforward of $\bigotimes_{i\in I}\zeta_i$ under a retained statistic and $\mathcal{K}_{iI}$ is the regular conditional law of child i given that statistic, then $\zeta_I\mathcal{K}_{iI}=\zeta_i$, the marginal of a disintegration. Downward action therefore needs a declared reverse channel that differs from the exact conditional, and its inheritance must be tested with the moving-map form of (6.4); no such construction is given here.

6.8 Statistical partitions and their limits

An independence partition is a property of a declared joint law. A tuple of individual agent laws does not itself specify a cross-agent joint law; taking their product would impose independence in the input. The following constructions apply when such a joint law has been supplied.

Proposition 6.24 (The finest independent partition). Let V be a finite nonempty index set and let $\mathbb{P}$ be a probability law on $\prod_{i\in V}\mathsf{Y}_i$, with standard Borel coordinate spaces. Call a partition π independent when

$$\mathbb{P}=\bigotimes_{B\in\pi}\mathbb{P}_B, \tag{6.42}$$

where $\mathbb{P}_B$ is its marginal on block B. There is a unique finest independent partition $\pi_*(\mathbb{P})$. Relabeling the coordinates relabels this partition, and separate bimeasurable changes of their presentations preserve it.

Proof. The one-block partition is independent. Suppose π and σ are independent. Marginalizing the σ factorization to a block $B\in\pi$ gives

$$\mathbb{P}_B=\bigotimes_{\substack{D\in\sigma\\ B\cap D\neq\varnothing}}\mathbb{P}_{B\cap D}. \tag{6.43}$$

Substituting these marginals into the π factorization proves independence of the common refinement consisting of the nonempty intersections $B\cap D$. There are finitely many partitions of V. Refining all independent ones therefore gives an independent partition finer than every one of them, which is necessarily unique. Relabelings and coordinatewise bimeasurable maps commute with taking marginals and product laws. Applying the inverse transformation gives the reverse implication, so the set of independent partitions and its unique finest member transform as stated. □

This statistical target is the finest-independent clustering problem [127]. Its existence does not require Gaussian laws or densities, but does not identify statistical independence with agency. In particular, it makes no statement about autonomous dynamics or future boundary responses.

Corollary 6.25 (Conditional independence partitions). Let the same coordinate tuple have a joint law with an already retained standard Borel variable C. Define

$$\mathcal{I}_C(\pi) = \int D_{\mathrm{KL}}\left(\mathbb{P}(dy_V \mid c) \middle\| \bigotimes_{B\in\pi} \mathbb{P}_B(dy_B \mid c)\right) \mathbb{P}_C(dc). \tag{6.44}$$

There is a unique finest partition for which $\mathcal{I}_C(\pi) = 0$.

Proof. Zero divergence means equality of the conditional joint and product laws for $\mathbb{P}_C$-almost every c. There are finitely many partitions. Outside the union of their relevant null sets, the common-refinement argument in proposition 6.24 applies pointwise to every conditionally independent partition. Their common refinement is therefore conditionally independent and uniquely finest. □

Choosing $C = Y_V$ makes every partition conditionally independent while retaining the whole tuple. The corollary does not choose the context that a useful parent should retain.

Proposition 6.26 (Weak coupling can destroy every independent split). For any even $N \geq 4$, let $y_i \in \{-1, 1\}$ and define the strictly positive finite law

$$\mathbb{P}_\epsilon(y) = Z_\epsilon^{-1} \exp\left(J \sum_{b=1}^{N/2} y_{2b-1}y_{2b} + \epsilon \sum_{b=1}^{N/2-1} y_{2b}y_{2b+1}\right), \qquad J > 0, \quad \epsilon \geq 0, \tag{6.45}$$

where Z_ϵ is the sum of the numerator over the finite alphabet. Then $\pi_*(\mathbb{P}_0)$ consists of the pairs $\{2b-1, 2b\}$, while $\pi_*(\mathbb{P}_\epsilon) = \{V\}$ for every $\epsilon > 0$, although $\mathbb{P}_\epsilon \to \mathbb{P}_0$ in total variation.

Proof. At $\epsilon = 0$, distinct pairs are independent and each pair is dependent: its odds ratio is $e^{4J} \neq 1$. Thus the finest partition consists of those pairs. For $\epsilon > 0$ the interaction path is connected. Any nontrivial partition cuts an edge. Fix all coordinates except the endpoints of one cut edge, whose coupling is $J_e \in \{J, \epsilon\}$. Positivity makes every such conditional law well defined. Its odds ratio is $e^{4J_e} \neq 1$, since all terms involving just one endpoint cancel in that ratio. Factorization across the proposed partition would make these endpoints conditionally independent after the other coordinates were fixed, forcing ratio one. No nontrivial independent partition exists. Continuity of each of the finitely many probabilities gives the convergence in total variation. □

Exact independence is therefore not a continuous soft-membership criterion.

Proposition 6.27 (Symmetry constrains a deterministic hard selector). Let all coordinate spaces equal Y, and suppose a deterministic partition selector Π reads only a law on Y^V. Assume relabeling naturality: $\Pi(\sigma_\#\mathbb{P}) = \sigma\Pi(\mathbb{P})$ for every permutation σ of V. Let $H_\mathbb{P} = \{\sigma : \sigma_\#\mathbb{P} = \mathbb{P}\}$ be the stabilizer of the law. Then $\Pi(\mathbb{P})$ is invariant under $H_\mathbb{P}$. If $H_\mathbb{P}$ is transitive, the blocks of $\Pi(\mathbb{P})$ form a block system of $H_\mathbb{P}$; if it is primitive, $\Pi(\mathbb{P})$ is either the singleton or the one-block partition. In particular this holds when $\mathbb{P}$ is invariant under all permutations.

Proof. For $\sigma \in H_\mathbb{P}$, naturality gives $\Pi(\mathbb{P}) = \Pi(\sigma_\#\mathbb{P}) = \sigma\Pi(\mathbb{P})$. An invariant partition of a transitive group is by definition a block system, and a transitive group is primitive when its only block systems are the two trivial partitions [48]. The full symmetric group is primitive. For $|V| \leq 2$ only the trivial partitions exist. For $|V| \geq 3$, if an invariant partition had a block B with $2 \leq |B| < |V|$, choose

distinct $i, j \in B$ and $k \notin B$. The transposition exchanging j and k sends B to a distinct block that still contains i, which is impossible for a partition. □

For a law on a four-cycle invariant under its dihedral group, the only nontrivial block system is the diagonal pairing $\{\{1,3\},\{2,4\}\}$, so a relabeling-natural law-only selector cannot output adjacent pairs; adjacent pairing needs asymmetric preparation data. A distribution over partitions, soft memberships, asymmetric realized histories, or additional geometric inputs are outside this proposition. An information-based hierarchy can also be retained before choosing a cut [29]. None of these possibilities supplies the required dynamics without a further construction. The symmetry result does not exclude symmetry breaking from asymmetric initial conditions.

The four results of this section need a cross-agent joint law, and the dynamics of chapter 4 evolves a tuple of agent laws with product recognition. The product of the agent laws has the singleton partition as its finest independent partition. The one joint law that the dynamics produces is the pushforward $(\Phi_t)_{\#}\mu_0$ of a preparation ensemble μ_0, and coupling makes it dependent across interacting agents: for $\dot{x} = Ax$ with $A = \left(\begin{smallmatrix} -1-\lambda & \lambda \\ \lambda & -1-\lambda \end{smallmatrix}\right)$ and $x(0) \sim \mathcal{N}(0, I)$, $\mathrm{Cov}(x_1(t), x_2(t)) = \frac{1}{2}(e^{-2t} - e^{-2(1+2\lambda)t}) > 0$ for $\lambda, t > 0$. These results are therefore constraints on admissible selectors rather than formation mechanisms; the selectors of the later chapters are declared thresholds for this reason.

6.9 Architectures whose selection cannot see the interaction

Two exact statements show that some architectures cannot satisfy the interaction-necessity clause of the formation standard, whatever selector they use within a natural class.

Proposition 6.28 (Scalar additive closure fixes the membership order). Let scalar parameters θ_i in an open interval J evolve by $\dot{\theta}_i = G(\theta_i, g(t))$, where g is a state common to all agents and G is continuous in its first argument. Suppose that for every pair of agents the sum retention $\theta_i + \theta_j$ intertwines with the flow, that is, $G(x, g) + G(y, g)$ depends on $x, y \in$ J only through $x + y$. Then $G(\theta, g) = a(g)\theta + c(g)$, and every difference $\theta_i - \theta_j$ and every difference of count-weighted block centroids equals its initial value times the common factor $m(t) = \exp \int_0^t a(g(\tau))d\tau$. Consider a selector that joins two current blocks when their centroid distance first reaches a fixed threshold from above, and suppose m is nonincreasing and the initial centroid distances have no ties. Then the sequence of selected partitions is an initial segment of a sequence determined by the initial data alone; the common state g determines only the event times and the length of the segment.

Proof. For fixed g put $\phi = G(\cdot, g)$. If $x + y = x' + y'$ then $\phi(x) + \phi(y) = \phi(x') + \phi(y')$; taking $x' = y' = (x + y)/2 \in$ J gives Jensen's equation $\phi((x + y)/2) = (\phi(x) + \phi(y))/2$, whose continuous solutions are affine. Hence $\frac{d}{dt}(\theta_i - \theta_j) = a(g)(\theta_i - \theta_j)$, and a count-weighted centroid obeys the same affine law, so centroid differences are also multiplied by $m(t)$. A pair of current blocks with initial centroid distance D is at distance $Dm(t)$ and first reaches the threshold δ when $m(t) = \delta/D$. Since m is common and nonincreasing, events occur in increasing order of D. After a join at level D, a pair involving the new block with initial centroid distance $D' \le D$ is already at or below the threshold and, m being nonincreasing, never reaches it from above; a pair with $D' > D$ joins later in order of D'. All of these comparisons involve only initial distances. Whether the event at level D occurs depends on whether m reaches δ/D, which is where g enters. □

Anisotropic linear laws escape this proposition. For vector parameters with $\dot{\theta}_i = A(g)\theta_i + c(g)$, differences evolve by a common propagator that need not preserve the ordering of distances: with $A = \mathrm{diag}(-1, -3)$, initial differences $(1, 0)$ and $(0, 1.5)$ become approximately $(0.37, 0)$ and $(0, 0.075)$ at $t = 1$, reversing which pair is closer. Whether a smooth reciprocal potential realizes an anisotropic, interaction-dependent propagator with exact additive closure is open.

Proposition 6.29 (Shared targets give a common clock). Work on finite positive tables as in theorem 4.6, with a common belief alphabet K. Suppose every evaluator is the same model-independent law, $G_i(m, \cdot) = G_0$; the belief source sets exclude self; and every belief channel is $\mathcal{K}^q_{ij}[\mathbf{q}](v, \cdot) = C[M]$ for all $j \neq i$ and all inputs v, where C is a twice continuously differentiable normalized law of the additive moment $M = \sum_i \mathbb{E}_{q_i}\mathsf{T}$ of a fixed statistic $\mathsf{T} : \mathsf{K} \to \mathbb{R}^p$. Then along the Fisher flow of eq. (4.4) (with dynamic rows) or of its reduced potential eq. (4.6),

$$\frac{d}{dt} \log \frac{q_i(k)}{q_j(k)} = -2 \log \frac{q_i(k)}{q_j(k)} + c_{ij}(t), \tag{6.46}$$

with c_{ij} independent of k. Along the null flow of definition 6.30 the coefficient -2 is replaced by -1.

Proof. With a model-independent evaluator the local term splits as $D_{\mathrm{KL}}(s_i \| r_i) + D_{\mathrm{KL}}(q_i \| G_0)$, whose belief derivative is $\log q_i - \log G_0$ plus a constant. Each peer cost $D_{\mathrm{KL}}(q_i \| C[M])$ contributes $\beta_{ij}(\log q_i - \log C[M])$ as receiver, and the rows of i sum to one over $j \neq i$. As a source, q_i enters every peer cost only through M, so it contributes $-\sum_l \sum_{j \neq l} \beta_{lj} \sum_u \frac{q_l(u)}{C[M](u)} \frac{\partial C[M](u)}{\partial M} \cdot \mathsf{T}(v)$, which is the same function of v for every i; with Gibbs rows the envelope identity eq. (4.8) gives the same expression with β^*. Hence $h_i = 2 \log q_i + V_t$ with V_t common to all agents, up to constants in v. The flow eq. (4.15) gives $\frac{d}{dt} \log q_i = -(h_i - \mathbb{E}_{q_i} h_i)$, and subtracting the equations for i and j proves (6.46). Deleting the peer costs leaves $h_i = \log q_i - \log G_0$. □

Hence $\log(q_i/q_j)$, modulo additive constants, equals e^{-2t} times its initial value under the full flow and e^{-t} times it under the null flow. Any relabeling-equivariant selector that reads only these pairwise log-ratios modulo constants produces under the full flow the membership history it produces under the null flow, with every event time halved. Its partition sequence is therefore that of the null flow, and it fails interaction necessity, which disregards event times. Interaction-sensitive selection under the reciprocal potential therefore requires receiver-dependent targets, such as selective channels with nonconstant Gibbs rows.

6.10 The formation standard

The thesis of the theory is that interactions between agents can produce further agents. A definition of formation must therefore exclude three ways in which the selection and inheritance clauses can be met without any such production. A parent that retains every child state compresses nothing: four agents with laws $\mathcal{N}(x_i, 1)$ under symmetric peer-KL consensus, a threshold selector, and the identity retention select two pairs and then their union with exact inheritance at every stage. Agents that do not interact can be grouped by any rule that reads their initial data and a common relaxation clock. A second stage that forms from every handoff whatever the first stage produced is a sequence of supplied scales rather than recursion. The standard below makes each exclusion a clause.

Definition 6.30 (Null flow). Let a construction specify a flow Φ generated by a potential of the form eq. (4.4) or by the per-agent dynamics of definition 4.19, together with references, evaluators, source priors, temperatures, a selector, and an admitted class of initial configurations. Its null flow Φ^0 is the flow obtained by deleting every peer cost D^q_{ij}, D^s_{ij} with $j \neq i$ and evaluating every law-dependent channel and evaluator at its value with the peer laws held at their initial values, with all other data unchanged. The membership history $H(x)$ of an initial configuration x is the sequence of selected partitions with their event times; $H^0(x)$ is the history produced by the same selector along Φ^0. The partition sequence $P(x)$ is $H(x)$ with its event times removed, and $P^0(x)$ is obtained from $H^0(x)$ in the same way.

Definition 6.31 (Inherited recursive formation). An inherited recursive formation is a construction satisfying all of the following clauses.

(F1) Admitted class. The initial configurations form a nonempty set that is open in a declared topology on a declared space of initially unpartitioned configurations, or a declared invariant family.

(F2) Selection. The dynamics selects nontrivial blocks with a declared persistence criterion, and each handoff is a measurable function of the state or of accessible history.

(F3) Inheritance. Each selected block carries inherited belief and model laws, evaluators, boundary responses, and dynamics in the sense of this chapter, with the class of admitted probes fixed before the retention is chosen.

(F4) Compression. Each handoff retention is non-injective on the post-handoff invariant set, and some admitted child-level probe distinguishes two configurations with the same retained value.

(F5) Interaction necessity. $P \neq P^0$ on a nonempty open subset of the admitted class. A difference only in event times does not count.

(F6) Recursion. The parents select a block containing at least two first-stage parents under the inherited interactions, without resetting their source data or supplying a newly tuned attraction, and this second-stage selection does not occur from every counterfactual handoff in the inherited class.

A construction satisfies the strong form of (F5) if its final partition is not a function of any statistic of the initial data whose evolution is the same under Φ and Φ^0. It is amplifying if, at each event, the ratio of the largest within-block diameter to the smallest between-block distance is smaller along Φ than along Φ^0 by a declared factor below one. Write (F6a) for the first sentence of (F6), a second selection among first-stage parents under the inherited interactions without reset or retuning, and (F6b) for its counterfactual part. A construction satisfying (F1), (F2), (F3), and (F6a) but failing at least one of (F4), (F5), and (F6b) is a two-stage selection with exact inheritance; a construction with only one selection stage that satisfies (F1) to (F3) is a single-stage selection with exact inheritance. The clauses that fail are always stated.

Persistence can mean an invariant block relation in deterministic dynamics or a specified path event in stochastic dynamics. Clause (F5) is a control experiment: it asks whether the grouping would occur with the interaction switched off. Clause (F4) excludes the identity retention described above, and (F6) is the counterfactual-handoff requirement. The finite construction of this chapter settles

inheritance for a supplied hierarchy and makes no selection claim. Chapter 7 gives a construction with explicit fusion events under a nonsmooth interaction; at zero coupling its agents never collide, so it meets (F5), but fused children are identical, so it fails (F4), and it fails (F6b); it is a two-stage selection with exact inheritance. The erasure construction of chapter 9 is a single-stage selection with exact inheritance. Under the original smooth potential, chapter 12 selects two parents and then their union on a prepared common-precision family, but its membership schedule is independent of the coupling (proposition 12.7), so it fails (F5); since the union also forms from every counterfactual handoff with sufficiently separated parents, it fails (F6b) as well, and it is a two-stage selection with exact inheritance. Chapter 13 gives a six-agent construction on a full open law neighborhood with native preferential selection; it meets (F4) and (F6a), but in the regime of its proof the null flow reproduces its membership history, so its theorem does not establish (F5), and (F6b) is not verified; it is a two-stage selection with exact inheritance. Both smooth constructions supply their compression architectures.

Open problem 6.32 (A recursive formation). The normalized per-agent construction of chapter 14 supplies a complementary one-stage result. It proves persistent interaction-dependent grouping with conflicting records and a physically noninjective parent preserving its declared port dynamics, finite adaptive histories, and nonflat geometry on a finite window. The sufficient phase interface is predeclared and the parent retains primitive ports. Joining already canonical parents loses no new coordinate, so neither a second compressive handoff nor (F6b) follows.

Construct a population dynamics under the reciprocal potential, the per-agent free energies, or a global free energy that satisfies (F1) to (F6), preferably in the strong and amplifying forms, and derive rather than declare its selector and its compression interface.

Chapter 7

Finite Fusion with Exact Statistical Inheritance

This chapter constructs a population whose dynamics makes agents coincide at finite times and whose coincident blocks carry exact inherited laws, evaluator, metric, and responses. The dynamics is a nonsmooth modification of the reciprocal law potential of definition 4.2: each peer divergence is replaced by the square root of twice that divergence, source rows are fixed, and the laws are restricted to a lognormal family with fixed variances. It is not a free energy of any normalized model, and its conclusions do not transfer to the smooth potential or to the per-agent and global free energies of chapter 4.

Pairwise fusion penalties occur in convex clustering [77], and suitable nonsmooth consensus protocols produce finite-time agreement [33]. On the restricted family the square-root interaction is exactly the absolute-value kernel, so in each sector the population is a weighted particle solution of the one-dimensional aggregation equation with attractive kernel $\lambda|x|$, whose sticky collisions and finite-time total collapse are known [19, 26]. What the construction adds is the inheritance bookkeeping: represented laws, model evaluator, transport, parent metric, and boundary kernels descend exactly through each collision. Its dynamical outcome is a finite coalescence tree that ends in global consensus for every initial datum (proposition 7.3); persistent heterogeneous organization does not occur. Section 7.7 reports the construction against the formation standard of definition 6.31.

7.1 A law family closed under geometric averaging

Work over one context, with state and model spaces $\mathsf{K} = \mathsf{M} = (0, \infty)$. Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of e^Z when $Z \sim N(\mu, \sigma^2)$. An agent of integer multiplicity $n \geq 1$ carries the product law

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad \zeta_n = q_n \otimes s_n. \tag{7.1}$$

Multiplicity is retained state information. The variances are fixed within each type; only the two log means evolve. This is a constraint on the dynamics, not an invariant family of a larger flow. If the common log variance v were released, the local divergence of proposition 7.1 below would have derivative $n(a-1)$ in v at $v = 1/n$, and the peer comparison between restored messages would have a nonzero source-variance derivative whenever the means differ. The Fisher flow with free variances therefore leaves the family, and releasing the variance destroys the encoding of multiplicity in the law. Laws are first written in a common reference coordinate. A fixed positive frame $U_i^q = U_i^s =: U_i \in \mathrm{GL}^+(1)$ represents both sectors locally by multiplication. Frames do not evolve in this example.

Choose constants $a > c > 0$ and define the normalized reference and evaluator

$$r_n = \mathrm{LN}\left(0, \frac{a}{n(a^2-c^2)}\right), \qquad G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{7.2}$$

The two occurrences of a lognormal expression in this equation denote ordinary probability laws and kernels. In logarithmic coordinates the joint reference $r_n(dm)G_n(m, dk)$ has precision

$$nA, \qquad A = \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}. \tag{7.3}$$

Its eigenvalues $n(a-c)$ and $n(a+c)$ are positive. The predictive law is

$$s_n G_n = \mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right). \tag{7.4}$$

Thus the evaluator depends on the represented model value. Physical references and evaluators are obtained by pushing both coordinates through the common local frame.

Proposition 7.1 (Local energy and metric). For every $n \geq 1$ and every pair of real means, let

$$C = a - 1 - \frac{1}{2}\log(a^2 - c^2), \qquad \mathcal{U}(\mu^q, \mu^s) = \frac{a}{2}\big((\mu^q)^2 + (\mu^s)^2\big) - c\mu^q\mu^s. \tag{7.5}$$

Then

$$E_{0,n} = D_{\mathrm{KL}}(q_n \otimes s_n \| r_n G_n) = C + n\mathcal{U}, \qquad g_n = n\big((d\mu^q)^2 + (d\mu^s)^2\big). \tag{7.6}$$

Proof. The logarithm is a measurable bijection, so it preserves KL divergence. The recognition law in log coordinates has covariance I_2/n, while the reference has covariance A^{-1}/n. The Gaussian KL formula gives

$$E_{0,n} = \frac{1}{2}\left[\operatorname{tr} A - 2 - \log\det A + n(\mu^q, \mu^s)A(\mu^q, \mu^s)^{\mathsf{T}}\right], \tag{7.7}$$

which is the stated expression. The score for either mean is $n(Z - \mu)$, whose squared expectation is n. The product law has zero cross score expectation, giving the displayed Fisher metric. □

For independent synchronized children of multiplicities n_j, set $n_I = \sum_{j\in I} n_j$. Their retained reference samples are

$$k_I = \prod_{j\in I} k_j^{n_j/n_I}, \qquad m_I = \prod_{j\in I} m_j^{n_j/n_I}. \tag{7.8}$$

Weighted log averages have variance $\sum_j (n_j/n_I)^2/n_j = 1/n_I$. The same calculation for the joint reference gives $r_{n_I}G_{n_I}$. Conditional log observation noise has variance $\sum_j (n_j/n_I)^2/(an_j) = 1/(an_I)$, so the evaluator itself is inherited. Physical samples are first returned through U_j^{-1} and the retained sample is expressed in a chosen parent frame U_I.

7.2 Messages and state-dependent scalar links

Define a normalized message kernel by

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw), \tag{7.9}$$

interpreting $B_1(k, dw) = \delta_k(dw)$. It restores unit log variance: $q_n B_n = \mathrm{LN}(\mu^q, 1)$ and $s_n B_n = \mathrm{LN}(\mu^s, 1)$. These messages give all multiplicities a common comparison family. The kernel has a statistical meaning. For n independent unit children with log samples $N(\mu, 1)$, the mean log sample is sufficient for μ and has the law of the parent's log sample; the conditional law of one child's log sample given that mean is $N(\bar{Z}, 1 - 1/n)$, which does not depend on μ. Thus B_n is the conditional law of one child sample given the parent sufficient statistic [5], and this parameter independence is why fresh messages descend exactly.

Model prediction also commutes with this interface. Put $h = (c/a)^2 + 1/a$ and define $B_n^h(k, dw) = \mathrm{LN}(\log k, h(1 - 1/n))(dw)$, again using the identity at $n = 1$. Both routes in the kernel identity

$$G_n B_n^h = B_n G_1 \tag{7.10}$$

have conditional log mean $(c/a) \log m$ and variance $h - (c/a)^2/n$. Integrating against s_n gives

$$(s_n B_n) G_1 = (s_n G_n) B_n^h = \mathrm{LN}\left(\frac{c}{a}\mu^s, h\right). \tag{7.11}$$

Thus a parent can emit the original predictive message by either normalized route.

For each sector $\alpha \in \{q, s\}$, choose a fixed $0 < \chi_\alpha < 1$ and set

$$L_{ij}^\alpha = \chi_\alpha(\mu_i^\alpha - \mu_j^\alpha), \qquad \Theta_{ij}^\alpha = U_i \exp(L_{ij}^\alpha) U_j^{-1}. \tag{7.12}$$

Under a local change of frame $U_i \mapsto h_i U_i$, the physical transport changes to $h_i \Theta_{ij}^\alpha h_j^{-1}$. The reference means and L_{ij}^α remain unchanged. These links are pure gauge. With $V_i^\alpha = U_i \exp(\chi_\alpha \mu_i^\alpha)$ one has $\Theta_{ij}^\alpha = V_i^\alpha (V_j^\alpha)^{-1}$, so every cycle product is trivial and $\Theta_{ji}^\alpha = (\Theta_{ij}^\alpha)^{-1}$: the transport is a law-dependent change of frame of the trivial connection.

Transport of the reference message from j shifts its log mean to $\mu_j^\alpha + \chi_\alpha(\mu_i^\alpha - \mu_j^\alpha)$. Consequently the directed comparison cost

$$d_{ij}^\alpha = \frac{\sqrt{2 D_{\mathrm{KL}}\big(\mathrm{LN}(\mu_i^\alpha, 1) \| (\exp L_{ij}^\alpha)_{\#} \mathrm{LN}(\mu_j^\alpha, 1)\big)}}{1 - \chi_\alpha} = |\mu_i^\alpha - \mu_j^\alpha| \tag{7.13}$$

is gauge invariant. The normalized cost does not depend on χ_α, so the potential and the flow below are the same for every $\chi_\alpha \in (0, 1)$, including the limit of identity transport. The transport therefore carries no gauge content beyond passive covariance in this example; a transport whose parameter survives the normalization would need, for example, a noncommuting generator on a vector log space. The equality includes the dependence of the link on both endpoint laws. For unequal means, differentiating its log-mean residual gives

$$\partial_{\mu_i^\alpha}\left[\mu_i^\alpha - \mu_j^\alpha - L_{ij}^\alpha\right] = 1 - \chi_\alpha. \tag{7.14}$$

This factor cancels the denominator in eq. (7.13). Holding the link fixed while differentiating would replace the coupling λ below by $\lambda/(1-\chi_\alpha)$.

7.3 The continuous subgradient dynamics

For multiplicities n_i with total $N=\sum_i n_i$, choose $\lambda>0$ and the potential

$$\mathcal{F}(\boldsymbol{\mu})=\sum_i n_i\big[C+\mathcal{U}(\mu_i^q,\mu_i^s)\big]+\lambda\sum_{i<j} n_i n_j\big(|\mu_i^q-\mu_j^q|+|\mu_i^s-\mu_j^s|\big). \tag{7.15}$$

All contacts are fixed and present. If source rows are recorded, take $\beta_{ij}=\gamma_{ij}=n_j/N$, including self entries. Summing child entries gives exactly the same parent row. Self comparison costs vanish, while their row mass records the corresponding contact multiplicity. There is no Gibbs row adaptation. Relative to definition 4.2, the local term is $\sum_i E_{0,n_i}$ up to the constant $\sum_i(n_i-1)C$, while the peer term $\sum_j \beta_{ij}D_{ij}^\alpha$ of agent i is replaced by $\frac{1}{2}\lambda N n_i\sum_j \beta_{ij}d_{ij}^\alpha$, where $d_{ij}^\alpha=\sqrt{2D_{ij}^\alpha}/(1-\chi_\alpha)$ is the normalized square-root cost of eq. (7.13); this is the nonsmooth modification. The local term has the stated KL origin; the flow below is a declared gradient flow, not inference in a normalized model, and no fixed sample-model ELBO identity is asserted for the entire potential.

With $W=\mathrm{diag}(n_1,n_1,n_2,n_2,\dots)$, define the flow on $\mathbb{R}^{2r}$, for r current agents, by

$$W\dot{\boldsymbol{\mu}}\in-\partial\mathcal{F}(\boldsymbol{\mu}). \tag{7.16}$$

Solutions are absolutely continuous in the external time $t\geq 0$ and satisfy this inclusion almost everywhere. There are no boundaries or stochastic driving terms. Both law sectors have the same basic mobility; the factor $1/n_i$ comes from their inherited Fisher metric. In coordinates,

$$\dot{\mu}_i^q=-a\mu_i^q+c\mu_i^s-\lambda\sum_{j\neq i} n_j\epsilon_{ij}^q, \tag{7.17}$$

$$\dot{\mu}_i^s=-a\mu_i^s+c\mu_i^q-\lambda\sum_{j\neq i} n_j\epsilon_{ij}^s, \tag{7.18}$$

where $\epsilon_{ij}^\alpha=-\epsilon_{ji}^\alpha$ belongs to $\mathrm{Sign}(\mu_i^\alpha-\mu_j^\alpha)$ and $\mathrm{Sign}(0)=[-1,1]$. The multiplicity cancels from an agent's own velocity, which depends only on the multiplicities of the others. The system is overdamped one-dimensional motion in a harmonic trap under the pair potential $n_i n_j|x_i-x_j|$, so a multiplicity acts as a friction weight and as an interaction charge; it is not an inertia.

Lemma 7.2 (Global well-posedness, contraction, and persistence). For every multiplicity vector and every initial datum, eq. (7.16) has a unique absolutely continuous solution on $[0,\infty)$. Two solutions with the same multiplicities obey

$$\|\boldsymbol{\mu}(t)-\widetilde{\boldsymbol{\mu}}(t)\|_W\leq e^{-(a-c)t}\|\boldsymbol{\mu}(0)-\widetilde{\boldsymbol{\mu}}(0)\|_W. \tag{7.19}$$

Synchronized unit-multiplicity children remain synchronized in both sectors.

Proof. The potential $\mathcal{F}$ is convex and finite on $\mathbb{R}^{2r}$, so $W^{-1}\partial\mathcal{F}$ is a maximal monotone operator in the W inner product. The theory of maximal monotone operators gives a unique global absolutely

continuous solution for every initial datum, whose right derivative is the element of minimal norm in $-W^{-1}\partial\mathcal{F}(\boldsymbol{\mu})$ [20]. The quadratic part is $(a-c)$-strongly convex in the W metric and the absolute-value part is convex. For subgradients $\xi, \widetilde{\xi}$ at two configurations, monotonicity gives

$$\langle \xi - \widetilde{\xi}, \boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}} \rangle \geq (a-c)\|\boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}}\|_W^2. \tag{7.20}$$

Substitution from eq. (7.16) and integration proves eq. (7.19). Swapping two equal unit-multiplicity children preserves the initial configuration, potential, and metric. Uniqueness therefore preserves their equality. This argument concerns the full two-sector system, not a restricted scalar equation. □

Because the subdifferential of a convex function is its Clarke generalized gradient, eq. (7.16) coincides with the Filippov regularization of the almost-everywhere gradient field $-W^{-1}\nabla\mathcal{F}$ [32, 57, 114], so no ambiguity of solution concept arises. Backward uniqueness fails: the flow is a contraction semigroup and not a group, which is what permits finite-time sticking.

Proposition 7.3 (Complete synchronization in finite time). Fix any multiplicity vector and any initial datum. Put $R_0 = \|\boldsymbol{\mu}(0)\|_W$, $\kappa = a - c$, and $t_0 = \kappa^{-1}\log\max\{1, 4cR_0/(\lambda N)\}$. Then $\max_{i,\alpha}|\mu_i^\alpha(t)| \leq e^{-\kappa t}R_0$, and there is a time

$$T_* \leq t_0 + \frac{4R_0 e^{-\kappa t_0}}{\lambda N} \tag{7.21}$$

such that all agents have equal means in both sectors for every $t \geq T_*$. Every solution therefore produces a finite sequence of synchronization events, at most $r-1$ of them, ending in a single block.

Proof. The zero configuration is stationary, because $0 \in \partial\mathcal{F}(0)$. Equation (7.19) with this stationary solution gives $\|\boldsymbol{\mu}(t)\|_W \leq e^{-\kappa t}R_0$, and $|\mu_i^\alpha| \leq \|\boldsymbol{\mu}\|_W$ because $n_i \geq 1$. Fix a sector α, write α' for the other sector, and let $D^\alpha = \max_i \mu_i^\alpha - \min_i \mu_i^\alpha$, which is absolutely continuous. At almost every t all coordinates and both extrema are differentiable. At such a time every maximizer i has $\dot{\mu}_i^\alpha$ equal to the derivative of the maximum, because the maximum minus μ_i^α is nonnegative and vanishes at t. Let T be the set of maximizers, with multiplicity n_T, and average the coordinate equation over T with weights n_i/n_T. Antisymmetry of ϵ_{ij}^α cancels the internal terms, and $\epsilon_{ij}^\alpha = 1$ for $j \notin T$. Hence

$$\frac{d}{dt}\max_i \mu_i^\alpha = -a\max_i \mu_i^\alpha + c\,\overline{\mu^{\alpha'}}_T - \lambda(N - n_T), \tag{7.22}$$

where $\overline{\mu^{\alpha'}}_T$ is the weighted average over T. The same argument for the minimizing set B gives the opposite sign of the last term. While $D^\alpha > 0$ the sets T and B are disjoint, so $n_T + n_B \leq N$ and

$$\dot{D}^\alpha \leq -aD^\alpha + 2cR_0e^{-\kappa t} - \lambda N \quad \text{almost everywhere on } \{D^\alpha > 0\}. \tag{7.23}$$

For $t \geq t_0$ the middle term is at most $\lambda N/2$, so $\dot{D}^\alpha \leq -\lambda N/2$ wherever $D^\alpha > 0$. An absolutely continuous nonnegative function with this property reaches zero within $2D^\alpha(t_0)/(\lambda N)$ and cannot become positive afterward. Since $D^\alpha(t_0) \leq 2R_0e^{-\kappa t_0}$, both sectors are synchronized by the stated time. Agents that coincide in both sectors remain coincident: expanding integer multiplicities into

unit children and applying lemma 7.2 proves this. The number of distinct agents therefore decreases at each event. □

The common-offset family studied next provides explicit event times; it is not needed for finite-time synchronization. The following estimate controls a single pair off that family.

Lemma 7.4 (Isolated pairs synchronize). Let $i \neq j$ and let J be a time interval on which, in each sector α, no third agent has a mean in the closed interval between μ_i^α and μ_j^α. Put $S = |\mu_i^q - \mu_j^q| + |\mu_i^s - \mu_j^s|$. Then, for $s < t$ in J,

$$S(t) \leq \max\{0, S(s) - \lambda(n_i + n_j)(t - s)\}. \tag{7.24}$$

Proof. For a third agent k, the hypothesis gives $\epsilon_{ik}^\alpha = \epsilon_{jk}^\alpha$, so its terms cancel from the equation for $\Delta^\alpha = \mu_i^\alpha - \mu_j^\alpha$, leaving $\dot{\Delta}^\alpha = -a\Delta^\alpha + c\Delta^{\alpha'} - \lambda(n_i + n_j)\epsilon_{ij}^\alpha$. Where $\Delta^\alpha \neq 0$, $\frac{d}{dt}|\Delta^\alpha| \leq -a|\Delta^\alpha| + c|\Delta^{\alpha'}| - \lambda(n_i + n_j)$; where $\Delta^\alpha = 0$ on a set of positive measure, the derivative of $|\Delta^\alpha|$ vanishes almost everywhere. Adding the two sectors gives $\dot{S} \leq -\kappa S - 2\lambda(n_i + n_j)$ when both differences are nonzero, and $\dot{S} \leq -\lambda(n_i + n_j)$ when exactly one is nonzero. Hence $\dot{S} \leq -\lambda(n_i + n_j)$ almost everywhere on $\{S > 0\}$, which proves eq. (7.24). □

7.4 An invariant class with explicit global solutions

Theorem 7.5 (Finite sticking on a common-offset family). Let all initial multiplicities be one and suppose

$$\mu_i^q(0) = u_i^0 + v_0, \qquad \mu_i^s(0) = u_i^0 - v_0, \tag{7.25}$$

where the u_i^0 are distinct real numbers and $v_0 \neq 0$. The unique global solution of the full inclusion for these initial data has, in both sectors, the form

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{7.26}$$

Blocks formed by equal u_i persist and follow the multiplicity-weighted version of the same equation.

Proof. Set $\kappa = a - c > 0$. Equal sector orderings reduce eq. (7.18) to

$$\dot{u}_i \in -\kappa u_i - \lambda \sum_{j \neq i} n_j \operatorname{Sign}(u_i - u_j), \qquad \dot{v} = -(a + c)v. \tag{7.27}$$

Introduce the increasing auxiliary time and rescaled positions

$$\tau(t) = \frac{e^{\kappa t} - 1}{\kappa}, \qquad u_i(t) = e^{-\kappa t} z_i(\tau(t)). \tag{7.28}$$

Since $\dot{\tau} = e^{\kappa t} > 0$, direct differentiation gives

$$z_i' \in -\lambda \sum_{j \neq i} n_j \operatorname{Sign}(z_i - z_j). \tag{7.29}$$

This is the sticky-particle solution of the one-dimensional aggregation equation with kernel $\lambda|z|$ on weighted atoms [19, 26]. An ordered block has constant velocity $\lambda(n_{\text{right}} - n_{\text{left}})$ between collisions,

where n_{right} and n_{left} are the total multiplicities to its right and left. Adjacent blocks approach with relative speed $\lambda(n_I + n_J) > 0$. At their collision, join their multiplicities, retain the common position, and use the new velocity $\lambda(n_{\mathrm{right}} - n_{\mathrm{left}})$. This equals the multiplicity-weighted average of their velocities just before collision. Choosing zero internal subgradients verifies the original inclusion for every constituent child after collision. Each event reduces the number of blocks, so at most $N - 1$ events occur, and the resulting piecewise-linear curve exists for all $\tau \geq 0$. Reversing eq. (7.28) constructs a global full-system solution. Lemma 7.2 makes it the solution. □

The class eq. (7.25) has a common difference between its two sector means. This is an initial-data restriction, not an imposed constraint force. It is an invariant affine subspace of codimension $r - 1$ in $\mathbb{R}^{2r}$. The two laws of each agent remain distinct at every finite time because $v(t) \neq 0$. The auxiliary τ is a monotone reparameterization of this solution, not a geometric or operational clock.

7.5 Two fusion stages without a supplied partition

Theorem 7.6 (Four agents fuse into two parents and then one). In theorem 7.5, take four agents with the unordered initial set

$$\{u_i^0\} = \{-3d, -2d, 2d, 3d\}, \qquad d > 0. \tag{7.30}$$

Two size-two blocks form at t_1 and remain distinct until $t_2 > t_1$, where they fuse under their inherited equations, with

$$t_1 = \frac{1}{\kappa} \log\left(1 + \frac{\kappa d}{2\lambda}\right), \qquad t_2 = \frac{1}{\kappa} \log\left(1 + \frac{5\kappa d}{4\lambda}\right). \tag{7.31}$$

Proof. In sorted order, the initial z velocities are $(3\lambda, \lambda, -\lambda, -3\lambda)$. The two outer gaps close at $\tau_1 = d/(2\lambda)$, leaving parent positions $-3d/2$ and $3d/2$. Their inherited multiplicities are two and their velocities are 2λ and -2λ. Their gap closes after another $3d/(4\lambda)$, giving $\tau_2 = 5d/(4\lambda)$. Inverting eq. (7.28) gives eq. (7.31). □

Exact reflection symmetry is unnecessary. For four sorted initial positions, let their adjacent gaps be $g_1, g_2, g_3 > 0$, and put $H = \max(g_1, g_3)$. The two outer pairs form first and then fuse if and only if

$$D = g_2 + \frac{g_1 + g_3}{2} - 2H > 0. \tag{7.32}$$

Suppose $g_1 \leq g_3$; the other case is the reflected calculation. Before the first outer collision the middle gap decreases at speed 2λ, so at $g_1/(2\lambda)$ it equals $g_2 - g_1$; afterward it decreases at speed 3λ until the other outer pair collides at $g_3/(2\lambda)$, where the same linear expression has the value D. If $D > 0$, then $g_2 > H$, the middle gap stays positive until both outer pairs have formed by $\tau_* = H/(2\lambda)$, and the two parent gap then decreases at speed 4λ, so the final time is

$$\tau_{\mathrm{final}} = \tau_* + \frac{D}{4\lambda} = \frac{2g_2 + g_1 + g_3}{8\lambda}. \tag{7.33}$$

If $D \leq 0$, the middle gap closes no later than the second outer collision, and the pattern fails; at $D = 0$ three or four agents merge at once. Condition eq. (7.32) is strict and defines an open set relative to the ordered common-offset family. Taking all label permutations gives an admitted class

with no assigned labels for blocks. The initial laws are unequal across agents, and the interaction rule contains no partition. Membership is selected by attained equality.

Corollary 7.7 (The two-stage pattern is open in the full space). Let $\bar{x} \in \mathbb{R}^8$ be a four-agent configuration of unit multiplicities in the common-offset class satisfying eq. (7.32), and let A and B be its two outer pairs. There is $\varepsilon > 0$ such that every initial configuration within distance ε of $\bar{x}$ has the following membership history: A and B synchronize in both sectors, in some order and before any other synchronization, and the two resulting parents then synchronize. The set of initial data with this history therefore contains an open subset of $\mathbb{R}^8$ that contains the open-gap set.

Proof. Let $\bar{\boldsymbol{\mu}}$ be the solution from $\bar{x}$, let t_* be its second outer collision time and $t_{\mathrm{f}} > t_*$ its final time, and choose $\eta > 0$ with $t_* + \eta < t_{\mathrm{f}}$. On $[0, t_* + \eta]$ every mean of A lies below every mean of B in each sector, with a margin $m > 0$. By eq. (7.19), a solution from a configuration within ε of $\bar{x}$ stays within ε of $\bar{\boldsymbol{\mu}}$ in every coordinate. If $2\varepsilon < m$, both pairs are isolated in the sense of lemma 7.4 on $[0, t_* + \eta]$, and no agent of A meets an agent of B in either sector there. At the collision time of A along $\bar{\boldsymbol{\mu}}$, the perturbed pair has $S \le 4\varepsilon$, so eq. (7.24) synchronizes it within $2\varepsilon/\lambda$; the same holds for B. If also $2\varepsilon/\lambda \le \eta$, both pairs synchronize before $t_* + \eta$ and no other event occurs before then. Lemma 7.2 keeps each pair synchronized, and proposition 7.3 synchronizes the two parents at a finite later time. □

Corollary 7.8 (Coalescence of any finite depth). Place $N = 2^k$ unit agents in the common-offset class with sorted initial positions u^0, and assign to the gap between sorted agents p and $p+1$ the level of the smallest dyadic block of indices containing both. If, for each level $\ell < k$,

$$\frac{N}{2} \max\{\text{gaps of level } \ell\} < \min\{\text{gaps of level } \ell + 1\}, \tag{7.34}$$

then all dyadic blocks of size 2^ℓ form before any event that joins two of them, for $\ell = 1, \dots, k$, and each stage is inherited exactly. The condition is strict, so the k-stage history holds on an open subset of the common-offset class.

Proof. In the z frame, adjacent blocks with multiplicities n_I, n_J approach at speed $\lambda(n_I + n_J) \in [2\lambda, \lambda N]$, whatever lies outside them. A gap of initial size g between two sorted agents therefore closes at an auxiliary time in $[g/(\lambda N), g/(2\lambda)]$. By eq. (7.34), every gap of level ℓ has closed before any gap of level $\ell + 1$ can close. Event times are $t = \kappa^{-1} \log(1 + \kappa\tau)$, and inheritance at every stage is the handoff identity of the next section. □

Every such hierarchy ends in one block, by proposition 7.3.

7.6 The inherited handoff and its scope

At a collision, restrict a block I to its common means. Summing its local energies and pulling back its metric gives

$$\sum_{i \in I} n_i (C + \mathcal{U}) = n_I (C + \mathcal{U}) = E_{0,n_I} + (n_I - 1)C, \qquad g_I = n_I \big((d\mu_I^q)^2 + (d\mu_I^s)^2 \big). \tag{7.35}$$

Every external pair term sums to $n_I n_J |\mu_I^\alpha - \mu_J^\alpha|$. Zero internal subgradients therefore lift each solution of the inherited equation to a fine solution. Uniqueness identifies that lift with the original

trajectory. This also proves persistence for inherited multiplicities by expanding a multiplicity into synchronized unit children. With R retaining block means, multiplicities, and frames, the corresponding parameter flows satisfy the following identity. For a fixed formed partition, keep both component index sets fixed even when further components coincide; a later merger is a separate retention composed with R. Then

$$R \circ \Phi_t = \Phi_t^c \circ R \tag{7.36}$$

on every synchronized stratum reached by the construction, for all future times. Weighted geometric averaging is associative, so later collisions give the same retained laws as direct averaging. The state is continuous at a collision; the velocity changes through the inherited multiplicity sum. Exact inheritance holds only for blocks of synchronized agents; for a partition into unsynchronized blocks the block average does not determine the external sign forces. In this construction, unlike an architecture in which inheritance holds for every partition, the selected partition is therefore distinguished by inheritance itself.

Along every solution, ordinary differentiation on each interval between collisions gives

$$\frac{d\mathcal{F}}{dt} = -\sum_i n_i \big((\dot{\mu}_i^q)^2 + (\dot{\mu}_i^s)^2\big) \quad \text{almost everywhere.} \tag{7.37}$$

The objective is continuous at collisions when the constants above are retained. By proposition 7.3, every solution converges to the stationary configuration with every mean zero, after complete synchronization at a finite time.

Remark 7.9 (A discrete update that keeps exact sticking). The natural time discretization is the proximal step

$$\boldsymbol{\mu}^{k+1} = \arg\min_x \left\{ \mathcal{F}(x) + \frac{1}{2h} \|x - \boldsymbol{\mu}^k\|_W^2 \right\}, \tag{7.38}$$

whose minimizer is unique because the objective is strongly convex. Its optimality condition $W(\boldsymbol{\mu}^k - \boldsymbol{\mu}^{k+1})/h \in \partial\mathcal{F}(\boldsymbol{\mu}^{k+1})$ and strong monotonicity give the discrete contraction $\|\boldsymbol{\mu}^{k+1} - \widetilde{\boldsymbol{\mu}}^{k+1}\|_W \le (1 + h\kappa)^{-1}\|\boldsymbol{\mu}^k - \widetilde{\boldsymbol{\mu}}^k\|_W$, and comparing the objective at its minimizer with its value at $\boldsymbol{\mu}^k$ gives $\mathcal{F}(\boldsymbol{\mu}^{k+1}) + (2h)^{-1}\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|_W^2 \le \mathcal{F}(\boldsymbol{\mu}^k)$. Uniqueness and the exchange symmetry of equal-multiplicity agents keep synchronized blocks synchronized, and the step creates exact ties: for two unit agents under the pair term alone, the step returns equal means exactly when the gap before the step is at most $2h\lambda$. The iterates converge to the solution of eq. (7.16) uniformly on bounded time intervals as $h \to 0$ [36]. An explicit Euler step with sign selections instead oscillates around coincidence and never sticks exactly, so equality-based membership is only meaningful for the proximal scheme or for an event-driven integration of eq. (7.29).

Sample retention is genuinely noninjective: different positive sample vectors have the same geometric mean. For n synchronized unit children, their log samples conditional on the retained log mean $\bar{Z}$ have law

$$(Z_1, \dots, Z_n) \mid \bar{Z} \sim N\left(\bar{Z}\mathbf{1}, I_n - \frac{\mathbf{1}\mathbf{1}^\mathsf{T}}{n}\right). \tag{7.39}$$

The residual is independent of the mean: their joint Gaussian cross covariance is zero. Exponentiation gives a normalized joint reconstruction kernel. Its one-child marginal is eq. (7.9). Independent residuals for all children would be incorrect because the off-diagonal conditional covariances are

$-1/n$. Generative joint samples similarly use the covariance of $(\log m, \log k)$ under $s_1 G_1$; their residual kernel is determined by that covariance, not by the recognition product law. The local KL residual $(n-1)C$ records the corresponding state-independent information loss under the reference comparison.

Remark 7.10 (Sufficiency beyond the lognormal family). The kernel eq. (7.39) is the conditional law of the children given their sufficient statistic, and it is free of the parameter for that reason. The same holds for any family with a sufficient statistic for independent children: the parent law is the law of the statistic, the Fisher metric of the children restricted to equal parameters is multiplied by the multiplicity, and the parameter-free conditional law supplies both the message kernel and the joint reconstruction. For Gamma children of shape k, for example, one child given the sum S of n children is S times a $\mathrm{Beta}(k,(n-1)k)$ variable. The ingredient specific to Gaussian location families is that the square-root divergence between restored messages is the convex function $|\Delta\mu|$; on a general regular family it equals the Fisher distance only to leading order near the diagonal, and global uniqueness would need semiconvexity estimates in place of convexity.

Parameter retention has a narrower meaning. On an exactly synchronized stratum, child means are redundant copies; retaining their common value does not identify different nonsynchronized configurations. Away from that stratum, a block average need not determine its external sign forces. Equation (7.36) makes no claim about such hidden parameter fibers.

The admitted boundary protocol draws fresh messages through B_n, fresh predictive messages through eq. (7.11), or complete fresh represented samples through the joint reconstruction. To type an accessible stored readout, augment the complete population configuration by $o \in \mathsf{O} = (0,\infty)$ and set $\rho(\boldsymbol{\mu}, \boldsymbol{n}, o) = o$. At declared epochs, the transition Γ advances the deterministic parameter flow and overwrites o using the selected normalized scalar output kernel. Between epochs, o is fixed, and it never enters the mean equation. Retention copies this accessible o exactly; it does not claim access to cached private residual samples. For vector readouts the same definition uses the corresponding finite product space and joint kernel. Equal parameter trajectories and conditional output kernels imply equal joint accessible readout laws for every finite sequence of such queries, by multiplying successive conditional kernels. Reusing persistent latent samples or introducing record feedback requires another state specification.

7.7 Status against the formation standard

We now check the clauses of definition 6.31. The admitted class of eq. (7.32), with all label permutations, is open in the declared invariant common-offset family, and by corollary 7.7 the same history holds on an open subset of the full space, so (F1) holds. Blocks are selected by attained equality in both sectors, they persist by lemma 7.2, and the handoff is the partition of the current state into coincident agents, so (F2) holds. Laws, evaluator, metric, rows, contacts, and the declared fresh-message, predictive, reconstruction, and stored-readout responses are inherited exactly on synchronized strata, with the probe class fixed before the retention, so (F3) holds.

The null flow of definition 6.30 deletes the pair terms, which is the case $\lambda = 0$. Each agent then follows the linear equation $\dot{\mu}_i = -A\mu_i$, whose flow is invertible, so distinct agents never coincide and H^0 contains no event. By proposition 7.3, every configuration with two distinct agents has at

least one event along Φ. Hence $H \neq H^0$ on the whole admitted class, and (F5) holds. The strong form of (F5) fails trivially, because the final partition is the single block for every initial datum. The construction is amplifying only in the degenerate sense that within-block diameters vanish at each event.

Clause (F4) fails. On the post-handoff invariant set the children of a block are identical copies, so the retention of common means and multiplicity is injective there; child frames are fixed passive data that no admitted probe reads in reference coordinates. The geometric mean is noninjective on sample vectors, but samples are not configurations, and the conditional law of the child samples given the retained sample is the parameter-free kernel eq. (7.39), so no admitted child-level probe distinguishes two configurations with the same retained value. Clause (F6a) holds: by theorem 7.6 and corollary 7.7, the two first-stage parents select their union under the inherited interaction, with no reset of their data and no retuned coupling. Clause (F6b) fails. By proposition 7.3 applied to multiplicities two, every configuration of two first-stage parents, whatever their means, fuses at a finite time, so the second-stage selection occurs from every counterfactual handoff in the inherited class; the order of the stages is the order of the initial gaps fixed by eq. (7.32).

The construction is therefore not a recursive formation. It satisfies (F1), (F2), (F3), (F5), and (F6a) and fails (F4) and (F6b), so it is a two-stage selection with exact inheritance: a finite coalescence in which the interaction is necessary for the grouping, the grouping is a coalescence tree ordered by the initial gaps, and the tree always ends in one block.

Remark 7.11 (Smooth peer KL dynamics and persistent parents). The finite mergers above depend on the nonsmooth interaction. Under the smooth reciprocal potential with a locally Lipschitz vector field, an interior trajectory cannot first reach an invariant synchronization manifold at finite time (proposition 6.22), so smooth constructions need a different selection criterion; the threshold selectors of chapters 12 and 13 are such criteria, and the remaining problem is open problem 6.32. Persistent parents are excluded here by proposition 7.3. A contact range, for example $\lambda_{ij} = \lambda \mathbf{1}\{|\mu_i - \mu_j| < \rho\}$ as in bounded-confidence dynamics [74], or a multiwell local reference would let separated blocks stop interacting. Both destroy the convexity on which uniqueness rests, and well-posedness and inheritance for such variants have not been established.

Chapter 8

Sufficient States for Smooth KL Evolution

This chapter works with the reciprocal law potential of definition 4.2 and its unit-mobility Fisher flow; its conclusions concern that potential, not the per-agent or global free energies of chapter 4. The question is which information a coarse description of an agent must keep when an exact reduction is wanted while agents remain different. We first identify the conditional information that enters a retained force, then construct an exact state completion made of general probability laws for one architecture: fixed independent private reference factors, and receiver reconstruction channels that discard the sender's private coordinate. These channels are additional constitutive data within the potential, not a consequence of an arbitrary law-dependent transport.

In this architecture the private sector is passive. Each private conditional relaxes along an explicit exponential arc toward its reference, independently of every public law, row, and other agent, and it acts on the public flow only as open-loop forcing through its conditional KL profile. The completion is therefore a solvable baseline. Its compression is the equimeasurability quotient of each private log density under its declared reference: it forgets which private point carries a log-density value, so it identifies distinct conditionals only when the reference has nontrivial symmetry, and it is not minimal for the public flow (proposition 8.10). It fails once a channel transmits the sender's private data (remark 8.12). The construction selects no group of agents, so the interaction-necessity clause of definition 6.31 does not apply to it; its private sector evolves identically under the null flow of definition 6.30, because the conditional equations contain no peer term.

8.1 The force that survives marginalization

Let Q be a positive law on a standard Borel space, let C be a measurable statistic, and write $\mu = C_{\#}Q$. Suppose a differentiable functional has variational derivative h_Q and Fisher velocity

$$\dot{Q} = -Q(h_Q - \mathbb{E}_Q h_Q). \tag{8.1}$$

All identities below are immediate for finite positive tables. On general spaces they need a dominated differentiable path, square-integrable scores, and integrability that permits the displayed conditional expectations and differentiations. Statements about flows on general spaces are made under the hypotheses of hypothesis 8.4, which give unique solutions in bounded log-density charts (proposition 8.8); they do not assert existence on every space of probability measures.

Proposition 8.1 (Conditional mean force). The retained velocity is

$$\dot{\mu} = -\mu\left(\mathbb{E}_Q[h_Q \mid C] - \mathbb{E}_Q h_Q\right). \tag{8.2}$$

Consequently, an autonomous velocity for μ exists on an admitted class precisely when the centered conditional mean on the right is determined by μ on that class. If, in addition, the admitted class is invariant under the fine flow and the resulting retained equation $\dot{\mu} = V(\mu)$ has unique solutions, then two fine trajectories in the class with the same initial marginal have the same marginal at all common times.

Proof. For every bounded measurable f, integrate $f(C)$ against eq. (8.1) and condition on C. This gives the weak form of eq. (8.2). Two laws with the same marginal have the same marginal velocity if and only if their centered conditional forces agree almost everywhere under that marginal. Invariance keeps both fine trajectories in the class, so each marginal solves $\dot{\mu} = V(\mu)$, and uniqueness identifies them. □

Equation eq. (8.2) is the class-mean aggregation of a replicator equation over types, familiar from the Price equation of selection [121]. For $Q(dz, dv) = \mu(dz)\kappa_z(dv)$ and a fixed reference $P(dz, dv) = \nu(dz)\pi_z(dv)$, the local KL force illustrates the missing information:

$$\mathbb{E}_Q[\log(dQ/dP) \mid z] = \log \frac{d\mu}{d\nu}(z) + H(z), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z \| \pi_z). \tag{8.3}$$

Thus a scalar total conditional loss does not usually determine the retained force. Its dependence on z matters. The Fisher metric makes the same separation: if a score is $f(z, v) = f_1(z) + f_2(z, v)$, with $\mathbb{E}_\mu f_1 = 0$ and $\mathbb{E}_{\kappa_z} f_2 = 0$, then

$$g_Q(fQ, fQ) = \int f_1(z)^2 \mu(dz) + \int \left[\int f_2(z, v)^2 \kappa_z(dv) \right] \mu(dz). \tag{8.4}$$

The cross term vanishes by conditional centering. This calculation is consistent with the statistical behavior of Fisher information under sufficient statistics [5]. Dynamical sufficiency asks a different question: it preserves the specified evolution and responses, and need not preserve inference about every discarded parameter.

Corollary 8.2 (Pushforward as a Fisher submersion). On a regular positive class, the differential of $Q \mapsto C_\# Q$ sends the tangent fQ to $\mathbb{E}_Q[f \mid C]\mu$. Its kernel consists of the scores with zero conditional mean given C, and it is isometric on scores that are functions of C. If the functional has the form $\mathcal{E} = \mathcal{E}_c \circ C_\#$, the retained marginal follows the Fisher gradient flow of $\mathcal{E}_c$ at the same time.

Proof. For bounded g, $\int g(C) f \, dQ = \int g \, \mathbb{E}_Q[f \mid C] \, d\mu$, which identifies the pushforward tangent. With $f_1 = \mathbb{E}_Q[f \mid C]$ and $f_2 = f - f_1$, eq. (8.4) gives the kernel and the isometry. If $\mathcal{E} = \mathcal{E}_c \circ C_\#$, the chain rule gives $h_Q = h_c \circ C$ up to a constant, so eq. (8.2) becomes $\dot{\mu} = -\mu(h_c - \mathbb{E}_\mu h_c)$. □

This is the equality case of the monotonicity of Fisher information under statistics [6]; the content of the present chapter is the case in which h_Q is not a function of C.

8.2 Conditional reconstruction within the original potential

For each agent take $\mathsf{K}_i = \mathsf{Z}_i \times \mathsf{V}_i$ and $\mathsf{M}_i = \mathsf{W}_i \times \mathsf{U}_i$, all standard Borel. The coordinates z, w are public in the sense that the channels below read them; the coordinates v, u are private in the sense

that no channel reads the sender's value of them, and the declared probes read them only through the marks of section 8.4. Use public densities against fixed reference probabilities ρ_i on Z_i and ϱ_i on W_i, so conditional versions share common full-measure public sets. Disintegrate

$$q_i(dz, dv) = \mu_i(dz)\kappa_{i,z}(dv), \qquad s_i(dw, du) = \sigma_i(dw)\varkappa_{i,w}(du). \tag{8.5}$$

Neither conditional need be independent of its retained coordinate. Choose fixed private reference probabilities A_i^q, A_i^s and a fixed normalized public reference

$$R_i^c(dz, dw) = r_i^c(dw)G_i^c(w, dz), \qquad R_i = R_i^c \otimes A_i^q \otimes A_i^s, \tag{8.6}$$

with the product reordered into state and model coordinates. The evaluator G_i^c can couple μ_i and σ_i. Independence of the private reference factors is an assumption of this construction, not a restriction of the general theory.

Let $B_{ij}^q[\boldsymbol{\mu}] : \mathsf{Z}_j \rightsquigarrow \mathsf{Z}_i$ be normalized channels, differentiable in the laws in the sense required by hypothesis 8.4 on general spaces. Define the full channels by

$$\mathcal{K}_{ij}^q[\mathbf{q}]((z_j, v_j), d(z_i, v_i)) = B_{ij}^q[\boldsymbol{\mu}](z_j, dz_i)\kappa_{i,z_i}(dv_i). \tag{8.7}$$

The model channel uses $B_{ij}^s[\boldsymbol{\sigma}]$ and $\varkappa_{i,w_i}$ in the same way. Source sets may exclude self edges; if self comparisons are included, use the primitive identity convention separately. Public channels may be informative and nonconstant functions of the retained laws. For example, on the public space $\{-1, 1\}$, the channel $B_\mu(z, \{z'\}) = (1 + \rho(\mu)zz')/2$, with $\rho(\mu) = (1 + (\mathbb{E}_\mu Z)^2)/4$, is strictly positive, transmits source information, and varies with the law. Public/private maps and reference factors are declared interface data; carrying them through a bimeasurable change of presentation preserves these constructions.

Proposition 8.3 (Exact cancellation of private peer information). Where the divergences are defined,

$$\begin{aligned} q_j\mathcal{K}_{ij}^q &= (\mu_j B_{ij}^q)\kappa_i, \\ D_{\mathrm{KL}}(q_i \| q_j\mathcal{K}_{ij}^q) &= D_{\mathrm{KL}}(\mu_i \| \mu_j B_{ij}^q). \end{aligned} \tag{8.8}$$

This is an identity on the full admitted class, including correlated q_i. Its derivative includes the sender, receiver, and channel derivatives.

Proof. Integration over the source conditional gives the first identity. The compared laws have the same receiver conditional $\kappa_{i,z}$, so their density ratio depends only on z. Integrating its logarithm proves the second identity. Differentiating the resulting functional identity on any regular path proves the derivative statement. Holding κ_i fixed only in the denominator would violate that identity and would define another force. □

Let $\mathcal{E}_c$ denote exactly eq. (4.4) applied to $\mu_i, \sigma_i, R_i^c, B_{ij}^q, B_{ij}^s$, with the same rows, source priors, and temperatures. Define

$$\begin{aligned} \ell_{i,z}^q &= \log(d\kappa_{i,z}/dA_i^q), \quad H_i^q(z) = \int \ell_{i,z}^q d\kappa_{i,z}, \\ \ell_{i,w}^s &= \log(d\varkappa_{i,w}/dA_i^s), \quad H_i^s(w) = \int \ell_{i,w}^s d\varkappa_{i,w}. \end{aligned} \tag{8.9}$$

On finite public alphabets with twice continuously differentiable channel entries it suffices that each private log density be bounded. On general public spaces we use the following hypotheses.

Hypothesis 8.4 (Regularity on general spaces). Use the bounded log-density chart for every public law: $\mu_i = e^{u_i}\rho_i/\mathbb{E}_{\rho_i}e^{u_i}$ with u_i in $L^\infty_0(\rho_i) = \{u \in L^\infty(\rho_i) : \mathbb{E}_{\rho_i}u = 0\}$, the same for σ_i against ϱ_i, and log-odds coordinates for interior rows.

(H1) The initial private log densities are bounded uniformly in the public coordinate: $\operatorname{ess\,sup}_{(z,v)} |\ell^q_{i,z}(v)| \le M < \infty$ with respect to $\rho_i \otimes A^q_i$ at time zero, and likewise in the model sector.

(H2) The public reference has bounded log densities: $G^c_i(w, dz) = g^c_i(w, z)\rho_i(dz)$ and $r^c_i = \tilde{r}_i\varrho_i$, with $\log g^c_i \in L^\infty(\varrho_i \otimes \rho_i)$ and $\log \tilde{r}_i \in L^\infty(\varrho_i)$.

(H3) On bounded subsets of the product of the charts, the public first variations $h^q_i = \delta\mathcal{E}_c/\delta\mu_i$ and $h^s_i = \delta\mathcal{E}_c/\delta\sigma_i$, including every interaction role and channel derivative, are Lipschitz into $L^\infty(\rho_i)$ and $L^\infty(\varrho_i)$, and the costs D^q_{ij}, D^s_{ij} are Lipschitz. Identity channels between agents sharing a public reference space satisfy this, as do channels with kernel densities whose logarithms are bounded on bounded subsets and which depend on the laws through a continuously differentiable map into L^∞ with locally Lipschitz derivative.

Positivity of the public densities alone does not suffice: a normalized, strictly positive, law-dependent channel whose first variation contains an unbounded function of z gives a public force outside L^∞, and the chart vector field is then undefined. A bound on each $\ell_{i,z}$ separately does not suffice either: a family of bounded conditionals whose bounds grow as z approaches a boundary point can make the public log density unbounded for every positive time. The class (H1) excludes conditionals whose density vanishes on a set of positive reference measure or is unbounded, such as Gaussian conditionals against a Gaussian reference; remark 8.11 treats those separately.

Theorem 8.5 (Exact marginal and conditional dynamics). For the architecture above the original energy is

$$\mathcal{E} = \mathcal{E}_c + \sum_i \int H^q_i(z)\mu_i(dz) + \sum_i \int H^s_i(w)\sigma_i(dw). \tag{8.10}$$

Write $h^q_i = \delta\mathcal{E}_c/\delta\mu_i$ and $h^s_i = \delta\mathcal{E}_c/\delta\sigma_i$, including every interaction role. The full unit-mobility Fisher dynamics gives

$$\begin{aligned} \dot{\mu}_i &= -\mu_i\left[h^q_i + H^q_i - \mathbb{E}_{\mu_i}(h^q_i + H^q_i)\right], \\ \partial_t\ell^q_{i,z} &= H^q_i(z) - \ell^q_{i,z}. \end{aligned} \tag{8.11}$$

The equations for σ_i, ℓ^s_i are obtained by exchanging sectors. Source rows evolve by their original Fisher equations with the public costs, or may be exactly optimized as in proposition 4.3.

Proof. Apply the relative-entropy chain rule to the local reference in eq. (8.6) and use eq. (8.8). This proves eq. (8.10). Up to a constant independent of (z, v), the full derivative with respect to q_i is $h^q_i(z) + \ell^q_{i,z}(v)$. This can be checked directly from the log-density ratio in the local KL; every peer contribution is a functional of μ alone. Conditional averaging gives the first equation of eq. (8.11). Subtract its logarithmic derivative from the full logarithmic derivative of $q_i = \mu_i\kappa_i$. The public score and global centering cancel, leaving the second equation. The private terms contain no source rows, so their variations do not change the row equations. □

The two row choices share stationary points but are different flows. At frozen costs the Fisher row equation relaxes log-ratio contrasts at the rate T_q (eq. (8.27)), so exactly optimized rows are the limit of infinite row mobility, and the two choices differ most in their transients when T_q is small.

The private equation in eq. (8.11) contains no public law, no row, and no other agent. Each private conditional is solved in closed form, independently of the public dynamics, by eq. (8.18) below: the exponential arc from $\kappa_{i,z}(0)$ to A_i^q. The public flow sees the private sector only through the function $H_i^q(z,t)$, which is fixed by the initial conditionals. The completion constructed next is therefore the public flow of $\mathcal{E}_c$ driven by the precomputable time-dependent potential $\int H_i^q(\cdot,t)d\mu_i$. Private information never influences an interaction in this architecture. The dependence of H_i^q on z is the only private contribution to the retained belief velocity, and it cannot generally be discarded.

8.3 An exact law-valued completion

Definition 8.6 (Conditional log-density spectrum). For a private reference A and conditional density κ_z, define

$$\xi_z = (\ell_z)_{\#}A, \qquad \ell_z = \log(d\kappa_z/dA). \tag{8.12}$$

This probability law records how much reference mass carries each log-density value. It forgets which private point carries that value. It obeys

$$\int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{8.13}$$

The spectrum is the law under A of the log-likelihood ratio of the dichotomy (κ_z, A). For mutually absolutely continuous pairs, equal spectra are equivalent to Blackwell equivalence of the two dichotomies [17, 137]. Two conditionals have the same spectrum exactly when their log densities are equimeasurable under A. The compression is therefore a property of the declared reference, not of the dynamics. On a finite private alphabet $\xi_z(\{r\}) = A(\{\ell_z = r\})$, so if no two distinct sets of atoms of A have equal mass, equal spectra force equal level sets and hence equal conditionals: the retention is injective. With $A = (0.6, 0.4)$, for example, the conditionals $(3/4, 1/4)$ and $(1/4, 3/4)$ have the different spectra with atoms $\log(5/4), \log(5/8)$ and $\log(5/12), \log(15/8)$ at masses $0.6, 0.4$. Noninjectivity requires a reference with equal-mass sets of atoms, such as a uniform finite reference, or an atomless reference. The log-Laplace transform of the spectrum is

$$\Lambda_z(b) = \log \int e^{br} \xi_z(dr) = (b-1) D_b(\kappa_z \| A), \tag{8.14}$$

where D_b is the Rényi divergence of order b [138]. Under (H1) the spectrum has compact support, so Λ_z on $(0, 1]$ determines ξ_z.

Theorem 8.7 (Closed conditional-spectrum evolution). For theorem 8.5, the velocity of the state consisting of all μ_i, σ_i, conditional spectra ξ_i^q, ξ_i^s, and source rows is a function of that state. Each spectrum satisfies, in weak form,

$$\frac{d}{dt} \int f(r) \xi_z(dr) = \int f'(r)(H(z) - r) \xi_z(dr), \tag{8.15}$$

for continuously differentiable test functions on the initial range $I = [m_0, M_0]$ of the log density, which contains the range at every later time. For a time interval t starting at zero, its exact solution is

$$\begin{aligned} a &= e^{-t}, & Z_z(a) &= \int e^{ar}\xi_z^0(dr), \\ \xi_z^t &= (r \mapsto ar - \log Z_z(a))_{\#}\xi_z^0, \end{aligned} \tag{8.16}$$

and it is the only narrowly continuous curve of probability laws supported in I that solves eq. (8.15) with $H(z)$ given by eq. (8.13) from the same initial law with $\int e^r \xi_z^0(dr) = 1$. The entropy profile obeys

$$\dot{H}(z) = -\operatorname{Var}_{e^r\xi_z}(r). \tag{8.17}$$

Proof. The reference A is fixed. Differentiate $\int f(\ell_z(v))A(dv)$ and substitute the conditional equation in eq. (8.11); this proves eq. (8.15). Direct differentiation shows that

$$\frac{d\kappa_z(t)}{dA}(v) = \frac{\exp(a\ell_z^0(v))}{Z_z(a)} \tag{8.18}$$

solves the normalized conditional equation; this is the explicit solution of the Fisher–Rao gradient flow of a relative entropy, the exponential arc toward the reference [30]. Pushing A through its logarithm gives eq. (8.16). For the range, Hölder's inequality gives $Z_z(a) \le 1$, and $e^{ar} \ge e^r e^{-(1-a)M_0}$ for $r \le M_0$ gives $Z_z(a) \ge e^{-(1-a)M_0}$; since $m_0 \le 0 \le M_0$ by normalization, $a\ell_z^0 - \log Z_z(a)$ takes values in $[m_0, M_0]$, and the width of its range is $a(M_0 - m_0)$.

For uniqueness, let ξ_t be a curve in the stated class and put $H(t) = \int re^r \xi_t(dr)$, which is continuous. Then ξ_t solves the linear continuity equation with the velocity field $H(t) - r$, which is Lipschitz in r, and continuity equations with such fields have unique narrowly continuous solutions [4, Sec. 8.1]. Hence $\xi_t = (X_t)_{\#}\xi_0$ with $X_t(r) = e^{-t}r + c(t)$ and $c(t) = \int_0^t e^{-(t-s)}H(s)ds$. Taking $f(r) = e^r$ in eq. (8.15) gives $\dot{N} = H(N-1)$ for $N = \int e^r \xi_t(dr)$, so $N \equiv 1$, which forces $e^{c(t)}Z(a) = 1$ and gives eq. (8.16).

Setting $f(r) = re^r$ in the weak equation gives $\dot{H} = H(H+1) - \int (r^2+r)e^r\xi_z(dr)$, which is eq. (8.17). The probability $e^r\xi_z(dr)$ is normalized by eq. (8.13). All private contributions to eq. (8.11) are now functions of the retained spectra. The remaining scores and row equations depend only on retained public data, so the retained velocity is a function of the retained state. □

That the retained state determines its own future additionally requires uniqueness of the public flow forced by $H_i(z,t)$. For finite public alphabets with twice continuously differentiable channels, the Picard–Lindelöf argument of theorem 4.6 applies to this forced system. On general public spaces the following proposition supplies it.

Proposition 8.8 (Well-posedness and exact lift on general spaces). Assume hypothesis 8.4. Then for every initial state in the charts:

(i) each private log density keeps its values in its initial essential range, $0 \le H_i^q(z,t) \le H_i^q(z,0) \le M$, and $|\partial_t H_i^q(z,t)| \le M^2$ uniformly in z, and likewise in the model sector;

(ii) the retained system of public laws, rows, and spectra has a unique maximal solution on an interval $[0, T^*)$, with public laws and rows continuously differentiable into the charts and spectra given by eq. (8.16);

(iii) the fine Fisher system has a unique maximal continuously differentiable solution in the charts $L_0^\infty(\rho_i \otimes A_i^q)$ and $L_0^\infty(\varrho_i \otimes A_i^s)$ on the same interval, and it equals the lift $q_i(t) = \mu_i(t)\kappa_i(t)$ with $\kappa_i(t)$ given by eq. (8.18), and likewise in the model sector;

(iv) if R sends a fine state to its public laws, spectra, and rows, then $R \circ \Phi_t = \Phi_t^c \circ R$ on $[0, T^*)$, where Φ^c is the solution map of (ii).

Proof. (i) The range statement is in the proof of theorem 8.7, and it holds uniformly in z by (H1). The function H is a relative entropy, hence nonnegative, and it is nonincreasing by eq. (8.17). A law on an interval of width at most $2M$ has variance at most M^2.

(ii) In the chart the public equation reads $\dot{u}_i = -(\phi_i - \mathbb{E}_{\rho_i}\phi_i)$ with $\phi_i = h_i^q + H_i^q(\cdot, t)$, because then $\partial_t \log(d\mu_i/d\rho_i) = -(\phi_i - \mathbb{E}_{\mu_i}\phi_i)$. By (i), $t \mapsto H_i^q(\cdot, t)$ is Lipschitz into $L^\infty(\rho_i)$. The normalizer and the centering are smooth on the chart, and (H2) and (H3) make h_i^q Lipschitz on bounded sets; the row block is a finite-dimensional equation whose costs are Lipschitz on bounded sets, and Gibbs rows are smooth functions of the costs. The Picard–Lindelöf theorem in Banach spaces [47] gives a unique maximal solution, with the alternative that it leaves every bounded set as $t \to T^*$ if $T^* < \infty$.

(iii) In the fine chart, the first variation of the energy with respect to q_i is $\varphi_i(z, v) = h_i^q(z) + \ell_{i,z}(v)$ up to a constant, by the proof of theorem 8.5. The public marginal map $l \mapsto \log \int e^{l(\cdot,v)} A_i^q(dv)$ is smooth from $L^\infty(\rho_i \otimes A_i^q)$ to $L^\infty(\rho_i)$, so the fine field is Lipschitz on bounded sets and its solutions are unique. For the lift, $\partial_t \log q_i = -(h_i^q + H_i^q - \mathbb{E}_{\mu_i}(h_i^q + H_i^q)) + (H_i^q - \ell_i) = -(\varphi_i - \mathbb{E}_{q_i}\varphi_i)$, because $\mathbb{E}_{q_i}\varphi_i = \mathbb{E}_{\mu_i}h_i^q + \mathbb{E}_{\mu_i}H_i^q$. The lift stays bounded on compact subintervals of $[0, T^*)$ by (i), so it is the fine solution there. Conversely, the projection of a fine solution solves the retained system, so a fine solution bounded beyond T^* would contradict the alternative in (ii).

(iv) Two fine states with the same retained value have the same public laws, rows, and spectra, hence the same forcing $H(t)$ and, by (ii), the same retained solution; by (iii) their projections are retained solutions. □

Under these hypotheses the retained state is therefore autonomous on the common maximal interval. Global existence, $T^* = \infty$, requires further estimates on the public sector, which are not given here.

This is a configuration retention. Its spectra can be infinite-dimensional, and there is no claim that they give a fixed finite number of parameters. If represented as a sample statistic, the map $v \mapsto \ell_z(v)$ depends on the current law; its motion is exactly the term differentiated in eq. (8.15). It must not be frozen while the law changes. The inherited spectral evolution is not a Fisher–Rao gradient flow on probability laws over (z, r). Such a flow $\dot{Q} = -Q(h - \mathbb{E}_Q h)$ keeps Q_t mutually absolutely continuous with Q_0, whereas the retained conditional $e^r \xi_z^t(dr)$ is supported on $a \operatorname{supp} \xi_z^0 - \log Z_z(a)$, which moves unless $\kappa_z = A$. The spectral evolution is a continuity equation in r with velocity $H(z) - r$.

Corollary 8.9 (Different internal conditionals with the original public flow). Suppose that, for each agent and each sector separately, the initial spectrum is independent of the public coordinate.

Then this property persists, and the public laws and source rows solve the original coarse KL/Fisher system generated by $\mathcal{E}_c$ on the common maximal interval.

Proof. The same initial spectrum gives the same solution of eq. (8.16) at every public coordinate. Consequently $H_i^q(z)$ is constant in z, and $H_i^s(w)$ is constant in w. Each constant disappears under its own marginal centering in eq. (8.11). □

Equal spectra do not require equal conditionals when the reference is uniform. With a private reference uniform on two points, the conditionals $(3/4, 1/4)$ and $(1/4, 3/4)$ share

$$\xi = \tfrac{1}{2}\delta_{\log(3/2)} + \tfrac{1}{2}\delta_{\log(1/2)}. \tag{8.19}$$

A public bit may select either conditional, producing correlation between public and private variables. Another law with the same public marginal may use the first conditional at both public values and be a product. When the original public flow from the common initial public state is unique, as it is for this finite public alphabet by theorem 4.6 and on general spaces under hypothesis 8.4, both have the same public trajectory in this architecture, although their internal laws differ. Different agents may have different public laws and different spectra. No equality between agents is imposed.

This class is invariant, but it is not reached for the first time at finite interior time. For any $a > 0$, the spectral update in eq. (8.16) is invertible on its image. Indeed, if ξ^t is given, the initial log value is recovered by the map

$$y \longmapsto \frac{y}{a} - \log \int e^{r/a} \xi^t(dr). \tag{8.20}$$

Substituting eq. (8.16) and using $\int e^r \xi^0(dr) = 1$ proves the assertion. Equal spectra at a finite time therefore had to be equal initially. The same inversion applied to eq. (8.18) proves that distinct conditionals cannot first become equal at finite time. Neither fact forbids an autonomous quotient that keeps the unequal spectra.

Proposition 8.10 (A coarser completion for the public flow). In the setting of proposition 8.8, the public laws and rows are determined, on the common maximal interval, by their initial values together with the classes of the maps $z \mapsto \Lambda^q_{i,z}|_{(0,1]}$ and $w \mapsto \Lambda^s_{i,w}|_{(0,1]}$ of eq. (8.14) modulo functions of b alone. These classes evolve autonomously by

$$\Lambda^t_z(b) = \Lambda^0_z(ab) - b\Lambda^0_z(a), \qquad a = e^{-t}. \tag{8.21}$$

The class is strictly coarser than the spectrum: if $\mathsf{V} = \mathsf{V}_1 \times \mathsf{V}_2$ with a product reference, the conditionals $\kappa_z^{(1)} \otimes \psi$ and $\kappa_z^{(1)} \otimes \psi'$ with $\psi \neq \psi'$ fixed have different spectra at every z whenever ψ and ψ' have different spectra, but the same class.

Proof. Substituting eq. (8.16) into the definition gives eq. (8.21). If $\Lambda^0_z = \widetilde{\Lambda}^0_z + k(b)$ with k independent of z, then eq. (8.21) adds $k(ab) - bk(a)$, again independent of z, so the flow acts on classes. The forcing is $H(z,t) = \partial_b \Lambda^t_z(1)$, which the class determines up to a function of t alone, and such a function disappears under marginal centering in eq. (8.11). Uniqueness in proposition 8.8 gives the conclusion. For the example, the log-Laplace transform of a product conditional against a product reference is the sum of the transforms of the factors, so the two maps differ by the z-independent

function $\Lambda_\psi - \Lambda_{\psi'}$, while the spectra are the convolutions of a common factor with the different spectra of ψ and ψ'. □

The full spectrum, or its marked version below, is needed only for records that read private data. Corollary 8.9 is the case in which the classes are trivial.

Remark 8.11 (Exponential families and finite-parameter private closure). Hölder's inequality gives $0 < Z_z(a) \le 1$ for every $\kappa_z(0)$ mutually absolutely continuous with A, so eq. (8.18) defines the private solution without (H1). If $\kappa_z(0) \propto \exp(\theta_z \cdot T)A$ for a statistic T with log-partition function ψ, then $\kappa_z(t) \propto \exp(e^{-t}\theta_z \cdot T)A$: the natural parameters contract linearly, and $H(z,t) = a\theta_z \cdot \nabla\psi(a\theta_z) - \psi(a\theta_z)$ with $a = e^{-t}$. The private sector then closes on finitely many parameters per public value, for instance on Gaussian conditionals relative to a Gaussian reference. Such families fall outside (H1) when T is unbounded, and the public part then needs its own justification; for the public equation alone, a bound on $\operatorname{ess\,sup}_z H(z,0)$ suffices for the forcing to be bounded, because H is nonincreasing in t. Nothing here makes the public laws finite-dimensional.

Remark 8.12 (Which channels permit the completion). Receiver reconstruction is not what produces closure. If every channel in a row is replaced by $B^q_{ij}(z_j, dz_i)A^q_i(dv_i)$, a fixed reconstruction independent of state and source, the chain rule gives $D^q_{ij} = D_{\mathrm{KL}}(\mu_i \| \mu_j B^q_{ij}) + \int H^q_i d\mu_i$. The added term is the same for every source, so it cancels from Gibbs rows and from the row velocities, the private energy doubles, and the conditional equation becomes $\partial_t \ell^q_{i,z} = 2(H^q_i(z) - \ell^q_{i,z})$; the spectrum still closes. What matters is that the channel discards the sender's private coordinate. A channel that transmits it fails. Take a trivial public space, $\mathsf{V} = \{1,2\}$ with uniform A, two agents attending to each other through the identity channel, and $\kappa_1 = (3/4, 1/4)$. With $\kappa_2 = \kappa_1$ the peer costs vanish, whereas with $\kappa_2 = (1/4, 3/4)$, which has the same spectrum, each peer cost is $\frac{1}{2}\log 3$ and the peer forces move κ_1 toward κ_2. The spectral velocity of agent 1 then differs between two configurations with the same per-agent spectra, so those spectra are not a closed state. For identity private channels on a common (V, A), every peer integrand is a function of the vector $(\ell_k(v))_k$ at one private point, which suggests the joint law of that vector under A as the retained state; its closure and its compression are not established here.

8.4 Boundary records and feedback

Public-only response kernels descend immediately: the channels B^q_{ij}, B^s_{ij} and evaluator G^c_i are determined by the retained public laws. A private-sensitive response needs more information. In eq. (8.19), let the probabilities of record 1 at the two private points be $(3/4, 1/4)$. The two conditionals yield record probabilities $5/8$ and $3/8$. After record 1, their private posteriors are $(9/10, 1/10)$ and $(1/2, 1/2)$. Thus an unmarked spectrum can fail both the boundary test and the next-update test.

There is an exact completion for a specified family of records. Fix a measurable mark $T : \mathsf{V} \to \mathsf{D}$, with D standard Borel, through which every allowed private readout factors. Retain

$$\eta_z = (\ell_z, T)_{\#} A. \tag{8.22}$$

The correlation between the log-density value and the mark is retained. For a normalized record kernel with strictly positive density $L(o \mid z, d)$, assume the following integrals are finite and nonzero and each updated law stays in the admitted regular domain. A bounded log likelihood is a sufficient local condition for the conditional log-density bounds. Put

$$\begin{aligned} N_z(o) &= \int e^r L(o \mid z, d)\eta_z(dr, dd), \\ N(o) &= \int N_z(o)\mu(dz). \end{aligned} \tag{8.23}$$

Proposition 8.13 (Exact marked feedback). The record density is $N(o)$. Exact conditioning of this sector gives the closed updates

$$\begin{aligned} \mu^+(dz) &= \frac{N_z(o)}{N(o)}\mu(dz), \\ \eta_z^+ &= \big((r, d) \mapsto (r + \log L(o \mid z, d) - \log N_z(o), d)\big)_{\#}\eta_z. \end{aligned} \tag{8.24}$$

Between records, eqs. (8.15) and (8.16) transport the first coordinate of η_z and leave its mark fixed. Hence the marked state determines any finite sequence of these records, feedback updates, and original smooth-flow intervals.

Proof. Since $d\kappa_z = e^{\ell_z} dA$, integration of the likelihood gives eq. (8.23). Bayes' rule gives the displayed marginal update and $d\kappa_z^+/dA = e^{\ell_z} L(o \mid z, T)/N_z(o)$. Taking the logarithm and pushing A gives the marked update. The fine reference remains fixed. The conditional equation after the update is therefore the same equation with this new conditional law. Inductively, the inherited flow gives the next retained state, the inherited record kernel gives the same next-record law, and the inherited update gives the same posterior retained state. Actions selected from retained histories preserve this induction. Any reference or likelihood parameters changed by records must also be retained and updated by a rule that depends only on this state. □

The composite of flow intervals and these updates is the record-interrupted dynamics of definition 4.24. The potential contains no record term and is not monotone across records. Records act only at their times: no rule changing the references is declared here, and between records each conditional relaxes toward its fixed reference by eq. (8.18), so the effect of a record on the private sector decays.

The update is sectorwise. A product of separate belief and model likelihoods preserves the recognition restriction $q_i \otimes s_i$. A likelihood jointly coupling the two sectors need not preserve that product; exact joint conditioning would require a joint recognition state, or a separately declared projection. Marked feedback can also destroy equality of unmarked spectra across public coordinates. The augmented marked state still closes, but the simpler conclusion of corollary 8.9 then need not hold.

For an explicit retained sample law, use

$$Q^c(dz, dr, dd) = \mu(dz)e^r\eta_z(dr, dd). \tag{8.25}$$

It is normalized and reproduces every declared marked record. It is the pushforward under the current, law-dependent statistic $(z, v) \mapsto (z, \ell_z(v), T(v))$. Different assignments of density values to private points within a mark can give the same retained state. If the permitted records distinguish every private point, the mark may have to retain that point, eliminating this compression. The boundary protocol therefore fixes what can be forgotten. Predictive-state constructions likewise make sufficiency relative to future observations [129]; the formulas here specify the state and update for this particular KL architecture.

8.5 What is still required for formation

Positive source temperature does not give exact isolation by itself. For optimized rows, every allowed finite-cost edge has positive weight. For a dynamic row, set

$$\theta_{jk} = \log \frac{\beta_j/\pi_j}{\beta_k/\pi_k}. \tag{8.26}$$

The Fisher row equation from eq. (4.4) gives

$$\dot{\theta}_{jk} = -(D_j - D_k) - T_q \theta_{jk}. \tag{8.27}$$

Multiplying by $e^{T_q t}$ and integrating shows that finite cost differences keep every such ratio finite on a regular finite interval. Initially positive weights therefore stay positive there. Uniformly bounded cost differences also bound these ratios for all time and prevent asymptotic disappearance of an allowed weight. A formed collective may nevertheless keep nonzero boundary interactions; isolation is not required for inheritance.

The construction above compresses private conditional information of each agent separately, and only up to the equimeasurability quotient of its declared reference. It does not combine several distinct public agent laws into one smaller collective state. Retaining their complete labeled tuple and calling it a parent would leave that aggregation problem untouched; whether a retention of such a tuple compresses is decided by clause (F4) of definition 6.31, which requires a noninjective retention whose fibers some admitted child-level probe distinguishes. Nor does recognizing a persistent source-preference group establish a compatible collective retention. The distinction between a coarse observable and an autonomous macroscopic evolution is also expressed by flow intertwining in Barnett and Seth [9].

Remark 8.14 (Aggregation with conditional information retained). The construction of this chapter selects no group and involves no grouping of agents, and its private sector is unchanged by deleting every peer cost. A recursive formation that uses the conditional-spectrum state as part of its retention is an instance of open problem 6.32. A coupled architecture under the reciprocal potential in which private conditionals feed back into public interactions, for example through private-transmitting channels or rows that read private data, and which still admits an exact compressing completion, is not available.

Chapter 9

State-Controlled Erasure and Persistent Source Preferences

This chapter concerns the reciprocal law potential of definition 4.2, in its exact Gibbs-reduced form eq. (4.6); nothing in it is a statement about the per-agent or global free-energy pictures. Inside that potential there is a conditional first-stage construction. A prescribed channel can lose its dependence on some internal laws when a dynamically evolving gate coordinate enters a flat part of that channel. The preferred sources of the remaining agents can change, and the retained dynamics can then become exactly autonomous. The mechanism is an engineered constitutive choice, and both of its conclusions follow from a declared ordering of scales. It supplies neither a general explanation of sufficient interfaces nor a second formation stage. Measured against the formation standard definition 6.31, it is a single-stage selection with exact inheritance by a decoupled-factor projection; section 9.5 records the clause-by-clause status and the null-flow control. The general laws of the preceding chapters remain the framework; the existence example below uses an explicitly invariant family of positive finite laws.

9.1 A homogeneous population with internal gate coordinates

Write $b_x(k) = (1 + kx)/2$ on $\{-1, 1\}$, set $d(x, z) = D_{\mathrm{KL}}(b_x \| b_z)$, and put $d_0(x) = d(x, 0)$. Four agents carry

$$q_i = b_{x_i} \otimes b_{z_i} \otimes b_{h_i}, \qquad s_i = b_{y_i} \otimes b_{w_i} \otimes b_{\widehat{h}_i}. \tag{9.1}$$

The first coordinate is called public, the second private, and the third is a gate coordinate. These words refer to a declared observable algebra: the peer channel below transmits samples of the public coordinate, it reads the private coordinate of a source only through that source's mean multiplied by the gate, and the admitted response class of proposition 9.4 reads public coordinates only. Their unit Fisher metric has diagonal entries $1/(1 - u^2)$ for each mean u. The product preparation within each sector is an assumption. It is invariant under the full table dynamics specified below, rather than a projection imposed after each step.

Take $J = \log 3$ and $h_* = -1/2$. Use the proper reference $r = b_0 \otimes b_0 \otimes b_{h_*}$ and normalized evaluator

$$G_J(k \mid m) = \frac{e^{Jkm}}{2 \cosh J}, \tag{9.2}$$

on the public coordinates, with independent private output b_0 and gate output b_{h_*}. The local term is exactly

$$\begin{aligned} E_{0,i} = d_0(x_i) + d_0(y_i) - Jx_iy_i + \log\cosh J \\ + d_0(z_i) + d_0(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*). \end{aligned} \tag{9.3}$$

All agents have the same reference and evaluator. Every nonself agent is an allowed source, with uniform prior and fixed temperature $T > 0$. We use the exact Gibbs-reduced version of the reciprocal potential. The bits are reference coordinates: a change of presentation carries the reference laws, channels, and readout maps with it, so the KL and Fisher covariance statements apply.

Define $\chi(r) = 0$ for $r \le 0$, $\chi(r) = 1$ for $r \ge 1$, and, for $0 < r < 1$,

$$\chi(r) = \frac{e^{-1/r}}{e^{-1/r} + e^{-1/(1-r)}}. \tag{9.4}$$

This increasing function is smooth, all its derivatives match at the endpoints, and $M_\chi = \max|\chi'| < \infty$. Fix $h_0 = 1/4$ and a gate width $\Delta > 0$, and set

$$H_q = \frac{1}{4}\sum_i h_i, \quad H_s = \frac{1}{4}\sum_i \widehat{h}_i, \quad \alpha(H) = \chi((H - h_0)/\Delta). \tag{9.5}$$

The gate is a nonlocal function of internal laws. Its threshold and flat region are constitutive data. There is no external time schedule; the schedule is internal, set by the initial gate positions and by h_*, as lemma 9.1 makes precise.

Choose gains $0 < \varepsilon, \varrho < 1$. On a belief edge $j \to i$, let the public output kernel be $(1-\varepsilon)b_{x_i} + \varepsilon\delta_{k_{\mathrm{pub}}}$, and independently draw private output from $b_{\alpha(H_q)\varrho z_j}$ and gate output from b_{h_*}. Its compared law is

$$\nu_{ij}^q = b_{(1-\varepsilon)x_i + \varepsilon x_j} \otimes b_{\alpha(H_q)\varrho z_j} \otimes b_{h_*}. \tag{9.6}$$

This normalized channel is strictly positive on the interior and depends smoothly on the current marginal means. It transmits public samples, mixed with the receiver's own public law, and uses the source's private mean rather than its private sample. At $\alpha = 0$ every private cost therefore equals $d_0(z_i)$ by construction. The model channel exchanges (x, z, h) with $(y, w, \widehat{h})$. Define

$$D_\varepsilon(x, v) = d(x, (1-\varepsilon)x + \varepsilon v). \tag{9.7}$$

The reciprocal peer cost becomes

$$D_{ij}^q = D_\varepsilon(x_i, x_j) + d(z_i, \alpha(H_q)\varrho z_j) + d(h_i, h_*). \tag{9.8}$$

No fusion cost has been added. Because the public kernel mixes in the receiver's own law, every public peer cost and force is of order ε^2.

The definitions extend to arbitrary positive tables on each three-bit space by taking the indicated marginal means. At a product table, the derivative of the receiver entropy is additive in its coordinates. Each compared law is a product, and every channel derivative is a function of a single

coordinate because the channel parameters depend on marginal means. The local evaluator also gives additive scores within each sector. The full Fisher score is consequently a sum of coordinate scores on eq. (9.1). Differentiating a product shows that the full table flow of theorem 4.6 preserves this family and induces the stated Bernoulli Fisher metric. This argument includes incoming and channel derivatives.

9.2 Finite entry into an exactly erasing region

Lemma 9.1 (Permanent gate closure under the reciprocal flow). For sufficiently small $\Delta > 0$ and then sufficiently small $\varrho > 0$ relative to Δ, there is an open class within eq. (9.1), with

$$|z_i|, |w_i| < \sigma_{\max} = 3/4, \qquad h_i, \widehat{h}_i \in (h_0 + \Delta/2, h_0 + 3\Delta/4), \tag{9.9}$$

and public means unrestricted, on which, for as long as the solution exists, both gates reach zero by the time $3\Delta/(2m_K)$ defined below and remain zero afterward. The gate means follow an autonomous relaxation toward h_* perturbed by a term of size at most $B\varrho M_\chi/\Delta$. On the admitted class of theorem 9.2 the solution exists for all time.

Proof. The exact envelope derivative eq. (4.8) gives

$$\dot{h}_i = -(1 - h_i^2)\left[2(\operatorname{atanh} h_i - \operatorname{atanh} h_*) + \frac{\alpha'(H_q)}{4}\sum_{a,j}\beta^*_{aj}\partial_\alpha d(z_a, \alpha\varrho z_j)\right]. \tag{9.10}$$

The factor two comes from the local term and the receiver's source-independent gate cost. For $|z_a|, |z_j| \le \sigma_{\max}$ and $\varrho \le 1/2$,

$$\partial_\alpha d(z_a, \alpha\varrho z_j) = \varrho z_j \frac{\alpha\varrho z_j - z_a}{1 - \alpha^2\varrho^2 z_j^2}. \tag{9.11}$$

Its absolute value is bounded by $B\varrho$, where one may take $B = 3\sigma_{\max}^2/[2(1 - \sigma_{\max}^2/4)]$. The four row sums equal one, so the correction to the gate force is bounded by $B\varrho M_\chi/\Delta$.

On $K = [1/8, 1/2]$, define the positive constants

$$\begin{aligned} m_K &= \min_{h\in K} 2(1 - h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \\ M_K &= \max_{h\in K} 2(1 - h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \qquad V_K = M_K + m_K/2. \end{aligned} \tag{9.12}$$

Choose $B\varrho M_\chi/\Delta \le m_K/2$. While its coordinates stay in K, each gate coordinate has $\dot{h}_i \le -m_K/2$ and $|\dot{h}_i| \le V_K$. Also choose Δ so that the initial interval lies in K and

$$\frac{3V_K\Delta}{2m_K} < h_0 - 1/8. \tag{9.13}$$

The sector mean reaches h_0 by time $3\Delta/(2m_K)$, before any coordinate can leave K. On $H_q = h_0$, both α and α' vanish. Thereafter

$$\dot{h}_i = -2(1 - h_i^2)(\operatorname{atanh} h_i - \operatorname{atanh} h_*). \tag{9.14}$$

All these coordinates start above h_* and decrease toward it. Their mean therefore cannot leave the closed-gate region. The model-sector proof is identical.

It remains to justify the private bounds used in the argument. At $\varrho = 0$ the private field is $\dot{z} = -2(1 - z^2)\,\mathrm{atanh}\, z$, strictly inward at $z = \pm\sigma_{\max}$. Private derivatives of eq. (9.8) converge uniformly to this field as $\varrho \to 0$, uniformly over normalized source rows and $0 \le \alpha \le 1$. The incoming terms are of order ϱ; differentiating a private mean does not differentiate α. Compactness therefore preserves the strict inward signs for sufficiently small ϱ. The same holds for w. This closes the interval argument. The private and gate means stay in compact subsets of $(-1, 1)$. On the admitted class of theorem 9.2 so do the public means, by the box invariance proved there, and the solution therefore continues for all time on that class. □

Let τ be the later of the two gate-closure times. The construction uses a smooth but nonanalytic channel: its exactly flat region makes finite entry into exact erasure possible. Decreasing a small nonzero analytic gate would not establish this conclusion. The gate mean is an internal clock: its motion is monotone and decoupled from the other coordinates up to a perturbation that the parameter choice makes subordinate.

9.3 A change of preferred sources without synchronization

Theorem 9.2 (Changing and persistent source preferences). For sufficiently small $\varepsilon > 0$, admit two public states in $(l, u)^2$ and two in $(-u, -l)^2$, with $l = 0.49$, $u = 0.51$, including every assignment of these signs to the four labels. These boxes are forward invariant. After τ, each agent's unique preferred source in both sectors is the other agent in its box. There are initial private states for which the preferred pairs are instead crossed between these boxes before closure. The public states within each final pair can remain distinct at every finite time.

Proof. For the local public flow, the inequalities

$$Jl - \mathrm{atanh}\, l > 0, \qquad \mathrm{atanh}\, u - Ju > 0, \tag{9.15}$$

give strict inward signs on every face of the two boxes. They follow by differentiating $Jv - \mathrm{atanh}\, v$ near its zero $v = 1/2$. Set $c = (1 - \varepsilon)x + \varepsilon v$. Direct differentiation gives

$$\begin{aligned} \partial_1 D_\varepsilon(x, v) &= \mathrm{atanh}\, x - \mathrm{atanh}\, c + (1 - \varepsilon)\frac{c - x}{1 - c^2}, \\ \partial_2 D_\varepsilon(x, v) &= \varepsilon \frac{c - x}{1 - c^2}. \end{aligned} \tag{9.16}$$

Both are uniformly $O(\varepsilon^2)$ on $[-u, u]^2$; the first-order terms cancel in the first line. The full public force has one outgoing row and three incoming sources, each weighted by a number in $[0, 1]$. A sufficiently small ε preserves every strict inward margin, independently of the private state and the gate.

After gate closure, the private and gate costs in a row do not depend on its source, so Gibbs ranking uses D_ε alone. Taylor's theorem in the first argument of d gives

$$\frac{1}{2}(x-c)^2 \le d(x,c) \le \frac{(x-c)^2}{2(1-u^2)}. \tag{9.17}$$

A same-box source has separation at most $u-l$, whereas an opposite-box source has separation at least $2l$. Since $(u-l)^2/(1-u^2) < (2l)^2$, the same-box partner is the unique preferred source throughout the invariant region. Cross-pair weights remain positive.

For an explicit initial reorganization, choose private means $\pm\sigma$, with $0 < \sigma < \sigma_{\max}$, in both sectors, assigning each same-private-sign pair one positive and one negative public member. Initially $\alpha \ge 1/2$. The private advantage of the same-sign partner over an opposite-sign source is

$$d(\sigma, -\alpha\varrho\sigma) - d(\sigma, \alpha\varrho\sigma) = 2\sigma\,\mathrm{atanh}(\alpha\varrho\sigma) \ge \varrho\sigma^2. \tag{9.18}$$

The public cost range is at most $2u^2\varepsilon^2/(1-u^2)$. Choosing this smaller than $\varrho\sigma^2$ makes the initial preferred pairs the crossed pairs. Strict inequalities persist under small perturbations of the initial private values. Thus the final persistent grouping is not an unchanged initial source assignment.

Choose initial public states within a box separated by a fixed positive distance. Their speeds are uniformly bounded, whereas $\tau \le 3\Delta/(2m_K)$. Taking Δ sufficiently small keeps them distinct through τ. Afterward the public field is autonomous, smooth, and permutation equivariant. Its equality diagonal for a pair is invariant. Backward uniqueness precludes first entry into that diagonal at a finite time. This proves finite-time distinctness. □

The preference change is a switch between two prescribed scales: before closure the private advantage $\varrho\sigma^2$ dominates the public cost range of order ε^2, and after closure only the public costs remain. The parameter choices are compatible: first choose a sufficiently small Δ, then ϱ sufficiently small relative to Δ, and finally ε sufficiently small that ε^2 is small relative to ϱ and the fixed public inward margins. All means stay in compact subsets of $(-1,1)$, giving global continuation in this admitted family. The two final groups are the two wells of the bistable single-agent energy $d_0(x) + d_0(y) - Jxy$, whose minima are $x = y = \pm 1/2$, and box membership is the supplied initial well.

The temperature enters this ordering. The proofs above use only that row weights lie in $[0,1]$, so $T > 0$ may be fixed before the gains are chosen. At fixed T every row cost is $O(\varrho + \varepsilon^2)$, so every Gibbs row is within $O((\varrho+\varepsilon^2)/T)$ of uniform, and a preferred source is then the largest entry of a nearly uniform row. Quantitatively, eq. (9.17) gives after closure

$$\beta^*_{i,\mathrm{same}} \ge \frac{1}{1+2e^{-\Delta_\varepsilon/T}}, \qquad \Delta_\varepsilon = \varepsilon^2\left[2l^2 - \frac{(u-l)^2}{2(1-u^2)}\right], \tag{9.19}$$

and before closure, with $\alpha \ge 1/2$, the crossed partner has weight at least $[1+2e^{-(\varrho\sigma^2 - 2u^2\varepsilon^2/(1-u^2))/T}]^{-1}$. Near one-hot selection therefore requires T of order ε^2 or smaller.

The heterogeneity of the final pairs is transient, as the next result shows.

Proposition 9.3 (Each preferred pair converges to a common state). Fix $T > 0$ and the parameters of theorem 9.2. There is $\varepsilon_T > 0$ such that for $0 < \varepsilon < \varepsilon_T$ the public system after τ has exactly one equilibrium in the closed boxes $Q = ([l, u]^2)^2 \times ([-u, -l]^2)^2$, ordered by box. At this equilibrium the two members of each box have equal public states, the two boxes carry opposite states $\pm(x_\varepsilon, y_\varepsilon)$, and $(x_\varepsilon, y_\varepsilon) \to (1/2, 1/2)$ as $\varepsilon \to 0$. Every admitted trajectory converges to it exponentially. In particular the in-pair public separation tends to zero, and each retained empirical measure η_I of eq. (9.24) converges to a Dirac mass.

Proof. After τ the public coordinates follow the unit Fisher gradient flow of the public part $\mathcal{E}_{\mathrm{pub}}$ of eq. (9.21), so

$$\frac{d\mathcal{E}_{\mathrm{pub}}}{dt} = -\sum_i (1 - x_i^2)(\partial_{x_i}\mathcal{E}_{\mathrm{pub}})^2 - \sum_i (1 - y_i^2)(\partial_{y_i}\mathcal{E}_{\mathrm{pub}})^2 \le 0, \tag{9.20}$$

with equality exactly at equilibria. The set Q is compact and forward invariant by theorem 9.2. By the invariance principle [92], the ω-limit set of each trajectory is a nonempty connected set of equilibria in Q.

At $\varepsilon = 0$ the public field decouples into single-agent fields. In the positive box their zeros solve $\operatorname{atanh} x = Jy$ and $\operatorname{atanh} y = Jx$. With the increasing map $g(t) = \operatorname{atanh}(t)/J$ this reads $x = g(y)$ and $y = g(x)$. If $x > g(x)$, then $y = g(x) < x$ and $x = g(y) < g(x)$, a contradiction; the case $x < g(x)$ is symmetric. Hence $\operatorname{atanh} x = Jx$, and since $\operatorname{atanh}(x)/x$ increases from one and $J > 1$, the only solution in $(0, 1)$ is $x = 1/2$. The Jacobian of $(\operatorname{atanh} x - Jy, \operatorname{atanh} y - Jx)$ there has diagonal entries $4/3$ and off-diagonal entries $-J$, and $16/9 - J^2 > 0$, so the zero is nondegenerate. The negative box is the mirror image. Thus at $\varepsilon = 0$ the field on Q has exactly one zero, and it is nondegenerate and interior.

The peer cost D_ε is a smooth function of (x, v, ε) that vanishes to second order in ε, so all its derivatives in (x, v) are $O(\varepsilon^2)$ on Q, and the derivatives of the Gibbs rows are $O(\varepsilon^2/T)$. For fixed T the public field therefore converges to the decoupled field in $C^1(Q)$ as $\varepsilon \to 0$. The decoupled field is bounded away from zero on Q outside any neighborhood of its zero, and the implicit function theorem gives a unique nondegenerate zero inside a small neighborhood. This proves uniqueness and the limit $(1/2, 1/2)$.

The post-closure public system has identical constitutive data for all agents and uniform nonself priors, so it is equivariant under permutations of the agents. It is also equivariant under the global reversal $(x, y) \mapsto (-x, -y)$, since d_0 is even, Jxy is invariant, and $d(-x, -c) = d(x, c)$. The transposition of two box-mates, and the composition of global reversal with the exchange of the two boxes, map Q to itself and therefore fix the unique equilibrium. This gives equal box-mates and opposite boxes.

Every ω-limit set is therefore this single equilibrium. It attracts every interior point of Q, so it is a local minimum of $\mathcal{E}_{\mathrm{pub}}$: a nondegenerate critical point of a gradient flow that is not a minimum has a stable set of positive codimension. The linearized gradient flow at a nondegenerate minimum has negative spectrum, which gives exponential convergence. □

Persistent heterogeneous pairs would require heterogeneous constitutive data. With identical agents each box carries a single attracting equilibrium, and proposition 9.3 forces its members together. If agent i instead carried its own coupling $J_i = \log 3 + \eta_i$, the single-agent equilibria near

$1/2$ would separate to first order by $|\eta_i - \eta_k|/(8/3 - 2\log 3)$ per coordinate, while the inward margins of eq. (9.15) survive for small $|\eta_i|$; box-mates would then be expected to converge to distinct states. We do not prove this variant here. A gate driven by a public statistic, such as the polarization $\frac{1}{4}\sum_i d_0(x_i)$, would make erasure depend on public relaxation rather than on a dedicated clock; its entry estimate is also not established.

9.4 The exact inherited state

For $t \geq \tau$ the reduced potential separates exactly:

$$\begin{aligned}\mathcal{E}_{\mathrm{red}} &= \mathcal{E}_{\mathrm{pub}}(\mathbf{x}, \mathbf{y}) \\ &\quad + 2\sum_i [d_0(z_i) + d_0(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*)].\end{aligned} \tag{9.21}$$

Here $\mathcal{E}_{\mathrm{pub}}$ is the coupled local KL and Gibbs-reduced reciprocal peer potential with costs D_ε. The product Fisher metric separates as well. After closure the state space is the product of eight public means and sixteen private and gate means, the flow is a product flow, and the second factor relaxes autonomously to $z = w = 0$ and $h = \widehat{h} = h_*$. Discarding that factor is projection onto a factor of a product flow, the most elementary exact lumping [79], and it gives an exact autonomous public system without fitting another potential or mobility.

This independence is generally absent before gate closure. To see that this is an actual change of the vector field, expand one reduced row with $\alpha > 0$ fixed. If $A_{ij} = (x_j - x_i)^2/[2(1 - x_i^2)]$, its first mixed public-private term is

$$\frac{\alpha\varrho\varepsilon^2}{T} z_i \operatorname{Cov}_{j\neq i}(A_{ij}, z_j). \tag{9.22}$$

This follows by expanding $D_\varepsilon = \varepsilon^2 A + O(\varepsilon^3)$, $d(z_i, \alpha\varrho z_j) = d_0(z_i) - \alpha\varrho z_i z_j + O(\varrho^2)$, and the logarithm of the row normalization in powers of the costs divided by T; the expansion is valid at fixed T as the gains tend to zero. The covariance is under the uniform source prior. It is not identically zero even within the crossed-preference preparation. Fix $v = (-b, -a, a, b)$ with $l < a < b < u$, keep $z = \sigma(1, -1, 1, -1)$ fixed, and vary only public $x_i = rv_i$ near $r = 1$. The summed coefficient is

$$\frac{2\sigma^2 r^2}{9}\left[\frac{a^2 - 3b^2 + 6ab}{1 - r^2 b^2} + \frac{b^2 - 3a^2 + 6ab}{1 - r^2 a^2}\right]. \tag{9.23}$$

Both numerator factors are positive on the stated intervals, so this expression has positive derivative in $r > 0$. Two fixed admissible values of σ therefore give different public directional gradients at leading order $\varrho\varepsilon^2$. The differentiated remainder is $O(\varrho\varepsilon^3 + \varrho^2\varepsilon^2)$ on these compact sets. For sufficiently small positive gains, equal public data thus have different public derivatives while the gate is open. This expansion establishes nonidentity of the preclosure force; eq. (9.21) is an exact postclosure identity. For T of order ε^2 or smaller the cumulant expansion is unavailable, and nonautonomy before closure would need a direct comparison of the forces under two private assignments.

For each selected pair I, retain its size and the empirical measure

$$\eta_I = \frac{1}{|I|}\sum_{i\in I}\delta_{(x_i,y_i)}, \qquad \mathsf{M} = \sum_I |I|\eta_I. \tag{9.24}$$

This is a deterministic configuration statistic, not an assumed probability distribution over uncertain population configurations. Let $V[v;\mathsf{M}]$ be the public Fisher vector field at the public state $v=(x,y)$, computed from $\mathcal{E}_{\mathrm{pub}}$. It is determined by v and the counting measure M: source sums use $\mathsf{M}-\delta_v$, and the derivative includes both receiver and incoming source terms from eq. (9.16). The normalized Gibbs denominator is computed from the same counting measure. Permutation equivariance makes this definition independent of labels, including multiplicities.

Proposition 9.4 (Exact parent evolution and public responses). The inherited parent equations are

$$\frac{d}{dt}\int f(v)\eta_I(dv) = \int \nabla f(v)\cdot V[v;\mathsf{M}]\eta_I(dv), \tag{9.25}$$

for every continuously differentiable test function. Among curves of realizable finite-atom states, that is, empirical measures of solutions of the public system, two curves with equal initial retained measures coincide at all later times. The retained measures also determine every declared fresh public response whose source selection depends only on public states, including joint responses with a shared sampled child.

Proof. Differentiate the finite sum in eq. (9.24) using the exact public equations. This gives eq. (9.25). Equal retained parent measures admit enumerations that differ only by within-pair permutations. Uniqueness and permutation equivariance of the public system give identical evolved measures, proving the uniqueness assertion among realizable curves. For a fresh response, select the child or children using the declared normalized public kernel, then integrate their public sample laws against the retained empirical measures. Both the selection probabilities and sample laws are determined by the retained states. For shared-source outputs, use one shared draw rather than multiplying separate marginals; the same calculation preserves their joint law. □

The proposition does not assert uniqueness for eq. (9.25) among general measure-valued curves; that would need a Lipschitz stability estimate for V in (v,M). A single η_I has no autonomous dynamics, since $V[v;\mathsf{M}]$ depends on the counting measure of both pairs; only the pair of parents is closed. The descent also holds for every partition of the four agents, so inheritance does not distinguish the selected pairs.

The parent can carry belief and model marginals $q_I = \int b_x \eta_I(d(x,y))$ and $s_I = \int b_y \eta_I(d(x,y))$, with normalized public evaluator G_J. These two marginals alone are insufficient: η_I retains the paired public roles and receiver-specific routing information. The inherited equations are eq. (9.25), not a Fisher flow of the reciprocal potential for the two mixture marginals. The parent is therefore not an agent of the children's type: no law pair with an evaluator and a reciprocal KL interaction carries its dynamics, and the construction cannot be iterated as it stands. A recursion would need a retention whose image is again a law pair with an inherited reciprocal interaction.

The compression here discards the private and gate laws, which no admitted response reads after closure: the admitted family has twenty-four continuous means, and the retained empirical state

keeps eight public means, up to within-pair permutations. It is a decoupled-factor projection followed by a finite label quotient, and it keeps every public child state. Cached private readouts, feedback that changes the erased channels, and protocols that later inspect discarded private coordinates are outside the response class. The reference, gate, initial sign-box preparation, and boundary restrictions are all supplied assumptions. The dynamics select changing source preferences and a persistent grouping within them, but do not discover the erasing architecture.

9.5 Status against the formation standard

Take as selector the map that assigns to each agent the source of smallest reciprocal row cost eq. (9.8) at the current state, and as handoff the closure time τ. The admitted class is open in the invariant family eq. (9.1), so (F1) holds in its invariant-family form. Selection with a persistence criterion, (F2), holds: the box pairs are an invariant block relation after τ, and τ is a function of the state. Inheritance, (F3), holds for the declared public response class through proposition 9.4, with the qualification that the parent state is an empirical measure rather than a law pair. Compression, (F4), fails: the retention is non-injective only on private and gate coordinates that no admitted probe reads, and it keeps every public child state. Recursion, (F6), fails, since there is a single stage and the two final boxes stay separated.

For the null-flow control of definition 6.30, delete every peer cost. The public coordinates then follow the single-agent flows, which keep the two sign boxes invariant by eq. (9.15); the private means relax monotonically toward zero without changing sign; and the gate means follow eq. (9.14) with the factor two replaced by one, so the gates still close. The same selector, evaluated along this flow, therefore reads the crossed pairs while the private advantage dominates and the box pairs after it is erased. The sequence of selected pairings coincides with that of the reciprocal flow, and only the switching time changes; the null gates relax at about half the speed. Clause (F5) is therefore met only through a shift of an event time, and its strong form fails: the final pairing is the sign pattern of the initial public states and the initial pairing is the sign pattern of the initial private states, and the null flow preserves both. The construction is a single-stage selection with exact inheritance by a decoupled-factor projection, not a two-stage selection and not a recursive formation.

Chapter 10

Public compression and inherited responses

This chapter concerns the reciprocal law potential of definition 4.2, in its exact Gibbs-reduced form eq. (4.6), with identity channels; it makes no claim about the per-agent or global free-energy pictures. A nonlinear projection of the reciprocal Fisher flow can discard continuous public differences without an erasure channel or finite-time equality. The projection is the asymptotic phase of the approach to a stable common reference, so it compresses the decaying part of the transient. This autonomous parent nevertheless fails to predict ordinary child sampling, and it is the only parent of its kind. A separate identifiability result explains why an unrestricted Gibbs interface retains every exposed child law, and how a fixed channel changes what it retains. The block is supplied and nothing is selected, so these are retention results rather than formation results; section 10.5 records their status against the formation standard definition 6.31.

10.1 An asymptotic collective phase near a stable reference

Let A and B be probability measures on standard Borel spaces K and M. Suppose there are measurable functions $\varphi : \mathsf{K} \to \{-1, 1\}$ and $\psi : \mathsf{M} \to \{-1, 1\}$ with zero reference means. For example, either space may be a balanced two-point factor times an arbitrary probability space. The agent laws below need not factor over these coordinates. Fix $|J| < 1$ and the normalized reference

$$R(dk, dm) = \frac{e^{J\varphi(k)\psi(m)}}{\cosh J} A(dk)B(dm). \tag{10.1}$$

Its model marginal is B, and its evaluator is $G(m, dk) = e^{J\varphi(k)\psi(m)}A(dk)/\cosh J$. Thus

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R) = D_{\mathrm{KL}}(q_i \| A) + D_{\mathrm{KL}}(s_i \| B) - J\mathbb{E}_{q_i}\varphi\mathbb{E}_{s_i}\psi + \log\cosh J. \tag{10.2}$$

For $J \neq 0$ the belief and model laws interact through the evaluator.

Take $N \geq 2$ identical agents, identity peer channels, all other agents as sources, uniform nonself source priors, and fixed positive temperatures. Use the exact Gibbs-reduced reciprocal potential of eq. (4.6), with unit Fisher mobility. No source or channel derivative is frozen. Introduce the Banach spaces

$$\begin{aligned} \mathsf{E}_q &= \{u \in L^\infty(A) : \mathbb{E}_A u = 0\}, \quad q_u = \frac{e^u}{\mathbb{E}_A e^u} A, \\ \mathsf{E}_s &= \{v \in L^\infty(B) : \mathbb{E}_B v = 0\}, \quad s_v = \frac{e^v}{\mathbb{E}_B e^v} B, \end{aligned} \tag{10.3}$$

and $\mathsf{E} = \mathsf{E}_q \times \mathsf{E}_s$, with the maximum norm. These are full bounded log-density charts, rather than a finite-parameter family. Since theorem 4.6 is stated for finite tables, we declare the flow in this chart directly. Let $h_i^q \in L^\infty(A)$ be the first variation of the reduced potential with respect to q_i, given by eq. (4.8) and eq. (4.10), and similarly h_i^s. The Fisher-Rao flow $\partial_t \log q_i = -(h_i^q - \mathbb{E}_{q_i} h_i^q)$ becomes, after centering against A,

$$\dot{u}_i = -(h_i^q - \mathbb{E}_A h_i^q), \qquad \dot{v}_i = -(h_i^s - \mathbb{E}_B h_i^s). \tag{10.4}$$

On every bounded ball of E^N the right side is a smooth map into E^N: exponentiation is analytic in the Banach algebra L^∞, the normalizers and compared densities are bounded away from zero, and the KL costs, Gibbs sums, and their first variations are finite combinations of integrals of such functions and of logarithms of positive bounded functions. The vector field is therefore locally Lipschitz, the Picard–Lindelöf theorem in Banach spaces gives a unique maximal solution for every initial point, and along it the reduced potential decreases at the rate $\sum_i [\mathrm{Var}_{q_i}(h_i^q) + \mathrm{Var}_{s_i}(h_i^s)]$. Set $z_i = (u_i, v_i)$, $\bar{z} = N^{-1}\sum_i z_i$, and $\delta_i = z_i - \bar{z}$. Write Φ_t for the resulting fine flow in E^N. The statements below are made for each fixed initial configuration; no measurability of Φ_t with respect to a sigma-algebra on E^N is used, so the norm topology of the nonseparable space L^∞ raises no measurability question here.

On the diagonal $z_1 = \cdots = z_N = z$, every peer force vanishes. The diagonal vector field is

$$f(u, v) = \big(-u + J\varphi \mathbb{E}_{s_v}\psi, -v + J\psi \mathbb{E}_{q_u}\varphi\big). \tag{10.5}$$

Denote its flow by ϕ_t. Its derivative is $Df(u,v)[h,k] = (-h + J\varphi\,\mathrm{Cov}_{s_v}(\psi, k), -k + J\psi\,\mathrm{Cov}_{q_u}(\varphi, h))$. If $\psi = \pm 1$ has mean m under s_v, then $|\mathrm{Cov}_{s_v}(\psi, k)| \le \|k\|\mathbb{E}_{s_v}|\psi - m| = (1 - m^2)\|k\| \le \|k\|$, and likewise for φ. Hence f is globally Lipschitz with constant

$$L = 1 + |J|, \tag{10.6}$$

has at most linear growth, and has a bounded second derivative. Its flow is defined for all $z \in \mathsf{E}$ and all positive and negative finite times. In particular, it can be used below without extending a locally defined diagonal dynamics.

Theorem 10.1 (An asymptotic-phase projection under the reciprocal KL flow). For the preceding data there is a forward-invariant neighborhood U of the common reference configuration in E^N on which

$$\Pi(x) = \lim_{t\to\infty} \phi_{-t}\big(\overline{\Phi_t x}\big), \tag{10.7}$$

exists in E and is continuously differentiable. It satisfies

$$\Pi(z, \dots, z) = z, \qquad \Pi(\Phi_t x) = \phi_t(\Pi x), \qquad D\Pi_{(z,\dots,z)}(h_1, \dots, h_N) = \frac{1}{N}\sum_i h_i. \tag{10.8}$$

After shrinking U if necessary, Π is a submersion with nontrivial fibers parametrized locally by $N - 1$ public log-density differences. The parent laws q_{u^c}, s_{v^c}, where $(u^c, v^c) = \Pi(x)$, follow eq. (10.5) exactly. If two fine child states (q_i, s_i) are initially different, they remain different at every finite time in this neighborhood. Every trajectory in U converges to the common reference.

Proof. By the preceding paragraph the reduced Fisher vector field V of eq. (10.4) is smooth on bounded log-density balls.

At the origin every peer KL cost has zero first variation, so the Gibbs rows equal $1/(N-1)$ and their first variations vanish. The receiver and sender derivatives each contribute one copy of the complete-graph difference operator. Therefore, with $\kappa_N = 2N/(N-1)$,

$$\begin{aligned}(DV(0)h)_i &= \mathcal{A}h_i - \kappa_N(h_i - \bar{h}),\\ \mathcal{A}(u,v) &= \big(-u + J\varphi\mathbb{E}_B[\psi v], -v + J\psi\mathbb{E}_A[\varphi u]\big).\end{aligned} \tag{10.9}$$

The off-diagonal operator in $\mathcal{A}$ has norm at most $|J|$. Hence its common and difference semigroups have bounds $e^{-(1-|J|)t}$ and $e^{-(1+\kappa_N-|J|)t}$ in the corresponding maximum norms. In the adapted norm $\max(\|\bar{z}\|, \max_i \|\delta_i\|)$ both linear semigroups are contractions with constant one; in the plain maximum norm on E^N the full linear semigroup has a constant up to three, which is why the adapted norm is used below.

Here are the local estimates needed for the limit, including its regularity. Choose $\mu > 0$ and ω strictly below these two decay rates, respectively, with

$$\omega > L = 1 + |J|. \tag{10.10}$$

This is possible because $1 + |J| < 1 + \kappa_N - |J|$ whenever $|J| < N/(N-1)$, which holds since $|J| < 1$; and μ exists because $|J| < 1$. In the adapted norm the nonlinear part of V is $o(r)$ on a ball of radius r, so for small r the one-sided Lipschitz constant of the full field on the ball stays below $-\mu$ and the ball is forward invariant. Taylor expansion and the variation-of-constants formula on such a ball give

$$\|\Phi_t x\| + \|D\Phi_t(x)\| \le Ce^{-\mu t}, \qquad \|\delta(t)\| + \|D_x\delta(t)\| \le Ce^{-\omega t}. \tag{10.11}$$

The constants are uniform on a smaller neighborhood. To detail the second estimate, the diagonal is invariant, being the fixed-point subspace of the permutation action [66], so the difference equation is linear in δ with coefficient $\mathcal{A} - \kappa_N I + O(\|x\|)$. Its differentiated equation has the same stable homogeneous part and an additional forcing bounded by $C\|\delta(t)\|\|D\Phi_t(x)\|$. Variation of constants bounds that forcing by $e^{-\omega t}$ times a convergent integral of $e^{-\mu t}$. Small losses in the two decay exponents absorb the coefficient perturbations. The first estimate follows in the same way from the strictly stable full linearization, after choosing the neighborhood small enough that its nonlinear derivative perturbation is less than the chosen decay margins. It also shows that every trajectory in U converges to the reference.

Permutation equivariance implies that the average derivative of V in a zero-sum direction vanishes at every diagonal point. Thus the exact average equation has the form

$$\dot{\bar{z}} = f(\bar{z}) + g(\bar{z}, \delta), \qquad g = O(\|\delta\|^2), \quad D_{\bar{z}}g = O(\|\delta\|^2), \quad D_\delta g = O(\|\delta\|). \tag{10.12}$$

It follows from eq. (10.11) that both $g(\Phi_t x)$ and its derivative with respect to x are bounded by $Ce^{-2\omega t}$.

By eq. (10.6) and the bounded second derivative of f, the complete flow obeys

$$\|D\phi_{-t}\| \le e^{Lt}, \qquad \|D^2\phi_{-t}\| \le Ce^{2Lt}, \qquad t \ge 0. \tag{10.13}$$

Differentiation in time gives the convergent integral

$$\Pi(x) = \bar{z}(0) + \int_0^\infty D\phi_{-t}\big|_{\bar{z}(t)} g(\bar{z}(t), \delta(t))dt. \tag{10.14}$$

The integrand is bounded by $Ce^{-(2\omega-L)t}$. Its derivative in x is the sum of a term with $D\phi_{-t}D_x g$ and one with $D^2\phi_{-t}[D_x\bar{z}, g]$. They are bounded by integrable multiples of $e^{-(2\omega-L)t}$ and $e^{-(2\omega-2L)t}$. Uniform convergence proves continuous differentiability.

On the diagonal the residual and its first derivative vanish, proving the retraction and derivative assertions. Shifting the limit in eq. (10.7) by a finite time proves the intertwining identity. The averaging derivative has the diagonal embedding as a bounded right inverse. The Banach implicit function theorem therefore gives the claimed local fibers, including continuously varying heterogeneous preparations. Finally, if two complete child states first became equal at a finite time, exchanging their labels would produce the same configuration at that time. Permutation equivariance and local uniqueness backward to the initial time would imply that they were initially equal, a contradiction. □

The limit does not depend on which average is used. Any smooth permutation-invariant retraction ρ of E^N onto the diagonal has the averaging map as its derivative at diagonal points, so $\rho(x) - \bar{z} = O(\|\delta\|^2)$, and $\|\phi_{-t}(\rho(\Phi_t x)) - \phi_{-t}(\overline{\Phi_t x})\| \le Ce^{-(2\omega-L)t} \to 0$. The same estimate gives a computable approximation. With $\Pi_{t_0}(x) = \phi_{-t_0}(\overline{\Phi_{t_0} x})$, eq. (10.14) gives

$$\|\Pi(x) - \Pi_{t_0}(x)\| \le \frac{C}{2\omega - L} e^{-(2\omega-L)t_0}, \tag{10.15}$$

so in a finite-dimensional subfamily the phase parent is evaluated, up to an exponentially small error, by one forward and one backward integration.

Corollary 10.2 (Uniqueness of the phase parent and extension to the basin). Let Π' be a map defined on U, Lipschitz on a neighborhood of the common reference, equal to the identity on the diagonal, and satisfying $\Pi' \circ \Phi_t = \phi_t \circ \Pi'$. Then $\Pi' = \Pi$. Moreover, for every configuration x whose trajectory enters U at some time $t_0 \ge 0$, the value $\widetilde{\Pi}(x) = \phi_{-t_0}(\Pi(\Phi_{t_0} x))$ does not depend on the choice of t_0, and $\widetilde{\Pi}$ is a continuously differentiable intertwining extension of Π to the basin of U; it is a submersion wherever $\Phi_{t_0} x$ lies in the neighborhood on which Π is.

Proof. Both maps are Lipschitz near the reference and equal the identity on the diagonal, so $\|\Pi(y) - \bar{y}\|$ and $\|\Pi'(y) - \bar{y}\|$ are bounded by $C\|\delta(y)\|$ there. Trajectories in U enter that neighborhood and stay in it. Hence $\|\phi_t(\Pi x) - \phi_t(\Pi' x)\| = \|\Pi(\Phi_t x) - \Pi'(\Phi_t x)\| \le Ce^{-\omega t}$ for large t, and eq. (10.13) gives $\|\Pi x - \Pi' x\| \le Ce^{-(\omega-L)t}$, which tends to zero by eq. (10.10). Independence of t_0 follows from the intertwining identity. Near x the extension is the composition $\phi_{-t_0} \circ \Pi \circ \Phi_{t_0}$ for one fixed t_0, because U is open and Φ_{t_0} is continuous; Φ_{t_0} and ϕ_{-t_0} are continuously differentiable with invertible derivatives, which proves the remaining assertions. □

The parent is a virtual collective state, generally different from either the arithmetic or the geometric pool [1, 63]. The restricted potential $ND_{\mathrm{KL}}(q^c \otimes s^c \| R)$ and metric Ng_{Fisher} give eq. (10.5). Fine states remain unchanged by the construction. They approach the common reference, so the theorem supplies neither persistent nonzero separation nor selected membership, and every fiber with its parent converges to one point.

The map Π is the asymptotic phase of the stable common reference relative to its invariant diagonal: it is the projection along the strong-stable foliation of the diagonal, whose existence is governed by the rate condition eq. (10.10) [55]. For periodic orbits this is the asymptotic phase of Guckenheimer [69], and for a stable equilibrium with a spectral gap its level sets are isostables [106]. The general theorems are stated for compact normally attracting invariant manifolds in finite dimensions [53] or for compact invariant manifolds of semiflows on Banach spaces [11]; the diagonal here is a noncompact subset of a Banach space, so the proof above is given directly in the chart rather than imported. The construction is an exact dynamical lumping [79]. What the reciprocal interaction contributes is the spectral gap: the difference rate exceeds the common rate by $\kappa_N = 2N/(N-1)$, which comes from the receiver and sender roles of the peer costs and does not depend on the temperatures at linear order, because row variations vanish at equality.

10.2 An ordinary response that the phase does not preserve

The failure of boundary sufficiency can be seen inside the preceding example. Take two binary agents, $q_i = s_i = \mathrm{Ber}_{\pm 1}(x_i)$, $0 < J < 1$, and set $x_1 = m + a$, $x_2 = m - a$. This is an invariant subfamily. The exact single-agent diagonal equation is

$$f_b(x) = -(1 - x^2)(\operatorname{atanh} x - Jx). \tag{10.16}$$

Both source rows have one entry. Including the incoming peer derivative gives

$$\dot{x}_i = -(1 - x_i^2)[2 \operatorname{atanh} x_i - \operatorname{atanh} x_j - Jx_i] - (x_i - x_j). \tag{10.17}$$

Expansion at the reference state yields

$$\begin{aligned} \dot{m} &= f_b(m) + (6 - 3J)ma^2 + O((|m| + |a|)^5), \\ \dot{a} &= -(5 - J)a + O((|m| + |a|)^3). \end{aligned} \tag{10.18}$$

Let $\bar{p}$ be the mean parameter of the phase law. By the remark after theorem 10.1, Π may be computed with the arithmetic mean m of the mean parameters, so eq. (10.14) reads $\bar{p} = m + \int_0^\infty D\phi^b_{-t}|_{m(t)} g_b(m(t), a(t))dt$, with ϕ^b the flow of f_b and $g_b = \dot{m} - f_b(m)$. Child exchange makes g_b even in a, simultaneous bit reversal makes it odd under $(m, a) \mapsto (-m, -a)$, and it vanishes at $a = 0$; hence $g_b = (6 - 3J)ma^2 + O(|m|a^2 r^2)$ with $r = |m| + |a|$. Along the flow, $m(t) = e^{-(1-J)t} m(1 + O(r^2))$, $a(t) = e^{-(5-J)t} a(1 + O(r^2))$, and $D\phi^b_{-t}|_{m(t)} = e^{(1-J)t}(1 + O(r^2))$, uniformly in $t \geq 0$, because the corrections to the linear rates are integrals of $O(m(s)^2 + a(s)^2)$ along exponentially decaying orbits. Integration gives

$$\bar{p} = m + \frac{3(2 - J)}{2(5 - J)} ma^2 + O(|m|a^2 r^2). \tag{10.19}$$

The coefficient is $(6-3J)/[2(5-J)]$, the time integral of $e^{-2(5-J)t}$ multiplied by the leading residual. Equivalently, substituting $\bar{p} = m + cma^2$ into the invariance equation $D\bar{p}(V) = f_b(\bar{p})$ gives $(6-3J) - 2(5-J)c = 0$. Along $m = a > 0$ sufficiently small, $\bar{p} \neq m$. The heterogeneous state and the diagonal state $(\bar{p}, \bar{p})$ have exactly the same phase parent, whereas choosing a child uniformly and sampling its public bit gives means m and $\bar{p}$, respectively. This response is not constant on the phase fibers. Replacing it by sampling the phase law would specify a new interface. By corollary 10.2, no Lipschitz intertwining parent that is a single-agent law and restricts to the identity on equal children can preserve this response.

The defect is of second order in the child differences. Let $\mathcal{R}$ be a permutation-invariant continuously differentiable response with $\mathcal{R}(z, \dots, z) = r(z)$, such as uniform child sampling. Permutation invariance gives $D\mathcal{R}_{(z,\dots,z)}(h) = Dr(z)\big(N^{-1}\sum_i h_i\big) = D(r \circ \Pi)_{(z,\dots,z)}(h)$, so $\mathcal{R} - r \circ \Pi$ vanishes together with its first derivative on the diagonal. Where both are twice differentiable it is $O(\|\delta\|^2)$, decaying like $e^{-2\omega t}$ along the flow while the parent signal decays like $e^{-(1-|J|)t}$; eq. (10.19) shows that the second order is attained.

10.3 What unrestricted Gibbs probes retain

Consider a block of exposed belief laws $q_j = f_j\lambda$ against a probability reference λ, with f_j bounded above and away from zero. Fix positive source weights w_j and a temperature $T > 0$. An identity-channel receiver with law $r\lambda$ sees the block normalizer

$$Z(r) = \sum_j w_j \exp[-D_{\mathrm{KL}}(r\lambda \| q_j)/T]. \tag{10.20}$$

The probability of selecting this block against a known outside source determines $Z(r)$, so the normalizer is an observable routing response without revealing a child label.

Proposition 10.3 (Unrestricted Gibbs response identifies the exposed child laws). If two finite source collections have equal eq. (10.20) for every positive receiver density in an L^∞ neighborhood of 1, then their weighted empirical measures of densities are equal:

$$\sum_j w_j \delta_{f_j} = \sum_k w'_k \delta_{f'_k}. \tag{10.21}$$

Repeated identical laws may be combined by adding their weights; distinct exposed laws cannot be forgotten by a retention preserving this entire response class.

Proof. Multiplication by the known receiver-entropy factor gives

$$e^{\int r \log r d\lambda/T} Z(r) = \sum_j w_j e^{\int r \log f_j d\lambda/T}. \tag{10.22}$$

List the distinct normalized densities in the union of the two collections. No two of their logarithms differ by a constant. There is therefore a bounded mean-zero function h for which the numbers $v_j = \int h \log f_j d\lambda$ are all different: each equality excludes one proper hyperplane in the finite-dimensional span of centered log-density differences, and finitely many such hyperplanes cannot

cover that span. Put $r = 1 + th$ for sufficiently small $|t|$. Subtracting the two expressions in eq. (10.22) gives $\sum_j c_j e^{tv_j/T} = 0$. Differentiation through orders zero to one less than the number of distinct laws gives an invertible Vandermonde system. Every c_j vanishes. Its positive factor $e^{\int \log f_j d\lambda/T}$ leaves exactly equality of the total weights on each density. □

The proof uses a single probe line $r = 1 + th$, with h outside a finite union of hyperplanes fixed by the children. Any receiver family that contains such a line therefore identifies the children, and a restricted family fails only on children whose centered log-density differences are annihilated by every probe direction. The Vandermonde nodes v_j/T approach one another as the children approach one another, so near the diagonal of theorem 10.1 the resolving power of the probe decays with the child differences.

In particular, a single ordinary law with one positive weight reproduces all these responses only if every exposed child law already equals it. The proposition concerns identity comparison channels and unrestricted receiver probes. It is not an impossibility result for arbitrary channels, restricted interfaces, or isolated dynamical factors such as theorem 10.1. For a fixed channel its scope is exactly the following.

Corollary 10.4 (Gibbs probes through a fixed channel). Let every comparison pass through one fixed Markov kernel $\mathcal{K}$, so that the compared laws are $\nu_j = q_j\mathcal{K}$, and suppose their densities against λ are bounded above and away from zero. Then the block normalizer $\sum_j w_j \exp[-D_{\mathrm{KL}}(r\lambda\|\nu_j)/T]$, for all positive receiver densities near 1, determines $\sum_j w_j \delta_{d\nu_j/d\lambda}$ and nothing more. Two collections with equal channelled laws give identical responses, so exact compression of this interface is possible precisely on the fibers of $q \mapsto q\mathcal{K}$ restricted to the exposed laws.

Proof. The normalizer depends on the collection only through the weighted channelled densities; apply proposition 10.3 to them. □

For a deterministic coarse-graining channel, the pushforward by a measurable map c, the incoming force also descends to the channelled laws. The incoming Fisher force of one Gibbs receiver with law r and weight β on source j is $\partial_t \log q_j = \beta(\mathcal{K}(r/\nu_j) - 1)$, and for $\mathcal{K}$ given by c this gives $\dot\nu_j = \beta(r - \nu_j)$, closed on the image. For a nondeterministic kernel two sources with equal channelled laws generally have different channelled velocities, so even the incoming force does not descend. The within-fiber conditional laws of a deterministic channel are hidden conditionals of the type treated in chapter 8; whether a deterministic comparison channel combined with that mechanism yields an exact parent for the full channelled Gibbs interface is not established here.

For a restricted affine receiver family $r_\theta = r_0 + \sum_a \theta_a h_a$, the identity-channel response instead identifies

$$\mu = \sum_j w_j e^{\iota_j/T}\delta_{v_j}, \qquad \iota_j = \int r_0 \log f_j d\lambda, \qquad (v_j)_a = \int h_a \log f_j d\lambda. \tag{10.23}$$

Equal slopes can be combined by adding the tilted weights. This possible boundary compression still requires a dynamical check. The incoming Fisher force from one Gibbs receiver is $\dot f_j = \alpha_j(r_\theta - f_j)$, and consequently

$$\dot\iota_j = \alpha_j\left(\int \frac{r_0 r_\theta}{f_j} d\lambda - 1\right), \qquad (\dot v_j)_a = \alpha_j \int \frac{h_a r_\theta}{f_j} d\lambda. \tag{10.24}$$

These reciprocal moments are additional information unless the admitted family supplies a closure identity. For an explicit witness, let λ be uniform on three points, take the extreme restricted family $\{r_0\}$ with $r_0 = 1$, and compare the children $(1/4, 1/4, 1/2)$ and $(c, c, 1-2c)$ with $c = (1+\sqrt{5})/8$. Their intercepts are equal, since $3c \cdot 3c \cdot 3(1-2c) = 27/32$, but with $\alpha_j = 1$ their intercept velocities are $1/9$ and $(5\sqrt{5}-10)/9$. The force $r_\theta - f_j$ is a mixture-geodesic term; it leaves every finite-dimensional exponential family in general, which is why these statements are made on full density charts.

10.4 The remaining recursive requirement

The phase construction retains one supplied block. A recursive use of any retention needs two further properties, which we state as a definition.

Definition 10.5 (Controlled projectability and descent of joining). Let a block I have fine state space X_I and a unique flow generated by a vector field $V_I(x_I; e)$ that depends on the state e of the agents outside I in a space $\mathsf{X}_{\mathrm{ext}}$. A retention $R_I : \mathsf{X}_I \to \mathsf{Y}_I$, together with an external retention $R_{\mathrm{ext}} : \mathsf{X}_{\mathrm{ext}} \to \mathsf{Y}_{\mathrm{ext}}$, is controlled projectable if there is a vector field $\bar{V}_I(y; \bar{e})$ on Y_I, parametrized by $\bar{e} \in \mathsf{Y}_{\mathrm{ext}}$, with

$$DR_I(x_I)V_I(x_I; e) = \bar{V}_I(R_I x_I; R_{\mathrm{ext}} e), \tag{10.25}$$

for all x_I and e, and if every declared boundary kernel of the block, including the memory of a reused child, factors through R_I. Within a family of blocks closed under joining, let $\mathcal{J}_{I,J} : \mathsf{X}_I \times \mathsf{X}_J \to \mathsf{X}_{I\cup J}$ identify the fine state of a union of disjoint blocks. Joining descends if there is a map $\bar{\mathcal{J}}_{I,J} : \mathsf{Y}_I \times \mathsf{Y}_J \to \mathsf{Y}_{I\cup J}$ with

$$R_{I\cup J} \circ \mathcal{J}_{I,J} = \bar{\mathcal{J}}_{I,J} \circ (R_I \times R_J). \tag{10.26}$$

Condition eq. (10.25) is the relatedness of control systems under a surjection, used in hierarchical control [117], with the external state as the control; for an empty exterior it is the intertwining identity of eq. (10.8). Condition eq. (10.26) says that the parent of a union is computed from the parents of its parts, which is what clause (F6) of definition 6.31 needs from a second stage that starts from first-stage parents. For the phase projection the block is the whole population, so only the intertwining case has been verified; for proper sub-blocks, where the external state enters V_I, neither condition has been checked, and the sampling response of section 10.2 already fails to factor through Π.

10.5 Status against the formation standard

The phase projection is a retention of a supplied block, the whole population, and the dynamics selects nothing, so clauses (F2), (F5), and (F6) of definition 6.31 are not addressed. The retention is non-injective, and an admitted child-level probe, uniform child sampling, distinguishes configurations in one fiber; but that probe is not inherited by the parent, so (F3) fails for it, and if the probe class is reduced to responses computed from the phase law, (F4) fails instead. The null-flow control of definition 6.30 shows where the reciprocal interaction enters. With every peer cost deleted, each agent follows the diagonal field eq. (10.5) independently, the linearized difference and common modes then have the same spectrum $\mathcal{A}$, and the rate condition eq. (10.10) fails, since the difference rate $1 - |J|$ is smaller than L. The spectral gap κ_N that makes the phase a compression of

differences is therefore produced by the peer costs, but no grouping is selected, and the results of this chapter are retention and identifiability statements rather than formation.

Chapter 11

Boundary Closure and Recursive Composition

This chapter concerns the reciprocal law potential of definition 4.2 and its Fisher flow, with identity peer channels, all-source uniform priors, and a coupled evaluator. It supplies a binary hierarchy together with an invariant family of agent laws, and proves that the family carries an exact hierarchy of public compressions. Each level preserves fresh uniform-descendant responses, their subsequent evolution, and a declared class of record assimilations. The chapter makes no selection claim. Membership and the hierarchy are supplied, so clauses (F2) and (F5) of definition 6.31 are not addressed, and the null flow of definition 6.30 has no selected membership history to reproduce. The mechanism is an equivariant symmetry reduction (section 11.3). The public partition data that the hierarchy discards are invisible to the fine energy, the Fisher metric, and the admitted records, and near the partition-independent laws their contrast decays exponentially under the same flow (proposition 11.12).

11.1 A general-law family on a supplied tree

Let A and B be probability references on standard Borel spaces with coordinates r and w. Choose a positive joint density $\rho(r, w)$ against $A \otimes B$, normalized to one, with bounded logarithm. Its disintegration supplies a proper model reference and normalized evaluator as in eq. (4.1). It may couple its arguments; for example, balanced signs $\phi(r), \psi(w)$ give $\rho = \exp(J\phi\psi)/\cosh J$.

Fix a binary tree of depth $d \geq 1$ with $N = 2^d$ leaves. Each leaf is a primitive agent. For every internal tree node v, append to the belief sample space an independent public coordinate θ_v with law λ_v, and append a separate set of such coordinates to the model sample space. The references extend by these product laws. Each node carries a measurable sign $\sigma_v(r, \theta_v) \in \{-1, 1\}$. The exact statements of sections 11.2 to 11.4 use only one property of these signs: conditional on the base, they are independent and fair. Two realizations are used. In the coin realization, λ_v is uniform on $\{-1, 1\}$ and $\sigma_v(r, \theta_v) = \theta_v s_v(r)$ with a measurable orientation $s_v(r) \in \{-1, 1\}$. In the circle realization, λ_v is Haar measure on a circle, and a measurable phase function $\alpha_v(r)$ defines

$$\sigma_v(r, \theta_v) = \operatorname{sign} \cos(\theta_v - \alpha_v(r)). \tag{11.1}$$

The sign on its null zero set is immaterial. The atomless circle realization is needed only where a continuum of balanced partitions is used, namely for the continuous phase families and the total-variation and Hellinger statements of section 11.5. In both realizations the coordinates θ_v are original public sample coordinates. Different phase functions or orientations change the laws in those same coordinates; they are held fixed along a trajectory.

Label the left and right edges by $\epsilon = +1, -1$. A leaf i defines a signed path

$$u_j(i,\theta) = \epsilon_j(i)\sigma_{v_{j-1}(i)}(r, \theta_{v_{j-1}(i)}), \qquad j = 1, \dots, d. \tag{11.2}$$

Let U_d be the uniform law on $\{-1,1\}^d$. Choose any positive bounded profile $f(r,u)$, with bounded logarithm and $\int f dA dU_d = 1$, and set

$$q_i(r,\theta) = f(r, u(i,\theta)), \qquad f_0(r) = \mathbb{E}_{U_d} f(r,u). \tag{11.3}$$

These are densities against A times all the node laws. Each leaf has the same base marginal $f_0 A$, but its full public law can differ from every other leaf law. Define model densities from an independent profile $g(w,z)$ and model coordinates in the same way. The base spaces and the profile functions are unrestricted within the stated regularity domain; no Gaussian or finite-parameter family is imposed.

All physical peer channels are identities, every other agent is an allowed source, and the original source priors are uniform. If two leaves share exactly k initial edges, there are $n_k = 2^{d-k-1}$ possible sources in that class for each receiver. Give each such source weight b_k/n_k, where b is a probability vector on $k = 0, \dots, d-1$. The belief and model vectors b^q, b^s evolve separately. Uniform original rows are included by taking

$$\pi_k = \frac{n_k}{N-1}, \qquad b^q_k = b^s_k = \pi_k. \tag{11.4}$$

The fine system is therefore the reciprocal potential eq. (4.4) with identity channels, the evaluator determined by ρ, and uniform all-source priors. The tree enters only the prepared relation among the agent laws and row classes. The fine potential and its uniform priors contain no additional preference for a chosen pair, and they are invariant under every permutation of the agents.

11.2 The exact inherited profile equations

For each k, define an operator P_k on bit profiles: keep the first k bits, reverse bit $k+1$, and average all later bits. It is a Markov operator, self-adjoint for U_d. It describes an average over a class of original source labels; it is not a replacement physical communication channel. Define

$$D_k(f) = \int f(r,u)[\log f(r,u) - (P_k \log f)(r,u)] dA(r) dU_d(u). \tag{11.5}$$

The Fisher flow of theorem 4.6 is proved on finite tables. The fine system here lives on general sample spaces, so we first state its chart and its well-posedness.

Lemma 11.1 (Fine identity-channel flow in bounded log-density charts). Let (X, λ) and (Y, μ) be standard Borel probability spaces whose coordinates include the base coordinates r and w, with base marginals A and B. Consider finitely many agents with normalized belief densities $q_i = e^{\ell_i}$ against λ and model densities $s_i = e^{m_i}$ against μ, where $\ell_i \in L^\infty(\lambda)$ and $m_i \in L^\infty(\mu)$. Suppose the peer channels are identities on a fixed directed edge set with nonempty source sets and positive priors, $T_q, T_s > 0$, and each local term is $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i)$, where R_i has a density $\rho_i(r,w)$ against $\lambda \otimes \mu$ that depends only on the base coordinates and has bounded logarithm. Up to additive

constants, the first variation of eq. (4.4) with respect to q_i is

$$\begin{aligned} h_i^q &= \log q_i + V_i^q + \sum_{j\in\mathcal{N}_i^q} \beta_{ij}(\log q_i - \log q_j) - \sum_{l:i\in\mathcal{N}_l^q} \beta_{li}\frac{q_l}{q_i}, \\ V_i^q(r) &= -\int s_{i,0}(w)\log\rho_i(r,w)dB(w), \end{aligned} \tag{11.6}$$

where $s_{i,0}$ is the base marginal density of s_i; the model variation h_i^s follows by exchanging the sectors. Then the system

$$\dot{\ell}_i = -(h_i^q - \mathbb{E}_{q_i} h_i^q), \qquad \dot{m}_i = -(h_i^s - \mathbb{E}_{s_i} h_i^s), \tag{11.7}$$

together with the row equations of eq. (4.15), has for every initial condition with interior rows a unique maximal solution in $L^\infty(\lambda)^N \times L^\infty(\mu)^N$ times the row coordinates, with interior rows. Along it the laws stay normalized, $\dot{q}_i = -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q)$, the right side is the Fisher gradient of $\mathcal{E}$ for the metric $\int(\delta q)^2/q$ in each block, and $\dot{\mathcal{E}} = -\sum \mathrm{Var} \le 0$ as in eq. (4.16).

Proof. For identity channels, eq. (4.10) gives the receiver variation $\log q_i - \log q_j + 1$ and the source variation $-q_i/q_j$ of $D_{\mathrm{KL}}(q_i\|q_j)$. The local term has variation $\log q_i + 1 + V_i^q$, because ρ_i reads only the base coordinates. This gives eq. (11.6). Define the field off the normalized set through $q_i = e^{\ell_i}/\int e^{\ell_i}d\lambda$. On a ball of radius L in the sup norm, every density lies between e^{-2L} and e^{2L}. The maps $\ell \mapsto \log q_i$, $\ell \mapsto q_l/q_i$, the base marginals $s_{i,0}$ as maps into $L^1(B)$, the linear map $s_{i,0} \mapsto V_i^q$ from $L^1(B)$ into $L^\infty(A)$ with norm at most $\|\log\rho_i\|_\infty$, the costs D_{ij}, and the expectations $\mathbb{E}_{q_i} h_i^q$ are therefore locally Lipschitz, and the row field is smooth on the interior of the simplices. Finite sums and products of bounded locally Lipschitz maps are locally Lipschitz, so the Picard–Lindelöf theorem in the Banach space gives a unique maximal solution. Since $\mathbb{E}_{q_i}\dot{\ell}_i = 0$, the integral of e^{ℓ_i} is constant, so the normalized set is invariant and $\dot{q}_i = q_i\dot{\ell}_i$. For a tangent vector v with $\int v d\lambda = 0$, the metric pairing of $q_i(h_i^q - \mathbb{E}_{q_i} h_i^q)$ with v is $\int h_i^q v d\lambda$, which identifies the Fisher gradient exactly as in the proof of theorem 4.6. The chain rule in the chart gives the dissipation identity. □

Theorem 11.2 (Exact coupled hierarchy dynamics). Consider the reciprocal potential eq. (4.4) with the data of section 11.1, in either realization of the signs, and its Fisher flow in the chart of lemma 11.1, including source-row evolution. The family eq. (11.3), with its model analogue and class-constant rows, is invariant under this flow. Its inherited profile energy and metric are the original restrictions divided by their common factor N. The complete profile equations retain the incoming source derivatives, and their solutions exist for all $t \ge 0$.

Proof. For any two leaves in class k, the first k path bits agree, bit $k+1$ is opposite, and later bits lie on disjoint branches. Integrating their independent signs gives exactly $D_{\mathrm{KL}}(q_i\|q_j) = D_k(f)$. Every leaf path has uniform bit law. Each local product law therefore has the same base marginals f_0, g_0 in its evaluator term. Direct substitution gives $\mathcal{E} = NF$, with

$$\begin{aligned} F = &\int f\log f dA dU_d + \int g\log g dB dU_d - \int f_0(r)g_0(w)\log\rho(r,w)dA dB \\ &+ \sum_{k=0}^{d-1} b_k^q D_k(f) + T_q D_{\mathrm{KL}}(b^q\|\pi) + \sum_{k=0}^{d-1} b_k^s D_k(g) + T_s D_{\mathrm{KL}}(b^s\|\pi). \end{aligned} \tag{11.8}$$

The metric restricts to N times

$$\mathcal{G}^c = \int \frac{(\delta f)^2}{f} dA dU_d + \int \frac{(\delta g)^2}{g} dB dU_d + \sum_{a=q,s} \sum_k \frac{(\delta b_k^a)^2}{b_k^a}. \tag{11.9}$$

Thus no new mobility is introduced.

Self-adjointness gives the exact derivative

$$\delta D_k(f) = \int \delta f \left[\log f + 1 - P_k \log f - \frac{P_k f}{f} \right] dA dU_d. \tag{11.10}$$

The last ratio is the incoming source contribution. Define

$$V^q(r) = -\int g_0(w) \log \rho(r,w) dB(w), \qquad V^s(w) = -\int f_0(r) \log \rho(r,w) dA(r). \tag{11.11}$$

Up to an irrelevant additive constant, the belief force and evolution are

$$\begin{aligned} h^q &= 2 \log f + V^q - \sum_k b_k^q \left[P_k \log f + \frac{P_k f}{f} \right], \\ c_q &= \int f h^q dA dU_d, \qquad \dot{f} = -f(h^q - c_q). \end{aligned} \tag{11.12}$$

Exchange the sectors for $\dot{g}$. With $D_k^q = D_k(f)$ and $D_k^s = D_k(g)$, the rows obey

$$\dot{b}_k^a = -b_k^a \left[D_k^a + T_a \log \frac{b_k^a}{\pi_k} - \sum_l b_l^a \left(D_l^a + T_a \log \frac{b_l^a}{\pi_l} \right) \right]. \tag{11.13}$$

To establish full-flow invariance, fix a receiver and a public sample point. The leaves in its class-k source subtree run through every suffix bit string once, regardless of the physical signs at that point. Their averages of $\log q_j$ and q_j are therefore $P_k \log f$ and $P_k f$. Incoming weights have the same class values. The fine force eq. (11.6) at that leaf is precisely eq. (11.12) evaluated at its path bits. All leaf normalizations agree because every path-bit vector is uniform under the reference. Source costs also agree within their classes, preserving the row form. Hence the full field is tangent to the profile family, and the restricted gradient is its actual evolution. The profile equations are locally Lipschitz in the bounded-profile chart by the argument of lemma 11.1, so they have a unique maximal solution. Its lift solves the fine equation by the preceding computation, and by the uniqueness in lemma 11.1 it is the fine solution. Global existence is lemma 11.3. □

In the coin realization with a finite base, the fine laws are positive tables on the base times $\{-1,1\}^{N-1}$, and theorem 4.6 applies to the fine system directly; the theorem is then a statement about finite tables that can be checked by exact computation.

Lemma 11.3 (Global bounded-log existence of the profile flow). Let $F(0)$ be the initial profile energy eq. (11.8) and $K = F(0) + 3\| \log \rho\|_\infty + 2$. Along the solution of theorem 11.2,

$$\min\left(\operatorname{ess\,inf} \log f(0), -K\right) \le \log f(t) \le \max\left(\operatorname{ess\,sup} \log f(0), K\right), \tag{11.14}$$

with the same bound for $\log g$, and $\log(b_k^a(t)/\pi_k) \geq \min(\log(b_k^a(0)/\pi_k), -C_a/T_a)$, where C_a is twice the bound eq. (11.14) on the absolute value of the corresponding log profile. The solution therefore exists for all $t \geq 0$.

Proof. Since $\sum_k b_k^q = 1$ and P_k preserves U_d, the normalizer in eq. (11.12) is $c_q = \int f \log f dA dU_d + \sum_k b_k^q D_k(f) + \int f_0 V^q dA - 1$. Every term of F except the evaluator term is nonnegative, the evaluator term is at least $-\|\log \rho\|_\infty$, and F decreases along the flow. Hence the first two terms of c_q lie between 0 and $F(0) + \|\log \rho\|_\infty$, and $|V^q| \leq \|\log \rho\|_\infty$ gives $|c_q| \leq F(0) + 2\|\log \rho\|_\infty + 1$. Given the continuous scalar inputs $c_q(t)$, $b^q(t)$ and the base function $V^q(t, \cdot)$, the profile equation is, for each base point, a finite system of ordinary differential equations for $l = \log f(r, \cdot) \in \mathbb{R}^{2^d}$:

$$\dot{l} = -2l - V^q + \sum_k b_k^q \left[P_k l + \frac{P_k e^l}{e^l} \right] + c_q.$$

At a bit string where l attains its maximum M_r, one has $P_k l \leq M_r$ and $P_k e^l \leq e^{M_r}$, so the upper Dini derivative of M_r is at most $-M_r + K$. At a minimizing bit string, $P_k l \geq m_r$ and $P_k e^l \geq e^{m_r}$, so the lower Dini derivative of the minimum m_r is at least $-m_r - K$. Comparison gives eq. (11.14) uniformly in r. For the rows, $y_k = \log(b_k/\pi_k)$ satisfies $\dot{y}_k = -D_k - T y_k + \sum_l b_l (D_l + T y_l)$, and the last sum is nonnegative because $D_l \geq 0$ and $\sum_l b_l y_l = D_{\mathrm{KL}}(b\|\pi) \geq 0$. Since $D_k \leq \sup l - \inf l \leq C$, we obtain $\dot{y}_k \geq -C - T y_k$. The model sector is identical. The solution thus stays in a bounded set of the chart on which the field is Lipschitz, with rows uniformly interior, and it extends to $[0, \infty)$. □

The same retentions apply to the original Gibbs-reduced flow. Replace each row by $b_k^{a,*} = \pi_k \exp(-D_k^a/T_a) / \sum_l \pi_l \exp(-D_l^a/T_a)$ and use the matter metric. The envelope identity in eq. (4.8) gives the displayed profile force with these weights, including its incoming term. All symmetry and response arguments below remain unchanged. The Gibbs-reduced flow and the flow with evolving rows are different flows. At frozen costs the row log-odds in eq. (11.13) relax at rate T_a, so their transients differ most when T_a is small; the retention statements hold for each of them.

The inherited cost is still a relative entropy. Let

$$J_k(dr, du, dv) = A(dr) f(r, u) U_d(du) P_k(u, dv), \tag{11.15}$$

and let J_k^{rev} exchange u, v. Reversibility gives $D_k(f) = D_{\mathrm{KL}}(J_k \| J_k^{\mathrm{rev}})$. Since J_k and J_k^{rev} are linear in f, joint convexity of relative entropy makes each D_k convex in f. The class cost differs from the divergence to the averaged source law by a Jensen term:

$$D_k(f) - D_{\mathrm{KL}}(fA \otimes U_d \| (P_k f) A \otimes U_d) = \int f[\log(P_k f) - P_k \log f] dA dU_d \geq 0. \tag{11.16}$$

Because f is positive and $P_k(u, \cdot)$ is uniform on its support, the right side vanishes exactly when $f(r, \cdot)$ does not depend on bits $k+2, \ldots, d$ for A-almost every r. It vanishes identically for $k = d-1$, where P_{d-1} is the deterministic reversal of the last bit, and is positive for $k < d-1$ whenever f depends on a later bit. Replacing the exact inherited cost by the averaged-source divergence would discard this term and change the dynamics.

11.3 The hierarchy as an equivariant symmetry reduction

The inheritance statements below are instances of classical facts about equivariant gradient flows. Let $\mathcal{S}$ be the group of pairs $\omega = (\Phi, \pi)$ in which $\Phi = (\Phi^q, \Phi^s)$ consists of fibered bimeasurable maps of the belief and model sample spaces, $\Phi^q(r, \theta) = (r, \varphi_r(\theta))$ and similarly for Φ^s, that preserve the references, and π is a permutation of the agents that preserves the edge set, the priors, and the evaluators. The group acts on configurations by $(\omega \cdot q)_i = q_{\pi(i)} \circ \Phi^q$, $(\omega \cdot s)_i = s_{\pi(i)} \circ \Phi^s$, and $(\omega \cdot \beta)_{ij} = \beta_{\pi(i)\pi(j)}$, with the same rule for model rows.

Lemma 11.4 (Equivariant reduction). In the setting of lemma 11.1, every $\omega \in \mathcal{S}$ acts on the chart by a linear isometry of the sup norm, preserves $\mathcal{E}$ and the Fisher metric, and commutes with the flow: if $x(t)$ solves the flow from x_0, then $\omega \cdot x(t)$ solves it from $\omega \cdot x_0$. Consequently the following hold. (i) For every subgroup $H \leq \mathcal{S}$, the fixed-point set $\mathrm{Fix}(H)$ is invariant under the flow, and on it the flow is the Fisher gradient flow of the restriction of $\mathcal{E}$ for the restricted metric. (ii) Every $\omega \in \mathcal{S}$ maps $\mathrm{Fix}(H)$ onto $\mathrm{Fix}(\omega H \omega^{-1})$ and conjugates the restricted flows. A subgroup $K \leq \mathcal{S}$ that maps $\mathrm{Fix}(H)$ into itself, for example a subgroup of the normalizer of H, acts on $\mathrm{Fix}(H)$ commuting with the flow, and the flow descends to the orbit space $\mathrm{Fix}(H)/K$. (iii) A response kernel Q from configurations to laws on an output space with $Q(\omega \cdot x) = Q(x)$ for $\omega \in K$ descends to $\mathrm{Fix}(H)/K$. If an assimilation map A_o of definition 4.24 satisfies $A_o(\omega \cdot x) = \omega \cdot A_o(x)$ for $\omega \in K$, and the law of its record is K-invariant, then every finite adaptive history law of the record-interrupted dynamics descends as well.

Proof. A reference-preserving bimeasurable Φ^q preserves relative entropy, so $D_{\mathrm{KL}}(q_{\pi(i)} \circ \Phi^q \| q_{\pi(j)} \circ \Phi^q) = D_{\mathrm{KL}}(q_{\pi(i)} \| q_{\pi(j)})$, and π preserves the edge set and priors, so the reindexed peer sums agree. The entropy part of each local term is preserved, and because Φ^q is fibered and reference-preserving, $q \circ \Phi^q$ has the same base marginal as q. The evaluator term reads only base marginals, and π preserves the evaluators, so $\mathcal{E}(\omega \cdot x) = \mathcal{E}(x)$. The metric is preserved since $\int (\delta q \circ \Phi^q)^2/(q \circ \Phi^q) = \int (\delta q)^2/q$. On log densities the action is $\ell_i \mapsto \ell_{\pi(i)} \circ \Phi^q$, a linear isometry of L^∞. Direct substitution in eq. (11.6), including the incoming term, gives $h^q_i(\omega \cdot x) = h^q_{\pi(i)}(x) \circ \Phi^q$, so the field is equivariant, and uniqueness in lemma 11.1 gives the commutation. For (i), if $x_0 \in \mathrm{Fix}(H)$ and $\eta \in H$, then $\eta \cdot x(t)$ is a solution from x_0 and equals $x(t)$. At a point of $\mathrm{Fix}(H)$ the field is H-fixed, hence tangent to $\mathrm{Fix}(H)$, and its metric pairing with every tangent vector v of $\mathrm{Fix}(H)$ equals the derivative of the restricted energy along v; it is therefore the restricted gradient. Statement (ii) follows from $\omega \cdot \mathrm{Fix}(H) = \mathrm{Fix}(\omega H \omega^{-1})$ and the commutation: if $y_0 = \omega \cdot x_0$ with $\omega \in K$, then $y(t) = \omega \cdot x(t)$, so orbits are carried to orbits. For (iii), $Q(y(t)) = Q(x(t))$, and an equivariant assimilation keeps related states related while assigning the same law to the record. Induction over the record times, with the flow between them, gives the history statement. □

Statement (i) is the flow form of the principle of symmetric criticality [116], and the invariance of fixed-point subspaces of equivariant vector fields is standard in equivariant bifurcation theory and dynamics [56, 67]. The lemma adds nothing beyond these classical facts; it fixes the form in which they are used here. The following proposition identifies the profile family as a fixed-point set.

Proposition 11.5 (The profile family is a fixed-point set). Fix a realization of the signs. Let $H_\alpha \leq \mathcal{S}$ consist of the pairs (Φ, π) such that π is an automorphism of the rooted binary tree acting on the leaves, and $u(\pi(i), \Phi^q(\xi)) = u(i, \xi)$ for every leaf i and almost every belief sample point ξ,

with the analogous condition on Φ^s. Then $\mathrm{Fix}(H_\alpha)$ consists exactly of the configurations of the form eq. (11.3) in both sectors, with class-constant rows $\beta_{ij} = b^q_k/n_k$ and $\gamma_{ij} = b^s_k/n_k$ for pairs in class k.

Proof. If $q_i = f(r, u(i,\xi))$ and $(\Phi,\pi) \in H_\alpha$, then $q_{\pi(i)}(\Phi^q\xi) = f(r, u(\pi(i), \Phi^q\xi)) = f(r, u(i,\xi)) = q_i(\xi)$, and class-constant rows are preserved because tree automorphisms preserve the common-prefix length. Conversely, let a belief configuration be fixed by H_α. Elements (Φ, id) whose Φ^q preserves every sign belong to H_α. In the coin realization the signs determine the node coordinates given r. In the circle realization, choose Φ^q to act on each cell $\{\sigma = \mathrm{s}\}$, in arc coordinates measured from $\alpha_v(r)$, by rotations of the arcs through rationally independent irrational fractions of their length. This map is ergodic on each cell for the normalized Haar product [141], so by Fubini's theorem each invariant q_i is almost everywhere a function $F_i(r,\sigma)$ of the base and the node signs.

For an internal node w, let π_w exchange the two subtrees below w by their mirror map, and let Φ^q_w reverse σ_w (a half-period shift of θ_w, or a coin flip) and exchange the coordinates of mirror nodes in the two subtrees, adjusted by their phases or orientations so that mirror signs are carried to mirror signs. Following the edge labels along each path shows that $(\Phi_w, \pi_w) \in H_\alpha$. For a leaf i not below w, $\pi_w(i) = i$, so F_i is invariant under reversing σ_w and permuting the signs strictly below w. Induction from the lowest internal nodes upward shows that F_i does not depend on the sign of any node that is not an ancestor of i. The ancestral signs of i determine $u(i,\xi)$ bijectively, so $q_i = f_i(r, u(i,\xi))$. For $(\Phi,\pi) \in H_\alpha$, fixedness gives $f_{\pi(i)}(r, u(i,\xi)) = f_i(r, u(i,\xi))$, and $u(i,\cdot)$ is uniform given r, so $f_{\pi(i)} = f_i$. The node elements generate a group that acts transitively on the leaves and on the pairs of leaves with a given common-prefix length. Hence $f_i = f$ for all i, and the rows are class constant. The model sector is identical. □

Two further subgroups give the quotient. For a measurable map $r \mapsto \tau(r)$ into the automorphisms of bit strings, let Φ_τ shift θ_v by a half period exactly when $\tau(r)$ reverses the bit that follows the original signed prefix of v. This map is fibered, bimeasurable, and reference-preserving, since each shift depends only on r and on ancestor coordinates, and $u(i, \Phi_\tau\xi) = \tau(r)u(i,\xi)$ for every leaf. Hence (Φ_τ, id) maps the family with profile f to the family with profile $f(r, \tau(r)u)$. Phase rotations $\theta_v \mapsto \theta_v + \alpha'_v(r) - \alpha_v(r)$ map $\mathrm{Fix}(H_\alpha)$ onto $\mathrm{Fix}(H_{\alpha'})$ with the profile unchanged, and orientation changes act in the same way in the coin realization. By lemma 11.4, the flow descends to the quotient by these maps, which is the unordered profile tree of section 11.4.

The data discarded by this quotient, namely the phases or orientations and the base-dependent ordering of branches, are orbit data of a group of exact symmetries of the fine energy and the Fisher metric. Relative to the observable algebra generated by the fine energy and metric, the fresh uniform-descendant responses, and the admitted base-only broadcast records, they are gauge: no agent, channel, record, or energy term of this construction reads them. They are resolved only by the protocols excluded in section 11.4, namely reused children, shared labels, and child-specific or unrestricted identity-channel probes. Symmetry does not supply the explicit class costs D_k or their pair-law form, the incoming term of eq. (11.12), the identification eq. (11.18) of a fresh response with a subtree mean, the contrast decay of proposition 11.12, or the observability obstruction of section 11.6; these are computed directly.

The root-level statement holds for every identity-channel configuration, not only on the profile family.

Proposition 11.6 (Descent of the joint conditional spectrum). In the setting of lemma 11.1, let μ^q be the law of $(r, q_1(\xi), \dots, q_N(\xi))$ for $\xi \sim \lambda$, and μ^s the law of $(w, s_1, \dots, s_N)$ under μ. For any edge set, priors, and initial rows, two configurations with equal μ^q, μ^s, and rows have solutions with equal $\mu^q(t)$, $\mu^s(t)$, and rows throughout their common existence interval. Response kernels that read only base coordinates of a fresh sample from a fixed agent, and assimilations by a base-only likelihood applied to every agent, descend to these spectra.

Proof. By eq. (11.6), the force h_i^q at ξ is a function of r, of the value vector $(q_1(\xi), \dots, q_N(\xi))$, and of quantities determined by the spectra and rows: V_i^q reads the base marginals $s_{i,0}(w) = \mathbb{E}[s_i \mid w]$, and the expectations $\mathbb{E}_{q_i} h_i^q$ and the costs $D_{ij} = \int q_i \log(q_i/q_j) d\lambda$ are integrals against μ^q. Fix the solution $x(t)$ from the first configuration, and let $\Psi_t(r, \cdot)$ be the flow of the finite-dimensional equations $\dot{v}_i = -v_i(h_i^q(r, v; t) - c_i(t))$ with the scalar inputs of $x(t)$, and similarly in the model sector. Applying Ψ_t pointwise to the second configuration gives a path whose spectra are the push-forwards of the common initial spectra, hence equal to those of $x(t)$. Its scalar inputs therefore coincide with those of $x(t)$, so it solves the fine flow, and by uniqueness it is the solution from the second configuration. The rows read only the costs, which agree. A base-only response or likelihood reads only (r, μ^q) and multiplies the value vector at r by a common factor, which acts on the spectra. □

On the profile family, the conditional law of the value vector given r is uniform on the orbit of the labeled vector $(f(r, u(i)))_i$ under the tree automorphisms, so μ^q and the unordered profile tree determine each other. The root case of theorem 11.9 is therefore an instance of proposition 11.6. What the hierarchy adds is that responses at intermediate nodes, which read physical subtree membership, also descend; there the relevant descriptors are the conditional spectra of subtrees given ancestor coordinates. This is the identity-channel analogue of the conditional spectrum of chapter 8.

Remark 11.7 (Compression requires the symmetry). Suppose the leaves of the two root branches carry different profiles f^L, f^R, a configuration outside $\mathrm{Fix}(H_\alpha)$. With $f_1^X(r, u_1) = \mathbb{E} f^X(r, u_1, \cdot)$, the fresh root response density is

$$\tfrac{1}{2}\left[f_1^L(r, \sigma_0) + f_1^R(r, -\sigma_0)\right], \tag{11.17}$$

which depends on the root sign, and hence on the root phase, unless $f_1^L(r, s) + f_1^R(r, -s)$ does not depend on s. Discarding the join phase under the fresh interface therefore needs this conjugacy condition, which holds automatically on the symmetric family. Whether such two-block families are invariant under the fine flow, and when the flow preserves the conjugacy condition, is not established here.

Remark 11.8 (Other symmetric hierarchies). The proofs use only a group of agent permutations preserving the edges, priors, and evaluator that acts transitively on each set of sibling subtrees, together with public coordinates that realize at each node a uniformly random arrangement of its children, fair given the base and ancestor coordinates. Rooted trees with larger branching, realized by uniform random child permutations, class-constant priors with an edge set that is a union of common-prefix classes, and phases that depend measurably on ancestor coordinates fall under lemma 11.4 in the same way. The operator P_k is then replaced by the average that keeps the first k

digits, replaces digit $k+1$ by a uniformly chosen different digit, and averages the rest. We do not write out these cases.

11.4 Recursive retention and the response process

At each base point, arrange the values of $f(r,u)$ in their binary profile tree. Retain this tree with unordered children at every internal node. Its leaves still carry density values, and its nested grouping is retained. Flattening it into a multiset would lose which values share a prefix and could change D_k.

For a physical subtree rooted at depth l, apply this retention only to its descendant values, keeping the dependence on the base and physical ancestor coordinates. This defines its parent descriptor. Include the corresponding model descriptor, the shared row-class data, primitive counts, and the remaining tree connections. At a lowest pair, the unordered values p_+, p_- can equivalently be written as a mean $(p_+ + p_-)/2$ and a squared contrast $((p_+ - p_-)/(p_+ + p_-))^2$. A first parent therefore retains more than its mixture law, but does not retain the two full child laws on their original node coordinates.

Theorem 11.9 (Recursive boundary inheritance on the supplied hierarchy). Under the hypotheses of theorem 11.2, these parent descriptors inherit the joint controlled dynamics at every level of the same fine system. Joining two sibling descriptors gives their unordered pair. The join commutes with evolution and with fresh uniform-descendant response kernels. Record likelihoods that depend only on the base and are applied to every agent, used as the assimilation events of definition 4.24, preserve these statements under arbitrary finite adaptive sequences, whenever the updates remain in the regular domain.

Proof. By proposition 11.5, the family is $\mathrm{Fix}(H_\alpha)$, and the maps (Φ_τ, id) and the phase rotations of section 11.3 act on it as stated there. At a fixed physical point, changing an internal sign exchanges the two branches below that node. Thus the map from physical leaf labels to their bit strings is an automorphism of the rooted binary tree. Forgetting its realization removes the phase at the joined node and leaves exactly the unordered profile subtree. As a direct check of lemma 11.4(ii), tree automorphisms preserve common-prefix length, U_d, and every P_k. They consequently preserve eqs. (11.8) and (11.9) and commute with eqs. (11.12) and (11.13), also when they depend measurably on the base, since the equations have no derivatives in the base coordinate, the local evaluator reads only f_0, g_0, and the global integrals are unchanged. The quotient flow is therefore independent of the chosen ordered representatives.

The quotient admits Borel representatives. Order $\mathbb{R}^{2^m}$ lexicographically. At each internal node, from the lowest level upward, place first the child subtree whose current vector of values is lexicographically smaller. Each step is a finite composition of Borel comparisons and coordinate permutations, and applying it at each base point to the measurable map $r \mapsto f(r,\cdot)$ gives a measurable canonical profile that depends only on the unordered tree. This also establishes measurability of the descended updates.

For partial retentions, the same argument applies to the product of automorphism groups of the retained subtrees. The complement supplies the retained sibling and ancestor data needed for their interactions; the joint parent network closes. Thus this is controlled closure for complements in the

stated prepared family, rather than autonomy under every external law. Combining sibling trees and then forgetting their ordering is the same operation as retaining their union directly. Repeated compatible joins give the same descriptor, and joins of disjoint subtrees commute. These are representation changes inside one fixed N-agent interaction graph. They do not add or delete fine edges, and do not authorize an arbitrary reassociation of the leaves into another hierarchy.

For a response at a node of depth l, choose a descendant freshly and uniformly, draw its full original public sample, and conceal its label. At every physical sample point the descendant labels run through every suffix bit string. The response density is exactly

$$f_l(r, u_1, \dots, u_l) = \mathbb{E}_{u_{l+1},\dots,u_d} f(r, u_1, \dots, u_d). \tag{11.18}$$

It is the mean of the retained subtree values, with all unused node factors left in the original output space, and it is invariant under the automorphisms acting below the node, as lemma 11.4(iii) requires. At the root it is $f_0(r)$. Choosing a sibling parent uniformly and then a descendant uniformly gives exactly the same full sample law. The model sector follows the same rule with a separate fresh label.

Fix a normalized positive record kernel $L_e^q(do \mid r)$ with bounded positive density on the admitted update domain, applied as a readout to the original samples of every agent. Every child has base marginal $f_0 A$, so a common record has the same component normalization

$$Z_o = \int f_0(r) L_e^q(o \mid r) dA(r). \tag{11.19}$$

Conditioning every mixture component preserves uniform labels and gives

$$f^+(r, u) = \frac{L_e^q(o \mid r) f(r, u)}{Z_o}. \tag{11.20}$$

This multiplies every value in a profile tree at the given base point by the same factor. It commutes with every element of $\mathcal{S}$ whose maps are fibered, and in particular with all retentions and joins, and its record law depends only on f_0. The model update is identical in form, and rows remain unchanged at the record update. Equality of retained states gives equality of the next response law and the retained update. Induction, using the inherited continuous flow between records, proves equality of every finite history law under policies choosing the next admitted experiment from the observed history. At a representation join, the already observed history is carried forward as controller state; no past record is erased. □

The records enter as assimilation events in the sense of definition 4.24. The potential is unchanged by them, and after an update the force eq. (11.12) depends on the record only through the updated profile. The admitted records are those whose likelihoods are invariant under the compression group and that are applied to every agent, which here means base-only and broadcast likelihoods; every admitted record history is a functional of the root trajectory $f_0(t)$. A likelihood that reads a public coordinate, for example $L(o \mid r, \sigma_0)$ at the root node, multiplies the profiles of the two root branches by $L(o \mid r, u_1)$ and $L(o \mid r, -u_1)$ respectively, and the next fresh root response eq. (11.17) then depends on the root phase. A base-only record delivered to one subtree produces the same

asymmetry. Records reading ancestor path bits would require retaining the ancestor phases and give a different descriptor.

Remark 11.10 (A persistent base-only evidence term). Add to the potential a fixed evidence term $\sum_i \int q_i(d\xi)[-\log L(o \mid r)]$ for a recorded base-only likelihood with bounded logarithm, and analogously in the model sector. This evidence-augmented potential is not the reciprocal potential of definition 4.2. Every leaf has base marginal $f_0 A$, so the term equals $N \int f_0(-\log L(o \mid r))dA$, and its force is the base-only function $-\log L(o \mid r)$, which enters h^q exactly as V^q does. The term is invariant under $\mathcal{S}$, so theorems 11.2 and 11.9 and lemma 11.3 hold with V^q replaced by $V^q - \log L(o \mid r)$ and $\|\log \rho\|_\infty$ increased by $\|\log L\|_\infty$. The stationary laws then depend on the accumulated evidence, while the compression is preserved.

The fresh-sampling condition matters. For a pair $q_\pm = p(1 \pm a\sigma)$, two samples ξ, ξ' from one reused child have joint density

$$p(\xi)p(\xi')[1 + a(\xi)a(\xi')\sigma(\xi)\sigma(\xi')], \tag{11.21}$$

which can reveal the discarded phase. A shared child label for a joint belief/model response can likewise expose a relative contrast. Unrestricted external identity-channel probes can distinguish these laws, as in chapter 10. These protocols are outside the stated inherited interface.

Open problem 11.11 (Minimality of the unordered profile tree). Decide whether the unordered profile tree is minimal for the fresh-response interface of the hierarchy: whether the response histories of all nodes, under admitted adaptive record policies, separate the base-dependent tree-automorphism orbits of profiles, at least generically. Proposition 11.13 proves local injectivity for a different two-agent binary model; for $d = 1$ and a two-point base, the analogous Jacobian computation for the root response and its time derivatives is a natural first case.

11.5 Public distinctions discarded at every depth

Choose, only as an initial witness,

$$f(r, u) = f_0(r) \prod_{j=1}^{d} [1 + a_j(r)u_j], \qquad 0 < a_j(r) < 1, \tag{11.22}$$

with regular bounds and $\int f_0 dA = 1$. The evolution subsequently uses the unrestricted positive profile f, and is not assumed to preserve this product. Before joining at depth l, the two child-parent response densities differ by their factor $1 \pm a_{l+1}\sigma_v$. Rotating α_v changes these full public laws, while their joined unordered profile tree stays fixed. Strict differences persist on a nonzero time interval by continuity. The same construction works at every depth and in both sectors. Under the same flow, however, these differences decay near the partition-independent profiles.

Proposition 11.12 (Decay of public contrast). Under the profile flow of theorem 11.2, the following hold in each sector. (i) The set of profiles that do not depend on u is invariant, and on it the rows converge to the class priors eq. (11.4). (ii) Along a solution in this set, write a nearby

profile as $f = f_0(1+\varphi)$ with $\mathbb{E}_{U_d}\varphi = 0$. To first order in φ,

$$\dot{\varphi} = -\left(3 - 2\sum_k b_k^q P_k\right)\varphi. \tag{11.23}$$

For the Walsh function $\chi_S(u) = \prod_{j\in S} u_j$ with largest index m, the rate is $\lambda_m = 3 + 2b_{m-1}^q - 2\sum_{k\geq m} b_k^q \geq 1$. At the class priors every rate equals $3 + 2/(N-1)$. (iii) Among profiles with a fixed base marginal, the belief part $\int f \log f dA dU_d + \sum_k b_k^q D_k(f)$ of the profile energy is minimized exactly by the profile that does not depend on u.

Proof. If f does not depend on u, then $P_k \log f = \log f$ and $P_k f/f = 1$, so h^q does not depend on u and the set is invariant. There $D_k = 0$, and eq. (11.13) is the Fisher flow of $T_a D_{\mathrm{KL}}(b^a\|\pi)$, whose log-odds relative to π decay at rate T_a. For (ii), $\log f = \log f_0 + \varphi + O(\varphi^2)$, $P_k \log f = \log f_0 + P_k\varphi + O(\varphi^2)$, and $P_k f/f = 1 + P_k\varphi - \varphi + O(\varphi^2)$. The u-dependent part of h^q is therefore $(3 - 2\sum_k b_k P_k)\varphi$, which has zero U_d-mean, while D_k is of second order and the rows and V^q are unchanged at first order. Writing $\dot{f} = \dot{f}_0(1+\varphi) + f_0\dot{\varphi}$ and taking the U_d-mean of $\dot{f} = -f(h^q - c_q)$ gives eq. (11.23). On χ_S, $P_k\chi_S = 0$ when $m > k+1$, $P_k\chi_S = -\chi_S$ when $m = k+1$, and $P_k\chi_S = \chi_S$ when $m \leq k$, which gives λ_m; since $\sum_k b_k = 1$, $\lambda_m \geq 1$. At the class priors, $\pi_{m-1} - \sum_{k\geq m}\pi_k = [2^{d-m} - (2^{d-m}-1)]/(N-1) = 1/(N-1)$. For (iii), $\int f \log f dA dU_d = \int f_0 \log f_0 dA + \int f_0 D_{\mathrm{KL}}(f(r,\cdot)/f_0(r)\|U_d)dA$, the conditional term vanishes exactly at the u-independent profile, and each D_k is nonnegative and vanishes there. □

The common rate $3+2/(N-1)$ equals $1+\kappa_N$ with $\kappa_N = 2N/(N-1)$, the rate of the complete-graph difference mode in eq. (10.9) at $J = 0$. The proposition is local: global convergence of the contrast to zero, and whether contrast can vanish at finite time, are not established here. The distinctions discarded at every depth are therefore the orbit data of lemma 11.4, and their effects through the excluded protocols decay near the partition-independent profiles.

In the circle realization these losses are continuous in total variation. For a phase shift δ with $|\delta| \leq \pi$, the sign disagreement has Haar mass $|\delta|/\pi$. If the two branch response densities at a node are p_+, p_-, then, including the base and ancestor integrations, the total variation distance induced in one child-parent response is

$$\frac{|\delta|}{2\pi}\int \mathbb{E}_{U_l}|p_+(r,U_l) - p_-(r,U_l)|dA(r). \tag{11.24}$$

For $\pi < |\delta| \leq 2\pi$, the factor $|\delta|$ is replaced by $2\pi - |\delta|$. With the squared Hellinger distance $H^2 = \frac{1}{2}\int(\sqrt{p} - \sqrt{p'})^2$, the same computation gives

$$H^2 = \frac{|\delta|}{2\pi}\int \mathbb{E}_{U_l}\left(\sqrt{p_+} - \sqrt{p_-}\right)^2 dA, \qquad |\delta| \leq \pi. \tag{11.25}$$

The Fisher–Rao distance between the two laws is $2\arccos(1-H^2)$, which lies between $2\sqrt{2H^2}$ and $\pi\sqrt{2H^2}$. Hence every phase path along which $p_+ \neq p_-$ on a set of positive measure is Lipschitz in total variation and Hölder of order one half in the Fisher–Rao distance, and it is not rectifiable in that distance: the chord sum over n equal steps of a fixed phase interval grows like $\sqrt{n}$. These

paths have no finite Fisher tangent and are not continuous in the bounded-log-density norm. In the coin realization, orientation changes are discrete.

Each phase can be an arbitrary measurable function of the base. The quotient therefore discards spatial patterns in the original public laws, beyond a permutation of primitive agents. It retains the amplitude profile tree, whose size can grow with the number of descendants; no fixed-size parent representation is asserted.

The sample-coordinate density is piecewise constant in the node coordinates, while the law-valued Fisher evolution remains smooth in time on the regular profile domain. The result does not give a smooth dimension reduction on an open law chart. It gives exact recursive public compression on a prepared invariant class, and that compression is a symmetry reduction. The general base laws and the coupled evaluator enter only through the base marginals f_0, g_0, which every level retains; the compressed data are orthogonal to them. The tree and the class are supplied, and no endogenous formation time has been derived.

The mechanism relies on complementary child laws whose equal mixture is flat in the public coordinate. Equal mixtures of Gaussian children do not have this property: with ϕ_d the standard normal density, $\frac{1}{2}[\phi_d(y-v)+\phi_d(y+v)] = \phi_d(y)e^{-|v|^2/2}\cosh(y\cdot v)$ depends on the direction of v unless $v=0$. A continuous phase of the kind used here therefore has no analogue for Gaussian child laws with a common isotropic covariance.

The reference coordinates r, w in this construction belong to represented sample spaces. They are not identified with the contextual base $\mathcal{C}$. Applying chapter 3 would additionally require an admitted smooth context-parameterized family and compatible connections. The measurable retention proved here does not supply the collective-connection condition in proposition 3.17.

11.6 Why ordinary response completion can remove all compression

The preceding restrictions cannot generally be removed by adding a few response moments to the phase parent. Observation derivatives can already recover a whole fine configuration, as in the local observability method of Hermann and Krener [75]. The following calculation uses only the inverse function theorem.

Proposition 11.13 (Fresh responses locally identify a binary pair). Consider two agents under the reciprocal potential with identity peer channels, binary state and model spaces, the uniform model reference, and the coupled binary evaluator $G_J(\kappa \mid \kappa') = e^{J\kappa\kappa'}/(2\cosh J)$ with $J \notin \{2, 6\}$; this includes the range $|J| < 1/4$ used in chapter 10. There is an open heterogeneous region where any retention preserving both fresh belief/model mixture responses and their future evolution is locally injective.

Proof. Write $q_i = p_{x_i}$, $s_i = p_{y_i}$, where $p_x(\kappa) = (1+\kappa x)/2$. The exact original equations, including incoming derivatives, are

$$\dot{x}_i = -(1-x_i^2)[2\,\mathrm{atanh}\,x_i - \mathrm{atanh}\,x_j - Jy_i] - (x_i - x_j), \tag{11.26}$$

and the exchanged equation for y; a nonuniform model reference would add its own term to the model equation. Set $x_{1,2} = m \pm a$, $y_{1,2} = n \pm e$. Fresh mixture responses are p_m, p_n. Their velocities

have the convergent local expansions

$$\begin{aligned} F_x &= -(1-m^2)(\operatorname{atanh} m - Jn) + (6m - Jn)a^2 - 2Jmae + O(5), \\ F_y &= -(1-n^2)(\operatorname{atanh} n - Jm) + (6n - Jm)e^2 - 2Jnae + O(5). \end{aligned} \tag{11.27}$$

Here $O(5)$ denotes terms of total degree at least five. To check the coefficient, the uncoupled part of F_x is

$$-\tfrac{1}{2}\sum_{\pm}[1-(m\pm a)^2]\operatorname{atanh}(m\pm a) + 2ma[\operatorname{atanh}(m+a) - \operatorname{atanh}(m-a)], \tag{11.28}$$

whose expansion is $-m + 2m^3/3 + 6ma^2 + O(5)$.

For $\mathcal{H} = (m, n, F_x, F_y)$, its Jacobian determinant at $m = n = \varepsilon c_1$, $a = e = \varepsilon c_2$, $c_1 c_2 \neq 0$, is

$$\det D\mathcal{H} = 4\varepsilon^4 c_1^2 c_2^2 (6-3J)(6-J) + O(\varepsilon^6). \tag{11.29}$$

Indeed, the lower-right block is $2\varepsilon^2 c_1 c_2$ times the matrix with diagonal $6-2J$ and off-diagonal $-J$, plus $O(\varepsilon^4)$. Its leading coefficient vanishes only at $J=2$ and $J=6$, so the determinant is nonzero for sufficiently small nonzero ε, and remains nonzero nearby. If two states in that neighborhood have the same inherited parent, their response curves agree. Equality at time zero and of their first derivatives gives equality of $\mathcal{H}$, hence equality of the states by local invertibility. No differentiability assumption on the retention is needed. □

This is a local obstruction in one original model. It neither removes the global child-exchange ambiguity nor excludes the selective protocols and invariant classes constructed above.

Chapter 12

Two-Stage Selection with Exact Inheritance on a Wishart Family

This chapter studies the reciprocal law potential of definition 4.2, with its exact Gibbs rows (proposition 4.3) and unit-Fisher dynamics, on an invariant family of matrix Wishart laws. It does not concern the global or per-agent free energies of chapter 4, and its conclusions are not transferred to them. Four initially singleton agents are grouped into two parents and then into their union by a declared threshold rule applied to their evolving laws. Every parent inherits the original-time dynamics, a declared joint response interface, and the collective connection and contextual tensor of its children. Every join discards a continuous distinction between model laws.

Measured against the formation standard definition 6.31, the construction is a two-stage selection with exact inheritance, not a recursive formation. The selector reads belief shapes, and the shape law $\dot{\alpha}_i = -2(\alpha_i - \alpha_0)$ contains neither the coupling, nor the precisions, nor the model sector, nor any other agent. Proposition 12.7 shows that the membership history is the count-weighted centroid dendrogram of the initial belief shapes, read on a clock whose origin is set by the threshold, and that the null flow of definition 6.30 reproduces it on the entire admitted class. Interaction necessity (F5) therefore fails, in its strong form as well. Proposition 12.3 identifies the structural cause, which every architecture with receiver-independent targets and additive moment channels shares. The discarded model-shape allocation is dynamically inert, and the inherited contextual geometry is prepared in the initial section, is present at zero coupling, and decays exponentially (proposition 12.11). What the construction establishes is exact inheritance: original-time parent dynamics with the complete channel derivative, extensive response histories under adaptive protocols, and preservation of the collective connection and contextual tensor through both joins.

The architecture is declared. Children have matrix Wishart laws with common precision in each sector, product recognition, a constant model-independent evaluator, and comparison channels depending on the population moment. Individual shapes are constant over the contextual base. Responses use fresh independent samples and return extensive sums. Membership is selected by a declared threshold rather than by a threshold derived from the potential. The preparation class is open within this invariant family. Each additive quotient is available initially, and inheritance holds for every partition at every time, so inheritance does not distinguish the selected partition; the rule selects when a quotient becomes the operative description. No law is reset at a join.

12.1 Matrix laws and the declared architecture

Let both represented spaces be the cone S_2^{++} of real symmetric positive-definite two-by-two matrices. A random matrix X or Y is represented content, distinct from a contextual coordinate.

Write P, Q for precision matrices, to distinguish them from the principal bundle P. For $\nu > 1/2$ and $\mathsf{P} \succ 0$, define

$$W_2(\nu, \mathsf{P})(dX) = \frac{\det(\mathsf{P})^\nu}{\Gamma_2(\nu)} \det(X)^{\nu - 3/2} e^{-\operatorname{tr}(\mathsf{P}X)} dX, \qquad \Gamma_2(\nu) = \sqrt{\pi}\Gamma(\nu)\Gamma(\nu - \tfrac{1}{2}). \tag{12.1}$$

Here $dX = dx_{11}dx_{12}dx_{22}$. This is the central Wishart law with real shape. Its normalizer is the multivariate gamma function, and congruence by B has Jacobian $\det(B)^3$ on this cone [110, Section 2.1 and Chapter 3]. The range $\nu > 1/2$ of real shapes for which eq. (12.1) is a probability law is Gindikin's set [65]; existence questions for the noncentral case are discussed by Letac and Massam [98]. No Gaussian representation or integer number of degrees of freedom is required. The proof below is self-contained.

Lemma 12.1 (Normalization and addition). The density in eq. (12.1) is normalized. Its mean is $\nu\mathsf{P}^{-1}$. Independent variables with laws $W_2(\nu, \mathsf{P})$ and $W_2(\mu, \mathsf{P})$ have sum law $W_2(\nu + \mu, \mathsf{P})$.

Proof. At $\mathsf{P} = I_2$, set $x_{11} = u > 0$, $x_{12} = \sqrt{u}z$, and $x_{22} = v + z^2$, with $v > 0$ and $z \in \mathbb{R}$. The determinant is uv, the trace is $u + v + z^2$, and the Jacobian is $\sqrt{u}$. The integral is $\Gamma(\nu)\Gamma(\nu - \frac{1}{2})\sqrt{\pi}$. Substitution $X = \mathsf{P}^{-1/2} Z \mathsf{P}^{-1/2}$ gives the general normalizer: congruence by a positive matrix B has Jacobian $\det(B)^3$ on this cone. Differentiating the normalizing integral with respect to P gives the stated mean.

The convolution density at $S \succ 0$ contains the integral

$$e^{-\operatorname{tr}(\mathsf{P}S)} \int_{0<X<S} \det(X)^{\nu - 3/2} \det(S - X)^{\mu - 3/2} dX. \tag{12.2}$$

The substitution $X = S^{1/2} U S^{1/2}$ makes it a constant times $e^{-\operatorname{tr}(\mathsf{P}S)} \det(S)^{\nu + \mu - 3/2}$. The constant is positive and finite by Tonelli's theorem and normalization of the convolution. Normalizing the resulting density determines it and proves the addition formula. □

Initially take $N = 4$ agents with

$$q_i = W_2(\alpha_i, \mathsf{P}), \qquad s_i = W_2(\eta_i, \mathsf{Q}), \qquad \zeta_i = q_i \otimes s_i, \tag{12.3}$$

where every shape exceeds $1/2$. The model shape η_i is unrelated to a source-row weight. Fix $\alpha_0, \eta_0 > 1/2$ and $\kappa > 0$. The local data are the proper reference $r_i = W_2(\eta_0, I_2)$ and the normalized constant evaluator $G_i(m, dX) = W_2(\alpha_0, I_2)(dX)$. For any admitted finite-moment configuration set

$$M_q = \sum_i \mathbb{E}_{q_i} X, \qquad \Lambda_q = I_2 + \frac{\kappa}{N} M_q, \qquad \mathcal{K}^q_{ij}[\mathbf{q}](k, dX) = W_2(\alpha_0, \Lambda_q)(dX). \tag{12.4}$$

The model sector uses $M_s = \sum_i \mathbb{E}_{s_i} Y$, $\Lambda_s = I_2 + \kappa M_s / N$, and target $W_2(\eta_0, \Lambda_s)$. Every source, including self, is allowed with prior $\pi_{ij} = 1/N$ and positive original temperatures. Each channel ignores its particular source sample k but reads the current laws through the total moment. It therefore depends on the whole configuration, not only on the two endpoint laws, as eq. (4.3) allows. The channel is a continuous function of M_q on the finite-moment configurations, so it is jointly

measurable in the configuration and the input. Independent draws across receiver children specify the joint channel experiment used when their outputs are added.

These primitive self channels are nonidentity channels by constitutive choice. With identity self channels, definition 4.2 would give $D^q_{ii} = 0$ while the peer costs are positive away from agreement, so the Gibbs rows would depend on each receiver's peer cost and would differ from the priors. The construction requires the nonidentity choice.

All source costs in a receiver row agree, so the exact Gibbs row equals its prior. For the same reason, promoting the source priors to state variables of the same potential would leave them fixed. The Gibbs-reduced reciprocal potential is $\mathcal{E}_q + \mathcal{E}_s$, where

$$\mathcal{E}_q = \sum_i \left[D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_q)) \right]. \tag{12.5}$$

The two sectors have separate population coupling. The evaluator is constant in its model input, and this assumption is load-bearing. By eq. (4.2), a model-dependent evaluator $G_i(m, \cdot) = W_2(\alpha_0, \Gamma(m))$ would add the term $\mathrm{tr}(\mathbb{E}_{s_i}[\Gamma(m)]X)$ to the belief force. Because the model laws differ across agents, the precisions would then cease to be common, and lemma 12.1 would no longer apply to the parents. The sum-pushforward of independent child evaluator outputs is a kernel from the retained model sum only because the evaluator ignores the model sample; for unequal precisions $\Gamma(m_i)$ the sum is not even a Wishart law. With the constant evaluator the belief and model sectors are uncoupled, and each sector is coupled across agents only through its own population moment. Outside the Wishart family the displayed functional requires finite relative entropies and moments; the differential also requires integrable log densities and log determinants, and differentiation under the integrals along the admitted paths.

12.2 The complete smooth Fisher flow

Proposition 12.2 (Invariant family and original-time evolution under the reciprocal potential). With the unit Fisher metric and all channel derivatives included, eq. (12.3) is invariant under eq. (12.5). In the belief sector,

$$C_q = \Lambda_q + \frac{\kappa}{N}(M_q - N\alpha_0\Lambda_q^{-1}) = I_2 + \frac{2\kappa}{N}M_q - \kappa\alpha_0\Lambda_q^{-1}, \tag{12.6}$$

$$\dot{\alpha}_i = -2(\alpha_i - \alpha_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q. \tag{12.7}$$

The model equations exchange $\alpha, \alpha_0, \mathsf{P}, M_q, C_q$ with $\eta, \eta_0, \mathsf{Q}, M_s, C_s$. These solutions are global and regular on the stated family. Distinct initial shapes remain distinct at every finite time.

Proof. Put $c_0 = \alpha_0 - 3/2$. Expanding the normalized densities gives

$$\begin{aligned}\mathcal{E}_q = 2\sum_i \int q_i \log q_i dX - 2c_0 \sum_i \mathbb{E}_{q_i} \log\det X + \mathrm{tr}\, M_q + \mathrm{tr}(\Lambda_q M_q) \\ - N\alpha_0 \log\det \Lambda_q + 2N \log\Gamma_2(\alpha_0).\end{aligned} \tag{12.8}$$

Since $d\Lambda_q = (\kappa/N)dM_q$, the differential of the last moment-dependent pair is $\mathrm{tr}(C_q dM_q)$. Thus the full variational force, up to a sample-independent scalar, is

$$h_i(X) = 2\log q_i(X) - 2c_0 \log\det X + \mathrm{tr}[(I_2 + C_q)X]. \tag{12.9}$$

The direct sender variation is zero because a mass-zero variation traversing a constant-in-input kernel gives zero. The channel variation is not zero: it is precisely the second term in the first expression for C_q. The unrestricted density equation is $\dot{q}_i = -q_i(h_i - \mathbb{E}_{q_i} h_i)$. Comparing the coefficients of $\log\det X$ and X in this equation with the derivative of eq. (12.1) gives eq. (12.7); normalization fixes the remaining scalar. This proves tangency of the full density field.

For regular Wishart data with possibly different precisions the same calculation gives $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + I_2 + C_q$. Consequently

$$\mathsf{P}_i(t) - \mathsf{P}_j(t) = e^{-2t}(\mathsf{P}_i(0) - \mathsf{P}_j(0)). \tag{12.10}$$

Initially common precision remains common. On that family put $A = \sum_i \alpha_i$ and $\bar{\alpha} = A/N$. Then $M_q = A\mathsf{P}^{-1}$ and $\bar{\alpha}(t) = \alpha_0 + e^{-2t}(\bar{\alpha}(0) - \alpha_0)$. The precision field commutes with P, so its initial eigenvectors remain fixed and each positive eigenvalue solves

$$\dot{p} = f(p, \bar{\alpha}(t)) = -2p + 2 + \frac{2\kappa\bar{\alpha}(t)}{p} - \frac{\kappa\alpha_0 p}{p + \kappa\bar{\alpha}(t)}. \tag{12.11}$$

Let $m_- = \min(\bar{\alpha}(0), \alpha_0)$ and $m_+ = \max(\bar{\alpha}(0), \alpha_0)$. For $\varepsilon = \min(1/2, m_-/\alpha_0)$, $f > 0$ when $0 < p \le \varepsilon$. For $L = 2 + \kappa m_+$, $f \le -2$ when $p \ge L$. Every eigenvalue therefore stays between $\min(p(0), \varepsilon) > 0$ and $\max(p(0), L) < \infty$. Shapes stay between their initial values and their reference shape, all strictly above $1/2$. The smooth finite-dimensional field consequently has a unique global solution on this invariant family.

Slightly enlarged compact parameter neighborhoods dominate polynomial moments and logarithmic determinant derivatives in eq. (12.1). They justify finite KL, square-integrable scores, and the differentiations above. Integrating the log-density equation with factor e^{2t} shows that any positive dominated regular solution with these initial laws has only the same log-determinant and linear matrix statistics; its parameters satisfy the same ODE. Finally, shape differences equal their initial differences times e^{-2t}. □

The log-density equation has the form $\partial_t \log q_i = -2\log q_i - V_t + c_i(t)$, with V_t common to all agents. For a fixed target this is the Fisher–Rao gradient flow of a relative entropy, whose explicit solution interpolates geometrically between the initial law and the target [30, Appendix B]; the rate is two here because two relative-entropy terms contain q_i. In the present construction the target moves through M_q, and the Wishart family carries the solution.

The precision flow is coupled across agents. For a symmetric moment variation H,

$$DC_q[H] = \frac{2\kappa}{N}H + \frac{\kappa^2\alpha_0}{N}\Lambda_q^{-1}H\Lambda_q^{-1}. \tag{12.12}$$

Its trace pairing with nonzero H is strictly positive. Varying another agent's shape produces $H = \mathsf{P}^{-1}d\alpha_j$ and changes the coefficient of the linear statistic X in the force on q_i, hence the precision velocity. It does not change any shape velocity: in eq. (12.9) the coupling enters only the

coefficient of X, while the coefficient of $\log\det X$ after substitution of $\log q_i$ is $2(\alpha_i-\alpha_0)$ for every $\kappa\geq 0$. The coupling comes from the declared population moment; the source rows remain at their priors and do not select preferred neighbors.

12.3 A common relaxation clock

The independence of the shape law from the interaction is an instance of a general identity for receiver-independent targets.

Proposition 12.3 (Shared targets impose a common relaxation clock). Consider the reciprocal potential for agents with common state and model spaces. Suppose every agent has the same model reference r, the same evaluator $G(m,dk)=g(dk)$, constant in m, and that every belief channel, for every receiver, source, and input, equals one normalized law $\nu^q[M_q]$ depending smoothly on $M_q=\sum_i \mathbb{E}_{q_i}T$ for a fixed measurable statistic $T:\mathsf{K}\to\mathbb{R}^d$; the model sector is analogous. Let all source priors and temperatures be positive. On a regular dominated family on which the unit-Fisher density equation of the Gibbs-reduced potential holds, with differentiation under the integrals, every pair of agents satisfies

$$\partial_t \log\frac{q_i}{q_j} = -2\log\frac{q_i}{q_j} + c_{ij}(t), \tag{12.13}$$

with c_{ij} independent of the sample. Hence $\log(q_i/q_j)(t)=e^{-2t}\log(q_i/q_j)(0)$ up to a sample-independent function of t, for every target map ν^q, in particular for every coupling strength and for a constant target. If the laws lie in an exponential family with natural parameter θ and affinely independent sufficient statistic, then $\theta_i(t)-\theta_j(t)=e^{-2t}(\theta_i(0)-\theta_j(0))$.

Proof. All costs in row i equal $D_i=D_{\mathrm{KL}}(q_i\|\nu^q[M_q])$, so by proposition 4.3 the Gibbs row equals the prior and $-T_q\log Z_i^q=D_i$. By eq. (4.2) with a constant evaluator, the part of the reduced potential that depends on the belief laws is $\sum_l[D_{\mathrm{KL}}(q_l\|g)+D_{\mathrm{KL}}(q_l\|\nu^q[M_q])]$. By eq. (4.10), a variation of q_i contributes $\log(q_i/g)+\log(q_i/\nu^q[M_q])$ through the two terms in which q_i is the receiver, and $\langle\Psi_t,T\rangle$ through M_q, where $\Psi_t=-\sum_l \mathbb{E}_{q_l}\partial_M\log\nu^q[M]$ at $M=M_q$. Since M_q is additive and every agent enters it through the same statistic, Ψ_t is the same for every i. Thus $h_i=2\log q_i+V_t$ up to a constant, with $V_t=-\log g-\log\nu^q[M_q]+\langle\Psi_t,T\rangle$ common to all agents. The density equation gives $\partial_t\log q_i=-2\log q_i-V_t+\mathbb{E}_{q_i}h_i$, and subtraction gives eq. (12.13). Integrating the linear equation proves the second statement. In an exponential family, $\log(q_i/q_j)=\langle\theta_i-\theta_j,S\rangle-\psi(\theta_i)+\psi(\theta_j)$, and affine independence of S identifies the coefficients. □

For the Wishart family the natural parameter is $(\alpha-3/2,-\mathrm{P})$, so proposition 12.3 contains both the shape law and eq. (12.10). Any relabeling-equivariant selector that reads only centered log-likelihood ratios between agents, for example natural-parameter differences, sees a rescaled copy of the initial data on a clock shared by every interaction in this class. Interaction-sensitive selection under the reciprocal potential therefore needs targets that depend on the receiver, for instance Gibbs rows that are not constant. Whether exact sum closure on an open fiber of child allocations is compatible with receiver-selective rows is not settled here.

A second obstruction concerns exact additive closure itself.

Proposition 12.4 (Scalar sum closure forces a common clock). Let scalar parameters θ_i in an open interval Θ evolve by $\dot\theta_i = G(\theta_i, g(t))$, where $g(t)$ is a state common to all agents and G is continuous in its first argument. Suppose that for every value of g the sum $G(x,g)+G(y,g)$ depends on $(x,y)\in\Theta^2$ only through $x+y$, as exact intertwining of sum retention for every two-element block requires. Then $G(\theta,g) = a(g)\theta + c(g)$, and every difference of count-weighted centroids of disjoint blocks is multiplied by the common factor $\exp\int_0^t a(g(u))du$. The ordering of all centroid distances is therefore that of the initial data at every time.

Proof. Fix g and write $G(\theta)$. For $x,y\in\Theta$ the midpoint $m=(x+y)/2$ lies in Θ and $x+y=m+m$, so $G(x)+G(y)=2G(m)$. Choose $u<v$ in Θ, and let ℓ be the affine function agreeing with G at u and v. Then $G-\ell$ satisfies the same midpoint equation and vanishes at u and v, hence at every dyadic point of $[u,v]$, hence on $[u,v]$ by continuity. Since u,v are arbitrary, G is affine on Θ. A count-weighted centroid $\theta_I = n_I^{-1}\sum_{i\in I}\theta_i$ then obeys $\dot\theta_I = a\theta_I + c$, so centroid differences obey a linear homogeneous equation with the common coefficient $a(g(t))$. □

The Wishart shapes are the instance $a=-2$, $c=2\alpha_0$, in which even a is independent of the interaction. Vector parameters with a population-dependent generator that is not a multiple of the identity escape proposition 12.4, because a common anisotropic propagator can change which of two differences is smaller; a smooth reciprocal-potential instance with exact closure is not constructed here.

12.4 Parent laws and exact joining

Theorem 12.5 (Additive parents with the original Fisher dynamics). For any disjoint current partition, retain

$$n_I = |I|, \qquad A_I = \sum_{i\in I}\alpha_i, \qquad B_I = \sum_{i\in I}\eta_i, \qquad q_I = W_2(A_I,\mathsf{P}), \quad s_I = W_2(B_I,\mathsf{Q}). \tag{12.14}$$

The parent evaluator, reference, and comparison channels are the corresponding sum-pushforwards of independent child outputs. Their shapes are $n_I\alpha_0, n_I\eta_0$, and their source priors are $\pi^c_{IJ} = n_J/N$. With the primitive $N=\sum_I n_I$ kept fixed, their ordinary unit-Fisher dynamics is

$$\dot A_I = -2(A_I - n_I\alpha_0), \qquad \dot B_I = -2(B_I - n_I\eta_0), \tag{12.15}$$

together with the same precision equations eq. (12.7). Joining disjoint blocks adds (n,A,B) and keeps the common precisions. It commutes with laws, channels, and evolution, and is associative.

Proof. Use the fixed sample map $(X_i,Y_i)_{i\in I}\mapsto(\sum_i X_i,\sum_i Y_i)$. Lemma 12.1 gives eq. (12.14). The same map on the independent reference or channel outputs gives $W_2(n_I\alpha_0, I_2)$ or $W_2(n_I\alpha_0,\Lambda_q)$, and the model exchange. The parent evaluator is constant in its model input, just as the fine evaluator is; its descent through the sum map uses this constancy. Source normalization gives $\pi^c_{IJ}=n_J/N$ by summing primitive sender weights. Its row is again the exact Gibbs prior.

Total moments satisfy $M_q = \sum_I \mathbb{E}_{q_I} X_I$ and $M_s = \sum_I \mathbb{E}_{s_I} Y_I$. The parent reciprocal energy in the belief sector is

$$\mathcal{E}_q^c = \sum_I \left[D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, I_2)) + D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, \Lambda_q)) \right]. \tag{12.16}$$

Its Λ_q terms are still $\operatorname{tr}(\Lambda_q M_q) - N\alpha_0 \log\det \Lambda_q$ because $\sum_I n_I = N$. Its complete moment derivative is therefore the same C_q. The density-force calculation in eq. (12.9), with reference shape $n_I\alpha_0$, gives eq. (12.15) and the original precision equation. This is the projected fine vector field at the same time, with no mobility correction.

Adding two parents preserves total moments, counts, and forcing. Shape equations add, and sample addition gives the joined law and normalized channel. Associativity follows from addition of the three retained scalars and of the sample matrices. □

The theorem holds for every partition at every time, so exact inheritance does not single out the partitions that the selector of the next section produces; selection and inheritance are logically independent here. The final single parent still has its inherited nonidentity internal-source channel. Replacing it by an identity self-comparison would delete the second KL term and change its flow. A parent source label is the pushforward of one primitive source-label draw; a batch exposing separate primitive labels would require their joint mark law. Those labels are not exposed by the interface below.

Exact evolution does not assert equality of fine and parent energies. For example, at $\mathsf{P} = I_2$ replace two shapes α_0, α_0 by $\alpha_0 + \epsilon, \alpha_0 - \epsilon$. The total moment and parent state are unchanged, but the fine energy increases by $2D_{\mathrm{KL}}(W_2(\alpha_0+\epsilon, I_2)\|W_2(\alpha_0, I_2)) + 2D_{\mathrm{KL}}(W_2(\alpha_0-\epsilon, I_2)\|W_2(\alpha_0, I_2)) > 0$. The vector-field calculation, rather than energy factorization, proves inheritance.

12.5 Selected membership

Definition 12.6 (Threshold membership rule). Fix a threshold $\delta > 0$ and begin with singletons whose initial belief shapes are pairwise farther than δ apart. For a current block read its mean belief-shape deviation

$$b_I = A_I/n_I - \alpha_0, \qquad \dot{b}_I = -2b_I, \qquad b_{I\cup J} = \frac{n_I b_I + n_J b_J}{n_I + n_J}. \tag{12.17}$$

Join two current blocks when their distance $|b_I - b_J|$ first reaches δ from above. Simultaneous threshold edges join their connected components. Recompute distances after each event.

This rule reads only retained laws and counts, is invariant under relabeling, and contains no supplied membership tree. The threshold specifies a persistence criterion; it is additional constitutive data. The separation hypothesis belongs to the domain of the rule: two shapes initially closer than δ never reach δ from above and are never joined to each other, even when they are nearest neighbors. Because every score is explicit, the event times below are computed in closed form, and no discretization or event-location scheme is involved.

Proposition 12.7 (Coupling-independent centroid dendrogram). On the invariant family of proposition 12.2, let $\kappa \geq 0$, all precisions, and all model data be arbitrary.

(i) Every shape obeys $\alpha_i(t) = \alpha_0 + e^{-2t}(\alpha_i(0) - \alpha_0)$, so every block score obeys $b_I(t) = e^{-2t} b_I(0)$, where $b_I(0)$ is the count-weighted centroid of the initial deviations $\alpha_i(0) - \alpha_0$, $i \in I$.

(ii) Let $N \geq 2$ initial belief shapes be pairwise farther than δ apart, and suppose that at every stage of the agglomeration below the closest pair of current centroids is unique. Then definition 12.6 makes $N-1$ binary joins and coincides with count-weighted centroid agglomerative clustering of the initial shapes: at stage k the two current blocks whose initial centroids are closest, at distance D_k, join at time $t_k = \frac{1}{2}\log(D_k/\delta)$. The heights D_k increase strictly; at each event every other current distance exceeds δ; and the scores of the two blocks joined at t_k differ by $\delta e^{-2(t-t_k)} < \delta$ at every later time. The membership tree depends only on the initial belief shapes. It is independent of κ, of the precisions, of the model sector, and of δ, which shifts every event time by $-\frac{1}{2}\log\delta$.

(iii) The null flow Φ^0 of definition 6.30, with every peer cost deleted, each self channel evaluated with the peer laws held at their initial values, and all other data fixed, exists globally on the Wishart family with individual precisions and has the same shape trajectories. Hence $P^0(x) = P(x)$, and indeed $H^0(x) = H(x)$ including event times, at every point x of the admitted class. The construction fails clause (F5) of definition 6.31 and its strong form. The null flow is not the flow at $\kappa = 0$: it keeps κ inside each self channel, so its precisions differ and need not stay common. Both flows nevertheless produce the same partition sequence and event times as the flow at every $\kappa > 0$.

Proof. (i) Substituting $\log q_i = (\alpha_i - \frac{3}{2})\log\det X - \mathrm{tr}(\mathsf{P}X) + \mathrm{const}$ into eq. (12.9) gives $h_i = 2(\alpha_i - \alpha_0)\log\det X + \mathrm{tr}[(I_2 + C_q - 2\mathsf{P})X] + \mathrm{const}$. The coupling, the precision, and the other agents enter only the coefficient of X; matching the coefficient of $\log\det X$ gives the shape law for every $\kappa \geq 0$, and at $\kappa = 0$ the channel target equals the reference. Block scores are count-weighted averages of agent scores, and averaging commutes with the common factor e^{-2t}.

(ii) By (i), every current distance is De^{-2t} with D a distance between initial centroids, so a pair reaches δ at $\frac{1}{2}\log(D/\delta)$, and the first pair to reach it is the pair with the smallest D. Since all initial distances exceed δ, every pair reaches δ from above. Suppose I and J are the closest current pair, at distance D, and let K be any other current block. The merged centroid lies strictly between the centroids of I and J. The centroid of K does not lie between them, since otherwise it would be closer than D to one of them. If it lies beyond the centroid of J, its distance to the merged centroid is its distance to J plus $n_I D/(n_I + n_J)$, which exceeds D; the other side is symmetric. Thus every distance after the join exceeds D, which proves that heights increase strictly, that centroids remain distinct, and that at the event time every other current distance is above δ, so there are no cascades and every later pair again reaches δ from above. The two joined blocks keep their scores along the fine flow, so their difference is $\delta e^{-2(t-t_k)}$. This is the agglomeration that joins the closest pair of current centroids and replaces it by its count-weighted centroid. The heights D_k are functions of the initial shapes alone.

(iii) The null flow deletes the peer costs and evaluates each self channel with the peer laws held at their initial values. Its belief potential is $\sum_i [D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_i^0))]$ with $\Lambda_i^0 = I_2 + (\kappa/N)(\mathbb{E}_{q_i} X + M_{-i}^0)$, where M_{-i}^0 is the initial moment of the other agents. The calculation of proposition 12.2 with this single-agent moment gives the same shape law and $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + I_2 + C_i^0$, where $C_i^0 = \Lambda_i^0 + (\kappa/N)(\alpha_i \mathsf{P}_i^{-1} - \alpha_0(\Lambda_i^0)^{-1})$. For a unit eigenvector v of the smallest eigenvalue p

of P_i, $\Lambda_i^0 \succeq I_2$ gives $v^T \dot{\mathsf{P}}_i v \geq -2p + 2 - \kappa\alpha_0/N + 2\kappa\alpha_i/(Np)$, which is positive for small p because α_i stays above $\min(\alpha_i(0), \alpha_0) > 1/2$. With the smallest eigenvalue bounded below, the largest eigenvalue p satisfies $\dot{p} \leq -2p + K$ for a constant K on each finite interval. The precisions therefore remain in compact subsets of the cone, and the solution is global and unique as in proposition 12.2. The selector reads only belief shapes, whose trajectories are the same under both flows, so the membership histories agree, and so do the partition sequences. At $\kappa = 0$ the channel target is the reference, the precision law is $\dot{\mathsf{P}} = -2\mathsf{P} + 2I_2$, and the shape law is again unchanged by (i). The final partition is a function of the initial shapes, whose evolution is the same under both flows, which is the negation of the strong form. □

Centroid clustering is one of the pair-group strategies of numerical taxonomy [134], unified with the other agglomerative strategies by Lance and Williams [90]. In one dimension it has no reversals, which is the monotonicity in (ii). The rule is not single-linkage clustering. For initial deviations $0, 1, 2.05, 3.5$, single linkage joins the third point to the first pair before the last point, whereas the rule forms the two pairs first, at heights 1 and 1.45, and their union at 2.275. The union time in eq. (12.20) below already carries the centroid distance $d_2 + (d_1 + d_3)/2$ rather than the single-linkage distance d_2. The construction also differs from bounded-confidence dynamics [74], in which the interaction determines which agents cluster. What it adds to centroid clustering is exact inheritance of laws, dynamics, responses, and geometry at each join.

Theorem 12.8 (Two selected stages on the four-agent family). Let $u_1 < u_2 < u_3 < u_4$ be the sorted initial belief shapes, with consecutive gaps d_1, d_2, d_3, and suppose

$$\delta < \min(d_1, d_3) < d_2, \qquad \max(d_1, d_3) < d_2 + \tfrac{1}{2}\min(d_1, d_3). \tag{12.18}$$

All four shapes exceed $1/2$, while model shapes vary independently above $1/2$. Under definition 12.6, the two lower and the two upper shapes are joined into parents at

$$t_L = \tfrac{1}{2}\log(d_1/\delta), \qquad t_R = \tfrac{1}{2}\log(d_3/\delta). \tag{12.19}$$

Their union is joined at

$$t_2 = \tfrac{1}{2}\log\left(\frac{d_2 + (d_1 + d_3)/2}{\delta}\right) > \max(t_L, t_R). \tag{12.20}$$

There are no intervening cross joins or cascades. Both first parents coexist on $[\max(t_L, t_R), t_2)$. Each joined child-block distance remains below threshold afterward, and all fine belief laws remain distinct at every finite time. When strict inequalities replace ties, eq. (12.18) is also necessary for this membership history. Each join forgets a continuous model-shape distinction even if all membership labels and event times are retained.

Proof. Apply proposition 12.7. Suppose $d_1 \leq d_3$. Since $d_1 < d_2$, the lower pair is the closest initial pair. After it joins, the distance of its centroid to u_3 is $d_2 + d_1/2$, which exceeds d_3 by eq. (12.18), so the upper pair joins next. The distance between the two parent centroids is $d_2 + (d_1 + d_3)/2$, giving eq. (12.20). The opposite order follows by exchanging upper and lower pairs. If $d_1 = d_3$, exactly the two disjoint outer edges reach threshold together, and the component rule forms the same pairs.

Conversely, if $d_2 < \min(d_1, d_3)$ the middle pair joins first, and if $\max(d_1, d_3) > d_2 + \frac{1}{2}\min(d_1, d_3)$ the first parent absorbs the nearer middle shape before the other pair joins.

The persistence statement is part of proposition 12.7; it follows from the equations rather than from storing a membership flag. Primitive shape differences never vanish at finite time. The two first-parent laws also remain distinct, because their mean belief shapes remain different.

For loss at a first join, fix all belief shapes and perturb two initial model shapes in that pair by $(\eta_i, \eta_j) \mapsto (\eta_i + \epsilon, \eta_j - \epsilon)$ for sufficiently small ϵ. The parent model sum, both population moments, all subsequent precision evolution, and every membership event and time are unchanged. Individual model laws change continuously. Distinct fixed belief shapes distinguish the labels, so this change is not a permutation. At the second join, transfer model shape between the two first-parent sums while keeping their union sum fixed. This gives a further continuous lost distinction and still leaves the belief-only selection schedule unchanged. □

The discarded allocation is a dynamical spectator in both descriptions. Individual model shapes obey $\dot{\eta}_i = -2(\eta_i - \eta_0)$, the model precision depends on the model shapes only through their total, the belief sector never reads the model laws because the evaluator is constant, and the selector reads only belief shapes. The witness therefore establishes noninjectivity of the retention, not loss of a coordinate to which any agent, channel, or selector responds. The belief allocation that the selector reads is not lost given the event record: at a join the two child scores differ by exactly δ, so the parent sum, the child counts, the join time, δ, and α_0 determine the child belief shapes up to exchange. The loss witness accordingly uses the model-shape allocation. Already observed sample records remain with the controller; they are not erased or asserted to have the same law across different initial preparations. Loss concerns the current parent state and the future interface specified next.

The hierarchy is transient. All shapes converge to α_0, η_0, and every precision eigenvalue converges to the unique positive zero of $f(\cdot, \alpha_0)$ or of its model analog, so every agent tends to one common law. The children are distinct at every finite time, and the consensus is asymptotic rather than reached in finite time.

The preparation class is nonempty and relatively open. For example take $\alpha_0 = \eta_0 = \kappa = 1$, $\delta = 1/10$, $(u_1, u_2, u_3, u_4) = (1, 6/5, 9/5, 21/10)$, and model shapes in a neighborhood of $(4/5, 11/10, 7/5, 17/10)$. The gaps are $(1/5, 3/5, 3/10)$, and the events occur at $\frac{1}{2}\log 2$, $\frac{1}{2}\log 3$, and $\frac{1}{2}\log(17/2)$. The two first parents coexist for a time $\frac{1}{2}\log(17/6)$. A sufficiently small neighborhood and every relabeling obey eq. (12.18). Openness is within the common-precision family, not among all positive density tuples. The smooth statistical retention applies on each fixed membership branch; no global smoothness across switching boundaries is required.

Against definition 6.31 the construction meets (F1) for the declared invariant family, open within it, (F2) with the declared threshold, and (F3) by theorem 12.5, corollary 12.9, and theorem 12.10. It meets (F4) in the sense that each join retention is noninjective and the singleton responses available before the join distinguish points of its fiber, although the distinguished coordinate is inert. It fails (F5) by proposition 12.7. In (F6), the parents select their union under the inherited dynamics without a reset or a new attraction, but because all scores contract on the common clock, the union also forms from every counterfactual handoff in which two parent scores are farther apart than δ;

the counterfactual part of (F6) is not met either. The construction is a two-stage selection with exact inheritance along a prepared contraction ordering.

12.6 Exact joint response histories

The response interface is noninvasive. At each query, draw fresh samples independently across children and sectors from $\bigotimes_i (q_i \otimes s_i)$. For the current disjoint partition Π, form the entire vector

$$Z_\Pi = ((X_I, Y_I))_{I\in\Pi}, \qquad X_I = \sum_{i\in I} X_i, \quad Y_I = \sum_{i\in I} Y_i. \tag{12.21}$$

An action may choose any normalized readout kernel on this vector. It can report overlapping unions of whole current blocks using the shared batch, so no independence between such overlapping reports is assumed. Query times and readouts may depend measurably on carried record history and retained state. Recording an output does not reset a belief or model law.

Corollary 12.9 (Adaptive extensive-response inheritance). For every admitted preparation and every finite sequence of finite query times selected by the preceding protocol, the fine and parent descriptions have the same joint law of records, actions, query times, and retained states. This includes the membership events of theorem 12.8.

Proof. Independence across disjoint blocks and lemma 12.1 give the conditional batch kernel

$$K_\Pi = \bigotimes_{I\in\Pi} [W_2(A_I, \mathsf{P}) \otimes W_2(B_I, \mathsf{Q})]. \tag{12.22}$$

Initialize the retained process from the same fine configuration. Condition on the retained state and carried history at a query. The same policy chooses the next waiting time and readout in both descriptions. Theorem 12.5 gives equal retained states after waiting, and definition 12.6 gives the same intervening events and addition handoffs. Freshness makes the next conditional batch law exactly eq. (12.22); composition with the same normalized readout gives the same next-record kernel. Induction proves equality of every finite joint history. Independent external randomization can be conditioned on in the same proof. □

The result is a corollary of two facts: the interface exposes only block sums, whose law depends on the fine allocation only through A_I and B_I, and the retained state is a deterministic function of the preparation and time. Records never feed back into laws, so adaptivity selects only query times and readouts, and the induction carries no further content. Every decision functional of the joint record law, such as the expected cost of an adaptive policy or the entropy of the record history, takes the same value in both descriptions; information gain about the discarded allocation does not transfer, because the parent has no such coordinate. The informative extension, in which records drive updates of the laws, is outside the declared interface.

The corollary compares two descriptions of each given preparation. A final aggregate is not claimed to reconstruct previously discarded records or the distribution of earlier records from a different preparation. Queries cutting through a current block need additional intersection data. For example,

same-batch sums over primitive subsets C, D have

$$\operatorname{Cov}(\operatorname{tr}(UX_C), \operatorname{tr}(VX_D)) = A_{C\cap D} \operatorname{tr}(\mathsf{P}^{-1}U\mathsf{P}^{-1}V), \tag{12.23}$$

which can expose a discarded shape sum. Repeated hidden-child sampling, child-specific Bayesian updates, and interventions changing the reference potential are outside the declared interface.

12.7 Inherited collective geometry

Use the trivial principal bundle $\mathcal{C}\times SO(2)$ and the common left sample action $X \mapsto gXg^T, Y \mapsto gYg^T$. It sends precisions to $g\mathsf{P}g^T, g\mathsf{Q}g^T$. The identity-precision references are invariant under this action. One rotation acting on belief and model samples simultaneously is a declared choice of collective group. Because the evaluator is constant, the constitutive data are also invariant under independent rotations of the two sectors, and nothing in the potential ties them. Use the collective product-law Fisher metric with unit weights, the common comparison frame, and the convention $D^A p = dp + J_p A$ of chapter 3. Let $\mathcal{C} = (a_-, a_+) \times (-\chi_0, \chi_0)$, with $0 < a_- < a_+$, and prepare the contextual section by

$$\mathsf{P}(0; a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a}) R_\chi^T, \qquad \mathsf{Q}(0) = \operatorname{diag}(e^{\tilde{a}}, e^{-\tilde{a}}), \qquad \tilde{a} > 0. \tag{12.24}$$

Here R_χ is plane rotation by χ. All individual shape parameters are constant over $\mathcal{C}$. Consequently the selection schedule is also constant over this contextual chart. The flow acts pointwise in the context, and no vector field, row, channel, or record depends on the connection: the connection below is a diagnostic of the evolved sections, and its curvature is the image under the flow of the prepared contextual dependence in eq. (12.24).

Theorem 12.10 (Exact connection and contextual-tensor inheritance for a prepared contextual family). For every block, before and after either join, sum retention preserves its collective minimizing connection class and minimized contextual tensor for the common $SO(2)$. For eq. (12.24), that tensor has rank two and the connection has nonzero curvature at every finite time, for every $\kappa \geq 0$.

Proof. Differentiating eq. (12.1) at fixed shape gives the precision score and Fisher metric

$$\ell_U(X) = \alpha \operatorname{tr}(\mathsf{P}^{-1}U) - \operatorname{tr}(UX), \qquad g^F(U, V) = \alpha \operatorname{tr}(\mathsf{P}^{-1}U\mathsf{P}^{-1}V). \tag{12.25}$$

The second identity follows by differentiating the normalizing integral twice. For common precision variations (U, V), the summed fine score is

$$A_I \operatorname{tr}(\mathsf{P}^{-1}U) - \operatorname{tr}(UX_I) + B_I \operatorname{tr}(\mathsf{Q}^{-1}V) - \operatorname{tr}(VY_I). \tag{12.26}$$

It equals the parent score and is measurable from the retained sums. Both contextual and common rotation-orbit directions have this form. Their Fisher pairings and all mixed pairings are therefore preserved. The infinitesimal stabilizers agree as well: in each description they are the rotations commuting with both precisions. Thus the normal equations for the minimizing connection have the same solutions modulo the same isotropy, and the residual tensors agree. This also verifies the

mixed-loss criterion in proposition 3.17. The sample addition channel is fixed, so it contributes no moving-channel derivative.

For explicit curvature, let $p_\pm(t,a)$ be the two eigenvalues of $\mathsf{P}(t;a,\chi)$, and let $\tilde{p}_\pm(t)$ be those of $\mathsf{Q}(t)$. The scalar field in eq. (12.11) has

$$f_p = -2 - \frac{2\kappa\bar{\alpha}}{p^2} - \frac{\kappa^2\alpha_0\bar{\alpha}}{(p+\kappa\bar{\alpha})^2} < 0. \tag{12.27}$$

Its solution map $\Phi_t(p_0)$ satisfies $\partial_{p_0}\Phi_t = \exp(\int_0^t f_p du) > 0$ for finite t. Hence $p_+ = \Phi_t(e^a) > p_- = \Phi_t(e^{-a}) > 0$, $\partial_a p_+ > 0$, and $\partial_a p_- < 0$. The model eigenvalues also remain distinct and have no a dependence. Define the rotational inertias

$$\mathcal{I}_1 = 2A_I\frac{(p_+ - p_-)^2}{p_+p_-}, \qquad \mathcal{I}_2 = 2B_I\frac{(\tilde{p}_+ - \tilde{p}_-)^2}{\tilde{p}_+\tilde{p}_-} > 0. \tag{12.28}$$

Writing $r = p_+/p_- > 1$ gives $\mathcal{I}_1 = 2A_I(r + r^{-1} - 2)$ and $\partial_a\mathcal{I}_1 > 0$. Radial precision variations are diagonal in an eigenframe, while orbit variations are off-diagonal, so their Fisher pairing vanishes. The collective normal equation then gives

$$A^F = -\frac{\mathcal{I}_1}{\mathcal{I}_1+\mathcal{I}_2}d\chi, \qquad h^{\min} = g_{aa}da^2 + \frac{\mathcal{I}_1\mathcal{I}_2}{\mathcal{I}_1+\mathcal{I}_2}d\chi^2, \tag{12.29}$$
$$g_{aa} = A_I\left[\left(\frac{\partial_a p_+}{p_+}\right)^2 + \left(\frac{\partial_a p_-}{p_-}\right)^2\right] > 0.$$

Identifying the one-dimensional Lie algebra with its rotation generator, the curvature is

$$F^F = dA^F = -\frac{(\partial_a\mathcal{I}_1)\mathcal{I}_2}{(\mathcal{I}_1+\mathcal{I}_2)^2}da\wedge d\chi \neq 0. \tag{12.30}$$

These computations hold for every retained block and its fine constituents with the same totals. None of them uses $\kappa > 0$. Under a passive local rotation by $\vartheta(c)$, the angular momentum one-form gains $(\mathcal{I}_1+\mathcal{I}_2)d\vartheta$, so $A^{F\prime} = A^F - d\vartheta$. The tensor is invariant, with exactly the affine connection law of the general construction. □

The minimizing connection is the mechanical connection of a two-component rotor with locked inertia $\mathcal{I}_1+\mathcal{I}_2$ [133], and $h^{\min}$ is the pullback of the quotient metric of the corresponding Riemannian submersion [113]. Its curvature measures how the ratio of belief to model rotational inertia varies along the context. Two features delimit it. First, the curvature belongs to the tied belief and model connection. For the independent group $SO(2)_q \times SO(2)_s$, which the constitutive data also admit, the Fisher cost of a connection (ξ_q,ξ_s) is $\mathcal{I}_1(d\chi+\xi_q)^2 + \mathcal{I}_2\xi_s^2 + g_{aa}da^2$, minimized by $\xi_q = -d\chi$ and $\xi_s = 0$; that connection is flat and the residual tensor $g_{aa}da^2$ has rank one. Second, the geometry is prepared rather than produced: it is present at $t = 0$ and at $\kappa = 0$, where the agents do not interact, and the next proposition shows that the flow erases it. At finite times $p_+ > p_-$, so the infinitesimal isotropy is zero and the constant-rank hypothesis of chapter 3 holds; agreement of the children's laws does not affect it, because the collective action is common.

Proposition 12.11 (Decay of the inherited contextual geometry). Let p_* be the unique positive zero of $f(\cdot, \alpha_0)$ in eq. (12.11), and put $\lambda = -f_p(p_*, \alpha_0)$, so that $\lambda > 2$ for $\kappa > 0$ and $\lambda = 2$ for $\kappa = 0$. For every context, $\mathsf{P}(t) \to p_* I_2$, and $e^{\lambda t}(p_+ - p_-)$, $e^{\lambda t}\partial_a p_+$, and $e^{\lambda t}\partial_a p_-$ converge to nonzero limits. Consequently $e^{2\lambda t} g_{aa}$ and $e^{2\lambda t}\mathcal{I}_1$ converge to positive limits, the minimized tensor tends to zero, and the limit laws are isotropic, with stabilizer all of $SO(2)$. For $\alpha_0 = \kappa = 1$, $p_* = \sqrt{2}$ and $2\lambda = 12 - 4\sqrt{2}$.

Proof. Since $f_p < 0$ and $f(p, \alpha_0)$ tends to $+\infty$ as $p \to 0$ and to $-\infty$ as $p \to \infty$, the zero p_* is unique. On the compact invariant region of the proof of proposition 12.2, f and f_p are Lipschitz. With $y = p - p_*$, write $\dot{y} = [f(p, \bar{\alpha}) - f(p_*, \bar{\alpha})] + [f(p_*, \bar{\alpha}) - f(p_*, \alpha_0)]$. Since $f_p \le -2$, the first bracket has the sign of $-y$ and modulus at least $2|y|$, and the second is bounded by a constant times $|\bar{\alpha} - \alpha_0| = O(e^{-2t})$. Hence $|y(t)| \le (|y(0)| + Kt)e^{-2t}$, uniformly for initial values in a compact set. Then $f_p(p(u), \bar{\alpha}(u)) + \lambda$ is integrable on $[0, \infty)$, so $\partial_{p_0}\Phi_t = e^{-\lambda t}(c(p_0) + o(1))$ with $c(p_0) > 0$ continuous, uniformly on compact sets of p_0. The identities $\partial_a p_\pm = \pm e^{\pm a}\partial_{p_0}\Phi_t(e^{\pm a})$ and $p_+ - p_- = \int_{e^{-a}}^{e^a} \partial_{p_0}\Phi_t(p_0)dp_0$ give the stated limits, and the formulas for g_{aa} and $\mathcal{I}_1$, with $A_I \to n_I\alpha_0$ and $p_\pm \to p_*$, give the rest. For $\alpha_0 = \kappa = 1$, substitution verifies $f(\sqrt{2}, 1) = 0$ and $f_p(\sqrt{2}, 1) = -6 + 2\sqrt{2}$. □

For the example constants and the block of the first two agents at $a = 3/10$, g_{aa} equals 4.40, 2.69×10^{-3}, and 5.20×10^{-6} at $t = 0, 1, 2$; by the root join at $t_2 \approx 1.07$ it is below 10^{-3} of its initial value. The model inertia decays at the rate given by the model analog of λ. Because A^F is a ratio of two vanishing inertias, the limit of the finite-time curvature depends on whether the two rates agree: for $\alpha_0 = \eta_0 = \kappa = 1$ it tends to approximately -1.81, while for $\alpha_0 = 1, \eta_0 = 2$ the connection tends to the flat value $-d\chi$ and the curvature to zero. In neither case is the limit a property of the limit laws, at which the connection is undetermined.

Remark 12.12 (Persistent contextual geometry from anisotropic references). Replace the belief evaluator by $W_2(\alpha_0, \Gamma(c))$ with $\Gamma(a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a})R_\chi^T$, the belief channel target by $W_2(\alpha_0, \Lambda_q)$ with $\Lambda_q = \Gamma + \kappa M_q/N$, and the model reference by $W_2(\eta_0, \operatorname{diag}(e^{\tilde{a}}, e^{-\tilde{a}}))$, with the analogous model channel. The calculation of proposition 12.2 gives $C_q = \Lambda_q + (\kappa/N)(M_q - N\alpha_0\Lambda_q^{-1})$ and $\dot{\mathsf{P}} = -2\mathsf{P} + \Gamma + C_q$, with the same shape law. Precisions remain common, evaluators and targets still add under sums, and theorem 12.5, corollary 12.9, proposition 12.7, and the selection schedule are unchanged. With the section $\mathsf{P}(0; a, \chi) = \Gamma(a, \chi)$, the eigenvalues converge to the zeros of $-2p + 2\gamma + 2\kappa\alpha_0/p - \kappa\alpha_0 p/(\gamma p + \kappa\alpha_0)$ with $\gamma = e^{\pm a}$, which differ and depend on a; the limit tensor and curvature are computed from these fixed points. For the example constants at $a = \tilde{a} = 3/10$ the block tensor tends to $\operatorname{diag}(1.168, 0.199)$ and the curvature to approximately -1.689, so rank two and nonzero curvature persist at infinite time. Both sectors need anisotropic local data; with an isotropic model reference the model sector isotropizes and the curvature tends to zero. The local data are then contextual sections that transform under the common action, so the statement above that the references are invariant no longer holds; the curvature remains a property of the declared contextual data, not of the interaction.

The context-constant allocation assumption is substantive. If $d\eta_1 = 1, d\eta_2 = -1$ while precisions and all other shapes are fixed, the parent model sum has zero derivative, whereas the fine score is the centered difference of two log determinants with positive variance. It is orthogonal to rotation scores because determinant is conjugation invariant. Its fine minimized length is positive and its retained length is zero. Such a contextual direction violates exact tensor inheritance.

Open problem 12.13 (Approximate inheritance off the common-precision family). The common-precision family has positive codimension, and exact sum laws fail off it. By eq. (12.10) and the shape law, the ratio of precision spread to shape spread is constant in time, so at a join the precision mismatch is δ times a preparation constant. Under moment matching, the first-order term of the log-Laplace transform of a sum of Wishart laws with nearby precisions cancels, so the sum law deviates from the Wishart family at second order in the precision spread. Prove that on an open set of Wishart preparations with unequal precisions the moment-matched parent inherits the law, the dynamics, and the extensive responses up to a stated relative-entropy defect at each join that decays afterward, and determine its order.

Chapter 13

Preferential Two-Stage Selection from Open Law Classes

This chapter concerns the reciprocal law potential of definition 4.2 with exact Gibbs rows (proposition 4.3); no statement below is transferred to the per-agent or global free energies of chapter 4. The common-precision construction selects an exact additive description within a prepared family. The construction given here instead admits initial laws ranging over a full open neighborhood. Native source rows, read by a fixed threshold selector, select three pairs and then a pair of their parents. The selection reads only a transmitted bit. The compression is an orientation symmetry of a fixed selective channel; it exists before the selected events and descends for every partition at every time. What the events change is the resolution of a declared external response interface, and the information discarded at a handoff is exactly the information that this interface stops resolving there. Measured against the formation standard definition 6.31, the main theorem establishes selection, exact inheritance, and compression on an open class over a finite window. It does not establish interaction necessity (F5): in the regime of its proof, integrations indicate that the null flow of definition 6.30 produces the same membership history (remark 13.6). Every block relation is transient, and the selector later merges the exterior pair (corollary 13.10). The construction is therefore a two-stage selection with exact inheritance, not a recursive formation.

13.1 A homogeneous selective-channel model

Every primitive belief and model space is $\mathsf{K} = \mathsf{M} = \{-1, 1\} \times S^1$. Use the probability measure

$$d\nu = \frac{1}{2} \sum_{z=\pm 1} \delta_z \otimes \frac{d\theta}{2\pi}. \tag{13.1}$$

All densities below are relative to this measure. The $2N$ normalized densities q_i, s_i vary independently over positive densities of class C^r in the angle, with $r \geq 3$ and logarithms bounded in C^r; this is the chart of the chapter. Internal bit-angle correlations are unrestricted. Recognition is the declared product $q_i \otimes s_i$. Take uniform proper references and evaluators, $r_i = G_i = 1$, equal unit Fisher mobilities, a fixed temperature $T > 0$, and all-source priors $\pi_{ij} = 1/N$, including self.

Use the same fixed normalized comparison channel in both sectors and on every edge, self edges included:

$$\mathcal{K}((z, \theta), d(z', \theta')) = \delta_z(dz') \frac{d\theta'}{2\pi}. \tag{13.2}$$

It transmits the bit and replaces the angle by an independent uniform draw. There is no channel switch at a selected event. In the terminology of chapter 8, the bit is the public coordinate, meaning

the coordinate that peer channels transmit, and the angle is a private coordinate with Haar private reference. The word public is used only in this sense below. The channel is the S^1 twirl of the angular coordinate, which averages a law over all rotations of its angle; it is the map that describes communication between parties that share no angular reference frame [10].

If

$$\mu_i(z) = \int q_i(z,\theta)\frac{d\theta}{2\pi}, \qquad \eta_i(w) = \int s_i(w,\psi)\frac{d\psi}{2\pi}, \tag{13.3}$$

then $q_j\mathcal{K}$ has density $\mu_j(z)$, and the model output has density $\eta_j(w)$. The chain rule gives

$$D_{\mathrm{KL}}(q_i\|\mu_j\otimes\mathrm{Haar}) = H_i^q + D_{\mathrm{KL}}(\mu_i\|\mu_j), \qquad H_i^q = D_{\mathrm{KL}}(q_i\|\mu_i\otimes\mathrm{Haar}). \tag{13.4}$$

The receiver term H_i^q cancels from the Gibbs normalization. Thus

$$\beta_{ij} = \frac{\exp[-D_{\mathrm{KL}}(\mu_i\|\mu_j)/T]}{\sum_k \exp[-D_{\mathrm{KL}}(\mu_i\|\mu_k)/T]}, \tag{13.5}$$

and γ is the same expression in η. All entries are positive. The cancellation uses the selective self channel. With the identity self channel of definition 4.2 one would have $D_{ii}^q = 0$ while every peer cost retains H_i^q, so each peer weight would carry the factor $e^{-H_i^q/T}$ relative to self; at the small temperatures used below this suppresses every peer weight unless H_i^q is of order T. The construction therefore requires the selective self channel, which is a constitutive choice. The angular term remains in the reduced energy:

$$\mathcal{E}_{\mathrm{red}} = \sum_i \big[D_{\mathrm{KL}}(q_i\|1) + D_{\mathrm{KL}}(s_i\|1)\big] - T\sum_i \log\left(\frac{1}{N}\sum_j e^{-D_{ij}^q/T}\right) - T\sum_i \log\left(\frac{1}{N}\sum_j e^{-D_{ij}^s/T}\right), \tag{13.6}$$

where D_{ij}^q is the left side of eq. (13.4).

Proposition 13.1 (Complete density force). On the positive chart, the exact unit-Fisher field is

$$\begin{aligned} h_i^q(z,\theta) &= 2\log q_i(z,\theta) - \sum_j \beta_{ij}\log\mu_j(z) - \sum_k \beta_{ki}\frac{\mu_k(z)}{\mu_i(z)}, \\ \partial_t q_i &= -q_i\big(h_i^q - \mathbb{E}_{q_i} h_i^q\big), \end{aligned} \tag{13.7}$$

with the exchanged formula for s, η, γ. Every sum includes self.

Proof. The local KL contributes $\log q_i$ modulo a constant. Differentiating a reduced row by its envelope identity gives its weighted divergence derivatives. The normalized receiver row contributes another $\log q_i - \sum_j \beta_{ij}\log\mu_j$. Varying a source marginal in every incoming row gives $-\sum_k \beta_{ki}\mu_k/\mu_i$. The fixed channel has no law derivative. Fisher inversion on normalized densities subtracts the q_i expectation and multiplies by $-q_i$.

For a self term the incoming ratio is one, but $D_{ii}^q = H_i^q$ need not vanish. Its constant incoming contribution and its angular entropy are both included. Logarithms and marginal ratios are smooth on a neighborhood bounded away from zero. Bounded integration, multiplication, and finite Gibbs normalization preserve C^r. The field contains no angular derivative, so it is locally Lipschitz both

for the C^r norm of log-densities and for their supremum norm, and it has a unique local interior flow on the chart. Its centering preserves normalization. □

13.2 Two stages selected by native routing

For a current partition $\mathcal{P}$, retain the actual block rows

$$B^q_{IJ} = \frac{1}{|I|}\sum_{i\in I}\sum_{j\in J}\beta_{ij}, \qquad B^s_{IJ} = \frac{1}{|I|}\sum_{i\in I}\sum_{j\in J}\gamma_{ij}. \tag{13.8}$$

Self mass remains in each normalized row. Its experiment chooses a receiver uniformly in I, draws its native source index, and reports the source block. If blocks are joined, finite summation gives

$$B^a_{CD} = \frac{1}{|C|}\sum_{I\subset C}|I|\sum_{J\subset D}B^a_{IJ}, \qquad a\in\{q,s\}. \tag{13.9}$$

This is associative aggregation with inherited primitive counts.

Definition 13.2 (Preferential pair rule). Fix $\varepsilon = 1/10$ and $\delta = 1/20$. Distinct current blocks I, J qualify when, in both sectors and directions,

$$B^a_{IJ} > \varepsilon, \qquad B^a_{IJ} - \sum_{K\in\mathcal{P}\setminus\{I,J\}} B^a_{IK} > \delta. \tag{13.10}$$

The exterior excludes self and the proposed partner. A partner need not beat the self loop. Merge qualifying disjoint pairs at their first closed-threshold contact when the strict inequalities hold immediately afterward: the event time is the first time at which the closed inequalities hold and the strict inequalities hold on a right neighborhood. In every event proved below the relevant crossing is transverse, so this time is the unique zero of a block-row difference with positive time derivative. A block cannot have two qualifying partners: each would have to exceed the other by more than δ. The rule is therefore unambiguous at simultaneous disjoint events and respects relabeling. Persistence below means strict contrast over the stated positive time windows.

The routing calculation first uses binary densities $\mu_i(z) = 1 + zx_i$, $\eta_i(w) = 1 + wy_i$, with $(x, y) \in (-1, 1)^{2N}$. Define $d(u, v) = D_{\mathrm{KL}}(\mu_u\|\mu_v)$. Its derivatives are

$$\partial_u d(u,v) = \operatorname{atanh} u - \operatorname{atanh} v, \qquad \partial_v d(u,v) = \frac{v-u}{1-v^2}. \tag{13.11}$$

The exact binary field is

$$\begin{aligned}\dot{x}_i = (1-x_i^2)\left[-\operatorname{atanh} x_i - \sum_j \beta_{ij}(\operatorname{atanh} x_i - \operatorname{atanh} x_j)\right] \\ + \sum_k \beta_{ki}(x_k - x_i).\end{aligned} \tag{13.12}$$

The model equation is identical and independent. Self summands vanish in this binary equation, consistently with $d(x_i, x_i) = 0$.

Lemma 13.3 (Interior binary evolution). Every binary initial configuration has a unique global interior solution, and all its means tend to zero.

Proof. At a maximum of $\vartheta_i = \operatorname{atanh} x_i$, both peer sums in the equation for $\dot{\vartheta}_i$ are nonpositive; at a minimum they are nonnegative. The finite maximum and minimum satisfy the corresponding one-sided differential inequalities. Scalar comparison yields

$$e^{-t} \min_i \vartheta_i(0) \le \vartheta_i(t) \le e^{-t} \max_i \vartheta_i(0). \tag{13.13}$$

The same holds for the model logits. The solution stays in a compact interior box, where the smooth field extends it past any proposed finite maximal time. The bounds imply convergence. This does not by itself establish the order of selection events. □

The perturbation steps below all use one elementary statement. It compares a reference trajectory with a perturbed one and tracks both the trajectory and the observables that define events, since the observables themselves change with each perturbation.

Lemma 13.4 (Persistence of strict windows). Let X solve $\dot{X} = \Phi(X)$ on $[0, L]$ in a finite-dimensional space with the supremum norm, and let K_ρ be the tube of closed balls of radius ρ about $X(t)$. On K_ρ suppose that a possibly time-dependent field Ψ satisfies $|\Psi - \Phi| \le e$, that Φ is C-Lipschitz on each slice, and that $|\Phi|, |\Psi| \le M$. Let O be a C^1 observable with $|\nabla O|_1 \le G$ and ∇O Lipschitz with constant L_1, and let $\widetilde{O}$ satisfy $|\widetilde{O} - O| \le \eta_0$ and $|\nabla \widetilde{O} - \nabla O|_1 \le \eta_1$ on K_ρ. If $\dot{Y} = \Psi(Y)$, $|Y(0) - X(0)| \le \delta_0$, and

$$\Delta = \delta_0 e^{CL} + e \frac{e^{CL} - 1}{C} < \rho, \tag{13.14}$$

then $Y(t) \in K_\rho$ on $[0, L]$, and

$$\begin{aligned} |\widetilde{O}(Y) - O(X)| &\le \eta_0 + G\Delta, \\ \left| \tfrac{d}{dt}\widetilde{O}(Y) - \tfrac{d}{dt} O(X) \right| &\le \eta_1 M + L_1 \Delta M + G(e + C\Delta). \end{aligned} \tag{13.15}$$

Suppose $|O(X) - \theta_0| \ge m$ outside an interval I, $O(X) - \theta_0$ has opposite signs at the ends of I, and $\frac{d}{dt} O(X) \ge s > 0$ on I. If the two bounds in eq. (13.15) are below m and s respectively, then $\widetilde{O}(Y) - \theta_0$ has exactly one zero in $[0, L]$, it lies in I, and it is transverse; strict windows of the reference persist.

Proof. While Y is in the tube, $|Y - X|' \le |\Psi(Y) - \Phi(Y)| + |\Phi(Y) - \Phi(X)| \le e + C|Y - X|$ because each slice is convex. Gronwall's inequality bounds $|Y - X|$ by $\Delta < \rho$, so the first exit time cannot occur. The first observable bound follows from $|\widetilde{O}(Y) - O(Y)| + |O(Y) - O(X)|$ and the mean value theorem on the convex slice. For the second, write $\frac{d}{dt}\widetilde{O}(Y) = \nabla \widetilde{O}(Y) \cdot \Psi(Y)$ and $\frac{d}{dt} O(X) = \nabla O(X) \cdot \Phi(X)$, and split the difference into the four terms $(\nabla \widetilde{O} - \nabla O)(Y) \cdot \Psi(Y)$, $(\nabla O(Y) - \nabla O(X)) \cdot \Psi(Y)$, $\nabla O(X) \cdot (\Psi - \Phi)(Y)$, and $\nabla O(X) \cdot (\Phi(Y) - \Phi(X))$. Outside I the perturbed observable keeps the reference sign; inside I it is strictly increasing and changes sign once. □

Proposition 13.5 (Two-stage preferential routing). For $N = 6$, some fixed $T > 0$ and a nonempty relabeling-invariant open set in $(-1, 1)^{12}$ have the following behavior. Every initial off-diagonal native weight is below $1/10$. Three preferential pairs form and coexist for a positive interval. Two of their parents subsequently join while the third remains an exterior with strictly smaller routing masses. The first pairs and the selected parent pair satisfy eq. (13.10) over positive windows. All events occur at finite interior times under eq. (13.12), with no parameter change.

Proof. Choose $T = h^2$ and write $x_i = hu_i$. The function $h^{-2}d(hu, hv)$ is even and analytic in h on $h \max(|u|, |v|) < 1$, with

$$h^{-2}d(hu, hv) = \frac{(u - v)^2}{2} + \frac{h^2}{12}(u - v)^2(u^2 + 2uv + 3v^2) + O(h^4). \tag{13.16}$$

Dividing eq. (13.12) by h, without changing time, gives a field and block rows that are analytic in (h^2, u) on $\{h|u|_\infty < 1\}$. At $h = 0$ the field is

$$\dot{u}_i = -u_i + \sum_j (w_{ij} + w_{ji})(u_j - u_i), \qquad w_{ij} = \frac{e^{-(u_i - u_j)^2/2}}{\sum_k e^{-(u_i - u_k)^2/2}}. \tag{13.17}$$

The self summand one bounds every denominator away from zero. Limit trajectories remain bounded on finite intervals by the same extrema comparison. The limit eq. (13.17) is the Euclidean gradient flow of the Gaussian log-sum-exp energy $\sum_i u_i^2/2 - \sum_i \log \sum_k e^{-(u_i - u_k)^2/2}$. Its pair formation is the clustering mechanism of Gaussian mean shift and its blurring variant [25, 31] and of self-attention particle dynamics [64]; here it is obtained from the original KL source rows at low temperature. The rest of the proof constructs a reference trajectory of eq. (13.17) with strict windows and transfers them three times by lemma 13.4: from decoupled clusters to the six-agent limit field, from the limit field to the exact scaled field, and from symmetric to open initial data.

For four agents first take the reflection-invariant configuration

$$u = (-a - b, -a + b, a - b, a + b), \qquad a(0) = A, \quad b(0) = 2. \tag{13.18}$$

Here a is half the distance between the two pair centers and b is the internal half-gap. Set $w = e^{-2b^2}$, $v_0 = e^{-2a^2}$, $v_\pm = e^{-2(a \pm b)^2}$, $Z_\pm = 1 + w + v_0 + v_\pm$, and

$$c_p = w(Z_+^{-1} + Z_-^{-1}), \qquad c_0 = v_0(Z_+^{-1} + Z_-^{-1}). \tag{13.19}$$

Substitution into the full field gives

$$\begin{aligned} \dot{a} &= -a - 2ac_0 - 2\left[\frac{(a + b)v_+}{Z_+} + \frac{(a - b)v_-}{Z_-}\right], \\ \dot{b} &= -b - 2bc_p - 2\left[\frac{(a + b)v_+}{Z_+} - \frac{(a - b)v_-}{Z_-}\right]. \end{aligned} \tag{13.20}$$

Both unequal row normalizers are retained.

For $a \geq 4$ and $0 \leq b \leq 2$, put $c = 1 + w + v_0 \geq 1$. The function $g_c(r) = re^{-2r^2}/(c + e^{-2r^2})$ satisfies

$$|g_c'(r)| \leq (1 + 4r^2)e^{-2r^2}. \tag{13.21}$$

This bound decreases for $r \geq 2$, so $|g_c(a+b) - g_c(a-b)| \leq 34be^{-8}$. Since $0 \leq c_p \leq 1$, $2ac_0 \leq 4ae^{-32}$, and $(a+b)v_+ + (a-b)v_- \leq 2av_- \leq 2ae^{-8}$, eq. (13.20) implies

$$-(3 + 68e^{-8})b \leq \dot{b} \leq -(1 - 68e^{-8})b, \qquad -(1 + 4e^{-32} + 4e^{-8})a \leq \dot{a} \leq -a. \tag{13.22}$$

Here $1 - 68e^{-8} > 0.977$ and $1 + 4e^{-32} + 4e^{-8} < 1.0014$. The positive coordinates cannot first reach their invariant zero axes at finite time.

Take $A \geq 34$. The first time τ with $b = 1/2$ satisfies $\tau \leq \log 4/0.977 < 1.42$, and $a(t) \geq Ae^{-1.0014 \cdot 1.42} > 0.239A \geq 8$ through this interval. Initial peer weights are at most e^{-8} and cross weights much smaller. A directed within-pair weight is $p_\pm = w/Z_\pm$. While $b \geq 1/2$, $\dot{w} \geq w/2$, whereas

$$|\dot{v}_0 + \dot{v}_\pm| \leq 8a^2e^{-2a^2} + 4(a+2)(2a+8)e^{-2(a-2)^2} < 1/100 \tag{13.23}$$

for $a \geq 8$. Both positive terms decrease on that range. Differentiating $p_\pm$ therefore gives $\dot{p}_\pm > 0$. At τ,

$$p_\pm \geq \frac{e^{-1/2}}{1 + e^{-1/2} + 2e^{-8}} > 1/4. \tag{13.24}$$

Each peer threshold is crossed transversely, with exterior mass less than $2e^{-8}$. The contrast is strict at entry.

Until the first time t_A with $a = 4$, the pair weights remain above $1/4$ and every cross weight is at most e^{-8}. An unwanted parent source column has at most two such entries, excluding a premature join even if one pair has formed before the other. Moreover,

$$\frac{\log(A/4)}{1.0014} \leq t_A \leq \log(A/4), \qquad b(t_A) \leq 2(A/4)^{-0.975}. \tag{13.25}$$

Thus the pairs coexist while their residual internal gaps become arbitrarily small.

Starting at $a = 4$, compare with the invariant trajectory $b = 0$. Its exact inherited cross mass and primitive peer weight are

$$B = \frac{e^{-2a^2}}{1 + e^{-2a^2}}, \qquad p = \frac{1 - B}{2}, \qquad \dot{a} = -a(1 + 4B). \tag{13.26}$$

Here $\dot{B} = 4a^2e^{-2a^2}(1 + 4B)/(1 + e^{-2a^2})^2 > 0$. The parent threshold is crossed at $a = \sqrt{\log 9/2}$. Stop this comparison when $B = 1/5$, at a finite time $S \leq \log(4/\sqrt{\log 4/2})$. Throughout it, $p \geq 2/5$ and the primitive exterior mass is at most $1/5$, so first-pair contrast is at least $1/5$. The parent crossing has a positive derivative. By lemma 13.4, with $\Psi = \Phi$ given by eq. (13.20) and the block rows as observables, these compact margins and the transverse crossing persist when the initial half-gap at $a = 4$ is below some $\varepsilon_b > 0$. This ε_b does not depend on A, since the comparison starts at $a = 4$. Choose one finite $A \geq 34$ with $2(A/4)^{-0.975} < \varepsilon_b$ using eq. (13.25), and fix $L = t_A + S$. The parent event then occurs between $\log(A/4)/1.0014$ and $\log(A/4) + S$: the second stage happens when the supplied core separation has contracted to order one.

A third pair makes the second event have a genuine exterior. Add initial coordinates $R-2, R+2$. In the decoupled comparison the two-agent interaction is antisymmetric, so their center is Re^{-t}, and the half-gap b_c obeys

$$\dot{b}_c = -b_c(1+4p_c), \qquad p_c = \frac{e^{-2b_c^2}}{1+e^{-2b_c^2}}, \qquad b_c(0) = 2. \tag{13.27}$$

Its threshold is transverse, and by $\log 4$ its peer weight exceeds $1/4$.

Let $X(t)$ be the decoupled reference, the four-agent core trajectory together with this two-agent trajectory, and compare it in the original coordinates with the six-agent limit field. On the radius-one supremum-norm tube about X, the core coordinates satisfy $|u_i| \le A+3$, because the extrema comparison keeps the reference core in $[-A-2, A+2]$, and the pair coordinates lie within 3 of Re^{-t}. Every core-pair separation is therefore at least $D_R = Re^{-L} - A - 6$. Let $\varphi(D) = e^{-D^2/2}$. For an agent in one cluster, the six-agent row differs from the decoupled row through the changed normalizer, $w_{ij} - \widetilde{w}_{ij} = -\widetilde{w}_{ij}\epsilon_i/Z_i$ with ϵ_i the cross-cluster numerator mass, in both the outgoing and incoming directions, and through the incoming and outgoing cross forces. Using $\widetilde{w}|x| \le e^{-1/2}$ and $|x|\varphi(|x|) \le D\varphi(D)$ for $|x| \ge D \ge 1$, these contributions give

$$|\Psi - \Phi|_\infty \le 8\big(e^{-1/2} + D_R\big)\varphi(D_R). \tag{13.28}$$

The same quotient-rule bookkeeping bounds the differences of the Jacobians and of the block-row gradients by absolute multiples of $(1+D_R^2)\varphi(D_R)$; for the core-parent row, $|B^{(6)} - B^{(4)}| \le 2\varphi(D_R)$. None of these constants depends on A or R. The decoupled field is the limit field on each component. Its interaction depends only on differences, and $w|x|$ and wx^2 are bounded, so its Lipschitz constant on the tube is bounded independently of A and R. Choosing R sufficiently large and finite, lemma 13.4 preserves every window and crossing of the reference and keeps every aggregated mass between the core and the third pair below 10^{-3} through L. At the core-parent crossing the contrast is then greater than $0.099 > 1/20$; the third pair cannot join either core block or their union.

Fix this R and let X now denote the six-agent limit trajectory on $[0, L]$. On the radius-$1/2$ tube about it, $|u|_\infty \le R+3$. For $h(R+3) < 1$, analyticity in (h^2, u) makes the exact scaled field and its block rows converge to the limit field and its rows in C^1, uniformly on this compact tube, as $h \to 0$. Lemma 13.4 with $\delta_0 = 0$ gives one sufficiently small fixed $h > 0$ for which the exact scaled binary field has the same strict windows and transverse crossings. Choose equal sector trajectories only as a reference, then perturb all twelve initial coordinates independently. Lemma 13.4 with $\Psi = \Phi$ preserves the windows and the positive crossing derivatives for sufficiently small initial perturbations. Each pair's event is the latest of its four directional-sector crossings, before the next separated window. The means can be initially distinct; their invariant equality diagonals and uniqueness preclude a first finite collision. Finally take the union of all population permutations of this open neighborhood. The constitutive data contain no block labels; the initial configuration chooses the eventual blocks through its scale hierarchy. □

Remark 13.6 (Null-flow control). The null flow of definition 6.30 deletes every peer cost. Here it keeps the angular self terms and gives, on the binary family, the decoupled field $\dot{x}_i = -(1-x_i^2)\operatorname{atanh} x_i$, whose scaled limit is $\dot{u} = -u$, so $u(t) = e^{-t}u(0)$. Read by the same rows and

definition 13.2, its membership history is a function of the initial positions alone: the rows at time t are a Gaussian kernel of bandwidth e^t on $u(0)$. Integrations of eq. (13.17) and of this null limit from the initial data of the proof, at $A = 8$, 64, and 4096 with a distant third pair, produce the same three pairs, the same core-parent join with the same exterior, and the same order of events; the pair events move from $t \approx 0.589$ to 0.647, and at $A = 4096$ the parent join moves from $t \approx 8.214$ to 8.271. These are integrations of the limit field, not of the exact scaled field at the fixed h of the proof, and they are not part of any statement proved here. What is proved is weaker and one-sided: while $a \geq 4$ the internal half-gap contracts along the full field at a rate between 0.977 and 3.03 by eq. (13.22), against exactly 1 along the null limit, so the peer forces tighten the selected pairs. Neither interaction necessity (F5) nor the amplification grade of definition 6.31 is established by proposition 13.5.

Remark 13.7 (Size of the constants). The witness is nonconstructive. With $A = 4096$ the core reference has $t_A \approx 6.93$ and $L \approx 8.36$, and a third pair separated by D_R of order ten requires $R \approx 1.8 \times 10^7$. Admissibility $h|u|_\infty < 1$ alone then forces T below about 3×10^{-15}. The tube step as written is more restrictive: for a far agent the scaled field is $-h^{-1}(1 - h^2u^2)\operatorname{atanh}(hu) = -u + \frac{2}{3}h^2u^3 + O(h^4u^5)$, so the uniform field error on the tube contains the common-mode term $\frac{2}{3}h^2R^3$, and keeping the far pair within a radius-$1/2$ tube needs $T = h^2$ below about 10^{-21} before any Gronwall amplification. Existence is unaffected, since h is only required to be sufficiently small. A comparison relative to each cluster's exact center would remove the common-mode error and need only hR small; we do not carry it out. The early phase above works from $A \geq 34$; a numerical interval enclosure of the late comparison indicates that $\varepsilon_b = 1$ suffices, which would make $A = 34$ admissible, but that enclosure is not part of the proof. The temperature of the theorem below is not quantified.

13.3 A full open domain of correlated laws

The binary result embeds exactly into eq. (13.7). The embedding extends to every correlated law, because the conditional angular laws evolve autonomously.

Proposition 13.8 (Exact reduction of the bit-circle flow). Write $q_i(z, \theta) = \mu_i(z) f_i(\theta \mid z)$ and $s_i(w, \psi) = \eta_i(w) g_i(\psi \mid w)$ for positive laws with bounded log-densities. Along eq. (13.7),

$$f_i(\theta \mid z, t) = \frac{f_i(\theta \mid z, 0)^{e^{-2t}}}{\int f_i(\psi \mid z, 0)^{e^{-2t}} d\psi/(2\pi)}, \tag{13.29}$$

for each z separately, and the bit means obey eq. (13.12) plus the exogenous forcing

$$-\frac{1}{2}(1 - x_i^2)\big[c_i(1, t) - c_i(-1, t)\big], \qquad c_i(z, t) = 2D_{\mathrm{KL}}\big(f_i(\cdot \mid z, t) \big\| \mathrm{Haar}\big), \tag{13.30}$$

with the exchanged statements in the model sector. The solution exists for all $t \geq 0$, and all bit and model means tend to zero. Rows, block rows, and the bit forcing are invariant under independent rotations of every belief and every model angle.

Proof. In eq. (13.7), $2 \log q_i = 2 \log \mu_i + 2 \log f_i$, so $h_i^q = 2 \log f_i(\theta \mid z) + \phi_i(z)$, where ϕ_i collects all terms depending on the bit alone. Integrating $\partial_t q_i = -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q)$ over the angle gives

$\partial_t \log \mu_i(z) = -(\mathbb{E}[h_i^q \mid z] - \mathbb{E}_{q_i} h_i^q)$. Subtracting, $\partial_t \log f_i = -2(\log f_i - \mathbb{E}[\log f_i \mid z])$, a separate Fisher-Rao relaxation toward Haar measure for each z, whose solution is eq. (13.29). Since $\mathbb{E}[h_i^q \mid z] = c_i(z,t) + \phi_i(z)$, the bit part reproduces the binary field and adds $-\frac{1}{2}\mu_i(1)\mu_i(-1)[c_i(1,t) - c_i(-1,t)]$ to $\dot{x}_i$, with $\mu_i(1)\mu_i(-1) = 1 - x_i^2$. In logit coordinates the forcing is bounded by $\sup_z c_i(z,0)$, because the relative entropy decreases along eq. (13.29); the extremum comparison of lemma 13.3 with this bounded forcing keeps the logits bounded on every finite interval, and since $c_i(z,t) \to 0$ it gives convergence to zero. The explicit solution satisfies eq. (13.7) and therefore coincides with its unique local solution. Rows depend only on bit marginals, and c_i is rotation invariant. □

The conditional relaxation eq. (13.29) is the explicit solution of the Fisher-Rao gradient flow of a relative entropy [30], and the reduction is the Haar-reference analogue of theorem 8.5, with relaxation rate two. The receiver term $H_i^q = D_{\mathrm{KL}}(q_i \| \mu_i \otimes \mathrm{Haar})$ is the μ_i-average of $c_i(z,t)/2$. For product laws $c_i(1,t) = c_i(-1,t)$ and the forcing vanishes. In particular, for arbitrary positive angular profiles, the product reference family

$$q_i(z,\theta) = \mu_i(z) f_i(\theta), \qquad s_i(w,\psi) = \eta_i(w) g_i(\psi) \tag{13.31}$$

is invariant, and its angular factors obey

$$\partial_t f_i = -2 f_i(\log f_i - \mathbb{E}_{f_i} \log f_i), \qquad f_i(\theta,t) = \frac{f_i(\theta,0)^{e^{-2t}}}{\int f_i(\psi,0)^{e^{-2t}} d\psi/(2\pi)}. \tag{13.32}$$

A von Mises factor $p_\kappa(\theta - a) = e^{\kappa\cos(\theta-a)}/I_0(\kappa)$ keeps its phase and has concentration κe^{-2t}.

Proposition 13.9 (Open-law transfer). The routing behavior of proposition 13.5 persists on a nonempty relabeling-invariant set of independently variable positive belief and model densities on $\{-1,1\} \times S^1$, including correlated bit-angle laws. The set is open for uniform convergence of log-densities, and hence in the C^r chart.

Proof. Let M bound the oscillation of every log-conditional $\log f_i(\cdot \mid z)$. For $\ell = \log f$ and $\rho = e^{-2t}$ write $\mathcal{R}_\rho(\ell)$ for the relative entropy to Haar measure of the law proportional to $e^{\rho\ell}$. Its derivative in a direction ς is $\rho^2 \operatorname{Cov}(\ell, \varsigma)$ under that law, which is at most $\rho^2 M \|\varsigma\|_\infty$ in absolute value along the segment between the two log-conditionals. Hence

$$|c_i(1,t) - c_i(-1,t)| \le 2e^{-4t} M \big\| \log f_i(\cdot \mid 1,0) - \log f_i(\cdot \mid -1,0) \big\|_\infty. \tag{13.33}$$

In the scaled coordinates of proposition 13.5, the forcing eq. (13.30) is therefore a time-dependent field perturbation of size at most $e^{-4t} M\epsilon/h$, where ϵ is the largest of these conditional differences over agents and sectors. Take the binary open set of proposition 13.5, shrunk so that its windows survive a field error $e_0 > 0$ by lemma 13.4. Every law whose bit and model means lie in it and whose conditionals satisfy $M\epsilon < h e_0$ has the routing behavior. This condition is open for uniform convergence of log-densities and contains every product law eq. (13.31); rows and block rows are the binary ones by proposition 13.8. Admissible bit-angle correlations are thus of order h. □

Corollary 13.10 (Transient relations and a forced final merge). For every initial law of proposition 13.9, every native row tends to $1/N$. Every first-stage contrast tends to $1/6 - 4/6 < 0$

and the core-parent contrast against the third pair tends to $0 < \delta$, so no block relation used in proposition 13.5 is forward invariant. After the core-parent join the partition has two blocks, the exterior sum in eq. (13.10) is empty, and the remaining block rows tend to $2/6$ and $4/6$. Under definition 13.2 every such trajectory therefore reaches the one-block partition at a finite time after L, with an empty exterior. Under the retention below, this last join discards one further relative orientation, since $\mathsf{H}_{I\cup J} \times \mathsf{H}_K \subsetneq \mathsf{H}_{\{1,\dots,6\}}$.

Proof. All means tend to zero by proposition 13.8, so every divergence in eq. (13.5) tends to zero. With two blocks the contrast condition reduces to $B^a > \varepsilon$ in both directions and sectors. The cross block rows are below 10^{-3} at the core-parent join and tend to limits above ε, so there is a finite time after which every strict inequality holds, and the event occurs no later than that time. □

The persistence in this chapter is therefore finite-window by construction. The integrations of remark 13.6 show the same final merge along the null limit flow.

13.4 Orientation compression and inherited dynamics

For an angle a, let $(\tau_a f)(z,\theta) = f(z,\theta - a)$. An active rotation of agent i applies τ_{a_i} to its belief and model laws in fixed sample coordinates. It changes the laws, unlike the passive changes of presentation of chapter 2. The energy, metric, rows, and eq. (13.7) are invariant or equivariant under independent active rotations of every belief and every model angle, a group $(S^1)^{2N}$. Below we use the diagonal subgroup $(S^1)^N$ that rotates both laws of one agent by the same angle. Whether an active rotation is detectable is a statement about a declared observable algebra (definition 13.14): no agent, channel, row, or evaluator of the population resolves it, and only the external response protocol of the next section can.

For a block I discard the subgroup

$$\mathsf{H}_I = \{(a_i)_{i\in I} : \sum_{i\in I} a_i = 0 \pmod{2\pi}\}. \tag{13.34}$$

Its dimension is $|I| - 1$. A quotient alone does not give the required contextual statistical realization, so retain a whole local contextual section with fixed anchor c_*. Suppose $u_i(c_*) = \mathbb{E}_{q_i(c_*)} e^{i\theta} \neq 0$, choose compatible local phase lifts, and define

$$\vartheta_i = \arg u_i(c_*), \qquad \phi_I = \frac{1}{|I|}\sum_{i\in I}\vartheta_i, \qquad a_i = \phi_I - \vartheta_i. \tag{13.35}$$

This normalization is a first-Fourier-mode slice [22] in the method of slices for equivariant evolution equations [14, 125], a cross-section in the sense of moving frames [54]; it is singular where an anchor moment vanishes. Apply a_i to the entire section, independently of evaluation context:

$$\widetilde{q}_i(c) = \tau_{a_i} q_i(c), \qquad \widetilde{s}_i(c) = \tau_{a_i} s_i(c), \qquad q_I = \bigotimes_{i\in I}\widetilde{q}_i, \quad s_I = \bigotimes_{i\in I}\widetilde{s}_i. \tag{13.36}$$

These are actual normalized parent laws. Their factors are recovered by marginalization. The relative phase profiles $\arg u_i(c) - \arg u_i(c_*)$ remain retained; only constant relative offsets are

discarded. On this chart the fibers are precisely the constant H_I orbits. Choosing the offsets separately at each context would instead remove contextual phase variation itself and change the contextual geometry; the anchored choice avoids this. The model offsets are rotated by the belief-anchored angles. Independent model offsets are also symmetries and could be discarded as well; we do not do so.

The parent retains every child law up to one constant rotation per child with zero sum, together with its primitive ports. Its state size grows with the number of children, and all public bit marginals are retained. In the sense of the closing discussion of chapter 8, it does not combine distinct public child laws into a smaller collective state.

Parent references and evaluators are products of their normalized primitive data. For each primitive port $i \in I, j \in J$, project a source-parent sample onto j, apply eq. (13.2), and compare with factor i. This is a normalized inherited channel. A full-parent version appends independent recipient factors on the other coordinates:

$$\mathcal{L}_{IJ}^{ij}(\boldsymbol{k}_J, d\boldsymbol{k}'_I) = \mathcal{K}(k_j, dk'_i) \bigotimes_{\ell \in I \setminus \{i\}} \widetilde{q}_\ell(dk'_\ell). \tag{13.37}$$

Consequently $D_{\mathrm{KL}}(q_I \| q_J \mathcal{L}_{IJ}^{ij}) = D_{ij}^q$ by cancellation of the matching normalized factors. If this representation is differentiated, its recipient-fill channel derivatives must be retained. It represents the original primitive sums and rows, not one newly fitted parent KL cost.

At a join of I, J, rotate their factors further by $\phi_{I\cup J} - \phi_I$ and $\phi_{I\cup J} - \phi_J$, where

$$\phi_{I\cup J} = \frac{|I|\phi_I + |J|\phi_J}{|I| + |J|}. \tag{13.38}$$

Each primitive total rotation becomes $\phi_{I\cup J} - \vartheta_i$, so direct and sequential retention agree. Primitive counts are inherited. Since $\mathsf{H}_I \times \mathsf{H}_J \subsetneq \mathsf{H}_{I\cup J}$, this join discards one further continuous relative block orientation. Native row trajectories and all selected event times are unchanged by these rotations, so recording the times does not recover the discarded offsets.

Remark 13.11 (Global representatives). For $n = |I| \geq 2$ no map $(S^1)^n \to S^1$ is both invariant under relabeling and equivariant under a common rotation. If v is the rotation by $2\pi/n$ and $x = (e, v, \ldots, v^{n-1})$, then vx is a cyclic relabeling of x, so such a map would give $\phi(x) = \phi(vx) = v\phi(x)$. The equal-phase anchor eq. (13.35) therefore exists only on local phase charts, with or without continuity. Dropping relabeling symmetry removes the obstruction: rotating every agent $j \geq 2$ of I to anchor phase zero and the first agent by the sum of their phases is a continuous H_I-canonical representative on the whole domain where the anchor moments are nonzero. Rows, events, response laws, and collective costs are H_I invariant, so they do not depend on which representative is used. The common rotation meets H_I in $\mathbb{Z}_n$; on parent orbits the common group acts through $S^1/\mathbb{Z}_n$, and the parent sees the holonomy of the common connection only modulo $\mathbb{Z}_n$.

Proposition 13.12 (Original-time parent flow). On the regular local chart the retained network of sections has an autonomous flow intertwining the original dynamics at the same time.

Proof. Equivariance and uniqueness imply that two sections related by a constant H_I rotation remain so for their common existence interval. This defines the quotient flow, the projectability

condition underlying exact lumpability [79]. An explicit representative equation is also available. Let

$$\omega_i = \operatorname{Im} \frac{\int e^{i\theta} V_i^q(c_*) d\nu}{u_i(c_*)}, \qquad \overline{\omega}_I = |I|^{-1} \sum_{i \in I} \omega_i. \tag{13.39}$$

This ratio is invariant under rotation and is computable from retained factors. Since $\dot{a}_i = \overline{\omega}_I - \omega_i$ and $\partial_a \tau_a f = -\partial_\theta \tau_a f$,

$$\partial_t \widetilde{q}_i = V_i^q(\widetilde{x}) - (\overline{\omega}_I - \omega_i) \partial_\theta \widetilde{q}_i, \tag{13.40}$$

with the same coefficient for $\widetilde{s}_i$. This is the in-slice equation of the freezing method [14]. The product rule gives the parent-law derivative. Every term is retained. Successive correction coefficients telescope by eq. (13.38). For finite C^r regularity the displayed derivative is in C^{r-1}; existence and uniqueness follow from the fine flow and equivariance, without assuming rotations are differentiable as maps $C^r \to C^r$. □

The Fisher metric of this canonical product slice need not be the kinetic quotient metric. For two location families with inertias $I_1, I_2 > 0$ and retained total phase F, minimizing lift norm subject to $\dot{\vartheta}_1 + \dot{\vartheta}_2 = \dot{F}$ gives

$$ds^2_{\text{quotient}} = \frac{dF^2}{I_1^{-1} + I_2^{-1}}, \qquad ds^2_{\text{slice}} = \frac{I_1 + I_2}{4} dF^2. \tag{13.41}$$

They differ unless the inertias agree. The first is the quotient metric of a Riemannian submersion [113]; the second is the metric induced on the section. The exact parent update is eq. (13.40), not an uncorrected unit-Fisher gradient of a new single parent KL. Its gradient structure is that of the quotient metric.

Proposition 13.13 (The parent flow and the quotient metric). Let V be the fine field of eq. (13.7) and let $\overline{g}$ be the quotient metric on H_I orbits, $\overline{g}(\pi_* v, \pi_* v) = \min_w g(v + w, v + w)$ over orbit directions w, where g is the unit Fisher metric. At every law of the regular chart, V is Fisher-orthogonal to the H_I orbit directions, the moving term in eq. (13.40) is the orbit component of the slice velocity, and the descended potential $\overline{\mathcal{E}}$ satisfies

$$\frac{d}{dt} \overline{\mathcal{E}} = -\overline{g}(\pi_* V, \pi_* V). \tag{13.42}$$

On the finite-dimensional invariant family of products of binary marginals and von Mises angular factors with positive concentrations, the H_I action is free, proper, and Fisher-isometric, and the parent flow is the gradient flow of $\overline{\mathcal{E}}$ for $\overline{g}$ at the original time.

Proof. The potential is C^1 for the supremum norm of positive densities, and for a density of class C^1 in the angle the orbit curve $a \mapsto \tau_a q$ is C^1 in that norm with tangent $-\partial_\theta q$. Invariance gives $d\mathcal{E}(J\xi) = 0$ for every orbit direction $J\xi$, hence $g(V, J\xi) = -d\mathcal{E}(J\xi) = 0$. The slice velocity in eq. (13.40) is V plus an orbit direction. Since V is orthogonal to orbits, $\overline{g}(\pi_* V, \pi_* V) = g(V, V)$, and $\frac{d}{dt} \overline{\mathcal{E}} = d\mathcal{E}(V) = -g(V, V)$. The family of the last statement is invariant by eq. (13.32); the full field is tangent to it, so it equals the Fisher gradient of the restricted potential for the induced metric. A rotation fixing a von Mises factor with positive concentration is trivial, the torus is compact, and rotations preserve the Fisher metric and the potential. The orbit map is then a Riemannian

submersion for $\overline{g}$, and proposition 6.5 with $d = 1$ and $\chi = \mathrm{id}$ gives same-time inheritance of the gradient flow. □

On the finite-r chart a rotation is not differentiable as a map $C^r \to C^r$, so no quotient Banach manifold is asserted there; the pointwise dissipation identity eq. (13.42) is what holds on the whole chart. On the von Mises family the phases are constant along the flow, so the moving term vanishes there and the last statement does not exercise it.

13.5 Responses and retained histories

Declare one recursive response rule before any selection. A query to current block I draws independent constituent samples $(Z_i, \Theta_i) \sim q_i$ and returns

$$\left((Z_i)_{i \in I},\ \sum_{i \in I} \Theta_i \pmod{2\pi} \right). \tag{13.43}$$

The model sector has its corresponding independent joint response. Native source draws, transmitted bit samples, and joint measurable functions of whole-block outputs are also admitted. A policy may depend on past records, but cannot split a compressed block to reveal an individual absolute angle. Queries are observational and do not reset the law state. The querier is not an agent of the population and does not act on its dynamics; it reads angles in a fixed angular frame that no agent holds, since every channel twirls the angle.

Definition 13.14 (Declared observable algebra). For a current partition $\mathcal{P}$, let $\mathcal{A}_{\mathrm{pop}}$ be the algebra of bounded measurable functions of the configuration generated by the quantities the population reads: rows, channel output laws, evaluators, and costs. The vector field is equivariant, so the population's evolution of these quantities does not depend on the orbit representative. Let $\mathcal{A}_{\mathcal{P}}$ be generated by $\mathcal{A}_{\mathrm{pop}}$ and by the probabilities of finite adaptive response histories under eq. (13.43) for $\mathcal{P}$. A distinction between two configurations is resolved by an algebra if some element takes different values on them.

Every element of $\mathcal{A}_{\mathrm{pop}}$ is invariant under $(S^1)^{2N}$, by proposition 13.8 and the invariance of the potential. Thus no active rotation is resolved within the population, and the words gauge and physical have no content here without the response protocol. The kernel in eq. (13.43) is a normalized pushforward of a product law. Since $\sum_i a_i = 0$, it is exactly the same pushforward of the parent law in eq. (13.36). Its first angular moment is $\prod_i u_i$, generally nonzero, so an aggregate angle is resolved by $\mathcal{A}_{\mathcal{P}}$. Joining concatenates bit vectors and adds angle sums. Thus the same experiment composes at each level.

Equal retained states have equal waiting-time evolution and equal next-record joint kernels. Conditioning successively on the realized history proves equality of every finite adaptive joint-history law. Prior records are carried forward. A finite collection of sampled angular records does not deterministically identify the erased offsets: all the densities are positive, leaving continuously many compatible configurations for the same finite history. An exact record of constituent laws would instead invalidate the loss assertion. An individual absolute-angle query also fails descent

immediately because its first moment changes under eq. (13.34). These experiments are outside the declared interface after the handoff.

Remark 13.15 (How much is lost). The loss is non-identifiability, not absence of information. Before a merge every agent is its own block, so eq. (13.43) returns samples of its angular marginal. For an angular marginal f, the Fisher information about its phase carried by one sample is the rotation inertia $\mathbb{E}_f[(\partial_\theta \log f)^2]$, which is $\mathcal{I}(\kappa)$ below for a von Mises factor. On the product reference family the phases are constant, so n_i pre-merge records of agent i carry information n_i times its inertia about the offsets that the merge discards, and their circular mean estimates those offsets with error of order $(n_i \mathbb{E}_f[(\partial_\theta \log f)^2])^{-1/2}$. This information is finite for every finite record set and grows linearly with the number of records. A uniform loss statement needs a protocol that bounds the number of pre-merge angle queries or discards records at merges; the present protocol does neither.

13.6 Collective geometry on actual parent laws

Take the common collective group to be S^1, acting on every belief and model angle simultaneously. This tie is a declared representation. The constitutive data are also invariant under independent belief and model rotations; for the untied group $S^1_q \times S^1_s$ the minimizing connection below is $A_q = -d\chi$, $A_s = 0$, which is flat, and the minimized tensor has rank one. The curvature computed below belongs to the tied representation. No row, channel, record, or vector field of this construction reads the connection; A^F is a diagnostic of the evolved sections.

The common rotation commutes with the independent active rotations. Use the original sum Fisher metric and convention $D^A p = dp + J_p A$. At fixed time the constants a_i in eq. (13.35) do not vary with evaluation context. Hence $d\widetilde{q}_i = \tau_{a_i} dq_i$ and similarly for s_i. A product score is a sum of centered independent factor scores, so all cross terms vanish. Rotation preserves the remaining integrals. For every contextual direction X and scalar gauge velocity ξ this proves

$$\overline{g}(dRpX + \overline{J}\xi, dRpX + \overline{J}\xi) = g(dpX + J\xi, dpX + J\xi). \tag{13.44}$$

Equality holds before minimization. The orbit inertia, contextual momentum, minimizing connection class, and minimized tensor are therefore identical. On a regular chart their curvature agrees. This is a calculation from normalized laws, rather than assigning the fine tensor to retained metadata. It is consistent with fixed-statistic Fisher sufficiency results [5], but the present retention is section valued.

Under a passive common rotation $\chi_0(c)$, each anchor phase shifts by $\chi_0(c_*)$, leaving every a_i unchanged. Parent sections transform by the same $\chi_0(c)$, and $A' = A - d\chi_0$ gives the usual covariant derivative. The local Lie-algebra stabilizer is preserved. Globally the total phase has $|I|$ roots; a common rotation $2\pi/|I|$ lies in the discarded group (remark 13.11). The construction stays on compatible local charts.

A nonflat family accompanies the routing trajectory. Define

$$p_\kappa(\theta) = \frac{e^{\kappa \cos\theta}}{I_0(\kappa)}, \qquad \mathcal{I}(\kappa) = \kappa \frac{I_1(\kappa)}{I_0(\kappa)}. \tag{13.45}$$

Here I_0 normalizes Haar measure and $I_1/I_0 = \mathbb{E}_{p_\kappa} \cos\theta$ [104]. Integrating the derivative of $\sin\theta p_\kappa(\theta)$ around the circle gives rotation inertia $\mathbb{E}_{p_\kappa}[\kappa^2 \sin^2\theta] = \mathcal{I}(\kappa)$. For $\kappa > 0$ this mean cosine is positive, and its derivative is $\mathrm{Var}_{p_\kappa}(\cos\theta) > 0$, so $\mathcal{I}, \mathcal{I}' > 0$.

Use contexts (κ, χ), belief factors $p_\kappa(\theta - \chi)$, and model factors $p_\lambda(\psi)$ with fixed $\lambda > 0$. Binary coordinates are initially context independent. Put $\rho_t = e^{-2t}$, $a_t = \mathcal{I}(\rho_t\kappa)$ and $b_t = \mathcal{I}(\rho_t\lambda)$. Parity makes concentration and rotation scores orthogonal. For a block of primitive count n the common-gauge cost in its angular direction is $n[a_t(d\chi + \xi)^2 + b_t\xi^2]$. Thus

$$A_t^F = -\frac{a_t}{a_t + b_t} d\chi, \qquad F_t^F = -\frac{b_t \rho_t \mathcal{I}'(\rho_t\kappa)}{(a_t + b_t)^2} d\kappa \wedge d\chi, \tag{13.46}$$

and

$$h_{\chi\chi} = n\frac{a_t b_t}{a_t + b_t} > 0, \qquad h_{\kappa\kappa} = n\rho_t^2 \,\mathrm{Var}_{p_{\rho_t\kappa}}(\cos\theta) > 0, \qquad h_{\kappa\chi} = 0. \tag{13.47}$$

Curvature is nonzero at every finite time. As $t \to \infty$, $\mathcal{I}(v) \sim v^2/2$ and $\mathcal{I}'(v) \sim v$ give

$$F_t^F \longrightarrow -\frac{2\lambda^2\kappa}{(\kappa^2 + \lambda^2)^2} d\kappa \wedge d\chi, \qquad e^{4t} h_{\chi\chi} \longrightarrow \frac{n\kappa^2\lambda^2}{2(\kappa^2 + \lambda^2)}, \qquad e^{4t} h_{\kappa\kappa} \longrightarrow \frac{n}{2}. \tag{13.48}$$

On the reference family the curvature coefficient is therefore bounded away from zero uniformly in $t \geq 0$ on compact context patches with $\kappa > 0$, while the tensor decays like e^{-4t}; the infinite-time degeneration is a loss of rank, not of curvature. The curvature is already present at $t = 0$ and does not depend on the selected partition. The retention preserves it because eq. (13.44) holds for every partition at every time (proposition 13.16). This geometry is supplied by the declared contextual family; it is not produced by the selection.

To extend this geometry through every nearby law value without encoding its erased orientation, fix $c_* = (\kappa_*, 0)$. For arbitrary nearby initial $x = (q_i, s_i)_i$, let $\vartheta_i(x) = \arg \mathbb{E}_{q_i} e^{i\theta}$ and assign

$$q_{i,x}(\kappa, \chi) = \tau_\chi \left[\frac{e^{(\kappa - \kappa_*)\cos(\theta - \vartheta_i(x))} q_i(z, \theta)}{\int e^{(\kappa - \kappa_*)\cos(\theta - \vartheta_i(x))} q_i(z, \theta) d\nu} \right], \qquad s_{i,x}(\kappa, \chi) = s_i. \tag{13.49}$$

This normalized section passes through x at the anchor and reproduces the preceding family at the product reference. An active rotation shifts both the density and its centering phase, so the entire section rotates by that same constant. Thus its retained profiles do not secretly store the discarded offsets.

On a compact context patch with $\kappa > 0$ and a finite time horizon, the reference densities, moments, inertia, tensor eigenvalues, and absolute curvature coefficient have positive lower bounds. Smooth dependence of the fine field and its contextual variational equations preserves these bounds on a sufficiently small full neighborhood of independent initial laws; this step uses the C^r chart, since the rotation scores are angular derivatives. The lift chooses the contextual jets on which geometry is asserted; it imposes no product or common-parameter restriction on their anchor values.

Select membership at the fixed anchor and hand off the whole section on that common time slice. This convention matters. If each context instead chooses its own birth time $\tau(c)$, the derivative is

$$d_c[p_{\tau(c)}(c)]X = (d_c p_t X)_{t=\tau(c)} + V(p_{\tau(c)}(c)) X\tau. \tag{13.50}$$

That distinct comparison includes the additional score and is not implied by eq. (13.44).

13.7 The combined existence statement

The inheritance statements above never use the selection. This is the content of the following proposition, which makes the division of labor in the main theorem explicit.

Proposition 13.16 (Symmetry reduction for every partition). Let $\mathcal{P}$ be any partition of the agents, let $t_0 \geq 0$, and suppose every anchor moment is nonzero at t_0. Put $\mathsf{H}_{\mathcal{P}} = \prod_{I \in \mathcal{P}} \mathsf{H}_I$. Then every row, block row, selector input, and event time is invariant under $(S^1)^{2N}$, hence under $\mathsf{H}_{\mathcal{P}}$; the anchored retention of $\mathcal{P}$ at t_0 has the original-time flow of proposition 13.12; every finite adaptive history of eq. (13.43) for the current partition $\mathcal{P}$ descends; and the collective minimizing costs satisfy eq. (13.44).

Proof. The first statement is proposition 13.8. The proofs of proposition 13.12, of response descent, and of eq. (13.44) use only equivariance under H_I, the anchor, and the product form of the parent laws, and none uses an event time or a selected block. □

The selected partition is therefore not distinguished by inheritance. Native routing contributes a law-determined and relabeling-natural choice of partition and of times. The response rule contributes the coarsening that makes the discarded directions unresolvable. Before an event the current partition is finer: a singleton query returns Θ_i, whose first moment changes under a rotation in $\mathsf{H}_{\{i,j\}}$, and a block query returns a sum whose first moment changes under a rotation in $\mathsf{H}_{I \cup J}$ that is not in $\mathsf{H}_I \times \mathsf{H}_J$. After the event these responses are no longer admitted. The distinction discarded at a handoff is exactly the one the declared interface stops resolving at that handoff; the handoff loss is created by the response interface, not by the selection.

Theorem 13.17 (Open-law two-stage selection with exact inheritance). Consider the reciprocal law potential definition 4.2 with the constitutive data of this chapter: six agents, bit-circle sample spaces, the selective channel eq. (13.2) on every edge including self, uniform references and evaluators, uniform all-source priors, unit Fisher mobilities, and one fixed temperature. There exist $T > 0$, a finite horizon, and a nonempty relabeling-invariant set of independently variable positive belief and model laws, open in the C^r chart, on which the following hold over that horizon and on compatible local phase charts.

(a) Selection. Definition 13.2 selects three pairs, which coexist over a positive window, and then joins two of their parents while the third pair is a nonempty exterior with strictly smaller routing masses. Rows and events depend only on the bit marginals.

(b) Interface coarsening. At each event the response interface eq. (13.43) passes to the coarser partition. A continuous distinction between laws, along the newly discarded subgroup, is resolved before the event by admitted queries and after it by no admitted query; no element of $\mathcal{A}_{\mathrm{pop}}$ resolves it at any time. Records acquired before the event carry finite information about it (remark 13.15).

(c) Exact inheritance. For the selected partitions, the anchored section-valued parent laws, inherited port channels, original-time parent flow, and every finite adaptive history of eq. (13.43) descend exactly.

(d) Geometry. On the declared contextual family eq. (13.49), the handoffs preserve the tied collective minimizing connection and a rank-two tensor with nonzero curvature.

The temperature is not quantified (remark 13.7). The theorem does not establish interaction necessity, and the block relations it selects are transient (remark 13.6 and corollary 13.10).

Proof. Use proposition 13.5 to fix finite A, R, h and its horizon, then proposition 13.9 to obtain a full positive-law neighborhood. Choose the angular profiles with nonzero first moments and the equivariant sections of eq. (13.49). Shrink this neighborhood in the C^r chart to preserve the geometric bounds, phase chart, and all routing windows. Intersection remains nonempty because the same product trajectory satisfies them with strict compact margins.

The primitive laws are distinct and stay distinct at finite regular times by uniqueness on the invariant equality diagonals. The selector and inherited rows are exactly definition 13.2 and eq. (13.9), and (a) follows. The first joins discard their H_I directions; the later join discards an additional relative parent direction. Small active rotations stay within the open neighborhood, and equivariance of the assigned sections preserves these fibers after all retained profile data are included. Rows and event times are invariant along them. Statement (b) is the paragraph after proposition 13.16, together with the invariance of $\mathcal{A}_{\mathrm{pop}}$ and positivity of the anchor moments. Parent normalization, port channels, recursive composition, original-time flow, and history descent in (c) follow from eqs. (13.36) and (13.38) and proposition 13.12 and proposition 13.16. Equation (13.44) preserves the actual minimizing costs, and eqs. (13.46) and (13.47) establish their nontrivial geometry throughout the same finite interval, proving (d). □

Against definition 6.31 the theorem gives (F1) with the C^r topology, (F2) with finite-window persistence, (F3) with the response class declared before selection, and (F4), since each retention is non-injective and a pre-handoff query distinguishes its fibers. It does not establish (F5) (remark 13.6). The first part of (F6) holds: the parent pair consists of first-stage parents under inherited rows, without reset or retuning. Its counterfactual part is not verified, since the second stage depends on the first only through the partner exclusion of definition 13.2 and occurs when the supplied core separation has contracted to order one. The construction is therefore a two-stage selection with exact inheritance in the sense of the standard, with that qualification, and not a recursive formation. The symmetry enabling compression is fixed in the selective architecture, and the parent is not a fixed-size ordinary KL agent. The result gives neither a preferred threshold nor generic selection for arbitrary channels or all initial laws.

Remark 13.18 (Finite depth by collapsed clusters). In eq. (13.17), two clusters of m coincident agents at $\pm a$ have inherited cross mass $B = v/(1+v)$ with $v = e^{-2a^2}$, and $\dot{a} = -a(1+4B)$, independently of m: each member's row has m unit entries and m entries v. The late comparison in the proof of proposition 13.5 therefore applies to collapsed clusters of every size. An induction over geometric scales would plausibly give any finite number of selected stages on a finite horizon, at a temperature that decreases with the depth. The uniformity of the tube estimates in the depth and the effect of the self mass on the contrast at large depth have not been checked, and we do not claim the result.

Open problem 13.19 (Interaction-necessary selection in the bit-circle architecture). Prove a version of theorem 13.17 on an open set where the null flow does not reproduce the membership

history, so that (F5) holds; integrations of the limit field at moderate separations suggest that such sets exist. Separately, find an architecture with a forward-invariant set on which a block relation holds and a later join of first-stage parents occurs; a compression symmetry produced by the dynamics rather than fixed by the channel, for example through an angle-transmitting channel of small concentration that makes the compression approximate on the selected blocks with a computable defect; a loss statement in which the information about discarded coordinates given all admitted records stays bounded; and explicit or validated witness constants.

13.8 Two tests of stronger inheritance claims

The first test makes the identity-channel response obstruction quantitative. For two binary source means $x_1 \neq x_2$ and fixed positive weights, an exterior receiver mean r determines the block normalizer. With $a_j = \operatorname{atanh} x_j$, $v_j = a_j/T$, and $\psi(r) = \frac{1+r}{2}\log(1+r) + \frac{1-r}{2}\log(1-r)$, removing its receiver entropy factor gives

$$F(r) = e^{\psi(r)/T} Z_I(r) = \sum_{j=1}^{2} b_j e^{v_j r}, \qquad b_j > 0. \tag{13.51}$$

At an interior r_0, set $m_k = F^{(k)}(r_0)$ and $c_j = b_j e^{v_j r_0}$. Then

$$H = m_0 m_2 - m_1^2 = c_1 c_2 (v_1 - v_2)^2 > 0. \tag{13.52}$$

The two scaled logits $v_j = a_j/T$ are the roots of

$$z^2 - \frac{m_0 m_3 - m_1 m_2}{H} z + \frac{m_1 m_3 - m_2^2}{H} = 0. \tag{13.53}$$

This is Prony's method for a two-term exponential sum [43]. Thus equality of response functions on any open receiver interval identifies the unordered source laws. Small within-pair KL makes this inverse ill-conditioned, not exactly redundant. This is a finite reconstruction within the unrestricted Gibbs-interface result; it does not obstruct the selective angular interface above.

The second test concerns propagation of Fisher sufficiency. Let a fixed statistic retain Z from positive densities $p_t(c; z, y) = \mu_t(c; z)\varkappa_t(c; y \mid z)$. Write P_t for conditional expectation given Z, let X be a contextual direction, and suppose the complete centered force is $\partial_t \log p_t = -(h_t - \mathbb{E}h_t)$. The hidden contextual score is $a_X = X \log \varkappa_t$. Conditional normalization gives

$$\partial_t a_X = -(I - P_t) X h_t + \operatorname{Cov}_{\varkappa_t}(h_t, a_X). \tag{13.54}$$

If $a_X(0) = 0$, the Fisher loss therefore starts as

$$g_t(dp_t X, dp_t X) - \overline{g}_t(dRp_t X, dRp_t X) = t^2 \mathbb{E}_0[((I - P_0) X h_0)^2] + o(t^2). \tag{13.55}$$

This follows by differentiating the conditional score and expanding its squared norm; on finite positive alphabets all operations are finite sums.

For two identity-coupled laws $q_i(z,y) = \mu_i(z)\varkappa_i(y \mid z)$ on a finite alphabet, written as probabilities with respect to counting measure, with local references r_i, the complete forces are $h_1 = 2\log q_1 - \log r_1 - \log q_2 - q_2/q_1$ and its exchanged expression. Initially vary only μ_2 with tangent v, keeping both conditionals fixed. Conditional centering of the incoming ratios gives the coefficient

$$\sum_z v_z^2 \left[\frac{\chi^2(\varkappa_2 \| \varkappa_1)}{\mu_1(z)} + \frac{\mu_1(z)^2}{\mu_2(z)^3} \chi^2(\varkappa_1 \| \varkappa_2) \right]. \tag{13.56}$$

Here the conditional divergences are evaluated at z. Taking $\mu_1 = \mu_2 = (1/2, 1/2)$, $v = (1, -1)$, and $\varkappa_1(y = 1 \mid z) = 1/4$, $\varkappa_2(y = 1 \mid z) = 3/4$ for both values of z gives $32/3$. Initial score sufficiency is lost immediately. This example uses a trivial collective group; with a nontrivial action one must test the actual minimizing residual rather than infer optimized loss from an arbitrary raw score. The anchored retention avoids this failure through the exact equality of its full minimizing costs.

Chapter 14

Per-Agent Grouping with Persistent Records

This chapter concerns the normalized per-agent model of definition 4.17, rather than the reciprocal potential used in chapters 7 and 13. Each agent changes its own recognition laws while treating the communicated laws of its peers as current model data. Records remain in its free energy. The first construction transfers a conditional channel to these dynamics and exhibits a grouping that its prescribed null never selects. A second construction gives persistent grouping and a physically noninjective parent with exact original-time dynamics, admitted response histories, and nonflat collective geometry on a declared family. Its sufficient interface is supplied before selection, and no second compressive handoff is established.

14.1 What changes under an own-objective derivative

All source priors below are positive and normalized. Source rows are minimized exactly at unit temperature; this is the Gibbs-reduced convention, not finite-mobility row evolution. At a fixed configuration let $D^q_{ii} = \int D_{\mathrm{KL}}(q_i \| G_i(m,\cdot)) s_i(dm)$ and $D^s_{ii} = D_{\mathrm{KL}}(s_i \| r_i)$, with the nonself costs defined by the communicated laws. Then

$$\beta_{ij} = \frac{\pi^q_{ij} e^{-D^q_{ij}}}{\sum_k \pi^q_{ik} e^{-D^q_{ik}}}, \qquad \gamma_{ij} = \frac{\pi^s_{ij} e^{-D^s_{ij}}}{\sum_k \pi^s_{ik} e^{-D^s_{ik}}}. \tag{14.1}$$

The envelope derivative is the row-weighted derivative of the costs. It does not differentiate the optimized row again. Each law follows its own agent's Fisher field, so derivatives of another agent's free energy are absent. The probability-coordinate form is the replicator equation associated with the Shahshahani metric [128]; the construction here concerns its particular normalized sources, records, and exact inheritance.

The null of definition 6.30 deletes the nonself costs, not their labels or priors. If $a = \pi^q_{ii}$, its content self row is

$$\beta^0_{ii} = \frac{a e^{-D^q_{ii}}}{a e^{-D^q_{ii}} + 1 - a}. \tag{14.2}$$

The model row has the same form. Replacing these rows by self probability one would specify a different control. Every example below uses the same row convention, records, selector, and initial state in the full and null flows.

14.1.1 A conditional four-state channel

Let $k = (a, x)$ and $m = (b, u)$ range over $\{-1, 1\}^2$ with counting measure, and let q_i, s_i be arbitrary positive normalized tables. Fix $J > 1$ and $0 < \alpha, \kappa < 1$. Take

$$r(b, u) = \frac{1}{4}, \qquad G((b, u), (a, x)) = \frac{e^{Jab}}{4 \cosh J}. \tag{14.3}$$

For a positive table p, define $\mu_p(x) = \sum_a p(a, x)$ and $L_p(b \mid x) = p(b, x)/\mu_p(x)$. The nonself kernel is

$$\begin{aligned} W_b(y \mid x) &= \frac{1 + \kappa b x y}{2}, \\ K[p]((a, x), (b, y)) &= \big[\alpha \mathbf{1}_{\{a=b\}} + (1 - \alpha) L_p(b \mid x)\big] W_b(y \mid x). \end{aligned} \tag{14.4}$$

Each row sums to one and is positive. The first operation has determinant α on each conditional two-state block, and the second has determinant κb on each content block, so $K[p]$ is invertible. Nevertheless its compared output simplifies exactly:

$$(pK[p])(b, y) = \sum_x p(b, x) W_b(y \mid x) =: (pW)(b, y), \qquad D(pK[p])[v] = vW. \tag{14.5}$$

The first identity follows by summing over a; its derivative includes the channel derivative. No loss of channel rank or frozen derivative is used.

Let the homogeneous self priors be $\rho_q, \rho_s \in (0, 1)$ and divide each remaining prior equally over the $N - 1$ peers. Write $m_i = \mathbb{E}_{q_i} a$, $n_i = \mathbb{E}_{s_i} b$, $\nu_j^q = q_j W$, and $\nu_j^s = s_j W$. One retained record $(o_q, o_s) \in \{-1, 1\}^2$ has normalized likelihood

$$\ell(o_q, o_s \mid a, x, b, u) = \frac{e^{h_q o_q a}}{2 \cosh h_q} \frac{e^{h_s o_s b}}{2 \cosh h_s}, \qquad h_q, h_s > 0. \tag{14.6}$$

For finitely many retained records put $H_i^q = \sum_t h_{q,t} o_{q,t}$ and $H_i^s = \sum_t h_{s,t} o_{s,t}$. Their negative expected log likelihood is a constant minus $H_i^q m_i + H_i^s n_i$. The ordered record word is retained whenever it remains an accessible response.

Proposition 14.1 (Transferred own-block and null fields). With product recognition $q_i \otimes s_i$, the normalized per-agent model with sources eqs. (14.3) and (14.4) has the original-time equations

$$\dot{q}_i = -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q), \qquad \dot{s}_i = -s_i(h_i^s - \mathbb{E}_{s_i} h_i^s), \tag{14.7}$$

where, modulo constants,

$$\begin{aligned} h_i^q(a, x) &= \log q_i(a, x) - \sum_{j \neq i} \beta_{ij} \log \nu_j^q(a, x) - \beta_{ii} J n_i a - H_i^q a, \\ h_i^s(b, u) &= \log s_i(b, u) - \sum_{j \neq i} \gamma_{ij} \log \nu_j^s(b, u) - \beta_{ii} J m_i b - H_i^s b. \end{aligned} \tag{14.8}$$

Writing $b_i^0 = \beta_{ii}^0$ and $g_i^0 = \gamma_{ii}^0$, the null potentials are

$$h_i^{q,0} = b_i^0(\log q_i - J n_i a) - H_i^q a, \qquad h_i^{s,0} = g_i^0 \log s_i - b_i^0 J m_i b - H_i^s b. \tag{14.9}$$

Proof. Integrating the record, content source, presentation source, and source priors in the local model proves normalization. Its content self cost is $\sum q_i \log q_i - Jm_i n_i + \log(4\cosh J)$, and its presentation self cost is $\sum s_i \log s_i + \log 4$. Every peer is fixed in the own-block variation. Differentiating the row-weighted costs and the persistent record term gives eq. (14.8); the coefficients of each full-law entropy sum to one. In particular both derivatives of the evaluator interaction carry β_{ii}, not γ_{ii}. After deletion only the self entropy coefficients remain, giving eq. (14.9). Centering these first variations in the Fisher metric gives eq. (14.7). □

On the synchronized family $q_i(a,x) = (1+ma)/4$, $s_i(b,u) = (1+nb)/4$, every peer cost and normalized peer tangent vanishes. Full and null fields coincide there. With no records, stationarity requires

$$\operatorname{atanh} m = Jn, \qquad \gamma_{ii} \operatorname{atanh} n = \beta_{ii} Jm. \tag{14.10}$$

Thus the reciprocal condition $m = n = r_*$, $\operatorname{atanh} r_* = Jr_*$ additionally requires equality of the two self rows. Synchronized bistability cannot by itself establish interaction necessity.

Proposition 14.2 (Interaction-dependent selection on a full open table class). For $N = 2$, $J = 3/2$, $\rho_s = 1/2$, and $\rho_q = \cosh J/(1+\cosh J)$, there are positive record strengths and a nonempty full open set of initial positive tables for which a permutation-covariant conditional-sign selector selects the pair for a nonzero time window, while its null never selects the pair.

Proof. At the record-free uniform table the self rows equal $1/2$ and their first variations vanish. The sign linearization is

$$\dot m_i = -m_i + \frac{1}{2} m_j + \frac{3}{4} n_i, \qquad \dot n_i = -n_i + \frac{1}{2} n_j + \frac{3}{4} m_i, \qquad j \neq i. \tag{14.11}$$

Its common sign eigenvalue is $1/4$ and the matching difference eigenvalue is $-3/4$. Other sign eigenvalues are $-5/4, -9/4$, and the content modes have eigenvalues $-1 \pm \kappa/2$, hence are stable. Start with $m = n = \eta(3,-1)$ and uniform conditional content. The relevant linear solution is

$$m_i(t) = n_i(t) = \eta\big[e^{t/4} + (a_i - 1)e^{-3t/4}\big], \qquad (a_1, a_2) = (3,-1). \tag{14.12}$$

At $t_* = \log 4$ its values are $\eta(3/\sqrt{2}, 1/\sqrt{2})$. The analytic field is its linearization plus a quadratic remainder. Variation of constants on a fixed finite interval gives an $O(\eta^2)$ state and velocity error for sufficiently small $\eta > 0$. Now fix such an η and add small records with $H_1^q = H_1^s = h > 0$ and $H_2^q = H_2^s = -h$. Continuous dependence preserves the strict positive crossing and a subsequent common interval, also under arbitrary small positive-table perturbations.

For the null, let $c_i^q(x) = \mathbb{E}[a \mid x]$ and $c_i^s(u) = \mathbb{E}[b \mid u]$. Conditional Fisher centering gives

$$\begin{aligned} \dot c_i^q(x) &= (1 - (c_i^q(x))^2)\{b_i^0[Jn_i - \operatorname{atanh} c_i^q(x)] + H_i^q\}, \\ \dot c_i^s(u) &= (1 - (c_i^s(u))^2)\{b_i^0 Jm_i - g_i^0 \operatorname{atanh} c_i^s(u) + H_i^s\}. \end{aligned} \tag{14.13}$$

At a first zero of a conditional initially aligned with its record, all other signed conditionals are nonnegative and the signed derivative is strictly positive. The two opposite cones are therefore invariant. All null log ratios have linear growth bounds because the self rows lie in $(0,1)$ and $|m_i|, |n_i| \le 1$, so this conclusion holds for every finite time.

Declare that the pair is selected when all eight conditional means exceed $\eta/(8\sqrt{2})$, or all are below its negative. The full flow meets this condition on the common interval; the null never does. Initially there is no selected pair. The condition is strict on a full open neighborhood of the chosen initial tables and is preserved by simultaneous relabeling of agents and records. The partition sequences differ after deleting their event times. □

14.2 A persistent group against conflicting evidence

A simpler local evaluator yields indefinite persistence with the initial records held fixed. Take three agents with positive Bernoulli belief and presentation laws of log odds l_i, n_i, uniform self reference and evaluator, and identity peer channels. Set $\pi_{ii} = a = 1/5$ and $\pi_{ij} = p = 2/5$ for $j \neq i$. Their common binary record likelihood, relative to counting measure, is

$$\ell_\epsilon(o \mid z) = \frac{e^{\epsilon o z/2}}{2\cosh(\epsilon/2)}, \qquad o, z \in \{-1, 1\}, \quad \epsilon > 0. \tag{14.14}$$

Condition on $o = (1, 1, -1)$ and retain it in the objective. The species are identical; the different observations are data.

Theorem 14.3 (Persistent interaction-dependent grouping). For all sufficiently small fixed $\epsilon > 0$, a nonempty open class of positive initial belief and presentation laws starts with three singleton blocks and selects a persistent three-agent block under the per-agent flow, whereas the prescribed null never selects that block.

Proof. The exact log-odds equations are

$$\dot{l}_i = -l_i + \sum_{j \neq i} \beta_{ij} l_j + \epsilon o_i, \qquad \dot{n}_i = -n_i + \sum_{j \neq i} \gamma_{ij} n_j. \tag{14.15}$$

For a Bernoulli law, the Fisher inverse cancels $dl/dq(+1)$, so the peer KL derivative contributes $l_i - l_j$ and the log-likelihood ratio contributes ϵo_i. The rows use the corresponding Bernoulli costs. The field is analytic with at most linear growth. The presentation sector is retained but does not enter the evaluator.

Put $l = \epsilon u$. Since every divergence has vanishing first variation at the uniform law, $\beta_{ij}(\epsilon u) = \pi_{ij} + O_{C^1}(\epsilon^2)$ on bounded sets, and

$$\dot{u} = Mu + o + O_{C^1}(\epsilon^2), \qquad M = -\frac{7}{5} I + \frac{2}{5} \mathbf{1}\mathbf{1}^\top, \qquad u_* = -M^{-1} o = \left(\frac{15}{7}, \frac{15}{7}, \frac{5}{7}\right). \tag{14.16}$$

The eigenvalues of M are $-1/5, -7/5, -7/5$. The implicit function theorem gives a nearby equilibrium $u_\epsilon = u_* + O(\epsilon^2)$. On a sufficiently small fixed neighborhood the symmetric part of the derivative stays negative definite. A smaller ball centered at u_ϵ is consequently forward invariant and attracting by differentiation of its squared radius. The limiting affine trajectory from $u^{\mathrm{in}} = (4, 1/2, -1)$ enters this ball after one finite time. Finite-time continuous dependence gives the same entry for the exact field and an open initial neighborhood.

Connect i, j when l_i, l_j have the same nonzero sign and

$$\frac{\min(|l_i|, |l_j|)}{\max(|l_i|, |l_j|)} > \frac{1}{4}, \tag{14.17}$$

and take connected components. Initially the only same-sign ratio is $1/8$, so there are no edges. At u_* the minimum ratio is $1/3$ and all signs are positive. Choose the attracting ball to preserve these margins.

The null instead has

$$\dot{l}_i^0 = -\beta_{ii}^0(l_i^0) l_i^0 + \epsilon o_i, \qquad \beta_{ii}^0 = \frac{a e^{-D_{\mathrm{KL}}(q_i \| q_0)}}{a e^{-D_{\mathrm{KL}}(q_i \| q_0)} + 2p}. \tag{14.18}$$

At $l_i = 0$ its derivative has the strict sign of o_i. The first two initial odds remain positive and the third remains negative for all finite null time. Thus the full block never appears under the same selector, irrespective of event timing. Independent presentation-law perturbations do not change the conclusion. □

This proves interaction necessity for a persistent state-selected group. It has not yet discarded any child law. The next construction supplies a conditional physical quotient while preserving that grouping mechanism.

14.3 A fixed-model angular extension

Let $\mathsf{K} = \{-1, 1\} \times S^1 \times S^1$, with uniform binary reference and independent Haar references H on both circles. Let $\mathsf{M} = \{-1, 1\}$ with counting measure. Write $e(\phi) = (\cos\phi, \sin\phi)$ and $h(A) = \log I_0(|A|)$, where $I_0(r) = \int e^{r\cos\phi} H(d\phi)$. The invariant belief family is

$$q_i(z, \phi, \psi) = \frac{\exp[l_i z/2 + A_i \cdot e(\phi) + z C_i \cdot e(\phi) + b_i \cos(\psi - \theta_i)]}{Z_i}, \tag{14.19}$$

where $A_i, C_i \in \mathbb{R}^2$, $b_i > 0$, $\theta_i \in S^1$, and

$$Z_i = I_0(b_i) \frac{e^{l_i/2} I_0(|A_i + C_i|) + e^{-l_i/2} I_0(|A_i - C_i|)}{2}. \tag{14.20}$$

The Cartesian hidden parameter $b_i e(\theta_i)$ is regular at zero; the phase chart used for compression is restricted to $b_i > 0$. Retain the independent Bernoulli presentation laws and priors of theorem 14.3.

Fix $J = \epsilon^3$ and the normalized self evaluator

$$G_i(m; z, \phi, \psi) = \frac{e^{Jz\cos\phi}}{I_0(J)}. \tag{14.21}$$

It is independent of m. Each nonself channel copies z and resamples both circles from Haar measure. Thus its output is $\mu_j \otimes H \otimes H$, where μ_j is the public binary marginal. Its density is

relative to the uniform binary reference, not counting measure:

$$\mu_i(z) = \frac{e^{L_i z/2}}{\cosh(L_i/2)}, \qquad L_i = l_i + h(A_i + C_i) - h(A_i - C_i). \tag{14.22}$$

The record space is a bit o with counting measure and an angle ω with Haar measure, with likelihood

$$\ell_\epsilon(o, \omega \mid z, \phi, \psi, m) = \frac{e^{\epsilon o z/2}}{2\cosh(\epsilon/2)} \frac{e^{\epsilon\cos(\omega-\phi)}}{I_0(\epsilon)}. \tag{14.23}$$

Integrating each circle and summing the record bit proves all normalizations. The channels, evaluator, and likelihood are fixed during an own-block variation. In particular this example does not reconstruct the receiver's changing conditional law inside its generative model.

Proposition 14.4 (Exact angular field). The full own-law Fisher field preserves eq. (14.19) and is

$$\begin{aligned} \dot{l}_i &= -l_i + \sum_{j\neq i} \beta_{ij} L_j + \epsilon o_i, \quad \dot{A}_i = -A_i + \epsilon e(\omega_i), \\ \dot{C}_i &= -C_i + \beta_{ii}\epsilon^3 e_1, \qquad\qquad \dot{b}_i = -b_i, \qquad \dot{\theta}_i = 0. \end{aligned} \tag{14.24}$$

The presentation equation remains the second equation of eq. (14.15). If $m_i = \tanh(L_i/2)$ and $t_i = \mathbb{E}_{q_i}[z\cos\phi]$, the unnormalized content row weights are

$$w_{ii} = \frac{a e^{J t_i}}{I_0(J)}, \qquad w_{ij} = \frac{p e^{L_j m_i/2}}{\cosh(L_j/2)}, \qquad \beta_{ij} = \frac{w_{ij}}{\sum_r w_{ir}}. \tag{14.25}$$

Proof. Put $S_i = \mathbb{E}_{q_i} \log q_i$. The costs are $D_{ii} = S_i - J t_i + \log I_0(J)$ and $D_{ij} = S_i - L_j m_i/2 + \log\cosh(L_j/2)$. The common entropy cancels in the Gibbs normalization, giving eq. (14.25). The centered negative own derivative is the logarithmic replicator velocity. It lies in the span of $z, e(\phi), ze(\phi), e(\psi)$ and constants. Matching coefficients gives eq. (14.24). The covariance of these seven sufficient statistics is positive definite: a linear combination constant under a positive density must be constant under the reference, and its Fourier modes and sign parity force every nonconstant coefficient to vanish. This is therefore the exact Fisher velocity on an invariant family, rather than a Euclidean update or a projected approximation. □

The marked sector participates in the public force. At $A = \alpha e_1$ and $C = c e_1$ with $c > 0$, $\partial_\alpha L = h_1'(\alpha + c) - h_1'(\alpha - c) > 0$, since $h_1''(r) = \mathrm{Var}_r(\cos\phi) > 0$. Starting with $C = 0$ generates positive c at positive time. Also $\partial_{L_j}\beta_{ij} = \beta_{ij}(1 - \beta_{ij})(m_i - m_j)/2$. Thus angular data can change source competition and the receiver's force. The separate hidden phase does not enter those quantities.

Theorem 14.5 (Grouping survives the angular extension). There is $\epsilon_0 > 0$ such that for every $0 < \epsilon < \epsilon_0$, the selector eq. (14.17) applied to the actual public odds L_i selects a persistent full block from a nonempty relatively open initial set in eq. (14.19), with records $o = (1, 1, -1)$ held fixed. Its prescribed null never selects that block. Initial marked and hidden concentrations can lie in a fixed small positive compact patch independent of ϵ, and the conclusion is uniform over the initial angular record values.

Proof. The null equations are

$$\begin{aligned} \dot{l}_i^0 &= -\beta_{ii}^0 l_i^0 + \epsilon o_i, \qquad \dot{A}_i^0 = -\beta_{ii}^0 A_i^0 + \epsilon e(\omega_i), \\ \dot{C}_i^0 &= \beta_{ii}^0(\epsilon^3 e_1 - C_i^0), \quad \dot{b}_i^0 = -\beta_{ii}^0 b_i^0, \qquad \dot{\theta}_i^0 = 0, \end{aligned} \tag{14.26}$$

with eq. (14.2) evaluated at the complete self cost. Its angular entropy cannot be canceled after deleting the nonself costs.

For $\epsilon \le 1/100$, consider

$$\frac{1}{4} \le \frac{o_i l_i}{\epsilon} \le 12, \qquad |A_i| \le \frac{1}{10}, \qquad |C_i| \le 2\epsilon^3, \qquad 0 \le b_i \le \frac{1}{10}. \tag{14.27}$$

In natural parameters $\eta = (l/2, A, C, be(\theta))$ the sufficient-statistic vector has squared norm four, so the log-partition Hessian has operator norm at most four. Taylor's integral remainder for the exponential-family divergence, relative to $\eta_G = (0, 0, \epsilon^3 e_1, 0)$, gives

$$D_{ii} \le 2|\eta - \eta_G|^2 \le 72\epsilon^2 + \frac{1}{25} + 18\epsilon^6 < \log 2, \qquad \frac{1}{9} \le \beta_{ii}^0 \le \frac{1}{5}. \tag{14.28}$$

For $r_i = o_i l_i/\epsilon$, $\dot{r}_i = 1 - \beta_{ii}^0 r_i$ points inward at both endpoints. The outward derivatives of the A and C norms are at most $\epsilon - 1/90 < 0$ and $-\beta_{ii}^0\epsilon^3 < 0$ on their respective boundaries, while b decreases. Thus eq. (14.27) is forward invariant. Since $\nabla h(0) = 0$ and $\|\nabla^2 h\| \le 1$,

$$|L_i - l_i| \le 2|C_i|(|A_i| + |C_i|) \le \frac{2}{5}\epsilon^3 + 8\epsilon^6 < \frac{\epsilon}{8}. \tag{14.29}$$

Consequently $o_i L_i \ge \epsilon/8$ for all finite null times.

For the full flow put $u = l/\epsilon$ and $c = C/\epsilon^3$, leaving A unscaled. On bounded sets, $L_i/\epsilon = u_i + O_{C^1}(\epsilon^2)$ and $m_i, t_i = O(\epsilon)$. Equation eq. (14.25) implies $\beta_{ij} = \pi_{ij} + O_{C^1}(\epsilon^2)$ even for fixed nonzero initial angular amplitudes. Hence

$$\dot{u} = Mu + o + O_{C^1}(\epsilon^2), \qquad \dot{c}_i = -c_i + ae_1 + O_{C^1}(\epsilon^2), \qquad \dot{A}_i = -A_i + \epsilon e(\omega_i). \tag{14.30}$$

At $\epsilon = 0$ this is a stable affine field with equilibrium $(u_*, c_i = ae_1, A_i = 0)$ and symmetric negative-definite derivative. The implicit function theorem supplies a nearby equilibrium with $u = u_* + O(\epsilon^2)$, $c_i = ae_1 + O(\epsilon^2)$, and $A_i = \epsilon e(\omega_i)$. On a sufficiently small common ball its symmetric derivative is at most $-I/10$. After recentering, the derivative of squared distance is at most minus one fifth of that distance squared. This proves an invariant attracting neighborhood with all public signs positive and all pair ratios above $1/4$.

Choose $u(0)$ near $(4, 1/2, -1)$, $|C_i(0)| < \epsilon^3$, and $|A_i(0)|, b_i(0) < 1/20$ with positive hidden amplitudes. The limiting affine flow enters the attraction neighborhood after a fixed finite time, uniformly over compact initial marked patches. Finite-time continuous dependence gives the same entry for small positive ϵ and a relatively open initial set. Initially $L_i = l_i + O(\epsilon^3)$, so the strict singleton margins persist. The angular records range over a compact torus and enter only through bounded constant forcings of size ϵ; the same entrance time and smallness threshold work uniformly over their values. The null retains opposite signs by eq. (14.29).

Finally A, C, b have finite-interval bounds directly from their equations, $|L - l| \leq 2|C|$, and both l and the presentation log odds have at most linear growth. Solutions therefore continue for every finite time. The subsequent group persistence uses the invariant attraction neighborhood, not indefinite extrapolation of finite-time continuity. □

14.4 A physical quotient with inherited records

Fix the operational protocol before selection. It admits fresh public and marked-coordinate queries on the primitive ports, normalized generative or recognition-predictive record transactions with the likelihood family eq. (14.23), and adaptive choices of ports, strengths and waiting times from the retained history. These two kinds of predictive law are declared separately and need not agree. A prediction is not automatically assimilated. An evidence transaction appends its actual outcome and adds its likelihood to the persistent objective. Every finite continuation has finite record coefficients. Hidden absolute-angle queries are excluded at all stages. An external source-label readout does not change the agent's conditioning convention; revealing such a label to the inferring agent would specify another source-row update.

Proposition 14.6 (Canonical phase parent). On the invariant family eq. (14.19), replace each hidden phase by zero and retain l_i, A_i, C_i, b_i, the presentation laws, primitive port labels, all constitutive data, event information, and the full record history. For a selected block take the product of these normalized canonical laws. This retention is physically noninjective at every finite time, intertwines the original-time full and null flows, and preserves every admitted finite adaptive joint response-record history, including its dependence with retained next states.

Proof. All full and null costs and retained velocities are independent of θ_i, and $\dot{\theta}_i = 0$. The canonical laws therefore obey the exact inherited port equations. Positive initial b_i remains positive at each finite time. The child diagnostic

$$\mathbb{E}_{q_i} e^{i\Psi} = \frac{I_1(b_i)}{I_0(b_i)} e^{i\theta_i} \tag{14.31}$$

distinguishes a phase fiber because $I_1(b)/I_0(b) > 0$ for $b > 0$. Thus the retention loses a continuous physical distinction after event data and records are included.

The current retained state determines every admitted next response law. Conditional on its actual outcome, the same likelihood adds the same binary and marked-vector coefficients to the retained objective and the same entry to history. The joint response and retained next-state kernel therefore factors through retention. Deterministic waiting also factors by the exact equations. Induction over the causal product of these kernels proves every finite adaptive joint history. This is the predictive-equivalence requirement familiar from causal-state constructions [129], proved here directly for the declared controlled protocol. □

The parent carries primitive ports and their directed own-objective field. No claim is made that it is a fixed-size agent with one new source row, or the Fisher gradient of the sum of its children's free energies. Such a sum would add source-role derivatives. Normalized inherited channels act by projecting to the selected primitive source port, applying the existing public channel, and appending normalized canonical factors on other output coordinates. No new parent force is fitted.

The diagnostic eq. (14.31) is a permitted child-level distinction for assessing physical loss, but it is not a future boundary response in the fixed protocol. If it is admitted there, the quotient fails. A persistent hidden-angle likelihood proportional to $e^{\lambda \cos(v-\psi)}$ also changes the hidden amplitude and phase:

$$\dot{b} = -b + \lambda \cos(v-\theta), \qquad \dot{\theta} = \frac{\lambda}{b} \sin(v-\theta) \tag{14.32}$$

in the full fixed-model example. Two states with the same retained b, v can then have different future b velocities. Finite-record uncertainty about θ does not remove this dynamical obstruction.

14.5 Collective geometry of the same parent

The marked comparison group is a common S^1 acting on ϕ within the selected block. Under a frame change g, transport ϕ and the record angle by addition of g, rotate A_i, C_i and the constitutive evaluator direction e_1 by the corresponding plane rotation, and leave z, b, ψ, θ unchanged. The presentation sector carries no marked orbit. This is passive covariance of the declared comparison data. Rotating only the beliefs while holding the numerical evaluator and record directions fixed is generally not a symmetry of the energy. The hidden phase rotations are separate physical symmetries, not the marked comparison gauge.

To include records in the actual statistical experiment, fix one finite initial episode and let $\Pi_c(dh)$ be its normalized generative predictive law at contextual point c. Conditional on the prepared configuration the agents' record kernels may be sampled independently; this defines an operational experiment and does not posit a global latent generative model. Let $Q^h_{c,t}$ be the product of fresh recognition and Gibbs-label laws conditional on the retained history h. The comparison law is

$$\mathcal{P}_{c,t}(dh, dy) = \Pi_c(dh) Q^h_{c,t}(dy), \qquad g^{\text{joint}} = g^{\text{history}} + \mathbb{E}_H[g^{\text{conditional}}]. \tag{14.33}$$

The metric identity follows because the conditional score has mean zero. For the grouping witness one may condition on the positive-probability public pattern $(+,+,-)$, keeping all angular records random and including the conditioning normalizer in the history score. Source-label scores remain part of Q^h.

Proposition 14.7 (Exact collective cost inheritance). On a regular contextual section with every discarded θ_i constant in context, the canonical parent of proposition 14.6 preserves the actual collective minimizing connection class and minimized contextual Fisher tensor for eq. (14.33).

Proof. The product shift $\psi_i \mapsto \psi_i - \theta_i$ is constant in context, time and history. It sends the fine joint law to the actual canonical parent law with unit Haar Jacobian and commutes with the marked action. Therefore it carries each contextual score s_X and marked orbit score s_{orb} isometrically. For every tangent X and comparison velocity ξ,

$$\mathcal{C}_{\text{fine}}(X, \xi) = \mathbb{E}_{\mathcal{P}}(s_X + \xi s_{\text{orb}})^2 = \mathcal{C}_{\text{parent}}(X, \xi). \tag{14.34}$$

The entire quadratic cost, its minimizer class modulo isotropy, and its minimum agree. The parent metric is computed from its normalized law rather than assigned by copying a tensor. □

If θ_i varies with context, the erased score contributes $b_i I_1(b_i)/I_0(b_i)(d\theta_i)^2$ and the proposition generally fails. It is a theorem on the declared family of sections. The common marked connection is a diagnostic of these evolved sections; it does not enter the present dynamical equations.

Theorem 14.8 (A common nonflat grouping window). The initial set in theorem 14.5 admits a compact rank-two contextual patch and a fixed finite post-selection window on which the actual collective connection has nonzero curvature, its minimized tensor has rank two, and the canonical parent has exactly the same quantities.

Proof. Choose fixed small positive $a_- < a_+$, d, and b_i, independent of ϵ. Use contextual coordinates (a, χ) with $A_1(0) = ae(\chi)$, $a \in [a_-, a_+]$, $A_2(0) = de_1$, and $C_i(0) = 0$. Other selected marked factors may have fixed positive concentrations. Keep hidden phases constant over context and take the public initial odds from theorem 14.5. Choose a finite T_+ containing its common entrance time and a positive subsequent interval, with no additional assimilation during this window.

At $\epsilon = 0$, the episode likelihood is uniform, public laws are uniform, and the full Gibbs rows equal the fixed priors. History and label scores vanish at this limiting parameter. The solution is $A_i(t) = e^{-t}A_i(0)$, $C_i(t) = 0$, $l_i(t) = 0$, and $b_i(t) = e^{-t}b_i(0)$. Put

$$I(x) = x\frac{I_1(x)}{I_0(x)}, \qquad V(x) = \mathrm{Var}_{v_x}(\cos\phi), \qquad B(t) = \sum_{i\neq 1} I(e^{-t}|A_i(0)|) > 0. \tag{14.35}$$

Here $v_x = e^{x\cos\phi}/I_0(x)$. Differentiating its partition function gives $I'(x) = I_1(x)/I_0(x) + xV(x) > 0$ for $x > 0$. Integration of the derivative of $\sin\phi e^{x\cos\phi}$ around the circle gives the phase Fisher identity $x^2\mathbb{E}\sin^2\phi = I(x)$. Minimizing the sum of marked phase-score costs gives

$$\begin{aligned}
\mathcal{A}_{0,t} &= -\frac{I(e^{-t}a)}{I(e^{-t}a) + B(t)}d\chi,\\
\mathcal{F}_{0,t} &= -\frac{B(t)e^{-t}I'(e^{-t}a)}{[I(e^{-t}a) + B(t)]^2}da \wedge d\chi,\\
h^{0,t}_{aa} &= e^{-2t}V(e^{-t}a), \qquad h^{0,t}_{\chi\chi} = \frac{I(e^{-t}a)B(t)}{I(e^{-t}a) + B(t)}, \qquad h^{0,t}_{a\chi} = 0.
\end{aligned} \tag{14.36}$$

The locked inertia, both tensor eigenvalues, and the magnitude of the curvature coefficient have positive minima on the compact contextual patch and $0 \le t \le T_+$.

The exact densities, episode kernels, and ODE coefficients are analytic on a common compact finite-time tube. Their first two contextual derivatives converge uniformly to the displayed limit as $\epsilon \to 0$: differentiating the flow gives $\dot{X}_c = DV(X)X_c$ and $\dot{X}_{cc} = DV(X)X_{cc} + D^2V(X)[X_c, X_c]$, whose integral inequalities bound the derivatives uniformly. Compact record spaces and positive bounded densities justify differentiating the Fisher integrals. Conditioning on the public pattern is regular because its limiting probability is positive. Thus the locked inertia K_ϵ, momentum j_ϵ, and unminimized contextual tensor B_ϵ converge with the derivatives required in $\mathcal{A}_\epsilon = -K_\epsilon^{-1}j_\epsilon$, $h_\epsilon = B_\epsilon - j_\epsilon \otimes j_\epsilon/K_\epsilon$, and $\mathcal{F}_\epsilon = d\mathcal{A}_\epsilon$. The positive limiting margins persist for sufficiently small positive ϵ, uniformly over the angular episode outcomes. The grouping threshold can be imposed simultaneously by taking the smaller admissible ϵ_0. Equation eq. (14.34) gives exact parent equality. □

The fixed positive initial marked amplitudes are needed in this finite-window argument: they keep the limiting inertia nonzero at the same finite entrance time as grouping. No numerical bound on ϵ_0 is asserted. The group persists indefinitely with its initial record episode fixed, while nonflatness is asserted on the declared finite window. Later admitted record continuations preserve quotient closure but may alter the grouping or geometry.

14.6 The remaining formation obligations

The construction meets the admitted-family, persistent-selection, conditional-inheritance, physical-compression, and interaction-necessity requirements at one stage. Its selector and phase-insensitive interface are constitutive choices. In particular the phase quotient is available before any group is selected and also under the null. Coupling the marked sector to public communication does not make the separate erased phase endogenously invisible.

Joining two already canonical parent products is associative and preserves their inherited port fields and histories. It erases no new phase. It therefore supplies neither a second noninjective handoff nor the counterfactual dependence required by (F6b). The next obligation is a genuinely collective or interaction-selected sufficient interface, followed by a further compressive inherited selection that depends on the first handoff. The present conditional theorem does not close open problem 6.32.

Chapter 15

Collective Compression and Conditional Local Parents

The phase parent of proposition 14.6 erases a distinction already invisible on each primitive port. A collective alternative transmits the second angular law between peers and admits relative-angle responses. Each proper block is then exposed by a response across its boundary, while the whole connected population can lose one common physical orientation. This chapter constructs that quotient and a local realization with finite reference memory. A finite preparation generator gives a second realization. A causal local phase memory removes the reference evaluation when the initial collective direction is coherent across context. That preparation condition is explicit; it is not generated by the recurrence. The unrestricted local-parent question remains inconclusive, with obstructions for specified memory classes rather than every possible mechanism.

15.1 A transmitted physical angle

There are three agents with the normalized per-agent model of definition 4.17. Relative to a uniform sign and two normalized Haar circles, write

$$q_i(z,\phi,\psi) = Z_i^{-1}\exp\{l_i z/2 + A_i\cdot e_\phi + zC_i\cdot e_\phi + B_i\cdot e_\psi\}, \tag{15.1}$$
$$Z_i = I_0(|B_i|)\frac{e^{l_i/2}I_0(|A_i+C_i|)+e^{-l_i/2}I_0(|A_i-C_i|)}{2}.$$

Here $e_\theta=(\cos\theta,\sin\theta)$ and $A_i,C_i,B_i\in\mathbb{R}^2$. The presentation probability is

$$s_i(w) = \frac{e^{n_i w/2}}{2\cosh(n_i/2)}, \qquad w\in\{-1,1\}. \tag{15.2}$$

The self evaluator is $G_i = e^{Jz\cos\phi}/I_0(J)$, with $J=\epsilon^3$, and the self presentation reference is uniform. Source priors in both sectors are $1/5$ for self and $2/5$ for each peer. Rows are reduced exactly at unit temperature.

The nonself content channel copies z,ψ and resamples ϕ from Haar measure. Define $h_0(B)=\log I_0(|B|)$ and $M(B)=\nabla h_0(B)$. The transmitted density is

$$\nu_j(z,\phi,\psi) = \frac{e^{L_j z/2}}{\cosh(L_j/2)}\frac{e^{B_j\cdot e_\psi}}{I_0(|B_j|)}, \qquad L_j = l_j + h_0(A_j+C_j) - h_0(A_j-C_j). \tag{15.3}$$

The sign reference is uniform, so its density differs by a factor of two from its probability mass. Presentation peers use the identity channel. Every source remains normalized.

Evidence uses the positive normalized likelihood

$$\ell_{r,\lambda}(o,\omega \mid z,\phi,\psi,w) = \frac{e^{roz/2}}{2\cosh(r/2)} \frac{e^{\lambda\cos(\omega-\phi)}}{I_0(\lambda)}, \tag{15.4}$$

relative to sign counting measure and angular Haar measure. Every admitted finite-strength evidence transaction appends its actual outcome and leaves its likelihood in the objective. Write $H_i = \sum ro$ and $\Lambda_i = \sum \lambda e_\omega$ for its persistent coefficients, retaining the entire ordered word as well. A passive marked frame change rotates the local embeddings of these fixed lab labels and the evaluator direction. There is one marked structure group $G = S^1$ with trivial representation on the presentation sector. The common physical rotation of ψ is a separate nongauge action.

Proposition 15.1 (The exact transmitted-angle field). On the positive family eq. (15.1), put $m_i = \tanh(L_i/2)$ and $t_i = \mathbb{E}_{q_i}[Z\cos\Phi]$. The content Gibbs weights and own-block Fisher equations are

$$\begin{aligned} w_{ii} &= \frac{e^{Jt_i}}{5I_0(J)}, \qquad w_{ij} = \frac{2e^{L_j m_i/2 + B_j\cdot M(B_i)}}{5\cosh(L_j/2)I_0(|B_j|)}, \\ \beta_{ij} &= w_{ij}/\sum_r w_{ir}, \qquad j \neq i, \end{aligned} \tag{15.5}$$

$$\begin{aligned} \dot l_i &= -l_i + \sum_{j\neq i}\beta_{ij}L_j + H_i, \quad \dot A_i = -A_i + \Lambda_i, \\ \dot C_i &= -C_i + \beta_{ii}Je_1, \qquad \dot B_i = -B_i + \sum_{j\neq i}\beta_{ij}B_j, \\ \dot n_i &= -n_i + \sum_{j\neq i}\gamma_{ij}n_j. \end{aligned} \tag{15.6}$$

Here γ is the original presentation Gibbs row. Every equation uses the original inference time.

Proof. Let $S_i = \mathbb{E}_{q_i}\log q_i$. The self cost is $S_i - Jt_i + \log I_0(J)$ and the peer cost is $S_i - L_j m_i/2 + \log\cosh(L_j/2) - B_j\cdot M(B_i) + h_0(B_j)$. Their common entropy cancels from the content row, giving eq. (15.5). The Gibbs envelope removes derivatives of the optimizing row from the own-objective first variation. The centered logarithmic Fisher velocity is a linear combination of $z, e_\phi, ze_\phi, e_\psi$. Matching these coefficients gives eq. (15.6). Their covariance is positive definite: a linear combination constant under a positive density has zero nonconstant Fourier and sign coefficients. This is an invariant family of the own-law Fisher field, without a Euclidean approximation or source-role forces from other agents' objectives. □

With the initial record pattern $(+,+,-)$ held fixed, this model has the interaction-dependent grouping window of theorem 14.5. To see why the transmitted angle does not invalidate that argument, take $B_i = \delta b_i$ with b_i in a compact pointed cone and nonzero. The new term in the row logarithm is $O_{C^1}(\delta^2)$ on bounded scaled charts. With $u = l/\epsilon$ and C/ϵ^3 scaled, the public limiting field is

$$\dot u = \left(-\frac{7}{5}I + \frac{2}{5}\mathbf{1}\mathbf{1}^\top\right)u + (1,1,-1)^\top + O_{C^1}(\epsilon^2+\delta^2), \qquad u_* = (15/7, 15/7, 5/7). \tag{15.7}$$

The remaining scaled blocks have a symmetric negative-definite limiting derivative. A sufficiently small attraction ball retains positive public signs and pairwise magnitude ratios above $1/4$, since

the smallest equilibrium ratio is $1/3$. Initial u near $(4, 1/2, -1)$ has strict singleton margins. Finite-time dependence followed by the attracting-ball estimate gives a common entrance time T and persistence with that initial episode fixed.

The prescribed null deletes nonself costs while retaining their source labels, priors and row entropy. Its surviving self weight is not one. On $|l_i| \le 12\epsilon$, $|A_i| \le 1/10$, $|C_i| \le 2\epsilon^3$, $|B_i| \le 1/10$, the self KL bound of eq. (14.28) still gives $1/9 \le \beta^0_{ii} \le 1/5$. Its equations include $\dot{l}^0_i = -\beta^0_{ii} l^0_i + \epsilon o_i$ and $\dot{B}^0_i = -\beta^0_{ii} B^0_i$. The signed strip $1/4 \le o_i l^0_i/\epsilon \le 12$ is invariant, while $|L_i - l_i| < \epsilon/8$. Thus the third public sign remains negative under the null and the full block is never selected. Sample the same state graph at fixed census times $0, T$, retaining both partition labels and times. Strict uniform margins make those outputs constant on the chosen contextual patch. No continuously reported parameter-dependent crossing time is inserted into a differentiable statistical experiment.

15.2 Collective rather than primitive physical loss

Declare the protocol before the handoff. It admits the public, marked-lab and presentation queries, original normalized evidence transactions and source-label readouts, and relative-angle queries on the complete graph. A joint relative query draws each involved primitive angle once and returns all requested differences. Shared vertices share samples; triangle outputs obey their cycle relation. Transactions use fresh samples unless reuse is explicitly recorded. A diagnostic appends history without becoming an assimilated likelihood. Policies use the complete history, fixed lab data and optional independent randomization. No absolute ψ response is admitted at any stage.

For $\zeta_i = [I_1(|B_i|)/I_0(|B_i|)]e^{\mathrm{i}\arg B_i}$, the pair moment is

$$c_{ij} = \mathbb{E}e^{\mathrm{i}(\Psi_i - \Psi_j)} = \zeta_i\overline{\zeta_j}, \qquad |\zeta_i| = \sqrt{\frac{|c_{ij}||c_{ik}|}{|c_{jk}|}}. \tag{15.8}$$

The function $I_1(b)/I_0(b)$ has strictly positive derivative $\mathrm{Var}_b(\cos\Psi)$ for $b > 0$. Thus the three pair laws identify the tuple up to one common rotation. With an exterior nonzero ζ_j fixed, $c_{ij}/\overline{\zeta_j}$ identifies every interior vector. Public and marked law queries identify l, A, C, and presentation queries identify n. The same erasure cannot therefore be performed independently on a proper block with its boundary retained. This is identification from exact response laws, not estimation from finitely many samples. The common symmetry preexists grouping; no interaction-created invisibility is asserted.

15.3 Finite reference memory and closed local jets

Let $x(c) = (l_i, A_i, C_i, B_i, n_i)_i \in \mathbb{R}^{24}$ be a C^2 section on a compact contextual patch, and fix c_0 before retention. At every context the same realized lab history is applied to eq. (15.6). Suppose the initial $B_i(c)$ lie in a common open pointed cone, with its rotational union admitted. For a vector v positive on that cone,

$$v \cdot B_i(c,t) \ge e^{-t} v \cdot B_i(c,0) > 0. \tag{15.9}$$

This follows by variation of constants in the cooperative B equation. Finite evidence words preserve it. The other natural parameters have finite-interval bounds from eq. (15.6), so every finite admitted branch remains regular.

Write $x_\circ = x(c_0)$, $S = \sum_i B_i(x_\circ)$ and

$$U(S) = \frac{1}{|S|}\begin{pmatrix} S_x & S_y \\ -S_y & S_x \end{pmatrix}, \qquad y = \mathsf{R}_U x(c), \qquad \widehat{x}_\circ = \mathsf{R}_U x_\circ, \tag{15.10}$$

where R_U rotates only the B blocks. This smooth formula uses no angle branch. A whole-section retention applies this one U at every c and removes precisely one context-constant physical angle. The local realization below retains only the data required at c and the reference.

Theorem 15.2 (Conditional evaluation-local realization). Retain $j_c^2 y$, the full finite native tuple $\widehat{x}_\circ$, all primitive ports, constitutive data, complete ordered history, census records and persistent likelihood coefficients. If record embeddings vary with context, retain their known two-jets. Give the past-law experiment its actual compatible initialization. These data have an autonomous original-time evolution and determine the exact local two-jet and every admitted future joint response/update kernel at c. They preserve the actual collective Fisher cost on the declared regular experiment. Initialization uses reference-context data and is not a rule from the germ at c alone.

Proof. Let $V(x, c, h)$ denote eq. (15.6), let $V_S = \sum_i V_{B_i}(\widehat{x}_\circ, c_0, h)$, and let $\mathcal{J}$ act by $J_2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on each B block and by zero elsewhere. Set

$$\Omega(\widehat{x}_\circ, h) = \frac{\det(S, V_S)}{|S|^2}, \qquad W(y, c, h; \widehat{x}_\circ) = V(y, c, h) - \Omega \mathcal{J} y. \tag{15.11}$$

Equivariance and $\dot{U} = -\Omega J_2 U$ give

$$\dot{y} = W(y, c, h; \widehat{x}_\circ), \qquad \dot{\widehat{x}}_\circ = V(\widehat{x}_\circ, c_0, h) - \Omega \mathcal{J} \widehat{x}_\circ. \tag{15.12}$$

The canonical anchor stays on $S_y = 0, S_x > 0$. The subtraction is the derivative of the retention, not a separately fitted parent force. This is the exact vector-field intertwining criterion of eq. (6.4), an instance of exact lumping [79].

At fixed t, h and prepared section, the reference is held fixed under evaluation differentiation in c. For contextual indices r, s,

$$\begin{aligned} \dot{y}_r &= W_y y_r + W_r, \\ \dot{y}_{rs} &= W_y y_{rs} + W_{yy}[y_r, y_s] + W_{yr} y_s + W_{ys} y_r + W_{rs}. \end{aligned} \tag{15.13}$$

The explicit partials hold $y, \widehat{x}_\circ, h$ fixed. Every state dependence of the Gibbs rows is differentiated. Since the native field reads no neighboring contexts or contextual derivatives, no third jet enters. Evidence adds the same saved likelihood at both evaluations without resetting either law. Diagnostics change only history. For a two-dimensional base the local law jet has $24(1 + 2 + 3) = 144$ coordinates, plus 24 reference coordinates before constraints. The full ordered history remains unbounded in length; this is not a fixed-memory claim.

The actual parent laws are the normalized products of eq. (15.1) in the y coordinates, together with their original presentation and Gibbs-label factors. The reference is deterministic internal memory, not an extra independent sample or agent. Common physical rotation preserves every joint relative response, every other response and its actual next retained state. Equal histories therefore supply the same next action and wait law. Induction over the joint transaction kernels proves all admitted finite adaptive histories, including response/next-state dependence.

For an unconditional history/current experiment, retain an initial local two-jet and initial reference tuple in one canonical frame as a replay seed, together with the full protocol and chronological word. Applying the original normalized kernels defines its actual history law Π_c. A current tuple and one observed word alone do not determine that past probability. Conditioning on the initial public pattern retains its positive probability normalizer. No distribution over law configurations is introduced.

Finally, for fixed t, h, rotating the fresh private samples by $U(S(x_\circ^h(t)))$ is independent of the evaluation coordinate c. The joint map $T(h, Y) = (h, T_h Y)$ preserves Haar measure and commutes with the marked action. Stored marked outcomes are fixed lab labels; their local direction embeddings transform. Hence, for

$$\mathcal{P}_{c,t}(dh, dY) = \Pi_c(dh) Q_{c,t}^h(dY), \qquad \overline{\mathcal{P}}_{c,t} = T_* \mathcal{P}_{c,t}, \tag{15.14}$$

the contextual score and marked orbit score are carried isometrically. For every X, ξ,

$$\overline{\mathcal{C}}(X, \xi) = \mathbb{E}_{\overline{\mathcal{P}}}(\overline{s}_X + \overline{\ell}_\xi)^2 = \mathbb{E}_{\mathcal{P}}(s_X + \ell_\xi)^2 = \mathcal{C}(X, \xi). \tag{15.15}$$

This proves actual-law cost inheritance before minimization, including history and label scores. It gives the same minimizing affine class, isotropy and minimized tensor. On a locally free patch their connection and curvature agree. □

The retained reference contains no inverse rotation: $U(R_\alpha S) R_\alpha = U(S)$, so simultaneous rotation changes none of the canonical local, reference or seed data. It still changes the excluded child absolute-angle diagnostic. The reference retains the phase difference between c and c_0, not the discarded global angle. For $B_i(c) = b_i e_{\chi(c)}$, the actual canonical density contains $b_i e_{\chi(c) - \chi(c_0)}$, and its private phase information is

$$\sum_i b_i \frac{I_1(b_i)}{I_0(b_i)} (d\chi)^2 > 0. \tag{15.16}$$

Pointwise canonicalization would delete this term. Copying it into a stored tensor would not restore it to a normalized parent law.

A local initial two-jet has an actual law realization: use its degree-two natural-parameter polynomial near the evaluation point and then evolve that preparation by the exact pointwise field, without truncating the evolved density. Its two-jet agrees with the original there throughout the common finite regular interval by eq. (15.13). This supplies actual law germs and their connection curvature, but does not reconstruct the unknown original section away from that point. Compatible initial data over all contexts remain a field. Marked overlap maps preserve this compatibility because the private canonicalization commutes with them.

For a common nonflat window take $A_1(c,0) = ae_\chi$, $A_2(c,0) = de_1$, with a, d in positive compact ranges independent of ϵ, δ, and small noncollinear cone vectors $B_i = \delta b_i$. Keep the initial episode fixed. At the uncanonicalized $\epsilon = \delta = 0$ comparison, put $v = e^{-t}a$ and $D(t) = \sum_{i\neq 1} \mathcal{I}(e^{-t}|A_i(0)|) > 0$, where $\mathcal{I}(v) = vI_1(v)/I_0(v)$. The same minimization as in eq. (14.36) gives

$$\begin{aligned} \mathcal{A}_{0,t} &= -\frac{\mathcal{I}(v)}{\mathcal{I}(v) + D(t)} d\chi, \\ \mathcal{F}_{0,t} &= -\frac{D(t)e^{-t}\mathcal{I}'(v)}{[\mathcal{I}(v) + D(t)]^2} da \wedge d\chi, \\ h^{0,t}_{aa} &= e^{-2t}\,\mathrm{Var}_v(\cos\Phi) > 0, \qquad h^{0,t}_{\chi\chi} = \frac{\mathcal{I}(v)D(t)}{\mathcal{I}(v) + D(t)} > 0. \end{aligned} \tag{15.17}$$

The cross coefficient vanishes and $\mathcal{I}'(v) > 0$. Compact finite-window margins persist for sufficiently small positive ϵ, δ: the exact finite-time flow, its two-jet, normalized history kernels and Fisher integrals depend continuously on those parameters on a common positive tube. Apply continuity to the fine laws, since the canonical anchor is singular at $\delta = 0$, and apply eq. (15.15) only at positive δ. This gives the same nonflat rank-two window containing T. Later records preserve inheritance, but need not preserve this grouping or nonflatness.

15.4 Handoff frames and finite preparation generators

Fix the census handoff T and freeze $U_T = U(S(x_\circ(T)))$ once. Then $z(c,t) = \mathsf{R}_{U_T} x(c,t)$ obeys $\dot z = V(z,c,h)$ for $t \geq T$, with no moving-frame term. Its actual costs still satisfy eq. (15.15). Future local evolution needs no live reference, but its initialization is reference-coherent memory, not an instantaneous quotient of $x(c,t)$ alone. Recovering the continuously reanchored representative later requires the reference evolution or its accumulated phase.

Proposition 15.3 (Conditional finite preparation-and-history realization). Suppose the prepared section has a known finite C^2 generator $x_0(c) = f(c;p)$, with $p \in P \subset \mathbb{R}^m$, equivariant under the common physical rotation and with nonzero reference anchor. Retain its canonical coefficients p_*, the full chronological history and original time. Then the current contextual laws and their two-jets have an exact whole-patch realization using these data, with all the preceding operational and geometric inheritance properties.

Proof. Let $\mathcal{E}_{t,h}(c)$ be the chronological composition of the unique native flows and original fixed-record coefficient appends. Evaluate

$$x^h_t(c;p_*) = \mathcal{E}_{t,h}(c) f(c;p_*). \tag{15.18}$$

Between transactions $\dot p_* = 0$, $\dot h = 0$ and $\dot t = 1$; cached local values satisfy the native field and eq. (15.13). At a transaction append its actual outcome and likelihood, if it is evidence. The original normalized response/update kernels evaluated on eq. (15.18) generate its full path law. Physical equivariance removes exactly the common seed orientation; it is not stored in the canonical seed. Evaluate the same generator at c_0 if a moving representative is wanted. The sample rotation is again independent of evaluation c, proving eq. (15.15). No projection onto a finite basis of evolved sections is used. □

For a nonempty family, take the marked patch used in eq. (15.17), constant initial l_i/ϵ near $(4, 1/2, -1)$, $C_i = 0$, small fixed presentation laws and three context-constant noncollinear b_i in the cone, with their rotational union. These are finite preparation coefficients, relatively open in their parameter manifold. They are not an open set of arbitrary C^2 functions. The same persistent grouping and nonflat window apply. The full word remains retained, and evaluation at a new context may require replay of its entire native history. Finite seed dimension gives neither constant evaluation cost nor bounded record memory.

15.5 A causal local phase memory

The reference frame can be replaced by an accumulated local phase on a restricted preparation family. Keep the same field V, physical action R, generator $\mathcal{J}$ and cone domain. At the actual initialization time t_* define, using only the value at the evaluation context,

$$S_{\mathrm{loc}}(x) = \sum_i B_i(x), \qquad \bar{x} = \mathsf{R}_{U(S_{\mathrm{loc}}(x))} x. \tag{15.19}$$

The sum of the B blocks of $\bar{x}$ lies on the positive first axis. Unlike the reference tuple, this slice uses no value at c_0. Define its local angular velocity by

$$\omega_{\mathrm{loc}}(\bar{x}, c, h) = \frac{\det(S_{\mathrm{loc}}(\bar{x}), \sum_i V_{B_i}(\bar{x}, c, h))}{|S_{\mathrm{loc}}(\bar{x})|^2}. \tag{15.20}$$

It uses the actual native Gibbs rows, including their dependence on the local public and marked laws. It is not a function of the sum alone.

Proposition 15.4 (Closed causal reconstruction in the original time). Starting at t_* with $\vartheta = 0$, evolve

$$\dot{\bar{x}} = W_{\mathrm{loc}}(\bar{x}, c, h) := V(\bar{x}, c, h) - \omega_{\mathrm{loc}} \mathcal{J} \bar{x}, \qquad \dot{\vartheta} = \omega_{\mathrm{loc}}. \tag{15.21}$$

The actual parent laws are the normalized native laws at

$$\widetilde{x} = \mathsf{R}_{R_\vartheta} \bar{x}. \tag{15.22}$$

They satisfy $\dot{\widetilde{x}} = V(\widetilde{x}, c, h)$ on every regular finite branch. Their local two-jet, together with that of ϑ, closes at order two. Every original finite adaptive joint response/update history, persistent likelihood and census record is preserved, with a compatible actual preparation and past-law initialization.

Proof. The cone estimate eq. (15.9) keeps S_{loc} nonzero. Differentiating its canonicalizing matrix gives $\dot{U} = -\omega_{\mathrm{loc}} J_2 U$. Physical equivariance therefore gives eq. (15.21). Differentiating the actual readout cancels its rotating-coordinate term:

$$\dot{\widetilde{x}} = \mathsf{R}_{R_\vartheta}(\dot{\bar{x}} + \dot{\vartheta} \mathcal{J} \bar{x}) = \mathsf{R}_{R_\vartheta} V(\bar{x}, c, h) = V(\widetilde{x}, c, h). \tag{15.23}$$

Thus no independently fitted force or time change is used. Uniqueness, with the canonical initial value, gives

$$\widetilde{x}(c, t) = \mathsf{R}_{U(S_{\mathrm{loc}}(x(c, t_*)))} x(c, t). \tag{15.24}$$

The memory may equivalently be an $\mathrm{SO}(2)$ variable satisfying $\dot{g} = \omega_{\mathrm{loc}} J_2 g$, $g(t_*) = 1$, so no global angle branch is required.

For explicit contextual derivatives a, b and state derivative D, the slice jets obey the prolongation eq. (15.13) with W_{loc} in place of W. The phase jets, initialized to zero, obey

$$\begin{aligned} \dot{\vartheta}_a &= D\omega_{\mathrm{loc}} \bar{x}_a + (\omega_{\mathrm{loc}})_a, \\ \dot{\vartheta}_{ab} &= D\omega_{\mathrm{loc}} \bar{x}_{ab} + D^2 \omega_{\mathrm{loc}}[\bar{x}_a, \bar{x}_b] \\ &\quad + D(\omega_{\mathrm{loc}})_a \bar{x}_b + D(\omega_{\mathrm{loc}})_b \bar{x}_a + (\omega_{\mathrm{loc}})_{ab}. \end{aligned} \tag{15.25}$$

All explicit partials hold $\bar{x}, h$ fixed. In contrast to the fixed-reference derivative, the contextual derivatives of ω_{loc} enter. The field has contextual order zero, so no third jet is needed. In each private block the actual law derivatives are

$$\begin{aligned} \widetilde{B}_{i,a} &= R_\vartheta (\bar{B}_{i,a} + \vartheta_a J_2 \bar{B}_i), \\ \widetilde{B}_{i,ab} &= R_\vartheta \big(\bar{B}_{i,ab} + \vartheta_a J_2 \bar{B}_{i,b} + \vartheta_b J_2 \bar{B}_{i,a} \\ &\quad + \vartheta_{ab} J_2 \bar{B}_i - \vartheta_a \vartheta_b \bar{B}_i \big). \end{aligned} \tag{15.26}$$

Every row and embedding derivative is included. Actual slice-valued initial families realize compatible initial jets, and their native evolution realizes these law germs.

At each fixed c, eq. (15.24) is a time-independent physical rotation. All original transaction kernels are invariant under it, including the entire joint relative response with one sample per involved primitive vertex. Induction on responses, waits and updates proves equality of every admitted adaptive joint history. Evidence appends its actual persistent likelihood without jumping the law; diagnostics append history without assimilation. The laws, slice and phase memory are continuous at these transactions. Both original census records remain retained. A local canonical preparation seed and full chronological word, or their compatible initialized past-law experiment, supply the unconditional history law. They do not introduce a prior over prepared sections or an independently sampled memory factor. □

The recurrence implements a frozen local initial frame, whereas the earlier handoff construction froze a reference frame. Retain its actual accumulated value at a later census T. Initializing $\vartheta = 0$ afresh at T instead changes eq. (15.24) to the frame at T; its compatibility with the original preparation is not automatic. Neither existing records nor elapsed phase memory may be reset. One finite state per context and its two-jet are still distinct from a finite global encoding of an unknown contextual function, and the complete word has unbounded length.

Theorem 15.5 (Actual geometric inheritance under initial coherence). In addition to proposition 15.4, suppose the actual preparation satisfies

$$\frac{S_{\mathrm{loc}}(x(c, t_*))}{|S_{\mathrm{loc}}(x(c, t_*))|} = e_* \in S^1 \qquad \text{for every } c \in \mathcal{C}, \tag{15.27}$$

where e_* is independent of subsequent history and its absolute angle is not retained. The causal parent preserves the actual complete Fisher cost eq. (15.15), hence the minimizing marked connection classes, isotropy, tensor and curvature. It removes one common physical orientation

after all memory is counted. There is a nonempty family satisfying this condition with the same interaction-dependent grouping and positive-rank nonflat window.

Proof. Every local initialization in eq. (15.19) applies the same matrix U_* under eq. (15.27). The map of current samples in eq. (15.24) is consequently independent of c, t, h, preserves Haar measure, leaves history and numerical lab labels fixed, and commutes with the marked action. Differentiating its normalized joint density proves eq. (15.15) for every contextual tangent and marked comparison velocity, including history and public-pattern normalization scores. The parent samples the laws at $\widetilde{x}$; the phase derivatives in eq. (15.26) therefore enter its actual scores. Sampling only the slice laws and attaching a stored tensor would not give this equality.

Take the marked family of eq. (15.17) with context-constant initial $B_i = \delta b_i$ in the pointed cone, allowing noncollinear b_i. It satisfies eq. (15.27); the published finite-window continuity argument then supplies positive rank and nonzero marked curvature for small positive ϵ, δ. The weak initial record episode is retained. The equality of actual costs transfers that same window. Later arbitrary evidence preserves exact dynamical and operational inheritance on each regular finite branch, while it need not preserve grouping or nonflatness.

The initial slice, its canonical replay seed, generated phase difference and complete history are invariant under a constant physical rotation of the original preparation. Conversely, equal canonical initial fields under eq. (15.27) determine preparations differing by one context-constant rotation. The erased angle is therefore not restored by memory. Relative probes still expose every proper block against its exterior; no independently invisible primitive factor is added. □

Initial coherence is a preparation restriction, not a consequence of causal updating or an open condition among arbitrary contextual sections. For a general preparation write locally $\alpha_0(c) = \arg S_{\mathrm{loc}}(x(c, t_*))$. The actual sample map now depends on c. At corresponding samples its contextual score is

$$\widetilde{s}_X(T_0 Y) = s_X(Y) - (X\alpha_0)\ell_{\mathrm{phys}}(Y), \tag{15.28}$$

where ℓ_{phys} is the score of positive common rotation of the B_i. Indeed $\psi_{\mathrm{parent}} = \psi_{\mathrm{fine}} - \alpha_0$ and $\ell_{\mathrm{phys}} = -\sum_i \partial_{\psi_i} \log q_i$. Thus operational history equality alone has not preserved the actual geometry. With $\alpha(c, t) = \arg S_{\mathrm{loc}}(x(c, t))$ locally,

$$d_c\alpha(c, t) = d_c\alpha_0(c) + d_c \int_{t_*}^{t} \omega_{\mathrm{loc}}(c, s)ds. \tag{15.29}$$

Temporal integration supplies the second term, not the initial contextual one-form. This exact physical-phase derivative is distinct from the marked minimizing connection, which can remain nonflat. No context-relocation action is part of the fixed protocol.

15.6 Scoped obstructions to stronger locality

The reference dependence cannot be removed by merely storing its sum. For $B_1 = B_3 = be_1$, $B_2 = be_2$, $A = C = 0$, $l_1 = l_3 = 0$ and $l_2 = v$, put

$$\begin{gathered} s = \frac{1}{5I_0(J)}, \quad k = \frac{2}{5I_0(b)}, \quad E = e^{bI_1(b)/I_0(b)}, \\ u = \operatorname{sech}(v/2), \qquad D = s + kE + ku. \end{gathered} \tag{15.30}$$

The exact rows give

$$\Omega = \frac{1}{5}\left(\frac{4ku - 2kE}{D} - \frac{2k}{s+2k}\right), \qquad \frac{d\Omega}{du} = \frac{4ks + 6k^2E}{5D^2} > 0. \tag{15.31}$$

Thus $S = (2b, b)$ stays fixed while its angular velocity changes. Also, for $c \neq c_0$, a section can be changed only near c_0 without changing its germ at c, yet its canonical value at c changes. These facts exclude sum-only closure and germ-only computation of this representative, not all alternative parent realizations.

Theorem 15.6 (Finite differential-order obstruction). Consider a rich smooth open class of sections in the positive chart, allowing independent finite-jet variations. Suppose the parent's actual law is a smooth rule $\bar{p}(c) = F(c, j_c^k x, h)$ of some finite order. Local native-history summaries may be included if they are themselves finite-jet functions on the fixed-history stratum. There is no separately retained phase-bearing field or reference-context memory. If the actual minimizing marked connection classes and minimized tensors agree for every section in that class, then the actual-law rule is independent of positive-order jets. It cannot also erase the common physical ψ angle while preserving that geometry.

Proof. For $k \geq 1$ fix a k-jet and vary the next jet in a contextual direction $X \neq 0$. Contraction of symmetric $(k+1)$-tensors with X is onto symmetric k-tensors; a supported polynomial primitive realizes each variation. At the point the parent law, orbit map $\bar{J}$, fine first score and fine minimizing class are fixed. The parent tangent has the form $\bar{u}_X(\tau) = \bar{u}_X(0) + \tau w$. Choose one representative $A_f(X)$ of the common minimizing class. Equality of tensors requires

$$\|\bar{u}_X(0) + \bar{J}A_f(X) + \tau w\|_{\bar{g}}^2 = h_f^{\min}(X, X) \tag{15.32}$$

on an interval. Its quadratic coefficient is $\bar{g}(w, w)$, so positivity forces $w = 0$. Descending in derivative order removes all positive-order dependence of the actual law.

For random histories make the argument in the joint L^2 space. Vary the highest initial derivative while holding lower initial jets fixed. The history law and fine first scores stay fixed, and the highest current variation is transported by the invertible fixed-record native-flow derivative on each regular branch. The quadratic coefficient is $\mathbb{E}_H\|w_H\|^2$, which must vanish. A finite basis of initial variations gives derivative independence almost surely on the actual history experiment. Smooth positive kernels give its full-support continuity extension where those hypotheses hold. No conditional equality is inferred merely from an equality of averaged minima.

An order-zero phase-erasing rule has zero derivative along a common physical rotation of the B_i. The fine score in that direction has squared norm $\sum_i |B_i| I_1(|B_i|)/I_0(|B_i|) > 0$. It is orthogonal to the marked orbit, which acts on ϕ, and the relative-history and row laws are unchanged. Its fine horizontal norm is therefore positive while the parent norm is zero. Inside a nonflat section patch one may instead hold its marked derivatives fixed and vary that physical phase derivative; the positive quadratic term gives the same contradiction. □

This theorem applies to the smooth subset of the admitted C^2 class. It does not assume that every C^2 section has arbitrarily high derivatives. Independent jet variation and equality of the connection classes are necessary hypotheses of the proof: tensor equality alone could hide a changing marked-orbit tangent. The result is not an application of fixed-statistic Fisher contraction to a moving law map; the latter has additional derivative terms, as in proposition 3.14. Score sufficiency for a fixed statistic is a separate statement [120]. Finite reference memory and constrained generator families are outside the theorem.

Proposition 15.7 (One regular finite encoding cannot hold arbitrary profiles). On a C^2-open class of sections, no single smooth state of dimension m can reproduce every direct public response law at every c, if retention and response evaluation are C^1 on finite-dimensional smooth section variations and all finite smooth memory is counted in m.

Proof. On one retained-history stratum choose $m+1$ disjoint contextual neighborhoods and points c_j in them. Perturb l_1 by $\sum_j a_j f_j(c)$ with $f_j(c_j) = 1$ and disjoint supports. At a regular finite time these current perturbations have admitted predecessors by local invertibility of the fixed-record flow. Strict census and window margins persist for small coefficients. The Jacobian of the $m+1$ direct sign probabilities is diagonal with positive entries, since $\Pr(Z = 1) = \mathrm{sigmoid}(L_1)$ and $\partial L_1/\partial l_1 = 1$. Its rank is $m+1$. Factoring through an m-dimensional state would bound the rank by m, a contradiction. □

This is a statement about one encoding of an indexed contextual experiment, not the addition of context-relocation actions to the protocol. It excludes neither one initialized local germ, compatible local fields, a finite generator family, nonfinite memory nor irregular encodings.

Theorem 15.8 (Local value-path memory cannot supply arbitrary initial phase). On the rich smooth open preparation class, let a strictly local causal retention read the entire value path $x(c,s)$, $t_* \le s \le t$, all actual records, persistent evidence, censuses and fixed templates. It reads no contextual derivatives, neighboring or reference contexts, or initialized contextual phase field. Require the complete retained state, including every memory and seed, to be invariant under constant physical rotation, and let the actual parent laws depend only on that state and fixed data. Such a retention cannot universally preserve the actual minimized Fisher tensor. There is no bound on its temporal horizon or memory dimension. The conclusion also holds for a regular stochastic retained-state kernel with this invariance and phase-independent randomization.

Proof. For any smooth time-independent $\kappa(c)$, pointwise equivariance gives

$$x^{\kappa,h}(c,t) = \mathsf{R}_{R_{\kappa(c)}} x^h(c,t), \qquad \Pi_c^\kappa = \Pi_c. \tag{15.33}$$

The second identity follows by coupling the original invariant transaction kernels and policy randomness; records and censuses coincide. At each fixed c the full local value paths differ by one constant rotation. Local rotation-invariant retention therefore gives identical parent law families for all κ, even with the complete temporal path available.

Choose $\kappa(c_*) = 0$ and $X\kappa(c_*) = \lambda$. The fine contextual score becomes $s_X + \lambda\ell_{\mathrm{phys}}$. The physical score is orthogonal to every marked orbit score: conditional on history the private factors are independent of the marked factors, have mean-zero scores, and neither the history nor the rows have a common-phase score. Its joint norm is

$$I_{\mathrm{phys}} := \|\ell_{\mathrm{phys}}\|^2 = \mathbb{E}_H \sum_i |B_i^h| \frac{I_1(|B_i^h|)}{I_0(|B_i^h|)} > 0. \tag{15.34}$$

The minimizing marked class at c_* is unchanged. With $s_X^{\mathrm{hor}} = s_X + \ell_{A_f(X)}$, the fine minimized cost is

$$h_f^\kappa(X, X) = h_f(X, X) + 2\lambda\mathbb{E}[s_X^{\mathrm{hor}}\ell_{\mathrm{phys}}] + \lambda^2 I_{\mathrm{phys}}. \tag{15.35}$$

It cannot be constant on an interval of λ, whereas the parent tensor is. Small supported smooth phase variations stay inside the strict common nonflat-window class and preserve the grouping exactly. This is a joint-experiment score identity; no conditional geometric equality is inferred from an averaged one. □

Corollary 15.9 (Smooth bounded-jet causal memories). Suppose a finite-dimensional deterministic memory is initialized smoothly from $c, j_c^k x(t_*)$ and the original initial records. Between transactions let $\dot{m} = G(m, j_c^k x, c, h)$; let record updates and the actual-law readout also be smooth functions of these data. Assume one finite order k bounds initialization, evolution and readout on every branch, with no reference seed or separately supplied contextual phase field. On a rich smooth open section class, such a parent cannot erase the common physical phase and inherit both the original minimizing marked affine classes and minimized tensors.

Proof. At a fixed finite word h, the native k-jet and memory form a closed smooth ordinary differential system. Its memory value is a smooth function of the initial k-jet. The native fixed-record flow is locally invertible on its reached regular image; record appends do not reset the law. Its jet prolongation is locally invertible as well. Hence the actual parent law has the finite-order form $F_{t,h}(c, j_c^k x(t))$. Apply theorem 15.6, including its joint-L^2 history argument. The chronological law remains initialized from the actual preparation; branchwise inversion does not infer an unconditional past law from a current snapshot or add a reverse-time protocol. □

The coherent preparation family of eq. (15.27) does not permit the independent phase-gradient variations used in these obstructions. Its causal construction therefore coexists with them. The unrestricted question remains inconclusive: an admissible causal source of the initial contextual phase relation outside these classes, or an obstruction covering it, is still needed. Genuinely unbounded contextual-order memory, differently specified infinite-dimensional memory and stochastic mechanisms outside theorem 15.8 are not classified by corollary 15.9. Temporal records do not supply an undeclared contextual calibration channel. The physical symmetry remains predeclared; interaction-created invisibility and a new second compressive handoff remain open.

Chapter 16

Contextual Coupling of Statistical Laws

The collective connection of proposition 3.9 acquires a dynamical role when its minimized contextual cost is added to the reciprocal law potential. This chapter gives two constructions on supplied contextual bases. On a finite graph, regular auxiliary alignment produces neighboring-context forces and a locally unique Fisher flow. On a flat torus, algebraic elimination of the spatial connection gives a shape Dirichlet energy and a coupled parabolic and ordinary differential system. The finite and continuum theorems use explicit regular-family hypotheses. Neither is a refinement theorem, and neither changes the formation results of the preceding chapters.

The evolving laws remain general normalized probability measures within declared finite-dimensional product families. Gaussian laws appear only in explicit examples. The selected potential is the reciprocal objective of chapter 4; its full derivative includes every receiver, source, and channel role. The normalized product experiments below define the Fisher metrics and do not turn this objective into a global ELBO. The finite contextual vertices and the continuum coordinates are separate supplied descriptions of the base.

16.1 Finite contextual graphs

16.1.1 Statistical and contextual data

Let V be a finite set of contexts and E a finite set of unoriented contextual edges. Choose one orientation $e : u \to v$ for each edge, positive vertex masses m_v, and positive edge weights w_e. The graph, masses, weights, agent census, and comparisons below are fixed. They are supplied contextual data. A change in the agent census is outside the evolution defined here.

At each context there are n agents with normalized laws on standard Borel spaces. For the reciprocal potential studied here impose the product restriction

$$\zeta_{iv}(dk, dm) = q_{iv}(dk)s_{iv}(dm). \tag{16.1}$$

This is an admitted family, not an assertion that unrestricted joint Fisher dynamics preserves independence. Each factor belongs to a finite-dimensional regular statistical manifold. Assume differentiability in quadratic mean with centered square-integrable scores and a smooth positive-definite Fisher tensor. Redundant parameters have been removed. No Gaussian assumption or finite sample alphabet is imposed.

Fixed bimeasurable comparisons S_{iv} place the joint laws in compatible target spaces, preserving the factorization when it is used. Write $p_{iv} = (\mathsf{S}_{iv})_{\#}\zeta_{iv}$ and $p_v = (p_{iv})_{i=1}^{n} \in \mathcal{N}$. The manifold $\mathcal{N}$ is

the declared product family, equipped with the metric of the normalized product experiment

$$P_{p_v} = \bigotimes_{i=1}^{n} p_{iv}, \qquad \mathsf{G}_{p_v}(a,b) = \sum_{i=1}^{n} g^F_{p_{iv}}(a_i, b_i), \qquad g^F_p(a,b) = \mathbb{E}_p[\ell_a \ell_b]. \tag{16.2}$$

The equality follows because the product score is the sum of the individual centered scores, so its cross terms vanish. For (16.1), $g^F_{p_{iv}}$ is the transported sum of the belief and model Fisher metrics. The product experiment specifies the metric; it is not a proposed population generative model for which the reciprocal potential is an ELBO. Unit agent weights are chosen to match the local Fisher flow. Other positive constant weights define a different declared metric and therefore different mobilities.

Let a finite-dimensional Lie group G act smoothly on $\mathcal{N}$ by diagonal pushforward through bimeasurable sample maps. One element acts on both sectors, through their respective representations when those are linear. The admitted family is stable under the group elements used. The action is isometric: a fixed sample bijection sends a score ℓ_a to $\ell_a \circ g^{-1}$, leaving its squared integral unchanged. Define the fundamental tangent

$$J_p \xi = \left.\frac{d}{dt}\right|_{t=0} \exp(t\xi) \cdot p, \qquad \xi \in \mathfrak{g}. \tag{16.3}$$

The notation $g \cdot p$ means pushforward, not multiplication of density values. A nontrivial continuous action on a finite sample alphabet is not assumed.

An agent-coordinate change g_{iv} and a common target change h_v act as

$$\zeta'_{iv} = (g_{iv})_{\#}\zeta_{iv}, \qquad \mathsf{S}'_{iv} = h_v \mathsf{S}_{iv} g_{iv}^{-1}, \qquad p'_v = h_v \cdot p_v. \tag{16.4}$$

These changes are fixed in evolution time. Comparisons that depend on the evolving laws would require their derivatives in both the metric pullback and the energy variation; they are not used here.

16.1.2 The chosen context-local potential

At context v, fix normalized references r_{iv}, normalized evaluator kernels $G_{iv}(m, dk)$, and the normalized joint references

$$R_{iv}(dk, dm) = r_{iv}(dm) G_{iv}(m, dk). \tag{16.5}$$

Choose finite nonempty source sets, positive normalized prior rows π^q_{iv}, π^s_{iv}, and temperatures $T_q, T_s > 0$. Belief rows β_{iv} and model rows γ_{iv} lie in the interiors of their probability simplexes. For normalized, possibly law-dependent, channels at that same context set

$$\nu^q_{ijv} = q_{jv} \mathcal{K}^q_{ijv}[p_v], \qquad D^q_{ijv} = D_{\mathrm{KL}}(q_{iv} \| \nu^q_{ijv}), \tag{16.6}$$

$$\nu^s_{ijv} = s_{jv} \mathcal{K}^s_{ijv}[p_v], \qquad D^s_{ijv} = D_{\mathrm{KL}}(s_{iv} \| \nu^s_{ijv}). \tag{16.7}$$

The local reciprocal law potential is

$$\begin{aligned}\mathcal{E}_v = &\sum_i D_{\mathrm{KL}}(q_{iv} \otimes s_{iv} \| R_{iv}) \\ &+ \sum_i \left[\sum_j \beta_{ijv} D^q_{ijv} + T_q D_{\mathrm{KL}}(\beta_{iv} \| \pi^q_{iv}) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ijv} D^s_{ijv} + T_s D_{\mathrm{KL}}(\gamma_{iv} \| \pi^s_{iv}) \right].\end{aligned} \tag{16.8}$$

Assume every term is finite and $\mathcal{E}_v$ is C^3 on the chosen law and row chart. Differentiability in quadratic mean alone does not imply this KL regularity. It is a separate hypothesis about the family, supports, references, and channels. Evaluators, references, and channel rules are fixed; their declared dependence on the evolving laws is differentiated. Links between agents at one context remain independent directed communication data.

This objective is a direct potential on laws. A fixed normalized global generative model with coordinate-descent recognition, and an agent's own normalized model treating peers as external sources, are different choices. The derivative of (16.8) contains all receiver, source, and channel roles. On a differentiable dominated subfamily with justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b) da - \int (a/b) db, \tag{16.9}$$

$$d\nu^q_{ijv} = (dq_{jv})\mathcal{K}^q_{ijv} + q_{jv} D\mathcal{K}^q_{ijv}(dp_v). \tag{16.10}$$

The absent $\int da$ term is zero by normalization. Thus differentiating a source law must include the costs of receivers attending to it. The coordinate derivative of (16.8) is used when a density formula is unavailable.

Under (16.4), carry references, evaluators, and channels with the same sample maps. For a sector channel this means

$$\mathcal{K}'_{ijv}(g_{jv} y, A) = \mathcal{K}_{ijv}(y, g_{iv}^{-1} A). \tag{16.11}$$

Relative entropy is then invariant by change of variables. Any law-dependent constitutive rule is required to obey this relation. Passive covariance does not imply active symmetry of arbitrarily fixed references.

16.1.3 Finite transports and their stationarity equation

For $e : u \to v$, let $U_e \in G$ send target coordinates at u to those at v. The other orientation of this geometric edge uses U_e^{-1}. This convention places no inverse constraint on independently attending agent links. Set

$$c_e(p_u, p_v, U_e) = \frac{1}{2} d_{\mathcal{N}}(p_v, U_e \cdot p_u)^2, \qquad \mathcal{C}(p, U) = \sum_{e \in E} w_e c_e(p_u, p_v, U_e). \tag{16.12}$$

Here $d_{\mathcal{N}}$ is the intrinsic Riemannian distance of the admitted family with metric (16.2), not necessarily the distance in the unrestricted space of all laws. We work where the compared pairs have a unique minimizing geodesic and squared distance is smooth. The edge transports transform by the endpoint law

$$U'_e = h_v U_e h_u^{-1}. \tag{16.13}$$

Indeed, $U'_e \cdot p'_u = h_v \cdot (U_e \cdot p_u)$, so isometry proves invariance of every c_e. Reversing an edge leaves its cost unchanged because $d(p_v, U_e p_u) = d(p_u, U_e^{-1} p_v)$.

Proposition 16.1 (Auxiliary transport equation and endpoint forces). Write $x = p_v$, $y = p_u$, and $z = U_e \cdot y$. At a smooth stationary transport,

$$\mathsf{G}_z(\operatorname{Log}_z x, J_z \xi) = 0 \quad \text{for every } \xi \in \mathfrak{g}. \tag{16.14}$$

Holding U_e fixed, the two endpoint gradients are

$$\operatorname{grad}_x c_e = -\operatorname{Log}_x(U_e \cdot y), \qquad \operatorname{grad}_y c_e = -\operatorname{Log}_y(U_e^{-1} \cdot x). \tag{16.15}$$

Proof. Let $\eta : [0,1] \to \mathcal{N}$ be the minimizing geodesic from x to z, with constant speed. Its energy is $d(x,z)^2/2$. For an endpoint variation, differentiating its energy and integrating the covariant derivative by parts gives $\langle \dot\eta(1), \delta z\rangle - \langle \dot\eta(0), \delta x\rangle$; the interior term vanishes because $\nabla_t \dot\eta = 0$. Since $\dot\eta(0) = \operatorname{Log}_x z$ and $\dot\eta(1) = -\operatorname{Log}_z x$, this gives both distance gradients. For $U_e(t) = \exp(t\xi) U_e$, the target velocity is $J_z \xi$. Setting its derivative to zero proves (16.14). For the source endpoint, pull the target gradient back by the isometry U_e. Isometries carry minimizing geodesics and their initial velocities to those of the transformed endpoints, giving the second formula in (16.15). □

Equation (16.14) is the exact finite connection equation of this construction. It is generally nonlinear in U_e. It selects comparisons between neighboring collective laws. There is no transport kinetic energy, curvature penalty, or independent evolution equation for U_e. Cyclic products of the selected transports can have holonomy; no flatness constraint has been imposed. Finite group elements on edges are not connection one-forms. Discrete connections and their relation to continuous connection data have a separate geometric theory [97]; here the edge cost and its elimination specify the finite model directly.

16.1.4 Elimination on a regular minimizing branch

Hypothesis 16.2 (Regular branch). On an open law domain $\mathcal{O} \subset \mathcal{N}^V$, each edge admits a chosen stationary point U_e^* at a reference configuration. Squared distance is smooth on a neighborhood of that point and those endpoints. In group coordinates a_e near U_e^*, the Hessian $\partial^2_{a_e a_e} c_e$ is positive definite. The local branch is continued only while these conditions hold.

Proposition 16.3 (Local elimination and envelope). After shrinking $\mathcal{O}$, hypothesis 16.2 determines a unique smooth local minimizing transport branch $U_e^*(p_u, p_v)$ through the specified point. Define its reduced energy by

$$\mathcal{C}_*(p) = \mathcal{C}(p, U^*(p)). \tag{16.16}$$

Then

$$d\mathcal{C}_*[\delta p] = \sum_e w_e \left[-\mathsf{G}_{p_v}(\mathrm{Log}_{p_v}(U_e^* p_u), \delta p_v) - \mathsf{G}_{p_u}(\mathrm{Log}_{p_u}((U_e^*)^{-1} p_v), \delta p_u) \right]. \tag{16.17}$$

Proof. The stationarity map $\partial_{a_e} c_e$ has invertible derivative in a_e. The implicit-function theorem gives a smooth critical branch; continuity of its positive-definite Hessian makes it a strict local minimum. Differentiate the composite energy. Its transport derivative is zero on the branch, so the term containing dU_e^* vanishes. The two remaining derivatives are (16.15). The smooth implicit-function theorem applies because the metric and action are smooth; C^3 data would already suffice for a C^2 reduced energy and a C^1 vector field. □

A local minimum is not automatically the global infimum over G. One sufficient global version assumes compact G, continuous cost on all of G, and a unique global minimizer at the initial pair, with the stated smoothness and positive Hessian there. Choose a small neighborhood on which the Hessian remains positive and the critical point is unique. Compactness and uniqueness give a positive cost gap on its complement at the initial pair. Uniform continuity preserves a smaller gap for nearby endpoints. Every global minimizer then lies in that neighborhood and equals the implicit branch. Under these added hypotheses $\mathcal{C}_*$ is the actual minimized energy locally. For noncompact groups, attainment and exclusion of escaping minimizing sequences require separate hypotheses.

These hypotheses hold on a nonempty neighborhood of agreement when G is compact and its action at a common law point p_0 is free. At (p_0, p_0), only the identity gives zero distance. Its group Hessian is $\mathsf{G}_{p_0}(J_{p_0}\cdot, J_{p_0}\cdot)$, which is positive definite because freeness makes J_{p_0} injective. Squared distance is smooth near the diagonal. The local implicit branch and compactness argument therefore give a unique smooth global minimizer for endpoint pairs sufficiently near (p_0, p_0). This is a local sufficient condition, not a statement about distant endpoint pairs.

Positive-definite group Hessians exclude infinitesimal stabilizers of the endpoints. With isotropy, the cost is invariant under

$$U_e \longmapsto h_v U_e h_u^{-1}, \qquad h_v \in G_{p_v}, \quad h_u \in G_{p_u}, \tag{16.18}$$

so transport representatives can be nonunique. A quotient version requires a specified smooth law stratum and a smooth bundle of minimizing classes, with positive normal Hessian in a smooth transverse slice. It also requires a smooth local representative and constancy of the cost along each minimizing fiber. On that stratum the same chain-rule proof gives a representative-independent differential. The gradient uses the induced Fisher metric, equivalently the Fisher-orthogonal projection of (16.15) onto the stratum tangent. These are additional hypotheses, not consequences of constant orbit rank. The main theorem below uses hypothesis 16.2 on an open chart and does not assert unrestricted passage through singular or competing-minimum configurations.

Under a fixed gauge change, transport the branch neighborhood and the chosen initial minimizer by (16.13). Isometry identifies the two minimization problems. Local uniqueness then gives

$$U_e^{*\prime}(p_u', p_v') = h_v U_e^*(p_u, p_v) h_u^{-1}. \tag{16.19}$$

A numerical branch rule defined by a preferred coordinate angle must satisfy this condition; coordinate-dependent branch switching need not.

16.1.5 The coupled Fisher field and its local solution

Let b_v denote all rows at v. On their interior simplexes use

$$g^{\mathrm{row}}_{\beta}(a,b)=\sum_j \frac{a_j b_j}{\beta_j}, \qquad \sum_j a_j=\sum_j b_j=0, \tag{16.20}$$

and the analogous metric for γ. On law and row configurations define

$$\mathbb{G}=\sum_v m_v\left(\mathsf{G}_{p_v}+\sum_i g^{\mathrm{row}}_{\beta_{iv}}+\sum_i g^{\mathrm{row}}_{\gamma_{iv}}\right), \tag{16.21}$$

$$\mathcal{F}_*=\sum_v m_v\mathcal{E}_v(p_v,b_v)+\lambda\mathcal{C}_*(p), \qquad \lambda\geq 0. \tag{16.22}$$

This is a Fisher metric up to a fixed overall scale. To see this, choose a context label with probabilities m_v/M_0, $M_0=\sum_v m_v$, then conditionally draw the product experiment (16.2) and independent categorical labels with the rows at that context. Its normalized joint law has Fisher metric $\mathbb{G}/M_0$. The overall scale only fixes the evolution time convention. This auxiliary sampling description does not identify $\mathcal{F}_*$ with a free energy.

For either orientation of an incident contextual edge write $U^*_{v\leftarrow u}$ for the transport into v, using its inverse when necessary, and put

$$B_v(p)=\sum_{e=\{u,v\}} w_e\,\mathrm{Log}_{p_v}(U^*_{v\leftarrow u}\cdot p_u). \tag{16.23}$$

The natural-gradient equation [3] is

$$\boxed{\dot p_v=-\operatorname{grad}_{\mathsf{G}}\mathcal{E}_v+\frac{\lambda}{m_v}B_v(p).} \tag{16.24}$$

In a chart θ_v, if $c_v=\partial_{\theta_v}\mathcal{C}_*$ is a covector, this is

$$\dot\theta^a_v=-(\mathsf{G}_v^{-1})^{ab}\left(\partial_{\theta^b_v}\mathcal{E}_v+\frac{\lambda}{m_v}c_{v,b}\right). \tag{16.25}$$

All local derivatives include (16.10). Define the row costs

$$h^q_{ijv}=D^q_{ijv}+T_q\log(\beta_{ijv}/\pi^q_{ijv}), \qquad \bar h^q_{iv}=\sum_j \beta_{ijv}h^q_{ijv}, \tag{16.26}$$

$$h^s_{ijv}=D^s_{ijv}+T_s\log(\gamma_{ijv}/\pi^s_{ijv}), \qquad \bar h^s_{iv}=\sum_j \gamma_{ijv}h^s_{ijv}. \tag{16.27}$$

Their equations remain

$$\dot\beta_{ijv}=-\beta_{ijv}(h^q_{ijv}-\bar h^q_{iv}), \qquad \dot\gamma_{ijv}=-\gamma_{ijv}(h^s_{ijv}-\bar h^s_{iv}). \tag{16.28}$$

The constant $+T_a$ in an unconstrained derivative cancels under centering. To derive the formula, pair its right side with any mass-zero row tangent using the row metric; the result is the negative

directional derivative. The factor m_v cancels between the local energy and its metric. A singleton row has no tangent variables and remains one. The contextual cost has no direct row derivative.

Theorem 16.4 (Finite-context local well-posedness and dissipation). Assume the stated smooth positive Fisher metric, C^3 finite local potentials on an open chart with positive rows, and hypothesis 16.2 for every edge. Through each configuration in a sufficiently small such domain, (16.24)–(16.28) have a unique local solution, continuously differentiable in the initial configuration. It extends uniquely to a maximal solution in the regular domain. Along it,

$$\frac{d\mathcal{F}_*}{dt} = -\sum_v m_v \Bigg[\Big\| \operatorname{grad}_{\mathsf{G}} \mathcal{E}_v - \frac{\lambda}{m_v} B_v \Big\|_{\mathsf{G}}^2 + \sum_{ij} \beta_{ijv} (h^q_{ijv} - \bar{h}^q_{iv})^2 + \sum_{ij} \gamma_{ijv} (h^s_{ijv} - \bar{h}^s_{iv})^2 \Bigg]. \tag{16.29}$$

Fixed gauge re-expression with transported constitutive data sends this solution to the solution of the re-expressed system. At $\lambda = 0$ the equations recover exactly the chosen context-local reciprocal Fisher flows.

Proof. Use ordinary coordinates for the law families and the first $r - 1$ entries of each interior row simplex. The Fisher matrices and their inverses are smooth and positive definite. Implicit elimination makes the reduced energy at least C^2, hence its metric gradient is C^1 and locally Lipschitz. For completeness, choose a closed coordinate ball inside the regular domain, on which the vector field f is bounded by K and has Lipschitz constant L. On a sufficiently short interval, the map $x(\cdot) \mapsto x_0 + \int_0^{\cdot} f(x(s))ds$ maps the closed continuous-path ball into itself and contracts by a factor at most $Lt < 1$. Its fixed point is a unique differentiable solution. The integral inequality for two solutions and Gronwall's inequality give continuous dependence. Differentiating this integral equation gives the variational equation and the corresponding differentiable dependence. Compatible local solutions concatenate by uniqueness, giving the maximal solution. Interior normalization is built into the coordinates; the displayed row equations also have zero sum.

The envelope identity shows that the full vector field is $-\operatorname{grad}_{\mathbb{G}} \mathcal{F}_*$. Pairing it with $d\mathcal{F}_*$ gives minus its squared metric norm. Substitution of the law and row components gives (16.29). Metric isometry, invariance of the transformed local potential, and (16.19) imply covariance of the gradient. Uniqueness then gives covariance of trajectories. Setting $\lambda = 0$ deletes precisely B_v and leaves (16.28) unchanged. □

The law and row solution can be continued whenever it remains in a compact subset of the regular domain. A finite endpoint therefore requires leaving every such compact subset: possibilities include a statistical-family boundary, a row approaching zero, unbounded coordinates, loss of KL regularity, a geodesic cut locus, or loss of the selected minimizing branch. Energy decrease alone excludes none of these in general. At $\lambda = 0$ total-energy stationarity places no condition on U at all. The law and row ODE is still unique; a separately chosen diagnostic transport does not become dynamically determined. Time-dependent frame changes add their coordinate-velocity term and are not covered by fixed-gauge covariance.

A coordinate-chart boundary alone can be crossed in another regular chart. Row positivity has a sharper criterion: for two entries of the same row,

$$\frac{d}{dt}\log\frac{\beta_j}{\beta_k} = -(D_j - D_k) - T_q\left(\log\frac{\beta_j}{\beta_k} - \log\frac{\pi_j}{\pi_k}\right). \tag{16.30}$$

Variation of constants bounds these log ratios on every finite interval with bounded costs, giving a positive row floor there. Thus a row cannot reach zero in finite time while all its costs remain bounded. This does not prevent the other regular-domain exits.

If the rows are instead minimized instantaneously, finite positive costs give

$$\beta^*_{ijv} = \frac{\pi^q_{ijv}e^{-D^q_{ijv}/T_q}}{Z^q_{iv}}, \qquad Z^q_{iv} = \sum_j \pi^q_{ijv}e^{-D^q_{ijv}/T_q}, \tag{16.31}$$

and analogously for γ^*. Replacing their row energies by $-T_q\log Z^q_{iv}$ and $-T_s\log Z^s_{iv}$ gives a different, reduced law-only flow. Direct differentiation yields $-T_q d\log Z^q_{iv} = \sum_j \beta^*_{ijv}dD^q_{ijv}$; there is no extra row-response term. The theorem applies to that smooth reduced potential, with the row dissipation terms absent. The main result above retains the rows as evolving variables.

For own-block per-agent free energies $\mathcal{F}_{iv}$, define $A_{iv} = \operatorname{grad}_{iv}\mathcal{F}_{iv}$ with peers held fixed. Adding the contextual force gives a locally well-posed vector field under the same local regularity assumptions, but it need not be a gradient of $\sum_{iv}\mathcal{F}_{iv} + \lambda\mathcal{C}_*$. If R denotes the omitted source-role and channel derivatives in that sum, its gradient is $A + \lambda\operatorname{grad}\mathcal{C}_* + R$ (with the chosen metric weights). Along the proposed own-block field the derivative is

$$-\|A + \lambda\operatorname{grad}\mathcal{C}_*\|^2 - \langle R, A + \lambda\operatorname{grad}\mathcal{C}_*\rangle. \tag{16.32}$$

The second term has no fixed sign. Thus (16.29) belongs to the reciprocal gradient construction.

16.1.6 Relation to the infinitesimal connection equation

Take a smooth local section $p(c)$ in a common frame, a base tangent X, and a displacement ϵX. Suppose the near-identity minimizing transport branch satisfies $U^*(0) = e$ and is smooth. Put

$$U^*(\epsilon) = \exp[-\epsilon A(X) + O(\epsilon^2)], \qquad p(c+\epsilon X) = p(c) + \epsilon dp(X) + O(\epsilon^2) \tag{16.33}$$

in local charts. The initial geodesic residual, identified to $T_{p(c)}\mathcal{N}$ by any smooth local trivialization, is

$$\operatorname{Log}_{U^*(\epsilon)p(c)}p(c+\epsilon X) = \epsilon\big(dp(X) + J_pA(X)\big) + O(\epsilon^2). \tag{16.34}$$

Dividing (16.14) by ϵ and taking its first-order coefficient gives

$$\boxed{\mathfrak{k}_p(A(X),\xi) = -\mathsf{G}_p(dp(X), J_p\xi), \qquad \mathfrak{k}_p(\eta,\xi) = \mathsf{G}_p(J_p\eta, J_p\xi).} \tag{16.35}$$

If J_p is injective, $\mathfrak{k}_p$ is positive definite. At coincident endpoints the group Hessian of c_e is exactly $\mathfrak{k}_p$, so the near-identity branch and its smooth dependence follow from the same implicit-function argument. Differentiating stationarity gives (16.35), and its solution is linear in X. With a kernel,

solutions form an affine class modulo $\ker J_p$ under the corresponding constant-rank hypotheses. The residual is the Fisher-orthogonal projection of $dp(X)$ off the orbit directions. This is the mechanical-connection construction expressed with a Fisher metric; the separation of rotation and shape in mechanics provides its classical antecedent [101].

Expanding the finite edge cost gives

$$c_e = \frac{\epsilon^2}{2}\|dp(X) + J_p A(X)\|_{\mathsf{G}}^2 + O(\epsilon^3). \tag{16.36}$$

Under a smooth base-dependent frame change, the endpoint law expands as

$$U'(c + \epsilon X, c) = h(c + \epsilon X)U(c + \epsilon X, c)h(c)^{-1}, \tag{16.37}$$

$$A'(X) = \mathrm{Ad}_h\, A(X) - dh(X)h^{-1}. \tag{16.38}$$

These are local consistency statements, not convergence of a refining graph or existence for a continuum equation. The familiar candidate continuum energy

$$\frac{1}{2}\int \gamma^{\mu\nu}\mathsf{G}(D_\mu^A p, D_\nu^A p)\mathrm{dvol}_\gamma \tag{16.39}$$

requires a supplied base metric, measure, functional domain, and boundary conditions. Finite graph weights do not supply those choices or their analytic estimates.

16.1.7 An explicit two-context witness

Use two contexts of unit mass, one edge of unit weight, and two agents. Both belief and model sample spaces are $\mathbb{R}^2$. At each context let

$$q_i = \mathcal{N}(\mu_i, I_2), \quad s_i = \mathcal{N}(\nu_i, I_2), \quad r_i = \mathcal{N}(0, I_2), \quad G_i(m, dk) = \mathcal{N}(m, I_2)(dk). \tag{16.40}$$

The evaluator depends on the model presentation, so both sectors contribute to the local dynamics. Direct integration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| r_i G_i) = \frac{1}{2}\|\mu_i - \nu_i\|^2 + \frac{1}{2}\|\nu_i\|^2 + 1. \tag{16.41}$$

Give each agent the other agent as its singleton source in each sector, with an identity channel. Rows and their priors equal one. The two directed belief divergences sum to $\|\mu_1 - \mu_2\|^2$, and likewise for models. Thus

$$\mathcal{E}(X) = \sum_{i=1}^{2}\left[\frac{1}{2}\|\mu_i - \nu_i\|^2 + \frac{1}{2}\|\nu_i\|^2 + 1\right] + \|\mu_1 - \mu_2\|^2 + \|\nu_1 - \nu_2\|^2, \tag{16.42}$$

where $X = (\mu_1, \nu_1, \mu_2, \nu_2) \in \mathbb{R}^{2\times 4}$. The Fisher metric is the Frobenius inner product on these means. It is the intrinsic metric of the fixed-covariance location family, whose full-law Fisher invariance is not asserted. A single $SO(2)$ rotation acts on every column, in both sectors. References, evaluator, and channels obey this symmetry.

For a source matrix X and a target matrix Y, write

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\theta = \cos\theta I + \sin\theta J, \quad C = \langle Y, X\rangle_F, \quad S = \langle Y, JX\rangle_F, \quad \rho = \sqrt{C^2 + S^2}. \tag{16.43}$$

Then

$$c(X, Y, R_\theta) = \tfrac{1}{2}(\|X\|_F^2 + \|Y\|_F^2) - C\cos\theta - S\sin\theta. \tag{16.44}$$

For $\rho > 0$ the unique minimizing group element and reduced energy are

$$R_* = (CI + SJ)/\rho, \qquad c_* = \tfrac{1}{2}(\|X\|_F^2 + \|Y\|_F^2) - \rho. \tag{16.45}$$

Stationarity is $C\sin\theta - S\cos\theta = 0$; the Hessian at this minimum is $\rho > 0$. The two envelope gradients are

$$\nabla_X c_* = X - R_*^T Y, \qquad \nabla_Y c_* = Y - R_* X. \tag{16.46}$$

The formula for R_* is smooth on $\rho > 0$, regardless of any angle-coordinate branch cut.

Choose the ordered columns

$$X = (e_1, e_1, e_2, e_2), \qquad Y = (2e_1, 3e_1, e_2, 2e_2). \tag{16.47}$$

Here $(C, S, \rho) = (8, 0, 8)$, $R_* = I$, and $c_* = 3$. From (16.42),

$$\nabla\mathcal{E}(X) = \begin{pmatrix} 2 & 3 & -2 & -2 \\ -2 & -2 & 2 & 3 \end{pmatrix}. \tag{16.48}$$

Consequently the source-context velocity is

$$\dot{X} = \begin{pmatrix} -2+\lambda & -3+2\lambda & 2 & 2 \\ 2 & 2 & -2 & -3+\lambda \end{pmatrix}. \tag{16.49}$$

Changing only the neighbor's first column from $2e_1$ to $3e_1$ keeps $R_* = I$ and adds exactly λe_1 to the local $\dot{\mu}_1$. This is a genuine contextual force on top of the reciprocal local flow. All laws, reference joints, and channels in the example are normalized. Gaussianity supplies the explicit formula; theorem 16.4 is a general regular-family result.

Free action does not prevent a branch singularity

Consider two Gaussian location factors with means $x_\epsilon = (e_1, (1+\epsilon)e_2)$ and $y = (e_1, -e_2)$ and the same diagonal rotation group. Both endpoints lie in the free-action region near $\epsilon = 0$. Nevertheless $C = -\epsilon$, $S = 0$, and

$$\min_R \tfrac{1}{2}\|x_\epsilon - Ry\|_F^2 = 2 + \epsilon + \tfrac{1}{2}\epsilon^2 - |\epsilon|. \tag{16.50}$$

For $\epsilon < 0$ the minimizer is I, and for $\epsilon > 0$ it is $-I$; at zero every rotation minimizes. The minimized energy is not differentiable at zero. The statistical metric is nonsingular and there is no geodesic cut locus. Thus constant orbit rank, even freeness, cannot replace the transport Hessian and branch hypotheses.

16.1.8 What the contextual energy controls

Auxiliary elimination penalizes variation of collective law shape after group alignment. Pure common frame variation can have zero cost. A single family on which G acts transitively has zero minimized contextual cost wherever alignment is possible; collective laws are needed if the common action otherwise removes every difference. The example retains shape mismatch despite optimizing the common rotation.

With the local potential absent, the contextual flow decreases $\mathcal{C}_*$ by its own negative squared gradient. With the local potential present, only the total dissipation (16.29) is asserted: local forces can increase contextual cost, and individual edge costs need not decrease even under the contextual flow. In a finite graph, smoothing means this aligned law-variation penalty and its gradient, not a continuum regularity theorem.

Independent rows have no contextual penalty in this model. Their local equations can react to changing laws, but this supplies no separate spatial regularity estimate for rows. Independent communication links and frames have no new evolution equation here. Law-dependent channels inherit whatever regularity follows from their declared constitutive maps, without becoming independent smoothed fields. The contextual graph, comparisons, reference data, and transport branch selection remain supplied; the laws and rows evolve, and the auxiliary transports follow their algebraic minimizers.

The continuum construction in section 16.2 supplies these data and a local theorem on a flat torus. It does not follow by taking a graph-refinement limit of this finite model. Chapter 17 separately constructs contextual displacement transport from regular law, interface, and inter-agent link data. Its comparison metric is a constitutive choice and need not be the law-Fisher metric minimized here. No formation, recursive inheritance, global convergence, or graph-refinement result follows from the local finite-system theorem.

16.2 A periodic continuum base

We now use x for the contextual coordinate and V for the context-local reciprocal potential. These symbols replace the vertex labels and $\mathcal{E}_v$ of the finite construction. The statistical target is again denoted $\mathcal{N}$ and carries the same unit-weight collective Fisher convention. The following hypotheses and proofs specify the continuum problem directly.

16.2.1 Declared geometry and normalized statistical data

Fix an integer $d \geq 1$ and the contextual base

$$C = \mathbb{T}^d = \mathbb{R}^d/(2\pi\mathbb{Z})^d, \qquad \gamma_{\mu\nu} = \delta_{\mu\nu}, \qquad \mathrm{d}\mu_C = (2\pi)^{-d}\mathrm{d}x^1 \cdots \mathrm{d}x^d. \tag{16.51}$$

Every field and constitutive datum is periodic. There is no boundary contribution. The base metric, measure, agent census, and contextual dimension are supplied and independent of evolution time. Work in the trivial principal bundle $C \times G$. This choice excludes the additional patching problem for nontrivial base bundles and avoids implicit boundary conditions.

At each x, agent i has normalized belief and model laws q_i and s_i on standard Borel sample spaces. Impose the admitted product restriction

$$\zeta_i(\mathrm{d}k, \mathrm{d}m) = q_i(\mathrm{d}k)s_i(\mathrm{d}m). \tag{16.52}$$

Each factor belongs to a finite-dimensional regular family: it is differentiable in quadratic mean, its scores are centered and square integrable, and its Fisher tensor is smooth and positive definite after redundant parameters are removed. This is a restriction of the evolving statistical model. Unrestricted joint Fisher dynamics need not preserve the restriction. The sample spaces need not be finite, and the general construction makes no Gaussian assumption.

Fixed bimeasurable comparisons identify the agents' product laws with compatible target spaces. These comparisons are independent of x in the chosen trivialization and of the evolving state. Write $p = (p_i)_i \in \mathcal{N}$ for the collective target configuration. Its metric is the Fisher metric of the normalized product experiment

$$P_p = \bigotimes_{i=1}^{n} p_i, \qquad \mathsf{G}_p(U, V) = \sum_i g^F_{p_i}(U_i, V_i), \qquad g^F_p(U, V) = \mathbb{E}_p[\ell_U \ell_V]. \tag{16.53}$$

The identity follows by adding product scores: centered independent scores have zero cross expectation. For (16.52), each summand is the transported sum of belief and model Fisher tensors. Unit agent weights are part of the declared metric. Other weights give other mobilities. This product experiment specifies a normalized statistical metric; it does not make the reciprocal energy below an ELBO of a fixed population generative model. Fisher gradient descent follows the metric principle of natural gradients [3].

Let a finite-dimensional Lie group G act on $\mathcal{N}$ by smooth diagonal pushforward. One group element acts on both sectors, through their respective sample actions. Fixed sample bijections preserve score integrals, so assume the admitted family is stable and the induced action is isometric. Define

$$J_p\xi = \frac{\mathrm{d}}{\mathrm{d}a}\bigg|_{a=0} \exp(a\xi) \cdot p, \qquad \xi \in \mathfrak{g}. \tag{16.54}$$

No nontrivial continuous action on a finite alphabet is presumed.

Hypothesis 16.5 (Regular quotient and one adapted chart). There is a G-invariant open region $\mathcal{N}_0 \subset \mathcal{N}$ on which the action is free and proper. Its smooth quotient $Q = \mathcal{N}_0/G$ has the submersion metric induced by G. The initial field lies compactly inside one adapted product coordinate chart

$$p = \Psi(u, r), \qquad u \in \mathbb{R}^{n_u}, \quad r \in \mathbb{R}^{n_r}, \quad n_r = \dim G, \tag{16.55}$$

where u coordinates the quotient and r the group fiber. All metric coefficients are smooth on a neighborhood of this chart.

Freeness makes the infinitesimal action injective. Properness and the isometric action supply a smooth local quotient and a positive quotient metric. These are substantive hypotheses. An action with stabilizers, a rank-changing orbit, or an initial image requiring several charts is outside the theorem as stated. The theorem is local in this regular region; it does not infer orbit regularity

from normalization alone. Riemannian submersion geometry provides the horizontal/vertical decomposition [113]; rotational shape geometry gives a classical mechanical-connection example [101].

16.2.2 The reciprocal local potential and the field metric

At each x, fix normalized references $r_i(x, \mathrm{d}m)$ and evaluator kernels $G_i(x, m, \mathrm{d}k)$, with normalized joint reference

$$R_i(x, \mathrm{d}k, \mathrm{d}m) = r_i(x, \mathrm{d}m) G_i(x, m, \mathrm{d}k). \tag{16.56}$$

The reference symbol r_i is distinct from the fiber coordinate r in (16.55). Fix finite nonempty source sets, positive normalized prior rows $\pi_i^q(x), \pi_i^s(x)$, and temperatures $T_q, T_s > 0$. The evolving rows β_i, γ_i are in their simplex interiors. Normalized channels may depend on p through fixed smooth constitutive rules:

$$\nu_{ij}^q = q_j \mathcal{K}_{ij}^q[x, p], \qquad D_{ij}^q = D_{\mathrm{KL}}(q_i \| \nu_{ij}^q), \tag{16.57}$$

$$\nu_{ij}^s = s_j \mathcal{K}_{ij}^s[x, p], \qquad D_{ij}^s = D_{\mathrm{KL}}(s_i \| \nu_{ij}^s). \tag{16.58}$$

There is no row dependence in these channel rules. Define

$$\begin{aligned} V(x, p, \beta, \gamma) = & \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i(x)) \\ & + \sum_i \left[\sum_j \beta_{ij} D_{ij}^q + T_q D_{\mathrm{KL}}(\beta_i \| \pi_i^q(x)) \right] \\ & + \sum_i \left[\sum_j \gamma_{ij} D_{ij}^s + T_s D_{\mathrm{KL}}(\gamma_i \| \pi_i^s(x)) \right]. \end{aligned} \tag{16.59}$$

Assume every term is finite and V is smooth jointly in x and the admitted law and row coordinates on a neighborhood of the initial image. Smooth Fisher geometry alone does not imply smooth relative entropy; support, integrability, and channel regularity are additional hypotheses here. All data are fixed in evolution time. Derivatives of V include every receiver role, source role, and prescribed law dependence of a channel. For example, when a dominated density calculation is justified,

$$\mathrm{d}D_{\mathrm{KL}}(a\|b) = \int \log(a/b) \mathrm{d}a - \int (a/b) \mathrm{d}b, \qquad \mathrm{d}(q_j \mathcal{K}[p]) = (\mathrm{d}q_j)\mathcal{K}[p] + q_j D\mathcal{K}[p](\mathrm{d}p). \tag{16.60}$$

Normalization removes the extra integral of $\mathrm{d}a$. Coordinate differentiation defines the derivative when this density formula is unavailable.

For a row with l entries, use log odds $b_j = \log(\beta_j/\beta_l)$, $1 \le j < l$. Collect all belief and model log odds in b. A singleton row contributes no coordinates. The categorical Fisher matrix in these coordinates is

$$B_{jk}(b) = \beta_j \delta_{jk} - \beta_j \beta_k, \qquad 1 \le j, k < l, \tag{16.61}$$

with one block per row. It is positive definite at every interior row. Thus the field metric is

$$\mathbb{G}_{p,b}((U,a),(W,c)) = \int_C [\mathsf{G}_p(U,W) + a^T B(b) c]\mathrm{d}\mu_C. \tag{16.62}$$

It is the Fisher metric of the normalized experiment that first samples $x \sim \mu_C$, then independently samples $P_{p(x)}$ and each categorical source row conditional on x. Differentiation under this fixed experiment gives the integral in (16.62). It is not a formal product of uncountably many probability laws.

Arbitrary fixed references need not be actively G-invariant. In adapted coordinates, $V(x,u,r,b)$ may depend on r. A passive re-expression transforms the reference laws, evaluators, and channels together with the laws. That covariance is distinct from a symmetry obtained while leaving the references fixed.

16.2.3 Auxiliary spatial connection and its elimination

For a Lie-algebra-valued periodic one-form $A = A_\mu \mathrm{d}x^\mu$, set

$$v_\mu = D^A_\mu p = \partial_\mu p + J_p A_\mu, \qquad \mathcal{F}[p,b,A] = \int_C \left[V(x,p,b) + \frac{\lambda}{2} \sum_\mu \mathsf{G}_p(v_\mu, v_\mu) \right] \mathrm{d}\mu_C, \tag{16.63}$$

where $\lambda > 0$ is fixed. The connection has no kinetic term, curvature energy, or independent evolution equation. Define the positive locked-inertia form

$$\mathfrak{k}_p(\xi,\eta) = \mathsf{G}_p(J_p\xi, J_p\eta). \tag{16.64}$$

At fixed p the integrand is a strictly convex quadratic in each A_μ. Its unique minimizer satisfies

$$\boxed{\mathfrak{k}_p(A^*_\mu, \xi) = -\mathsf{G}_p(\partial_\mu p, J_p\xi) \quad \text{for all } \xi.} \tag{16.65}$$

Let the mechanical connection $\theta_p : T_p\mathcal{N}_0 \to \mathfrak{g}$ be defined by

$$\mathfrak{k}_p(\theta_p U, \xi) = \mathsf{G}_p(U, J_p\xi), \qquad P_H U = U - J_p\theta_p U. \tag{16.66}$$

Then

$$A^* = -p^*\theta, \qquad D^{A^*}_\mu p = P_H \partial_\mu p. \tag{16.67}$$

The notation $p^*\theta$ denotes pullback along the field $p : C \to \mathcal{N}_0$. Smooth positive inertia makes this elimination smooth on the regular domain, unlike a possibly nonunique finite-distance minimizer over group elements.

In the adapted chart the metric has the form

$$\mathsf{G} = h_{ab}(u)\mathrm{d}u^a\mathrm{d}u^b + k_{\alpha\beta}(u,r)(\mathrm{d}r^\alpha + C^\alpha_a(u,r)\mathrm{d}u^a)(\mathrm{d}r^\beta + C^\beta_b(u,r)\mathrm{d}u^b). \tag{16.68}$$

Here h and k are positive definite. Orthogonal complement gives horizontal coordinate vectors $H_a = \partial_{u^a} - C^\alpha_a \partial_{r^\alpha}$. The quotient metric h is independent of r. The coefficient C represents this horizontal distribution in a fiber coordinate chart; it is not itself the Lie-algebra-valued field A^*.

Proposition 16.6 (Reduced energy). Under hypothesis 16.5, auxiliary minimization gives

$$\boxed{\mathcal{F}_*[u,r,b] = \int_C \left[V(x,u,r,b) + \frac{\lambda}{2} h_{ab}(u)\partial_\mu u^a \partial_\mu u^b \right] \mathrm{d}\mu_C.} \tag{16.69}$$

Repeated base and coordinate indices are summed. Its contextual term is the Dirichlet energy of the shape map $u : C \to Q$.

Proof. Equation (16.65) makes every residual orthogonal to the group orbit. The horizontal component of $\partial_\mu p$ is $H_a \partial_\mu u^a$. Pairing with (16.68) gives $\mathsf{G}(P_H \partial_\mu p, P_H \partial_\mu p) = h_{ab}\partial_\mu u^a \partial_\mu u^b$. The quadratic reaches its unique minimum there. Substitution proves the formula. □

Pure common frame variation has zero contextual cost. If the group is transitive on the admitted family, there are no shape variables and the reduced contextual term vanishes. The theorem below includes that degenerate case as a local ODE, but actual contextual smoothing requires retained shape directions.

16.2.4 Derivation of the coupled Fisher flow

Let Γ^a_{bc} be the Levi-Civita Christoffel symbols of h, and define the quotient tension

$$\tau_h(u)^a = \Delta u^a + \Gamma^a_{bc}(u)\partial_\mu u^b \partial_\mu u^c. \tag{16.70}$$

For a periodic variation δu, integration by parts in the Dirichlet term gives $-\int h_{ab}\tau_h^b \delta u^a \mathrm{d}\mu_C$. Thus

$$\frac{\delta \mathcal{F}_*}{\delta u} = V_u - \lambda h \tau_h, \qquad \frac{\delta \mathcal{F}_*}{\delta r} = V_r, \qquad \frac{\delta \mathcal{F}_*}{\delta b} = V_b. \tag{16.71}$$

The kinetic norm induced by (16.62) is

$$\|\partial_t(u,r,b)\|^2_{\mathbb{G}} = \int_C \left[u_t^T h u_t + (r_t + Cu_t)^T k (r_t + Cu_t) + b_t^T B b_t \right] \mathrm{d}\mu_C. \tag{16.72}$$

Pairing the negative gradient condition with independent coordinate variations yields

$$hu_t + C^T k(r_t + Cu_t) = -V_u + \lambda h \tau_h, \tag{16.73}$$

$$k(r_t + Cu_t) = -V_r, \qquad Bb_t = -V_b. \tag{16.74}$$

Solving this block system gives the coupled equations

$$\boxed{u_t = \lambda \Delta u + \lambda \Gamma(u)(\partial_\mu u, \partial_\mu u) - h^{-1}(V_u - C^T V_r),} \tag{16.75}$$

$$\boxed{r_t = -C(u,r)u_t - k(u,r)^{-1}V_r, \qquad b_t = -B(b)^{-1}V_b.} \tag{16.76}$$

The $C^T V_r$ term is required when fixed references give a nonbasic potential. Setting it to zero without $V_r = 0$ would change the Fisher gradient.

For the potential (16.59), the row equation has the explicit form

$$H_{ij}^q = D_{ij}^q + T_q \log(\beta_{ij}/\pi_{ij}^q), \qquad \overline{H}_i^q = \sum_j \beta_{ij} H_{ij}^q, \tag{16.77}$$

$$\partial_t \beta_{ij} = -\beta_{ij}(H_{ij}^q - \overline{H}_i^q), \tag{16.78}$$

and analogously for γ. Equivalently, for each nonreference entry of a row,

$$\partial_t b_j = -(D_j - D_l) - T\left(b_j - \log\frac{\pi_j}{\pi_l}\right). \tag{16.79}$$

The context penalty has no direct row derivative. Positive normalized rows are reconstructed by softmax from b. Bounded costs on a finite time interval bound all log odds by variation of constants in (16.79), hence give a positive row floor. There is no row Laplacian or separate spatial regularization estimate.

Intrinsic variation and gauge covariance

At fixed A, set $X_\mu(p) = J_p A_\mu(x)$. The derivative of v_μ in a law variation U is

$$\nabla_U v_\mu = \nabla_\mu U + \nabla_U X_\mu. \tag{16.80}$$

Because the group action is isometric, each X_μ is Killing. Its covariant derivative is skew adjoint. Periodic integration by parts therefore gives the negative gradient of the contextual term as

$$\tau_A(p) = \sum_\mu \left[\nabla_\mu v_\mu + \nabla_{v_\mu} X_\mu\right]. \tag{16.81}$$

The second summand accounts for the state dependence of the fundamental vector field. The envelope derivative from varying $A^*(p, \mathrm{d}p)$ vanishes by (16.65). Since the reduced contextual energy depends only on u, its metric gradient is horizontal. Comparing its first variation with (16.71) gives

$$\tau_{A^*}(p) = H_a \tau_h(u)^a, \qquad p_t = -\operatorname{grad}_{\mathsf{G}} V + \lambda \tau_{A^*}(p). \tag{16.82}$$

The full flow need not be horizontal, since $\operatorname{grad} V$ may have a vertical component.

For a smooth frame re-expression $g : C \to G$ fixed in evolution time,

$$p' = g \cdot p, \qquad A' = \mathrm{Ad}_g\, A - (\mathrm{d}g)g^{-1}, \qquad D^{A'} p' = g_* D^A p. \tag{16.83}$$

Push references and evaluators forward as well, and transform each sector channel by $\mathcal{K}'_{ij}(g_j y, S) = \mathcal{K}_{ij}(y, g_i^{-1} S)$ in its corresponding agent coordinates. Relative entropy and the Fisher norm are unchanged. The auxiliary minimizer transforms by (16.83), by uniqueness, and the gradient equations are covariant. This statement applies wherever both field descriptions admit the regular charts in the theorem. A time-dependent $g(t, x)$ adds $J_{p'}((\partial_t g)g^{-1})$ to the velocity. An associated temporal connection or modified evolution would be required; it is not part of the present system.

The principal second-order operator in law coordinates is a horizontal projection of the Laplacian. It is degenerate in frame directions. Harmonic-map heat flow supplies the shape operator [52], but a

uniformly parabolic theorem for all law coordinates cannot be invoked. The next proof retains the frame equation coupled to u_t.

A check with a fixed reference that breaks active symmetry

For a unit-covariance Gaussian location law with nonzero mean $p \in \mathbb{R}^2$, let $SO(2)$ rotate the mean. In a local chart choose

$$p = uR_{r+\eta u}e_1, \qquad u > 0, \qquad \mathsf{G} = \mathrm{d}u^2 + u^2(\mathrm{d}r + \eta \mathrm{d}u)^2, \tag{16.84}$$

where η is a fixed real constant. A fixed reference with mean ae_1, $a \neq 0$, gives the local KL term

$$V = \tfrac{1}{2}(u^2 + a^2 - 2au\cos\varphi), \qquad \varphi = r + \eta u. \tag{16.85}$$

Here $V_r = au\sin\varphi$ and $V_u = u - a\cos\varphi + a\eta u\sin\varphi$. With $C = \eta$ and $k = u^2$, the coupled equations are

$$u_t = \lambda\Delta u - u + a\cos\varphi, \qquad r_t = -\eta u_t - \frac{a}{u}\sin\varphi. \tag{16.86}$$

At zero coupling they reconstruct the Cartesian Fisher equation $p_t = -(p - ae_1)$. Omitting $C^T V_r$ would incorrectly change the radial velocity by $-a\eta u\sin\varphi$. This local term illustrates why passive coordinate covariance cannot replace the missing active symmetry assumption.

16.2.5 Local well-posedness in an explicit Sobolev class

Choose an integer $s > d/2 + 2$ and put $m = s - 1 > d/2 + 1$. All Sobolev spaces below are periodic, vector valued, and use the normalized measure (16.51). For $z = (u, r, b)$ write $CH^j = C([0,T]; H^j(C))$, and likewise for time-integrated spaces.

Theorem 16.7 (Continuum local existence, uniqueness, and continuation). Assume (16.51)–(16.59) and hypothesis 16.5, with smooth coefficients and finite smooth V on an open chart. Fix $\lambda > 0$. Suppose

$$u_0 \in H^s(C), \qquad r_0, b_0 \in H^m(C), \tag{16.87}$$

and the pointwise image of z_0 is contained in a compact subset of that regular interior chart. Then there is $T > 0$ and a unique solution of (16.75)–(16.76), remaining in the chart, with

$$u \in CH^s \cap L^2(0,T; H^{s+1}) \cap H^1(0,T; H^m), \tag{16.88}$$

$$r, b \in CH^m \cap H^1(0,T; H^m). \tag{16.89}$$

In fact $b \in C^1([0,T]; H^m)$. The solution depends locally Lipschitz-continuously on the initial data, in these norms, on a common sufficiently short interval near each admissible datum. It extends uniquely while

$$\|u(t)\|_{H^s} + \|r(t)\|_{H^m} + \|b(t)\|_{H^m} \tag{16.90}$$

stays bounded and its image remains in a fixed compact subset of the regular coordinate domain.

The asymmetric regularity is intentional. Only the shape coordinate is assigned the extra spatial derivative. Smooth initial data satisfy these assumptions; for $d = 1$, one admissible choice is $s = 3$,

$m = 2$. The law map itself will generally have only H^m regularity. The theorem does not claim instantaneous smoothing of the frame coordinate or rows.

Lemma 16.8 (Heat estimate on the declared base). For $0 < T \le 1$, $v_0 \in H^s$, and $f \in L^2(0,T;H^{s-1})$, the equation

$$v_t - \lambda \Delta v = f, \qquad v(0) = v_0, \tag{16.91}$$

has a unique solution with

$$\|v\|_{CH^s} + \|v\|_{L^2H^{s+1}} + \|v_t\|_{L^2H^{s-1}} \le K_\lambda \left(\|v_0\|_{H^s} + \|f\|_{L^2H^{s-1}} \right). \tag{16.92}$$

The constant is independent of $T \le 1$. The same estimate for a difference with zero initial value omits the initial term. Also,

$$\|(e^{\lambda t \Delta} - I)v_0\|_{H^{s-1}} \le \sqrt{\lambda t}\|v_0\|_{H^s}. \tag{16.93}$$

Proof. For Fourier modes $k \in \mathbb{Z}^d$, solve $\partial_t \widehat{v}_k + \lambda |k|^2 \widehat{v}_k = \widehat{f}_k$. For nonzero modes, multiplication by $(1+|k|^2)^s \overline{\widehat{v}_k}$, summation, and Young's inequality give

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|P_{\neq 0} v\|_{H^s}^2 + \frac{\lambda}{2}\sum_{k \neq 0} |k|^2 (1+|k|^2)^s |\widehat{v}_k|^2 \le \frac{c}{\lambda}\|f\|_{H^{s-1}}^2. \tag{16.94}$$

Indeed, $(1+|k|^2)/|k|^2 \le 2$ for nonzero integer modes, which is the required weight comparison. The dissipative term controls the nonzero-mode H^{s+1} norm. The zero mode satisfies $\widehat{v}_0(t) = \widehat{v}_0(0) + \int_0^t \widehat{f}_0(a)\mathrm{d}a$, so Cauchy–Schwarz controls its remaining norms for $T \le 1$. Finally $v_t = \lambda \Delta v + f$ gives the derivative estimate. Fourier truncations converge by this same bound. Approximation by smooth data gives continuity into H^s, including at zero. The homogeneous estimate proves uniqueness. For (16.93), use $(1-e^{-a})^2 \le a$ for $a \ge 0$ on each Fourier coefficient and sum. □

Proof of theorem 16.7. Write the system as

$$u_t - \lambda \Delta u = F(z), \qquad F(z) = \lambda \Gamma(u)(\partial_\mu u, \partial_\mu u) - h^{-1}(V_u - C^T V_r), \tag{16.95}$$

$$r_t = -C(z)u_t + Q(z), \qquad Q(z) = -k^{-1}V_r, \tag{16.96}$$

$$b_t = L(z), \qquad L(z) = -B^{-1}V_b. \tag{16.97}$$

The maps include the fixed smooth x dependence of V and of the prior rows. Since $m > d/2$, $H^m(C)$ is a Banach algebra. For clarity, its product bound follows from Fourier convolution, the weight inequality

$$\langle k \rangle^m \le c_m(\langle l \rangle^m + \langle k-l \rangle^m), \qquad \|\widehat{a}\|_{\ell^1} \le \left(\sum_k \langle k \rangle^{-2m} \right)^{1/2} \|a\|_{H^m}, \tag{16.98}$$

where the displayed sum is finite. Young's convolution inequality gives

$$\|ab\|_{H^m} \le c_m \|a\|_{H^m}\|b\|_{H^m}. \tag{16.99}$$

The associated smooth-composition estimate on bounded H^m sets with compact image follows by the chain rule, Leibniz's formula, Sobolev interpolation for its derivative products, and the identity

$$N(z) - N(\widetilde{z}) = \int_0^1 DN(\widetilde{z} + a(z - \widetilde{z}))(z - \widetilde{z})\mathrm{d}a. \tag{16.100}$$

One can first extend the smooth coefficients to a compact neighborhood of the initial image; their derivatives are bounded there. The small pointwise neighborhoods used below ensure that the needed segments remain in that extension. These estimates apply to smooth periodic x dependence as well.

On a bounded admissible set in $H^s \times H^m \times H^m$, there are constants M_N, L_N for $N = F, C, Q, L$ such that

$$\|N(z)\|_{H^m} \le M_N, \tag{16.101}$$

$$\|N(z) - N(\widetilde{z})\|_{H^m} \le L_N\big(\|u - \widetilde{u}\|_{H^s} + \|r - \widetilde{r}\|_{H^m} + \|b - \widetilde{b}\|_{H^m}\big). \tag{16.102}$$

For F, the gradient product is harmless in this estimate because $\partial_\mu u \in H^m$. No derivative of r or b occurs in F.

Use the complete path space

$$X_T = CH^s \times CH^m \times CH^m \tag{16.103}$$

with fixed initial values. On a closed subset with bounded norms and small CH^m displacement from z_0, define a map $z \mapsto z^+ = (u^+, r^+, b^+)$ by solving

$$u_t^+ - \lambda \Delta u^+ = F(z), \qquad u^+(0) = u_0, \tag{16.104}$$

and then setting

$$r^+(t) = r_0 + \int_0^t [-C(z)u_t^+ + Q(z)]\mathrm{d}a, \tag{16.105}$$

$$b^+(t) = b_0 + \int_0^t L(z)\mathrm{d}a. \tag{16.106}$$

The new heat solution's derivative is used. Thus an input path needs no time derivative. Choose the norm bounds larger than the initial norms. Lemma 16.8 and contraction of the homogeneous heat semigroup on H^s give

$$\|u^+\|_{CH^s} \le \|u_0\|_{H^s} + K_\lambda \sqrt{T} M_F, \tag{16.107}$$

$$\|u_t^+\|_{L^2 H^m} \le K_\lambda(\|u_0\|_{H^s} + \sqrt{T} M_F) =: A_T. \tag{16.108}$$

The displacement estimates are

$$\|u^+ - u_0\|_{CH^m} \le \sqrt{\lambda T}\|u_0\|_{H^s} + T M_F, \tag{16.109}$$

$$\|r^+ - r_0\|_{CH^m} \le c_m \sqrt{T} M_C A_T + T M_Q, \tag{16.110}$$

$$\|b^+ - b_0\|_{CH^m} \le T M_L. \tag{16.111}$$

The first follows from (16.93) and the Duhamel formula using heat contraction on H^m. Sobolev embedding $H^m \hookrightarrow C^0$ makes a sufficiently small displacement preserve the chart and its positive metric bounds. All right sides tend to zero with T. Hence the map preserves the chosen closed set for sufficiently small T.

For two input paths let $D = \|z - \widetilde{z}\|_{X_T}$. The zero-initial heat estimate gives

$$\|u^+ - \widetilde{u}^+\|_{CH^s} + \|u_t^+ - \widetilde{u}_t^+\|_{L^2H^m} \le K_\lambda L_F \sqrt{T} D. \tag{16.112}$$

Split the fiber integrand difference into $C(z)(u_t^+ - \widetilde{u}_t^+) + (C(z) - C(\widetilde{z}))\widetilde{u}_t^+$ and the Q difference. The algebra bound gives

$$\|r^+ - \widetilde{r}^+\|_{CH^m} \le [c_m M_C K_\lambda L_F T + c_m L_C A_T \sqrt{T} + L_Q T] D, \tag{16.113}$$

$$\|b^+ - \widetilde{b}^+\|_{CH^m} \le L_L T D. \tag{16.114}$$

Together with (16.112), this is a contraction for small T. Its fixed point solves the system. Lemma 16.8 supplies (16.88); the products in the fiber equation belong to L^2H^m, giving (16.89). The row right side is continuous into H^m, giving the stronger row regularity.

Retaining initial-data differences in the same estimates proves locally Lipschitz dependence in X_T and the heat regularity norms. Substitution into the fiber equation then proves it also for the H^1H^m norms of r, b. Uniqueness among solutions in the stated class follows from these difference estimates on a sufficiently short common interval; concatenation extends it to their common interval of existence.

Finally, a bound on (16.90) and a fixed compact interior image give uniform coefficient bounds, norm radii, and a positive distance to the chart boundary. The displacement estimates (16.109)–(16.111) make the restart time uniform. This uses continuity of the heat semigroup from H^s into H^{s-1} with a uniform rate on bounded sets, rather than a false operator-norm continuity claim on H^s. Restarting at times approaching a finite endpoint extends the solution beyond it. This proves the continuation alternative. □

Dissipation and reconstructed-law regularity

Proposition 16.9 (Exact energy identity). For the solution of theorem 16.7, $\mathcal{F}_*(t)$ is absolutely continuous and, for almost every t,

$$\begin{aligned}\frac{\mathrm{d}\mathcal{F}_*}{\mathrm{d}t} &= -\int_C \left[u_t^T h u_t + (r_t + Cu_t)^T k (r_t + Cu_t) + b_t^T B b_t\right] \mathrm{d}\mu_C \\ &= -\int_C \left[u_t^T h u_t + V_r^T k^{-1} V_r + V_b^T B^{-1} V_b\right] \mathrm{d}\mu_C.\end{aligned} \tag{16.115}$$

Consequently

$$\mathcal{F}_*(t) + \int_0^t \|\partial_a(u, r, b)\|_{\mathbb{G}}^2 \mathrm{d}a = \mathcal{F}_*(0). \tag{16.116}$$

Proof. The regularity $m > d/2 + 1$ makes the composition and product chain rules valid, with spatial gradients and their time derivatives integrable in the needed norms. Approximation by smooth

periodic fields, or direct Sobolev integration by parts, gives

$$\frac{\mathrm{d}\mathcal{F}_*}{\mathrm{d}t} = \int_C [(V_u - \lambda h \tau_h)^T u_t + V_r^T r_t + V_b^T b_t] \mathrm{d}\mu_C. \tag{16.117}$$

Every term is integrable in time by (16.88)–(16.89) and the compact-image coefficient bounds. Insert the metric-gradient block equations preceding (16.75). This gives the first identity; the fiber and row equations give the second. Integration in time proves the assertion, including absolute continuity. □

The reconstructed law parameters satisfy

$$p = \Psi(u, r) \in CH^m \cap H^1(0, T; H^m), \qquad A^* = -p^* \theta \in CH^{m-1} \cap H^1(0, T; H^{m-1}). \tag{16.118}$$

The latter follows by differentiating p once in space and using smooth-composition and product estimates; its time derivative contains $p_t \partial_\mu p$ and $\partial_\mu p_t$, both in the stated class. The cancellation in (16.67) gives the better residual regularity $D^{A^*} p = H_a \partial_\mu u^a \in CH^m$. There is no separate connection evolution hidden in these claims.

Dissipation controls the total energy, not each summand separately. It gives no general bound on the Sobolev norms in (16.90) and no exclusion of a statistical-family boundary, loss of finite smooth KL costs, or approach to a singular orbit. A coordinate-chart exit may be removable by another chart, but such patching is not needed for this local theorem. Positive target curvature is not an obstruction to this local proof. Global harmonic-map results with additional curvature hypotheses [52] cannot be imported as global conclusions for this coupled nonbasic system. Related sigma-flow work studies statistical-manifold energies but leaves its general existence question open [27]; the present result uses the declared Levi-Civita quotient metric, fixed flat base, and explicit regularity class.

Zero coupling is a separate local ODE theorem

Setting $\lambda = 0$ in the metric-gradient equations removes exactly the contextual term and recovers the chosen pointwise reciprocal Fisher flow. Smooth composition makes this an ordinary differential equation in the Banach space $(H^m)^{n_u + n_r + n_b}$, locally Lipschitz on bounded compact-image subsets. The integral contraction therefore gives a unique local $C_t^1 H_x^m$ solution. The energy identity holds with the contextual term absent. The energy at zero coupling does not select A.

The constants in Lemma 16.8 are not uniform as $\lambda \downarrow 0$. Nor is the extra H^s regularity of u automatic for the zero-coupling equation. For example, Euclidean Fisher coordinates of Gaussian location laws with a correlated Gaussian reference can give

$$V(u, r) = \tfrac{1}{2}(u^2 - 2aur + r^2) + \text{constant}, \qquad 0 < a < 1. \tag{16.119}$$

Take the isometric translation action in r, so u is the quotient coordinate. The zero-coupling solution is

$$u(t) = e^{-t}[\cosh(at) u_0 + \sinh(at) r_0]. \tag{16.120}$$

If $u_0 \in H^s$ and $r_0 \in H^m \setminus H^s$, then $u(t) \notin H^s$ for every $t > 0$. This example concerns the regular statistical-gradient class underlying the proof; it is not needed to specialize the multi-agent potential. Algebraic zero-coupling recovery is established, while a uniform singular-limit theorem is not asserted.

16.2.6 An exact two-sector continuum witness

Take $C = S^1$ with measure $\mathrm{d}x/(2\pi)$ and two agents, with

$$q_i = \mathcal{N}(\mu_i, I_2), \quad s_i = \mathcal{N}(\nu_i, I_2), \quad r_i = \mathcal{N}(0, I_2), \quad G_i(m, \mathrm{d}k) = \mathcal{N}(m, I_2)(\mathrm{d}k). \tag{16.121}$$

Each agent attends to the other through an identity channel in both sectors. These singleton rows have no tangent variables. Direct integration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| r_i G_i) = \tfrac{1}{2}\|\mu_i - \nu_i\|^2 + \tfrac{1}{2}\|\nu_i\|^2 + 1. \tag{16.122}$$

Thus both sectors participate in the evaluator term, and both directed peer interactions participate in the reciprocal potential. With columns $P = (\mu_1, \nu_1, \mu_2, \nu_2)$, the Fisher metric is Frobenius and

$$V(P) = 2 + \tfrac{1}{2}\langle P, PH\rangle_F, \qquad H = \begin{pmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & 0 & -2 \\ -2 & 0 & 3 & -1 \\ 0 & -2 & -1 & 4 \end{pmatrix}. \tag{16.123}$$

Equivalently, the peer terms are $\|\mu_1 - \mu_2\|^2 + \|\nu_1 - \nu_2\|^2$. A single $SO(2)$ rotation acts on every column. The action is free for $P \neq 0$, is proper, and preserves these constitutive data. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R_\varphi = \cos\varphi I + \sin\varphi J, \qquad A_x = \alpha J. \tag{16.124}$$

The connection equation and reduced energy are

$$\alpha_* = -\frac{\langle P_x, JP\rangle_F}{\|P\|_F^2}, \tag{16.125}$$

$$\mathcal{F}_*[P] = \int_{S^1} \left[2 + \tfrac{1}{2}\langle P, PH\rangle_F + \frac{\lambda}{2}\left(\|P_x\|_F^2 - \frac{\langle P_x, JP\rangle_F^2}{\|P\|_F^2}\right)\right] \mathrm{d}\mu_C. \tag{16.126}$$

For fixed α, $D_x = \partial_x + \alpha J$ is skew adjoint on periodic matrix fields. Its envelope variation therefore gives

$$\boxed{P_t = -PH + \lambda D_x D_x P, \qquad \alpha = \alpha_*.} \tag{16.127}$$

In expanded form the second-order term is $P_{xx} + \alpha_x JP + 2\alpha JP_x - \alpha^2 P$. Its principal part after eliminating α is $P_{xx} - \langle P_{xx}, JP\rangle_F JP/\|P\|_F^2$, which displays the missing vertical diffusion. The vector field in this particular symmetric example is horizontal: $H = H^T$ gives $\langle PH, JP\rangle_F = 0$, and differentiating $\langle D_x P, JP\rangle_F = 0$ covariantly gives $\langle D_x D_x P, JP\rangle_F = 0$.

Set

$$\tau = \frac{\sqrt{5}-1}{2}, \qquad \kappa = 5 - \tau = \frac{11-\sqrt{5}}{2}, \qquad v = (1, \tau, -1, -\tau)^T, \qquad V_0 = e_1 v^T. \tag{16.128}$$

The identity $\tau^2 + \tau = 1$ gives $Hv = \kappa v$ and $w := \|V_0\|_F^2 = 5 - \sqrt{5}$. For any smooth periodic real phase $\varphi(x)$, take

$$P(t,x) = R_{\varphi(x)} a(t,x) V_0, \qquad a(t,x) > 0. \tag{16.129}$$

Direct substitution yields

$$\alpha_* = -\varphi_x, \qquad D_x P = R_\varphi a_x V_0, \qquad D_x D_x P = R_\varphi a_{xx} V_0. \tag{16.130}$$

The full matrix equation reduces exactly to

$$a_t = \lambda a_{xx} - \kappa a, \qquad \varphi_t = 0. \tag{16.131}$$

For $n \geq 1$ and $a_0 > |a_1| > 0$, an exact solution is

$$\boxed{a(t,x) = a_0 e^{-\kappa t} + a_1 e^{-(\kappa + \lambda n^2)t} \cos(nx).} \tag{16.132}$$

It is positive at every finite $t \geq 0$, since $a(t,x) \geq e^{-\kappa t}(a_0 - |a_1|) > 0$. All four means are nonzero, and the collective action remains free at every finite time. The zero-mean singular orbit is approached only as $t \to \infty$ for this solution. With $\varphi(x) = \varepsilon \sin x$, $\varepsilon \neq 0$, the nonzero connection is $A_x = -\varepsilon \cos x J$. It is pure gauge; a one-dimensional base gives no curvature witness.

The amplitude $a = \|P\|_F / \sqrt{w}$ is gauge invariant. Its spatial-mode-to-constant-mode ratio is

$$\frac{a_1}{a_0} e^{-\lambda n^2 t}. \tag{16.133}$$

At zero coupling this ratio is constant; at positive coupling it decays faster for higher spatial frequency. This is an exact contextual effect in shape, with a nontrivial reciprocal local reaction. Initial fields with $n = 1$ and $n = 2$ have the same laws and first derivative at $x = 0$, but

$$P_t^{(2)}(0,0) - P_t^{(1)}(0,0) = -3\lambda a_1 R_{\varphi(0)} V_0 \neq 0. \tag{16.134}$$

Thus the local velocity depends on neighboring contextual variation.

Using the normalized base measure, the exact energy is

$$\mathcal{F}_*(t) = 2 + \frac{w\kappa a_0^2}{2} e^{-2\kappa t} + \frac{w(\kappa + \lambda n^2) a_1^2}{4} e^{-2(\kappa + \lambda n^2)t}. \tag{16.135}$$

Its derivative is

$$\frac{\mathrm{d}\mathcal{F}_*}{\mathrm{d}t} = -w\kappa^2 a_0^2 e^{-2\kappa t} - \frac{w}{2}(\kappa + \lambda n^2)^2 a_1^2 e^{-2(\kappa + \lambda n^2)t} = -\int_{S^1} \|P_t\|_F^2 \mathrm{d}\mu_C. \tag{16.136}$$

The existence for every finite time in this explicit sector is a property of its formula. It does not turn the general local theorem into a global theorem, and singleton source rows do not test the nontrivial row part of that theorem.

16.2.7 Scope and remaining obligations

The theorem establishes local well-posedness on a supplied flat torus and a regular finite-dimensional family of general normalized laws. It retains nonbasic fixed-reference forces. Global continuation requires the stated Sobolev and compact-image bounds. Unrestricted infinite-dimensional law dynamics, changes of orbit type, nontrivial base bundles, multiple-chart solutions, physical boundaries, an evolving base metric, and alternative statistical connections require additional analysis.

The construction does not derive its base, agent census, references, or channel data. It proves no graph-refinement limit, formation, or recursive inheritance. Deriving displacement transport from peer links is the separate construction of chapter 17; substituting its selector into this energy would require its full constitutive variation and a new evolution theorem. Independent frames and source rows acquire no separate diffusion. The exact Gaussian sector illustrates the continuum coupling within these boundaries.

Chapter 17

Base Transport from Contextual Agent Data

A smooth family of agent interactions can determine a connection once its cross-context comparison rule is specified. This chapter constructs such rules from law moments, receiving interfaces, and directed links. A full-rank message matrix gives displacement maps with identity at zero displacement; their first derivative gives the connection, and products along refining paths give its parallel transport. A second construction aligns the contextual link data directly and admits nonzero curvature even when every agent backtrack is the identity.

The comparison data are constitutive choices. The message metric and an augmented link metric need not be the Fisher metric of the joint law in proposition 3.9. The new selectors therefore need not equal that chapter's collective Fisher connection. They supply a conditional construction for open problem C.4; they add no evolution equation or formation result to the contextual-flow extension in chapter 16.

17.1 Contextual data and their transformation laws

Let $\mathcal{C}$ be a smooth finite-dimensional contextual base, $P \to \mathcal{C}$ a principal matrix-group bundle, and $E = P \times_G V$ an associated vector bundle. Fix one sector temporarily. The explicit construction first uses $G = \mathrm{GL}(K, \mathbb{R})$ and $V = \mathbb{R}^K$. Section 17.7 treats a smaller declared matrix group. A context is denoted c; agent labels i, j are not base coordinates. All derivatives below are contextual derivatives at fixed evolution time.

On a patch choose a reference presentation of E. Agent i displays vectors in a coordinate copy E_i through an invertible map $U_i : V \to E_i$. Its normalized law ν_i may be a belief or model marginal. A passive change of displayed coordinates and a change of common reference act differently:

$$U_i' = g_i U_i, \qquad \nu_i' = (g_i)_{\#}\nu_i, \tag{17.1}$$

$$U_i' = U_i h, \qquad \bar{\nu}_i' = (h^{-1})_{\#}\bar{\nu}_i, \qquad \bar{\nu}_i = (U_i^{-1})_{\#}\nu_i. \tag{17.2}$$

The presentation maps and coordinate changes are C^2 and may vary with context. Reference-independent statements must satisfy both rules.

A directed, invertible interaction at one context is

$$\Theta_{ij} = U_i K_{ij} U_j^{-1} : E_j \longrightarrow E_i, \qquad K_{ij}' = h^{-1} K_{ij} h. \tag{17.3}$$

For example, $K_{ij} = \exp(L_{ij})$, where $L_{ij} = O_i T_j$ is a receiving–sending product on V that may depend on the current laws. General normalized channels need not be invertible; the factorization

arguments here apply to invertible links only. Neither Θ_{ij} nor its exponential generator is already a displacement map between c and $c + \delta c$.

Our convention for a reference connection is $D^a = d + a$, with

$$a' = h^{-1}ah + h^{-1}dh. \tag{17.4}$$

In agent i's displayed coordinates the same connection is

$$B_i = U_i a U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i F^a U_i^{-1}, \qquad F^a = da + a \wedge a. \tag{17.5}$$

Thus $B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}$ under eq. (17.1), and B_i is unchanged by eq. (17.2) together with eq. (17.4). Products of matrix-valued forms include matrix multiplication; $(a \wedge a)(Y, Z) = [a(Y), a(Z)]$.

The issue is to derive a from stated contextual agent data. Merely declaring a_μ and inserting $\exp(-a_\mu \delta c^\mu)$ supplies the answer as additional data. The construction below begins with finite cross-context messages and differentiates their normalized alignment.

17.2 What an agent frame does determine

Proposition 17.1 (A square frame selects flat local transport). Let $Q(c)$ be a C^2 invertible matrix whose columns are declared marked frame vectors, expressed in one reference coordinate system. Put

$$a_Q = -dQQ^{-1}, \qquad T_Q(c', c) = Q(c')Q(c)^{-1}. \tag{17.6}$$

Then $T_Q(c, c) = I$, $D^{a_Q} Q = 0$, $F^{a_Q} = 0$, and $T_Q(c'', c')T_Q(c', c) = T_Q(c'', c)$. If $Q' = h^{-1}Q$, then a_Q obeys eq. (17.4) and $T_Q' = h(c')^{-1} T_Q h(c)$.

Proof. Identity, composition and covariance follow by multiplication. Taylor expansion gives $T_Q(c + \delta c, c) = I - a_Q(\delta c) + O(|\delta c|^2)$. The identity $dQ^{-1} = -Q^{-1}(dQ)Q^{-1}$ gives $da_Q = -a_Q \wedge a_Q$, proving flatness. □

An agent frame can therefore select a connection if declaring that frame parallel is part of the model. When the frame is only a coordinate choice, that declaration changes the model under an arbitrary change of frame unless the connection is transformed with it. In particular $B_i = -dU_i U_i^{-1}$, with U_i as in eq. (17.2), holds the reference connection $a = 0$ fixed. Replacing U_i by $U_i h(c)$ while again setting $a = 0$ changes B_i by $-U_i dh h^{-1} U_i^{-1}$. This rule is not independent of the common reference.

Several individually flat connections can give a curved affine average. They must already be connections on the same bundle. For two such connections a_1, a_2, put $\Delta = a_1 - a_2$ and let λ be a smooth scalar. Then $a = a_2 + \lambda\Delta$ has

$$F^a = d\lambda \wedge \Delta - \lambda(1 - \lambda)\Delta \wedge \Delta. \tag{17.7}$$

Flatness gives $D^{a_2}\Delta = -\Delta \wedge \Delta$, which substituted into $F^{a_2 + \lambda\Delta}$ proves the formula. Weight variation can produce Abelian curvature, and noncommuting disagreements can produce non-Abelian curvature even at constant weight. The weights and identification of the individual connections

remain constitutive inputs. Averaging square frames and applying eq. (17.6) still gives a flat connection.

17.3 Retaining reciprocal and backtrack information

Suppose each backtrack $N_{ij,j} = \Theta_{ji}\Theta_{ij}$ has spectrum disjoint from $(-\infty, 0]$. The ordered-pair subscripts distinguish loops from different neighbors. Define

$$S_{ij,j} = N_{ij,j}^{1/2}, \qquad \Omega_{ij} = \Theta_{ij} S_{ij,j}^{-1}, \qquad \Theta_{ij} = \Omega_{ij} S_{ij,j}. \tag{17.8}$$

The principal real matrix root is smooth on this spectral domain and commutes with similarity [76]. Since $N_{ij,i} = \Theta_{ij} N_{ij,j} \Theta_{ij}^{-1}$, multiplication gives

$$\Omega_{ji} = \Omega_{ij}^{-1}, \qquad \Omega'_{ij} = g_i \Omega_{ij} g_j^{-1}, \qquad S'_{ij,j} = g_j S_{ij,j} g_j^{-1}. \tag{17.9}$$

This factorization retains the original directed pair exactly. It does not impose inverse relations on the original Θ fields. An interaction loop at one context is also distinct from a loop in the contextual base.

The construction may use raw Θ fields directly, without any square-root branch. Where the branch exists, including both Ω-transformed and Θ-transformed messages makes their different roles explicit. Storing Ω and S retains all directed links, even though their eventual connection need not encode them injectively.

17.4 A full-rank contextual message construction

Fix a finite census and persistent message labels. Their message space is $H = \mathbb{R}^r$, with a stated, context-independent Euclidean inner product. A fixed positive message metric can be absorbed into the columns. Identifying these labels between contexts is an assumption; agent labels or fixed receiving-interface columns give one realization. Section 17.10 treats changing message coordinates.

For a normalized law with a finite first moment, let $\mu_i = \mathbb{E}_{\nu_i}[v] \in E_i$ and $\bar{\mu}_i = U_i^{-1}\mu_i$. The law need not be Gaussian. Smoothness of its moments is an assumption on the law family, not a consequence of normalization. Define

$$R_{ij} = U_i^{-1}\Omega_{ij}U_j, \qquad K_{ij} = U_i^{-1}\Theta_{ij}U_j. \tag{17.10}$$

One permissible collective message matrix is a chosen concatenation

$$X(c) = \left[\{\bar{\mu}_i\}_i,\ \{R_{ij}\bar{\mu}_j\}_{(i,j)},\ \{K_{ij}\bar{\mu}_j\}_{(i,j)}\right] : H \longrightarrow V. \tag{17.11}$$

Each column has a stated label. Receiving-interface columns $O_i : H_i \to V$, or their link-transformed columns, can replace or supplement the means. Gauge-invariant scalar column weights may depend smoothly on laws and rows; their derivatives are included in dX. Column selection, weights and message metric specify the rule. The identity self link does not imply that these columns span V.

Under passive agent re-expression every reference column is unchanged. Under reference rechoice it obeys

$$X' = h^{-1}X. \tag{17.12}$$

For example $(h^{-1}R_{ij}h)(h^{-1}\bar{\mu}_j) = h^{-1}R_{ij}\bar{\mu}_j$. The same calculation applies to raw link and interface columns. Thus the construction is reference consistent when its original link and interface rules are reference consistent.

Hypothesis 17.2 (Regular message region). On an open region $\mathcal{U} \subset \mathcal{C}$, X is C^2 with row rank K. Its columns refer to one fixed labeled message space. Local expressions on overlaps obey eq. (17.12). No spatial connection on E is included in these data.

Put

$$H_X = XX^T, \qquad Q_X = X^T H_X^{-1}, \qquad P_X = Q_X X. \tag{17.13}$$

Here H_X is a positive-definite $K \times K$ Gram matrix, $Q_X : V \to H$ is a right inverse of X, and P_X is the orthogonal projector in message space onto $\operatorname{im} X^T$.

Theorem 17.3 (Displacement links and their connection). Under the regular message assumption, the map

$$C(c', c) = X(c')Q_X(c) \tag{17.14}$$

equals I at $c' = c$, is invertible on a neighborhood of the diagonal, and obeys

$$C'(c', c) = h(c')^{-1}C(c', c)h(c). \tag{17.15}$$

Its first derivative determines a C^1 connection:

$$C(c + \delta c, c) = I - a_c(\delta c) + O(|\delta c|^2), \qquad a = -dXQ_X. \tag{17.16}$$

The derivative is linear in the tangent displacement. The local forms obey eq. (17.4) and define a connection wherever the data have the stated overlap laws. In agent coordinates

$$C_i(c', c) = U_i(c')C(c', c)U_i(c)^{-1} \tag{17.17}$$

has connection eq. (17.5), is reference independent, and transforms by $g_i(c')$ and $g_i(c)^{-1}$ at its two endpoints.

Proof. Full row rank gives $XQ_X = I$. Continuity of the determinant makes C invertible near each diagonal point. Under eq. (17.12),

$$H'_X = h^{-1}H_X h^{-T}, \qquad Q'_X = Q_X h, \qquad P'_X = P_X. \tag{17.18}$$

This proves the endpoint law. Differentiating the first argument of eq. (17.14) gives eq. (17.16); Taylor's theorem gives its remainder locally uniformly on compact subsets. Moreover,

$$-d(h^{-1}X)Q_X h = h^{-1}(-dXQ_X)h + h^{-1}dh. \tag{17.19}$$

The affine law and the bundle transition cocycle glue the local covariant derivatives. Expanding eq. (17.17) gives B_i; substitution proves passive covariance and reference independence. □

The displacement maps are locally invertible, but the connection can be global on a nontrivial bundle if a global surjection $\mathcal{C} \times \mathbb{R}^r \to E$ with these data is supplied. When $r > K$, such a surjection need not trivialize E: its projector describes a moving subspace of the message space. A global square frame is the more restrictive case $r = K$.

A receiver's displayed columns $\mu_i, \Omega_{ij}\mu_j, \Theta_{ij}\mu_j$ also transform by left multiplication by g_i. Applying the formula gives a receiver-selected connection. Different receivers' selections need not agree. The collective reference construction eq. (17.11) instead selects one a, displayed by all agents through eq. (17.5).

17.5 Reversal, finite paths, and zero displacement

The cross-map C generally fails both reversal and composition. These failures do not invalidate its first jet as a connection, but matter before calling a finite map parallel transport. Discrete connections and their infinitesimal limits are classical geometric constructions [97].

Proposition 17.4 (Reciprocal displacement maps with the same derivative). After shrinking the neighborhood of the diagonal, define

$$N(c', c) = C(c, c')C(c', c), \qquad T(c', c) = C(c', c)N(c', c)^{-1/2}. \tag{17.20}$$

Then $T(c, c) = I$, $T(c, c') = T(c', c)^{-1}$, and T obeys eq. (17.15). Furthermore,

$$N(c + \delta c, c) = I + O(|\delta c|^2), \qquad T(c + \delta c, c) = I - a_c(\delta c) + O(|\delta c|^2). \tag{17.21}$$

Proof. At the diagonal $N = I$, so its principal root is smooth nearby. The calculation in eq. (17.9) proves reversal and endpoint covariance. Differentiating $XQ_X = I$ gives $XdQ_X = -dXQ_X$, so the two factors of N have opposite linear terms. Its root is $I + O(|\delta c|^2)$, proving the expansion. □

The new $N(c', c)$ is a cross-context reconstruction defect. It differs from the agent backtrack $N_{ij,j}(c)$, retained in the input state through eq. (17.8). Reciprocalizing a displacement map does not delete the backtrack data that determined X.

Theorem 17.5 (Parallel transport from the derived first jet). For a piecewise C^2 path $\gamma : [0, 1] \to \mathcal{U}$, the equation

$$\dot{\mathcal{T}}(t) = -a_{\gamma(t)}(\dot{\gamma}(t))\mathcal{T}(t), \qquad \mathcal{T}(0) = I \tag{17.22}$$

has a unique invertible solution. It composes under concatenation, inverts under reversal, and has the endpoint law. Ordered products of C or T along partitions of γ converge to $\mathcal{T}(1)$ as the maximum time step tends to zero.

Proof. On each smooth segment the coefficient in eq. (17.22) is continuous and bounded. Picard iteration on a sufficiently short interval is a contraction, and finitely many intervals cover the path. The solution of $\dot{S} = Sa(\dot{\gamma})$, $S(0) = I$, satisfies $\frac{d}{dt}(S\mathcal{T}) = 0$, proving invertibility. Uniqueness proves concatenation and reversal. Differentiating $h(\gamma(t))^{-1}\mathcal{T}(t)h(\gamma(0))$ verifies its transformed equation.

Include the finitely many corners among partition points. On each smooth segment the exact short propagator and discrete factor equal $I - a(\dot\gamma)\Delta t + O(\Delta t^2)$ uniformly on the compact path image. Their norms are bounded by $1 + M\Delta t$. Telescoping the product difference bounds it by $C\exp(2M)\sum_k \Delta t_k^2$, which tends to zero because $\sum_k \Delta t_k^2 \le \max_k \Delta t_k$. This proves convergence for both discrete factors. Local trivializations can be joined by transition maps; the endpoint laws make the result independent of this description. □

An unrefined $C(c', c)$ or $T(c', c)$ need not equal exact transport along a selected path. Its first jet determines eq. (17.22), while the finite endpoint rule contains additional, nonunique higher-order data.

17.6 Curvature and a curved law-dependent example

Proposition 17.6 (Curvature of the rectangular construction). For eq. (17.16),

$$\begin{aligned} F^a &= dX \wedge dP_X Q_X = dX \wedge (I - P_X) dQ_X \\ &= dX \wedge (I - P_X) dX^T H_X^{-1}, \end{aligned} \tag{17.23}$$

$$dH_X + aH_X + H_X a^T = 0, \qquad \operatorname{tr} a = -\tfrac{1}{2} d \log\det H_X, \qquad \operatorname{tr} F^a = 0. \tag{17.24}$$

The connection preserves the fiber metric H_X^{-1}. When $r = K$, $P_X = I$ and the curvature vanishes. Redundant columns permit nonzero curvature.

Proof. Exterior differentiation of $a = -dXQ_X$ gives $da = dX \wedge dQ_X$. Differentiating $P_X = Q_X X$ and multiplying by Q_X gives $dP_X Q_X = dQ_X + Q_X dX Q_X$. This proves the first equality after adding $a \wedge a$. Differentiating $P_X Q_X = Q_X$ proves the second. In $dQ_X = dX^T H_X^{-1} + X^T dH_X^{-1}$ the second term is killed by $I - P_X$, giving the last expression. Finally $aH_X = -dXX^T$ and $H_X a^T = -XdX^T$, proving metric compatibility. Taking its trace after multiplying by H_X^{-1} gives $2\operatorname{tr} a = -d\log\det H_X$. The trace of a commutator vanishes, so $\operatorname{tr} F^a = d\operatorname{tr} a = 0$. □

This projector construction is an instance of induced subbundle geometry, related to the classical universal connection on a Stiefel bundle [105, 111]. No universality claim for this particular real message rule is made: eq. (17.24) restricts its curvature. The choice here is how agent laws and directed links supply the columns.

On $\mathbb{R}^2$ with coordinates (x, y) take

$$X = \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \end{pmatrix}, \qquad v = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad s = 1 + x^2 + y^2. \tag{17.25}$$

It has rank two everywhere. Substitution gives

$$a = -\frac{(dv)v^T}{s}, \tag{17.26}$$

$$F^a = \frac{1}{s^2} \begin{pmatrix} -xy & 1 + x^2 \\ -(1 + y^2) & xy \end{pmatrix} dx \wedge dy. \tag{17.27}$$

At the origin $a = 0$ but $F^a_{xy} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq 0$.

An explicit realization uses normalized non-Gaussian laws and an invertible link. Let a receiver carry receiving columns e_1, e_2. Let a source law be the product Laplace law of unit coordinate variances on $\mathbb{R}^2$, translated to mean $(x, 1)^T$. Choose

$$K_{ij} = \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix} = \exp(yE_{21}), \qquad K_{ji} = K_{ij}^{-1}. \tag{17.28}$$

The linked mean is $(x, 1 + xy)^T$. Replacing v in eq. (17.25) by this vector gives a full-rank, link-dependent matrix. Its curvature is the pullback of eq. (17.27) under $(x, y) \mapsto (x, 1 + xy)$, nonzero wherever $x \neq 0$ because the Jacobian determinant is x. The labels and receiving axes are declared interface data. No Gaussian closure or evolving distributional ansatz is used.

Backtracks can affect the connection while the reciprocal agent link stays fixed. In a scalar example take a receiver seed column 1, source mean 1, and $\Theta_{ij} = \Theta_{ji} = s(c) > 0$. Then $\Omega_{ij} = 1$ and $N_{ij,j} = s^2$. The raw linked column gives $X = [1, s]$ and

$$a = -\frac{sds}{1 + s^2}, \tag{17.29}$$

whereas using only the reciprocal column gives $X = [1, 1]$ and $a = 0$. Both scalar connections are flat, but their transports differ for varying s. This proves a backtrack contribution can survive the selector, without asserting that every backtrack distinction is recoverable from one connection.

17.7 A declared structure group and joint sector data

The explicit matrix eq. (17.16) need not lie in an arbitrarily chosen subalgebra $\mathfrak{g} \subset \mathfrak{gl}(K)$. A $\mathrm{GL}(E)$ connection is not automatically a connection on a prescribed G-reduction.

Let $M(c)$ be a C^1 positive-definite fiber metric with $M' = h^T M h$, and let X have full row rank and eq. (17.12). The choice $M = H_X^{-1}$ is available. For $Y \in T_c\mathcal{C}$, minimize over $\xi \in \mathfrak{g}$

$$\mathcal{J}_Y(\xi) = \tfrac{1}{2} \operatorname{tr}\big[M(dX(Y) + \xi X)(dX(Y) + \xi X)^T\big]. \tag{17.30}$$

Proposition 17.7 (A selector in the declared Lie algebra). For a closed matrix group G, eq. (17.30) has a unique minimizer $a(Y) \in \mathfrak{g}$, linear in Y, C^1 in c, and satisfying eq. (17.4). It solves

$$\operatorname{tr}[M(a(Y)X)(\eta X)^T] = -\operatorname{tr}[M dX(Y)(\eta X)^T] \quad (\eta \in \mathfrak{g}). \tag{17.31}$$

For $\mathfrak{g} = \mathfrak{gl}(K)$ the solution is eq. (17.16).

Proof. The Hessian is positive: $\xi X = 0$ implies $\xi = 0$ after multiplying by Q_X. Its matrix in any basis of $\mathfrak{g}$ is invertible, giving existence, uniqueness and linear, smooth dependence. Under reference rechoice the candidate $\xi' = h^{-1}\xi h + h^{-1}dh(Y)$ belongs to $\mathfrak{g}$ and $dX'(Y) + \xi' X' = h^{-1}(dX(Y) + \xi X)$.

The norm is unchanged; uniqueness gives the affine law. In the unrestricted algebra the normal equation is $(dX + aX)X^T = 0$, yielding eq. (17.16). □

For a proper subalgebra, the induced-metric and explicit curvature formulas of proposition 17.6 need not hold for the constrained minimizer. Finite G-valued transport follows by eq. (17.22) along declared short paths with this derived a. Belief and model sectors carry representations ϱ_q, ϱ_s of one group. To select one principal connection from both, sum their costs, replacing ξX by $d\varrho_a(\xi)X^a$, and minimize over the same $\xi \in \mathfrak{g}$. The required nondegeneracy is

$$\{\xi : d\varrho_q(\xi)X^q = 0,\ d\varrho_s(\xi)X^s = 0\} = \{0\}. \tag{17.32}$$

The same proof applies to the summed normal equation. Two unrelated sector connections do not automatically arise from one principal connection. The two interaction link fields can remain distinct; equality of the belief and model links is a separate hypothesis.

17.8 Direct alignment of contextual link data

The rectangular construction is sufficient, but a full-rank vector collection is not necessary when the group action on links is itself identifiable. Let G be a finite-dimensional Lie group and let $\mathcal{Z}$ be a finite-dimensional Riemannian target of laws, interfaces and links with a smooth isometric left G-action and metric $\mathcal{H}$. Under reference rechoice the target obeys $Z'(c) = h(c)^{-1} \cdot Z(c)$. Write $Z(c) \in \mathcal{Z}$ and $J_Z\xi = \frac{d}{dt}\big|_0 \exp(t\xi) \cdot Z$. Suppose Z is smooth and J_Z is injective on the region considered.

Proposition 17.8 (Local link alignment). Near each diagonal point there is a unique critical branch near e that is a local minimizer of

$$t(c', c) = \operatorname*{argmin}_{g \text{ near } e} \tfrac{1}{2} d_{\mathcal{H}}(Z(c'), g \cdot Z(c))^2. \tag{17.33}$$

It satisfies $t(c, c) = e$, reversal by inversion, and the endpoint law. Its connection solves

$$\mathcal{H}(J_Z a(Y), J_Z \eta) = -\mathcal{H}(dZ(Y), J_Z \eta). \tag{17.34}$$

Proof. Use squared distance inside a normal neighborhood. At a diagonal point the group gradient vanishes and its Hessian is the positive locked form $\mathcal{H}(J_Z\xi, J_Z\eta)$. The implicit function theorem gives a unique smooth critical branch, whose Hessian remains positive nearby. Expanding $t(c + \delta c, c) = e - a(\delta c) + O(|\delta c|^2)$ gives residual $dZ(Y) + J_Z a(Y)$ and hence eq. (17.34). Isometry makes a reference rechoice carry the selected branch to $h(c')^{-1}t(c', c)h(c)$. Symmetry of distance and isometry identify the reversed objective at g^{-1} with the original at g; local uniqueness proves inversion. □

This is the mechanical connection of the declared target metric, as in the rotation–shape separation of Littlejohn and Reinsch [101]. The statement concerns the branch continuous from the identity. Infinitesimal identifiability does not exclude distant competing minima. On overlaps the local identity branches agree after restriction by uniqueness. Its first jet gives path transport by the preceding ODE argument.

The target can retain reciprocal link matrices and backtrack factors, both transforming by conjugation, as well as laws and interfaces. Its invariant metric is an additional comparison choice: it is one smooth metric on the augmented target, including any dependence on the law or interface coordinates of that target. For example on a full-rank matrix region the terms

$$\operatorname{tr}(H_X^{-1}\delta X\delta X^T)+\sum_e \operatorname{tr}(H_X^{-1}\delta K_e H_X \delta K_e^T) \tag{17.35}$$

are positive and invariant under $X\mapsto h^{-1}X$, $K_e\mapsto h^{-1}K_e h$. One can include Fisher terms for regular statistical families. An interface or link norm is not automatically the Fisher metric of a normalized statistical experiment.

17.8.1 A curved connection from reciprocal links alone

There is a concrete link-only example for $G=SO(3)$. Identify its Lie algebra with vectors by $\widehat{v}w=v\times w$, and use the product bi-invariant metric on $SO(3)^2$, normalized by $\langle\widehat{v},\widehat{w}\rangle=v\cdot w$. On $(x,y)\in(0,\pi)^2$, fix $\beta\in(0,\pi)$ and set

$$Q_1=R_x(x),\qquad Q_2=R_z(y)R_x(\beta)R_z(-y). \tag{17.36}$$

These can be the two edges of a three-agent star, with the reverse links set to their inverses. All agent backtracks are the identity. Under reference change the pair transforms by simultaneous conjugation.

Let $C_r=I-\mathrm{Ad}_{Q_r}$ and $V_r=dQ_rQ_r^{-1}$. The right-trivialized infinitesimal conjugation is $C_r\xi$, so eq. (17.34) reads

$$\mathcal{K}a=-\sum_{r=1}^{2}C_r^*V_r,\qquad \mathcal{K}=\sum_{r=1}^{2}C_r^*C_r. \tag{17.37}$$

For a rotation through angle α about unit axis n, $(I-Q)^T(I-Q)=4\sin^2(\alpha/2)(I-nn^T)$. Here the axes are e_1 and $n(y)=(\cos y,\sin y,0)^T$, which are nonparallel on the stated region. Thus $\mathcal{K}$ is positive definite. Write $w_x=4\sin^2(x/2)$ and $w_\beta=4\sin^2(\beta/2)$. The identities $V_1=\widehat{e}_1dx$ and $V_2=C_2\widehat{e}_3dy$ imply $C_1^*V_1=0$ and $C_2^*V_2=w_\beta\widehat{e}_3dy$. Since $\mathcal{K}\widehat{e}_3=(w_x+w_\beta)\widehat{e}_3$, one obtains

$$a=-\frac{w_\beta}{w_x+w_\beta}\widehat{e}_3dy,\qquad F^a=\frac{2w_\beta\sin x}{(w_x+w_\beta)^2}\widehat{e}_3dx\wedge dy\neq 0. \tag{17.38}$$

The wedge square vanishes because a has only one base component. This proves nonzero curvature from smoothly varying reciprocal agent links and a declared comparison metric, without adding directional generators or a spatial connection as input. Normalized rotationally invariant agent laws may be attached; their orbit terms are zero and do not alter this example. The axes are selected by the links, not by those invariant laws.

17.9 What pointwise links do not determine

Proposition 17.9 (An invisible gauge subgroup). Suppose the only input fields are reference links and their contextual derivatives, with links transforming by simultaneous conjugation. If a

nonzero $Z \in \mathfrak{g}$ commutes with every admitted link field, then no full connection selector from these inputs alone can obey eq. (17.4) for every smooth reference change $h(c) = \exp(f(c)Z)$.

Proof. Every link field is unchanged as a function of c, so all its derivatives are also unchanged. A selector of those data returns the same a. Choose a point c_0 and f with $f(c_0) = 0$ and $df(c_0) \neq 0$. At that point the affine law requires $a' = a + Z df$, a contradiction. □

For group-valued links the Lie algebra of the center is always invisible to conjugation. For arbitrary endomorphism-valued links, the relevant ambiguity is their actual common commutant within the represented group; a central group element need not commute with arbitrary maps between inequivalent representation blocks. In the defining $\mathrm{GL}(K)$ representation scalar matrices are invisible to every conjugation link. An Abelian group acting on its own group-valued links is entirely invisible. The $SO(3)$ example avoids this obstruction because the two nonparallel axes have zero common infinitesimal stabilizer.

Laws, labeled vectors, or interfaces can resolve an ambiguity if their combined infinitesimal action is identifiable. Full row rank of X is sufficient in the fundamental representation. It is not implied by many links: centered invariant laws and deficient interfaces can fail it.

Even after identifiability is supplied, several covariant comparison rules can select different connections. Column sets, weights, message metrics, and augmented target metrics affect the selector. Gauge covariance does not choose these data. A finite link collection at one context supplies no contextual derivative; the constructions require a smooth contextual field of such data.

17.10 Message coordinates, dynamics, and remaining scope

The fixed message space assumption has an intrinsic formulation. If X maps a message bundle into E and the message bundle has connection Γ, use

$$a = -(dX - X\Gamma)Q_X. \tag{17.39}$$

For a change $w' = s(c)w$ of message coordinates, $X' = Xs^{-1}$, $\Gamma' = s\Gamma s^{-1} - dss^{-1}$ and $Q'_X = sQ_X$, provided the message metric is transformed as well. Hence $dX' - X'\Gamma' = (dX - X\Gamma)s^{-1}$, and eq. (17.39) is unchanged. Fixed labeled coordinates with $\Gamma = 0$ give eq. (17.16). Resetting Γ to zero after a context-dependent message re-expression changes the model. A supplied nonflat message connection can contribute additional curvature beyond eq. (17.23).

The contextual laws, rows, interfaces and links remain supplied fields; no evolution theorem is implied. If the selected connection is inserted into a contextual energy, its complete constitutive dependence must be varied. An envelope simplification requires stationarity for that same energy. Agreement with a law-Fisher selector requires matching its target, action, metric and inputs; the shared affine law alone transfers no Fisher-flow theorem.

Dynamic selection of the comparison rule, preservation or recovery of rank, singular continuation, and inheritance of the link sector remain open. The base, its dimension and topology are supplied; no formation result follows.

Chapter 18

Communication Codes and Learned Alignment

A communication channel can preserve a transmitted marginal while leaving the receiver to supply its conditional interpretation. An invertible dictionary instead aligns complete content spaces. Keeping these objects separate permits a precise account of lost information, learned alignment, and the distinctions that a loop of communicating agents can resolve.

The constructions take place at one context of the principal bundle P of chapter 1. One group G acts on belief and model contents through ϱ_q and ϱ_s. A frame change is one group element acting in both sectors. General-law reconstruction uses standard Borel spaces and normalized kernels. For the matrix calculations, write the belief representation as $V = \mathbb{R}^K$ with $G \subseteq \mathrm{GL}(V)$ and suppress the sector superscript. The model-sector calculation uses its own representation and codes; using one dictionary for both sectors requires the tied-link hypothesis. Opposite attending dictionaries remain independent.

For sender j and receiver i, selective transmission has the form

$$q_j \longrightarrow (T_j)_{\#} q_j \longrightarrow \text{transmission noise} \longrightarrow \text{receiver conditional law} \longrightarrow q_i'. \tag{18.1}$$

This is a law-reconstruction protocol when the message law is supplied. Conditioning on one realized message is a different operation. The general-law information identities below apply to both finite alphabets and continuous contents. Gaussian assumptions enter only the explicit decoder and tangent posterior calculations.

18.1 Codes and dictionaries

Definition 18.1 (Code of a modality). For general laws, a modality is a normalized Markov kernel from a standard Borel content space to a standard Borel message space. A noiseless modality is a measurable map. The linear realization used for identifiability below is more restrictive. A modality c, such as speech, drawing or gesture, has a public message space $H_c = \mathbb{R}^{r_c}$ and a noise covariance R_c, either positive definite or zero for a noiseless code. The message space carries no frame of its own: its coordinates are public conventions. Sender j's code in modality c is a linear map $\bar{T}_j^c : V \to H_c$ of rank $r_c \le K$ on reference coordinates. Its displayed form is $T_j^c = \bar{T}_j^c U_j^{-1}$. A message produced from displayed content y_j is

$$h = T_j^c y_j + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_c). \tag{18.2}$$

Under a passive change $U_j \mapsto g_j U_j$ the displayed content becomes $g_j y_j$ and the displayed code becomes $T_j^c g_j^{-1}$, so the message is unchanged. Under a common reference rechoice $U_j \mapsto U_j h$, the reference code becomes $\bar{T}_j^c h$ and the reference dictionary becomes $h^{-1}\bar{\theta}_{ij} h$. Re-expressing these constitutive data is the reference-consistency requirement. Public messages therefore do not depend on the sender's frame, and the reference code $\bar{T}_j^c$ is unchanged by passive display changes at fixed reference coordinates. Codes are agent specific. A shared language is part of a shared generative model in the sense of section 1.4: members of a species use codes related by the frame map of their shared model.

Definition 18.2 (Dictionary). Receiver i's dictionary for sender j is

$$\theta_{ij} = U_i \bar{\theta}_{ij} U_j^{-1} \in G, \qquad \bar{\theta}_{ij} \in G. \tag{18.3}$$

The general dictionary is unrestricted in G. The local parameterization $\bar{\theta}_{ij} = \exp(\omega_{ij})$ with $\omega_{ij} \in \mathfrak{g}$ is used for tangent learning. It can cover less than G: a single real matrix exponential need not cover even $\mathrm{GL}^+(K, \mathbb{R})$ [39]. The global coset and exact-identification results below concern the stated group, and a restriction to the exponential image retains only the solutions lying in that image.

It is owned by the receiver: it is i's hypothesis about how j's internal reference coordinates sit relative to its own. Content with reference coordinates $\bar{y}$ for i corresponds, according to i, to $\bar{\theta}_{ij}^{-1}\bar{y}$ for j. The effective code that i attributes to j is therefore

$$\hat{A}_{ij}^c = \bar{T}_j^c \bar{\theta}_{ij}^{-1}. \tag{18.4}$$

The dictionary θ_{ji} is j's hypothesis about i. It is an independent datum, learned by a different agent, so θ_{ji} need not equal θ_{ij}^{-1}. The product $\bar{\theta}_{ij}\bar{\theta}_{ji}$ is the holonomy of the two-cycle $i \to j \to i$: the composite of what i thinks j means with what j thinks i means. It is the identity exactly when the two agents' translations of each other are mutually inverse; otherwise it measures mutual misunderstanding. Geometric reversal of one particular dictionary is its inverse; the separately learned opposite attending edge is independent.

Remark 18.3 (Why the dictionary is group valued). A model that i holds of j need not in general be a group element, and the group-valued form is a hypothesis with a stated scope. Four considerations support it within a species. First, the two agents' internal spaces are copies of one G-space, and the relative position of two frames of one principal bundle at a context is a group element; partial access is carried by the code, not by the dictionary. Second, one dictionary serves every modality, which is what lets several codes pool their evidence in corollary 18.11. Third, invertibility is used: the consensus theorem theorem 18.16 needs inverses to show that transported beliefs are well defined along different paths, and group-valued holonomy composes invertible links; singular path products can still have trace invariants, but do not define transport by group elements. Fourth, i's knowledge of j is a probability law over G, or over ω_{ij} in a declared exponential chart; uncertainty about the dictionary is carried by that law. For agents of different kinds, whose internal spaces differ in dimension or representation, an invertible dictionary need not exist. The natural replacement is a linear or Markov map between different spaces, with consistency measured by global sections of a sheaf rather than by holonomy [72].

18.2 Reception through a code

Fix a receiver i, a sender j and one modality, and suppress the indices. In i's reference coordinates let p be i's prior law of the content and let $q = (\bar{\theta})_{\#}\bar{q}_j$ be the sender's belief carried into i's coordinates by the dictionary. Write $A = \hat{A}$ for the effective code.

A conditional law decodes a message under a receiver prior. Reconstructing an entire sender law uses that decoder for every possible message. For a measurable code T and laws q, p on a standard Borel space, choose a regular conditional law $p(dy \mid h)$ of p given $T = h$, and define the reconstruction

$$\hat{q}(dy) = \int p(dy \mid h)(T_{\#}q)(dh). \tag{18.5}$$

It keeps exactly what the code transmits, the message law $T_{\#}q$, and fills every discarded direction with the receiver's own conditional belief.

Corollary 18.4 (Meaning lost through a noiseless code). If $D_{\mathrm{KL}}(q\|p) < \infty$, then

$$D_{\mathrm{KL}}(q\|\hat{q}) = D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(T_{\#}q\|T_{\#}p). \tag{18.6}$$

This is the conditional-lift construction of proposition 5.2 after transporting the sender into the receiver's reference coordinates. The decisive proof is the chain-rule argument in corollary 5.3: its two nonnegative terms are finite under the hypothesis here, so subtraction gives eq. (18.6). The code is sufficient for the pair precisely when the discarded conditional laws agree.

In words, the meaning lost when a belief is sent through a code and rebuilt by a receiver equals the full disagreement between sender and receiver minus the disagreement that the code makes visible. It vanishes exactly when the code is sufficient for the pair, that is, when the sender's conditional belief given the message already equals the receiver's.

Corollary 18.5 (Noisy codes). Let a noisy code be a Markov kernel $\mathcal{K}$ from the content space to a standard Borel message space, for example the Gaussian kernel $\mathcal{K}(\bar{y}, \cdot) = \mathcal{N}(A\bar{y}, R)$. Let p^h and q^h be regular versions of the posteriors of p and q given h, and define the reconstruction $\hat{q} = \int p^h(q\mathcal{K})(dh)$. If $D_{\mathrm{KL}}(q\|p) < \infty$, then

$$D_{\mathrm{KL}}(q\|\hat{q}) \le D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}). \tag{18.7}$$

Proof. The chain rule applied to the joint laws of $(\bar{y}, h)$ under $q \otimes \mathcal{K}$ and $p \otimes \mathcal{K}$, which share the kernel, gives $D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}) + \int D_{\mathrm{KL}}(q^h\|p^h)(q\mathcal{K})(dh)$. Since $q = \int q^h(q\mathcal{K})(dh)$, joint convexity of relative entropy bounds $D_{\mathrm{KL}}(q\|\hat{q})$ by the integral. □

The proof uses only the chain rule and joint convexity, so the bound holds for every kernel and every pair of laws with $D_{\mathrm{KL}}(q\|p) < \infty$. For $q = \mathcal{N}(a, \Sigma_q)$, $p = \mathcal{N}(m, \Sigma_p)$, and the Gaussian kernel, $\hat{q}$ has mean $m + OA(a - m)$ and covariance $\Sigma_p - OA\Sigma_p + O(A\Sigma_q A^T + R)O^T$ with O from eq. (18.8).

Proposition 18.6 (Reception of one message). Let $p = \mathcal{N}(m, \Sigma_p)$ with Σ_p positive definite, let A have full row rank r, and let R be positive definite. Given a message h from the model $h = A\bar{y} + \varepsilon$,

the posterior of $\bar{y}$ is $\mathcal{N}(m + O(h - Am), \Sigma_p - OA\Sigma_p)$ with

$$O = \Sigma_p A^T (A\Sigma_p A^T + R)^{-1}. \tag{18.8}$$

The posterior precision is $\Sigma_p^{-1} + A^T R^{-1} A$, a correction of rank at most r, and the information carried by one message is

$$I(\bar{y}; h) = \tfrac{1}{2} \log\det\big(I_r + R^{-1} A \Sigma_p A^T\big). \tag{18.9}$$

Proof. The pair $(\bar{y}, h)$ is jointly Gaussian with $\mathrm{Cov}(\bar{y}, h) = \Sigma_p A^T$ and $\mathrm{Cov}(h) = A\Sigma_p A^T + R$, and Gaussian conditioning gives the posterior [83]. The precision form follows from the Woodbury identity. The mutual information is the difference of the entropies of $\bar{y}$ and $\bar{y}$ given h, $\frac{1}{2}\log\det\Sigma_p - \frac{1}{2}\log\det(\Sigma_p - OA\Sigma_p)$, and the matrix determinant lemma turns this into eq. (18.9). □

The receiving map is therefore derived. It is the gain O, fixed by the receiver's own uncertainty Σ_p and by its model A of the sender's code. The rank bound makes precise the loss in the motivating picture: one message changes the receiver's precision in at most r directions, however large K is.

This is the Gaussian instance of conditional decoding. That the decoder is a linear map, that the posterior covariance does not depend on the message received, and that one message changes the precision in at most r directions are features of the Gaussian family, not of decoding in general.

Proposition 18.7 (Gaussian reconstruction). Let $q = \mathcal{N}(a, \Sigma_q)$, $p = \mathcal{N}(m, \Sigma_p)$ with positive-definite covariances, and let A have full row rank. Put

$$G = \Sigma_p A^T (A\Sigma_p A^T)^{-1}, \qquad P = GA. \tag{18.10}$$

Then $AG = I_r$, $P^2 = P$, $\ker P = \ker A$, the range of P is the range of $\Sigma_p A^T$, and the reconstruction through the noiseless code A is

$$\hat{q} = \mathcal{N}\big((I - P)m + Pa,\ (I - P)\Sigma_p (I - P)^T + P\Sigma_q P^T\big). \tag{18.11}$$

Proof. The identities for G and P are immediate. Conditioning p on $A\bar{y} = h$ gives $\mathcal{N}(m + G(h - Am), \Sigma_p - GA\Sigma_p)$. Mixing over $h \sim \mathcal{N}(Aa, A\Sigma_q A^T)$ gives mean $m + GA(a - m) = (I - P)m + Pa$ and covariance $\Sigma_p - P\Sigma_p + GA\Sigma_q A^T G^T$. The last term is $P\Sigma_q P^T$. The matrix $P\Sigma_p = \Sigma_p A^T (A\Sigma_p A^T)^{-1} A\Sigma_p$ is symmetric and satisfies $P\Sigma_p P^T = P\Sigma_p$, so $\Sigma_p - P\Sigma_p = (I - P)\Sigma_p (I - P)^T$. □

The reconstructed belief adopts the sender's belief along the communicated directions and keeps the receiver's own belief along the discarded ones. Its update of the mean,

$$\mu_i' = \mu_i + G\hat{A}_{ij}\big(\bar{\theta}_{ij}\bar{\mu}_j - \mu_i\big), \tag{18.12}$$

is a projected message of the kind computed by attention. The direction in which a communicated coordinate is lifted, the range of $\Sigma_p A^T$, is set by the receiver's own correlations: two receivers with different priors read the same message as meaning different things. This is the formal content of the statement that the receiver internalizes the message through its own conceptual structure. The loss eq. (18.6) evaluates in closed form for eq. (18.11).

Remark 18.8 (The link with a code-shaped generator). For matrices O of size $K \times r$ and T of size $r \times K$, $e^{OT} = I + O\varphi_1(TO)T$ with $\varphi_1(z) = (e^z - 1)/z$ [76, Section 1.8], and $\det e^{OT} = e^{\mathrm{tr}(OT)} > 0$. With the gain of eq. (18.10), $AG = I_r$ gives $e^P = I + (e-1)P$: the invertible link whose generator is the reconstruction projector is a dilation along the communicated directions, not the reconstruction. Reconstruction is the affine use $(I - P)m + Pa$ of the projector; as a map of the sender mean a with fixed receiver mean m, it is noninvertible when $r < K$. For a square full-rank code, $P = I$ and this particular loss disappears. This is the precise sense in which no group element factors through a lossy code and the two are related only through the Lie algebra.

18.3 What messages reveal about the dictionary

A dictionary can be tested only against something both agents refer to. Let a referent have law ρ on V in i's reference coordinates, with finite positive-definite second moment $M = \mathbb{E}_\rho[\bar{y}\bar{y}^T]$. The sender encodes the referent in its own coordinates, and the receiver models the message as

$$h \mid \bar{y} \sim \mathcal{N}\big(\bar{T}\bar{\theta}^{-1}\bar{y}, R\big), \tag{18.13}$$

with R positive definite.

Theorem 18.9 (Identifiability up to the code stabilizer). Let $\mathrm{Stab}_G(\bar{T}) = \{g \in G : \bar{T}g = \bar{T}\}$. For parts (i) and (iii), allow any fixed additive-noise law on $\mathbb{R}^r$, independent of the referent; part (ii) assumes the Gaussian noise of eq. (18.13).

(i) Two dictionaries $\bar{\theta}', \bar{\theta}''$ give the same conditional message law for every referent if and only if $\bar{\theta}'^{-1}\bar{\theta}'' \in \mathrm{Stab}_G(\bar{T})$.

(ii) For the perturbation $\bar{\theta}e^\delta$, $\delta \in \mathfrak{g}$, the Fisher information of eq. (18.13), averaged over ρ, is

$$F(\delta, \delta) = \mathrm{tr}\big(R^{-1}\bar{T}\delta C \delta^T \bar{T}^T\big), \qquad C = \bar{\theta}^{-1} M \bar{\theta}^{-T}, \tag{18.14}$$

and its kernel is $\{\delta \in \mathfrak{g} : \bar{T}\delta = 0\}$, the Lie algebra of $\mathrm{Stab}_G(\bar{T})$.

(iii) For $G = \mathrm{GL}(K, \mathbb{R})$ and $\bar{T}$ of rank r, $\mathrm{Stab}_G(\bar{T})$ has dimension $K(K - r)$, and the identifiable part of the dictionary has dimension Kr.

Proof. (i) A probability law on $\mathbb{R}^r$ has no nonzero translation symmetry: its characteristic function is nonzero near the origin, where translation by a would require $e^{\mathrm{i}\langle t, a\rangle} = 1$ and hence $a = 0$. Thus two translates of the same noise law agree exactly when their locations agree, and $\bar{T}\bar{\theta}'^{-1}\bar{y} = \bar{T}\bar{\theta}''^{-1}\bar{y}$ for every $\bar{y}$ means $\bar{T}\bar{\theta}'^{-1} = \bar{T}\bar{\theta}''^{-1}$, that is, $\bar{T}\bar{\theta}'^{-1}\bar{\theta}'' = \bar{T}$. The set $\mathrm{Stab}_G(\bar{T})$ is a subgroup because $\bar{T}g_1g_2 = \bar{T}g_2 = \bar{T}$. (ii) The mean $\bar{T}e^{-\delta}\bar{\theta}^{-1}\bar{y}$ has derivative $-\bar{T}\delta\bar{\theta}^{-1}\bar{y}$ at $\delta = 0$, and the Fisher information of a Gaussian mean with fixed covariance R is $\mathbb{E}_\rho[(\bar{T}\delta\bar{\theta}^{-1}\bar{y})^T R^{-1}(\bar{T}\delta\bar{\theta}^{-1}\bar{y})]$, which is eq. (18.14). Since C and R^{-1} are positive definite, $F(\delta, \delta) = 0$ exactly when $R^{-1/2}\bar{T}\delta C^{1/2} = 0$, that is, when $\bar{T}\delta = 0$. For such δ every power of δ is annihilated by $\bar{T}$, so $\bar{T}e^{t\delta} = \bar{T}$ for all t. (iii) $\bar{T}g = \bar{T}$ means that $g - I$ maps V into $\ker \bar{T}$, which has dimension $K - r$; such g form an open subset of an affine space of dimension $K(K - r)$ containing the identity. □

Messages never test the part of the dictionary that the code discards. Relative to the observable algebra of coded messages that part is gauge in the sense of the observable-algebra convention

of chapter 1: no message distinguishes two dictionaries that differ by an element of the stabilizer. The stated full identification uses shared referents with a positive-definite second moment. If only one referent vector were supplied, the conditional means would determine $\bar{T}\bar{\theta}^{-1}$ only on one vector. Noise rescales the Fisher information but does not change its kernel.

Remark 18.10 (Noise laws beyond the Gaussian). Gaussian noise is a computational simplification here. Let the grounded model be the location family $h = \bar{T}\bar{\theta}^{-1}\bar{y} + \varepsilon$ with ε independent of $\bar{y}$ and of a fixed law ν on $\mathbb{R}^r$. Parts (i) and (iii) of theorem 18.9 hold for every ν, because a probability law on $\mathbb{R}^r$ differs from each of its nonzero translates: its characteristic function φ is nonzero near the origin, and $e^{\mathrm{i}\langle t,a\rangle}\varphi(t) = \varphi(t)$ there forces $a = 0$. If ν has a continuously differentiable density f with finite Fisher information $J_f = \mathbb{E}[\nabla \log f(\varepsilon) \nabla \log f(\varepsilon)^T]$, positive definite, then part (ii) holds with R^{-1} replaced by J_f,

$$F(\delta, \delta) = \operatorname{tr}\big(J_f \bar{T}\delta C \delta^T \bar{T}^T\big), \tag{18.15}$$

with the same kernel, since the score of the perturbation is $\nabla \log f(\varepsilon)^T \bar{T}\delta\bar{\theta}^{-1}\bar{y}$ and ε is independent of $\bar{y}$. Corollaries 18.11 and 18.12 and the structural results of section 18.7 therefore hold for every such noise law. The Gaussian value $J_f = R^{-1}$ is the instance with the least Fisher information at a given covariance, by the Cramér–Rao inequality for location families. Codes whose messages are not a location family, such as codes into finite alphabets, are open problem 18.34.

18.4 Several modalities

Corollary 18.11 (Pooling modalities through one dictionary). Let modalities $c = 1, \dots, C$ have codes $\bar{T}^c$ and noises R_c, all read through the same dictionary $\bar{\theta}$. Assume the messages are conditionally independent given their referents and dictionary. Then the dictionary is identified up to $\bigcap_c \operatorname{Stab}_G(\bar{T}^c)$, and the Fisher information is $\sum_c F_c$ with kernel $\{\delta \in \mathfrak{g} : \bar{T}^c\delta = 0 \text{ for all } c\}$. For $G = \mathrm{GL}(K, \mathbb{R})$ the identifiable dimension is

$$K \dim\Big(\sum_c \operatorname{row} \bar{T}^c\Big) \le K \min\Big(K, \sum_c r_c\Big), \tag{18.16}$$

If $\sum_c r_c \le K$, equality in the bound holds exactly when the row spaces form a direct sum. If $\sum_c r_c \ge K$, equality holds exactly when their sum is V^*; independence of the row spaces is then unnecessary. The dictionary is fully identified exactly when the row spaces span V^*.

Proof. Identity of the conditional laws of all messages requires identity for each modality, which gives the intersection by theorem 18.9. Fisher information adds over conditionally independent messages, and a sum of positive-semidefinite forms vanishes exactly where each does. For $\mathrm{GL}(K, \mathbb{R})$, $\bar{T}^c\delta = 0$ for all c means that the range of δ lies in $\bigcap_c \ker \bar{T}^c$, whose dimension is K minus the dimension of the sum of the row spaces. □

Modalities that overlap, in the sense that their row spaces share directions, add less than their widths suggest; complementary modalities, such as speech and drawing when they convey different aspects, add fully. The corollary depends on the dictionary being shared across modalities. Without it each modality would have its own effective code to learn, and the modalities could not pool their evidence about the other agent.

Corollary 18.11 assumes that receiver i hears sender j in every modality. A receiver with several senders instead chooses, separately in each modality, whom to listen to. Let $\beta_{ij}^{(c)}$ be the probability that modality c of receiver i carries a message from j, and let $F_c^{(j)}$ denote the form eq. (18.14) for the code $\bar{T}_j^c$, the noise R_c and the dictionary $\bar{\theta}_{ij}$.

Corollary 18.12 (Routed modalities). In each exchange let a referent $\bar{y} \sim \rho$ be drawn, and for each modality c independently let a source label J_c be drawn with $\mathbb{P}(J_c = k) = \beta_{ik}^{(c)}$ and a message with law $\mathcal{N}(\bar{T}_k^c \bar{\theta}_{ik}^{-1} \bar{y}, R_c)$ given $J_c = k$ and $\bar{y}$. Consider the dictionary $\bar{\theta}_{ij} e^{\delta}$ of one edge $j \to i$.

(i) If the labels are observed, the Fisher information about δ per exchange is $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$.

(ii) If the labels are latent, the Fisher information is at most $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$ in the Loewner order.

(iii) With observed labels the information vanishes exactly on $\{\delta \in \mathfrak{g} : \bar{T}_j^c \delta = 0$ for every c with $\beta_{ij}^{(c)} > 0\}$.

Proof. (i) Given $\bar{y}$ and the labels, the messages of different modalities are independent, so the complete-data log likelihood is a sum over c. A term with $J_c = k \neq j$ does not depend on $\bar{\theta}_{ij}$ and contributes nothing. A term with $J_c = j$ contributes the information of theorem 18.9. Averaging the indicator of $J_c = j$ gives the weight $\beta_{ij}^{(c)}$. (ii) By the missing-information principle [103], the information of the observed data equals the complete-data information minus the expected information about δ carried by the conditional law of the labels given the messages, which is positive semidefinite. (iii) A sum of positive-semidefinite forms with positive weights vanishes exactly where every form vanishes, and theorem 18.9 identifies the kernel of each. □

A modality that never listens to j teaches i nothing about its dictionary for j, and the directions that the attending modalities' codes discard stay invisible. When the receiver must infer who was heard, it learns less than the routing-weighted sum.

Proposition 18.13 (A receiver's reading of a code is a dictionary modulo the stabilizer). Let $\bar{T}$ and S be $r \times K$ matrices of full row rank r. Then $\{\theta \in \mathrm{GL}(K, \mathbb{R}) : \bar{T}\theta^{-1} = S\}$ is a single left coset $\theta_0 \, \mathrm{Stab}_{\mathrm{GL}(K,\mathbb{R})}(\bar{T})$, and for $r < K$ it meets $\mathrm{GL}^+(K, \mathbb{R})$.

Proof. Complete the rows of $\bar{T}$ to a basis of the dual space and let P be the invertible matrix whose first r rows are those of $\bar{T}$; define Q likewise from S. With $E = (I_r \; 0)$, $\bar{T} = EP$ and $S = EQ$, so $\theta_0^{-1} = P^{-1}Q$ satisfies $\bar{T}\theta_0^{-1} = S$. For $r < K$, changing the sign of a completing row of Q changes the sign of $\det \theta_0$ without changing EQ. Finally $\bar{T}\theta^{-1} = \bar{T}\theta_0^{-1}$ exactly when $\bar{T}\theta_0^{-1}\theta = \bar{T}$, that is, when $\theta_0^{-1}\theta \in \mathrm{Stab}(\bar{T})$. □

A receiver that reads a sender's code through an arbitrary full-rank map S therefore holds exactly the identifiable part of a dictionary, a point of the rK-dimensional quotient of theorem 18.9. The proposition also applies to the stacked code $(\bar{T}^1; \ldots; \bar{T}^C)$ and the stacked readings $(S_1; \ldots; S_C)$. When both stacks have full row rank $\sum_c r_c \leq K$, there is one dictionary $\bar{\theta}$ with $S_c = \bar{T}^c \bar{\theta}^{-1}$ for every c, unique modulo $\bigcap_c \mathrm{Stab}(\bar{T}^c)$: readings learned independently for each modality are then automatically readings of one shared dictionary. For arbitrary stacks, equal ranks alone do not establish compatibility.

Proposition 18.14 (Compatibility of stacked readings). Let T_*, S_* be $m \times K$ matrices, with no full-rank assumption. There is a dictionary $\theta \in \mathrm{GL}(K, \mathbb{R})$ satisfying $T_*\theta^{-1} = S_*$ exactly when

$$\ker(T_*^T) = \ker(S_*^T). \tag{18.17}$$

When this holds, the solution set is $\theta_0 \, \mathrm{Stab}_{\mathrm{GL}(K,\mathbb{R})}(T_*)$ for any one solution θ_0. If the common rank is less than K, the solution set meets $\mathrm{GL}^+(K, \mathbb{R})$.

Proof. Right multiplication by an invertible matrix preserves every linear relation among rows, proving necessity. Conversely, equality of the kernels makes the map $a^T T_* \mapsto a^T S_*$ a well-defined linear isomorphism between the row spaces. Extend bases of these row spaces to bases of $(\mathbb{R}^K)^*$. Extending that isomorphism to the full dual space gives right multiplication by an invertible matrix A, with $T_*A = S_*$. Put $\theta = A^{-1}$. The coset statement follows as in proposition 18.13. When the common rank is less than K, the extension can reverse one unused basis vector, changing the determinant sign without changing the rows of S_*. □

For example, $T_1 = T_2 = (1, 0)$, $S_1 = (1, 0)$, and $S_2 = (2, 0)$ give two rank-one stacks but no common dictionary: the relation $T_1 - T_2 = 0$ fails for the readings. Sharing also constrains the readings when one dictionary must serve routing and content roles, as discussed in remark 18.15.

Remark 18.15 (Attention heads as modalities). The separation of a sending map, a receiver reading, a decoder, and a routing row provides a structural comparison with multi-head attention [139]. A head reads a finite-width code and routes sources, but the usual query, key, value, and output maps are independently parameterized. Such an architecture is not automatically the Bayesian message model above. Under the full-row-rank hypothesis of proposition 18.13, a reading map can be expressed as a dictionary modulo its stabilizer; several readings share a dictionary precisely under eq. (18.17). Requiring that same dictionary for source selection and content interpretation, and deriving the decoder from the receiver prior as in eq. (18.8), imposes additional restrictions. A learned output matrix alone does not establish those restrictions.

18.5 Curvature as dissonance

Let the communication graph Γ have a directed edge $j \to i$ when i receives from j, with dictionaries $\bar{\theta}_{ij}$. For a directed closed walk $\gamma = (i_0 \to i_1 \to \cdots \to i_k = i_0)$ define its holonomy $H_\gamma = \bar{\theta}_{i_k i_{k-1}} \cdots \bar{\theta}_{i_1 i_0}$, the composite of the receivers' dictionaries along the walk. Reference belief means $(\mu_i)_i$ are in consensus when

$$\mu_i = \bar{\theta}_{ij} \mu_j \qquad \text{for every edge } j \to i, \tag{18.18}$$

so that every receiver, translating each sender's belief through its dictionary, finds agreement.

Theorem 18.16 (Consensus and holonomy invariance). Let Γ be strongly connected and fix i_0. Suppose the dictionaries act as bimeasurable bijections on a standard Borel content space. The map $(q_i)_i \mapsto q_{i_0}$ is a bijection from transported consensus laws, satisfying $q_i = (\bar{\theta}_{ij})_\# q_j$ on every edge, onto laws invariant under every directed closed-walk holonomy based at i_0. In the

linear realization, the map $(\mu_i)_i \mapsto \mu_{i_0}$ is a linear bijection from the solutions of eq. (18.18) onto $\bigcap_\gamma \mathrm{Fix}(H_\gamma)$.

Proof. Composing the edge relations around a closed walk proves that a consensus base law is invariant under its holonomy. Conversely, let q_0 have that invariance. For each vertex i, choose a directed path p from i_0 to i and set $q_i = \Pi(p)_\# q_0$, where $\Pi(p)$ is the ordered product of the dictionaries. If p' is another such path, strong connectivity supplies a directed return path s from i to i_0. Both $\Pi(s)\Pi(p)$ and $\Pi(s)\Pi(p')$ are loop holonomies and preserve q_0. Pushforward by the bimeasurable inverse $\Pi(s)^{-1}$ gives $\Pi(p)_\# q_0 = \Pi(p')_\# q_0$. The definition is independent of the path. Appending an edge to a path gives the required edge relation, and the base law determines the entire family. For vectors, replace pushforward by matrix multiplication in the same argument. This gives path-independent values $\mu_i = \Pi(p)v$ for every common fixed vector v, and both directions are linear. □

The theorem uses the invertibility of the dictionaries and the directed return paths. A two-cycle is included among its loops and constrains consensus alongside longer loops. If the matrix holonomies have no common nonzero fixed vector, the only consensus mean is zero. This does not exclude a nontrivial invariant law: a centered isotropic Gaussian, for example, is invariant under rotation holonomies. For Gaussian laws, consensus constrains both mean and covariance; for general laws, it constrains every feature moved by the holonomies.

Corollary 18.17 (Frustration). For positive weights w_{ij}, the consensus energy $E(\mu) = \sum_{j\to i} w_{ij}\|\mu_i - \bar\theta_{ij}\mu_j\|^2$ is a positive-semidefinite quadratic form whose kernel is the solution set of eq. (18.18). Its least eigenvalue on configurations of unit norm is positive exactly when the holonomies have no common nonzero fixed vector.

Proof. $E(\mu) = \|B\mu\|^2$ for the weighted incidence operator B with blocks $\sqrt{w_{ij}}(I, -\bar\theta_{ij})$, so $E = \mu^T B^T B\mu$ and its kernel is $\ker B$. The least eigenvalue of $B^T B$ is positive exactly when $\ker B = 0$, and theorem 18.16 identifies $\ker B$. □

For orthogonal links on undirected graphs with the inverse convention this is the connection Laplacian, and its least eigenvalue as a measure of frustration was studied by Singer and Wu [131] and Bandeira et al. [8]. The directed version above admits independent opposite dictionaries.

Definition 18.18 (Dissonance of a loop). The dissonance of a directed closed walk γ is the conjugacy class of H_γ. A declared elementary cycle, or plaquette, is consonant when its holonomy is the identity and dissonant otherwise. A quantitative label is any conjugation-invariant defect of H_γ, for example $\|H_\gamma - I\|_F$ for orthogonal dictionaries; for general dictionaries the covariance-dressed loop defect of proposition 2.13 supplies an invariant measure.

Remark 18.19 (A law comparison and a dressed link comparison). The covariance-dressed defect

$$Q(H;\Sigma) = \tfrac{1}{2}\|\Sigma^{-1/2}(I-H)\Sigma^{1/2}\|_F^2, \qquad \Sigma > 0, \tag{18.19}$$

vanishes only at $H = I$ and has a positive-definite tangent form on every matrix Lie algebra. Its nonlinear use does not introduce law-stabilizer directions. In contrast, the law-comparison penalty $D_{\mathrm{KL}}(q\|H_\# q)$ vanishes whenever $H_\# q = q$ and can have a larger null space. Its local information

geometry requires a declared regular family of laws. For a quarter-turn H and $q = \mathcal{N}(0, I_2)$, the law-comparison penalty is zero while $Q(H; I_2) = 2$.

Observability depends on the record experiment. Comparing coded means tests the weaker condition $\bar{T}_j^c(\mu_j - \bar{\theta}_{ij}^{-1}\mu_i) = 0$ on each edge and modality. Grounded messages with a positive-definite referent second moment instead identify the whole effective code, modulo its stabilizer, as in theorem 18.9. The tangent statement proposition 18.24 concerns this latter experiment. A perturbation can fix one operating content while changing the code on other referents, so invisibility at one content does not imply invisibility to grounded messages. Separately, the law-comparison cost $D_{\mathrm{KL}}(q\|H_{\#}q)$ is blind to any holonomy stabilizing q. The covariance-dressed link defect eq. (18.19) is an observable of the link and covariance together and vanishes only at identity. These are different comparisons.

18.6 Bandwidth and source rows

Proposition 18.20 (Coded messages give Gibbs rows at temperature one over the message count). Let receiver i consider finitely many candidate sources j with prior $\pi_j > 0$, each modeled as emitting independent messages with density ν_j on the message space. Let $h_1, h_2, \dots$ be independent with density ν, and suppose $\mathbb{E}_\nu|\log \nu_j| < \infty$ for every j. Then, almost surely,

$$\frac{1}{n}\log\frac{\mathbb{P}(J = j \mid h_{1:n})}{\mathbb{P}(J = k \mid h_{1:n})} \longrightarrow \mathbb{E}_\nu[\log \nu_j - \log \nu_k]. \tag{18.20}$$

If both $D_{\mathrm{KL}}(\nu\|\nu_j)$ and $D_{\mathrm{KL}}(\nu\|\nu_k)$ are finite, this limit is $D_{\mathrm{KL}}(\nu\|\nu_k) - D_{\mathrm{KL}}(\nu\|\nu_j)$. Integrability of $\log \nu$ under ν, in addition to the candidate-log assumptions, is a sufficient condition for that representation for every pair.

Proof. By Bayes' rule the left side is $\frac{1}{n}\log(\pi_j/\pi_k) + \frac{1}{n}\sum_{t\le n}[\log \nu_j(h_t) - \log \nu_k(h_t)]$. The strong law of large numbers sends the average to the integrable expected log-likelihood ratio in eq. (18.20). If both relative entropies are finite, subtract their defining logarithmic integrals; the common $\log \nu$ term cancels. Without that finiteness, subtracting two infinite relative entropies is undefined. □

The finite-sample posterior is exactly the Gibbs row $\pi_j \exp(-nC_{j,n})/Z_n$, with empirical cost $C_{j,n} = -n^{-1}\sum_{t=1}^n \log \nu_j(h_t)$. When all candidate relative entropies are finite, its limiting relative exponential rates are those of $\pi_j e^{-D_{\mathrm{KL}}(\nu\|\nu_j)/T}/Z$ at temperature $T = 1/n$, the posterior concentration of a possibly misspecified model [13]. The costs are divergences between message laws, which are code-visible divergences: by data processing they are at most the corresponding divergences between full beliefs, with equality exactly when the code is sufficient. When a common noiseless injective code is used, its message-law divergences equal the full-law divergences by invariance under a measurable injection. This identifies a cost, not a population dynamics: the normalized global model, the per-agent model, and the reciprocal potential remain distinct objectives. Finite message count and selective coding control different aspects of the source posterior.

18.7 Learning the dictionaries

Dictionary learning requires a likelihood and a prior in addition to the communication maps. Theorem 18.9 says what messages can teach: a coset of the stabilizer of the code. This section asks what a prior that rewards agreement around loops adds to that, and answers the question exactly in the tangent model, the linearization at the identity dictionaries. Centering the prior at identity is a declared alignment hypothesis: before observing messages, the receiver expects the sender to use its own reference coordinates.

The tangent model is a computational simplification in two respects: it linearizes at the identity dictionaries, and it takes Gaussian noise and a Gaussian anchor. Its conclusions split accordingly. The invisible directions, the unreachable directions and their sheaf count, the split of the loop space into filled and detectable directions as a dimension count, and the exact identification results propositions 18.28 and 18.29 depend only on the identifiability kernel and on the loops; by remark 18.10 they hold for every location family of noise and every prior. The limiting law as a conditioned Gaussian, the shrinkage and bias formulas, and the variances of remark 18.30 are features of the Gaussian family.

18.7.1 The tangent model

Definition 18.21 (Tangent learning model). Let Γ be a finite communication graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, and let the edge $e : j \to i$ carry receiver i's generator $\omega_e = \omega_{ij} \in \mathfrak{g} \subseteq \mathfrak{gl}(K, \mathbb{R})$, so that $\omega = (\omega_e)_e \in X = \mathfrak{g}^{\mathcal{E}}$. Write $\bar{T}_e$ for the stacked code of the modalities with $\beta_{ij}^{(c)} > 0$ in corollary 18.12. Let $\mathcal{L}$ be a finite set of directed closed walks of Γ that contains every two-cycle, and define the linearized holonomy $B : X \to \mathfrak{g}^{\mathcal{L}}$ by $(B\omega)_\gamma = \sum_{e \in \gamma} \omega_e$, an edge being counted as often as γ traverses it. The tangent learning model consists of the prior law with density proportional to

$$\exp\Big(-\tfrac{1}{2}\omega^T \Pi_0 \omega - \tfrac{\lambda}{2} \sum_{\gamma \in \mathcal{L}} \|(B\omega)_\gamma\|^2_{W_\gamma}\Big), \tag{18.21}$$

where Π_0 is positive definite on X, $\lambda \geq 0$, and each W_γ is a positive-definite quadratic form on $\mathfrak{g}$, with $W = \bigoplus_\gamma W_\gamma$; and of the messages that i attributes to j, with observed labels as in corollary 18.12(i), each produced from a referent $\bar{y} \sim \rho$ with finite second moment M positive definite as

$$h = \bar{T}_e^c (I - \omega_e)\bar{y} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_c). \tag{18.22}$$

The prior is Gaussian with precision $\Lambda_0 = \Pi_0 + \lambda B^T W B$, which is positive definite, so the prior is normalized for every $\lambda \geq 0$. The message model is the first-order form of eq. (18.13) at $\bar{\theta}_e = e^{\omega_e}$, and $(B\omega)_\gamma$ is the first-order form of $\log H_\gamma$. By the Baker–Campbell–Hausdorff formula both errors are of second order in ω. Inside the tangent model the likelihood is Gaussian and linear in ω. The recognition law, here the exact posterior, is therefore Gaussian after any number of messages and is updated one message at a time by the information filter

$$\Lambda \leftarrow \Lambda + J_t^T R_c^{-1} J_t, \qquad \eta \leftarrow \eta + J_t^T R_c^{-1}\big(h_t - \bar{T}_e^c \bar{y}_t\big), \qquad J_t\omega = -\bar{T}_e^c \omega_e \bar{y}_t, \tag{18.23}$$

started at $(\Lambda_0, 0)$, with mean $\Lambda^{-1}\eta$ and covariance Λ^{-1}.

A change of the common reference coordinates by $h \in G$ acts by $\omega_e \mapsto h^{-1}\omega_e h$ under reference consistency, and then $(B\omega)_\gamma \mapsto h^{-1}(B\omega)_\gamma h$. The loop term is invariant when each W_γ is. The Frobenius norm is invariant under orthogonal h. For $\mathrm{GL}(K,\mathbb{R})$ the dressed norm $\|X\|_\Sigma^2 = \mathrm{tr}(X^T\Sigma^{-1}X\Sigma)$, with Σ the reference covariance of the belief of the agent at which γ is based, satisfies $\|h^{-1}Xh\|^2_{h^{-1}\Sigma h^{-T}} = \|X\|^2_\Sigma$. A law is needed here because $\mathfrak{gl}(K,\mathbb{R})$ carries no conjugation-invariant positive-definite form for $K \geq 2$: conjugation by $\mathrm{diag}(t,1,\dots,1)$ multiplies E_{12} by t. The anchor Π_0 need not be invariant, since it expresses the receiver's expectation in its own reference coordinates.

For the statements below, let $N_e = \{\delta \in \mathfrak{g} : \bar{T}_e\delta = 0\}$, the Lie algebra of the stabilizer of the stacked code, and let $N = \bigoplus_e N_e \subseteq X$, the invisible directions. Let Z be a matrix whose columns form an orthonormal basis of N for the Frobenius inner product on X, and Y one for $N^\perp$. Put

$$S = Z^T\Lambda_0 Z, \qquad \Pi_N = ZS^{-1}Z^T\Lambda_0, \qquad L_N = Z^TB^TWBZ, \tag{18.24}$$

so that Π_N is the Λ_0-orthogonal projection of X onto N and L_N is the loop operator on the invisible directions. Let $\bar{F}_e = \sum_c \beta_{ij}^{(c)} F_c^{(j)}$ be the information per exchange of corollary 18.12(i), evaluated at $\bar{\theta}_e = I$, where $C = M$.

18.7.2 Concentration and selection

Theorem 18.22 (Learning with a loop prior). In the tangent learning model let the truth be $\omega^\star \in X$, let each edge e receive $n_e = \lceil \kappa_e n \rceil$ independent exchanges with $\kappa_e > 0$, and write $z^\star = Z^T\omega^\star$.

(i) For every set of messages the likelihood is constant on each coset $\omega + N$, so the density of the recognition law relative to the prior depends on ω only through $Y^T\omega$.

(ii) Almost surely, as $n \to \infty$, the mean of the recognition law converges to

$$\omega_\infty = (I - \Pi_N)\omega^\star, \tag{18.25}$$

its covariance is $ZS^{-1}Z^T + O(1/n)$, and n times its covariance in the coordinates $Y^T\omega$ converges to $(Y^TF_\kappa Y)^{-1}$ with $F_\kappa = \bigoplus_e \kappa_e \bar{F}_e$. Equivalently, the recognition law converges to the prior conditioned on the coset $\omega^\star + N$. Its precision grows at rate n in the identified directions, and within the coset it is the loop prior restricted to the coset.

(iii) Let $\Pi_0 = \sigma^{-2}I$. In the coordinates $Z^T\omega$ the limiting covariance is $\sigma^2(I + \lambda\sigma^2 L_N)^{-1}$. It equals the prior covariance on $\ker L_N = Z^T(N \cap \ker B)$, is strictly smaller on the orthogonal complement, and tends to zero there as $\lambda \to \infty$. The rank of L_N is $\dim B(N) = \dim N - \dim(N \cap \ker B)$.

(iv) If in addition the truth is flat, $B\omega^\star = 0$, then

$$Z^T(\omega_\infty - \omega^\star) = -(I + \lambda\sigma^2 L_N)^{-1} z^\star. \tag{18.26}$$

The loop term recovers the invisible part of the truth along an eigenvector of L_N with eigenvalue $\mu > 0$ up to the factor $\lambda\sigma^2\mu/(1 + \lambda\sigma^2\mu)$, and as $\lambda \to \infty$ it recovers all of it except the component in $\ker L_N$, which stays at the prior mean zero.

(v) For a general truth, as $\lambda \to \infty$,

$$Z^T(\omega_\infty - \omega^\star) \longrightarrow -P_{\ker} z^\star - L_N^+ Z^T B^T W B \omega^\star, \tag{18.27}$$

with $P_{\ker}$ the orthogonal projection onto $\ker L_N$ and L_N^+ the pseudoinverse.

Proof. (i) By eq. (18.22) the law of a message on edge e depends on ω only through $\bar{T}_e^c \omega_e$ for attending modalities c, which does not change when ω_e changes by an element of N_e. Bayes' rule gives the second statement.

(ii) After the messages the precision is $\Lambda_n = \Lambda_0 + nF_n$ with $nF_n = \sum_t J_t^T R_c^{-1} J_t$, and since $h_t - \bar{T}_e^c \bar{y}_t = J_t \omega^\star + \varepsilon_t$, the information vector is $\eta_n = nF_n \omega^\star + \xi_n$ with $\xi_n = \sum_t J_t^T R_c^{-1} \varepsilon_t$. Every J_t vanishes on N, so $F_n = Y F_{Y,n} Y^T$ and $Z^T \xi_n = 0$. By the strong law of large numbers $F_{Y,n} \to Y^T F_\kappa Y$ almost surely, and this limit is positive definite because the kernel of F_κ is N by corollary 18.12(iii). In the basis (Y, Z) the matrix Λ_n has blocks $A_n = Y^T \Lambda_0 Y + nF_{Y,n}$, $D = Y^T \Lambda_0 Z$ and S, and block inversion gives

$$\Lambda_n^{-1} = Z S^{-1} Z^T + (Y - Z S^{-1} D^T)(A_n - D S^{-1} D^T)^{-1} (Y - Z S^{-1} D^T)^T, \tag{18.28}$$

with $n(A_n - D S^{-1} D^T)^{-1} \to (Y^T F_\kappa Y)^{-1}$. This proves the statements about the covariance, since $Y^T \Lambda_n^{-1} Y = (A_n - D S^{-1} D^T)^{-1}$. For the mean, $nF_n \omega^\star = \Lambda_n \omega^\star - \Lambda_0 \omega^\star$ gives

$$\Lambda_n^{-1} \eta_n = \omega^\star - \Lambda_n^{-1} \Lambda_0 \omega^\star + \Lambda_n^{-1} \xi_n. \tag{18.29}$$

The second term converges to $Z S^{-1} Z^T \Lambda_0 \omega^\star = \Pi_N \omega^\star$. Because $Z^T \xi_n = 0$, the third term equals $(Y - Z S^{-1} D^T)(A_n - D S^{-1} D^T)^{-1} Y^T \xi_n$, which tends to zero since $\xi_n / n \to 0$ almost surely by the strong law. Finally, the Gaussian law with precision Λ_0 conditioned on the affine set $\omega^\star + N$, in the coordinate z of $\omega^\star + Zz$, has precision S and the mean that minimizes $\tfrac{1}{2}(\omega^\star + Zz)^T \Lambda_0 (\omega^\star + Zz)$, which is $z = -S^{-1} Z^T \Lambda_0 \omega^\star$. Its mean is ω_∞ and its covariance is $Z S^{-1} Z^T$.

(iii) Here $S = \sigma^{-2} I + \lambda L_N$, and the eigenvalues of S^{-1} are $\sigma^2/(1 + \lambda \sigma^2 \mu)$ for the eigenvalues $\mu \geq 0$ of L_N. Since W is positive definite, $L_N z = 0$ exactly when $BZz = 0$, so $\ker L_N = Z^T(N \cap \ker B)$ and the rank of L_N is that of BZ.

(iv) From (ii) and $Z^T \Lambda_0 \omega^\star = \sigma^{-2} z^\star + \lambda Z^T B^T W B \omega^\star$,

$$Z^T(\omega_\infty - \omega^\star) = -S^{-1}\big(\sigma^{-2} z^\star + \lambda Z^T B^T W B \omega^\star\big), \tag{18.30}$$

which is eq. (18.26) when $B\omega^\star = 0$.

(v) The vector $Z^T B^T W B \omega^\star$ lies in the range of $(W^{1/2} B Z)^T$, which is the range of $L_N = (W^{1/2} B Z)^T (W^{1/2} B Z)$. As $\lambda \to \infty$, $\lambda S^{-1} \to L_N^+$ and $S^{-1} \to 0$ on the range of L_N, while $\sigma^{-2} S^{-1} = I$ on $\ker L_N$. Applying this to eq. (18.30) gives eq. (18.27). □

With an isotropic anchor and no loop term, the recognition law on the invisible coordinates stays at its anchor marginal. For a general anchor, messages can condition these coordinates through prior correlations, while leaving their conditional law given the identified coordinates unchanged. The loop term changes the geometry in which the identified part of the truth is completed, from the anchor to the anchor plus the loop energy, and eq. (18.25) is the resulting projection.

18.7.3 Filling and detection

Definition 18.23 (Detectable dissonance). The detectable dissonance of a configuration ω is

$$\mathcal{D}(\omega) = \min_{\nu \in N} \|B(\omega + \nu)\|_W^2, \tag{18.31}$$

the least linearized loop energy among the configurations that no message distinguishes from ω.

Proposition 18.24 (Filling and detection are complementary). In the tangent learning model let Q be the W-orthogonal projection of $\mathfrak{g}^{\mathcal{L}}$ onto the W-orthogonal complement of $B(N)$.

(i) $\mathcal{D}(\omega) = \|QB\omega\|_W^2$. In particular $\mathcal{D}$ is constant on cosets of N, so messages determine it.

(ii) With $\Pi_0 = \sigma^{-2} I$, as $\lambda \to \infty$ the limit mean ω_∞ satisfies $B\omega_\infty \to QB\omega^\star$, so its loop energy tends to $\mathcal{D}(\omega^\star)$.

(iii) The loop space $B(X)$ is the W-orthogonal sum of $B(N)$, of dimension $\dim N - \dim(N \cap \ker B)$, and of $B(X) \cap B(N)^{\perp_W}$, of dimension $\operatorname{rank} B - \dim B(N)$. The loop energy splits as $\|B\omega\|_W^2 = \|(I-Q)B\omega\|_W^2 + \|QB\omega\|_W^2$. The first term is the only one that depends on the invisible coordinates, and a dissonance in $B(N)$ is not detectable; the second is a function of the identified coordinates alone.

Proof. (i) Since $B(\omega + Zz) = B\omega + BZz$, the minimum over z leaves the W-orthogonal residual of $B\omega$ from the range of BZ, which is $B(N)$; this residual is $QB\omega$. For $\nu \in N$, $QB\nu = 0$. (ii) By eq. (18.27), $\omega_\infty - \omega^\star$ tends to $-Z(P_{\ker} z^\star + L_N^+ Z^T B^T W B \omega^\star)$. Since $BZP_{\ker} = 0$, $B\omega_\infty$ tends to $(I - BZL_N^+ (BZ)^T W) B\omega^\star$, and $BZ((BZ)^T W BZ)^+ (BZ)^T W$ is the W-orthogonal projection onto the range of BZ, which is $I - Q$. (iii) The dimension of $B(N)$ is the rank of BZ, computed in theorem 18.22(iii). The energy splits because Q is a W-orthogonal projection, and $QB\omega$ does not depend on the component of ω in N by (i). □

The proposition states the cost of the loop prior. Every loop direction in which agreement around loops completes a dictionary is a direction in which the community assumes an agreement it cannot check. If the truth is dissonant there, a strong loop prior can select a flat representative of the same message laws, as quantified by the bias term of eq. (18.27). Dissonance is detectable only in the loop directions that the invisible directions cannot reach.

18.7.4 Which directions loops cannot reach

Proposition 18.25 (The unreachable directions). Let Γ be connected, and suppose that the chains of the walks in $\mathcal{L}$ span the cycle space of Γ, regarded as an undirected multigraph in which the edge $e : j \to i$ is oriented from j to i. Then

$$\ker B = \big\{ (a_i - a_j)_{e : j \to i} : a \in \mathfrak{g}^{\mathcal{V}} \big\}, \tag{18.32}$$

and $N \cap \ker B$ consists of the configurations in eq. (18.32) with $\bar{T}_e(a_i - a_j) = 0$ on every edge $e : j \to i$. For $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$,

$$\begin{aligned} \dim(N \cap \ker B) &= K\big(\dim H^0 - K\big), \\ H^0 &= \big\{ u \in (\mathbb{R}^K)^{\mathcal{V}} : \bar{T}_e(u_i - u_j) = 0 \text{ for every edge } e : j \to i \big\}. \end{aligned} \tag{18.33}$$

Proof. The value $(B\omega)_\gamma$ is the pairing of ω, read as a $\mathfrak{g}$-valued cochain on edges, with the chain of γ, and that chain is a cycle because a closed walk leaves each vertex as often as it enters it. If these chains span the cycle space, $\omega \in \ker B$ exactly when ω vanishes on every cycle. Choose a spanning tree and a root, let a_v be the sum of ω along the tree path from the root to v, each edge taken with the sign of its orientation, and put a equal to zero at the root. Then $\omega_e - (a_i - a_j)$ vanishes on tree edges and, on any other edge, equals the value of ω on its fundamental cycle, which is zero. Conversely every configuration of the form eq. (18.32) vanishes on cycles. The condition for membership in N is the definition of N_e. For $\mathfrak{gl}(K, \mathbb{R})$ the condition $\bar{T}_e(a_i - a_j) = 0$ holds column by column, so $a \mapsto (a_i - a_j)_e$ maps the families whose K columns lie in H^0, a space of dimension $K \dim H^0$, onto $N \cap \ker B$. Its kernel consists of the constant families, of dimension K^2, because Γ is connected. □

The configurations in eq. (18.32) are the infinitesimal pure-gauge configurations, what the dictionaries would become if each agent's reference coordinates were moved by e^{a_v}. The space $N \cap \ker B$ consists of the re-readings of this kind that no attending code registers. They create no holonomy, so no loop term reaches them, and no message reveals them. The space H^0 is the space of global sections of a cellular sheaf on Γ whose stalk at an agent is $\mathbb{R}^K$, whose stalk at an edge is the message space, and whose two restriction maps at an edge are both the attending code; it is a sheaf of the kind that Hansen and Ghrist [72] use for discourse, arising here from identifiability.

Corollary 18.26 (Complete exchange). Let every pair of agents exchange in both directions, and let $\mathcal{L}$ contain every two-cycle and every directed triangle. Then the hypothesis of proposition 18.25 holds and

$$H^0 = \big\{u : u_i - u_j \in \ker \bar{T}_{j\to i} \cap \ker \bar{T}_{i\to j} \text{ for every pair } \{i, j\}\big\}, \tag{18.34}$$

where $\bar{T}_{j\to i}$ is the code on the edge $j \to i$. If the pairs with $\ker \bar{T}_{j\to i} \cap \ker \bar{T}_{i\to j} = 0$ connect all agents, then H^0 consists of constant sections, $N \cap \ker B = 0$, and for a flat truth in the Gaussian tangent model the recognition law recovers every dictionary in the limits $n \to \infty$ and then $\lambda \to \infty$.

Proof. Fix an agent 0. The edges $0 \to i$ form a spanning tree. The fundamental cycle of $i \to 0$ is the two-cycle through 0 and i, and that of $j \to i$ with $i, j \neq 0$ is the directed triangle $0 \to j \to i \to 0$ minus the two-cycle through 0 and i. The two edges between i and j give $\bar{T}_{j\to i}(u_i - u_j) = 0$ and $\bar{T}_{i\to j}(u_j - u_i) = 0$. If the pairs with trivial intersection connect the agents, u is constant along each of them, hence constant. The last statement is theorem 18.22(iv). □

The sufficient condition is not necessary. In $K = 3$, suppose the pairwise intersections of the two code kernels are three linearly independent lines. A section in H^0 has its three successive vertex differences on those lines, and their sum is zero. Linear independence forces every difference to vanish, although no pairwise kernel intersection is zero. Thus triangles can remove invisible flat directions that remain for each pair separately.

Proposition 18.27 (Reciprocal pair). Let two agents exchange in both directions with $\mathcal{L}$ the two-cycle, W the Frobenius norm and $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$. Write $\bar{T}_1$ for the code on $1 \to 0$ and $\bar{T}_0$ for the code on $0 \to 1$, $\mathcal{K}_a = \ker \bar{T}_a$, $\mathcal{R}_a = \mathcal{K}_a^\perp$ for the row space of $\bar{T}_a$, and $P_\cap$ for the orthogonal projection

onto $\mathcal{R}_0 \cap \mathcal{R}_1$. Then

$$\begin{aligned} \dim B(N) &= K \dim(\mathcal{K}_0 + \mathcal{K}_1), \qquad \dim(N \cap \ker B) = K \dim(\mathcal{K}_0 \cap \mathcal{K}_1), \\ \operatorname{rank} B - \dim B(N) &= K \dim(\mathcal{R}_0 \cap \mathcal{R}_1), \end{aligned} \tag{18.35}$$

and $\mathcal{D}(\omega) = \|P_\cap(\omega_{01} + \omega_{10})\|_F^2$.

Proof. N consists of the pairs (ν_{01}, ν_{10}) with the range of ν_{01} in $\mathcal{K}_1$ and the range of ν_{10} in $\mathcal{K}_0$, and $B\nu = \nu_{01} + \nu_{10}$. Hence $B(N)$ is the set of matrices with range in $\mathcal{K}_0 + \mathcal{K}_1$, of dimension $K \dim(\mathcal{K}_0 + \mathcal{K}_1)$. Its Frobenius-orthogonal complement in $B(X) = \mathfrak{gl}(K, \mathbb{R})$ is the set of matrices with range in $(\mathcal{K}_0 + \mathcal{K}_1)^\perp = \mathcal{R}_0 \cap \mathcal{R}_1$, and the orthogonal projection onto it is $X \mapsto P_\cap X$. Thus $Q = P_\cap$ and proposition 18.24(i) gives $\mathcal{D}$. The space $N \cap \ker B$ consists of the pairs with $\nu_{01} = -\nu_{10}$ and range in $\mathcal{K}_0 \cap \mathcal{K}_1$. □

A pair detects its mutual misunderstanding only in what both can say, repairs its dictionaries through reciprocity in what at least one of them can say, and leaves what neither can say at the prior. When the two row spaces are independent and together have dimension K, the invisible flat-truth coordinates are recovered as the loop weight grows, while every tangent loop discrepancy can also be absorbed by an invisible change. Recovery under the flat-truth hypothesis therefore differs from detecting whether that hypothesis holds.

18.7.5 Without linearization

The next results use unrestricted group-valued dictionaries and are exact without linearization. They identify the configurations consistent with grounded messages and exactly flat loops. Restricting dictionaries to a single-exponential image requires intersecting the stated solution sets with that image. The following two statements hold for the grounded model eq. (18.13) itself, in the constraint form of the loop term.

Proposition 18.28 (Exact reciprocal identification). Let $G = \mathrm{GL}(K, \mathbb{R})$ and let the true dictionaries of two agents be reciprocal, $\bar\theta^\star_{10} = (\bar\theta^\star_{01})^{-1}$. The pairs $(\bar\theta_{01}, \bar\theta_{10})$ that reproduce the message laws of both edges for every referent and satisfy $\bar\theta_{01}\bar\theta_{10} = I$ are exactly

$$\bar\theta_{01} = \bar\theta^\star_{01} s, \qquad \bar\theta_{10} = s^{-1}\bar\theta^\star_{10}, \qquad s \in \mathrm{Stab}(\bar T_1) \cap \mathrm{Stab}(\bar T_0 \bar\theta^\star_{01}). \tag{18.36}$$

The pair is determined uniquely exactly when the stacked map $(\bar T_1; \bar T_0\bar\theta^\star_{01})$ is injective; otherwise the solutions form a set of dimension $K \dim\ker(\bar T_1; \bar T_0\bar\theta^\star_{01})$.

Proof. By theorem 18.9(i), the edge $1 \to 0$ reproduces its law exactly when $\bar\theta_{01} = \bar\theta^\star_{01} s$ with $s \in \mathrm{Stab}(\bar T_1)$. Reciprocity forces $\bar\theta_{10} = s^{-1}\bar\theta^\star_{10}$, and the edge $0 \to 1$ reproduces its law exactly when $(\bar\theta^\star_{10})^{-1} s^{-1} \bar\theta^\star_{10} = \bar\theta^\star_{01} s^{-1} (\bar\theta^\star_{01})^{-1} \in \mathrm{Stab}(\bar T_0)$, that is, when $\bar T_0 \bar\theta^\star_{01} s^{-1} = \bar T_0 \bar\theta^\star_{01}$. The intersection of the two stabilizers is the stabilizer of the stacked map, the set of s with $s - I$ mapping $\mathbb{R}^K$ into its kernel, which is open in an affine space of dimension K times the dimension of that kernel. □

The map $\bar T_0\bar\theta^\star_{01}$ is agent 0's code expressed in agent 1's reference coordinates, so the pair is identified exactly when, between them, the two agents can say everything.

Proposition 18.29 (Exact flat identification). Let $G = \mathrm{GL}(K, \mathbb{R})$ and Γ be connected, and suppose that every configuration whose holonomies on $\mathcal{L}$ are the identity is pure gauge, $\bar\theta_{ij} = k_i k_j^{-1}$ for some $k \in G^{\mathcal{V}}$. Let the truth be pure gauge, $\bar\theta^\star_{ij} = g_i g_j^{-1}$, and put $\bar T'_e = \bar T_e g_j$ for $e : j \to i$. The configurations that reproduce every message law and have identity holonomy on $\mathcal{L}$ are exactly

$$\bar\theta_{ij} = g_i u_i u_j^{-1} g_j^{-1}, \qquad u \in G^{\mathcal{V}}, \qquad \bar T'_e(u_i - u_j) = 0 \text{ for every edge } e : j \to i, \tag{18.37}$$

and two families u, u' give the same configuration exactly when $u'_v = u_v c$ for all v and one $c \in G$. The truth is the only such configuration exactly when the space H^0 of eq. (18.33), formed with the codes $\bar T'_e$, consists of constant sections. The hypothesis holds for the complete exchange with every two-cycle and every directed triangle.

Proof. By hypothesis $\bar\theta_{ij} = k_i k_j^{-1}$; write $k_v = g_v u_v$. By theorem 18.9(i), the edge e reproduces its law exactly when $(\bar\theta^\star_{ij})^{-1}\bar\theta_{ij} = g_j u_i u_j^{-1} g_j^{-1} \in \mathrm{Stab}(\bar T_e)$, that is, when $\bar T'_e u_i u_j^{-1} = \bar T'_e$, that is, when $\bar T'_e u_i = \bar T'_e u_j$. Two families give the same configuration exactly when $(u'_i)^{-1}u_i = (u'_j)^{-1}u_j$ on every edge, which on a connected graph means $u' = uc$. If H^0 consists of constant sections, every column of u is constant, so $u_v = u_0$ for all v and the configuration is the truth. If $s \in H^0$ is not constant, $u_v = I + t s_v w^T$ with $w \neq 0$ and small $t \neq 0$ is an invertible family satisfying the condition, and it is not of the form $(c)_v$, so its configuration differs from the truth. For the complete exchange put $k_0 = I$ and $k_i = \bar\theta_{i0}$. The two-cycle through 0 and i gives $\bar\theta_{0i} = k_i^{-1}$, and the directed triangle $0 \to j \to i \to 0$, whose holonomy is $\bar\theta_{0i}\bar\theta_{ij}\bar\theta_{j0}$, gives $\bar\theta_{ij} = \bar\theta_{0i}^{-1}\bar\theta_{j0}^{-1} = k_i k_j^{-1}$. □

The residual set of eq. (18.37) has dimension $K(\dim H^0 - K)$ at the truth, the count of proposition 18.25.

18.7.6 Receiver-owned recognition laws

Remark 18.30 (Ownership and the loop term). Under a variational restriction with factors owned by receivers, each receiver infers its own dictionaries, so its dictionary recognition law is one factor $\mathfrak{q}_i(\omega_{i\cdot})$ of a law factorized over receivers. A two-cycle couples two receivers and a directed triangle three, so the loop term couples the factors. In the tangent model the factorized law closest to the posterior in $D_{\mathrm{KL}}(\mathfrak{q}\|\text{posterior})$ has the exact posterior means and, as covariances, the inverses of the diagonal blocks of the posterior precision [15, Section 10.1.2]. These are smaller in the Loewner order than the exact marginal covariances. For this one Gaussian target, the factorization therefore leaves the estimates of the dictionaries unchanged and makes each receiver overconfident about them. Exact means are a feature of Gaussian targets: for a non-Gaussian posterior a factorized law generally has means different from the exact ones, so for general laws the factorization can change the estimates as well. To update its factor, receiver i needs the current means of the dictionaries that other agents hold along its loops: a loop prior can be used only if agents communicate about how they read each other, in addition to communicating about the world.

18.8 Extensions beyond the declared experiments

Open problem 18.31 (Learning the dictionaries beyond the tangent model). Theorem 18.22 and proposition 18.24 settle the learning problem in the tangent model, and propositions 18.28

and 18.29 settle the constraint form of the loop term for a flat truth without linearization. The remaining questions have separate scopes. For the full grounded model eq. (18.13) with a prior on G or on a declared exponential chart, establish the limiting conditional law on the identified fiber $\prod_e \bar{\theta}_e^{\star}\,\mathrm{Stab}(\bar{T}_e)$, or on its preimage in the chart, and determine the selection within the fiber for a dissonant truth and a finite loop weight. The nonlinear covariance-dressed defect and the law-comparison penalty defined above lead to different learning problems; the latter requires its own regularity and stabilizer analysis. Another extension treats latent routing labels, which lower the information below the routed sum of corollary 18.12, and loop weights built from the routing weights. For location-family noise and a non-Gaussian anchor, determine the limit of the recognition law on the fiber, which is no longer a conditioned Gaussian, and whether the filled directions are still recovered exactly as the loop weight grows.

Open problem 18.32 (Codes as part of the species model). Let codes be learned as well, so that a shared language is a joint fixed point of code and dictionary learning. The expression-map dynamics of Hansen and Ghrist [72] is the vector-valued analogue on a sheaf. Determine when the learned codes of a species become related by its frame map, and when learning drives code-visible dissonance to zero while leaving dissonance in unread directions.

Open problem 18.33 (Dictionaries in the free-energy pictures). The Gaussian tangent learning calculation fixes a normalized likelihood and an anchored loop prior. Extending it to fully nonlinear receiver-owned dictionaries and learned codes requires separate global, per-agent, and reciprocal objectives, with every law dependence differentiated in the chosen objective. Whether those dynamics reduce code-visible dissonance is a further dynamical question; posterior conditioning within the tangent model alone does not establish it.

Open problem 18.34 (Beyond Gaussian codes). Corollaries 18.4 and 18.5 and proposition 18.20 already hold for general laws and measurable codes, including codes into finite alphabets such as words, and the consensus theorem holds for laws as well as means. Remark 18.10 extends identifiability to every location family of noise. Extend the identifiability, learning and dissonance results to codes whose message laws are not a location family, such as codes into finite alphabets, where the Fisher form is replaced by the information in the induced discrete channel and the stabilizer by the subgroup that leaves the channel unchanged, and to agents of different kinds, whose dictionaries are not invertible.

The relation to discourse sheaves is the use of private content spaces and restriction maps into shared message spaces [72]. The reconstruction step additionally specifies a receiver conditional law, while invertible dictionaries supply loop products. Unknown frame conventions have also been studied as communication constraints [10]. These comparisons do not identify graph loops at one context with transport between contexts.

Chapter 19

Learning Directed Links under Three Objectives

The receiver-owned dictionaries of definition 18.2 can be uncertain and learned together with the contents they interpret. Their dynamics depends on which of the three objectives in chapter 4 is chosen. A global model differentiates every comparison record at both endpoints. A per-agent model updates only the receiver's variables while treating the source law as data. A reciprocal potential differentiates both arguments of a divergence between communicated laws. These distinctions concern the objective; the recognition family supplies the Fisher metric that turns its derivatives into velocities.

The general dependency statements and the pointwise record invariance below do not require Gaussian laws. Explicit mean flows, grounding rates, and confidence formulas use a linear-Gaussian tangent family with fixed operating contents. The fixed-covariance reciprocal mean theorem is followed by the coupled flow of all content and dictionary means and covariances, with either relaxing or Gibbs-optimized source rows. Its global-existence theorem and exact stability example concern the restricted family of definition 4.10; they do not assert that a full law-space flow preserves Gaussianity.

19.1 The tangent communication model

Two agents who exchange messages about something they both represent can register whether a message matches what the receiver expected. A comparison record is a noisy linear function $A_i y_i - B_{ij} y_j + \epsilon$ of the two agents' contents. With codes and receiver-owned dictionaries the natural choice is $B_{ij} = \bar{T}_e$, the sender's code, and $A_i = \bar{T}_e \bar{\theta}_e^{-1}$, the receiver's prediction of the message from its own content translated into the sender's coordinates. The following construction linearizes that record and places it in the two free energies.

Definition 19.1 (Tangent communication model). Let $\mathcal{V}$ be a finite set of agents and $\mathcal{E}$ a set of directed edges $e : j \to i$ with receiver i and sender j, and write $\mathrm{in}(i)$ for the edges received by i. Agent i has content $y_i \in \mathbb{R}^K$ in its reference coordinates. Edge e carries the receiver's dictionary $\bar{\theta}_e = e^{\omega_e}$ with $\omega_e \in \mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$, the sender's attending code $\bar{T}_e : \mathbb{R}^K \to H_e = \mathbb{R}^{n_e}$ of full row rank $n_e \le K$ and a positive-definite noise covariance R_e. Fix operating contents $c_i \in \mathbb{R}^K$ and write $\kappa_i \omega = \omega c_i$. The dictionary is a receiver-owned hypothesis about the sender's coordinates; the independently owned opposite dictionary need not be its inverse. Define $N_e = \{\delta \in \mathfrak{g} : \bar{T}_e \delta = 0\}$, the directions invisible to grounded referents that span the content space. These can be fewer than the directions invisible at a single operating content, which satisfy only $\bar{T}_e \delta c_i = 0$. The comparison

record of e is

$$o_e = \bar{T}_e\big(y_i - y_j - \kappa_i\omega_e\big) + \varepsilon_e, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e), \tag{19.1}$$

the first-order form of $\bar{T}_e\bar{\theta}_e^{-1}y_i - \bar{T}_e y_j + \varepsilon_e$ at $\bar{\theta}_e = I$ and $y_i = c_i$. Each agent has a self term $\mathcal{N}(m_i, P_i^{-1})$ with P_i positive definite, the Gaussian form at the operating point of its own generative model and its own records. Each edge has a grounded term $\exp(-\frac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e})$ with G_e positive semidefinite, the information of grounded messages, with $\ker G_e = N_e$ when their referents identify every coded direction. The dictionaries have an anchored loop prior with anchor $\sigma^{-2}I$ on each edge, a finite set $\mathcal{L}$ of simple directed loops containing every two-cycle, linearized holonomy $(B\omega)_\gamma = \sum_{e\in\gamma}\omega_e$, positive-definite forms W_γ and weight $\lambda \geq 0$. Put $W = \bigoplus_\gamma W_\gamma$ and use $\|v\|^2_A = v^T A v$. All codes, operating contents, grounding data, and prior forms are fixed. The Frobenius anchor and loop forms are declared reference-coordinate data, not an assertion of a positive-definite conjugation-invariant form on $\mathfrak{gl}(K, \mathbb{R})$.

Definition 19.2 (Global joint). Write $x = (y, \omega)$ and $L_e x = \bar{T}_e(y_i - y_j - \kappa_i\omega_e)$. The global joint has posterior density proportional to $e^{-E(x)}$ with

$$\begin{aligned} E(x) = \sum_i \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_e \Big(\tfrac{1}{2\sigma^2}\|\omega_e\|^2 + \tfrac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e} + \tfrac{1}{2}\|o_e - L_e x\|^2_{R_e^{-1}} \Big) \\ + \tfrac{\lambda}{2}\sum_{\gamma\in\mathcal{L}} \|(B\omega)_\gamma\|^2_{W_\gamma}. \end{aligned} \tag{19.2}$$

Its precision J is positive definite because the self terms and the anchor are, so the joint is normalized for every value of the registered records. Write η for its information vector and $x^\star = J^{-1}\eta$ for the posterior mean.

The linearization drops the product $\omega_e(y_i - c_i)$, which is of second order in the deviations from the operating point. Registering $o_e = 0$ registers agreement. The record is invariant under passive changes of frame when the displayed contents, codes, and dictionaries are transformed together, because the dictionary sits in the receiver's matrix.

Definition 19.3 (Per-agent tangent model). Fix agent i and the current peer laws, $q_k = \mathcal{N}(\mu_k, \Sigma_k)$ for contents and $r_f = \mathcal{N}(a_f, V_f)$ for the dictionaries that i does not own. Agent i's model has the variables $x_i = (y_i, \omega_{\mathrm{in}(i)})$ and negative log density, up to a constant,

$$\begin{aligned} E_i = \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_{e\in\mathrm{in}(i)} \Big(\tfrac{1}{2\sigma^2}\|\omega_e\|^2 + \tfrac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e} + \tfrac{1}{2}\big\|o_e - \bar{T}_e(y_i - \mu_j - \kappa_i\omega_e)\big\|^2_{\tilde{R}_e^{-1}} \Big) \\ + \sum_{\gamma\in\mathcal{L}_i} \tfrac{1}{2}\Big\| \sum_{e\in\gamma\cap\mathrm{in}(i)} \omega_e + \sum_{f\in\gamma\setminus\mathrm{in}(i)} a_f \Big\|^2_{\Omega_{i\gamma}}, \end{aligned} \tag{19.3}$$

where $\mathcal{L}_i$ is the set of loops through an edge of i and

$$\tilde{R}_e = R_e + \bar{T}_e\Sigma_j\bar{T}_e^T, \qquad \Omega_{i\gamma} = \Big((\lambda W_\gamma)^{-1} + \sum_{f\in\gamma\setminus\mathrm{in}(i)} V_f\Big)^{-1}. \tag{19.4}$$

For $\lambda > 0$, these are the record eq. (19.1) and the loop factor integrated against the peers' laws; we call this the law-level model. At $\lambda = 0$ the loop terms are omitted, equivalently $\Omega_{i\gamma} = 0$ by

continuity. The convolution uses the declared product of peer dictionary marginals for each loop factor. Replacing $\tilde{R}_e$ by R_e and $\Omega_{i\gamma}$ by λW_γ gives the plug-in model, in which peers enter only through their means.

The integration is Gaussian convolution: $\int \mathcal{N}(o;\bar{T}(y_i - y_j - \kappa_i\omega), R)q_j(dy_j) = \mathcal{N}(o;\bar{T}(y_i - \mu_j - \kappa_i\omega), \tilde{R})$, and $\int e^{-\frac{\lambda}{2}\|u+s\|_W^2}\mathcal{N}(s;\bar{a},\bar{V})ds$ is proportional to $e^{-\frac{1}{2}\|u+\bar{a}\|_\Omega^2}$ with $\Omega = ((\lambda W)^{-1} + \bar{V})^{-1}$. The model is normalized for every value of the peer laws, which are parameters of i's model and not variables over which it infers, as in a per-agent generative model. The records with peers enter through the record term of the per-agent free energy, and the loop factors enter i's prior over its own dictionaries. Agent i's model contains no record in which i is the sender; such records belong to the receivers' models.

The recognition laws are Gaussian. In the global picture they are factorized by agent, $Q = \bigotimes_i Q_i(x_i)$, each agent's factor holding its content and the dictionaries it owns; each local update varies one factor of a single global free energy. In the per-agent picture agent i holds a Gaussian Q_i on x_i and lowers its own free energy $\mathcal{F}_i$.

19.2 What conversation determines

Remark 19.4 (The interpretation gauge without linearization or Gaussian noise). Let the records be $o_e = \bar{T}_e(\bar{\theta}_e^{-1}y_i - y_j) + \varepsilon_e$ with noise of any law, independent of everything else, or more generally let each record kernel read $(y,\bar{\theta})$ only through the coded discrepancy $\bar{T}_e(\bar{\theta}_e^{-1}y_i - y_j)$. Let $K \geq 2$ and let G contain $\mathrm{GL}^+(K,\mathbb{R})$. For every configuration $(y,\bar{\theta})$ and every belief displacement $y \mapsto y'$ with $y'_i \neq 0$ and $v_e = \bar{\theta}_e^{-1}y_i + y'_j - y_j \neq 0$ on every edge $e : j \to i$, there are dictionaries $\bar{\theta}'_e \in \mathrm{GL}^+(K,\mathbb{R})$ with $\bar{\theta}'_e v_e = y'_i$, because $\mathrm{GL}^+(K,\mathbb{R})$ acts transitively on $\mathbb{R}^K \setminus \{0\}$ for $K \geq 2$. Then $\bar{\theta}'^{-1}_e y'_i - y'_j = v_e - y'_j = \bar{\theta}_e^{-1}y_i - y_j$, so $(y',\bar{\theta}')$ gives every record the same law as $(y,\bar{\theta})$. The interdependence of belief and meaning is therefore not a feature of the Gaussian tangent model: it holds pointwise, without linearization, for every record kernel of this form, and only terms other than the records can separate the two readings.

Theorem 19.5 (Belief and meaning are interdependent). In the tangent communication model the law of the comparison records depends on (y,ω) only through the coded discrepancies $L_e x = \bar{T}_e(y_i - y_j - \kappa_i\omega_e)$.

(i) The displacements $(\delta y, \delta\omega)$ that leave the law of every record unchanged form the subspace

$$\mathcal{G} = \big\{(\delta y,\delta\omega) : \bar{T}_e(\delta y_i - \delta y_j - \kappa_i\delta\omega_e) = 0 \text{ for every edge } e : j \to i\big\}. \tag{19.5}$$

If $c_i \neq 0$ for every receiver, the projection of $\mathcal{G}$ onto belief displacements is all of $(\mathbb{R}^K)^{\mathcal{V}}$: for every δy, the choice $\delta\omega_e = (\delta y_i - \delta y_j)c_i^T/|c_i|^2$ completes it to an element of $\mathcal{G}$.

(ii) The Fisher information of the records about (y,ω) vanishes exactly on $\mathcal{G}$, so along every direction of $\mathcal{G}$ the posterior precision is the sum of the contributions of the self terms, the grounded terms and the dictionary prior.

(iii) Among the displacements with $\delta\omega_e \in N_e$ for every edge, the belief components of $\mathcal{G}$ are exactly the sections $\{\delta y : \bar{T}_e(\delta y_i - \delta y_j) = 0 \text{ for every edge}\}$ of the code sheaf, whose edge condition is the displayed equality. They contain the common shifts $\delta y_i = v$.

Proof. (i) Each record is $L_e x$ plus independent noise, so its law is unchanged exactly when L_e annihilates the displacement. For the stated choice, $\kappa_i \delta\omega_e = (\delta y_i - \delta y_j) c_i^T c_i / |c_i|^2 = \delta y_i - \delta y_j$. (ii) The Fisher information of Gaussian records with fixed covariances is $\sum_e L_e^T R_e^{-1} L_e$, whose kernel is $\bigcap_e \ker L_e = \mathcal{G}$; the posterior precision is J, the sum of this form and of the other terms of eq. (19.2). (iii) For $\delta\omega_e \in N_e$, $\bar{T}_e \kappa_i \delta\omega_e = \bar{T}_e \delta\omega_e c_i = 0$, so the condition reduces to $\bar{T}_e(\delta y_i - \delta y_j) = 0$. □

An exchange that disagrees with the receiver's belief can be explained by changing the belief or by changing the dictionary. The comparison record alone does not choose between these responses. The self terms, grounded records, and anchored prior supply distinct information. Even when grounding removes dictionary freedom modulo N_e, common content shifts remain invisible to every pairwise discrepancy.

Proposition 19.6 (Persuasion and reinterpretation). Take one edge $e : j \to i$, independent Gaussian priors $y_i \sim \mathcal{N}(m_i, \Sigma_i^0)$, $y_j \sim \mathcal{N}(m_j, \Sigma_j^0)$ and $\omega_e \sim \mathcal{N}(\bar{\omega}_e, V_e)$, and one record o_e. Put

$$r = o_e - \bar{T}_e(m_i - m_j - \kappa_i \bar{\omega}_e), \qquad C_\rho = \kappa_i V_e \kappa_i^T. \tag{19.6}$$

(i) In the global joint, with $S = \bar{T}_e(\Sigma_i^0 + \Sigma_j^0 + C_\rho)\bar{T}_e^T + R_e$, the posterior mean moves by

$$\delta y_i = \Sigma_i^0 \bar{T}_e^T S^{-1} r, \qquad \delta y_j = -\Sigma_j^0 \bar{T}_e^T S^{-1} r, \qquad \kappa_i \delta\omega_e = -C_\rho \bar{T}_e^T S^{-1} r. \tag{19.7}$$

(ii) In the per-agent model of the receiver, with the sender's law $\mathcal{N}(m_j, \Sigma_j^0)$ as data and $S' = \bar{T}_e(\Sigma_i^0 + C_\rho)\bar{T}_e^T + R_e + \bar{T}_e \Sigma_j^0 \bar{T}_e^T$, the receiver's mean moves by $\delta y_i = \Sigma_i^0 \bar{T}_e^T S'^{-1} r$ and $\kappa_i \delta\omega_e = -C_\rho \bar{T}_e^T S'^{-1} r$, and the sender does not move.

(iii) For the isotropic anchor $V_e = \sigma^2 I$, $C_\rho = \sigma^2 |c_i|^2 I$.

Proof. The quantity $u = y_i - y_j - \kappa_i \omega_e$ has prior mean $m_i - m_j - \kappa_i \bar{\omega}_e$ and covariance $\Sigma_i^0 + \Sigma_j^0 + C_\rho$, and $o_e = \bar{T}_e u + \varepsilon_e$. The covariances of y_i, y_j and $\kappa_i \omega_e$ with o_e are $\Sigma_i^0 \bar{T}_e^T$, $-\Sigma_j^0 \bar{T}_e^T$ and $-C_\rho \bar{T}_e^T$, and that of o_e is S, so Gaussian conditioning gives (i). In (ii) the term $\bar{T}_e y_j$ is distributed as $\mathcal{N}(\bar{T}_e m_j, \bar{T}_e \Sigma_j^0 \bar{T}_e^T)$ and is absorbed into the noise, which gives S'. For (iii), $\kappa_i^T v = v c_i^T$ and $\kappa_i (v c_i^T) = |c_i|^2 v$. □

The Gaussian conditional update assigns a discrepancy to receiver adjustment, sender adjustment, and reinterpretation through the three cross-covariances in eq. (19.7). Observation noise leaves a residual gap, so these adjustments need not reproduce the record exactly. In the per-agent model the sender's uncertainty widens the receiver's likelihood while the sender remains fixed. The interpretation variance grows as $|c_i|^2$ under the isotropic anchor; at $c_i = 0$ the first-order dictionary term cannot absorb a discrepancy. These linear shares are features of the Gaussian family.

Corollary 19.7 (Grounding turns reinterpretation into persuasion). Let Π_e be the orthogonal projection of $\mathfrak{g}$ onto $N_e^\perp$. Then $\bar{T}_e C_\rho \bar{T}_e^T = \bar{T}_e \kappa_i \Pi_e V_e \Pi_e \kappa_i^T \bar{T}_e^T$, so only the uncertainty of the identified part of the dictionary can absorb a disagreement. If the grounded information on edge e is $g\bar{G}_e$ with $\ker \bar{G}_e = N_e$, then $\Pi_e V_e \Pi_e = O(1/g)$ and the reinterpretation share of a record is $O(1/g)$.

Proof. For $\nu \in N_e$, $\bar{T}_e \kappa_i \nu = \bar{T}_e \nu c_i = 0$, so $\bar{T}_e \kappa_i = \bar{T}_e \kappa_i \Pi_e$. With a positive-definite prior precision Λ on ω_e and grounded precision $g\bar{G}_e$, block inversion in the orthonormal bases of $N_e^\perp$ and N_e shows,

by taking the Schur complement of the fixed invisible block, that the covariance restricted to $N_e^{\perp}$ is $O(1/g)$. □

Grounding therefore suppresses the part of the response attributable to uncertainty about the identified dictionary coordinates.

19.3 The two flows

Proposition 19.8 (The global flow). Let the global recognition be $Q = \bigotimes_i \mathcal{N}(\mu_i, S_i)$ on the agent blocks $x_i = (y_i, \omega_{\mathrm{in}(i)})$, let J_{ii} be the diagonal blocks of J and $D = \bigoplus_i J_{ii}$.

(i) Up to a constant, $\mathcal{F}_G(Q) = \frac{1}{2}(\mu - x^\star)^T J(\mu - x^\star) + \frac{1}{2}\sum_i[\mathrm{tr}(J_{ii}S_i) - \log\det S_i]$, which is minimized by $S_i = J_{ii}^{-1}$ and $\mu = x^\star$.

(ii) At these covariances the Fisher-metric gradient flow of the means is $\dot\mu = -D^{-1}J(\mu - x^\star)$. The matrix $D^{-1}J$ is similar to the positive-definite matrix $D^{-1/2}JD^{-1/2}$, so its spectrum is real and positive, $\mathcal{F}_G$ decreases strictly away from $x^\star$, and $\mu \to x^\star$ exponentially.

(iii) Each agent's block of the force is the gradient of its local free energy, the global free energy as a function of its own factor. A record therefore moves the receiver's content, the sender's content and the receiver's dictionary through one functional.

Proof. (i) Since the factors are independent, $\mathbb{E}_Q[E] = \frac{1}{2}(\mu - x^\star)^T J(\mu - x^\star) + \frac{1}{2}\sum_i \mathrm{tr}(J_{ii}S_i) + \mathrm{const}$, and the entropy is $\frac{1}{2}\sum_i \log\det S_i + \mathrm{const}$. (ii) The Fisher metric of $\mathcal{N}(\mu_i, S_i)$ in μ_i is $S_i^{-1} = J_{ii}$, and $\nabla_\mu \mathcal{F}_G = J(\mu - x^\star)$. The similarity is $D^{-1}J = D^{-1/2}(D^{-1/2}JD^{-1/2})D^{1/2}$, and $\frac{d}{dt}\mathcal{F}_G = -\nabla\mathcal{F}_G^T D^{-1}\nabla\mathcal{F}_G$. (iii) follows by varying one factor in the same global functional; every incident likelihood contributes to that variation. □

The optimized factor means equal the Gaussian posterior mean, whereas J_{ii}^{-1} is no larger in the Loewner order than the exact marginal covariance. Indeed, the exact marginal precision is the Schur complement $J_{ii} - J_{i,-i}J_{-i,-i}^{-1}J_{-i,i} \preceq J_{ii}$. Exact factor means are a Gaussian property; the coordinate-descent identity holds more generally, without implying exact moments [15].

Proposition 19.9 (The per-agent flow). Hold the peers' covariances at a fixed point of the per-agent covariance updates, and let agent i hold $\mathcal{N}(\mu_i, S_i)$ on x_i.

(i) Up to a constant, $\mathcal{F}_i = \frac{1}{2}(\mu_i - x_i^\dagger)^T A_i(\mu_i - x_i^\dagger) + \frac{1}{2}[\mathrm{tr}(A_iS_i) - \log\det S_i]$, where $A_i = \nabla_{x_i}^2 E_i$ depends on the peers' covariances but not on their means, and $A_i x_i^\dagger = b_i + \sum_{k\neq i} C_{ik}\mu_k$ is affine in the peers' means.

(ii) At $S_i = A_i^{-1}$ the Fisher flow of the means is $\dot\mu = -A^{-1}(H\mu - b)$ with $A = \bigoplus_i A_i$ and $H = A - C$. Its fixed points solve $H\mu = b$.

(iii) In the plug-in model, the rows of H and b that belong to dictionaries equal the corresponding rows of J and η. This equality concerns objective derivatives before Fisher preconditioning. The rows that belong to the content of agent i equal those of J and η minus the contributions $L_e^T R_e^{-1} L_e$ and $L_e^T R_e^{-1} o_e$ of the records e in which i is the sender. In the law-level model R_e is in addition replaced by $\tilde R_e$ and λW_γ by $\Omega_{i\gamma}$.

(iv) For an edge $e : j \to i$ the block $H_{y_i y_j}$ contains $-\bar{T}_e^T \tilde{R}_e^{-1} \bar{T}_e$, while e contributes nothing to $H_{y_j y_i}$. Hence H need not be symmetric, and its fixed point need not equal $x^\star$.

Proof. (i) and (ii) follow as in proposition 19.8, with the peers' means as data in E_i. For (iii), compare the terms of eqs. (19.2) and (19.3). A dictionary ω_e appears in the anchor, in its grounded term, in the record of e and in the loop factors, and all of these belong to the model of its owner, with the same precisions in the plug-in model; the loop factor of γ in the owner's model has the same gradient in ω_e as the single loop term of the global energy. The content y_i appears in its self term, in the records of $\mathrm{in}(i)$, which are in i's model, and in the records in which i is the sender, which are in the receivers' models only. (iv) is read off from (iii). □

Ownership removes sender-role content derivatives from the per-agent update. The equality of dictionary derivative rows holds only in the plug-in model: the law-level model uses both $\tilde{R}_e$ and $\Omega_{i\gamma}$. These derivatives are covectors. Their Fisher velocities are obtained using the block mobilities J_{ii}^{-1} and A_i^{-1}, which can differ and mix content with owned dictionary coordinates. Equal dictionary derivative rows consequently do not establish equal dictionary velocities. The dependency statement survives for general laws whenever the declared model contains only received records and the dictionary appears only in its owner's factors; equality of derivatives after marginalizing peers does not follow from ownership.

Proposition 19.10 (A directed ring). Let $K = 1$ and let $\ell \geq 3$ agents form a directed ring in which agent i receives from agent $i + 1$, indices modulo ℓ, with $\bar{T}_e = 1$, record precision $\tilde{q}$, self precision p, anchor precision $s = \sigma^{-2}$, common operating content c, and no grounded or loop term. With the own-block Gaussian covariances held at their optima, the per-agent mean flow has the eigenvalue -1 with multiplicity ℓ and the eigenvalues $-(1 - \zeta w)$ for the ℓ-th roots of unity ζ, where

$$w = \frac{\tilde{q}s}{ps + p\tilde{q}c^2 + \tilde{q}s} = \frac{\tilde{q}_{\mathrm{eff}}}{p + \tilde{q}_{\mathrm{eff}}}, \qquad \tilde{q}_{\mathrm{eff}} = \Big(\frac{1}{\tilde{q}} + \frac{c^2}{s}\Big)^{-1}. \tag{19.8}$$

The eigenvalues with $\zeta \notin \mathbb{R}$ are not real, so the per-agent flow is not a gradient flow for any Riemannian metric, whereas the global flow of the same ring has real spectrum. As $s \to \infty$, $w \to \tilde{q}/(p + \tilde{q})$; as $s \to 0$ with $c \neq 0$, $w \to 0$. If $c = 0$, then $w = \tilde{q}/(p + \tilde{q})$ for every $s > 0$.

Proof. Agent i's energy is $\frac{1}{2}p(y_i - m_i)^2 + \frac{1}{2}s\omega_i^2 + \frac{1}{2}\tilde{q}(o_i - y_i + y_{i+1} + c\omega_i)^2$. In the variables (y_i, ω_i) its Hessian is $A_i = \left(\begin{smallmatrix} p+\tilde{q} & -\tilde{q}c \\ -\tilde{q}c & s+\tilde{q}c^2 \end{smallmatrix}\right)$ with $\det A_i = ps + p\tilde{q}c^2 + \tilde{q}s$, and its coupling to (y_{i+1}, ω_{i+1}) is $C' = \left(\begin{smallmatrix} -\tilde{q} & 0 \\ \tilde{q}c & 0 \end{smallmatrix}\right)$. The flow matrix is block circulant, so on the vectors $x_k = \zeta^k v$ it acts as $-(I + \zeta A_i^{-1} C')$. The matrix $A_i^{-1}C'$ has zero second column, and its first diagonal entry is $\det A_i^{-1}[(s + \tilde{q}c^2)(-\tilde{q}) + (\tilde{q}c)^2] = -w$. Hence the eigenvalues are -1 and $-(1 - \zeta w)$. The linearization of a gradient flow at an equilibrium is self-adjoint for the metric there and has real spectrum, since its Jacobian is the negative metric inverse times the symmetric objective Hessian. The global statement is proposition 19.8(ii). The second form of w and the limits are algebra. □

For $c \neq 0$, uncertainty about the dictionary acts in the mean equations as additional record noise of variance c^2/s. The directed oscillatory modes retain angular frequency $w \operatorname{Im} \zeta$, while $0 < w < 1$ keeps every mode stable. In the limit $s \downarrow 0$ the receiver can absorb the discrepancy through reinterpretation and the coupling vanishes. At $c = 0$ the dictionary does not enter the tangent

record, so weakening its anchor does not weaken persuasion. Here $\tilde{q} = R^{-1}$ in the plug-in model and $\tilde{q} = (R + \Sigma)^{-1}$ in the law-level model at a fixed symmetric content variance Σ. The complex spectrum is one explicit instance of non-reciprocal dynamics [62]; it rules out a general gradient-flow interpretation of the per-agent picture.

19.4 The grounded limit

Theorem 19.11 (Separation in the grounded limit). Let $G_e = g\bar{G}_e$ with $\bar{G}_e$ positive semidefinite and $\ker \bar{G}_e = N_e$ for every edge, keep the grounded centers $\hat{\omega}_e$ fixed, and let $g \to \infty$.

(i) In the global joint the posterior mean converges to (y^G, ω^∞), where y^G minimizes

$$E_{\text{known}}(y) = \sum_i \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_e \tfrac{1}{2}\big\|o_e + \bar{T}_e\kappa_i\hat{\omega}_e - \bar{T}_e(y_i - y_j)\big\|^2_{R_e^{-1}}, \tag{19.9}$$

the identified part of ω_e^∞ is that of $\hat{\omega}_e$, and the invisible part minimizes the dictionary prior on the coset $\hat{\omega} + \bigoplus_e N_e$. In the limit the beliefs do not depend on the dictionary prior.

(ii) In the plug-in per-agent model, if the content equations

$$P_i(y_i - m_i) - \sum_{e\in\text{in}(i)} \bar{T}_e^T R_e^{-1}\big(o_e + \bar{T}_e\kappa_i\hat{\omega}_e - \bar{T}_e(y_i - y_j)\big) = 0 \qquad (i \in \mathcal{V}) \tag{19.10}$$

have a unique solution y^P, the fixed point converges to it in the contents, and the identified dictionary parts converge to those of $\hat{\omega}_e$.

(iii) In both cases the error is $O(1/g)$.

Proof. Write $\omega_e = \hat{\omega}_e + Y_e\alpha_e + Z_e\nu_e$ in orthonormal bases Y_e of $N_e^\perp$ and Z_e of N_e. Since $\bar{T}_e\kappa_i Z_e = 0$, the records depend on (y, α) only. The energy is $\frac{g}{2}\sum_e \alpha_e^T\Gamma_e\alpha_e + \tilde{E}(y, \alpha, \nu)$ with $\Gamma_e = Y_e^T\bar{G}_eY_e$ positive definite and $\tilde{E}$ independent of g. In both pictures the stationarity equations are linear and have the block form $(g\Gamma + H_{\alpha\alpha})\alpha + H_{\alpha z}z = b_\alpha$ and $H_{z\alpha}\alpha + H_{zz}z = b_z$ with $z = (y, \nu)$ and H, b independent of g. Eliminating $\alpha = (g\Gamma + H_{\alpha\alpha})^{-1}(b_\alpha - H_{\alpha z}z) = O(1/g)$ gives $(H_{zz} + O(1/g))z = b_z + O(1/g)$, so $z \to H_{zz}^{-1}b_z$ at rate $1/g$ whenever H_{zz} is invertible. At $\alpha = 0$ the records involve y alone and the dictionary prior involves ν alone, so H_{zz} is block diagonal. Its content block gives eq. (19.9) in the global joint, where J is positive definite, and eq. (19.10) in the plug-in model, invertible by hypothesis. Its invisible-dictionary block is the restriction of the dictionary prior precision $\sigma^{-2}I + \lambda B^TWB$ to the invisible directions, positive definite, in both pictures by proposition 19.9(iii). □

The theorem identifies two different known-dictionary limits. The global mean minimizes a symmetric quadratic comparison energy, while the plug-in per-agent fixed point solves receiver-only equations. The identified dictionaries agree in the limit, but the contents need not. The theorem makes no claim about law-level covariance updates that vary with g; controlling those updates is an additional obligation.

19.5 Confidence under ownership

Mean agreement does not imply agreement about uncertainty. The following example isolates one coordinate that records and grounding do not read, so all information coupling it to other dictionaries comes from one declared loop factor.

Proposition 19.12 (Echo around a loop). Let $\lambda > 0$ and let $\ell \geq 2$ edges form one loop, with distinct receivers and one coordinate z_e per edge that no record or grounded term reads, with anchor $\mathcal{N}(0, \sigma^2)$ and loop factor $\exp(-\frac{\lambda}{2}(\sum_e z_e)^2)$. Every recognition below has mean zero, and the variances of z_e are as follows.

(i) Exact joint posterior: $\sigma^2(1 + (\ell - 1)\lambda\sigma^2)/(1 + \ell\lambda\sigma^2)$, which tends to $\sigma^2(\ell - 1)/\ell$ as $\lambda \to \infty$.

(ii) Receiver-factorized global recognition: $(\sigma^{-2} + \lambda)^{-1}$, which tends to zero.

(iii) Per-agent law-level recognition at its symmetric covariance fixed point: the unique positive solution v of

$$v^{-1} = \sigma^{-2} + \big(\lambda^{-1} + (\ell - 1)v\big)^{-1}, \tag{19.11}$$

which tends to $\sigma^2(\ell - 2)/(\ell - 1)$ for $\ell \geq 3$ and satisfies $v = \sigma\lambda^{-1/2}(1 + o(1))$ for $\ell = 2$.

(iv) Cavity recognition, in which each receiver's loop source uses its partners' laws with its own contribution removed, so that those laws are the anchors: $(\sigma^{-2} + (\lambda^{-1} + (\ell - 1)\sigma^2)^{-1})^{-1}$, which equals (i) for every λ.

Proof. All terms are even in z, so the means vanish. (i) The precision is $\sigma^{-2}I + \lambda\mathbf{1}\mathbf{1}^T$, and the Sherman–Morrison formula gives the diagonal entry $\sigma^2 - \sigma^4\lambda/(1 + \ell\lambda\sigma^2)$. (ii) The factorized covariance is the inverse of the diagonal of the precision. (iii) By symmetry each receiver's loop source is the convolution of the loop factor with $\ell - 1$ partner laws of variance v, which gives eq. (19.11). With $\epsilon = \lambda^{-1}$, clearing denominators gives $(\ell-1)\sigma^{-2}v^2 + (\epsilon\sigma^{-2} + 2 - \ell)v - \epsilon = 0$, whose roots have product $-\epsilon\sigma^2/(\ell - 1) < 0$, so exactly one is positive. As $\epsilon \to 0$ it tends to $\sigma^2(\ell - 2)/(\ell - 1)$ for $\ell \geq 3$, and for $\ell = 2$ it satisfies $v^2 + \epsilon v - \epsilon\sigma^2 = 0$, so $v = \sigma\epsilon^{1/2}(1 + o(1))$. (iv) With $\epsilon = \lambda^{-1}$, $(\sigma^{-2} + (\epsilon + (\ell - 1)\sigma^2)^{-1})^{-1} = \sigma^2(\epsilon + (\ell - 1)\sigma^2)/(\epsilon + \ell\sigma^2)$, which is (i) after multiplying numerator and denominator by λ. □

All four constructions have the same zero mean and different variances. Factorized reverse-KL recognition suppresses the correlations by which the exact law satisfies the sum constraint. The per-agent update instead treats a partner's current marginal as new independent information, even though that marginal already contains information received from the same loop. For a two-cycle its fixed-point variance tends to zero while the exact marginal tends to $\sigma^2/2$. Removing the loop contribution from the peer messages gives the anchor laws and restores the exact marginal here. This is belief propagation on the tree consisting of one factor connected to ℓ variables [118]; it is not a general exactness theorem for overlapping loop factors.

The recycled-information mechanism can occur for non-Gaussian laws as well. The displayed fixed points, rates, and equality of factor means are conclusions only for this Gaussian single-loop model.

19.6 The reciprocal potential

The reciprocal potential is a function of agent laws. It moves each source through the costs of receivers that attend to it. The following extension includes dictionary laws as configuration variables. Registered comparison records are absent from this potential; adding persistent likelihood terms would define another objective.

Definition 19.13 (Dictionary laws and source rows). Consider one content sector, with singleton model-presentation spaces. Let q_i and r_e be laws on standard Borel content and dictionary spaces, let p_i be fixed proper content references, and let π_ω be a fixed proper joint dictionary reference. Recognition is restricted to the product $r = \bigotimes_e r_e$, even when π_ω correlates edges. Fixed normalized kernels give laws $A_e(\omega_e) = q_i \mathcal{K}_e^{\mathrm{rec}}(\omega_e)$ and $B_e = q_j \mathcal{K}_e^{\mathrm{src}}$ on a common message space for each $e : j \to i$. Put

$$\begin{aligned} C_e &= \int \Delta_e(\omega_e) r_e(d\omega_e), & \Delta_e(\omega_e) &= D_{\mathrm{KL}}(A_e(\omega_e) \| B_e), \\ F_0 &= \sum_i D_{\mathrm{KL}}(q_i \| p_i) + D_{\mathrm{KL}}(r \| \pi_\omega). \end{aligned} \tag{19.12}$$

All displayed relative entropies and expected costs are required to be finite. Each nonempty candidate set $\mathcal{I}_i = \mathrm{in}(i)$ has a positive normalized prior $\pi_i = (\pi_e)_{e \in \mathcal{I}_i}$, a temperature $T_i > 0$, and a positive normalized row β_i. Rows with no candidates are omitted. The variable-row potential and its minimum over rows are

$$\begin{aligned} \mathcal{E}_T &= F_0 + \sum_i \left[\sum_{e \in \mathcal{I}_i} \beta_e C_e + T_i D_{\mathrm{KL}}(\beta_i \| \pi_i) \right], \\ Z_i &= \sum_{e \in \mathcal{I}_i} \pi_e e^{-C_e/T_i}, \qquad \beta_e^* = \frac{\pi_e e^{-C_e/T_i}}{Z_i}, \qquad E_T = F_0 - \sum_i T_i \log Z_i. \end{aligned} \tag{19.13}$$

All reference laws, kernels, priors, and temperatures are fixed in the variation. The dictionary expectation is outside the message divergence. Averaging receiver laws before taking the divergence defines a different cost.

On regular dominated charts with positive message densities and integrable envelopes justifying differentiation, eq. (4.10) gives

$$dC_e = \int \Delta_e dr_e + \int r_e(d\omega_e) \left[\int \log(A_e/B_e) dA_e - \int (A_e/B_e) dB_e \right]. \tag{19.14}$$

Here $dA_e = (dq_i)\mathcal{K}_e^{\mathrm{rec}}(\omega_e)$ and $dB_e = (dq_j)\mathcal{K}_e^{\mathrm{src}}$. The term in dr_e accounts for changing dictionary recognition. In density notation its own-block first variation is $\delta \mathcal{E}_T / \delta r_e = \log r_e - \mathbb{E}_{r_{-e}} \log \pi_\omega + \beta_e \Delta_e$, up to a sample-independent constant, whenever the displayed conditional expectation and split variation are integrable. This follows by differentiating the product entropy and the expected log joint prior, and retains the prior's correlations. For a parametric Fisher flow the family must additionally have finite positive-definite Fisher tensor. None of these regularity conditions follows from standard Borel measurability alone.

Proposition 4.3 applies to these expected costs and gives

$$\mathcal{E}_T = E_T + \sum_i T_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*), \qquad dE_T = dF_0 + \sum_e \beta_e^* dC_e. \tag{19.15}$$

The second identity differentiates the complete row objective, including entropy. Substitution of $\beta^*(x)$ into the entropy-free weighted sum does not give this derivative.

Definition 19.14 (Tangent reciprocal potential). Let a configuration consist of content laws $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and a factorized dictionary law $r = \bigotimes_e \mathcal{N}(a_e, V_e)$. For an edge $e : j \to i$ put $S_{i,e} = \bar{T}_e \Sigma_i \bar{T}_e^T + R_e$ and $S_{j,e} = \bar{T}_e \Sigma_j \bar{T}_e^T + R_e$, and define the message laws

$$m_e^{\mathrm{rec}}(\omega_e) = \mathcal{N}\big(\bar{T}_e(\mu_i - \kappa_i \omega_e), S_{i,e}\big), \qquad m_e^{\mathrm{src}} = \mathcal{N}\big(\bar{T}_e \mu_j, S_{j,e}\big), \tag{19.16}$$

the law of the message the receiver would send about its own content read in the sender's coordinates, and the law of the sender's message. With fixed positive rows β_e, the tangent reciprocal potential is

$$\mathcal{E} = \sum_i D_{\mathrm{KL}}\big(q_i \| \mathcal{N}(m_i, P_i^{-1})\big) + \sum_e \beta_e \mathbb{E}_r\big[D_{\mathrm{KL}}\big(m_e^{\mathrm{rec}}(\omega_e) \| m_e^{\mathrm{src}}\big)\big] + D_{\mathrm{KL}}(r \| \pi_\omega), \tag{19.17}$$

where π_ω is the normalized dictionary prior with density proportional to $\exp(-\frac{1}{2}\omega^T(\sigma^{-2}I + \lambda B^T W B)\omega - \frac{1}{2}\sum_e \|\omega_e - \hat{\omega}_e\|^2_{G_e})$.

The receiver's law comes first and the source's second, as in a directed peer divergence, and the costs are divergences between message laws, rather than comparisons between independent content samples. The self term replaces the local term $\mathcal{E}_0$ by its tangent form. The last term makes the dictionary laws configuration variables whose part of the potential is a recognition free energy over dictionaries. For normalized variable rows, use $\mathcal{E}_T = \mathcal{E} + \sum_i T_i D_{\mathrm{KL}}(\beta_i \| \pi_i)$ from eq. (19.13). Its Gibbs reduction has the pointwise derivative eq. (19.15), so the derivative statements below apply with the current Gibbs weights inserted. State-dependent weights do not preserve the constant-coefficient mean-flow theorem.

Proposition 19.15 (Forces in the reciprocal potential). Hold the covariances Σ_i and V_e fixed. Let E^S be the global energy eq. (19.2) with every R_e replaced by $S_{j,e}/\beta_e$ and every registered value o_e by zero, with precision J^S and information vector η^S.

(i) $\mathcal{E} = E^S(\mu, a) + c(\Sigma, V)$, where c does not depend on the means.

(ii) The force on a dictionary mean a_e comes only from the cost of its own edge and from π_ω, because no other cost contains ω_e. There is no additional sender-role edge derivative for the dictionary. The force on a content mean contains the terms of the edges on which the agent is the sender, as in the global joint.

(iii) The Fisher gradient flow of the means is $(\dot{\mu}, \dot{a}) = -M J^S\big((\mu, a) - x^S\big)$ with $M = (\bigoplus_i \Sigma_i) \oplus (\bigoplus_e V_e)$ and $x^S = (J^S)^{-1}\eta^S$. Its spectrum is real and negative, and its stationary point is the posterior mean of the global joint in which every record has noise $S_{j,e}/\beta_e$ and registered value zero.

(iv) For $\beta_e = 1$, the content rows of J^S minus the sender terms are the content rows of the per-agent law-level model with $o_e = 0$ at the same covariances. The dictionary rows of J^S differ from the per-agent law-level ones only in that the loop sources of the latter are widened as in eq. (19.4).

Proof. (i) For Gaussian laws $D_{\mathrm{KL}}(\mathcal{N}(\alpha, A)\|\mathcal{N}(\beta', B)) = \frac{1}{2}[\mathrm{tr}(B^{-1}A) - r + (\alpha-\beta')^T B^{-1}(\alpha-\beta') + \log\det B - \log\det A]$. Under r the mean of m_e^{rec} is $\bar{T}_e(\mu_i - \kappa_i a_e) - \bar{T}_e\kappa_i(\omega_e - a_e)$, so the averaged cost is

$$\tfrac{1}{2}\big[\mathrm{tr}(S_{j,e}^{-1}S_{i,e}) - n_e + d_e^T S_{j,e}^{-1} d_e + \mathrm{tr}(S_{j,e}^{-1}\bar{T}_e\kappa_i V_e \kappa_i^T \bar{T}_e^T) + \log\det S_{j,e} - \log\det S_{i,e}\big] \tag{19.18}$$

with $d_e = \bar{T}_e(\mu_i - \mu_j - \kappa_i a_e) = L_e(\mu, a)$. The self term is $\frac{1}{2}(\mu_i - m_i)^T P_i(\mu_i - m_i)$ plus covariance terms. Since r is a product and $\mathbb{E}_r[\omega^T A\omega] = a^T Aa + \mathrm{tr}(AV)$, $D_{\mathrm{KL}}(r\|\pi_\omega)$ is $\frac{1}{2}a^T(\sigma^{-2}I + \lambda B^T W B)a + \frac{1}{2}\sum_e (a_e - \hat{\omega}_e)^T G_e (a_e - \hat{\omega}_e)$ plus covariance terms and a constant. The mean-dependent terms sum to E^S. (ii) In eq. (19.17) the variable ω_e appears only in the cost of e and in π_ω. (iii) The Fisher metric of q_i in μ_i is Σ_i^{-1} and that of $\mathcal{N}(a_e, V_e)$ in a_e is V_e^{-1}, so the Fisher gradient is $M\nabla E^S = MJ^S((\mu, a) - x^S)$, and MJ^S is similar to $M^{1/2}J^S M^{1/2}$, which is positive definite because the self terms and the anchor are. (iv) In the per-agent law-level model $\tilde{R}_e = R_e + \bar{T}_e\Sigma_j\bar{T}_e^T = S_{j,e}$; the rest is proposition 19.9(iii). □

At fixed covariances and fixed rows, the reciprocal potential has the same mean-dependent quadratic as a particular widened Gaussian global model. This identity does not identify their covariance objectives, their recognition families, or their full dynamics. A dictionary occurs only in its own edge cost and in the declared prior, so other dictionary means enter its derivative only through the loop term. This locality is a dependency property of the declared fixed codes and prior; it persists for other laws with the same factor structure. The explicit quadratic and spectral conclusions use Gaussian laws and fixed covariances.

Proposition 19.16 (Disagreement and certainty).

(i) In the Gaussian tangent forms of the global joint and of both per-agent models, the optimizing recognition covariances do not depend on the registered records or on any mean. This is a feature of linear-Gaussian models, whose posterior covariance never depends on the data, and not a property of the free energies.

(ii) In the tangent reciprocal potential, the derivative of the unweighted cost of an edge $e : j \to k$ with respect to the sender's covariance Σ_j is

$$\tfrac{1}{2}\bar{T}_e^T S_{j,e}^{-1}\big[S_{j,e} - S_{k,e} - d_e d_e^T - \bar{T}_e\kappa_k V_e \kappa_k^T \bar{T}_e^T\big] S_{j,e}^{-1}\bar{T}_e, \tag{19.19}$$

so the source-role force on the sender's covariance vanishes exactly when the sender's message covariance equals the receiver's predicted message covariance plus the outer product of the coded disagreement and the reinterpretation covariance.

(iii) Let $K = 1$, $\bar{T}_e = 1$, $\beta_e = 1$, and hold the dictionary deterministic, so $V_e = 0$. Hold the means, $R > 0$, and receiver variance v_k fixed, and let the sender's variance-dependent objective consist only of its self term with precision p_j and one outgoing edge cost. The stationary

sender variance v solves $p_j - v^{-1} + (v+R)^{-1} - (v_k + R + d^2)(v+R)^{-2} = 0$, and at a nondegenerate minimum it increases strictly with d^2.

Proof. (i) The global covariances are J_{ii}^{-1} and the per-agent ones are A_i^{-1}; these matrices depend on the precisions and on the peers' covariances only. (ii) The terms of the averaged cost that contain $S_{j,e}$ are $\frac{1}{2}[\operatorname{tr}(S_{j,e}^{-1}C) + \log\det S_{j,e}]$ with $C = S_{k,e} + d_e d_e^T + \bar{T}_e \kappa_k V_e \kappa_k^T \bar{T}_e^T$; their derivative with respect to $S_{j,e}$ is $\frac{1}{2}(S_{j,e}^{-1} - S_{j,e}^{-1} C S_{j,e}^{-1})$, and $S_{j,e} = \bar{T}_e \Sigma_j \bar{T}_e^T + R_e$ gives eq. (19.19) by the chain rule. (iii) The displayed function is twice the derivative of $\mathcal{E}$ in v; its derivative with respect to d^2 is $-(v+R)^{-2} < 0$, and at a nondegenerate minimum its derivative in v is positive, so the implicit function theorem gives $dv/d(d^2) > 0$. □

For a regular exponential-family source q_η with sufficient statistic t, differentiation gives $\partial_\eta D_{\mathrm{KL}}(p\|q_\eta) = \mathbb{E}_{q_\eta} t - \mathbb{E}_p t$ whenever the moments and derivatives exist. Thus source-role stationarity is a moment-matching condition. The Gaussian covariance formula is its second-moment instance, including the squared discrepancy about the sender's mean. Its contribution is multiplied by β_e in the full potential. The scalar result proves increasing stationary variance with disagreement only while the other specified quantities are frozen. Competing self terms, receiver roles, other source roles, moving means, and changing rows determine the full covariance velocity, so no monotone loss of certainty is asserted for that flow. General non-Gaussian Bayesian posteriors may also change their uncertainty when records change.

For product dictionary laws in definition 19.14, covariance stationarity gives $V_e^{-1} = \sigma^{-2} I + \lambda (B^T W B)_{ee} + G_e + \beta_e \kappa_i^T \bar{T}_e^T S_{j,e}^{-1} \bar{T}_e \kappa_i$. The grounded and record information vanish on N_e. When $W_\gamma = I$, the loop diagonal block is the number of simple loops containing e times the identity, so N_e is invariant under this precision. In the single-loop setting of proposition 19.12, the variance on an unread coordinate is $(\sigma^{-2} + \lambda)^{-1}$, the same value as factorized global recognition. No general marginal-variance formula for overlapping loops is implied.

19.7 Coupled means, covariances, and rows

All content and dictionary parameters now vary within the Gaussian product family. The codes and operating contents remain fixed. Identify $\mathfrak{g}$ with its $d_\omega = K^2$ reference-coordinate vector space and write

$$\begin{aligned} &A_\omega = \sigma^{-2} I + \lambda B^T W B + \operatorname{diag}(G_e), \quad b_\omega = (G_e \hat{\omega}_e)_e, \quad a_0 = A_\omega^{-1} b_\omega, \\ &K_e = \bar{T}_e \kappa_i, \qquad\qquad J_e = S_{j,e}^{-1}, \qquad Q_e = S_{i,e} + d_e d_e^T + K_e V_e K_e^T. \end{aligned} \tag{19.20}$$

Thus $\pi_\omega = \mathcal{N}(a_0, A_\omega^{-1})$ is a normalized joint prior. Its recognition covariance is $V = \operatorname{diag}(V_e)$ and its mean is $a = (a_e)_e$. The expected edge cost and dictionary KL are

$$\begin{aligned} C_e &= \tfrac{1}{2}[\operatorname{tr}(J_e Q_e) - n_e + \log\det S_{j,e} - \log\det S_{i,e}], \\ D_{\mathrm{KL}}(r\|\pi_\omega) &= \tfrac{1}{2}[(a-a_0)^T A_\omega (a-a_0) + \operatorname{tr}(A_\omega V) - \log\det(A_\omega V) - d_{\mathrm{tot}}], \end{aligned} \tag{19.21}$$

where $d_{\mathrm{tot}} = |\mathcal{E}| d_\omega$. Since V is block diagonal, $\operatorname{tr}(A_\omega V) = \sum_e \operatorname{tr}((A_\omega)_{ee} V_e)$ and $\log\det(A_\omega V) = \log\det A_\omega + \sum_e \log\det V_e$. Off-diagonal prior blocks still couple the dictionary means.

Proposition 19.17 (Complete Gaussian Fisher equations). Use the fixed data of definition 19.1 and positive rows and temperatures of definition 19.13. Let g_{μ_i}, g_{a_e} be mean derivatives and G_{Σ_i}, G_{V_e} symmetric covariance derivatives under the trace pairing. In the full flow β is independent; in the reduced flow it is $\beta^*(x)$. Then

$$\begin{aligned} g_{\mu_i} &= P_i(\mu_i - m_i) + \sum_{e\in \mathrm{in}(i)} \beta_e \bar{T}_e^T J_e d_e - \sum_{e\in \mathrm{out}(i)} \beta_e \bar{T}_e^T J_e d_e, \\ g_{a_e} &= (A_\omega a - b_\omega)_e - \beta_e K_e^T J_e d_e, \\ 2G_{\Sigma_i} &= P_i - \Sigma_i^{-1} + \sum_{e\in \mathrm{in}(i)} \beta_e \bar{T}_e^T (J_e - S_{i,e}^{-1}) \bar{T}_e \\ &\quad + \sum_{e\in \mathrm{out}(i)} \beta_e \bar{T}_e^T (J_e - J_e Q_e J_e) \bar{T}_e, \\ 2G_{V_e} &= (A_\omega)_{ee} - V_e^{-1} + \beta_e K_e^T J_e K_e. \end{aligned} \tag{19.22}$$

Here $\mathrm{out}(i)$ consists of edges on which i is the source. The unit-mobility product Fisher equations are

$$\begin{aligned} \dot{\mu}_i &= -\Sigma_i g_{\mu_i}, \quad \dot{\Sigma}_i = -2\Sigma_i G_{\Sigma_i} \Sigma_i, \\ \dot{a}_e &= -V_e g_{a_e}, \quad \dot{V}_e = -2V_e G_{V_e} V_e. \end{aligned} \tag{19.23}$$

For full row relaxation, append

$$h_e = C_e + T_i(\log(\beta_e/\pi_e) + 1), \qquad \dot{\beta}_e = -\beta_e \left(h_e - \sum_{f\in \mathcal{I}_i} \beta_f h_f \right). \tag{19.24}$$

For Gibbs reduction this row ODE is absent. Its kinetic metric is the matter-family Fisher metric, not the pullback of the full metric to $x \mapsto (x, \beta^*(x))$.

Proof. Differentiate eq. (19.21). Its mean derivatives are $\bar{T}_e^T J_e d_e$, $-\bar{T}_e^T J_e d_e$, and $-K_e^T J_e d_e$ for receiver, source, and dictionary. Varying $S_{i,e}$ gives $(J_e - S_{i,e}^{-1})/2$. Using $dJ_e = -J_e(dS_{j,e})J_e$ gives $(J_e - J_e Q_e J_e)/2$ in the source covariance; varying V_e gives $K_e^T J_e K_e/2$. Add the self KL derivatives and the prior derivatives $A_\omega a - b_\omega$ and $((A_\omega)_{ee} - V_e^{-1})/2$ to obtain eq. (19.22). For a Gaussian perturbation (u, U) the Fisher quadratic form is $u^T\Sigma^{-1}u + \mathrm{tr}(\Sigma^{-1}U\Sigma^{-1}U)/2$, obtained by averaging the product of the Gaussian scores. Inverting these blocks gives eq. (19.23), as in eq. (4.17). The categorical score metric is $\sum_e (d\beta_e)^2/\beta_e$ on zero-sum tangents, whose gradient centers h_e as in eq. (19.24). Finally eq. (19.15) gives the reduced derivatives without an extra $d\beta^*$ term. □

The full energy dissipation is

$$\begin{aligned} -\dot{\mathcal{E}}_T = &\sum_i [g_{\mu_i}^T \Sigma_i g_{\mu_i} + 2\,\mathrm{tr}(\Sigma_i G_{\Sigma_i} \Sigma_i G_{\Sigma_i})] \\ &+ \sum_e [g_{a_e}^T V_e g_{a_e} + 2\,\mathrm{tr}(V_e G_{V_e} V_e G_{V_e})] + \sum_i \mathrm{Var}_{\beta_i}(h). \end{aligned} \tag{19.25}$$

Each covariance trace is nonnegative: $\operatorname{tr}(\Sigma G \Sigma G) = \operatorname{tr}((\Sigma^{1/2} G \Sigma^{1/2})^2)$. For E_T the same identity omits the last sum and uses the current Gibbs weights. This does not give a sign to an individual edge cost or covariance velocity.

Since every Fisher block is positive definite, stationarity sets all four derivatives in eq. (19.22) to zero, and full row stationarity also imposes $\beta = \beta^*$. In particular,

$$\begin{aligned} V_e^{-1} &= (A_\omega)_{ee} + \beta_e K_e^T J_e K_e, \\ [A_\omega + \operatorname{diag}(\beta_e K_e^T J_e K_e)]a &= b_\omega + (\beta_e K_e^T J_e \bar{T}_e(\mu_i - \mu_j))_e. \end{aligned} \tag{19.26}$$

The latter system determines a unique dictionary mean for fixed content means, content covariances, and rows. It does not prove uniqueness of the complete stationary configuration, because the rows and source precisions depend on that configuration.

Theorem 19.18 (Global flow in the Gaussian tangent family). Fix a finite graph, fixed codes and operating maps κ_i, proper Gaussian content references with $P_i \succ 0$, a fixed normalized dictionary prior with $A_\omega \succ 0$, message noises $R_e \succ 0$, and positive fixed normalized source priors and temperatures. From every finite initial mean, positive-definite content and dictionary covariance, and interior initial row, eqs. (19.23) and (19.24) has a unique solution for every $t \geq 0$. The means remain bounded, all covariance eigenvalues remain bounded above and away from zero, and all row entries have positive uniform lower bounds depending on the initial state and fixed data. The Gibbs-reduced convention has the same conclusions. Both potentials have an interior stationary point; along either flow the squared Fisher-gradient norm is integrable and tends to zero, and every accumulation point is stationary.

Proof. The vector fields are smooth on the product of mean spaces, positive-definite covariance cones, and simplex interiors, so they have unique maximal interior solutions. All terms of $\mathcal{E}_T$ are nonnegative. Since $C_e \geq 0$ and the priors are normalized, $0 < Z_i \leq 1$, so $E_T \geq F_0 \geq 0$ as well. Energy dissipation therefore bounds each content self KL and the dictionary KL by the initial energy.

If λ_k are the eigenvalues of $P_i^{1/2} \Sigma_i P_i^{1/2}$, the content KL is

$$\tfrac{1}{2}\|\mu_i - m_i\|_{P_i}^2 + \tfrac{1}{2} \sum_k (\lambda_k - \log \lambda_k - 1). \tag{19.27}$$

The scalar summand is nonnegative and diverges at zero and infinity. Thus the means are bounded and the covariances are spectrally bounded in the interior. The same argument applies to the eigenvalues of $A_\omega^{1/2} V A_\omega^{1/2}$ and the mean $a - a_0$ in eq. (19.21). Fixed positive-definite A_ω converts these to bounds on V, and hence on every V_e. The product-covariance restriction is closed within these bounds. The matter variables therefore stay in a compact interior set $\mathcal{K}$, on which every continuous cost lies in $[0, M]$ for some finite M.

For Gibbs rows, $\beta_e^* \geq \pi_e e^{-M/T_i}$. For a full row put $\eta_{ef} = \log(\beta_e/\pi_e) - \log(\beta_f/\pi_f)$. Subtracting eq. (19.24) in logarithmic form gives

$$\dot{\eta}_{ef} = -T_i \eta_{ef} - (C_e - C_f), \qquad |\eta_{ef}(t)| \leq \max\{|\eta_{ef}(0)|, M/T_i\}. \tag{19.28}$$

The bound follows by variation of constants. Let L_i be the maximum of these bounds over the row. With $u_e = \beta_e/\pi_e$, one has $u_f \le e^{L_i} u_e$ and $\sum_f \pi_f u_f = 1$, so $\beta_e \ge \pi_e e^{-L_i} > 0$. Singleton rows are identically one. The full trajectory stays in a compact subset of its smooth domain; ordinary finite-dimensional continuation excludes a finite maximal time. The reduced field is smooth on $\mathcal{K}$ and has the same conclusion.

The reduced potential attains a global minimum on a nonempty compact sublevel by eq. (19.27); the minimum lies in the interior. Its Gibbs rows give a full stationary point by eq. (19.15). Finally let $\mathcal{D}$ be the right side of eq. (19.25), with the row term omitted in the reduced convention. Integrating dissipation gives $\int_0^\infty \mathcal{D}(t)dt \le E_{\mathrm{init}}$. On the compact trajectory closure $\mathcal{D}$ and its derivative along the smooth vector field are bounded, so $\mathcal{D}$ is uniformly continuous. A nonnegative uniformly continuous integrable function tends to zero: otherwise disjoint intervals of fixed positive width would each contribute a fixed positive amount to its integral. Continuity and the positive-definite Fisher metric then make every accumulation point stationary. □

These bounds need not be uniform as temperature tends to zero or reference precisions become singular. The theorem concerns a fixed finite Gaussian family; it neither proves uniqueness of the stationary point nor asserts convergence to one particular point. Its hypotheses exclude moving operating contents and changes of the reference laws.

Proposition 19.19 (Source adaptation and the complete Hessian). On any regular finite-dimensional matter family where F_0 and all C_e are twice differentiable, let $c_e = \nabla C_e$ and $\bar{c}_i = \sum_{e\in\mathcal{I}_i} \beta_e^* c_e$. The reduced Hessian is

$$H_{\mathrm{red}} = \nabla^2 F_0 + \sum_e \beta_e^* \nabla^2 C_e - \sum_i T_i^{-1} \left[\sum_{e\in\mathcal{I}_i} \beta_e^* c_e c_e^T - \bar{c}_i \bar{c}_i^T \right]. \tag{19.29}$$

At a critical point the full and reduced Hessians have the same negative and zero indices. With either positive-definite Fisher metric, a positive-definite Hessian gives local exponential stability, while a negative direction gives instability. The linearized eigenvalues are real; a zero Hessian mode requires a separate nonlinear analysis.

Proof. Differentiating the normalized Gibbs row gives $d\beta_e^*[\xi] = -\beta_e^*(dC_e[\xi] - \sum_f \beta_f^* dC_f[\xi])/T_i$. Differentiate eq. (19.15) once more to obtain eq. (19.29). For a row tangent v_i with $\sum_e v_e = 0$, write $z_e = dC_e[\xi]$ and $\bar{z}_i = \sum_e \beta_e^* z_e$. At a critical point, direct completion of squares gives

$$\delta^2 \mathcal{E}_T[(\xi, v), (\xi, v)] = \xi^T H_{\mathrm{red}} \xi + \sum_i T_i \sum_{e\in\mathcal{I}_i} \frac{[v_e + \beta_e^*(z_e - \bar{z}_i)/T_i]^2}{\beta_e^*}. \tag{19.30}$$

The shift of v_i has zero sum, so this is an invertible tangent change of variables plus a positive row form. It is the Schur-complement identity behind proposition 4.9 and Haynsworth's inertia relation [73]. At a stationary point the flow Jacobian is $-g^{-1}H$, because the metric derivative multiplies a vanished gradient. It is similar to the symmetric matrix $-g^{-1/2} H g^{-1/2}$. Positive definiteness gives a negative spectrum and local exponential stability; a negative Hessian direction gives a positive eigenvalue and instability. A zero eigenvalue is not classified by this argument. □

The last term of eq. (19.29) lowers the quadratic form relative to rows frozen at the same state and weights, by $-\sum_i T_i^{-1}\operatorname{Var}_{\beta_i^*}(dC_e[\xi])$ in direction ξ. This is not a general ordering in temperature: changing temperature usually changes the weights and the stationary point. Nor does finite row relaxation create a Hopf bifurcation at an interior stationary point of this exact gradient system.

19.8 An exact stability threshold with changing uncertainty

The following example keeps every content and dictionary mean and variance dynamic. It shows why the fixed-covariance mean theorem does not decide stability after source selection and uncertainty are allowed to respond.

Theorem 19.20 (A three-agent coupled threshold). Take three scalar content laws $q_i = \mathcal{N}(\mu_i, v_i)$, $i = 0, 1, 2$, and two scalar dictionary laws $r_j = \mathcal{N}(a_j, V_j)$ on edges $j \to 0$, $j = 1, 2$. Set $\bar{T}_e = 1$, $R_e = 1$, $\kappa_0 = 1$, and $\lambda = 0$. Use content reference precisions $(P_0, P_1, P_2) = (1, 5/4, 5/4)$ and means $(m_0, m_1, m_2) = (0, -23/15, 23/15)$, and independent dictionary references $\mathcal{N}(0, 4/3)$. Receiver 0 has exactly the two peer candidates, with prior $(1/2, 1/2)$ and temperature $T > 0$; no identity self candidate is added to this row. The source agents may have singleton identity self rows, with zero contribution. In the coordinate order

$$x = (\mu_0, \mu_1, \mu_2, a_1, a_2, v_0, v_1, v_2, V_1, V_2), \tag{19.31}$$

the point

$$x_* = (0, -4/3, 4/3, 1/3, -1/3, 1, 1, 1, 1, 1), \qquad \beta_* = (1/2, 1/2) \tag{19.32}$$

is stationary for every $T > 0$. For both the Gibbs-reduced flow and full unit-mobility row relaxation it is locally exponentially stable when $T > T_c = 3269/10118$, and a hyperbolic saddle with one unstable direction when $0 < T < T_c$. At $T = T_c$ the Hessian has exactly one zero mode. If all five variances are instead held at one, the mean-only stability threshold is $T_{\text{mean}} = 31/122 < T_c$.

Proof. Put $S_i = v_i + 1$ and $d_j = \mu_0 - \mu_j - a_j$. The costs and full objective are

$$\begin{aligned} C_j &= \tfrac{1}{2}[(S_0 + d_j^2 + V_j)/S_j - 1 + \log S_j - \log S_0], \\ \mathcal{E}_T(x, b) &= F_0(x) + bC_1 + (1-b)C_2 + T[b\log(2b) + (1-b)\log(2(1-b))], \end{aligned} \tag{19.33}$$

where F_0 is the sum of the five Gaussian self KL terms with the references just specified. At x_*, $S_i = 2$, $d_1 = 1$, $d_2 = -1$, and $C_1 = C_2 = 1/2$, so $b_* = 1/2$. The two source self mean derivatives are $(1/4, -1/4)$ and their weighted edge derivatives are $(-1/4, 1/4)$; the dictionary mean derivatives have the same cancellation, and the two receiver mean forces cancel each other. Each source variance has self and weighted edge derivatives $1/8$ and $-1/8$, each dictionary variance has $-1/8$ and $1/8$, and the receiver variance derivatives vanish. Thus all ten matter derivatives vanish.

Let $H = \nabla^2(F_0 + (C_1 + C_2)/2)(x_*)$ and $g = \nabla(C_1 - C_2)(x_*)$. Separate the first and last five coordinates by a semicolon; then

$$g = (1, -1/2, -1/2, -1/2, -1/2; 0, -1/4, 1/4, 1/4, -1/4)^T. \tag{19.34}$$

For an arbitrary perturbation $\xi = (u_0, u_1, u_2, b_1, b_2, h_0, h_1, h_2, k_1, k_2)$, set $z_j = u_0 - u_j - b_j$ and $\epsilon_1 = 1$, $\epsilon_2 = -1$. Direct differentiation of eq. (19.33) gives

$$\begin{aligned}\xi^T H \xi = u_0^2 + \tfrac{5}{4}(u_1^2 + u_2^2) + \tfrac{3}{4}(b_1^2 + b_2^2) + \tfrac{1}{2}h_0^2 \\ + \sum_{j=1}^{2} \left[\tfrac{1}{4}(z_j - \epsilon_j h_j/2)^2 + \tfrac{1}{16}(h_0 - h_j)^2 + \tfrac{71}{128}h_j^2 + \tfrac{1}{2}(k_j - h_j/8)^2\right].\end{aligned} \tag{19.35}$$

For completeness, one weighted edge contributes $z_j^2/4 - \epsilon_j z_j h_j/4 + h_0^2/16 - h_0 h_j/8 + 3h_j^2/16 - h_j k_j/8$. Adding the self mean terms and the five self variance terms $\sum_i h_i^2/2 + \sum_j k_j^2/2$ and completing squares gives eq. (19.35). Its vanishing forces all ten components of ξ to vanish, proving $H \succ 0$ without restricting the tangent space.

Solving the linear equations specified by this quadratic form gives, in the same two blocks,

$$\begin{aligned}(H^{-1}g)_{\mu,a} &= (2400, -960, -960, -1600, -1600)^T/5059, \\ (H^{-1}g)_{v,V} &= (0, -716, 716, 2440, -2440)^T/5059, \\ g^T H^{-1} g &= 6538/5059.\end{aligned} \tag{19.36}$$

These rational values are checked by multiplying with the symmetric matrix of eq. (19.35). At the equal-cost point, eq. (19.29) becomes $H_T = H - gg^T/(4T)$. Congruence by $H^{-1/2}$ gives the identity minus one rank-one matrix, with nine positive eigenvalues and the remaining sign determined by $1 - g^T H^{-1} g/(4T)$. Hence $T_c = g^T H^{-1} g/4 = 3269/10118$, with the stated positive, negative, and zero indices.

The inverse matter Fisher metric is

$$M(x) = \operatorname{diag}(v_0, v_1, v_2, V_1, V_2, 2v_0^2, 2v_1^2, 2v_2^2, 2V_1^2, 2V_2^2). \tag{19.37}$$

For full row relaxation the inverse metric is $\operatorname{diag}(M(x), b(1-b))$ and the stationary Hessian is

$$\mathbb{H}_T = \begin{pmatrix} H & g \\ g^T & 4T \end{pmatrix}, \qquad (\xi, \delta b)^T \mathbb{H}_T (\xi, \delta b) = \xi^T H_T \xi + 4T(\delta b + g^T \xi/(4T))^2. \tag{19.38}$$

It has the same negative and zero indices, with an additional positive direction. Proposition 19.19 therefore gives the asserted stability classifications in all ten or eleven coordinates.

To freeze the variances, set $h_i = k_j = 0$ in eq. (19.35). Put $t = u_0 - (u_1 + u_2 + b_1 + b_2)/2 = g_{\text{mean}}^T \xi_{\text{mean}}$. Convexity and exchange of the two sources put the minimum of the mean quadratic at fixed t in $u_1 = u_2 = y$, $b_1 = b_2 = z$. Minimizing there gives $(u_0, y, z) = (15, -6, -10)t/31$ and value $61t^2/62$. The minimum of $\xi^T H_{\text{mean}} \xi$ under $g_{\text{mean}}^T \xi = t$ is also $t^2/(g_{\text{mean}}^T H_{\text{mean}}^{-1} g_{\text{mean}})$ by completing a positive quadratic square. Thus the inverse quadratic form is $62/61$ and $T_{\text{mean}} = 31/122$. □

The exact inequality $31/122 < 3/10 < 3269/10118$ means that at $T = 3/10$ the frozen-variance mean system is stable while both complete coupled systems have one unstable direction. At T_c the reduced null vector is $H^{-1}g$, and the full null vector is $(H^{-1}g, -1)$. Source and dictionary uncertainty participate in the zero linear mode. The following nonlinear calculation classifies that mode within this Gaussian tangent model.

19.9 The local subcritical pitchfork

Retain all hypotheses, references, coordinates, and Fisher mobilities of theorem 19.20. Write $s = b - 1/2$, $\delta = T - T_c$, and

$$\begin{aligned} A(x) &= F_0(x) + (C_1(x) + C_2(x))/2, \qquad \Delta(x) = C_1(x) - C_2(x), \\ G(x,s) &= A(x) + s\Delta(x), \\ h(s) &= (1/2+s)\log(1+2s) + (1/2-s)\log(1-2s). \end{aligned} \tag{19.39}$$

Thus $\mathcal{E}_T = G + Th$. The scalar s is the actual row imbalance for full relaxation, and the Gibbs row imbalance $s = b^*(x,T) - 1/2$ for the reduced flow. Both reductions below use the original time variable.

Theorem 19.21 (Critical dynamics and nearby branches). There are local parameter-dependent invariant center manifolds on which

$$\dot{s} = -\frac{4\delta s + cs^3}{d} + O(\delta^2|s| + |\delta||s|^3 + |s|^5), \qquad d = \begin{cases} D, & \text{Gibbs reduction,} \\ D+4, & \text{full row relaxation,} \end{cases} \tag{19.40}$$

where the remainder is uniform near $(s,\delta) = (0,0)$ and

$$c = -\frac{657534393590529472}{6223404588394626861} < 0, \qquad D = \frac{19189456}{25593481} > 0. \tag{19.41}$$

In sufficiently small state and temperature neighborhoods, the symmetric equilibrium is the only equilibrium for $\delta \le 0$. For each sufficiently small $\delta > 0$ there are exactly two additional equilibria, common to the full and reduced systems, with

$$s_\pm(\delta) = \pm\sqrt{-4\delta/c} + O(\delta^{3/2}). \tag{19.42}$$

Each asymmetric equilibrium has one unstable direction, with eigenvalue $8\delta/d + O(\delta^2)$, and nine reduced or ten full stable directions. At $T = T_c$ the symmetric equilibrium is nonlinearly unstable. With destabilizing parameter $T_c - T$, this is a subcritical pitchfork; it creates no locally stable asymmetric branch.

Proof. The reflection and source exchange

$$Rx = (-\mu_0, -\mu_2, -\mu_1, -a_2, -a_1, v_0, v_2, v_1, V_2, V_1) \tag{19.43}$$

fixes x_*, preserves A, and sends Δ to $-\Delta$. The mobility satisfies $M(Rx) = RM(x)R^T$. Since $H = A_{xx}(x_*) \succ 0$, the implicit-function theorem gives a unique analytic stationary matter curve $X(s)$ near x_* with

$$G_x(X(s),s) = 0, \qquad X(0) = x_*, \qquad p = X'(0) = -H^{-1}g. \tag{19.44}$$

Uniqueness gives $X(-s) = RX(s)$. The stationary scalar potential $\Phi(s,T) = G(X(s),s) + Th(s)$ is therefore even. This curve solves the matter stationary equations at each fixed row; invariance under either flow has not been asserted.

All tensor derivatives in the next calculation are at x_*. Put $k = A_{xxx}[p,p] + 2\Delta_{xx}p$. Differentiating eq. (19.44) twice gives $X''(0) = -H^{-1}k$. Four derivatives of Φ then give

$$\Phi_{ssss}(0,T_c) = A_{xxxx}[p,p,p,p] + 4\Delta_{xxx}[p,p,p] - 3k^T H^{-1}k + 32T_c = 6c. \tag{19.45}$$

For the explicit costs in eq. (19.33) and the tangent p in eq. (19.36), the three matter contractions are

$$\begin{aligned} A_{xxxx}[p,p,p,p] &= \frac{236914859731152}{655026269697361}, \\ \Delta_{xxx}[p,p,p] &= -\frac{31633266640}{129477420379}, \\ 3k^T H^{-1}k &= \frac{21485600170644766720}{2074468196131542287}. \end{aligned} \tag{19.46}$$

These values follow by differentiating the rational costs and Gaussian self terms; together with $32T_c$ they give eq. (19.41). The negative stable-response term in eq. (19.45) would be absent if one restricted the potential to the straight line $x_* + ps$. Since $\Phi_{ss}(0,T) = 4(T - T_c)$ and h is even,

$$\Phi_s(s, T_c + \delta) = 4\delta s + cs^3 + O(|\delta||s|^3 + |s|^5). \tag{19.47}$$

Analyticity and evenness give $\Phi_s = sQ(s^2,\delta)$ with $Q(0,\delta) = 4\delta$ and $Q_z(0,0) = c < 0$. A second implicit-function argument gives the unique small root $z(\delta) = -4\delta/c + O(\delta^2)$. It is positive precisely for small positive δ, proving the complete local equilibrium count and eq. (19.42). At these points G_{xx} remains positive definite, whereas $\Phi_{ss} = -8\delta + O(\delta^2) < 0$. The latter is its Schur complement in the full stationary Hessian. Thus that Hessian has one negative and ten positive directions. Eliminating the positive row block gives one negative and nine positive directions in the reduced Hessian. Positive Fisher mobilities preserve these stability indices.

It remains to compute the actual center velocity. The fields are analytic on the interior and equivariant under the orthogonal reflection above. At T_c each has a simple zero eigenvalue and all remaining eigenvalues are negative. The parameter-dependent equivariant center-manifold theorem [126, Theorem 1.2] therefore supplies a local C^7 invariant reduction containing the nearby equilibria; no analytic center manifold is required. The full coordinate $s = b - 1/2$ is regular on its center tangent $(p, 1)$. For the reduced flow,

$$D_x s^*(x_*, T_c)p = -g^T p/(4T_c) = 1, \tag{19.48}$$

so its Gibbs imbalance also supplies a regular coordinate. Reflection makes the scalar field $f(s,\delta) = \dot{s}$ odd in s.

Write $Y(s,\delta)$ for the matter coordinates on either invariant manifold. Its matter equation is

$$Y_s f = -M(Y)G_x(Y,s). \tag{19.49}$$

At $\delta = 0$, $Y_s(0,0) = p$ and $f = O(s^3)$, so comparison through quadratic order gives $Y_{ss}(0,0) = X''(0)$. Set $B = H^{-1}M_*^{-1}p$, with $M_* = M(x_*)$, and let $w = Y - X$. Then

$$D = p^T M_*^{-1} p = \frac{19189456}{25593481}, \qquad g^T B = -D. \tag{19.50}$$

If $f = \alpha\delta s + \eta s^3 + O(\delta^2|s| + |\delta||s|^3 + |s|^5)$, the coefficients of eq. (19.49) give $w_{\delta s} = -\alpha B$ and $w_{s^3} = -\eta B$. Variation of M contributes only at higher orders. Consequently the row force $\Delta(Y) + (T_c + \delta)h'(s)$ has coefficients $4 + D\alpha$ and $c + D\eta$ at δs and s^3.

For the reduced flow this force is identically zero, so $\alpha = -4/D$ and $\eta = -c/D$. Full relaxation instead obeys

$$f = -(1/4 - s^2)[\Delta(Y) + (T_c + \delta)h'(s)], \tag{19.51}$$

giving $\alpha = -(4 + D\alpha)/4$ and $\eta = -(c + D\eta)/4$. These yield the denominator $D + 4$ in eq. (19.40). Odd C^7 regularity gives its stated uniform remainder. In particular,

$$Y(s, 0) - X(s) = \frac{c}{d} B s^3 + O(s^4), \tag{19.52}$$

which exhibits the nonzero difference between stationary elimination and dynamical invariance.

At either asymmetric branch, differentiation of the center field gives $8\delta/d + O(\delta^2) > 0$. At criticality it gives $\dot{s} = (-c/d)s^3 + O(s^5)$, pointing away from zero on both sides. Arbitrarily small nonzero center data therefore leave a fixed small interval, proving nonlinear instability at the threshold. The stable transverse spectrum persists from the Hessian calculation. □

The additional 4 is the Fisher squared length of the row tangent at $b = 1/2$. Thus full relaxation has $D/(D + 4) = 4797364/30390845$ times the leading center speed of matter-metric Gibbs reduction, or a leading time scale about 6.34 times longer. This comparison identifies the leading scalar polynomials after a constant time rescaling; it does not identify higher-order trajectories or stable transients. Distant stable equilibria, folds, and the destinations of trajectories leaving this neighborhood require separate arguments. The local result establishes neither monotone uncertainty growth, hysteresis, grouping, nor formation of an inherited parent.

19.10 Scope and open problems

Open problem 19.22 (Coupled dynamics beyond the finite example). Proposition 19.17 and theorem 19.18 give the complete Gaussian tangent flow and global forward existence, and theorem 19.21 resolves the local critical mode of the three-agent example as a subcritical pitchfork with nearby asymmetric saddles. Determine its distant stationary branches and long-time destinations, the corresponding behavior on graphs with cycles, and well-posedness beyond regular finite-dimensional law families. Neither the scalar certainty result nor this local instability establishes grouping or formation.

Open problem 19.23 (Convergence of per-agent flows). Find conditions on the self, record, and loop precisions under which the per-agent fixed point is unique and its mean flow converges. Determine whether the law-level covariance updates have a unique fixed point. The directed ring supplies one stable non-gradient example, not a general convergence theorem.

Open problem 19.24 (Beyond the tangent model). Keep the product $\omega_e(y_i - c_i)$ and let the operating contents follow the beliefs. Factorized recognition then acquires second-moment terms, and the interpretation gauge of theorem 19.5 becomes a nonlinear orbit. Determine whether grounding still separates belief from meaning at the nonlinear level.

Open problem 19.25 (General laws). The Gaussian tangent model is a computational simplification. Extend its conclusions to general laws, with record kernels that read the coded discrepancy and location-family noise. Four parts are open: the attribution of a coded disagreement among receiver persuasion, sender persuasion and reinterpretation when contents and dictionary priors are not Gaussian; the grounded limit, in which the reinterpretation share should vanish as grounding identifies the dictionaries; whether disagreement changes certainty in the two free energies, and how that Bayesian effect compares with the source-role moment matching of the reciprocal potential; and the echo and cavity statements for non-Gaussian loop factors, where cavity messages are belief propagation without Gaussian closure and the factorized means are no longer exact.

Open problem 19.26 (What agents must exchange). A receiver using a loop prior needs information about dictionaries owned elsewhere. An echo-free protocol must specify which prior contributions are removed from each transmitted law. Formulate that exchange through declared codes and record kernels, and determine the loop information it makes observable.

Chapter 20

Curvature Costs and Supplied Hierarchies

Integrating out contents can couple dictionaries even when the model contains no loop prior. The resulting action is the negative log evidence of the observed records at fixed dictionaries. In a Gaussian tangent model its quadratic part measures coded reinterpretations that changes of contents cannot absorb. Its linear part depends on the records, so this calculation does not imply that flat dictionaries are always preferred. A second construction restricts a supplied hierarchy to rigid clusters. It gives a parent density under explicit normalization hypotheses and a Gaussian minimizer limit, without selecting the clusters or establishing inherited responses and dynamics.

20.1 Contents, records, and supplied parents

Work in one linear sector with reference content space $\mathbb{R}^K$. The graph has finite vertex set V and directed edge set E. An edge $e : j \to i$ has receiver-owned dictionary generator $\omega_e \in \mathfrak{gl}(K, \mathbb{R})$, code $\bar{T}_e : \mathbb{R}^K \to H_e$, and positive-definite noise covariance R_e on the finite-dimensional message space H_e. Fix operating contents c_i and put $\kappa_i \omega = \omega c_i$. The contents $y_i \in \mathbb{R}^K$ have independent self laws $\mathcal{N}(m_i, P_i^{-1})$ with P_i positive definite. These self laws summarize the declared local models and records in this Gaussian example. They are not general definitions of an agent.

The comparison records and global posterior energy are

$$o_e = \bar{T}_e(y_i - y_j - \kappa_i \omega_e) + \varepsilon_e, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e), \tag{20.1}$$

$$\begin{aligned} E(y, \omega) = {} & \frac{1}{2} \sum_i \|y_i - m_i\|^2_{P_i} + \frac{1}{2\sigma^2} \sum_e \|\omega_e\|^2_F \\ & + \frac{1}{2} \sum_e \|o_e - \bar{T}_e(y_i - y_j - \kappa_i \omega_e)\|^2_{R_e^{-1}}, \end{aligned} \tag{20.2}$$

where $\sigma > 0$ and the noises are independent. The matrix anchor is isotropic for the declared reference inner product. Coordinate changes carry this metric with the other constitutive data. There are no grounded dictionary factors and no explicit loop factor: the loop weight is $\lambda = 0$. The tangent approximation drops the product $\omega_e(y_i - c_i)$ in $\bar{T}_e(e^{-\omega_e} y_i - y_j)$. Its operating contents remain fixed.

Let $Y = (\mathbb{R}^K)^{\mathsf{V}}$, $X = \mathfrak{gl}(K, \mathbb{R})^{\mathsf{E}}$, and $H = \bigoplus_e H_e$. Define

$$(\delta y)_e = \bar{T}_e(y_i - y_j), \qquad (\mathcal{V}\omega)_e = \bar{T}_e \omega_e c_i, \qquad P = \bigoplus_i P_i, \qquad R = \bigoplus_e R_e. \tag{20.3}$$

The record term is $\frac{1}{2}\|o-\delta y+\mathcal{V}\omega\|^2_{R^{-1}}$. The map $\delta: Y \to H$ is the coboundary of the code sheaf: the two restrictions to the edge stalk H_e are the same code $\bar{T}_e$. Its first cohomology on the graph, which has no two-cells, is $H^1 = H/\operatorname{im}\delta$ [72].

For a directed closed walk γ, let $(B\omega)_\gamma = \sum_{e\in\gamma}\omega_e$, counting traversals with multiplicity. The corresponding product of dictionaries $\bar{\theta}_e = e^{\omega_e}$ is ordered along the walk, and $B\omega$ is its first-order term at the identity. Opposite attending edges remain independent; traversing a particular transport in geometric reverse would instead use its inverse. Write $\Delta_e = y_i - y_j$. Then $\sum_{e\in\gamma}\Delta_e = 0$ on a directed closed walk.

Definition 20.1 (A supplied hierarchy). Partition the child agents into clusters I, introduce a parent A for each cluster, and give it a self law $\mathcal{N}(m_A, P_A^{-1})$. Add the cross-scale comparison factor

$$\exp\left(-\frac{\tau}{2}\|y_i - y_A - \omega_{iA}c_A\|^2\right), \qquad i\in I, \qquad \tau>0, \tag{20.4}$$

with an isotropic dictionary anchor of variance σ^2. This is the tangent form of the conditional law $y_i \mid y_A, \omega_{iA} \sim \mathcal{N}(e^{\omega_{iA}}y_A, \tau^{-1}I)$ at $y_A = c_A$. It is a comparison factor with identity code and fixed operating vector c_A. Edges between different child clusters are called bridges. The partition, parents, and cross-scale factors are supplied model data.

20.2 The action induced by contents

Remark 20.2 (Evidence gives an induced action for general laws). Let $\pi(d\omega)$ be a dictionary prior, let $p(dy)$ be a content law on a standard Borel space, and let $\ell(o \mid y,\omega)$ be a chosen density of the record kernel at the registered record o. Suppose

$$Z(\omega) = \int \ell(o \mid y,\omega)p(dy), \qquad 0 < \int Z(\omega)\pi(d\omega) < \infty. \tag{20.5}$$

Disintegration of the joint density gives the posterior

$$\pi(d\omega \mid o) = \frac{Z(\omega)\pi(d\omega)}{\int Z(\omega')\pi(d\omega')}. \tag{20.6}$$

Where $0 < Z(\omega) < \infty$, the contents induce the action $-\log Z(\omega)$. This identity requires no Gaussian family. If the record kernels use the dictionaries only through coded discrepancies, that dependence is inherited by Z. Neither monotonicity in a loop defect nor invariance under arbitrary re-readings of contents follows: the content law and the registered records also enter the integral.

Theorem 20.3 (The Gaussian action and its cohomological quadratic part). In the model of eq. (20.2), integrating out y leaves the complete dictionary action

$$S_{\mathrm{marg}}(\omega) = \frac{1}{2\sigma^2}\|\omega\|^2 + \frac{1}{2}\|o+\mathcal{V}\omega-\delta m\|^2_{C_P^{-1}} + \mathrm{const}, \qquad C_P = R + \delta P^{-1}\delta^T. \tag{20.7}$$

Thus the dictionary marginal is Gaussian with precision

$$\Lambda_\omega = \sigma^{-2}I + \mathcal{V}^T C_P^{-1}\mathcal{V}. \tag{20.8}$$

If $P = \epsilon \bar{P}$ with fixed positive-definite $\bar{P}$ and $\epsilon \downarrow 0$, its record precision tends to $\mathcal{V}^T \Pi_H \mathcal{V}$, where

$$\Pi_H = R^{-1} - R^{-1}\delta(\delta^T R^{-1}\delta)^+ \delta^T R^{-1}. \tag{20.9}$$

The superscript $+$ denotes the Moore–Penrose inverse. For every ω,

$$(\mathcal{V}\omega)^T \Pi_H \mathcal{V}\omega = \min_{a \in Y} \|\mathcal{V}\omega - \delta a\|^2_{R^{-1}}. \tag{20.10}$$

This is the squared quotient norm of the class of $\mathcal{V}\omega$ in H^1. It vanishes exactly when $\bar{T}_e \omega_e c_i = \bar{T}_e(a_i - a_j)$ for one $a \in Y$ on every edge.

Suppose in addition that every code is injective, the underlying undirected multigraph is connected, all operating contents equal one $c \neq 0$, and the chains of the declared directed closed walks span its real cycle space. Opposite attending edges count as separate edges of that multigraph. Then the limiting quadratic part vanishes exactly when $(B\omega)_\gamma c = 0$ for every declared walk, and its rank on X is $K(|\mathrm{E}| - |\mathrm{V}| + 1)$. For positive P, an absorbable reinterpretation obeys

$$(\delta a)^T C_P^{-1} \delta a \leq a^T P a. \tag{20.11}$$

Proof. Under the content prior, $\delta y + \varepsilon$ has mean δm and covariance C_P. The registered value in that Gaussian experiment is $o + \mathcal{V}\omega$. Its density gives eq. (20.7), including the affine term, and hence eq. (20.8). The determinant of C_P is independent of ω in this tangent model.

Set $A = R^{-1/2}\delta \bar{P}^{-1/2}$. Then $C_P^{-1} = R^{-1/2}(I + \epsilon^{-1} A A^T)^{-1} R^{-1/2}$. On the range of A the middle factor tends to zero; on its orthogonal complement it is identity. The limit is eq. (20.9), the residual quadratic form for weighted least squares. Minimizing over a proves eq. (20.10) and its zero criterion.

For injective codes that criterion becomes $\omega_e c = a_i - a_j$. A vector-valued edge cochain has this form exactly when it pairs to zero with every cycle. To see the converse directly, choose a spanning tree and define a_i by signed sums along its paths from a root. Each remaining edge closes a fundamental cycle, whose zero sum gives its required difference. The cycle-spanning hypothesis makes the declared directed-walk tests sufficient for these signed cycle tests; it does not identify an opposite attending edge with geometric reversal. Since $\omega_e \mapsto \omega_e c$ is onto $\mathbb{R}^K$ for $c \neq 0$, the rank is the edge-cochain dimension $K|\mathrm{E}|$ minus the coboundary dimension $K(|\mathrm{V}| - 1)$.

Finally, Gaussian elimination, or completion of the square, gives $u^T C_P^{-1} u = \min_y [y^T P y + \|u - \delta y\|^2_{R^{-1}}]$. Set $u = \delta a$ and evaluate the right side at $y = a$ to obtain eq. (20.11). □

The cycle-spanning hypothesis is stronger than connectedness. Strong connectivity, with all directed closed walks admitted, is sufficient: a strictly positive circulation is obtained by summing directed cycles containing each edge. Adding a sufficiently large multiple of that circulation to any signed cycle makes it nonnegative. Both nonnegative circulations decompose into directed cycles, so their difference lies in the directed-cycle span.

For a connected graph where the hypothesis fails, take scalar edges $1 \to 2$, $2 \to 3$, $1 \to 3$, unit codes and noise, $c = 1$, and $\mathcal{V}\omega = (0, 0, 1)$. There is no nonconstant directed closed walk, but

$$\delta y = (y_2 - y_1, y_3 - y_2, y_3 - y_1), \qquad \min_a \|(0, 0, 1) - \delta a\|^2 = \frac{1}{3}. \tag{20.12}$$

Indeed, $(1, 1, -1)$ spans the orthogonal complement of $\operatorname{im}\delta$, and the squared projection onto it is $1/3$. The coded obstruction exists even though this directed graph has no loop product that detects it.

The theorem describes the Hessian of an evidence cost, not a record-independent preference for flatness. With weak anchors the full record action tends to $\frac{1}{2}(o + \mathcal{V}\omega)^T \Pi_H (o + \mathcal{V}\omega)$. At zero records it charges the cohomology class of the coded reinterpretation. At nonzero records it can prefer a nonzero class that compensates the observed residual. For finite anchors, the shift by δm in eq. (20.7) must also be retained.

Remark 20.4 (Which experiment sees a dictionary). The matter term at one fixed operating content is blind to a perturbation A when $\bar{T}_e A c_i = 0$. Grounded messages with a referent law of positive-definite second moment instead test $\bar{T}_e A$, as their means range over the referents. These are different kernels. With $K = 2$, $\bar{T} = I_2$, $c = (1, 0)^T$, and $A = E_{22}$, the first experiment sees no change, whereas grounded referents identify A. A loop prior can constrain directions left free by a particular matter experiment, but those directions need not be untestable in another admitted experiment.

There is a further distinction between law comparisons and link observables. For a loop map H and law q, $D_{\mathrm{KL}}(q \| H_{\#} q)$ vanishes whenever H preserves q. For positive-definite Σ, the covariance-dressed link defect

$$Q(H; \Sigma) = \frac{1}{2} \|\Sigma^{-1/2}(I - H)\Sigma^{1/2}\|_F^2 \tag{20.13}$$

vanishes exactly at $H = I$, since the similarity transform is invertible. A quarter-turn preserves $q = \mathcal{N}(0, I_2)$, giving zero law divergence, while $Q(H; I_2) = 2$. The choice of defect therefore does not inherit the law stabilizer automatically.

A shared-sample relay around a loop is one way to probe its action. Another is to identify the constituent dictionaries from sufficiently informative grounded edge records and compose them. The evidence calculation requires neither direct loop observations nor such an identification procedure. Its conclusion is that records couple dictionaries through shared contents in the declared joint.

20.3 Stationary reinterpretation

Proposition 20.5 (One operating content gives a rank-one stationary mean). At a stationary point of eq. (20.2), each dictionary satisfies

$$\omega_e = \sigma^2 \bar{T}_e^T \big(R_e + \sigma^2 |c_i|^2 \bar{T}_e \bar{T}_e^T\big)^{-1} (\bar{T}_e \Delta_e - o_e) c_i^T. \tag{20.14}$$

It has rank at most one. For registered agreement, $o_e = 0$,

$$\omega_e c_i = F_e \Delta_e, \qquad F_e = \sigma^2 |c_i|^2 \bar{T}_e^T \big(R_e + \sigma^2 |c_i|^2 \bar{T}_e \bar{T}_e^T\big)^{-1} \bar{T}_e. \tag{20.15}$$

These are also equations for the stationary Gaussian recognition means. In the plug-in per-agent model the same equation holds with $\Delta_e = \mu_i - \mu_j$. In its law-level version, with peer covariance Σ_j fixed at the considered state, replace R_e by $\widetilde{R}_e = R_e + \bar{T}_e \Sigma_j \bar{T}_e^T$. For the tangent reciprocal potential at fixed positive covariances and positive row weights β_e, use $o_e = 0$ and replace R_e by $(R_e + \bar{T}_e \Sigma_j \bar{T}_e^T)/\beta_e$. In all cases the operating contents are fixed, the anchor is isotropic, and there are no loop or grounded terms. The replacements do not assert equality of dictionary forces across these objectives.

Proof. Only the anchor and the edge record depend on ω_e. Its normal equation is

$$(\sigma^{-2} I + \kappa_i^T \bar{T}_e^T R_e^{-1} \bar{T}_e \kappa_i)\omega_e = \kappa_i^T \bar{T}_e^T R_e^{-1}(\bar{T}_e \Delta_e - o_e). \tag{20.16}$$

The adjoint is $\kappa_i^T v = v c_i^T$ for the Frobenius inner product, and $\kappa_i \kappa_i^T = |c_i|^2 I$. Applying the push-through identity gives eq. (20.14); multiplication by c_i gives eq. (20.15). A quadratic Gaussian free energy has the same mean normal equation as its energy. In the per-agent objectives only the owner's record and anchor contain this dictionary, with the stated convolution covariance in the law-level case. In the reciprocal objective its mean-dependent edge cost is $\frac{\beta_e}{2} \|\bar{T}_e(\mu_i - \mu_j - \omega_e c_i)\|^2_{(R_e + \bar{T}_e \Sigma_j \bar{T}_e^T)^{-1}}$, which gives the last replacement. □

Corollary 20.6 (Equal gains cancel the linearized loop sum). Let the operating content be a common $c \neq 0$, register $o = 0$, and suppose the stationary equations of proposition 20.5 hold, with the effective noise of the chosen objective. Then $(B\omega)_\gamma = g_\gamma c^T$ for a vector g_γ, and

$$(B\omega)_\gamma c = \sum_{e \in \gamma} F_e \Delta_e = \sum_{e \in \gamma} (F_e - \bar{F}) \Delta_e \tag{20.17}$$

for every fixed matrix $\bar{F}$. Thus the linearized loop map annihilates $c^\perp$. It vanishes when all gains on the loop agree, or when every disagreement on the loop is zero. On a two-cycle $e : j \to i$, $e' : i \to j$, its action on c is $(F_e - F_{e'})\Delta_e$.

Proof. Every term of eq. (20.14) has right factor c^T. Summing the equations and applying them to c gives the first equality in eq. (20.17). The second uses $\sum_\gamma \Delta_e = 0$. A matrix $g c^T$ with $c \neq 0$ is zero exactly when its action on c is zero, which proves the cancellation statements. □

Nonzero stationary loop action therefore requires both disagreement and variation of the gains in this fixed-topic tangent experiment; those conditions are not sufficient. The gains depend on code, noise, and anchor strength. If all self means coincide and all records register agreement, the strictly convex global energy has the common-content, zero-dictionary minimizer. Conflicting self means or other registered records change this conclusion. For general laws the evidence depends on the complete content distributions, and one operating vector need not determine a stationary dictionary. Even within the Gaussian example, cancellation of $B\omega$ is a first-order statement: higher terms in the ordered product of exponentials are not removed by this argument.

20.4 Rigid clusters and their boundary dictionaries

Remark 20.7 (Reinterpretation limits the effective top-down precision). Integrating out the free Gaussian cross-scale dictionary in eq. (20.4) leaves a factor in $y_i - y_A$ with precision

$$(\tau^{-1} + \sigma^2|c_A|^2)^{-1} I. \tag{20.18}$$

This follows because $\omega_{iA} c_A$ has covariance $\sigma^2|c_A|^2 I$ and adds to the cross-scale record noise. For $c_A \neq 0$ the precision is bounded above by $(\sigma^2|c_A|^2)^{-1} I$ as τ increases. For $c_A = 0$ it equals τI and there is no such finite cap. Pinning the cross-scale dictionaries removes this source of reinterpretation. This elimination identity alone does not prove that disagreement concentrates on a cluster boundary.

Proposition 20.8 (A normalized rigid restriction of a supplied hierarchy). Let child and parent contents lie in one standard Borel space E with a common sigma-finite reference measure μ, and let G be a measurable dictionary group with identity I. Specify measurable self densities p_i, p_A with respect to μ. Let the record likelihoods have the form $\ell_e(o_e \mid d_e(z_i, z_j, \theta_e))$, where $d_e : E \times E \times G \to H_e$ is a declared measurable discrepancy with $d_e(z, z, I) = 0$. Fix internal and cross-scale dictionaries at the identity. Suppose every internal or cross-scale likelihood at its registered record and zero discrepancy is a finite positive constant. For every supplied cluster require

$$0 < Z_A = \int_E p_A(z) \prod_{i \in I} p_i(z) \mu(dz) < \infty. \tag{20.19}$$

Evaluate the joint density on $y_i = y_A$ and use μ once for each parent coordinate, together with the declared bridge-dictionary prior. Assume that the remaining bridge factors have a finite positive normalizer against this product measure. Then the resulting normalized rigid restriction is a communication model with parent self density

$$p'_A(z) = Z_A^{-1} p_A(z) \prod_{i \in I} p_i(z) \tag{20.20}$$

and one record factor for each original bridge, retaining its code, dictionary, and registered value.

For the Gaussian tangent family its parent self parameters are

$$P'_A = P_A + \sum_{i \in I} P_i, \qquad P'_A m'_A = P_A m_A + \sum_{i \in I} P_i m_i. \tag{20.21}$$

A bridge retains its fixed operating vector c_i; these vectors can be bridge dependent at the parent level unless they are common within each receiving cluster. With positive-definite self precisions, identity internal and cross-scale dictionaries, and the Gaussian cross-scale factors of precision τ, the global stationary minimizer tends to the minimizer of this rigid parent model at rate $O(\tau^{-1})$.

Proof. On equal contents an internal discrepancy is zero, and the hypotheses replace each internal and cross-scale record by a finite positive constant. The self factors at a cluster multiply at their common argument. Dividing by Z_A gives eq. (20.20); the assumed remaining normalizer makes the entire parent law a probability law. Each bridge retains its original likelihood evaluated at the

two parent contents. In the tangent model this argument is $\bar{T}_e(y_{A(i)} - y_{A(j)} - \omega_e c_i)$. Completing the square in the Gaussian self factors gives eq. (20.21).

For the minimizer limit collect all free Gaussian variables, including bridge dictionaries, in x. The energy is $E_\tau(x) = \frac{\tau}{2}\|Cx\|^2 + \frac{1}{2}x^T H x - h^T x + \text{const}$, where Cx lists the child-parent differences and H is positive definite. Choose orthonormal bases U for $\ker C$ and V for its orthogonal complement, and write $x = Ua + Vb$. With $H_{aa} = U^T H U$ and the analogous blocks, the first normal equation gives $a = a_0 - H_{aa}^{-1} H_{ab} b$, where $a_0 = H_{aa}^{-1} U^T h$ is the constrained minimizer. The second gives

$$\left[\tau V^T C^T C V + H_{bb} - H_{ba} H_{aa}^{-1} H_{ab}\right] b = V^T h - H_{ba} a_0. \tag{20.22}$$

Both the first coefficient and the Schur complement are positive definite on the b space. Hence $b = O(\tau^{-1})$ and $a - a_0 = O(\tau^{-1})$, proving the claim. □

The reference measure and normalizers are part of this restriction. It is neither conditioning a continuous law on a probability-zero diagonal without a specified convention nor integrating out children. Individually normalized self densities do not guarantee eq. (20.19): two copies of $p(z) = \frac{1}{3} z^{-2/3}$ on $(0, 1)$ have a product proportional to $z^{-4/3}$, whose integral is infinite. Disjoint supports can instead make the product integral zero. The Gaussian minimizer theorem does not establish convergence of normalized soft hierarchy laws to the rigid restriction for arbitrary laws.

Proposition 20.9 (Flat wall walks and a common boundary dictionary). Consider two clusters I, J whose internal dictionaries are identity. Assume that bridges exist in both directions. For every bridge $e : j \to i$ from J to I and every bridge $g : i' \to j'$ from I to J, require directed internal paths $i \to i'$ in I and $j' \to j$ in J. Internal strong connectivity is sufficient. Cross-scale edges may be included only when they supply the required directed paths. Let the bridge dictionaries be arbitrary elements of a group G.

Every directed walk that uses e, such internal paths, and g has holonomy $\bar{\theta}_g \bar{\theta}_e$. All these walks are flat exactly when there is one $\bar{\theta}_D \in G$ such that all bridges $J \to I$ carry $\bar{\theta}_D$ and all bridges $I \to J$ carry $\bar{\theta}_D^{-1}$. At first order the corresponding condition is $\omega_e = D$, $\omega_g = -D$.

For rigid clusters with common operating content c and zero bridge records, if the free bridge dictionaries satisfy proposition 20.5, the linearized wall loop acts on c by $(F_e - F_g)(y_I - y_J)$. More generally, with identity internal dictionaries but nonzero internal disagreements, that action is

$$(\omega_e + \omega_g)c = (F_e - F_g)\Delta_e + F_g(\Delta_e + \Delta_g). \tag{20.23}$$

Proof. The internal factors are identity, so the ordered product is $\bar{\theta}_g \bar{\theta}_e$. The path hypothesis realizes every opposite bridge pair as a walk of this kind. Fix one bridge in each direction, which is possible by nonemptiness. The equations $\bar{\theta}_g \bar{\theta}_e = I$ for all pairs imply that every forward bridge is the inverse of the fixed backward bridge, and every backward bridge is the inverse of that common forward value. The converse follows by multiplication. Replacing products by their first-order sums gives the generator condition. The stationary bridge equations give $\omega_e c + \omega_g c = F_e \Delta_e + F_g \Delta_g$, which is eq. (20.23); rigidity makes $\Delta_g = -\Delta_e$. □

Without the path hypothesis, separate reciprocal bridge pairs can each be flat while carrying different dictionaries. For $I = \{a_1, a_2\}$ and $J = \{b_1, b_2\}$ with no internal directed paths, let $b_1 \leftrightarrow a_1$

carry scalar dictionaries $1, 1$ and $b_2 \leftrightarrow a_2$ carry $2, 1/2$. Every available directed loop is flat, but the bridges from J to I disagree. Parent-to-child edges alone supply no return paths between these pairs.

When the proposition applies, all records in one bridge direction read a common dictionary through their own codes. Grounded referents with positive-definite second moment identify it up to the intersection of those code stabilizers; an injective stacked code identifies it completely for the location-family experiments. The group-level flatness statement is independent of Gaussianity. Its stationary gain formula is a Gaussian tangent statement. Neither licenses replacing distinct dictionaries by one merely because a single-topic experiment fails to distinguish them.

Remark 20.10 (Repeating a supplied rigid restriction). The density factorization can be repeated along a supplied hierarchy whenever the required measures and normalizers exist. In the Gaussian family the parent self term pools child information through eq. (20.21), while the bridge factors retain the boundary records. At finite coupling, integrating out children instead gives a Schur complement; representing it within the prescribed comparison-record family is an additional closure requirement.

These statements do not select a partition, prove its persistence, or construct an inherited response and update kernel for a predeclared probe class. They also do not establish compression on a post-handoff invariant set, interaction necessity against a null flow, or a second selection depending on the first. Those are separate requirements of the formation standard. A supplied rigid parent density therefore does not by itself establish recursive formation or complete dynamical inheritance.

20.5 Remaining extensions

Open problem 20.11 (Formation with top-down feedback). Determine whether a specified dynamics can select persistent blocks and create parents whose feedback supports further selection. The construction must retain the laws, records, evaluators, and complete response and update data required by its admitted protocols, and must meet the compression, null-flow, and counterfactual-recursion conditions. The supplied partitions and minimizer limits above do not settle these questions.

Open problem 20.12 (Fluctuations beyond the fixed tangent experiment). In the present tangent model C_P is dictionary independent, so its determinant contributes no dictionary force. For a nonlinear link model with Gaussian conditional content precision $J(\omega)$, information vector $b(\omega)$, and scalar energy term $k(\omega)$, Gaussian integration gives $k(\omega) - \frac{1}{2} b(\omega)^T J(\omega)^{-1} b(\omega) + \frac{1}{2} \log\det J(\omega)$ up to a constant, provided $J(\omega)$ is positive definite. Determine the complete action in a declared nonlinear communication model, including its off-topic directions and the signs of the resulting loop terms. The quadratic penalty of theorem 20.3 alone does not supply that calculation.

Open problem 20.13 (General content laws). Identify hypotheses under which the evidence action for general laws admits a useful decomposition into absorbable reinterpretations and residual incompatibility. Such a result must preserve record dependence and distinguish an invariance of the likelihood from an invariance of the content measure. Establishing a normalized soft-to-rigid limit also requires more than the finite product normalizers used for the static restriction. The stationary rank-one and gain formulas are not claimed outside the Gaussian tangent family.

Open problem 20.14 (Learning gains from loop records). For a declared grounded or relay experiment, determine which combinations of code, noise, and dictionary-anchor strength can be inferred from observed loop effects. The gain-weighted formula is an identity at a stationary tangent state; an identifiability theorem requires the full admitted record experiment and cannot be inferred from that identity alone.

Chapter 21

Beyond the Existence Constructions

The communication and link constructions of chapters 18 to 20 add a declared observation model to the dictionary sector. General-law reconstruction and identifiability statements distinguish a code, its stabilizer, and the receiver's conditional reconstruction. The normalized tangent prior and Gaussian recognition formulas give a bounded model of learning those dictionaries. The global, plug-in per-agent, law-level per-agent, and reciprocal calculations retain different forces and metrics; equality of one derivative does not identify their flows. The induced curvature cost includes its record-dependent affine term. Rigid-cluster calculations assume a supplied partition and do not establish its dynamical formation. The Gaussian tangent extension in proposition 19.17 and theorem 19.18 includes joint covariance and source-row evolution with global forward existence under fixed proper references. Its three-agent example has the local subcritical pitchfork of theorem 19.21, with nearby asymmetric saddles. Nonlinear dictionary learning, distant branches and destinations on general graphs, unrestricted law-space dynamics, and inheritance of a learned link sector remain open.

The theory distinguishes three objectives for interacting agents. In the global picture one normalized joint covers the contents and records of every agent, and local updates are coordinate descent of one free energy (proposition 4.14). In the per-agent picture each agent has its own generative model in which its own evaluator and its peers' communicated laws compete as sources of prior information (proposition 4.18). The reciprocal law potential is a third object, and proposition 4.21 shows that it is not the free energy of any fixed normalized model in which the agent laws are recognition laws. Every construction of chapter 7 to chapter 13 concerns the reciprocal potential or, in chapter 7, a nonsmooth modification of it. Chapter 14 separately derives the own-block forces and null control of normalized per-agent examples. It establishes persistent interaction-dependent grouping and a conditional compressive parent; it does not transfer the earlier reciprocal theorems wholesale.

Measured against the formation standard definition 6.31, no construction in this manuscript is yet a recursive formation under the smooth potential. The Wishart construction of chapter 12 selects two parents and then their union with exact original-time inheritance, but its membership schedule is the centroid dendrogram of the initial belief shapes and does not depend on the coupling; it is a two-stage selection with exact inheritance. The six-agent construction of chapter 13 admits a full open neighborhood of positive laws and inherits a moving-representative flow, but in the regime of its proof the null flow reproduces its membership history, and its handoff loss is created by the response interface rather than by the selection. Both smooth constructions supply their compression architectures. Their quotients exist before membership occurs, so the dynamics selects when a description is used without creating a new operational equivalence. The nonsmooth fusion construction of chapter 7 meets interaction necessity at the cost of changing the interaction, and its fused parents compress nothing. Every construction is therefore a single-stage or two-stage selection with exact inheritance, and each chapter states which clauses of the standard fail.

The supplied-architecture results retain their scopes. The profile hierarchy of chapter 11 is an equivariant symmetry reduction with a supplied tree; the data it discards are invisible to the fine energy, metric, and admitted records. The collective phase of chapter 10 is an asymptotic-phase projection that loses ordinary child sampling. The erasure construction of chapter 9 keeps public child states and its heterogeneity is transient. The per-agent parent of proposition 14.6 preserves complete admitted histories and persistent records, while proposition 14.7 and theorem 14.8 preserve the actual collective geometry on sections with context-constant discarded phases. Its phase quotient is also supplied before grouping. Adding a future hidden-angle probe destroys that quotient, and joining already canonical parents erases no new coordinate. Response sufficiency depends on the interface: an unrestricted identity-channel Gibbs interface identifies distinct exposed laws, and child-specific probes, exact constituent-law records, or feedback can require discarded distinctions.

The collective alternative of chapter 15 exposes every proper block through fixed relative-angle probes and permits one common physical orientation to be lost only collectively. Its reference-memory realization closes a local two-jet in the original time and preserves the actual normalized-law Fisher cost. A finite preparation generator yields a complete conditional whole-patch realization, with the full chronological record retained. The causal accumulator of proposition 15.4 instead integrates the local angular velocity from the actual preparation time. It preserves the original native dynamics and every admitted joint history, while theorem 15.5 requires initial collective-phase coherence for actual geometric inheritance. That condition is not generated by the memory. Equation (15.29) identifies the missing initial contextual phase relation for arbitrary preparations. Theorem 15.8 and corollary 15.9 exclude specified value-path and smooth bounded-jet causal-memory classes, in addition to the finite-order and finite global-encoding obstructions. They leave the unrestricted causal-local-parent question inconclusive. More general admissible memory mechanisms, interaction-created invisibility and a new second compressive handoff remain unresolved.

The remaining problems are more specific than an existence question. Open problem 6.32 asks for a construction that meets every clause of the standard with a derived selector and interface. The per-agent example meets interaction necessity at one stage, but still needs an interaction-selected sufficient interface and a second compressive handoff with counterfactual dependence. Under the reciprocal potential, interaction-sensitive selection requires receiver-dependent targets, since shared references with additive moment channels make every pairwise law comparison relax on a common clock. Open problem 4.26 asks how the three objectives are related and which conclusions survive a change of objective. The connection remains diagnostic in the formation constructions. Chapter 16 separately gives it a dynamical role through auxiliary alignment: the finite graph has a regular-branch Fisher ODE, and a flat periodic base has a locally well-posed shape PDE coupled to frame and source-row ODEs. Independent smoothing of those rows, frames, and communication links, global continuation, and graph refinement remain open. Chapter 17 supplies displacement maps and their connection from regular contextual agent data, including a curved reciprocal-link example. The message or target metric remains a declared comparison rule. Selecting that rule, continuing across rank or orbit degeneracy, and proving dynamical or parent inheritance of the resulting link sector remain open. Full permutation symmetry still constrains deterministic law-only selection as in proposition 6.27. Persistent membership under the fixed

per-agent record episode does not extend its geometric theorem beyond the declared finite window. None of the results establishes an indefinite spacetime metric, a thermodynamic limit, typicality under random initialization, or unbounded recursive formation.

Appendix A

A Normalized Internal Record Process

This appendix constructs a finite-history law from an internal interaction rule. The rule is supplied; normalization does not select it. Let S and O be standard Borel spaces. A state x includes an accessible record register. Write $\rho : \mathsf{S} \to \mathsf{O}$ for its readout.

Definition A.1 (Event and registration). Choose a normalized kernel $R : \mathsf{S} \rightsquigarrow \mathsf{O}$ and a measurable update $A : \mathsf{S} \times \mathsf{O} \to \mathsf{S}$ satisfying

$$\rho(A(x, o)) = o, \tag{A.1}$$

for $R(x, \cdot)$-almost every o, for every x in a Borel admitted set $\mathsf{S}_{\mathrm{adm}} \subseteq \mathsf{S}$. Define

$$J(x, dx', do) = R(x, do)\delta_{A(x,o)}(dx'). \tag{A.2}$$

If necessary, the state can be explicitly augmented by a register so that eq. (A.1) holds. The admitted set is assumed to contain μ_0-almost every initial state and to be invariant, $J(x, \mathsf{S}_{\mathrm{adm}} \times \mathsf{O}) = 1$ for $x \in \mathsf{S}_{\mathrm{adm}}$, so that the registration condition holds at every visited state. The map $x \mapsto J(x, B)$ is measurable for every measurable $B \subseteq \mathsf{S} \times \mathsf{O}$, because $x \mapsto \int f(x, o)R(x, do)$ is measurable for every bounded jointly measurable f, by a monotone-class argument from products. When the event kernel is generated by an interaction transition Γ and the readout, R is the record kernel of proposition 1.8, and the state marginal of J reproduces it: $R(x, B) = \int \mathbf{1}_B(\rho(A(x, o)))R(x, do)$.

Proposition A.2 (Consistent finite histories). For an initial law μ_0 on S, the iterated kernel product

$$\mathbb{P}_n(dx_0, do_1, dx_1, \ldots, do_n, dx_n) = \mu_0(dx_0) \prod_{t=1}^{n} J(x_{t-1}, dx_t, do_t), \tag{A.3}$$

is a probability law. Marginalizing its last pair gives $\mathbb{P}_{n-1}$, and every record agrees with the readout of the resulting state almost surely.

Proof. The product denotes successive kernel integration, not independent factors. Integrating the last pair gives one because $J(x, \mathsf{S} \times \mathsf{O}) = 1$. Repeating this argument gives total mass one and the stated consistency. Each transition is supported on $x' = A(x, o)$ and hence on $\rho(x') = o$. For a finite history, the intersection of these probability-one events still has probability one. □

By the Ionescu-Tulcea theorem [82], the consistent family $(\mathbb{P}_n)_n$ extends to a unique law on infinite histories, on which persistence statements about later readouts can be posed. When each record is binary, the records of a finite history carry at most $n \log 2$ nats about any variable, since $I(X; O_{1:n}) \le H(O_{1:n}) \le n \log 2$.

Nothing in this construction requires a variable external to the agents. It does require a declared accessible register and an update. A stochastic A is handled by replacing its Dirac kernel with a

normalized kernel supported on the registration condition. A deterministic transition is recovered when R is a Dirac kernel as well.

A.1 Prediction followed by assimilation

One possible internal state contains a model law s and a record register. Fix a normalized density $k(o \mid m)$ with respect to a reference measure on O. Let

$$R(s, do) = \left[\int k(o \mid m)s(dm)\right] do, \qquad s^o(dm) = \frac{k(o \mid m)s(dm)}{\int k(o \mid m')s(dm')}. \tag{A.4}$$

The update replaces s by s^o and stores o. The denominator is positive and finite for predictive-almost every record. Defining an arbitrary probability law on the remaining null set completes the measurable rule under the usual conditional-law hypotheses.

Proposition A.3 (The Bayesian history identity). For a fixed normalized observation density k and initial law s_0, sequential prediction and Bayesian assimilation give the record density

$$p(o_1, \ldots, o_n) = \int \prod_{t=1}^{n} k(o_t \mid m)s_0(dm). \tag{A.5}$$

Proof. After $t-1$ records, the posterior is proportional to $s_0(dm)\prod_{r<t} k(o_r \mid m)$. The next predictive density is the integral with the product extended to t, divided by the integral up to $t-1$. Multiplying these conditional densities cancels every intermediate normalizer and leaves eq. (A.5). □

Let the presentations m be the laws on a finite record alphabet, let $k(o \mid m) = m(o)$, and let s_0 be a Dirichlet law with parameter α and total concentration A. The predictive law after n records is $\bar{m}_n = (\alpha + c_n)/(A + n)$, with c_n the record counts. Then $\bar{m}_n - \bar{m}_{n-1} = (e_{o_n} - \bar{m}_{n-1})/(A + n)$, so the total-variation step is at most $1/(A + n)$: a slow model sector produced by accumulated records rather than by an inserted rate.

This identity applies to the stated fixed observation model. If the likelihood itself is changed using the current state, the product of the actual predictive kernels still defines a normalized process, but it need not equal the fixed-latent expression. Likewise, replacing Bayes' rule by a finite natural-gradient step defines a different assimilation mechanism unless the updates coincide. The normalization of a process and its interpretation as Bayesian conditioning are separate statements. Their difference has an exact measure.

Proposition A.4 (Departure from Bayesian assimilation). Let $\mathbb{P}_n$ be the record law of n steps of eq. (A.4) and let $\tilde{\mathbb{P}}_n$ be the record law generated from the same s_0 and k by any measurable assimilation rule, with model laws s_{t-1} and $\tilde{s}_{t-1}$ after the same history $o_{1:t-1}$. Then

$$D_{\mathrm{KL}}(\mathbb{P}_n \| \tilde{\mathbb{P}}_n) = \sum_{t=1}^{n} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(s_{t-1}k \| \tilde{s}_{t-1}k) \le \sum_{t=1}^{n} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(s_{t-1} \| \tilde{s}_{t-1}), \tag{A.6}$$

where $s_{t-1}k$ denotes the predictive record law.

Proof. Both processes predict the next record by $s_{t-1}k$ and $\tilde{s}_{t-1}k$ respectively, so the chain rule for relative entropy over the successive records gives the equality, and the data-processing inequality for the kernel k gives each bound. □

The identity applies to record laws. The joint laws of states and records of the two processes are generally mutually singular, because their state updates are different deterministic functions of the same history.

Appendix B

Conditional Prediction and Generative Memory

A generative model can relate a cue to a different result. Evidence about the cue then changes the predicted result through that relation. This appendix realizes the general agent model of definition 1.10 with correlated recognition, derives its exact local inference, and separates that inference from a subsequent compression of the posterior. The construction begins with general probability laws. Gaussian mixture regression supplies an explicit finite-dimensional example [24]; its use here makes the model-presentation sector and the retained information explicit.

B.1 A normalized cue and result model

Definition B.1 (Conditional prediction model). Fix a context. Let $\mathsf{M}, \mathsf{X}, \mathsf{Y}, \mathsf{O}$ be standard Borel spaces. Choose a proper presentation reference $r(dm)$, a normalized cue kernel $\mu_m(du)$, a normalized result kernel $K_m(u, dv)$, and a normalized record kernel $\mathcal{L}(u, do)$. All kernels are jointly measurable in their conditioning variables. Define

$$P(dm, du, dv, do) = r(dm)\mu_m(du)K_m(u, dv)\mathcal{L}(u, do). \tag{B.1}$$

Successive integration gives total mass one. In the agent notation, take $\mathsf{K} = \mathsf{X} \times \mathsf{Y}$, evaluator $G(m, d(u, v)) = \mu_m(du)K_m(u, dv)$, and record kernel $\ell((u, v), m; do) = \mathcal{L}(u, do)$. Thus eq. (B.1) is precisely $rG\ell$. It declares conditional independence of the result and record given the cue and presentation.

Assume $\mathcal{L}(u, do) = l(o \mid u)\lambda(do)$ for a fixed reference measure. At a specified record o, suppose

$$\begin{aligned} 0 < Z_m(o) &:= \int l(o \mid u)\mu_m(du) < \infty \quad r\text{-almost everywhere}, \\ 0 < Z(o) &:= \int Z_m(o)r(dm) < \infty. \end{aligned} \tag{B.2}$$

Bayes' rule gives the presentation posterior and component conditional

$$w^o(dm) = \frac{Z_m(o)}{Z(o)}r(dm), \qquad \Pi^o_m(du, dv) = \frac{l(o \mid u)\mu_m(du)K_m(u, dv)}{Z_m(o)}. \tag{B.3}$$

The joint recognition law is $\zeta^o(dm, du, dv) = w^o(dm)\Pi^o_m(du, dv)$. It has the general disintegration of eq. (1.3). Dependence between presentation and contents is retained. Since $K_m(u, \mathsf{Y}) = 1$, the presentation evidence depends on the cue and record mechanism, while the conditional result kernel determines what each selected presentation predicts.

The represented content can instead be the result alone. Disintegrate $\mu_m(du)K_m(u,dv) = G_Y(m,dv)R_m(v,du)$ and set

$$\ell_Y(v,m;do) = \int \mathcal{L}(u,do)R_m(v,du). \tag{B.4}$$

Disintegration on standard Borel spaces supplies measurable versions, with arbitrary extensions on null sets [82]. The kernel ℓ_Y is normalized, and $rG_Y\ell_Y$ is the (M,Y,O) marginal of eq. (B.1). Both representations therefore give the same result posterior. A presentation-dependent likelihood is allowed by definition 1.10.

Proposition B.2 (Correlated local free energy). At a record satisfying eq. (B.2), let a candidate joint recognition law be $\mathcal{Q}(dm,dk) = s(dm)q_m(dk)$, where $k = (u,v)$. Assume the absolute continuity, finite relative entropies, and integrable log likelihood needed below. Then

$$\mathcal{F}(\mathcal{Q};o) = D_{\mathrm{KL}}(s\|w^o) + \int D_{\mathrm{KL}}(q_m\|\Pi_m^o)s(dm) - \log Z(o). \tag{B.5}$$

Consequently the unrestricted minimum is $-\log Z(o)$, attained uniquely by the posterior up to null sets.

Proof. The unnormalized joint at the observed record is $Z(o)\zeta^o$. Substituting this into the logarithmic density ratio defining free energy gives $\mathcal{F} = D_{\mathrm{KL}}(\mathcal{Q}\|\zeta^o) - \log Z(o)$. Disintegration splits the joint relative entropy into the two terms of eq. (B.5); their nonnegativity and equality conditions prove the minimum statement. □

For a finite presentation space with $r(a) = \hat{\pi}_a > 0$, fixing the content conditionals gives

$$s_a \propto \hat{\pi}_a e^{-f_a}, \qquad f_a = D_{\mathrm{KL}}(q_a\|G_a) - \mathbb{E}_{q_a}\log l(o \mid X). \tag{B.6}$$

At the exact conditionals, $f_a = -\log Z_a(o)$. These are Gibbs weights for optimized component free energies. They are distinct from the peer-source rows of chapter 4: the label here chooses a model presentation.

The product restriction changes the inference problem. For the structured family $s_a q_X(du)K_a(u,dv)$, the result conditional cancels from the density ratio, and the label update is

$$s_a \propto \hat{\pi}_a \exp[-D_{\mathrm{KL}}(q_X\|\mu_a)]. \tag{B.7}$$

The record term is common to the labels at fixed q_X. With cue densities $\mu_a(u)$, varying q_X gives $q_X(u) \propto l(o \mid u)\exp(\sum_a s_a \log \mu_a(u))$, whenever the normalizer is finite. This is a product over (A,X), while retaining the result conditional; it is not a product over $(A,(X,Y))$. It illustrates the expected-log restriction described in remark 4.1 and classical mean-field inference [18]. Exact conditioning, this structured approximation, and the product-family Fisher flow are separately specified operations.

B.2 A Gaussian realization and its response

Let $\mathsf{M} = \{1, \dots, M\}$ and $\mathsf{X} = \mathsf{Y} = \mathsf{O} = \mathbb{R}^d$. Fix $\hat{\pi}_a > 0$, $\sum_a \hat{\pi}_a = 1$, vectors ν_a, ω_a, positive-definite matrices L_a, Q_a, C, and arbitrary matrices F_a. Set

$$\begin{aligned} X \mid A = a &\sim \mathcal{N}(\nu_a, L_a), \\ Y \mid X = u, A = a &\sim \mathcal{N}(\omega_a + F_a(u - \nu_a), Q_a), \\ O \mid X = u &\sim \mathcal{N}(u, C). \end{aligned} \tag{B.8}$$

The record covariance is fixed side information for this model. Each evaluator is the joint Gaussian with mean (ν_a, ω_a) and covariance

$$\begin{pmatrix} L_a & L_a F_a^T \\ F_a L_a & F_a L_a F_a^T + Q_a \end{pmatrix}. \tag{B.9}$$

Its Schur complement is $Q_a \succ 0$, so arbitrary, including singular, regressions are allowed.

Proposition B.3 (Exact component and result posteriors). At $O = \xi$, put $W_a = L_a + C$. Then

$$\begin{aligned} Z_a(\xi) &= \mathcal{N}(\xi; \nu_a, W_a), & w_a &= \frac{\hat{\pi}_a Z_a(\xi)}{\sum_b \hat{\pi}_b Z_b(\xi)}, \\ m_{X,a} &= \nu_a + L_a W_a^{-1}(\xi - \nu_a), & P_{X,a} &= (L_a^{-1} + C^{-1})^{-1}, \\ m_{Y,a} &= \omega_a + F_a(m_{X,a} - \nu_a), & P_{Y,a} &= Q_a + F_a P_{X,a} F_a^T. \end{aligned} \tag{B.10}$$

The result marginal is $p_Y^\xi = \sum_a w_a \mathcal{N}(m_{Y,a}, P_{Y,a})$, with moments

$$\bar{m} = \sum_a w_a m_{Y,a}, \qquad S = \sum_a w_a \left[P_{Y,a} + (m_{Y,a} - \bar{m})(m_{Y,a} - \bar{m})^T \right]. \tag{B.11}$$

Proof. The cue and independent record noise sum to a Gaussian with covariance W_a, giving Z_a. Completing the cue's quadratic log density gives precision $L_a^{-1} + C^{-1}$ and information vector $L_a^{-1}\nu_a + C^{-1}\xi$. Solving for the mean gives the displayed gain. Applying the affine result kernel gives its conditional moments and cross-covariance $\mathrm{Cov}(X, Y \mid \xi, a) = P_{X,a} F_a^T$. Total expectation and total covariance give eq. (B.11). These are conditional-mixture identities, as in Gaussian mixture regression [24]. □

For the result-only representation eq. (B.4), let $V_a = F_a L_a F_a^T + Q_a$, $B_a = L_a F_a^T V_a^{-1}$, and $S_a = L_a - L_a F_a^T V_a^{-1} F_a L_a$. Then

$$\begin{aligned} G_Y(a, dv) &= \mathcal{N}(dv; \omega_a, V_a), \\ \ell_Y(v, a; do) &= \mathcal{N}(do; \nu_a + B_a(v - \omega_a), C + S_a). \end{aligned} \tag{B.12}$$

Conditioning the joint evaluator on $Y = v$ gives cue covariance $S_a \succ 0$, and adding independent record noise proves the likelihood formula.

Proposition B.4 (Cross-prediction response and posterior Fisher tensor). Hold the prior joint law of (A, X, Y) and $C \succ 0$ fixed, with record $O = X + \epsilon$ for independent $\epsilon \sim \mathcal{N}(0, C)$. Suppose the

posterior moments used below exist and differentiation can be passed through their integrals on an open set of records. Then

$$D_\xi \mathbb{E}[Y \mid \xi] = \operatorname{Cov}(Y, X \mid \xi) C^{-1}, \qquad I_\xi = C^{-1} \operatorname{Cov}(X \mid \xi) C^{-1}, \tag{B.13}$$

where I_ξ is the Fisher tensor of the complete posterior family in record coordinates. These assumptions hold for the finite Gaussian model above.

Proof. The gradient of the log record likelihood is $C^{-1}(X - \xi)$. Subtracting its posterior expectation, the gradient of the log normalizer, gives posterior score $C^{-1}(X - \mathbb{E}[X \mid \xi])$. Multiplying the score by Y and integrating proves the response identity. Its score second moment gives I_ξ. Finite Gaussian mixtures have smooth finite component moments and locally dominated Gaussian integrands, which justify these operations. □

For same-state prediction $Y = X$, the whitened Jacobian is $C^{-1/2} \operatorname{Cov}(X \mid \xi) C^{-1/2}$, positive definite in the Gaussian example. For a distinct result the cross-covariance can have either sign and need not be symmetric. Already with one component, $D_\xi \mathbb{E}[Y \mid \xi] = FL(L + C)^{-1}$. For any prescribed dense response H, choosing $F = H(L + C)L^{-1}$ gives H exactly. The result center sets the affine intercept. This conditional response is not a gradient step on the cue, and its sign places no conflict with the positive Fisher tensor. The identity is a partial derivative at fixed bank and record covariance; it does not differentiate a larger computation in which those data also vary.

B.3 Retained moments and statistical geometry

Proposition B.5 (Gaussian projection and the loss of dependence). For the positive finite Gaussian mixture p_Y^ξ above, $q_G = \mathcal{N}(\bar{m}, S)$ uniquely minimizes $D_{\mathrm{KL}}(p_Y^\xi \| q)$ over nondegenerate Gaussians. If $\Pi_{A,Y}^\xi$ is the exact label/result joint, then

$$D_{\mathrm{KL}}(\Pi_{A,Y}^\xi \| w \otimes q_G) = I_{\Pi^\xi}(A; Y) + D_{\mathrm{KL}}(p_Y^\xi \| q_G). \tag{B.14}$$

For the diagonal carrier $q_D = \mathcal{N}(\bar{m}, \operatorname{diag} S)$,

$$D_{\mathrm{KL}}(p_Y^\xi \| q_D) - D_{\mathrm{KL}}(p_Y^\xi \| q_G) = \frac{1}{2} \log \frac{\prod_k S_{kk}}{\det S}. \tag{B.15}$$

Proof. The expectation of a Gaussian log density depends only on the mean and covariance of the sampling law. Consequently, for every nondegenerate Gaussian q, $D_{\mathrm{KL}}(p_Y^\xi \| q) - D_{\mathrm{KL}}(p_Y^\xi \| q_G) = D_{\mathrm{KL}}(q_G \| q)$. This proves the minimum and its uniqueness. Inserting $w \otimes p_Y^\xi$ into the joint logarithmic ratio proves eq. (B.14). Subtracting the two Gaussian log densities and using $\mathbb{E}[(Y - \bar{m})(Y - \bar{m})^T] = S$ gives eq. (B.15). All entropies and relative entropies here are finite for the stated positive finite mixtures. □

This inclusive-divergence projection is the local moment-matching operation used in assumed-density filtering [107]. It generally differs from minimizing the reverse divergence of variational free energy. For example, let $p_a = \frac{1}{2}\mathcal{N}(-a, 1) + \frac{1}{2}\mathcal{N}(a, 1)$ and $q_a = \mathcal{N}(0, 1 + a^2)$. The component $\mathcal{N}(a, 1)$

has reverse divergence at most $\log 2$ from p_a, whereas

$$D_{\mathrm{KL}}(q_a\|p_a) \geq a^2 - a\sqrt{\frac{2(1+a^2)}{\pi}} - \frac{1}{2}\log(1+a^2) \longrightarrow \infty. \tag{B.16}$$

Indeed, $p_a(y) \leq (2\pi)^{-1/2}e^{-(|y|-a)^2/2}$; integrate its negative logarithm under q_a and subtract the Gaussian entropy. Thus moment matching does not even minimize the reverse divergence for sufficiently separated components.

Proposition B.6 (Moments need not preserve later Gaussian records). For $0 < t < 1$, the admissible result mixture

$$p_t = \tfrac{1}{2}\mathcal{N}(t, 1-t^2) + \tfrac{1}{2}\mathcal{N}(-t, 1-t^2) \tag{B.17}$$

has constant Gaussian carrier $q_G = \mathcal{N}(0,1)$, but a later record kernel $K_c(y, do) = \mathcal{N}(do; y, c)$, $c > 0$, distinguishes p_t from q_G.

Proof. Identical cue components, equal weights, $F_1 = F_2 = 0$, result centers $\pm t$, and result noise $1 - t^2$ realize p_t. Its mean is zero and variance one. Convolution with the record noise gives densities at zero

$$(p_tK_c)(0) = \frac{e^{-t^2/[2(1-t^2+c)]}}{\sqrt{2\pi(1-t^2+c)}}, \qquad (q_GK_c)(0) = \frac{1}{\sqrt{2\pi(1+c)}}. \tag{B.18}$$

They agree at $t = 0$, while the derivative of the logarithm of the first is $-t^3/(1-t^2+c)^2 < 0$. Their densities therefore differ for every $t > 0$; continuity gives a differing probability on a small interval about zero. □

Moments preserve expectations of affine and quadratic readouts. Proposition B.6 shows that this does not imply preservation of every later observation law. It concerns the chosen projection, not an obstruction to richer retained states or to general-law parents. Exact inheritance still requires the complete configuration retention and interface of definition 6.1 and proposition 6.2.

For a regular family $\zeta_\theta(a, dv) = w_a\mathcal{N}(dv; m_a, P_a)$, the score split underlying proposition 3.2 gives

$$\begin{aligned} g^F_\zeta(u,v) = &\sum_a \frac{(d_uw_a)(d_vw_a)}{w_a} \\ &+ \sum_a w_a\left[(d_um_a)^TP_a^{-1}(d_vm_a) + \frac{1}{2}\operatorname{tr}(P_a^{-1}d_uP_aP_a^{-1}d_vP_a)\right]. \end{aligned} \tag{B.19}$$

The mixed term vanishes because each conditional Gaussian score has conditional mean zero. Null parameter directions must be removed before calling this tensor a metric. For the label/result family producing eq. (B.17), its coefficient along t is $(1+t^2)/(1-t^2)^2 > 0$, while the Gaussian carrier has zero coefficient. The result marginal itself also varies, with fourth moment $3 - 2t^4$. Joint statistical geometry therefore cannot be recovered from the moment carrier in general. These are admissible model families, not assertions about a particular evolution.

B.4 Frames, sharing, and sequential records

One group $\mathrm{GL}(d)$ acts on the doubled content by $h \cdot (u, v) = (hu, hv)$ and trivially on the finite labels. Re-expressing the evaluator and record data requires

$$\begin{gathered} \nu'_a = h\nu_a, \quad \omega'_a = h\omega_a, \quad L'_a = hL_a h^T, \quad Q'_a = hQ_a h^T, \\ F'_a = hF_a h^{-1}, \quad C' = hCh^T, \quad \xi' = h\xi. \end{gathered} \tag{B.20}$$

The evidence densities acquire the same factor $|\det h|^{-1}$, so the weights are invariant; the conditional means and covariances transform by h and congruence. This follows directly by substitution into eq. (B.10). In local token frames U_i, the reference cue $(U_i^{-1}\mu_i, U_i^{-1}\Sigma_i U_i^{-T})$ is invariant under a simultaneous passive change of frame and local moments. A singular F_a still defines a normalized response kernel; it is not a group-valued dictionary.

Sharing a bank supplies common reference and evaluator data. Sharing a generative model in the sense of definition 1.11 also requires matched record kernels under the declared frame maps. Different covariances C_i instead give a shared family of conditional mechanisms indexed by supplied side information. The distinction must be stated if uncertainty varies between agents. A diagonal reference parametrization is not closed under arbitrary rebasing: if hDh^T is diagonal for every positive diagonal D, its off-diagonal entries force $h_{ik}h_{jk} = 0$ for $i \neq j$, so invertibility makes h a permutation times an invertible diagonal matrix. This restriction on the reference parametrization does not remove local passive covariance with that reference fixed.

Definition B.7 (An exact carried-presentation filter). Let the finite label evolve by a fixed row-stochastic matrix T. At step l, draw a fresh cue from μ_{A_l}, a result from K_{A_l}, and a record from $\mathcal{L}_{C_l}$, with draws at distinct steps conditionally independent given the label sequence. For fixed positive record covariances, the finite law is

$$\begin{aligned} &P(a_{0:n}, du_{0:n}, dv_{0:n}, do_{0:n}) \\ &\quad = \hat{\pi}_{a_0} \prod_{l=1}^{n} T_{a_{l-1}a_l} \prod_{l=0}^{n} \left[\mu_{a_l}(du_l) K_{a_l}(u_l, dv_l) \mathcal{L}_{C_l}(u_l, do_l)\right]. \end{aligned} \tag{B.21}$$

Integration and summation of the last step give one conditional on the previous label, proving normalization and consistency. If $w_l(a) = P(A_l = a \mid O_{0:l})$, marginalization and Bayes' rule give

$$v_l(b) = \sum_a w_{l-1}(a) T_{ab}, \qquad w_l(b) = \frac{v_l(b) Z_b(o_l; C_l)}{\sum_c v_l(c) Z_c(o_l; C_l)}. \tag{B.22}$$

This is a hidden Markov emission model. The label vector suffices for its next label prediction because the next cue is fresh given the next label. A later interaction that reads the current result or its dependence on the label requires a different sufficiency argument. The continuous result law can still be needed for current readouts even in this model.

A modified prior proportional to $\hat{\pi}_b v_l(b)^\vartheta$ has a separate one-step interpretation. With baseline prior $\hat{\pi}$ and a record $o \mid B = b \sim \mathrm{Dir}(\alpha \mathbf{1} + \vartheta e_b)$, where $\alpha > 0$ and $\vartheta \geq 0$, its normalizer is independent of b, and the density at an interior simplex value $o = v_l$ is proportional to $v_l(b)^\vartheta$. Conditioning gives the modified prior. It equals the prediction in eq. (B.22) for uniform $\hat{\pi}$ and $\vartheta = 1$. Combining this

record with a cue requires declaring their joint conditional law; the algebra alone does not supply conditional independence.

Appendix A distinguishes normalization of a history from exact Bayesian assimilation under its record mechanism. That distinction applies here. If $O_2 = f(O_1)$ and O_1 is already registered, conditioning on (O_1, O_2) is the same as conditioning on O_1; an independent-replicate likelihood would count the evidence again. Communication may bring information that is new to one receiver even in a deterministic population computation. A repeated conditional-prediction agent must therefore specify its accessible history, communication-generated record, retained joint law, and any projection. The local identities above establish neither a common free energy for an arbitrary composition nor a new formation result.

Appendix C

Connections in a Moving Presentation

Links between agents at a single context do not determine transport between neighboring contexts. A connection requires a smooth dependence on the direction and size of a base displacement. This appendix records, in matrix form, the transformation law of connection data supplied in a moving presentation: given a presentation U_i and a one-form a_i, it computes the displayed connection and curvature and their behavior under changes of coordinates. The one-form a_i is supplied for the transformation calculation in this appendix. Chapter 17 separately constructs it from regular contextual laws, interfaces, and links. The remaining selection and continuation questions are open problem C.4. Collective least-squares selection in proposition 3.9 is a separate construction. The context-local and formation flows use it diagnostically. The added energy in chapter 16 instead makes collective alignment enter the law velocity through a contextual force.

Fix one sector and one agent, suppressing the sector superscript. On a coordinate neighborhood of $\mathcal{C}$, let $U_i(c)$ be a twice continuously differentiable invertible presentation and let $a_i = a_{i,\mu}(c)dc^\mu$ be a continuously differentiable matrix-valued one-form. For a principal G-connection its values must lie in the represented Lie algebra. General endomorphism-valued forms instead describe connections for the extended general linear frame bundle.

Definition C.1 (Infinitesimal directional transport). The proposed transport family has first-order behavior

$$\Theta_i(c+\delta c, c) = U_i(c+\delta c)\exp[-a_{i,\mu}(c)\delta c^\mu]U_i(c)^{-1} + O(|\delta c|^2). \tag{C.1}$$

The first-order generator is linear in the tangent displacement. A single finite matrix at c supplies neither this linear dependence nor the identity value at zero displacement.

Proposition C.2 (Connection and curvature). With the convention $D_i = d + B_i$, eq. (C.1) has connection and curvature

$$B_i = U_i a_i U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i(da_i + a_i \wedge a_i)U_i^{-1}. \tag{C.2}$$

Proof. Expanding the two factors at c gives $\Theta_i = I + (dU_iU_i^{-1} - U_ia_iU_i^{-1})(\delta c) + O(|\delta c|^2)$, which identifies B_i. Acting on a vector-valued function gives the operator identity $d + B_i = U_i(d + a_i)U_i^{-1}$. Squaring it yields $(d + B_i)^2 = U_i(d + a_i)^2U_i^{-1}$, and the squares act by the curvature two-forms in eq. (C.2). □

For an explicit curved example, take $U_i = I$ over an open subset of $\mathbb{R}^2$ and set

$$a_{i,x} = E_{12}, \qquad a_{i,y} = E_{21}. \tag{C.3}$$

Both are supplied rank-one receiving–sending products, while their derivatives vanish. Nevertheless,

$$F_{i,xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0. \tag{C.4}$$

Thus even constant directional interfaces can give curvature through noncommutativity. By contrast, replacing a finite pointwise map by a new frame $V(c)$ and setting $B = -dVV^{-1}$ gives zero curvature: it is the case $a = 0$ of eq. (C.2).

Finite paths use the solution of

$$\frac{d}{dt}T_i(t) = -B_i(\dot{\gamma}(t))T_i(t), \qquad T_i(0) = I. \tag{C.5}$$

Uniqueness of this linear equation gives composition under concatenation and inversion under path reversal. The solution is the path-ordered exponential; the usual link and Wilson-line constructions use this ordering [136]. Independently declared opposite attending edges are not two orientations of this same path and need not be inverses.

Proposition C.3 (Compatibility under re-expression). Under an agent-coordinate change $U_i' = g_i U_i$ with the same reference form,

$$B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}. \tag{C.6}$$

Under a change of reference presentation $U_i' = U_i h$, keeping the same intrinsic transport instead requires

$$a_i' = h^{-1} a_i h + h^{-1} dh. \tag{C.7}$$

Proof. Insert $g_i U_i$ into eq. (C.2) and apply the product rule. Inserting $U_i h$ and eq. (C.7) cancels the two terms containing dh, leaving the original B_i. □

The same affine law must hold between overlapping coordinate patches. A directional interface product transforming only by conjugation is a tensorial one-form. It can describe the difference between two connections, but it is not by itself the full affine connection data. Constructing displacement links with the endpoint transformation law and then differentiating them is one way to obtain the required inhomogeneous term.

If patch coordinates satisfy $v_\alpha = g_{\alpha\beta} v_\beta$, their transition functions must obey $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. The displayed affine law then makes the local connections compatible. A nontrivial bundle need not admit a single global presentation U_i. Even on a trivial bundle, zero curvature does not imply path independence: the scalar connection $B = b\,d\theta$ on the circle is flat but has holonomy $e^{-2\pi b}$, which differs from one for real $b \neq 0$. Global path independence requires trivial holonomy as well as local flatness.

Different agents may be assigned different connection data in their own presentations. Since all agents are sections of one associated bundle, these data define one connection on P only when, after re-expression in a common frame, their local forms agree and satisfy its overlap relations; otherwise they are several connections on the same bundle, each used by one agent. Neither case requires an averaging construction.

Open problem C.4 (Selecting and continuing interaction-derived transport). The full-rank construction in theorems 17.3 and 17.5 supplies displacement identity, a linear first jet, compatible endpoint laws, and path transport from contextual law and link messages. The regular alignment construction in proposition 17.8 gives another selector, and its reciprocal-link example has nonzero curvature. Both require a declared comparison rule. The directed backtrack sector can enter the data and is not identified with the reversal of a contextual path.

The remaining problem is to select the message labels, column weights, or augmented comparison metric from a stated agent model, and to establish continuation when rank or infinitesimal identifiability fails. Link-only data cannot identify a continuous gauge subgroup that fixes every input, as proposition 17.9 proves. A selector also needs new dynamical analysis before it can replace the law-Fisher auxiliary field in chapter 16: its full constitutive derivative must enter the energy variation. Neither construction derives the contextual base, its topology, or its dimension.

Appendix D

A Conditional Determinant Geometry

The collective construction of proposition 3.9 does not require a Lorentzian target. This appendix gives one additional choice for a complex rank-two representation. Its indefinite form is supplied by a determinant; the Fisher metric used to choose the connection remains positive.

Let $W \to \mathcal{U}$ be a complex rank-two vector bundle with structure group $\mathrm{GL}(2,\mathbb{C})$. In a frame, identify the real rank-four bundle $\mathrm{Herm}(W)$ with Hermitian matrices transforming by $H' = gHg^\dagger$. Suppose the collective target laws p_i on W have finite second moments and that their moment maps are smooth on the admitted regular families. With the same declared positive weights as in the collective metric, set

$$M = \mathcal{M}(\boldsymbol{p}) = \sum_i w_i \int zz^\dagger p_i(dz), \qquad e(X) = d\mathcal{M}_{\boldsymbol{p}} r_X. \tag{D.1}$$

Equivariance of the moment map gives

$$e = dM + A^F M + M(A^F)^\dagger, \qquad e' = geg^\dagger. \tag{D.2}$$

The first expression is independent of the stabilizer-valued representative of A^F by proposition 3.12.

Proposition D.1 (Determinant metric). Assume $\mathcal{U}$ is four dimensional and the actual one-form $e : T\mathcal{U} \to \mathrm{Herm}(W)$ is a bundle isomorphism. Supply a smooth positive determinant-scale section $\kappa_{\det}$ with local transformation $\kappa'_{\det} = |\det g|^2 \kappa_{\det}$. Then

$$\eta_{\det}(H,K) = \tfrac{1}{2}[\mathrm{tr}(HK) - \mathrm{tr}(H)\,\mathrm{tr}(K)], \qquad \gamma(X,Y) = \kappa_{\det}^{-1}\eta_{\det}(eX,eY). \tag{D.3}$$

defines a gauge-invariant metric with one negative and three positive directions. The positive-definite Hermitian cone determines a time orientation through e.

Proof. For a Hermitian matrix $H = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}$, direct evaluation gives

$$\eta_{\det}(H,H) = -\det H = -t^2 + x^2 + y^2 + z^2. \tag{D.4}$$

Multiplicativity of the determinant under congruence gives $-\det(gHg^\dagger) = |\det g|^2(-\det H)$. Polarizing proves the same scaling law for $\eta_{\det}(H,K)$. The compensator cancels that factor in eq. (D.3), and the isomorphism e preserves its inertia. Positive Hermitian matrices have $t > 0$ and $t^2 > x^2 + y^2 + z^2$, so they form one timelike cone, preserved by congruence. □

The positive determinant scale has a canonical choice built from the same data. Where M is positive definite, $\det(gMg^\dagger) = |\det g|^2 \det M$, so $\kappa_{\det} = \det M$ obeys the required transformation

law. For 2×2 matrices $\operatorname{tr}(B^2)=(\operatorname{tr}B)^2-2\det B$; applying this to $B=M^{-1}H$ and polarizing gives

$$\frac{\eta_{\det}(H,K)}{\det M}=\tfrac{1}{2}\big[\operatorname{tr}(M^{-1}HM^{-1}K)-\operatorname{tr}(M^{-1}H)\operatorname{tr}(M^{-1}K)\big]. \tag{D.5}$$

With this scale,

$$\gamma=\tfrac{1}{2}\big(h^{\mathcal{CN}}-\vartheta\otimes\vartheta\big),\qquad h^{\mathcal{CN}}(X,Y)=\operatorname{tr}\big(M^{-1}e(X)M^{-1}e(Y)\big),\qquad \vartheta(X)=\operatorname{tr}\big(M^{-1}e(X)\big). \tag{D.6}$$

Here $h^{\mathcal{CN}}$ is the Fisher metric of the proper complex Gaussian law with covariance M, pulled back by e, and $\vartheta=d\log\det M+2\operatorname{Re}\operatorname{tr}A^F$ is the covariant derivative of its entropy $\log\det(\pi eM)$. The indefinite signature is a rank-one subtraction from a Fisher-type metric: a direction is timelike exactly when the squared entropy rate exceeds the squared Fisher speed, and the dilation $e=cM$ has $\gamma=-c^2$. The form is invariant under constant rescaling of M, so any physical scale remains a calibration. For non-Gaussian targets $h^{\mathcal{CN}}$ is the Fisher metric of the moment projection, not of the actual laws. The scale also changes the geometry. For an exact soldering $e=dM$ with constant $\kappa_{\det}$, γ is the pullback of a flat Minkowski form. With $\kappa_{\det}=\det M$ it is the pullback of the $\mathrm{GL}(2,\mathbb{C})$-invariant form $-\det(dM)/\det M$ on the positive Hermitian cone, which is a Lorentzian product of a line with hyperbolic three-space and has scalar curvature -6.

The rank hypothesis is imposed on the actual contextual first jet. It constrains the number and configuration of the target laws.

Proposition D.2 (Rank of the aggregate jet for proper complex Gaussian targets). Let p_i be centered proper complex Gaussian laws with positive-definite covariances M_i, $i=1,\dots,n$, with the collective metric $\sum_i w_i\operatorname{tr}(M_i^{-1}h_iM_i^{-1}k_i)$ and the action $M\mapsto gMg^\dagger$. Then

$$H=\Big\{(h_i)_i:\ \sum_i w_iM_i^{-1}h_i=0\Big\},\qquad d\mathcal{M}(h)=\sum_i w_ih_i. \tag{D.7}$$

For $n=1$, $H=0$. For $n=3$ at triples whose orbit has the maximal dimension seven, the rank of $d\mathcal{M}|_H$ is at most three, so e cannot be an isomorphism on a four-dimensional base there. For $n=4$, at $M_1=I$, $M_2=2I$, $M_3=\operatorname{diag}(1,2)$, $M_4=\begin{pmatrix}2&1\\1&2\end{pmatrix}$ with unit weights, the orbit has dimension seven and $d\mathcal{M}|_H$ is surjective.

Proof. An orbit direction is $(\xi M_i+M_i\xi^\dagger)_i$ for $\xi\in\mathfrak{gl}(2,\mathbb{C})$. By cyclicity of the trace its pairing with $(h_i)_i$ is $2\operatorname{Re}\operatorname{tr}(B\xi)$ with $B=\sum_i w_ih_iM_i^{-1}$. This vanishes for all ξ exactly when $B=0$, equivalently when $B^\dagger=\sum_i w_iM_i^{-1}h_i=0$; this proves eq. (D.7). For $n=1$ it forces $h_1=0$. Put $k_i=w_ih_i$. For $n=3$, $\dim H=12-7=5$, and the rank of $d\mathcal{M}|_H$ is 5 minus the dimension of $\{k:\sum_i k_i=0,\ \sum_i M_i^{-1}k_i=0\}$. Eliminating k_3 leaves $\mathsf{A}k_1+\mathsf{B}k_2=0$ with $\mathsf{A}=M_1^{-1}-M_3^{-1}$ and $\mathsf{B}=M_2^{-1}-M_3^{-1}$. When A,B are invertible, $(k_1,k_2)=(\mathsf{A}^{-1},-\mathsf{B}^{-1})$ and $(\mathsf{A}^{-1}\mathsf{B}\mathsf{A}^{-1},-\mathsf{A}^{-1})$ are Hermitian solutions, independent unless B is a multiple of A. So the rank is at most three on the complement of a proper real-algebraic subset, which is dense in the open set of triples with seven-dimensional orbits. The rank of a smoothly varying linear map is lower semicontinuous, so the set where it is at least four is open; being disjoint from a dense set, it is empty. For the four-agent tuple, a unitary g fixing $\operatorname{diag}(1,2)$ is diagonal, and a diagonal unitary fixing M_4 is scalar,

so the stabilizer is the central circle and the orbit has dimension seven. Given a Hermitian S, the choice $k_1 = -S$, $k_2 = 2S$, $k_3 = k_4 = 0$ satisfies $\sum_i M_i^{-1} k_i = 0$ and $\sum_i k_i = S$. □

The bound for three agents holds only on the stratum of maximal orbit dimension. Off it the horizontal space grows: two agents with proportional covariances M and $2M$ have a four-dimensional orbit, H of dimension four, and $d\mathcal{M}|_H$ surjective by the same choice of k. A section supplying an invertible e is still required; the proposition shows only where one can exist. For improper complex laws the central phase acts nontrivially on the pseudo-covariance $\mathbb{E}[zz^T]$, which transforms as gPg^T, so the normal equation can then determine the phase component of the connection; we do not develop that case. Surjectivity of the horizontal differential of $\mathcal{M}$ establishes the availability of such a jet at a law tuple, but does not select a section or prove persistence under inference. If the base has dimension greater than four and e is surjective, the same formula gives a nondegenerate form on the quotient vector bundle by $\ker e$. A quotient-manifold interpretation still requires a regular leaf quotient and basicness, as in proposition 3.6.

Proposition D.3 (Circle completion). Let $\mathscr{Q}_{\det} \to \mathcal{U}$ be the unit circle bundle of $\det W$ for the line metric whose local coefficient is $\kappa_{\det}^{-1}$. Write $\pi_{\det}$ for its projection. Supply a real principal circle connection $\omega_{\det}$ normalized to one on the vertical generator of period 2π, and a smooth positive radius $r_{\det} : \mathcal{U} \to (0, \infty)$. Then

$$G_{\mathrm{KK}} = \pi_{\det}^* \gamma + (r_{\det} \circ \pi_{\det})^2 \omega_{\det} \otimes \omega_{\det} \tag{D.8}$$

is circle invariant and has one negative and four positive directions.

Proof. The connection splits each total tangent space into its horizontal subspace and the one-dimensional vertical subspace. On the first, eq. (D.8) is the pullback of γ; on the second, it is positive because $r_{\det} > 0$. The two subspaces are orthogonal. The base pullback, radius, and principal connection are invariant under the circle action, proving invariance of the total metric. □

A supplied global $\mathrm{GL}(2, \mathbb{C})$ connection A gives the local expression $\omega_{\det} = d\theta_{\det} + \operatorname{Im} \operatorname{tr} A$ on this circle bundle. The coordinate transformation $\theta'_{\det} = \theta_{\det} + \arg \det g$ cancels $\operatorname{Im} \operatorname{tr} A' = \operatorname{Im} \operatorname{tr} A - d \arg \det g$. When the collective minimization determines only a class, its lift to a full connection is further data. For instance, central phase rotations act trivially on centered proper complex Gaussian laws and their covariance matrices, so that family cannot fix the corresponding phase connection from its Fisher normal equations. Neither the positive circle radius nor equations of motion for γ follow from the construction. The determinant and circle choices give this conditional geometric example without identifying it with physical spacetime dynamics.

Appendix E

Notation

Symbol	Meaning
$\mathcal{C}, c_*$	Context base and a selected context.
P, G	Principal bundle and its structure group, one group acting on both sectors.
ϱ_q, ϱ_s	Representations of G on the belief space V_q (dimension K) and the model space V_s (dimension $\tilde{K}$).
$\mathcal{E}_q, \mathcal{E}_s$	Associated content bundles $P \times_{\varrho_q} V_q$ and $P \times_{\varrho_s} V_s$.
$\Phi, \tilde{\Phi}$	Belief and model content-bundle morphisms induced by one principal bundle map.
$\theta_{ij}, \tilde{\theta}_{ij}$	Group-valued belief and model links, with $\Theta^q_{ij} = \varrho_q(\theta_{ij})$ and $\Theta^s_{ij} = \varrho_s(\tilde{\theta}_{ij})$; tied when equal.
$\omega_{ij}, \tilde{\omega}_{ij}$	Their generators in the Lie algebra $\mathfrak{g}$.
Ξ_{ij}	Sector mismatch $\tilde{\theta}_{ij}\theta_{ij}^{-1}$, covariant under conjugation at the receiver.
$\mathsf{K}_i, \mathsf{M}_i, \mathsf{Y}_i$	Represented state contents, model presentations, and $\mathsf{Y}_i = \mathsf{K}_i \times \mathsf{M}_i$ of agent i.
$\mathcal{B}_i, E_i$	Statistical bundle $P \times_G \mathcal{P}(\mathsf{Y}_i)$ and sample bundle $P \times_G \mathsf{Y}_i$.
$\mathcal{P}(X)$	Probability measures on a measurable space X.
ζ_i, q_i, s_i	Joint agent law (recognition law) and its belief and model marginals.
r_i, G_i, ℓ_i	Prior over presentations, evaluator, and record kernel of agent i's generative model.
$P_i, \mathcal{F}_i$	Normalized generative model of agent i and its single-agent or per-agent free energy.
$R_i, \mathcal{E}_0$	Reference joint $r_i G_i$ and the sum of local terms $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i)$.
ϕ, ψ	Presentation and content maps relating two agents that share a generative model.
Z, P_0, Q	Coefficient space, prior, and recognition law of a content law on fields.
f_i, F_D, Q_D	Sample-section maps, a finite-design evaluation map, and its law.
Π_o, Z_o	Posterior and evidence of a fixed normalized model at record value o.
g^F, ℓ_v	Fisher metric of an admitted regular family and the score in tangent direction v.
$D^\omega\zeta, h^\omega_\zeta$	Covariant statistical derivative and its Fisher pullback on the context base.
$J_{\boldsymbol{p}}, H_{\boldsymbol{p}}$	Infinitesimal collective group action and its Fisher-orthogonal complement.
$\mathfrak{k}, A^F$	Collective orbit metric and a representative of the minimizing connection class.
u_i, σ, U_i	Frame of agent i in P_{c_*}, a reference point, and U_i with $u_i = \sigma U_i^{-1}$.
V_a, E^a_i	Reference vector space of sector $a \in \{q, s\}$ and agent i's coordinate copy of it.
U^q_i, U^s_i	Frame representatives in the two sectors.
$\bar{q}_i, \bar{s}_i$	Laws in common reference coordinates on a trivializing patch.
L^q_{ij}, L^s_{ij}	Law-dependent interaction generators in the vector realization.
$\Theta^q_{ij}, \Theta^s_{ij}$	Invertible source-to-receiver interaction maps.
$\mathcal{K}^q_{ij}, \mathcal{K}^s_{ij}$	Normalized source-to-receiver comparison channels.
ν^q_{ij}, D^q_{ij}	Compared law $q_j\mathcal{K}^q_{ij}$ and peer divergence $D_{\mathrm{KL}}(q_i \| \nu^q_{ij})$; model sector analogous.
T^a_j, O^a_i	Sending and receiving maps in sector $a \in \{q, s\}$.
μ_j	Message law $(T_j)_\# q_j$.
β_i, γ_i	Belief-sector and model-sector source rows.
$\pi^q_i, \pi^s_i, T_q, T_s$	Source priors and row temperatures.
H_i	State-to-record kernel.

Symbol	Meaning
K_i^{rec}	Normalized model-to-record predictive evaluator.
S, x	Complete agent-configuration space and a configuration.
O_i, ρ_i	Record alphabet and internal record readout.
$O_{ij,t}, o_{ij,t}$	Record random variable and one realized value.
$\Gamma_{ij}, \mathcal{R}_{ij}$	Interaction transition on configurations and the record kernel it induces.
$\mathcal{E}, \mathcal{E}_{\mathrm{red}}$	Reciprocal law potential and its row-reduced form.
$\mathcal{E}_o$	Evidence-augmented reciprocal potential.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model.
Φ, Φ^0	A flow of a construction and its null flow, with peer costs deleted.
$H(x), P(x)$	Membership history with event times and the partition sequence without them; H^0, P^0 along the null flow.
(F1)–(F6)	Clauses of the formation standard; (F5) is interaction necessity, $P \neq P^0$ on an open set.
$W_2(\nu, \mathsf{P})$	Matrix Wishart law with shape $\nu > 1/2$ and precision P in the conditional formation example.
n_I, A_I, B_I	Primitive count and belief/model shape sums of a retained block in that example.
Λ_q, C_q	Moment-dependent comparison precision and its complete force coefficient; the model sector uses Λ_s, C_s.
B_{IJ}^q, B_{IJ}^s	Exact receiver-averaged, source-summed native rows of current blocks in the open-law construction.
H_I, ϕ_I	Discarded constant relative rotations and the retained local mean anchor phase.
F, R, F^c	Fine deterministic update, retention map, and inherited update, with $R \circ F = F^c \circ R$.
a_i, B_i, F_i	Reference connection form, its local representative, and curvature.
X, U, Z, Y	Source, encoded representation, decoder information, and prediction target in the coding construction.
$\mathcal{T}, \Delta$	Encoder kernel and predictive decoder in the coding construction.

The symbol F for a discrete update and F_i for curvature are used in separate constructions, and the calligraphic $\mathcal{F}_i$ denotes a free energy. The letter B denotes a connection representative B_i, block shape sums B_I, and native rows B_{IJ}^q, each in its own construction. The symbol ν denotes a compared law ν_{ij} in the dynamics, a sector marginal ν_i^a in the interaction chapter, and a Wishart shape. A frame U_i^a is an invertible map; the coding variable U is a random representation. A Markov kernel returns a probability measure, whereas an ordinary map returns one value. The symbol $\circ$ composes maps and juxtaposition composes kernels, in the order in which a law traverses them.

Bibliography

[1] Ali E. Abbas. A Kullback-Leibler view of linear and log-linear pools. *Decision Analysis*, 6(1):25–37, 2009. doi: 10.1287/deca.1080.0133.

[2] P.-A. Absil, R. Mahony, and B. Andrews. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization*, 16(2):531–547, 2005.

[3] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998. doi: 10.1162/089976698300017746.

[4] Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2 edition, 2008.

[5] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162:327–364, 2015. doi: 10.1007/s00440-014-0574-8.

[6] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Parametrized measure models. *Bernoulli*, 24(3):1692–1725, 2018. doi: 10.3150/16-BEJ910. URL https://arxiv.org/abs/1510.07305.

[7] Paul Baird and John C. Wood. *Harmonic Morphisms Between Riemannian Manifolds*, volume 29 of *London Mathematical Society Monographs, New Series*. Oxford University Press, Oxford, 2003.

[8] Afonso S. Bandeira, Amit Singer, and Daniel A. Spielman. A Cheeger inequality for the graph connection Laplacian. *SIAM Journal on Matrix Analysis and Applications*, 34(4):1611–1630, 2013.

[9] Lionel Barnett and Anil K. Seth. Dynamical independence: Discovering emergent macroscopic processes in complex dynamical systems. *Physical Review E*, 108:014304, 2023. doi: 10.1103/PhysRevE.108.014304.

[10] Stephen D. Bartlett, Terry Rudolph, and Robert W. Spekkens. Reference frames, superselection rules, and quantum information. *Reviews of Modern Physics*, 79(2):555–609, 2007. doi: 10.1103/RevModPhys.79.555.

[11] Peter W. Bates, Kening Lu, and Chongchun Zeng. Invariant foliations near normally hyperbolic invariant manifolds for semiflows. *Transactions of the American Mathematical Society*, 352(10):4641–4676, 2000. doi: 10.1090/S0002-9947-00-02503-4.

[12] Giorgio Battistelli and Luigi Chisci. Kullback–Leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability. *Automatica*, 50(3):707–718, 2014.

[13] Robert H. Berk. Limiting behavior of posterior distributions when the model is incorrect. *Annals of Mathematical Statistics*, 37(1):51–58, 1966.

[14] Wolf-Jürgen Beyn and Vera Thümmler. Freezing solutions of equivariant evolution equations. *SIAM Journal on Applied Dynamical Systems*, 3(2):85–116, 2004. doi: 10.1137/030600515.

[15] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

[16] Pier Giovanni Bissiri, Chris C. Holmes, and Stephen G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130, 2016.

[17] David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24 (2):265–272, 1953. doi: 10.1214/aoms/1177729032.

[18] David M. Blei, Alp Kucukelbir, and Jon D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.

[19] Giovanni A. Bonaschi, José A. Carrillo, Marco Di Francesco, and Mark A. Peletier. Equivalence of gradient flows and entropy solutions for singular nonlocal interaction equations in 1D. *ESAIM: Control, Optimisation and Calculus of Variations*, 21(2):414–441, 2015. doi: 10.1051/cocv/2014032.

[20] Haïm Brézis. *Opérateurs Maximaux Monotones et Semi-Groupes de Contractions dans les Espaces de Hilbert*, volume 5 of *North-Holland Mathematics Studies*. North-Holland, Amsterdam, 1973.

[21] Peter Buchholz. Exact and ordinary lumpability in finite Markov chains. *Journal of Applied Probability*, 31(1):59–75, 1994. doi: 10.2307/3215235.

[22] Nazmi Burak Budanur, Predrag Cvitanović, Ruslan L. Davidchack, and Evangelos Siminos. Reduction of SO(2) symmetry for spatially extended dynamical systems. *Physical Review Letters*, 114:084102, 2015. doi: 10.1103/PhysRevLett.114.084102.

[23] C. J. Burke and M. Rosenblatt. A Markovian function of a Markov chain. *The Annals of Mathematical Statistics*, 29(4):1112–1122, 1958. doi: 10.1214/aoms/1177706444.

[24] Sylvain Calinon. A tutorial on task-parameterized movement learning and retrieval. *Intelligent Service Robotics*, 9(1):1–29, 2016. doi: 10.1007/s11370-015-0187-9. URL https://calinon.ch/papers/Calinon-JIST2015.pdf.

[25] Miguel Á. Carreira-Perpiñán. Fast nonparametric clustering with Gaussian blurring mean-shift. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 153–160, 2006. doi: 10.1145/1143844.1143864.

[26] José A. Carrillo, Marco Di Francesco, Alessio Figalli, Thomas Laurent, and Dejan Slepčev. Global-in-time weak measure solutions and finite-time aggregation for nonlocal interaction equations. *Duke Mathematical Journal*, 156:229–271, 2011. doi: 10.1215/00127094-2010-211.

[27] Jonas Cassel, Bastian Boll, Stefania Petra, Peter Albers, and Christoph Schnörr. Sigma flows for image and data labeling and learning structured prediction, 2024. URL https://arxiv.org/abs/2408.15946v1. Version 1.

[28] N. N. Čencov. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1982.

[29] Chung Chan, Ali Al-Bashabsheh, Qiaoqiao Zhou, Tarik Kaced, and Tie Liu. Info-clustering: A mathematical theory for data clustering, 2016. URL https://arxiv.org/abs/1605.01233.

[30] Yifan Chen, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, and Andrew M. Stuart. Sampling via gradient flows in the space of probability measures, 2023.

[31] Yizong Cheng. Mean shift, mode seeking, and clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(8):790–799, 1995. doi: 10.1109/34.400568.

[32] Frank H. Clarke. *Optimization and Nonsmooth Analysis*. Wiley, New York, 1983.

[33] Jorge Cortés. Finite-time convergent gradient flows with applications to network consensus. *Automatica*, 42(11):1993–2000, 2006. doi: 10.1016/j.automatica.2006.06.015.

[34] Thomas A. Courtade and Tsachy Weissman. Multiterminal source coding under logarithmic loss. *IEEE Transactions on Information Theory*, 60(1):740–761, 2014. doi: 10.1109/TIT.2013.2288257.

[35] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, Hoboken, NJ, second edition, 2006.

[36] Michael G. Crandall and Thomas M. Liggett. Generation of semi-groups of nonlinear transformations on general Banach spaces. *American Journal of Mathematics*, 93(2):265–298, 1971. doi: 10.2307/2373376.

[37] Imre Csiszár. I-divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975.

[38] Imre Csiszár. Sanov property, generalized I-projection and a conditional limit theorem. *The Annals of Probability*, 12(3):768–793, 1984.

[39] Walter J. Culver. On the existence and uniqueness of the real logarithm of a matrix. *Proceedings of the American Mathematical Society*, 17(5):1146–1151, 1966. doi: 10.1090/S0002-9939-1966-0202740-6.

[40] A. D'Adda, M. Lüscher, and P. Di Vecchia. A $1/n$ expandable series of non-linear σ models with instantons. *Nuclear Physics B*, 146(1):63–76, 1978. doi: 10.1016/0550-3213(78)90432-7.

[41] M. H. A. Davis. Piecewise-deterministic Markov processes: a general class of non-diffusion stochastic models. *Journal of the Royal Statistical Society: Series B*, 46(3):353–388, 1984.

[42] A. P. Dawid. Conditional independence in statistical theory. *Journal of the Royal Statistical Society: Series B (Methodological)*, 41(1):1–31, 1979. doi: 10.1111/j.2517-6161.1979.tb01052.x.

[43] Gaspard Riche de Prony. Essai expérimental et analytique: sur les lois de la dilatabilité des fluides élastiques et sur celles de la force expansive de la vapeur de l'alkool, à différentes températures. *Journal de l'École Polytechnique*, 1(22):24–76, 1795.

[44] Morris H. DeGroot. Reaching a consensus. *Journal of the American Statistical Association*, 69(345): 118–121, 1974.

[45] Salem Derisavi, Holger Hermanns, and William H. Sanders. Optimal state-space lumping in Markov chains. *Information Processing Letters*, 87(6):309–315, 2003. doi: 10.1016/S0020-0190(03)00343-0.

[46] Persi Diaconis and Sandy L. Zabell. Updating subjective probability. *Journal of the American Statistical Association*, 77(380):822–830, 1982.

[47] Jean Dieudonné. *Foundations of Modern Analysis*. Academic Press, New York, 1960.

[48] John D. Dixon and Brian Mortimer. *Permutation Groups*, volume 163 of *Graduate Texts in Mathematics*. Springer, New York, 1996. doi: 10.1007/978-1-4612-0731-3.

[49] Monroe D. Donsker and S. R. Srinivasa Varadhan. Asymptotic evaluation of certain Markov process expectations for large time, I. *Communications on Pure and Applied Mathematics*, 28(1):1–47, 1975.

[50] Paul Dupuis and Richard S. Ellis. *A Weak Convergence Approach to the Theory of Large Deviations*. Wiley, New York, 1997.

[51] E. B. Dynkin. *Markov Processes, Volume I*, volume 121 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 1965.

[52] Jr. Eells, James and J. H. Sampson. Harmonic mappings of riemannian manifolds. *American Journal of Mathematics*, 86(1):109–160, 1964. doi: 10.2307/2373037.

[53] Jaap Eldering, Matthew Kvalheim, and Shai Revzen. Global linearization and fiber bundle structure of invariant manifolds. *Nonlinearity*, 31(9):4202–4245, 2018. doi: 10.1088/1361-6544/aaca8d. URL https://arxiv.org/abs/1711.03646.

[54] Mark Fels and Peter J. Olver. Moving coframes: I. A practical algorithm. *Acta Applicandae Mathematicae*, 51(2):161–213, 1998. doi: 10.1023/A:1005878210297.

[55] Neil Fenichel. Asymptotic stability with rate conditions. *Indiana University Mathematics Journal*, 23 (12):1109–1137, 1974. doi: 10.1512/iumj.1974.23.23090.

[56] Michael J. Field. *Dynamics and Symmetry*, volume 3 of *ICP Advanced Texts in Mathematics*. Imperial College Press, London, 2007.

[57] Aleksei F. Filippov. *Differential Equations with Discontinuous Righthand Sides*, volume 18 of *Mathematics and Its Applications (Soviet Series)*. Kluwer Academic Publishers, Dordrecht, 1988.

[58] Cédric Fournel, Jordan François, Serge Lazzarini, and Thierry Masson. Gauge invariant composite fields out of connections, with examples. *International Journal of Geometric Methods in Modern Physics*, 11(1):1450016, 2014.

[59] Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2): 127–138, 2010.

[60] Karl J. Friston, Erik D. Fagerholm, Tahereh S. Zarghami, Thomas Parr, Inês Hipólito, Loïc Magrou, and Adeel Razi. Parcels and particles: Markov blankets in the brain. *Network Neuroscience*, 5(1):211–251, 2021. doi: 10.1162/netn_a_00175.

[61] Tobias Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, 2020.

[62] Michel Fruchart, Ryo Hanai, Peter B. Littlewood, and Vincenzo Vitelli. Non-reciprocal phase transitions. *Nature*, 592:363–369, 2021.

[63] Christian Genest and James V. Zidek. Combining probability distributions: A critique and an annotated bibliography. *Statistical Science*, 1(1):114–135, 1986. doi: 10.1214/ss/1177013825.

[64] Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. A mathematical perspective on Transformers. *Bulletin of the American Mathematical Society*, 62(3):427–479, 2025. arXiv:2312.10794.

[65] S. G. Gindikin. Invariant generalized functions in homogeneous domains. *Functional Analysis and Its Applications*, 9(1):50–52, 1975. doi: 10.1007/BF01078179.

[66] Martin Golubitsky and Ian Stewart. *The Symmetry Perspective: From Equilibrium to Chaos in Phase Space and Physical Space*, volume 200 of *Progress in Mathematics*. Birkhäuser, Basel, 2002. doi: 10.1007/978-3-0348-8167-8.

[67] Martin Golubitsky, Ian Stewart, and David G. Schaeffer. *Singularities and Groups in Bifurcation Theory, Volume II*, volume 69 of *Applied Mathematical Sciences*. Springer-Verlag, New York, 1988.

[68] Robert M. Gray. *Entropy and Information Theory*. Springer-Verlag, New York, 1990. URL https://www-ee.stanford.edu/~gray/it.pdf. Corrected first edition, revised June 26, 2023.

[69] John Guckenheimer. Isochrons and phaseless sets. *Journal of Mathematical Biology*, 1(3):259–273, 1975. doi: 10.1007/BF01273747.

[70] A. Guichardet. On rotation and vibration motions of molecules. *Annales de l'I.H.P. Physique théorique*, 40(3):329–342, 1984. URL https://www.numdam.org/item/AIHPA_1984__40_3_329_0/.

[71] Paul R. Halmos and L. J. Savage. Application of the Radon–Nikodym theorem to the theory of sufficient statistics. *The Annals of Mathematical Statistics*, 20(2):225–241, 1949.

[72] Jakob Hansen and Robert Ghrist. Opinion dynamics on discourse sheaves. *SIAM Journal on Applied Mathematics*, 81(5):2033–2060, 2021.

[73] Emilie V. Haynsworth. Determination of the inertia of a partitioned Hermitian matrix. *Linear Algebra and its Applications*, 1(1):73–81, 1968.

[74] Rainer Hegselmann and Ulrich Krause. Opinion dynamics and bounded confidence: models, analysis and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3), 2002.

[75] Robert Hermann and Arthur J. Krener. Nonlinear controllability and observability. *IEEE Transactions on Automatic Control*, 22(5):728–740, 1977. doi: 10.1109/TAC.1977.1101601.

[76] Nicholas J. Higham. *Functions of Matrices: Theory and Computation*. Society for Industrial and Applied Mathematics, Philadelphia, 2008. doi: 10.1137/1.9780898717778.

[77] Toby Dylan Hocking, Armand Joulin, Francis Bach, and Jean-Philippe Vert. Clusterpath: An algorithm for clustering using convex fusion penalties. In *Proceedings of the 28th International Conference on Machine Learning*, pages 745–752, 2011. URL https://icml.cc/2011/papers/419_icmlpaper.pdf.

[78] Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, 1998.

[79] Leonhard Horstmeyer and Fatihcan M. Atay. Characterization of exact lumpability for vector fields on smooth manifolds. *Differential Geometry and its Applications*, 48:46–60, 2016. doi: 10.1016/j.difgeo.2016.06.001.

[80] Michael I. Jordan, Zoubin Ghahramani, Tommi S. Jaakkola, and Lawrence K. Saul. An introduction to variational methods for graphical models. *Machine Learning*, 37(2):183–233, 1999.

[81] Abram Kagan and Lawrence A. Shepp. A sufficiency paradox: An insufficient statistic preserving the Fisher information. *The American Statistician*, 59(1):54–56, 2005. doi: 10.1198/000313005X21041.

[82] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

[83] R. E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

[84] Alexander S. Kechris. *Classical Descriptive Set Theory*, volume 156 of *Graduate Texts in Mathematics*. Springer, New York, 1995.

[85] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, New York, 1976. ISBN 9780387901923. Reprint of the 1960 edition, with a new appendix.

[86] Michael Kirchhoff, Thomas Parr, Ensor Palacios, Karl Friston, and Julian Kiverstein. The Markov blankets of life: Autonomy, active inference and the free energy principle. *Journal of the Royal Society Interface*, 15(138):20170792, 2018. doi: 10.1098/rsif.2017.0792.

[87] Shoshichi Kobayashi and Katsumi Nomizu. *Foundations of Differential Geometry, Volume I*. Interscience Publishers, New York, 1963.

[88] Solomon Kullback. *Information Theory and Statistics*. Wiley, New York, 1959.

[89] Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi: 10.1214/aoms/1177729694.

[90] G. N. Lance and W. T. Williams. A general theory of classificatory sorting strategies. I. Hierarchical systems. *The Computer Journal*, 9(4):373–380, 1967. doi: 10.1093/comjnl/9.4.373.

[91] Kim G. Larsen and Arne Skou. Bisimulation through probabilistic testing. *Information and Computation*, 94(1):1–28, 1991. doi: 10.1016/0890-5401(91)90030-6.

[92] Joseph P. LaSalle. Some extensions of Liapunov's second method. *IRE Transactions on Circuit Theory*, 7(4):520–527, 1960. doi: 10.1109/TCT.1960.1086720.

[93] Steffen L. Lauritzen. *Graphical Models*, volume 17 of *Oxford Statistical Science Series*. Clarendon Press, Oxford, 1996.

[94] Lieven Le Bruyn and Claudio Procesi. Semisimple representations of quivers. *Transactions of the American Mathematical Society*, 317(2):585–598, 1990. doi: 10.1090/S0002-9947-1990-0958897-0.

[95] Lucien Le Cam and Grace Lo Yang. *Asymptotics in Statistics: Some Basic Concepts*. Springer, New York, second edition, 2000. doi: 10.1007/978-1-4612-1166-2.

[96] John M. Lee. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 2013. doi: 10.1007/978-1-4419-9982-5.

[97] Melvin Leok, Jerrold E. Marsden, and Alan D. Weinstein. A discrete theory of connections on principal bundles, 2005. URL https://arxiv.org/abs/math/0508338.

[98] Gérard Letac and Hélène Massam. Existence and non-existence of the non-central Wishart distributions, 2017. URL https://arxiv.org/abs/1108.2849v2. Version 2.

[99] Debra Lewis, Nilima Nigam, and Peter J. Olver. Connections for general group actions. *Communications in Contemporary Mathematics*, 7(3):341–374, 2005. doi: 10.1142/S0219199705001787.

[100] Genyuan Li and Herschel Rabitz. A general analysis of exact lumping in chemical kinetics. *Chemical Engineering Science*, 44(6):1413–1430, 1989. doi: 10.1016/0009-2509(89)85014-6.

[101] Robert G. Littlejohn and Matthias Reinsch. Gauge fields in the separation of rotations and internal motions in the n-body problem. *Reviews of Modern Physics*, 69(1):213–275, 1997. doi: 10.1103/RevModPhys.69.213.

[102] Stanisław Łojasiewicz. Une propriété topologique des sous-ensembles analytiques réels. In *Les Équations aux Dérivées Partielles*, pages 87–89. Éditions du Centre National de la Recherche Scientifique, Paris, 1963.

[103] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 44(2):226–233, 1982.

[104] Kanti V. Mardia and Peter E. Jupp. *Directional Statistics*. Wiley, Chichester, 2000.

[105] Fortune Massamba and George Thompson. The universal connection and metrics on moduli spaces, 2003. URL https://arxiv.org/abs/math/0311198.

[106] Alexandre Mauroy, Igor Mezić, and Jeff Moehlis. Isostables, isochrons, and Koopman spectrum for the action-angle representation of stable fixed point dynamics. *Physica D: Nonlinear Phenomena*, 261:19–30, 2013. doi: 10.1016/j.physd.2013.06.004.

[107] Thomas P. Minka. Expectation propagation for approximate bayesian inference. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, pages 362–369. Morgan Kaufmann, 2001. URL https://tminka.github.io/papers/ep/minka-ep-uai.pdf.

[108] Pierre Molino. *Riemannian Foliations*, volume 73 of *Progress in Mathematics*. Birkhäuser, Boston, 1988. doi: 10.1007/978-1-4684-8670-4.

[109] Dov Monderer and Lloyd S. Shapley. Potential games. *Games and Economic Behavior*, 14(1):124–143, 1996.

[110] Robb J. Muirhead. *Aspects of Multivariate Statistical Theory*. John Wiley & Sons, New York, 1982.

[111] M. S. Narasimhan. Existence of universal connections. In *General Topology and its Relations to Modern Analysis and Algebra*, page 286, Prague, 1962. Academia. URL https://dml.cz/handle/10338.dmlcz/700905. Report on joint work with S. Ramanan.

[112] Radford M. Neal and Geoffrey E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In Michael I. Jordan, editor, *Learning in Graphical Models*, pages 355–368. Kluwer, Dordrecht, 1998.

[113] Barrett O'Neill. The fundamental equations of a submersion. *Michigan Mathematical Journal*, 13(4):459–469, 1966. doi: 10.1307/mmj/1028999604.

[114] Brad E. Paden and Shankar S. Sastry. A calculus for computing Filippov's differential inclusion with application to the variable structure control of robot manipulators. *IEEE Transactions on Circuits and Systems*, 34(1):73–82, 1987. doi: 10.1109/TCS.1987.1086038.

[115] Ensor Rafael Palacios, Adeel Razi, Thomas Parr, Michael Kirchhoff, and Karl Friston. On Markov blankets and hierarchical self-organisation. *Journal of Theoretical Biology*, 486:110089, 2020. doi: 10.1016/j.jtbi.2019.110089.

[116] Richard S. Palais. The principle of symmetric criticality. *Communications in Mathematical Physics*, 69 (1):19–30, 1979. doi: 10.1007/BF01941322.

[117] George J. Pappas, Gerardo Lafferriere, and Shankar Sastry. Hierarchically consistent control systems. *IEEE Transactions on Automatic Control*, 45(6):1144–1160, 2000. doi: 10.1109/9.863598.

[118] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA, 1988.

[119] Dénes Petz. Sufficiency of channels over von Neumann algebras. *The Quarterly Journal of Mathematics*, 39(1):97–108, 1988.

[120] David Pollard. A note on insufficiency and the preservation of Fisher information. In *From Probability to Statistics and Back: High-Dimensional Models and Processes*, volume 9 of *IMS Collections*, pages 266–275. Institute of Mathematical Statistics, 2013. doi: 10.1214/12-IMSCOLL919. URL https://arxiv.org/abs/1107.3797.

[121] George R. Price. Selection and covariance. *Nature*, 227:520–521, 1970. doi: 10.1038/227520a0.

[122] Claudio Procesi. The invariant theory of $n \times n$ matrices. *Advances in Mathematics*, 19(3):306–381, 1976. doi: 10.1016/0001-8708(76)90027-X.

[123] Bruce L. Reinhart. Foliated manifolds with bundle-like metrics. *Annals of Mathematics*, 69(1):119–132, 1959. doi: 10.2307/1970097.

[124] L. C. G. Rogers and J. W. Pitman. Markov functions. *The Annals of Probability*, 9(4):573–582, 1981. doi: 10.1214/aop/1176994363.

[125] Clarence W. Rowley, Ioannis G. Kevrekidis, Jerrold E. Marsden, and Kurt Lust. Reduction and reconstruction for self-similar dynamical systems. *Nonlinearity*, 16(4):1257–1275, 2003. doi: 10.1088/0951-7715/16/4/304.

[126] David Ruelle. Bifurcations in the presence of a symmetry group. *Archive for Rational Mechanics and Analysis*, 51:136–152, 1973. URL https://www.ihes.fr/~ruelle/PUBLICATIONS/[39].pdf.

[127] Daniil Ryabko. Independence clustering (without a matrix), 2017. URL https://arxiv.org/abs/1703.06700.

[128] Siavash Shahshahani. *A New Mathematical Framework for the Study of Linkage and Selection*, volume 17 of *Memoirs of the American Mathematical Society*. American Mathematical Society, Providence, RI, 1979.

[129] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001. URL https://arxiv.org/abs/cond-mat/9907176.

[130] Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE International Convention Record*, 7(4):142–163, 1959. URL https://gwern.net/doc/cs/algorithm/information/1959-shannon.pdf.

[131] Amit Singer and Hau-Tieng Wu. Vector diffusion maps and the connection Laplacian. *Communications on Pure and Applied Mathematics*, 65(8):1067–1144, 2012.

[132] Morris Skibinsky. Adequate subfields and sufficiency. *The Annals of Mathematical Statistics*, 38(1): 155–161, 1967.

[133] Stephen Smale. Topology and mechanics. I. *Inventiones Mathematicae*, 10:305–331, 1970. doi: 10.1007/BF01418778.

[134] Robert R. Sokal and Charles D. Michener. A statistical method for evaluating systematic relationships. *The University of Kansas Science Bulletin*, 38(22):1409–1438, 1958.

[135] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method, 1999. URL https://arxiv.org/abs/physics/0004057. Manuscript dated September 30, 1999; arXiv:physics/0004057, posted 2000.

[136] David Tong. Gauge theory. University of Cambridge lecture notes, 2018. URL https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf.

[137] Erik Torgersen. *Comparison of Statistical Experiments*, volume 36 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1991.

[138] Tim van Erven and Peter Harremoës. Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory*, 60(7):3797–3820, 2014. doi: 10.1109/TIT.2014.2320500.

[139] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[140] N. N. Vorob'ev. Consistent families of measures and their extensions. *Theory of Probability and Its Applications*, 7(2):147–163, 1962. doi: 10.1137/1107014.

[141] Peter Walters. *An Introduction to Ergodic Theory*, volume 79 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1982.

[142] James Wei and James C. W. Kuo. A lumping analysis in monomolecular reaction systems: Analysis of the exactly lumpable system. *Industrial & Engineering Chemistry Fundamentals*, 8(1):114–123, 1969. doi: 10.1021/i160029a019.

[143] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445.

[144] Aaron D. Wyner and Jacob Ziv. The rate-distortion function for source coding with side information at the decoder. *IEEE Transactions on Information Theory*, 22(1):1–10, 1976. doi: 10.1109/TIT.1976.1055508.

[145] Ram Zamir. A proof of the Fisher information inequality via a data processing argument. *IEEE Transactions on Information Theory*, 44(3):1246–1250, 1998. doi: 10.1109/18.669301.